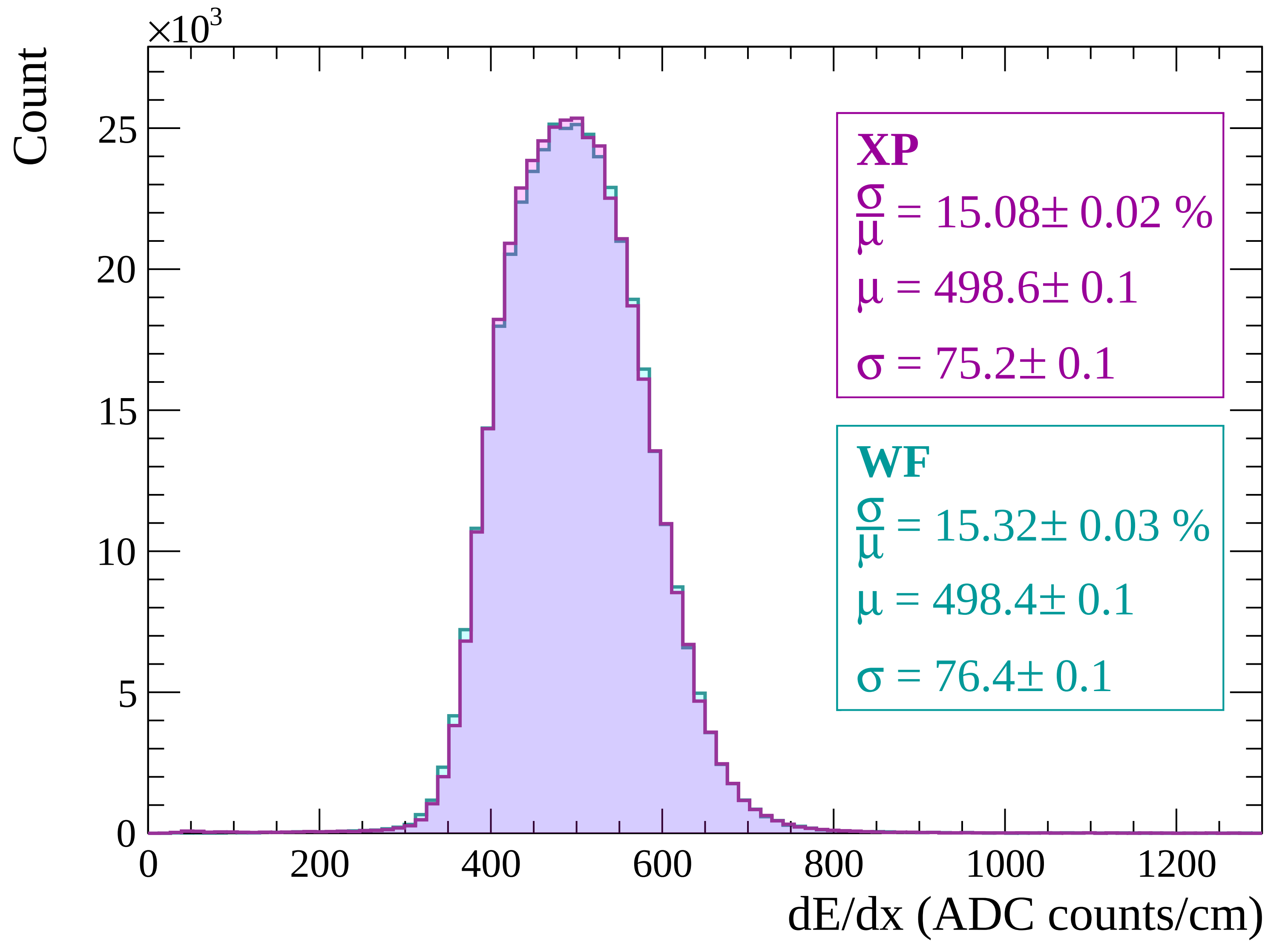
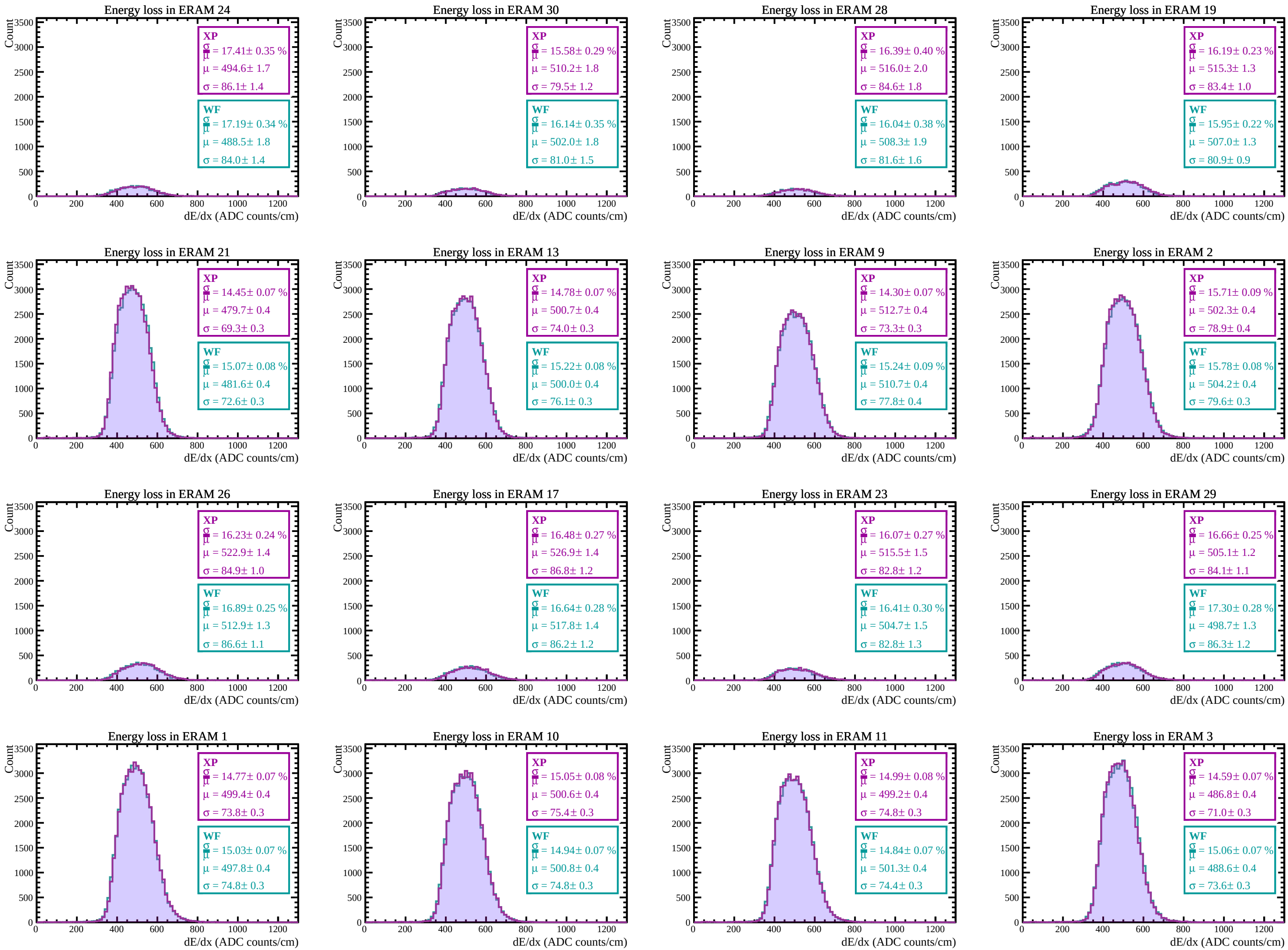
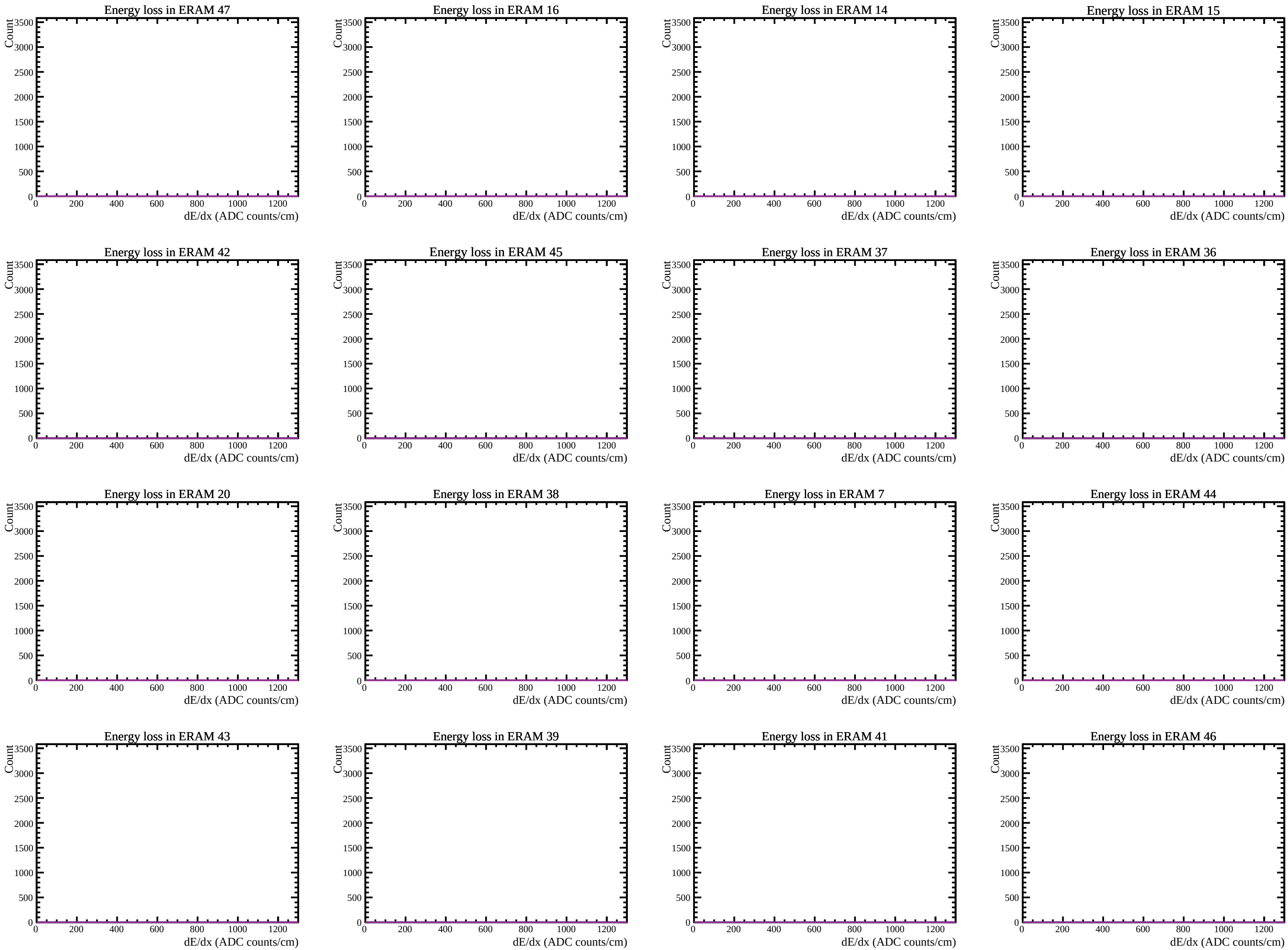
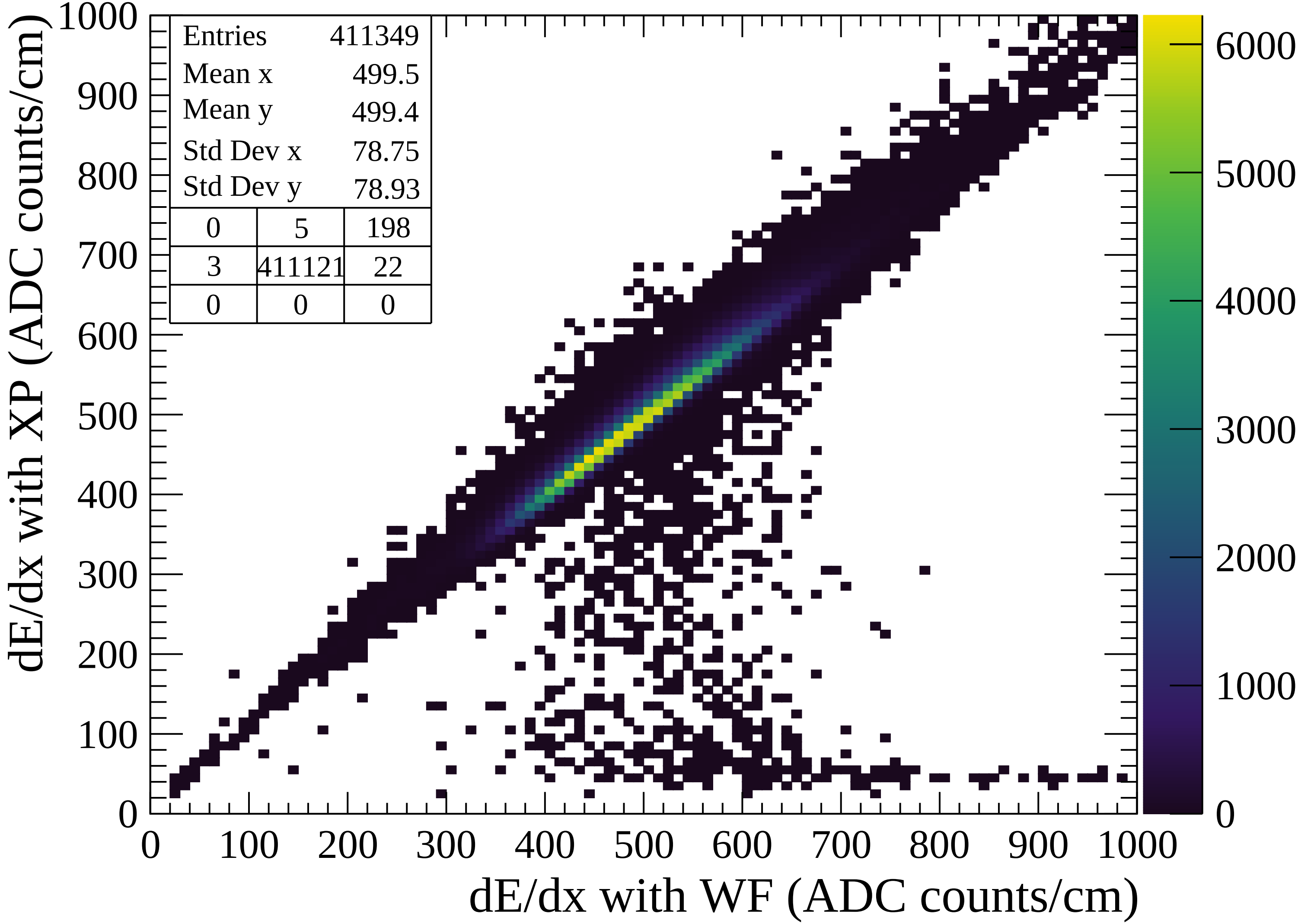

Count
×10^3
25
20
15
10
5
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 15.08± 0.02 %
μ = 498.6± 0.1
σ = 75.2± 0.1
WF
σ/μ = 15.32± 0.03 %
μ = 498.4± 0.1
σ = 76.4± 0.1

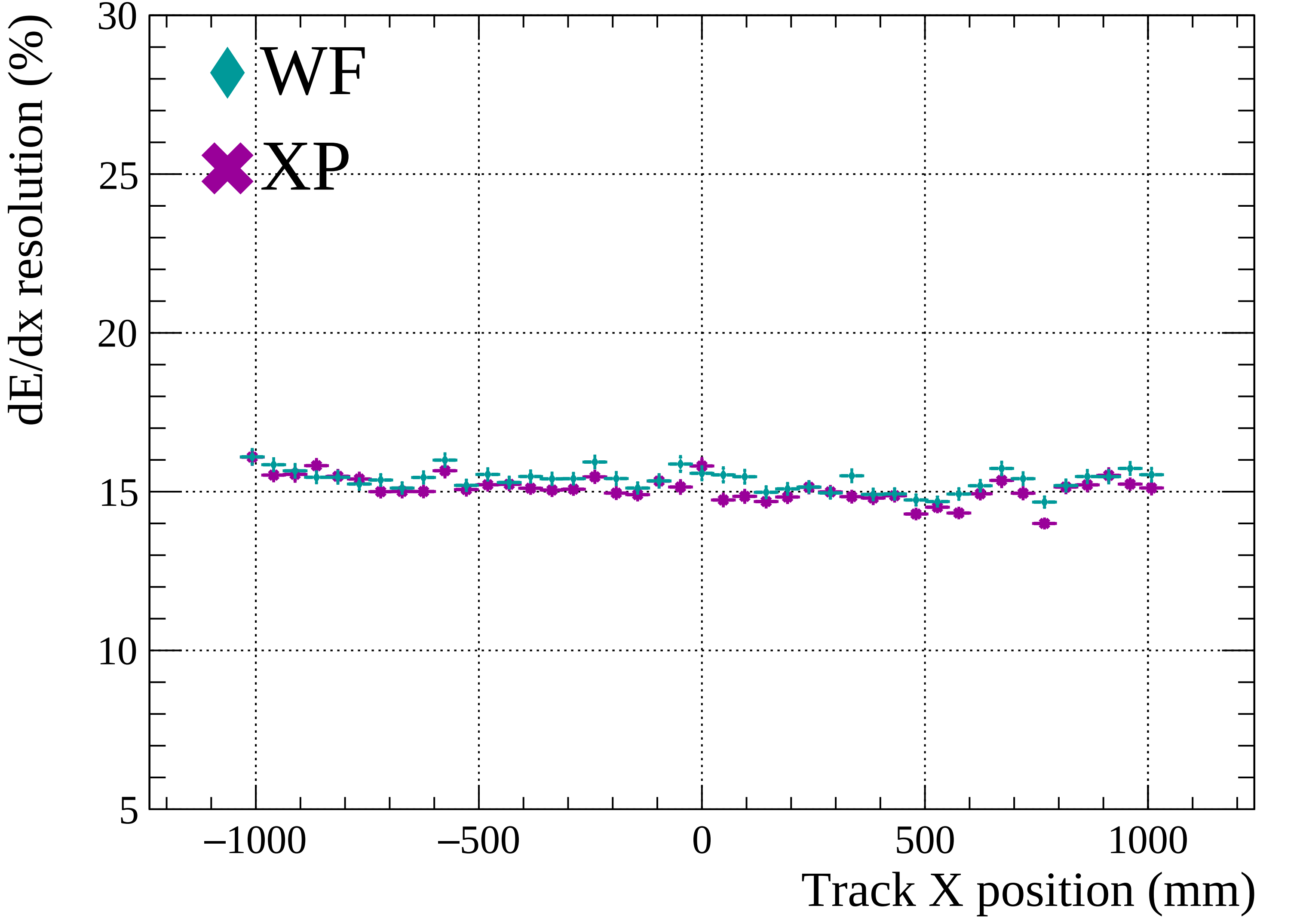


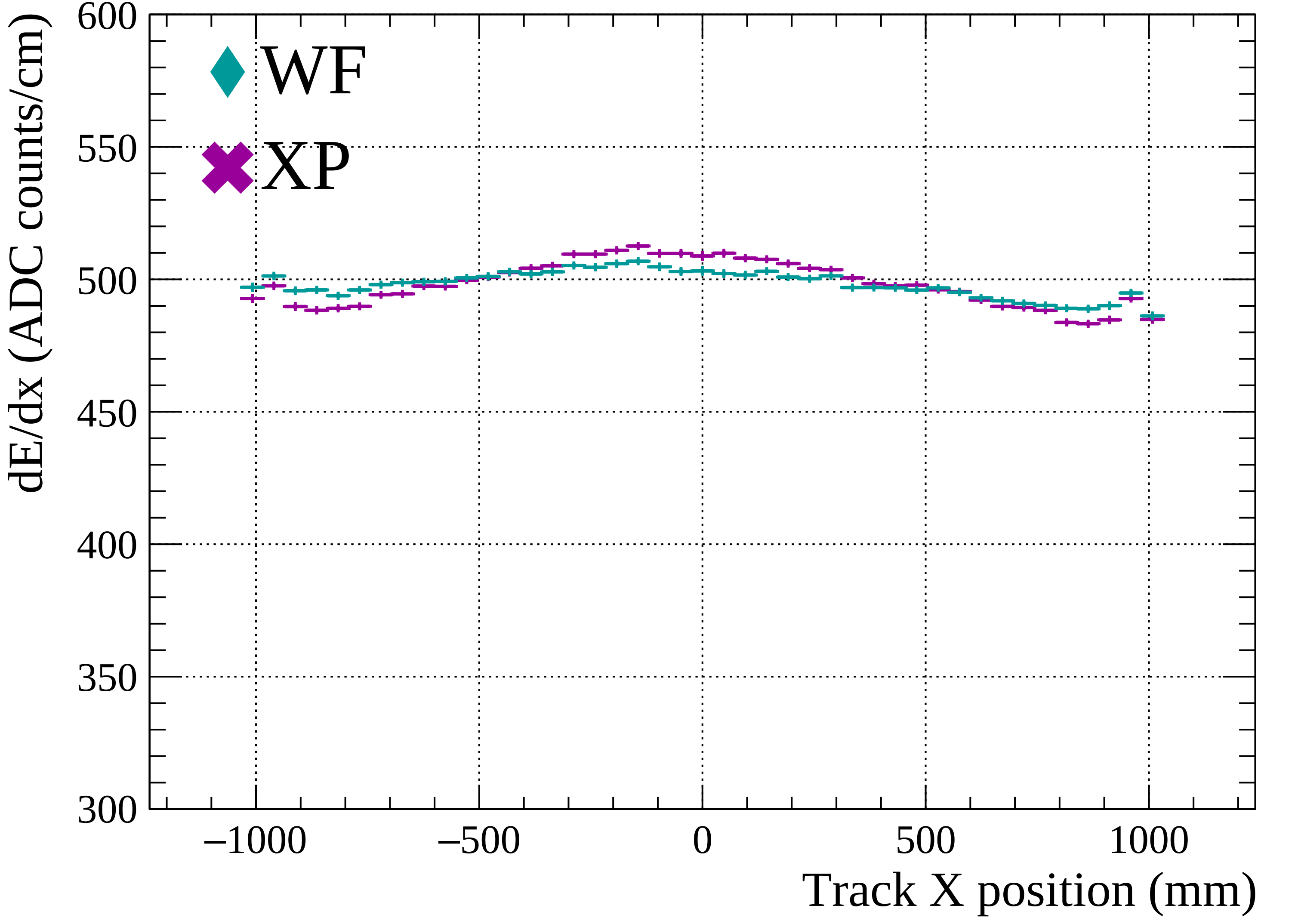
Energy loss in ERAM 24
XP σ/μ = 17.41± 0.35 % μ = 494.6± 1.7 σ = 86.1± 1.4
WF σ/μ = 17.19± 0.34 % μ = 488.5± 1.8 σ = 84.0± 1.4
Energy loss in ERAM 30
XP σ/μ = 15.58± 0.29 % μ = 510.2± 1.8 σ = 79.5± 1.2
WF σ/μ = 16.14± 0.35 % μ = 502.0± 1.8 σ = 81.0± 1.5
Energy loss in ERAM 28
XP σ/μ = 16.39± 0.40 % μ = 516.0± 2.0 σ = 84.6± 1.8
WF σ/μ = 16.04± 0.38 % μ = 508.3± 1.9 σ = 81.6± 1.6
Energy loss in ERAM 19
XP σ/μ = 16.19± 0.23 % μ = 515.3± 1.3 σ = 83.4± 1.0
WF σ/μ = 15.95± 0.22 % μ = 507.0± 1.3 σ = 80.9± 0.9
Energy loss in ERAM 21
XP σ/μ = 14.45± 0.07 % μ = 479.7± 0.4 σ = 69.3± 0.3
WF σ/μ = 15.07± 0.08 % μ = 481.6± 0.4 σ = 72.6± 0.3
Energy loss in ERAM 13
XP σ/μ = 14.78± 0.07 % μ = 500.7± 0.4 σ = 74.0± 0.3
WF σ/μ = 15.22± 0.08 % μ = 500.0± 0.4 σ = 76.1± 0.3
Energy loss in ERAM 9
XP σ/μ = 14.30± 0.07 % μ = 512.7± 0.4 σ = 73.3± 0.3
WF σ/μ = 15.24± 0.09 % μ = 510.7± 0.4 σ = 77.8± 0.4
Energy loss in ERAM 2
XP σ/μ = 15.71± 0.09 % μ = 502.3± 0.4 σ = 78.9± 0.4
WF σ/μ = 15.78± 0.08 % μ = 504.2± 0.4 σ = 79.6± 0.3
Energy loss in ERAM 26
XP σ/μ = 16.23± 0.24 % μ = 522.9± 1.4 σ = 84.9± 1.0
WF σ/μ = 16.89± 0.25 % μ = 512.9± 1.3 σ = 86.6± 1.1
Energy loss in ERAM 17
XP σ/μ = 16.48± 0.27 % μ = 526.9± 1.4 σ = 86.8± 1.2
WF σ/μ = 16.64± 0.28 % μ = 517.8± 1.4 σ = 86.2± 1.2
Energy loss in ERAM 23
XP σ/μ = 16.07± 0.27 % μ = 515.5± 1.5 σ = 82.8± 1.2
WF σ/μ = 16.41± 0.30 % μ = 504.7± 1.5 σ = 82.8± 1.3
Energy loss in ERAM 29
XP σ/μ = 16.66± 0.25 % μ = 505.1± 1.2 σ = 84.1± 1.1
WF σ/μ = 17.30± 0.28 % μ = 498.7± 1.3 σ = 86.3± 1.2
Energy loss in ERAM 1
XP σ/μ = 14.77± 0.07 % μ = 499.4± 0.4 σ = 73.8± 0.3
WF σ/μ = 15.03± 0.07 % μ = 497.8± 0.4 σ = 74.8± 0.3
Energy loss in ERAM 10
XP σ/μ = 15.05± 0.08 % μ = 500.6± 0.4 σ = 75.4± 0.3
WF σ/μ = 14.94± 0.07 % μ = 500.8± 0.4 σ = 74.8± 0.3
Energy loss in ERAM 11
XP σ/μ = 14.99± 0.08 % μ = 499.2± 0.4 σ = 74.8± 0.3
WF σ/μ = 14.84± 0.07 % μ = 501.3± 0.4 σ = 74.4± 0.3
Energy loss in ERAM 3
XP σ/μ = 14.59± 0.07 % μ = 486.8± 0.4 σ = 71.0± 0.3
WF σ/μ = 15.06± 0.07 % μ = 488.6± 0.4 σ = 73.6± 0.3
Count
dE/dx (ADC counts/cm)

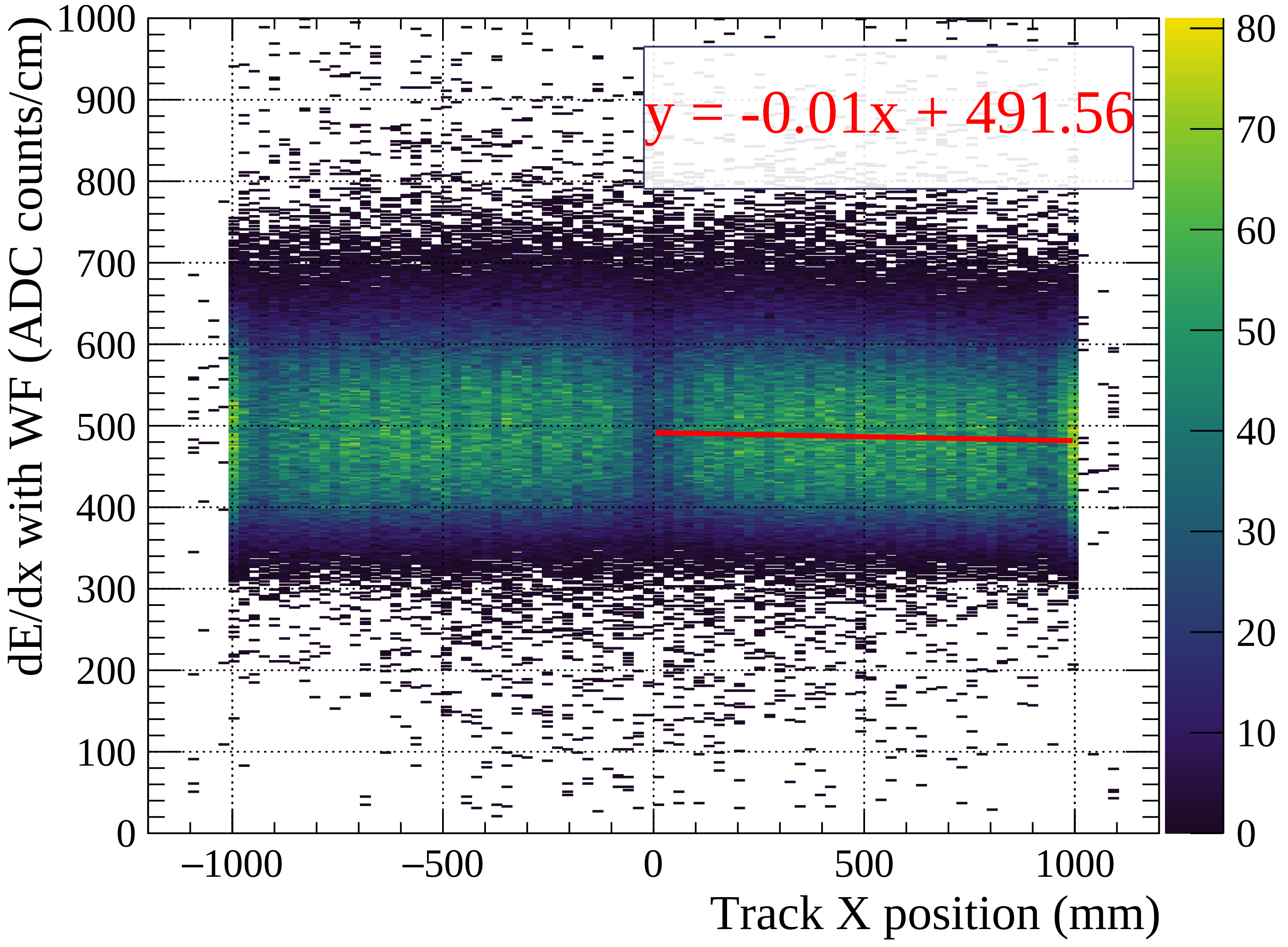

Energy loss in ERAM 47
Energy loss in ERAM 16
Energy loss in ERAM 14
Energy loss in ERAM 15
Energy loss in ERAM 42
Energy loss in ERAM 45
Energy loss in ERAM 37
Energy loss in ERAM 36
Energy loss in ERAM 20
Energy loss in ERAM 38
Energy loss in ERAM 7
Energy loss in ERAM 44
Energy loss in ERAM 43
Energy loss in ERAM 39
Energy loss in ERAM 41
Energy loss in ERAM 46
Count
dE/dx (ADC counts/cm)



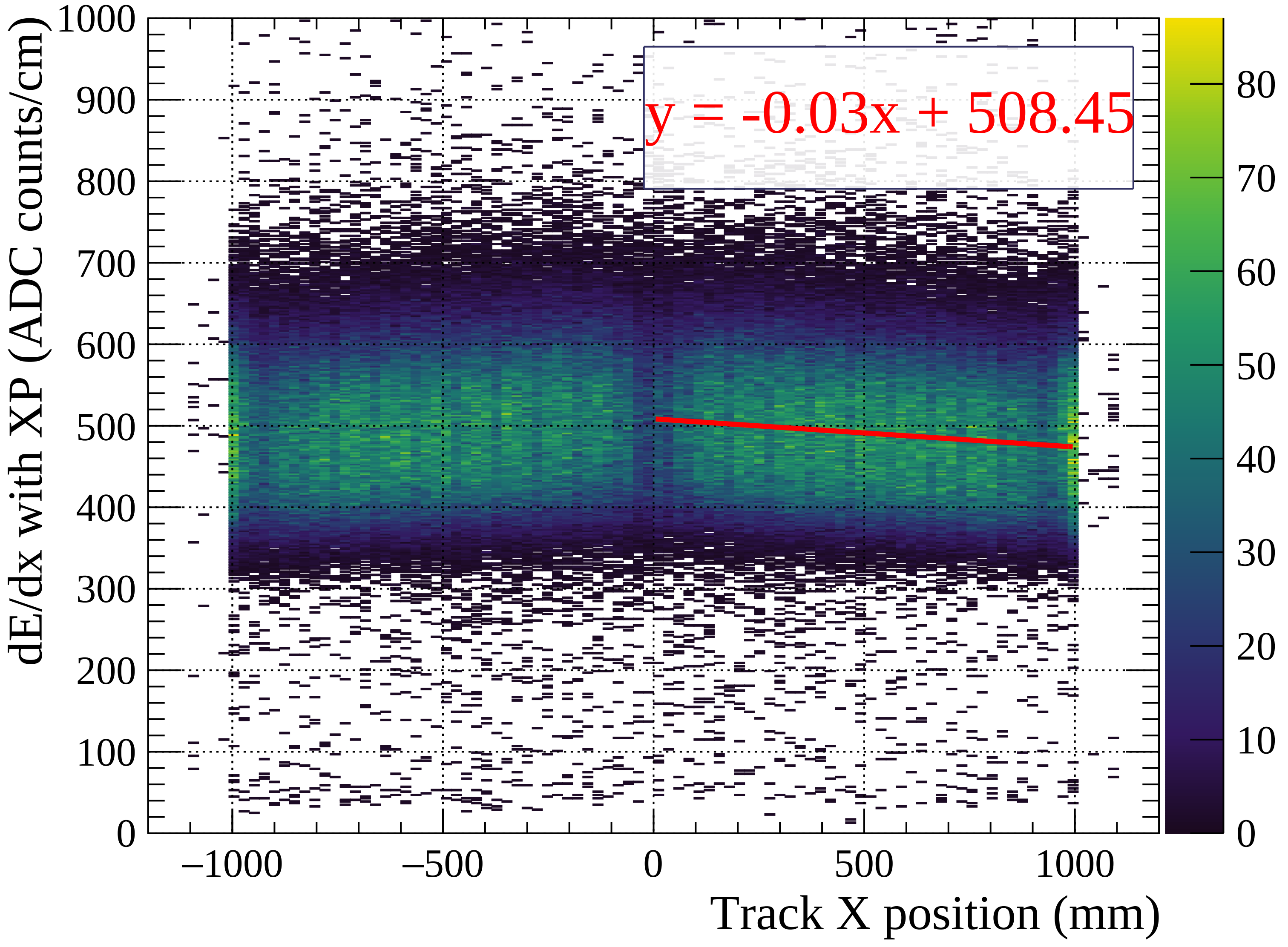
Entries 411349
Mean x 499.5
Mean y 499.4
Std Dev x 78.75
Std Dev y 78.93
0 5 198
3 411121 22
0 0 0
dE/dx with XP (ADC counts/cm)
dE/dx with WF (ADC counts/cm)
0
100
200
300
400
500
600
700
800
900
1000
1000
2000
3000
4000
5000
6000

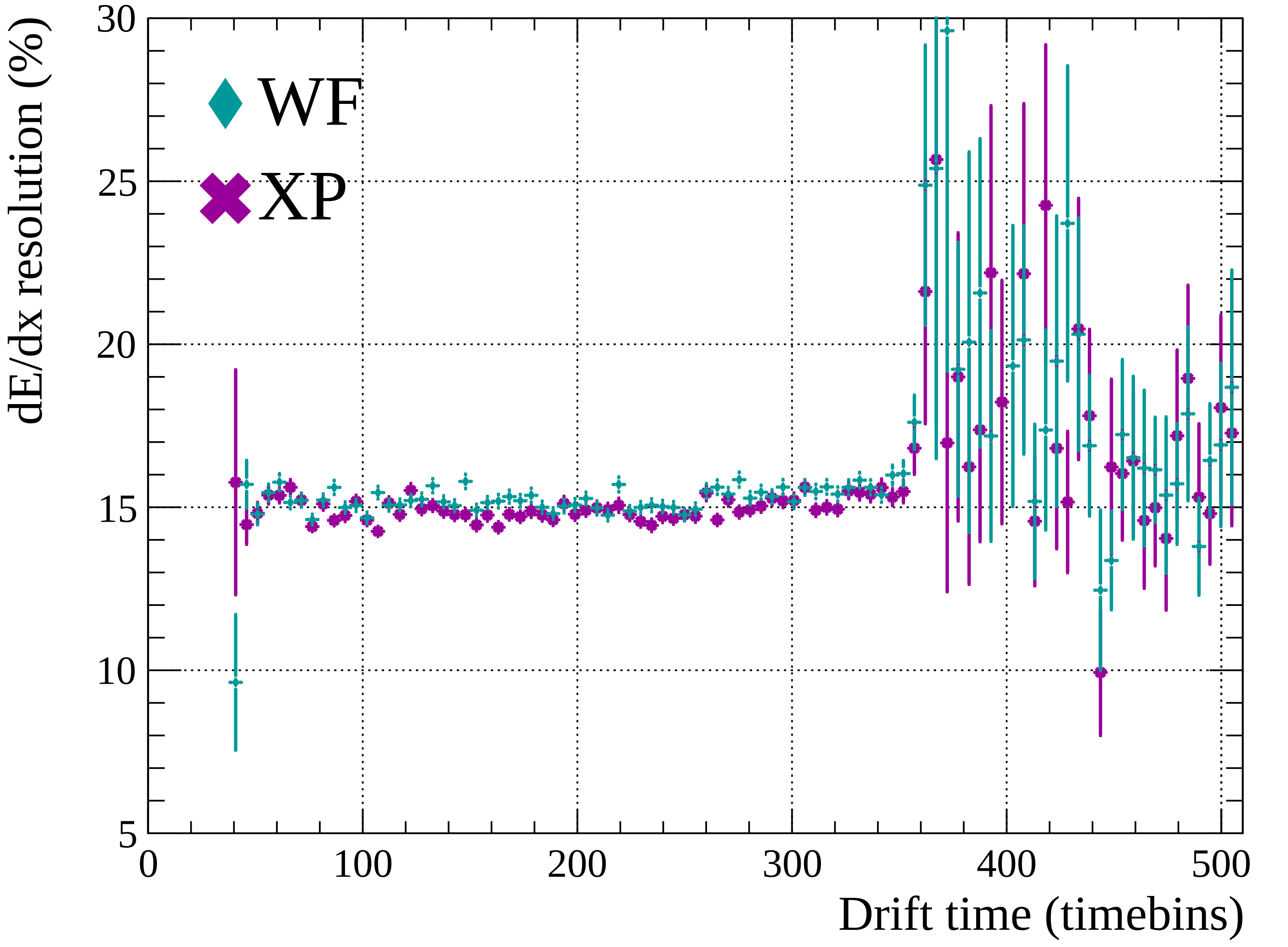

WF
XP
dE/dx resolution (%)
Track X position (mm)
30
25
20
15
10
5
−1000
−500
0
500
1000


WF
XP
dE/dx (ADC counts/cm)
Track X position (mm)
600
550
500
450
400
350
300
−1000
−500
0
500
1000

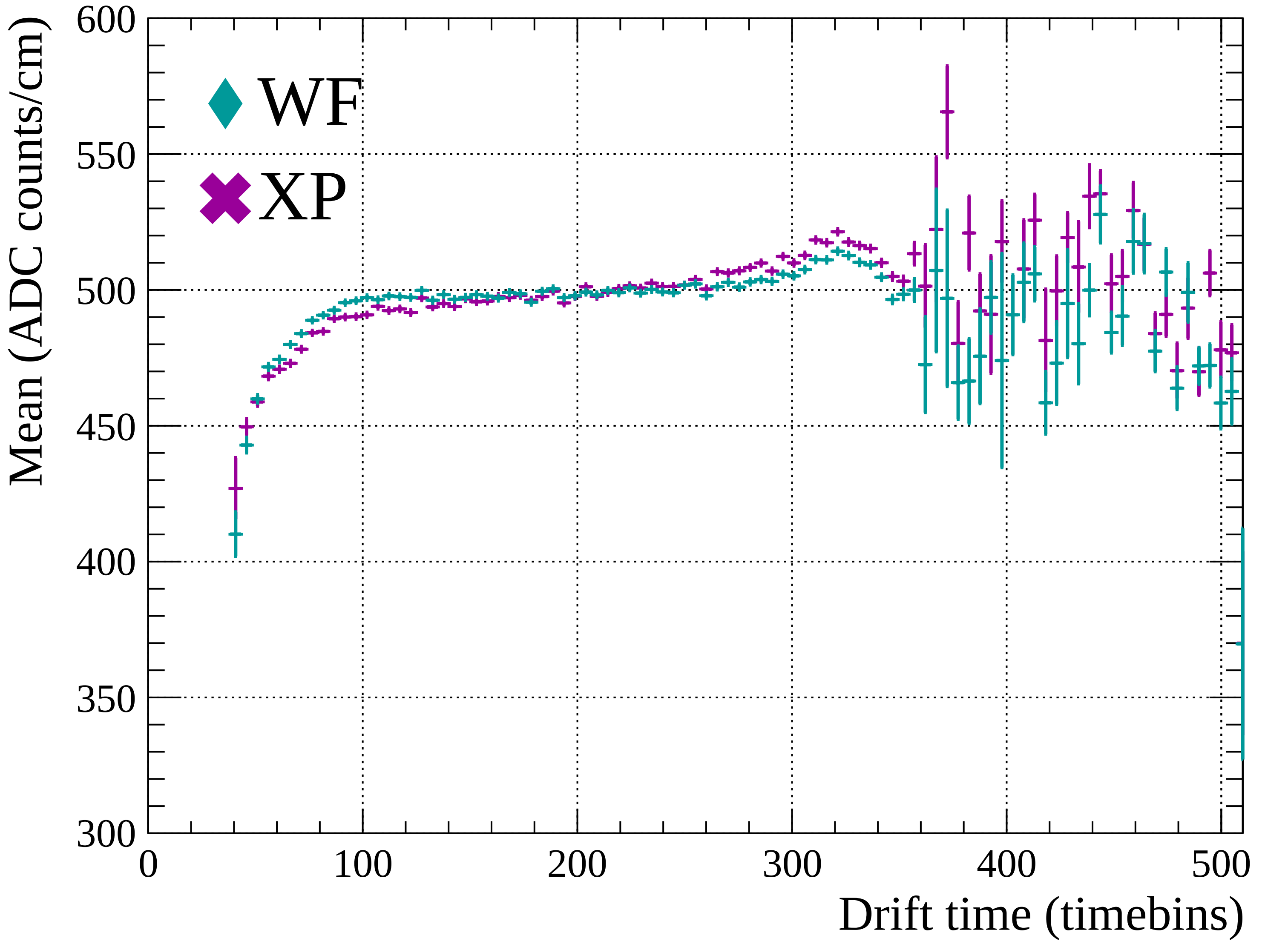


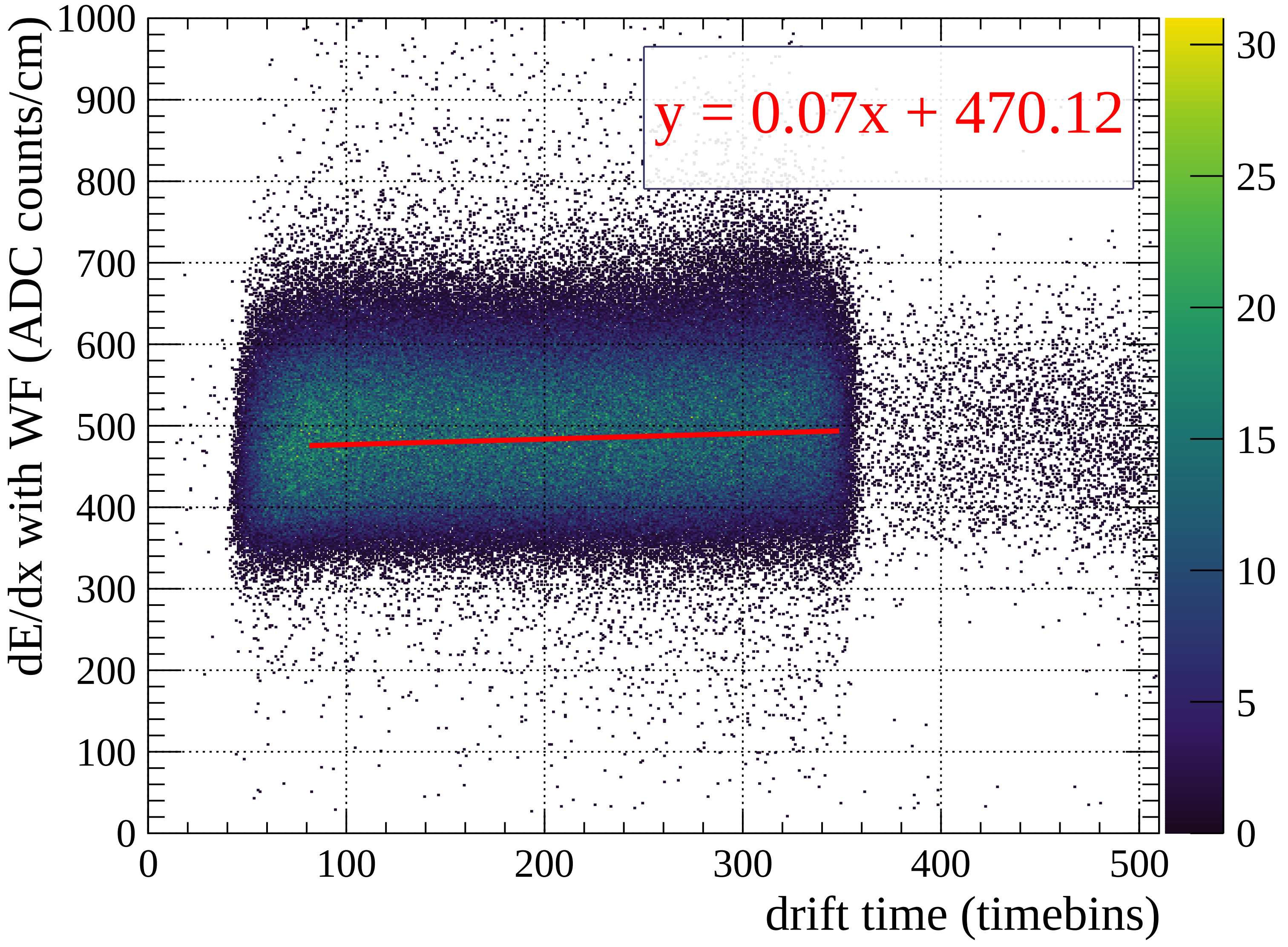
y = -0.01x + 491.56
dE/dx with WF (ADC counts/cm)
Track X position (mm)
0
100
200
300
400
500
600
700
800
900
1000
−1000
−500
0
500
1000
0
10
20
30
40
50
60
70
80


y = -0.03x + 508.45
dE/dx with XP (ADC counts/cm)
Track X position (mm)
0
100
200
300
400
500
600
700
800
900
1000
−1000
−500
0
500
1000
0
10
20
30
40
50
60
70
80

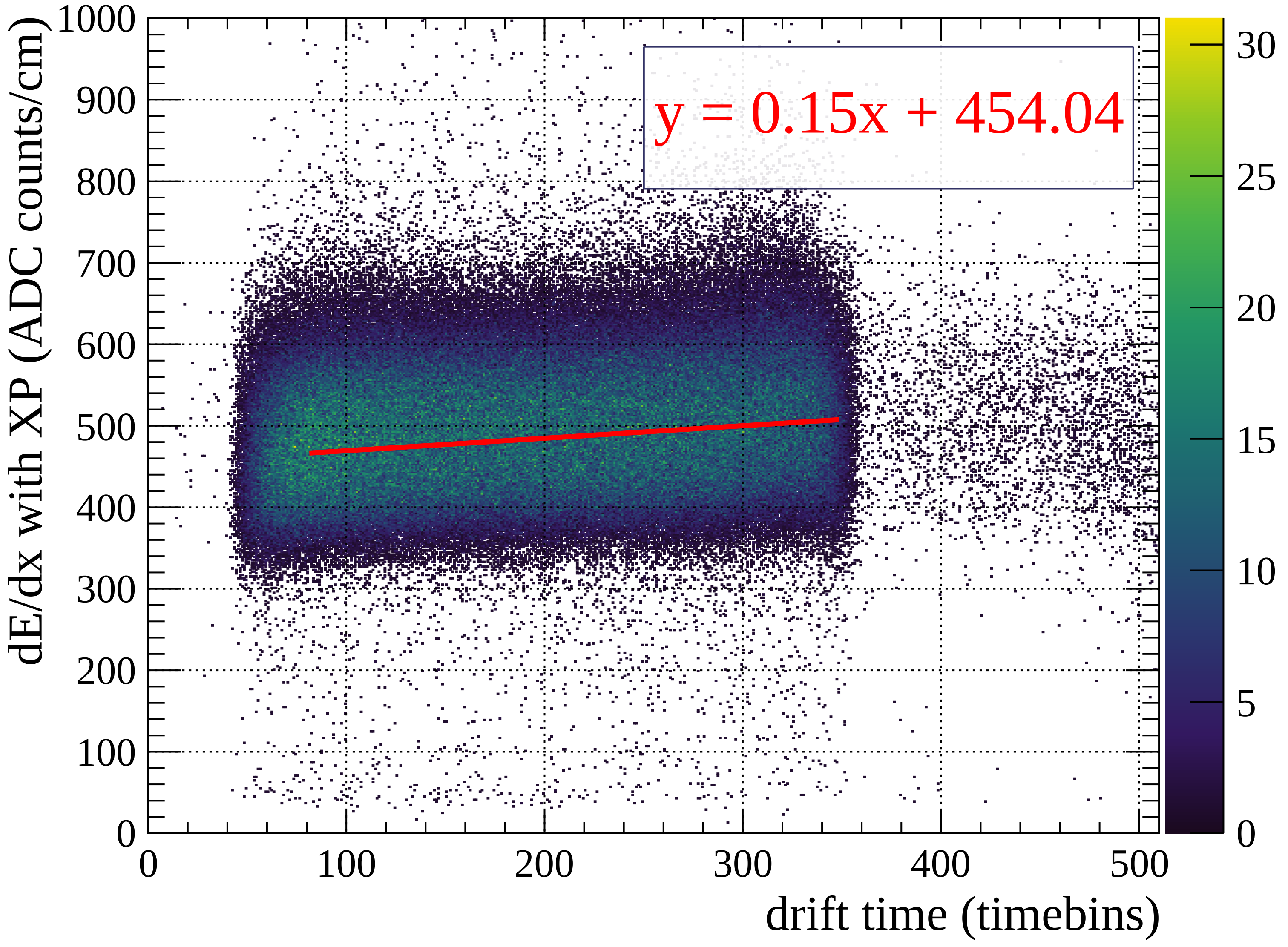


WF
XP
dE/dx resolution (%)
Drift time (timebins)
5
10
15
20
25
30
0
100
200
300
400
500

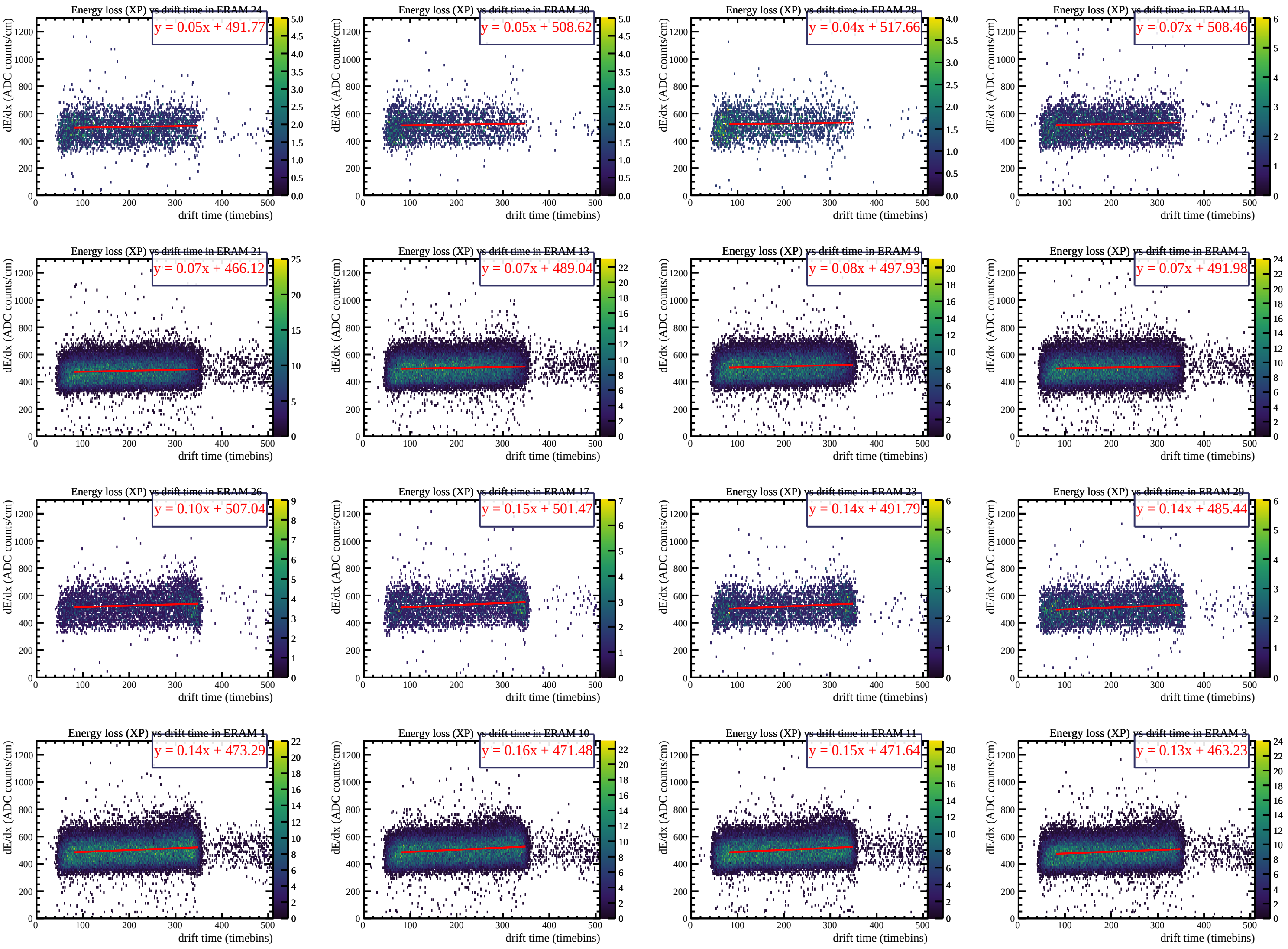

Mean (ADC counts/cm)
Drift time (timebins)
WF
XP
600
550
500
450
400
350
300
0
100
200
300
400
500

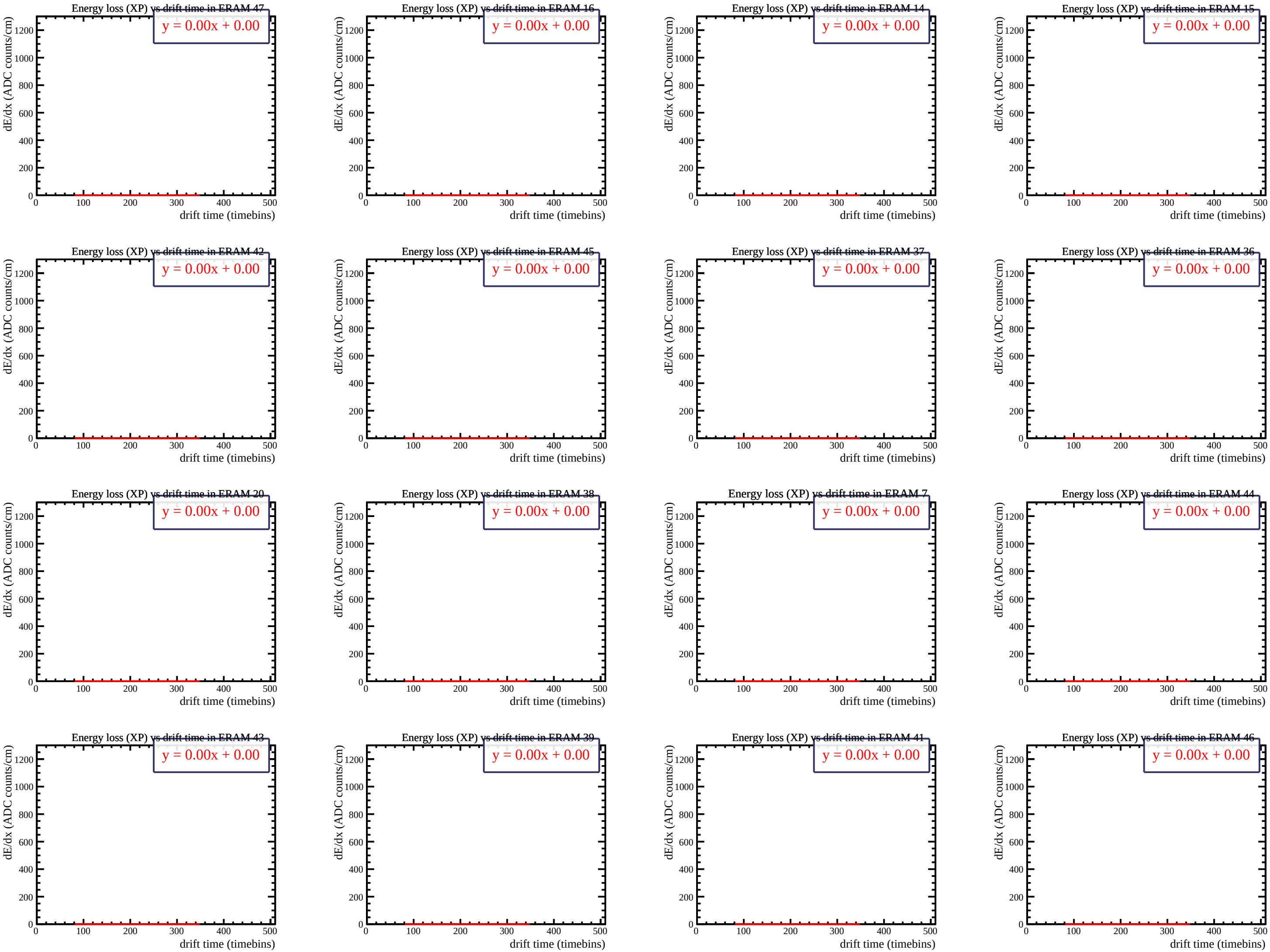


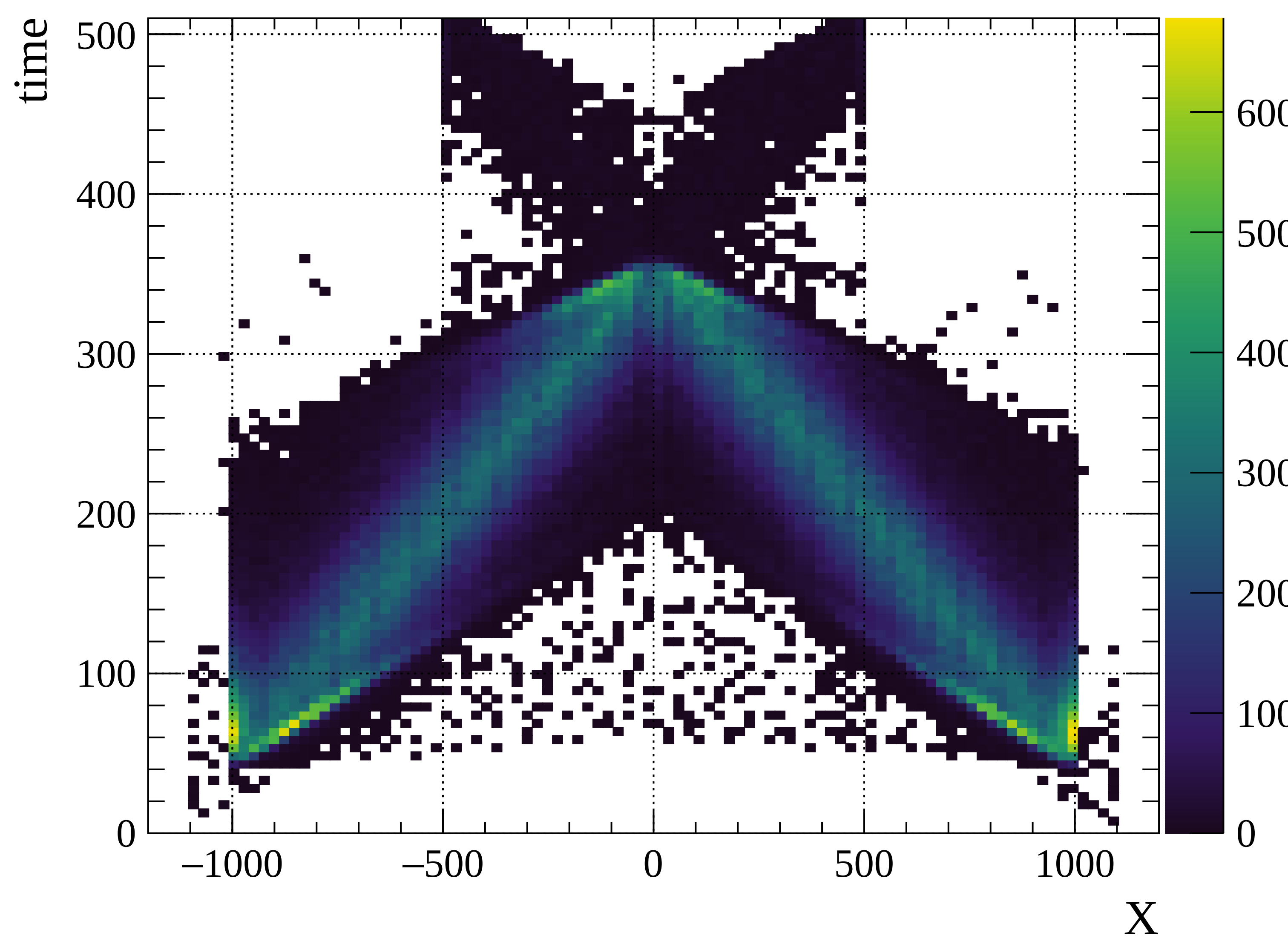
y = 0.07x + 470.12
dE/dx with WF (ADC counts/cm)
drift time (timebins)
0
100
200
300
400
500
600
700
800
900
1000
0
100
200
300
400
500
0
5
10
15
20
25
30

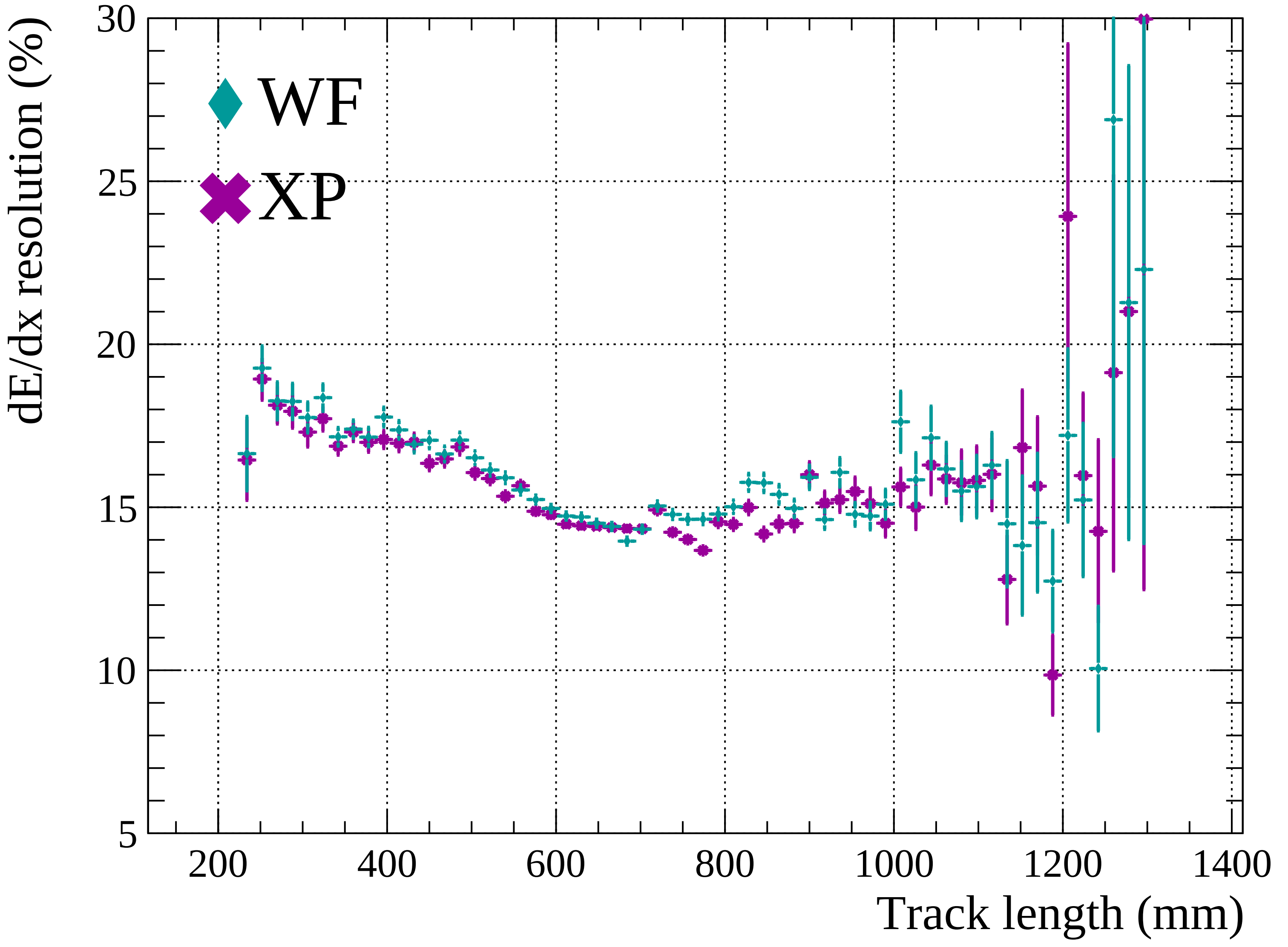


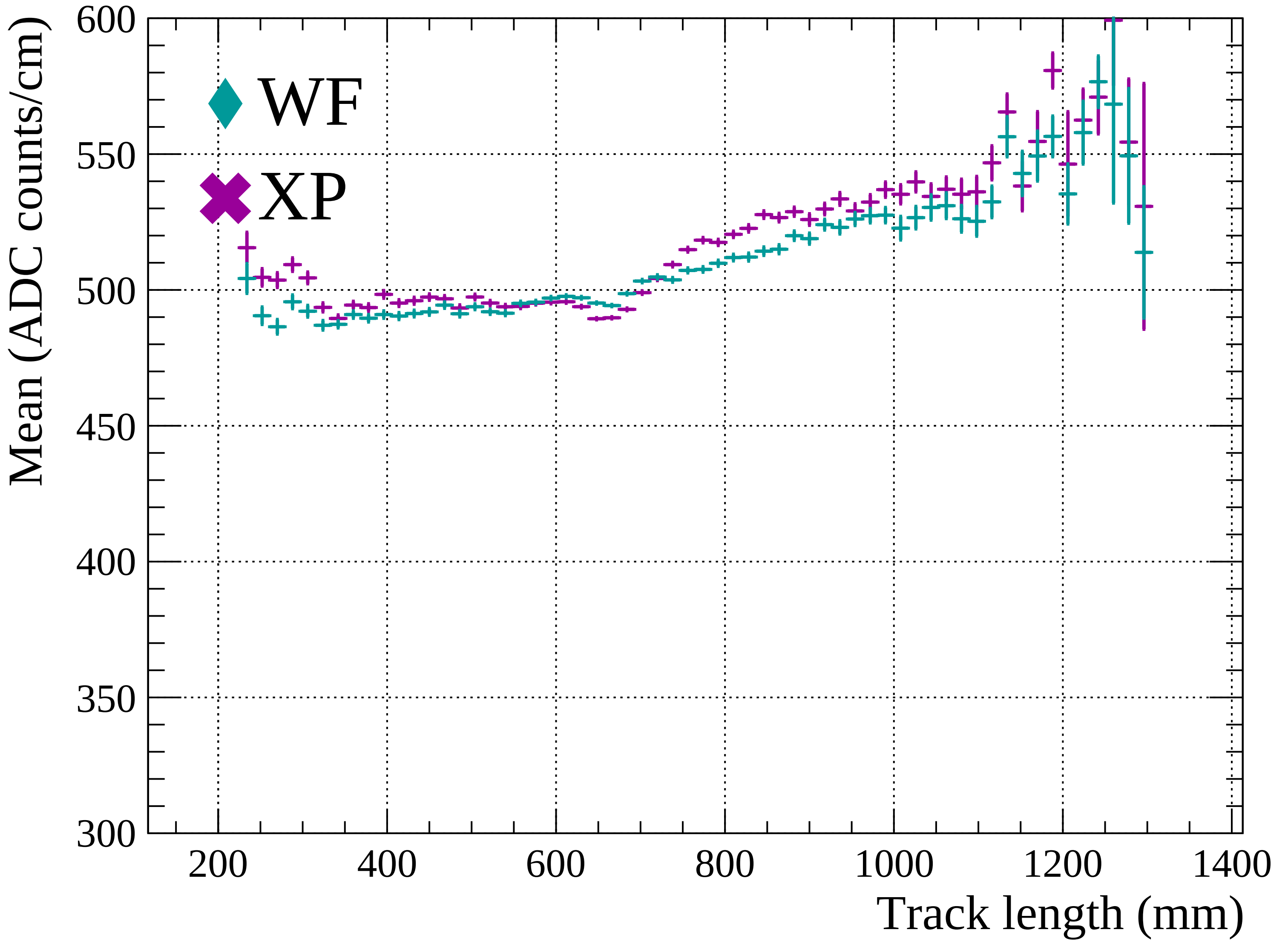
y = 0.15x + 454.04
dE/dx with XP (ADC counts/cm)
drift time (timebins)
0
100
200
300
400
500
600
700
800
900
1000
0
100
200
300
400
500
0
5
10
15
20
25
30

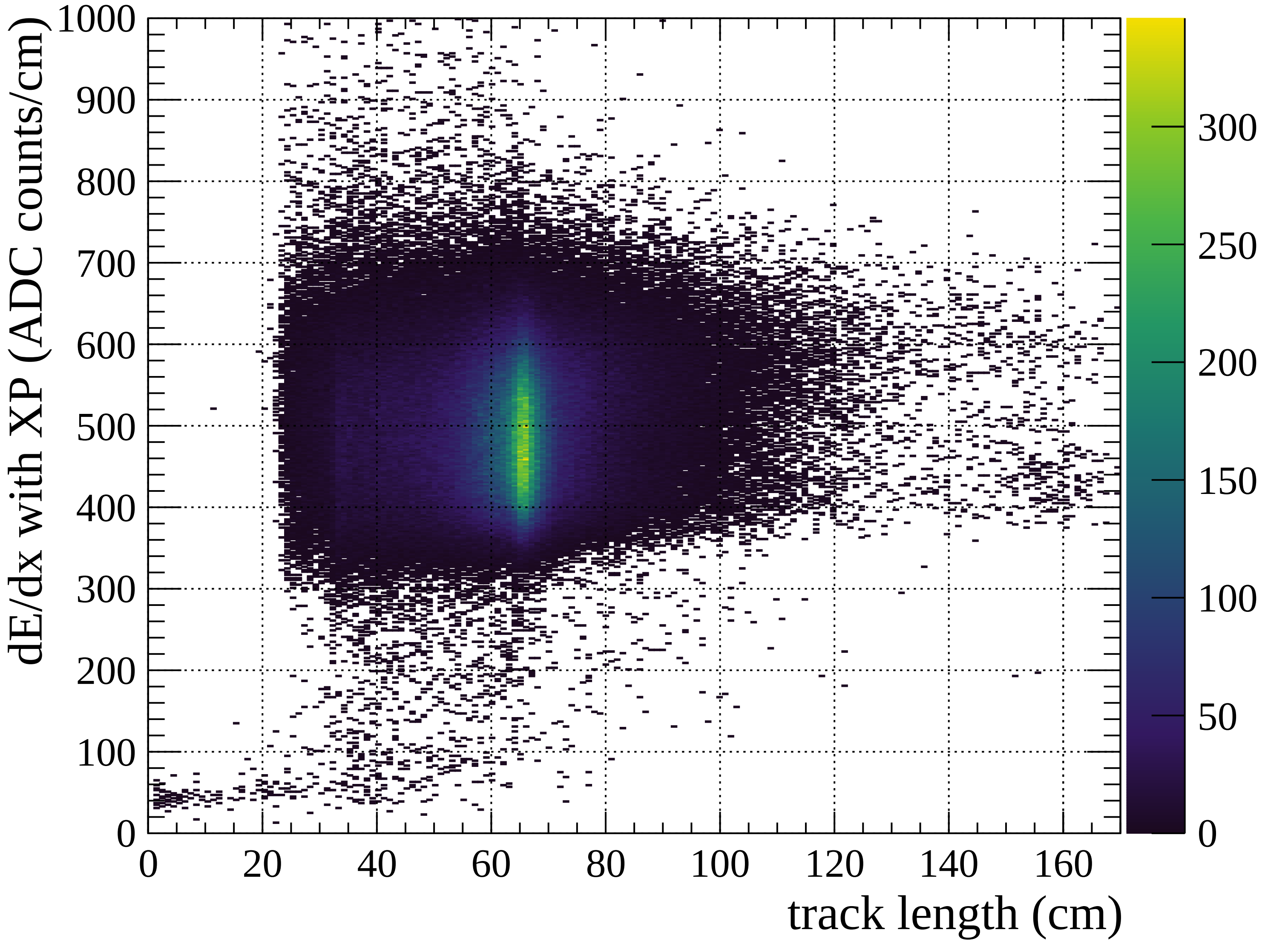

Energy loss (XP) vs drift time in ERAM 24
y = 0.05x + 491.77
Energy loss (XP) vs drift time in ERAM 30
y = 0.05x + 508.62
Energy loss (XP) vs drift time in ERAM 28
y = 0.04x + 517.66
Energy loss (XP) vs drift time in ERAM 19
y = 0.07x + 508.46
Energy loss (XP) vs drift time in ERAM 21
y = 0.07x + 466.12
Energy loss (XP) vs drift time in ERAM 13
y = 0.07x + 489.04
Energy loss (XP) vs drift time in ERAM 9
y = 0.08x + 497.93
Energy loss (XP) vs drift time in ERAM 2
y = 0.07x + 491.98
Energy loss (XP) vs drift time in ERAM 26
y = 0.10x + 507.04
Energy loss (XP) vs drift time in ERAM 17
y = 0.15x + 501.47
Energy loss (XP) vs drift time in ERAM 23
y = 0.14x + 491.79
Energy loss (XP) vs drift time in ERAM 29
y = 0.14x + 485.44
Energy loss (XP) vs drift time in ERAM 1
y = 0.14x + 473.29
Energy loss (XP) vs drift time in ERAM 10
y = 0.16x + 471.48
Energy loss (XP) vs drift time in ERAM 11
y = 0.15x + 471.64
Energy loss (XP) vs drift time in ERAM 3
y = 0.13x + 463.23
dE/dx (ADC counts/cm)
drift time (timebins)

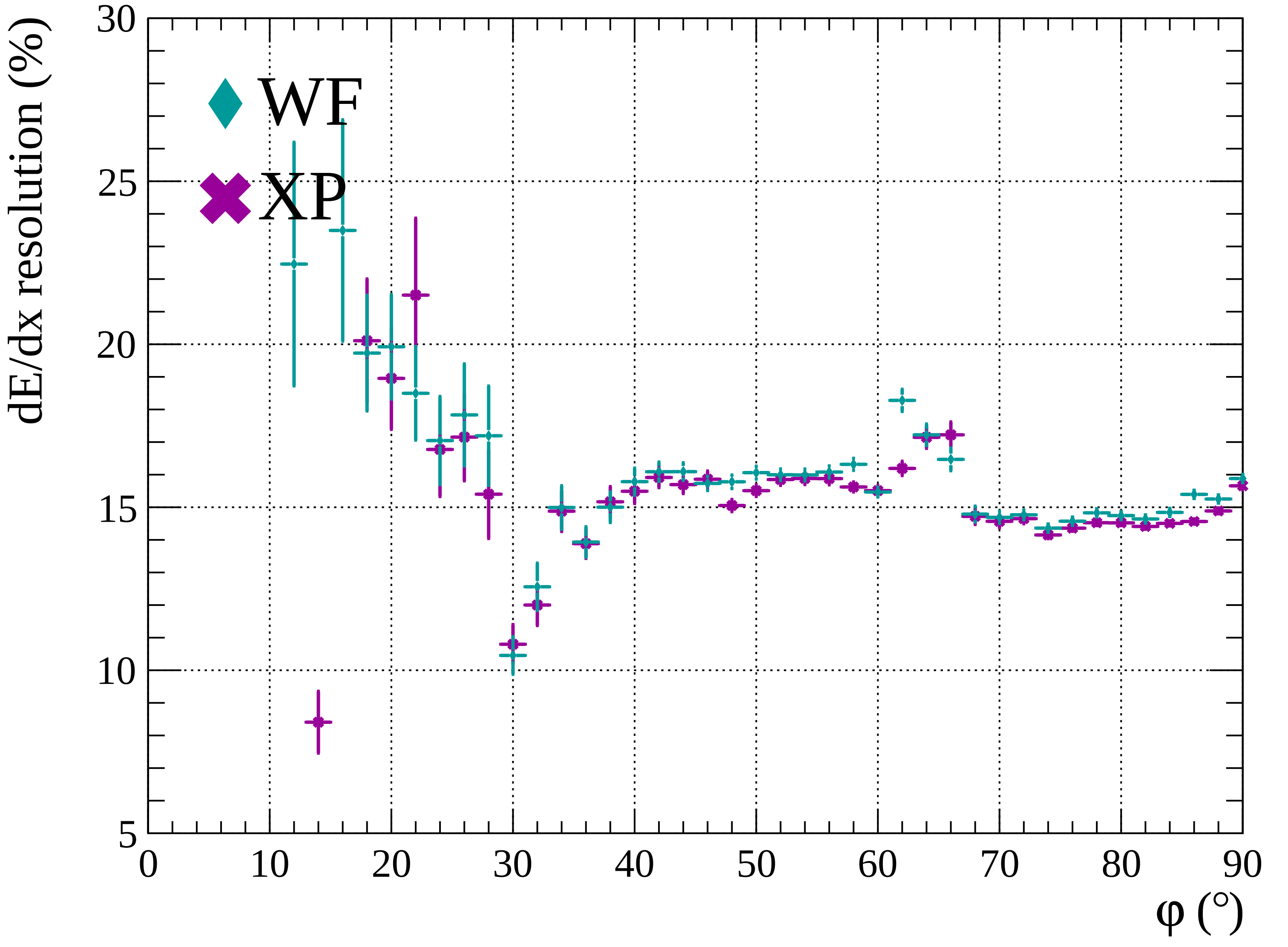

Energy loss (XP) vs drift time in ERAM 47
Energy loss (XP) vs drift time in ERAM 16
Energy loss (XP) vs drift time in ERAM 14
Energy loss (XP) vs drift time in ERAM 15
Energy loss (XP) vs drift time in ERAM 42
Energy loss (XP) vs drift time in ERAM 45
Energy loss (XP) vs drift time in ERAM 37
Energy loss (XP) vs drift time in ERAM 36
Energy loss (XP) vs drift time in ERAM 20
Energy loss (XP) vs drift time in ERAM 38
Energy loss (XP) vs drift time in ERAM 7
Energy loss (XP) vs drift time in ERAM 44
Energy loss (XP) vs drift time in ERAM 43
Energy loss (XP) vs drift time in ERAM 39
Energy loss (XP) vs drift time in ERAM 41
Energy loss (XP) vs drift time in ERAM 46
y = 0.00x + 0.00
dE/dx (ADC counts/cm)
drift time (timebins)

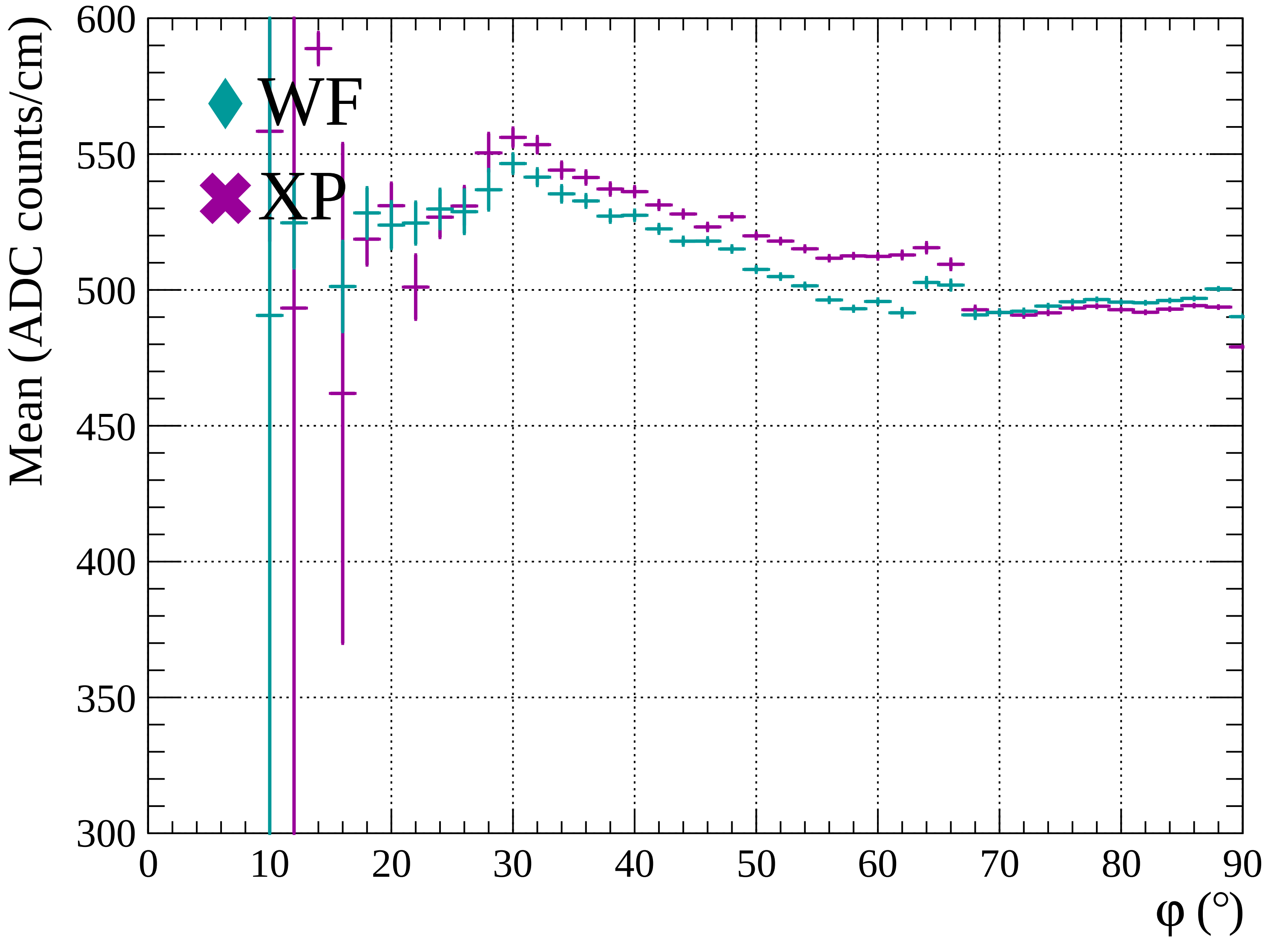

time
500
400
300
200
100
0
−1000
−500
0
500
1000
X
600
500
400
300
200
100
0


WF
XP
dE/dx resolution (%)
Track length (mm)
5
10
15
20
25
30
200
400
600
800
1000
1200
1400

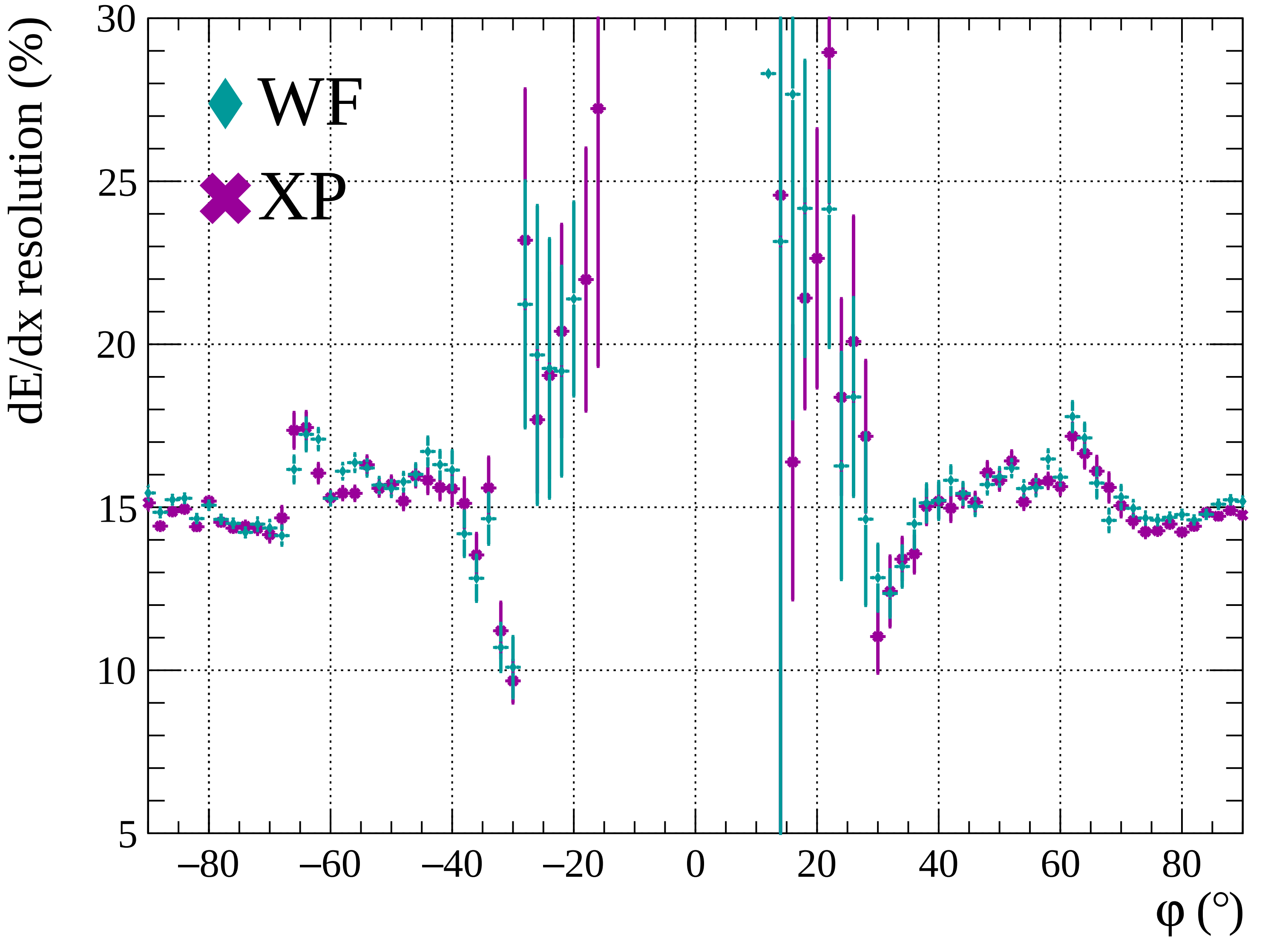

WF
XP
Mean (ADC counts/cm)
Track length (mm)
600
550
500
450
400
350
300
200
400
600
800
1000
1200
1400


dE/dx with XP (ADC counts/cm)
track length (cm)
0
100
200
300
400
500
600
700
800
900
1000
0
20
40
60
80
100
120
140
160
0
50
100
150
200
250
300

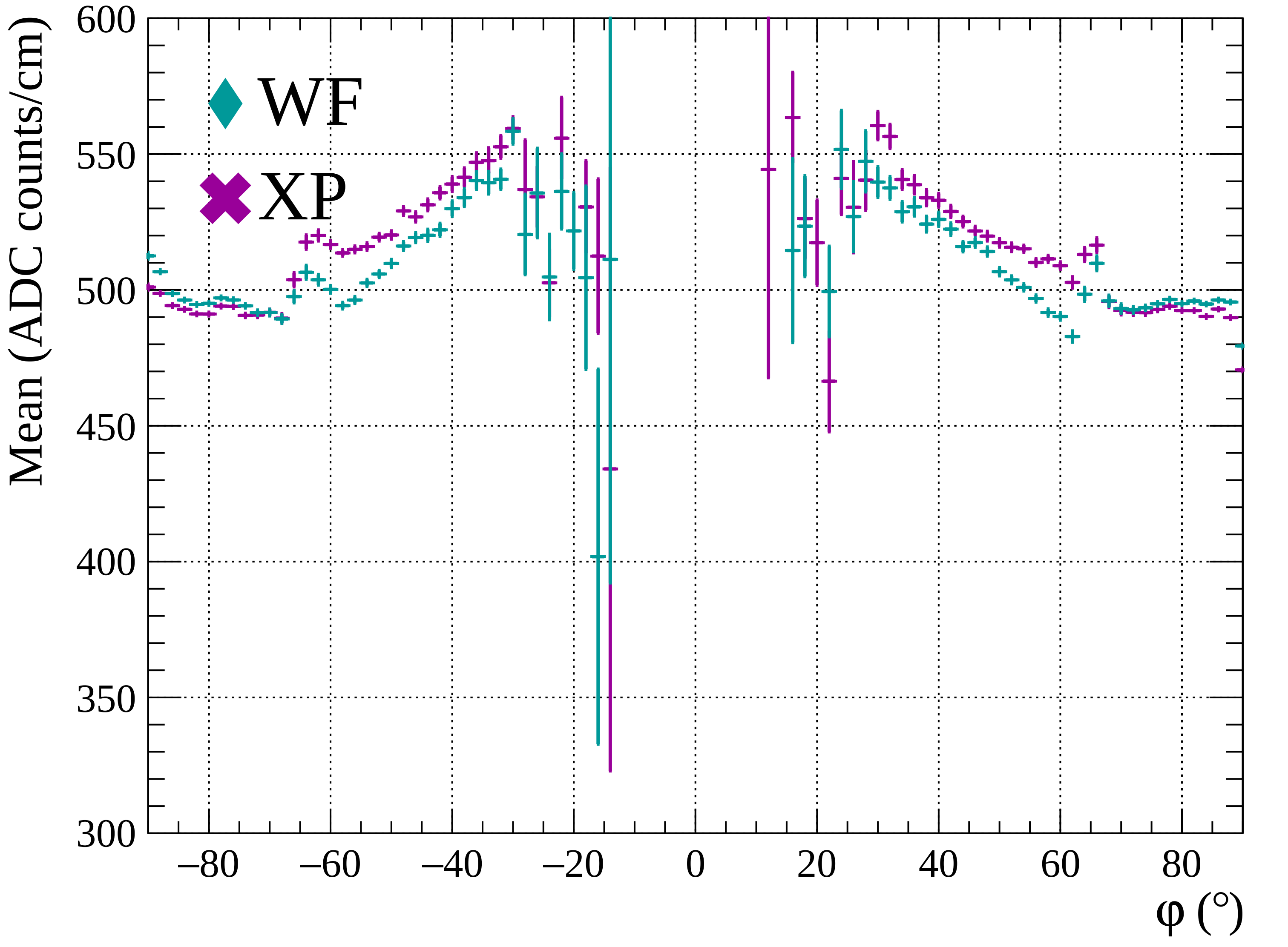

WF
XP
dE/dx resolution (%)
φ (°)
0
10
20
30
40
50
60
70
80
90
5
10
15
20
25
30



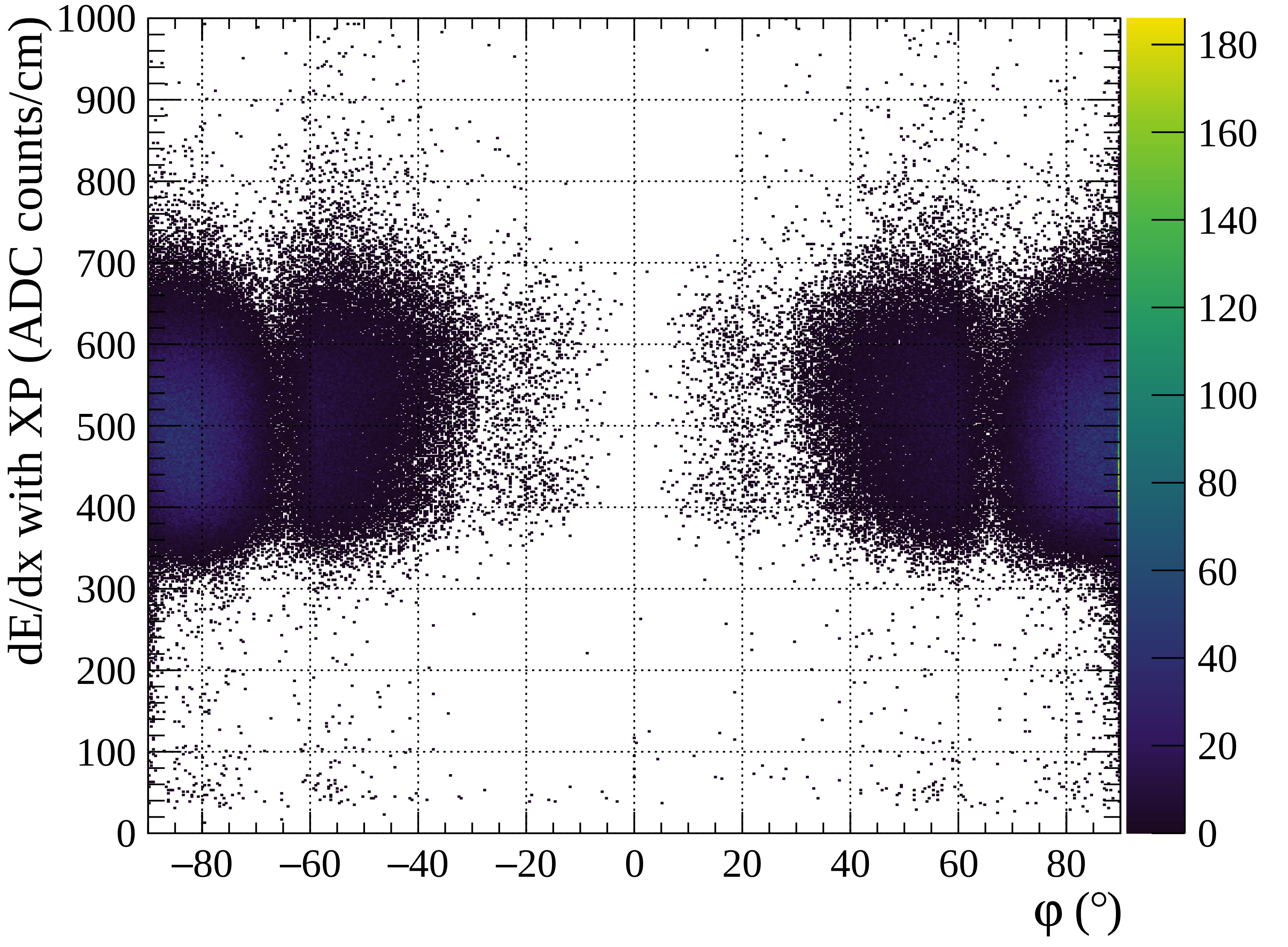
Mean (ADC counts/cm)
WF
XP
600
550
500
450
400
350
300
0
10
20
30
40
50
60
70
80
90
φ (°)

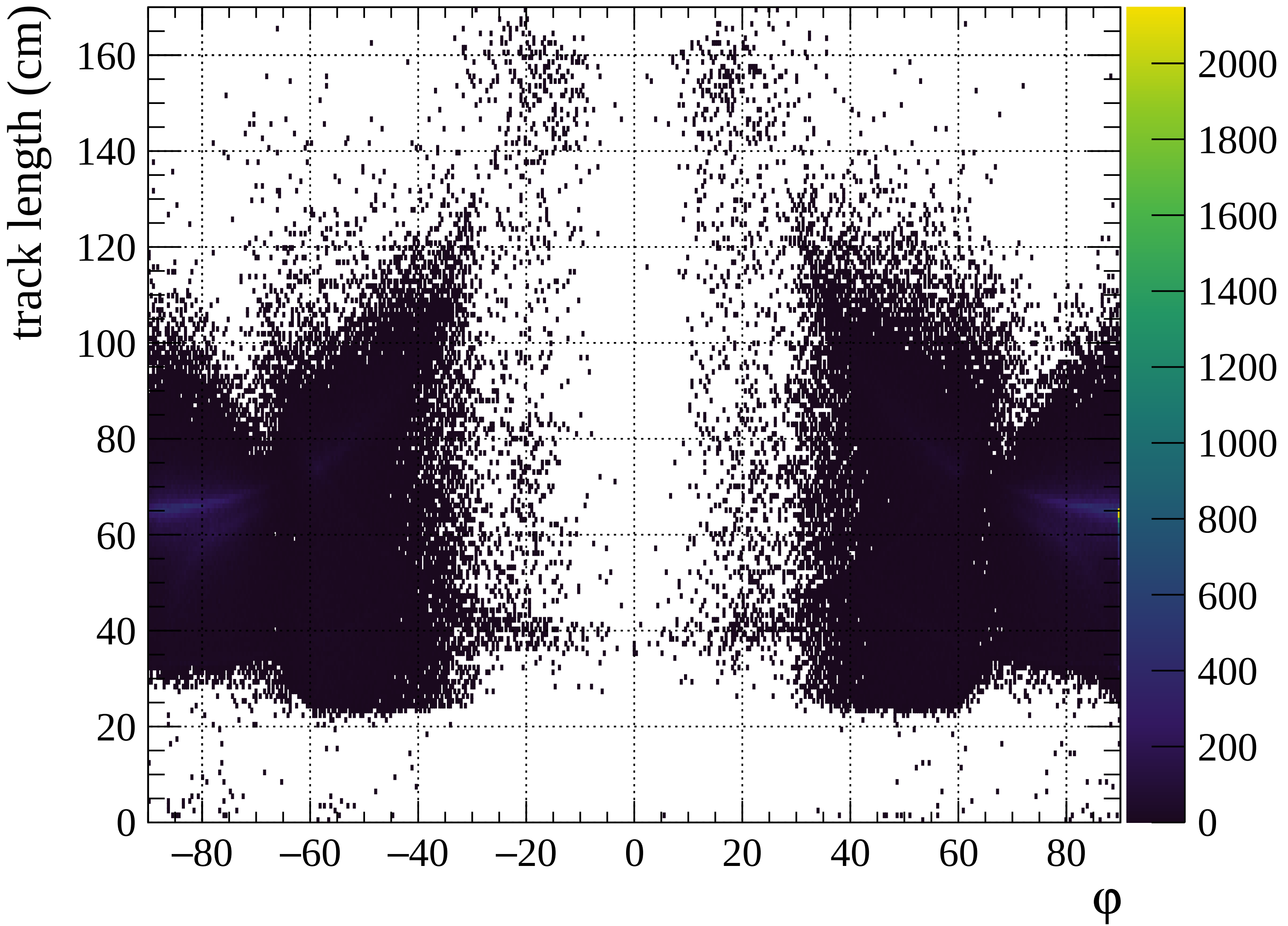


dE/dx resolution (%)
30
25
20
15
10
5
−80
−60
−40
−20
0
20
40
60
80
φ (°)
WF
XP

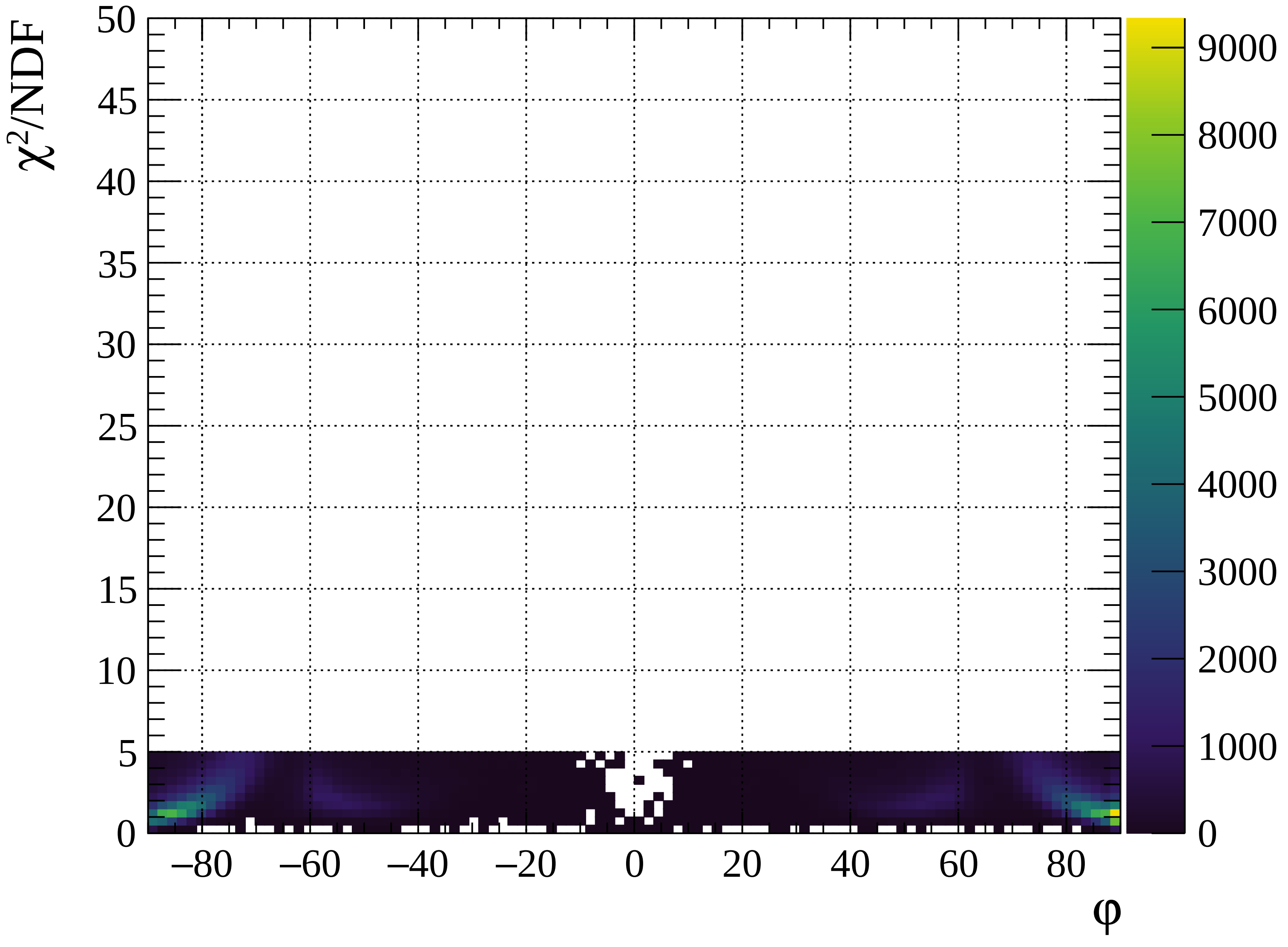


WF
XP
Mean (ADC counts/cm)
φ (°)
600
550
500
450
400
350
300
−80
−60
−40
−20
0
20
40
60
80

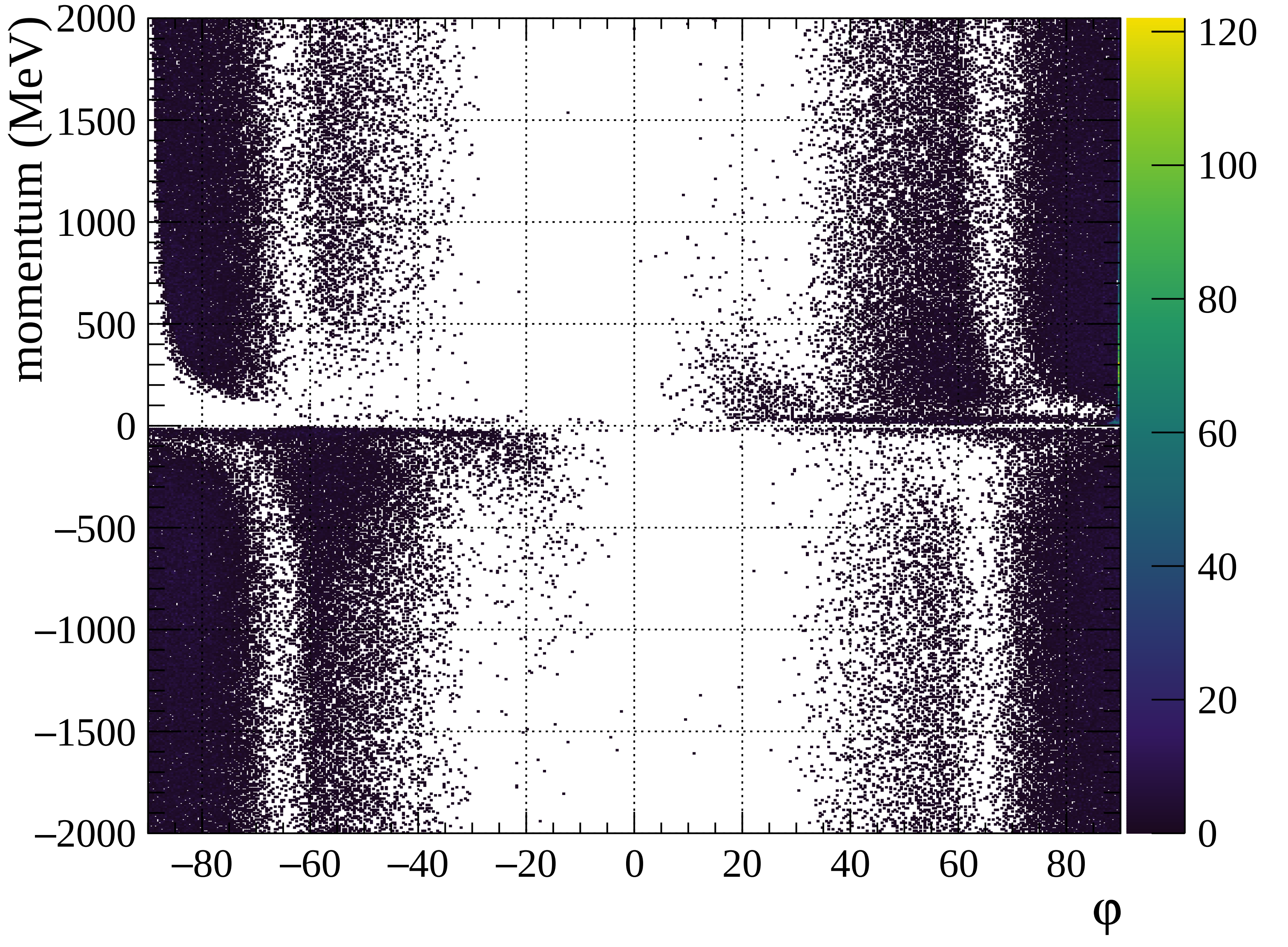

dE/dx with XP (ADC counts/cm)
1000
900
800
700
600
500
400
300
200
100
0
−80
−60
−40
−20
0
20
40
60
80
φ (°)
180
160
140
120
100
80
60
40
20
0



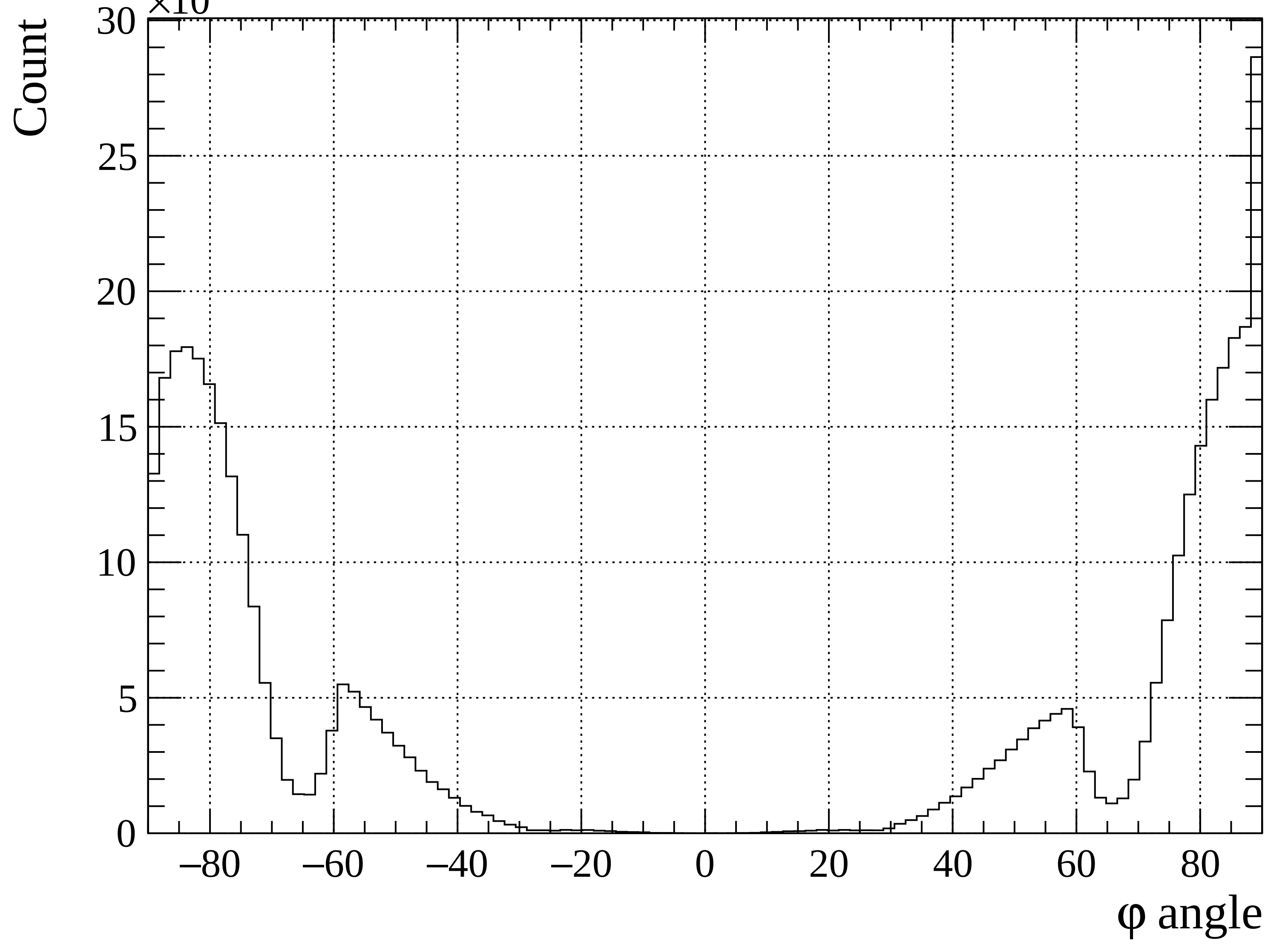
track length (cm)
0
20
40
60
80
100
120
140
160
−80
−60
−40
−20
0
20
40
60
80
φ
0
200
400
600
800
1000
1200
1400
1600
1800
2000



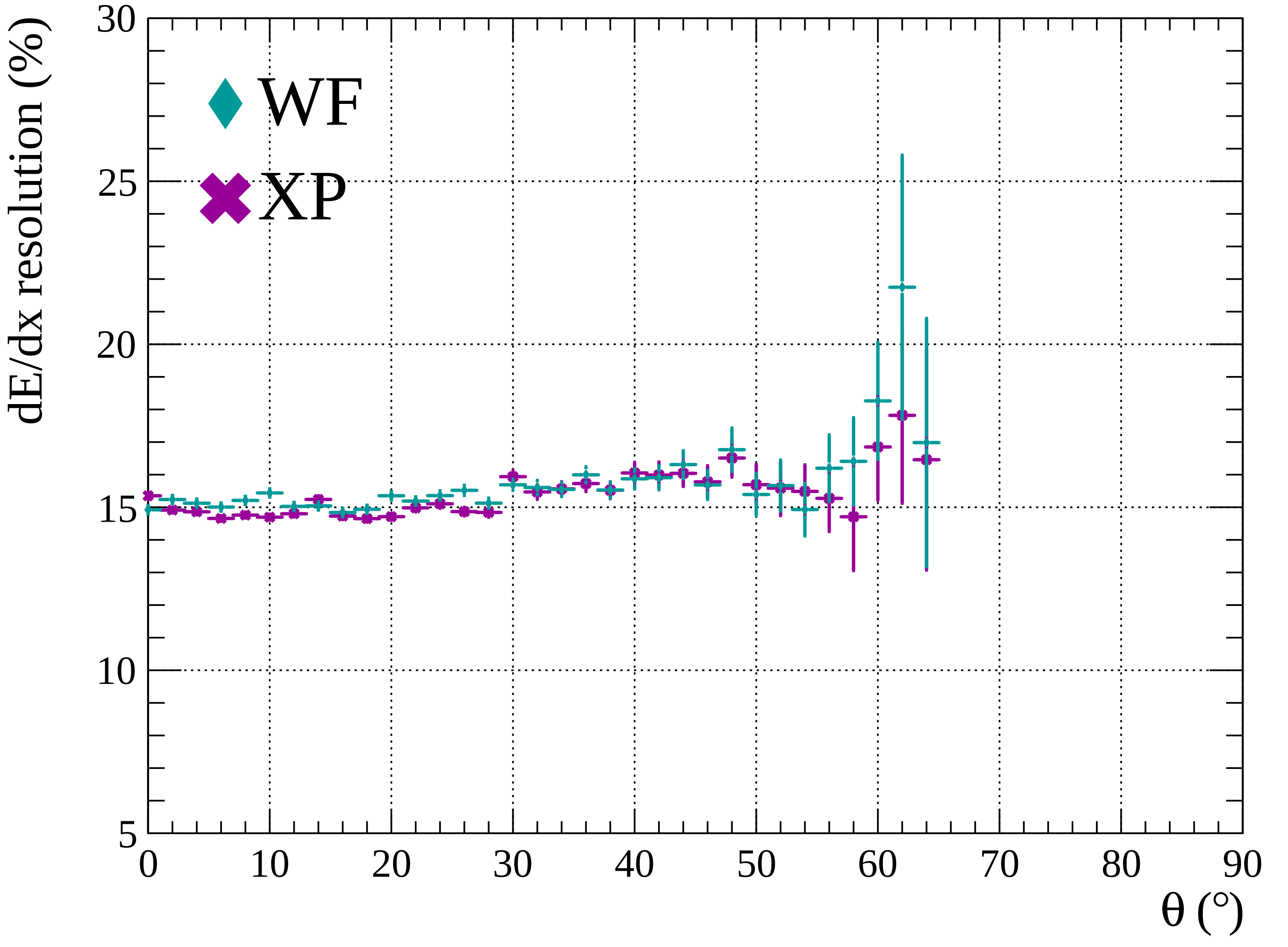
χ^2/NDF
50
45
40
35
30
25
20
15
10
5
0
−80
−60
−40
−20
0
20
40
60
80
φ
9000
8000
7000
6000
5000
4000
3000
2000
1000
0



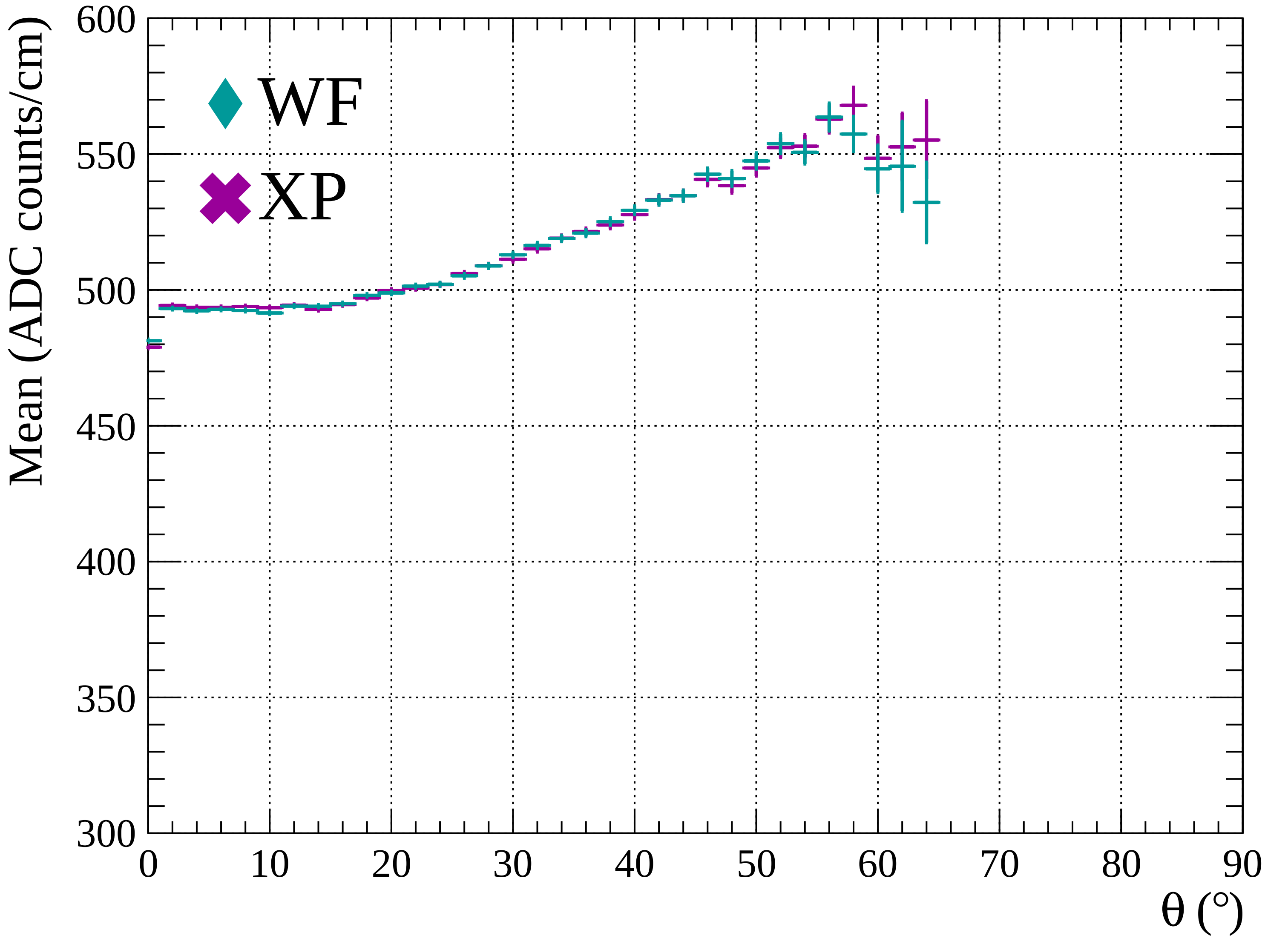
momentum (MeV)
2000
1500
1000
500
0
−500
−1000
−1500
−2000
−80
−60
−40
−20
0
20
40
60
80
φ
120
100
80
60
40
20
0

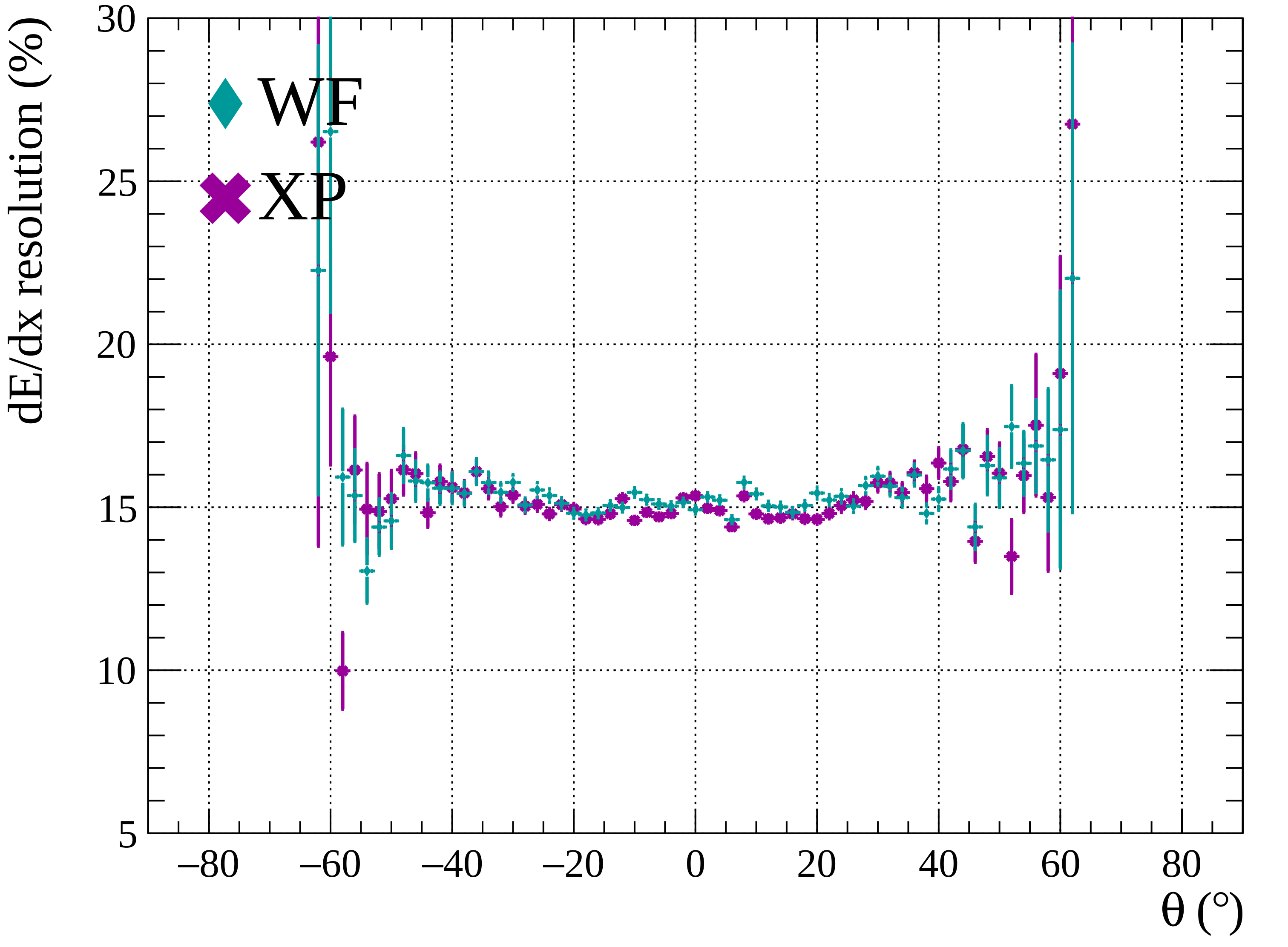


Count
×10
30
25
20
15
10
5
0
−80
−60
−40
−20
0
20
40
60
80
φ angle

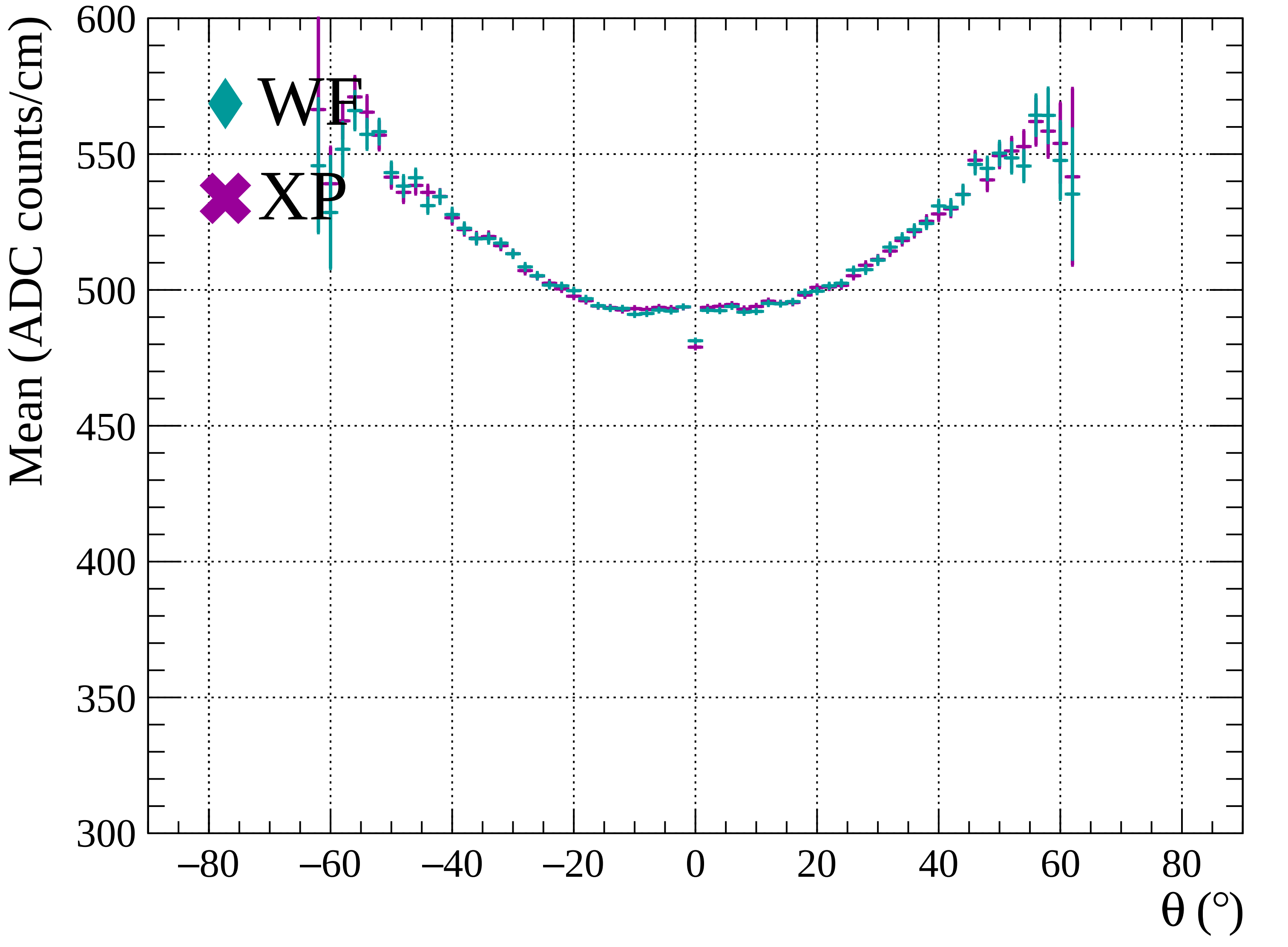


WF
XP
dE/dx resolution (%)
θ (°)
5
10
15
20
25
30
0
10
20
30
40
50
60
70
80
90

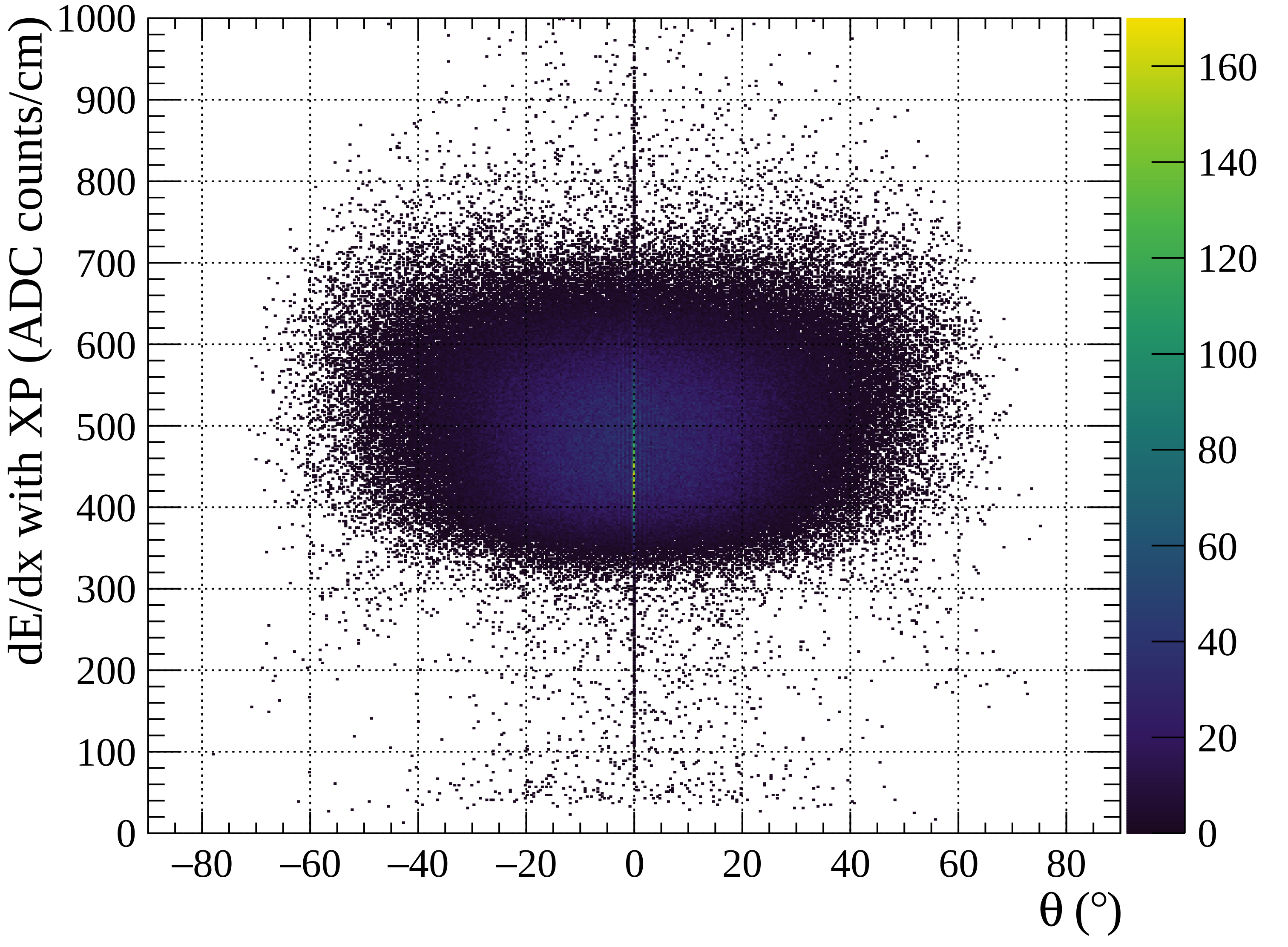

Mean (ADC counts/cm)
600
550
500
450
400
350
300
0
10
20
30
40
50
60
70
80
90
θ (°)
WF
XP

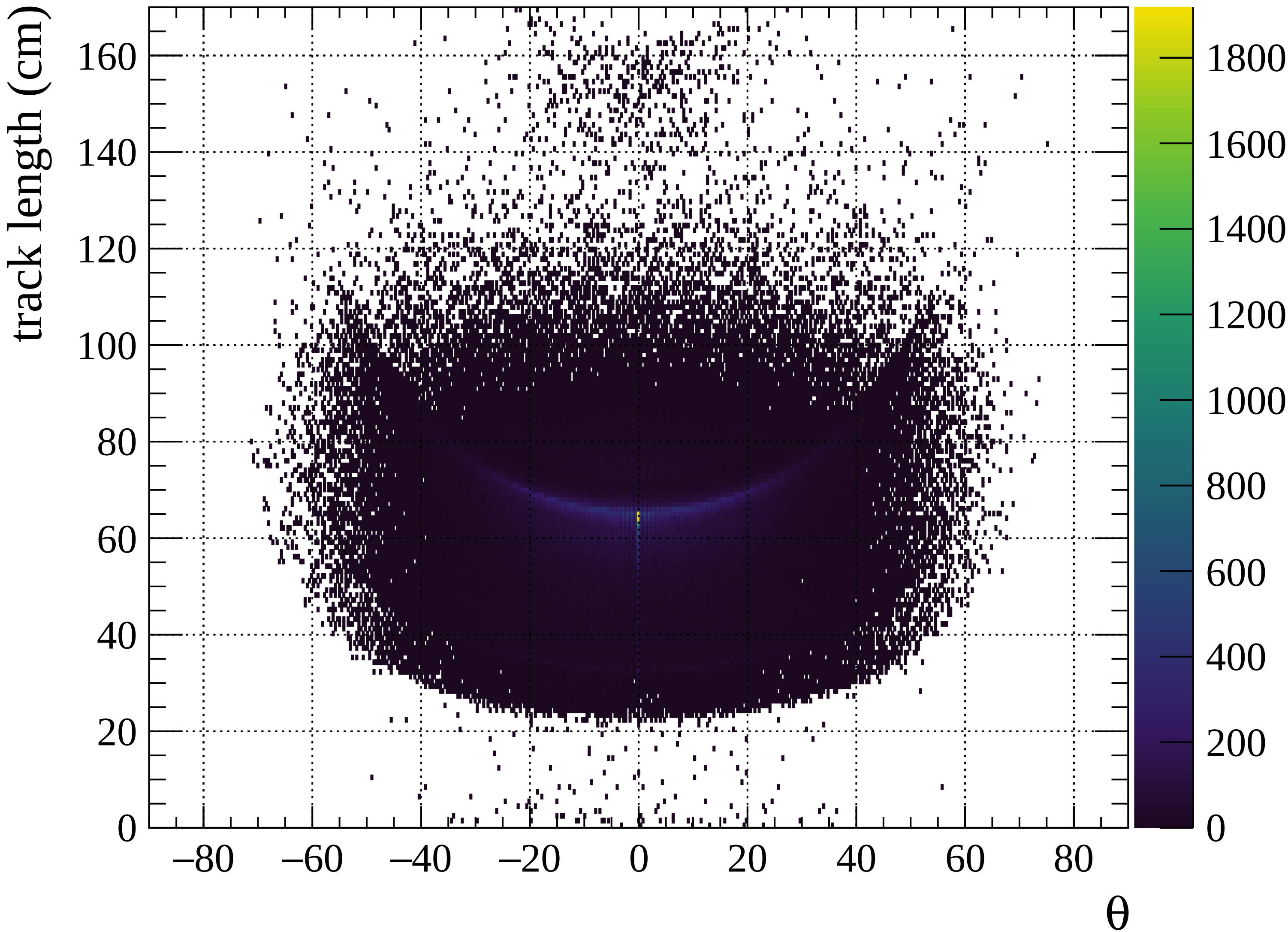


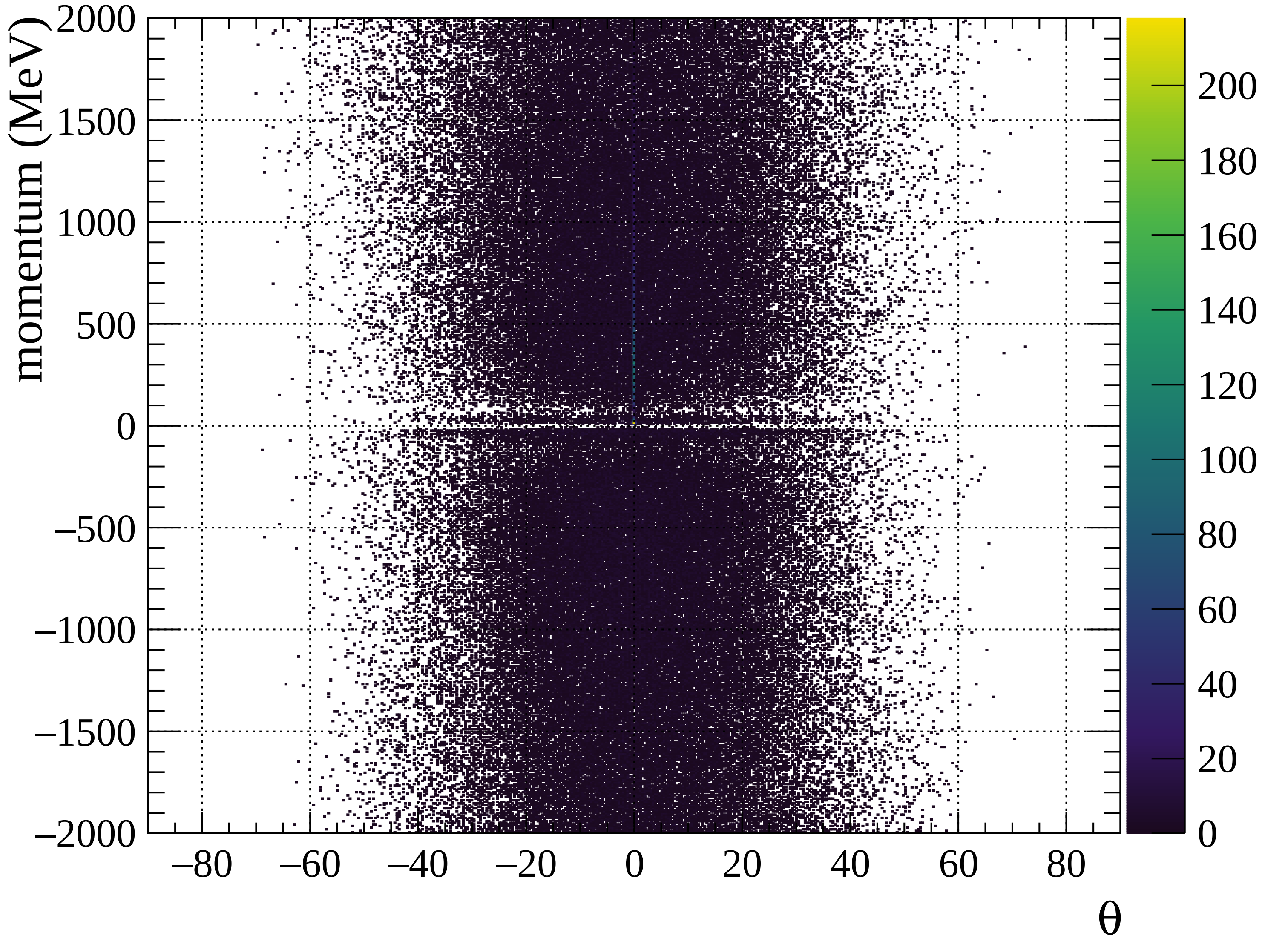
WF
XP
dE/dx resolution (%)
θ (°)
30
25
20
15
10
5
−80
−60
−40
−20
0
20
40
60
80

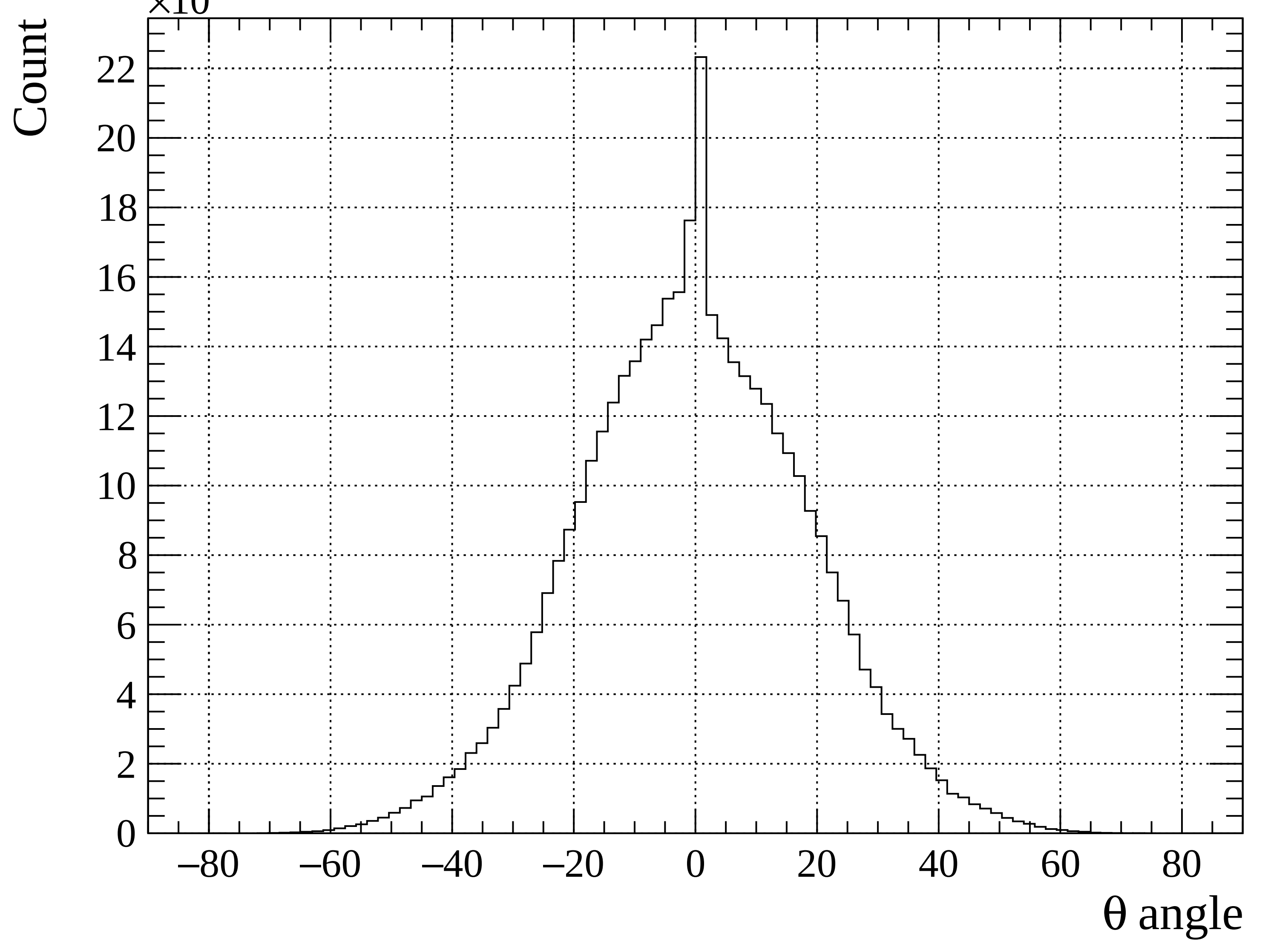


WF
XP
Mean (ADC counts/cm)
θ (°)
600
550
500
450
400
350
300
−80
−60
−40
−20
0
20
40
60
80

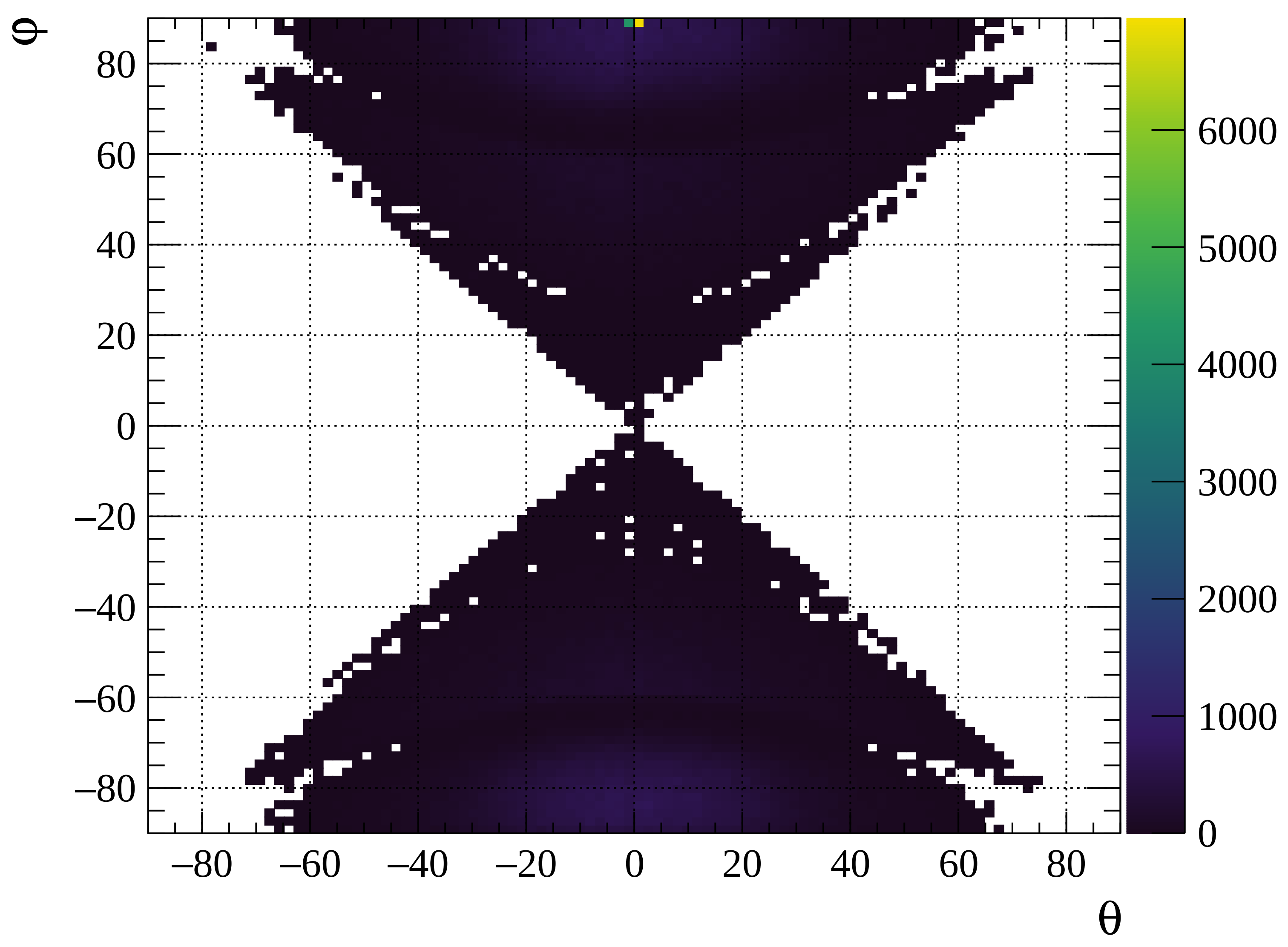

dE/dx with XP (ADC counts/cm)
θ (°)
1000
900
800
700
600
500
400
300
200
100
0
−80
−60
−40
−20
0
20
40
60
80
160
140
120
100
80
60
40
20
0

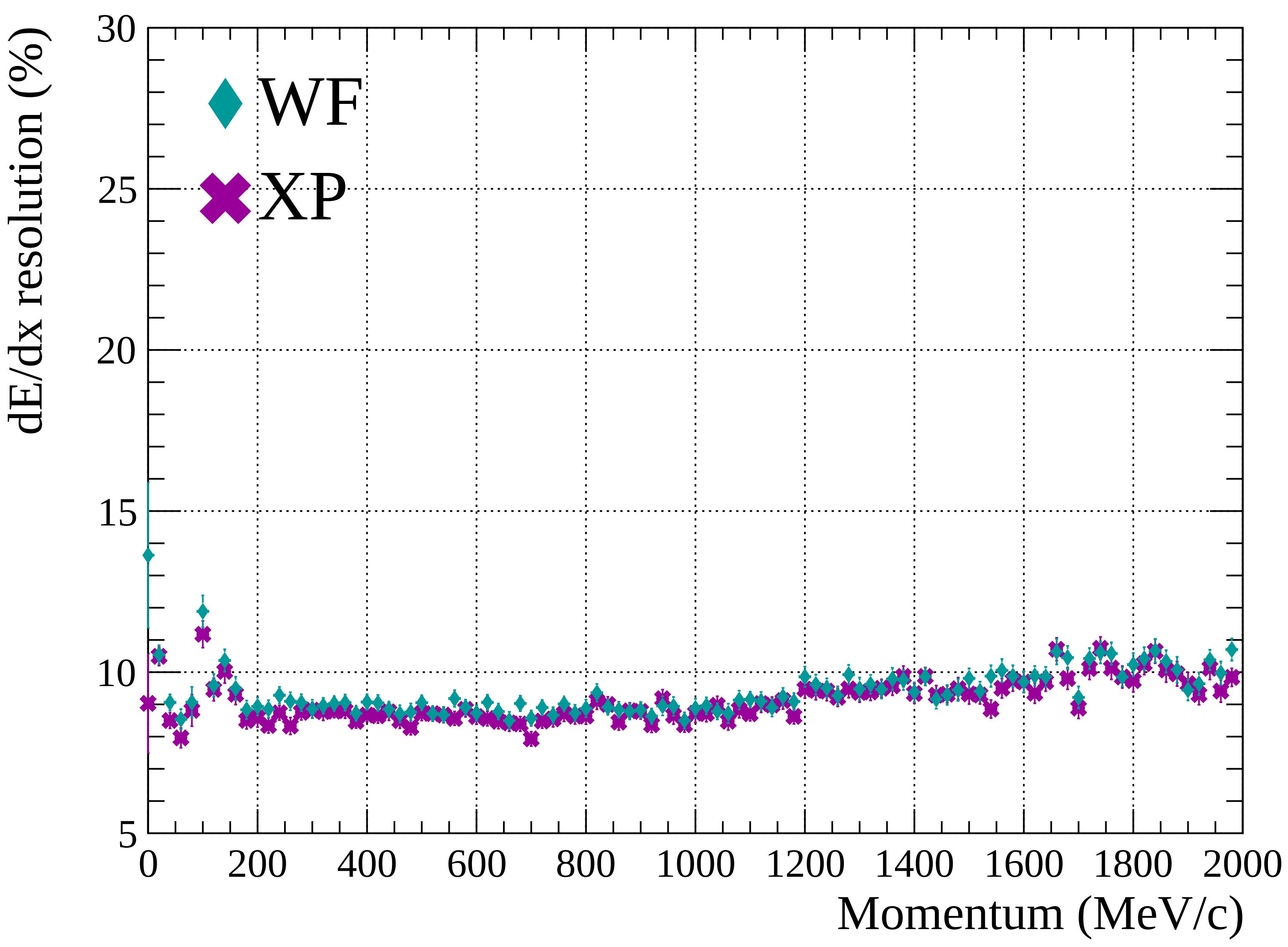

track length (cm)
160
140
120
100
80
60
40
20
0
−80
−60
−40
−20
0
20
40
60
80
θ
1800
1600
1400
1200
1000
800
600
400
200
0

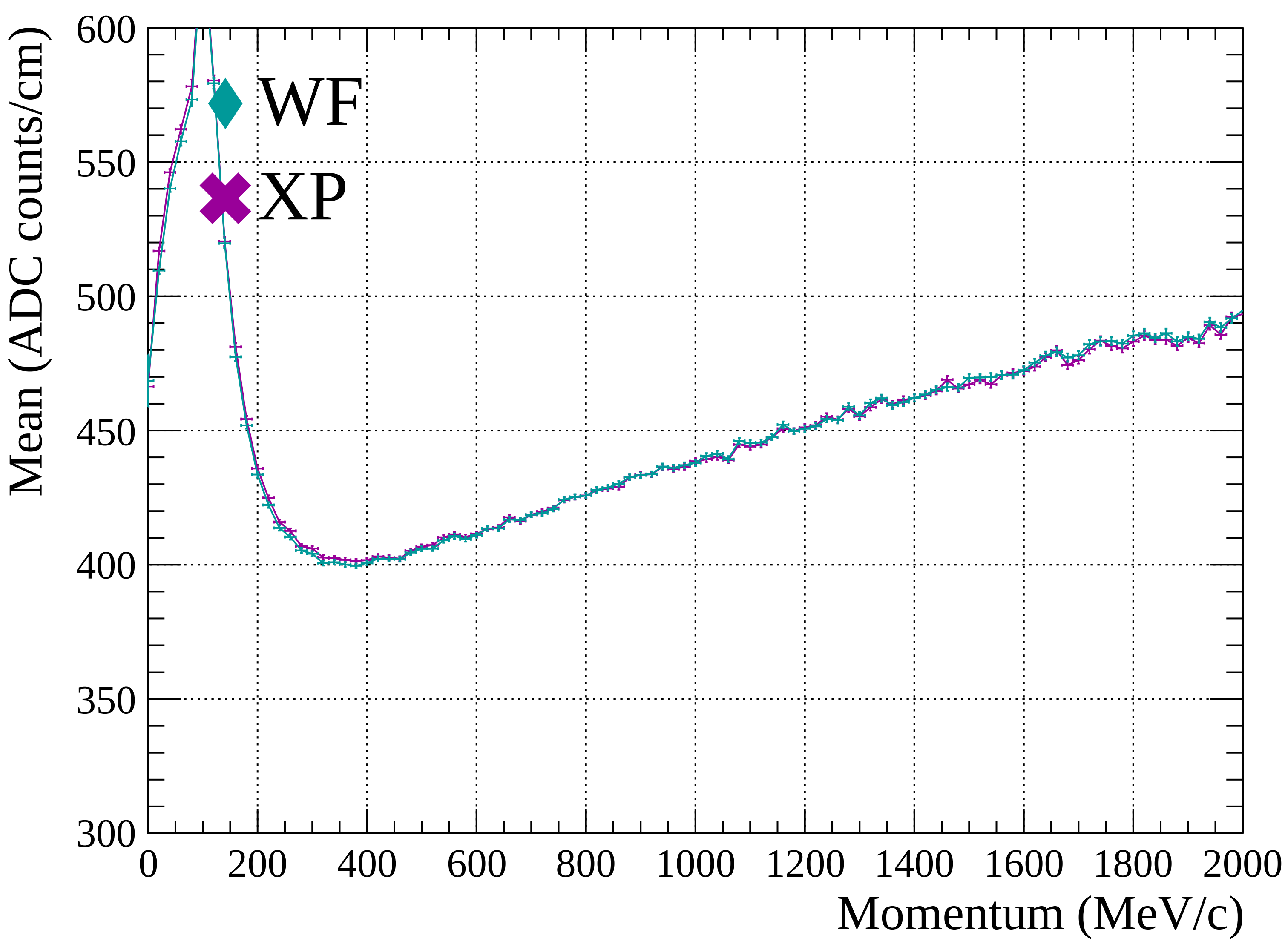

momentum (MeV)
2000
1500
1000
500
0
−500
−1000
−1500
−2000
−80
−60
−40
−20
0
20
40
60
80
θ
200
180
160
140
120
100
80
60
40
20
0

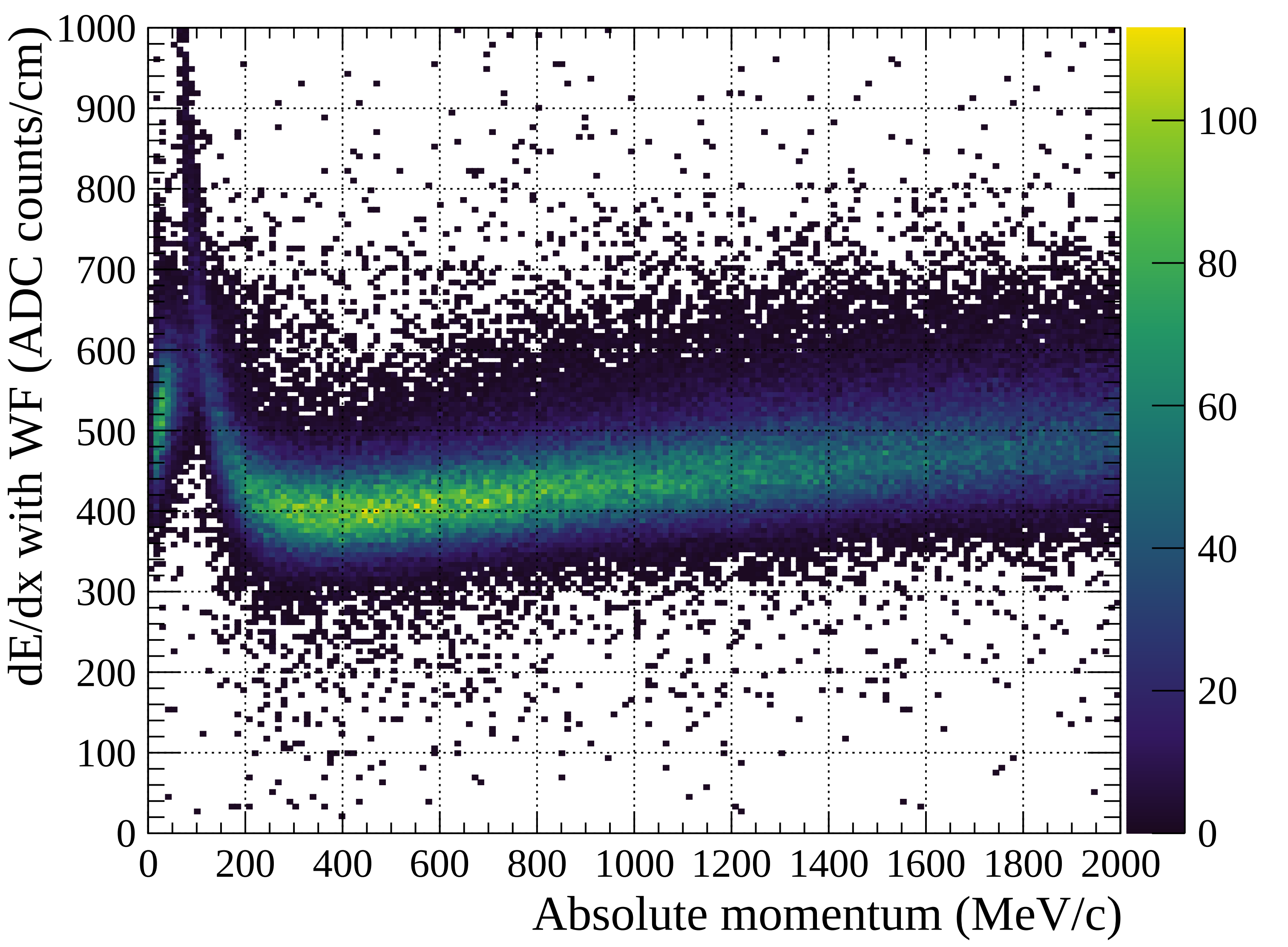


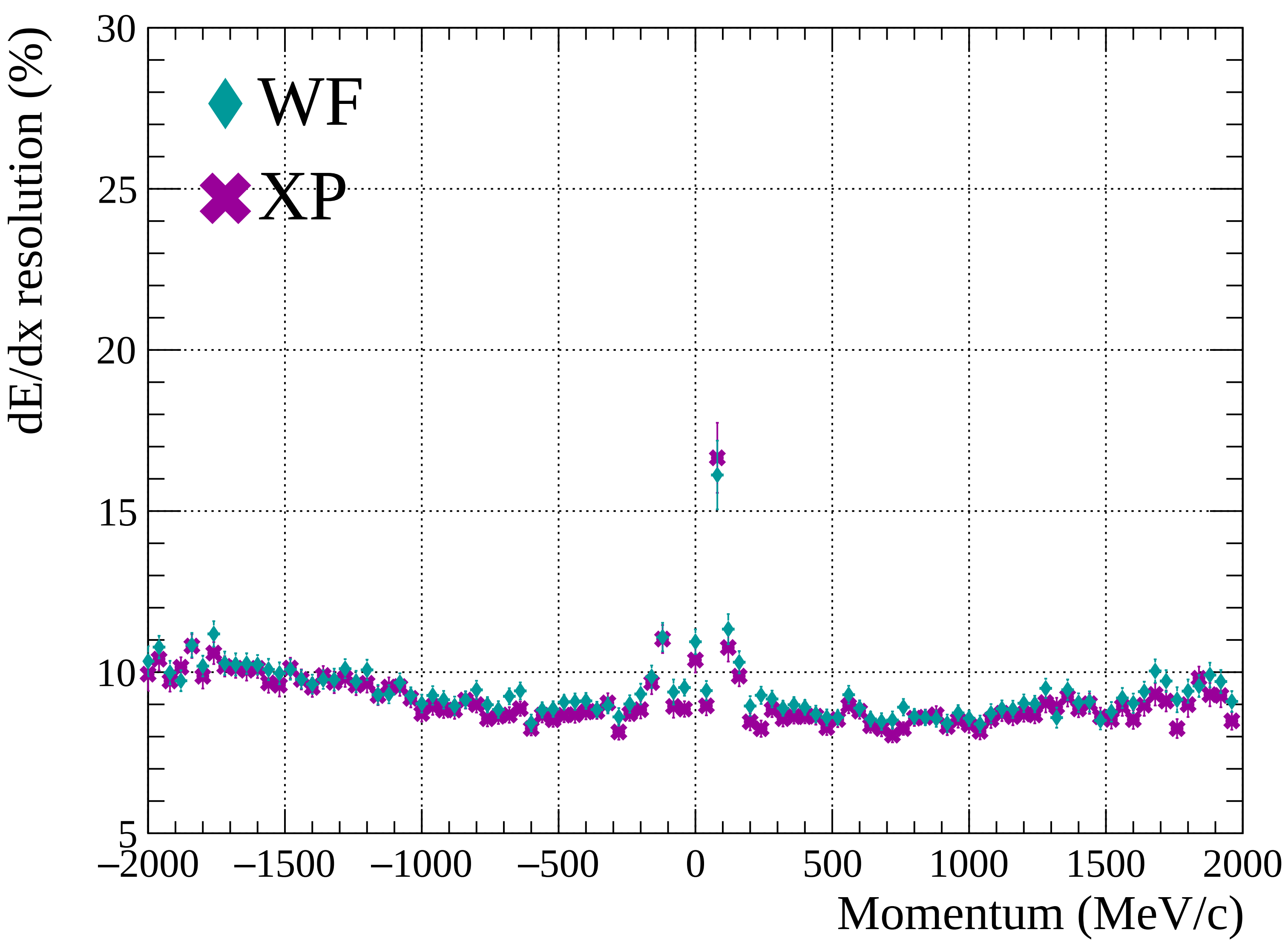
×10
Count
22
20
18
16
14
12
10
8
6
4
2
0
−80
−60
−40
−20
0
20
40
60
80
θ angle


φ
80
60
40
20
0
−20
−40
−60
−80
−80
−60
−40
−20
0
20
40
60
80
θ
6000
5000
4000
3000
2000
1000
0

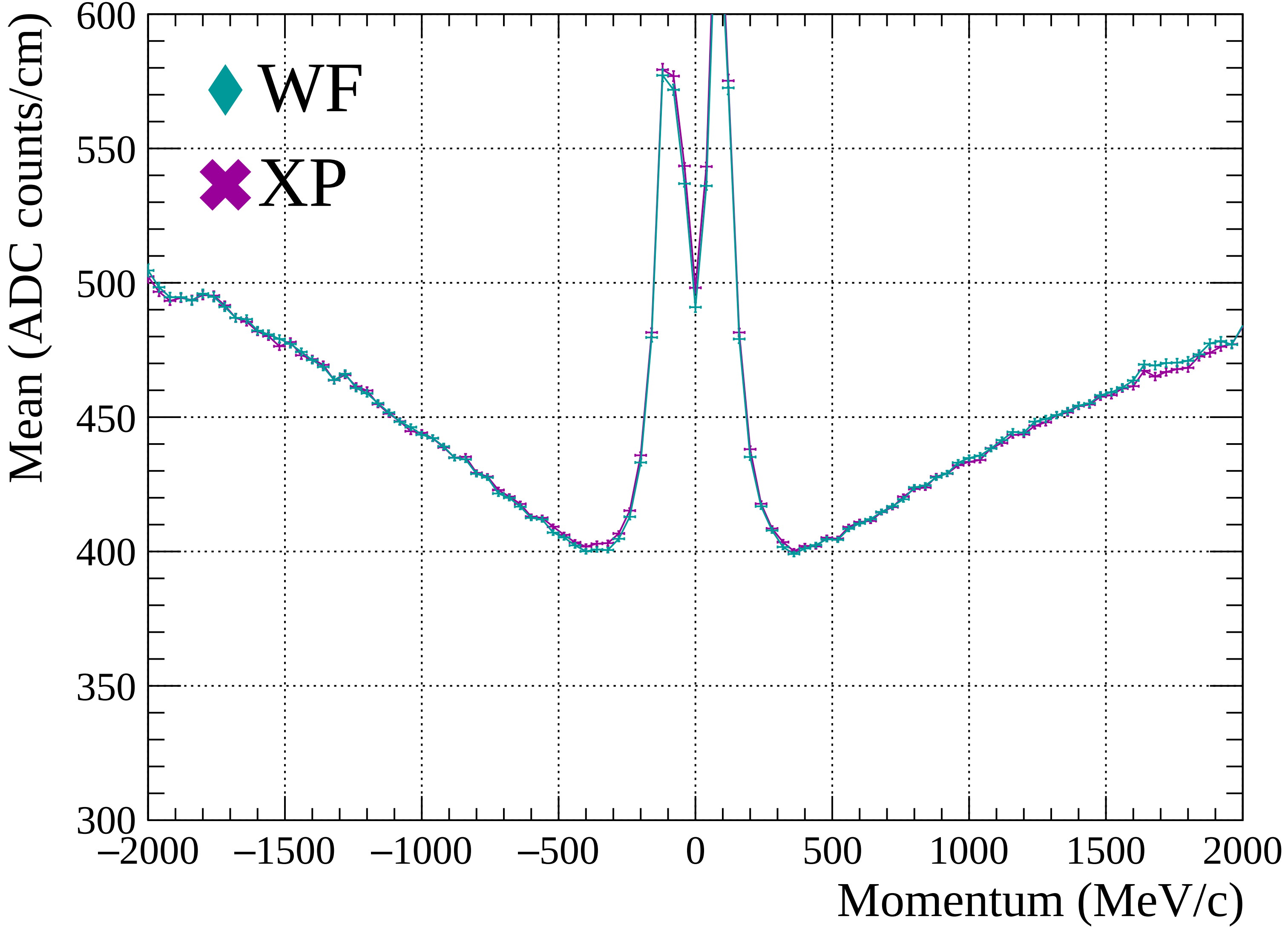

dE/dx resolution (%)
Momentum (MeV/c)
WF
XP
30
25
20
15
10
5
0
200
400
600
800
1000
1200
1400
1600
1800
2000

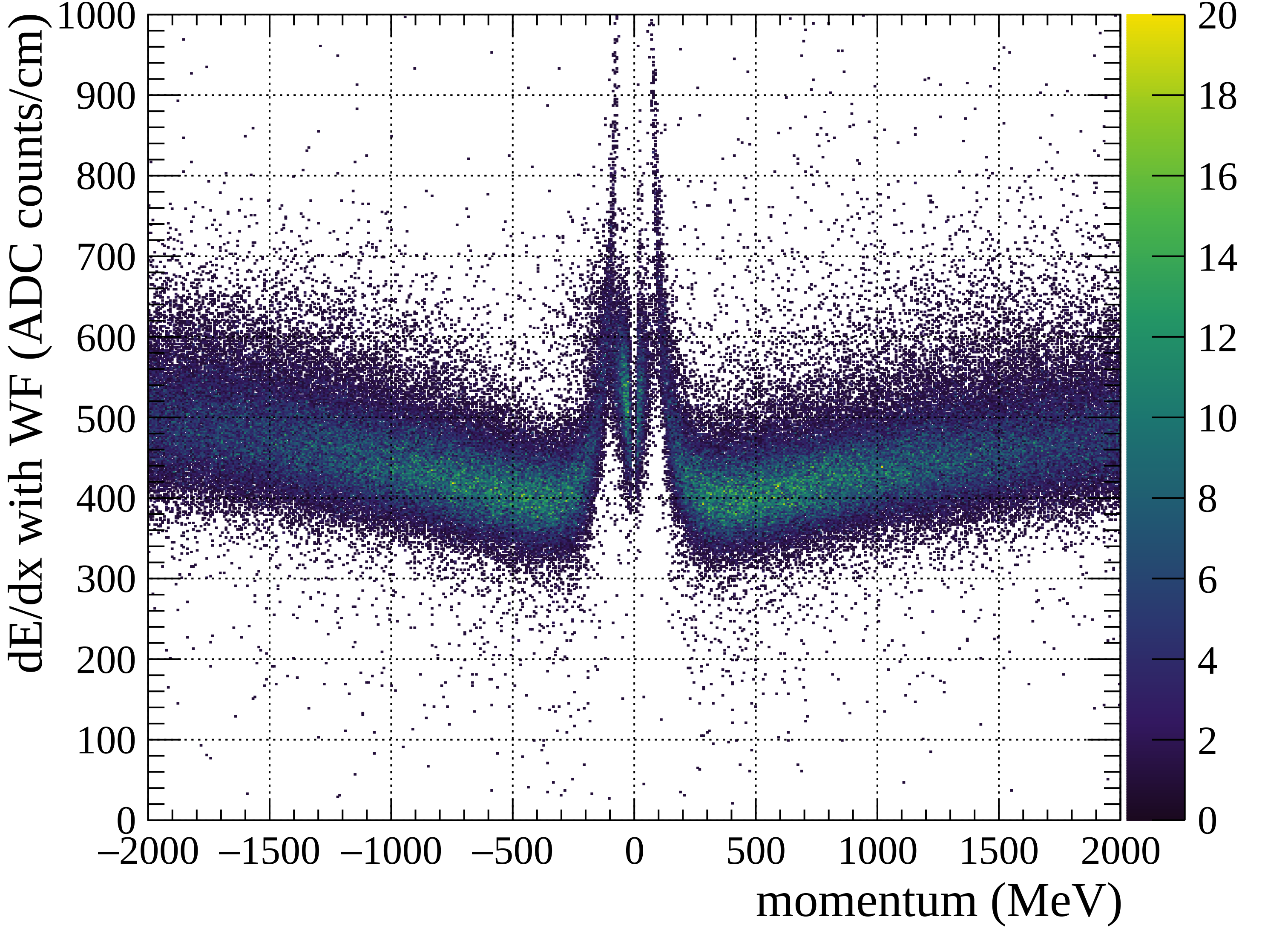

Mean (ADC counts/cm)
600
550
500
450
400
350
300
WF
XP
0
200
400
600
800
1000
1200
1400
1600
1800
2000
Momentum (MeV/c)

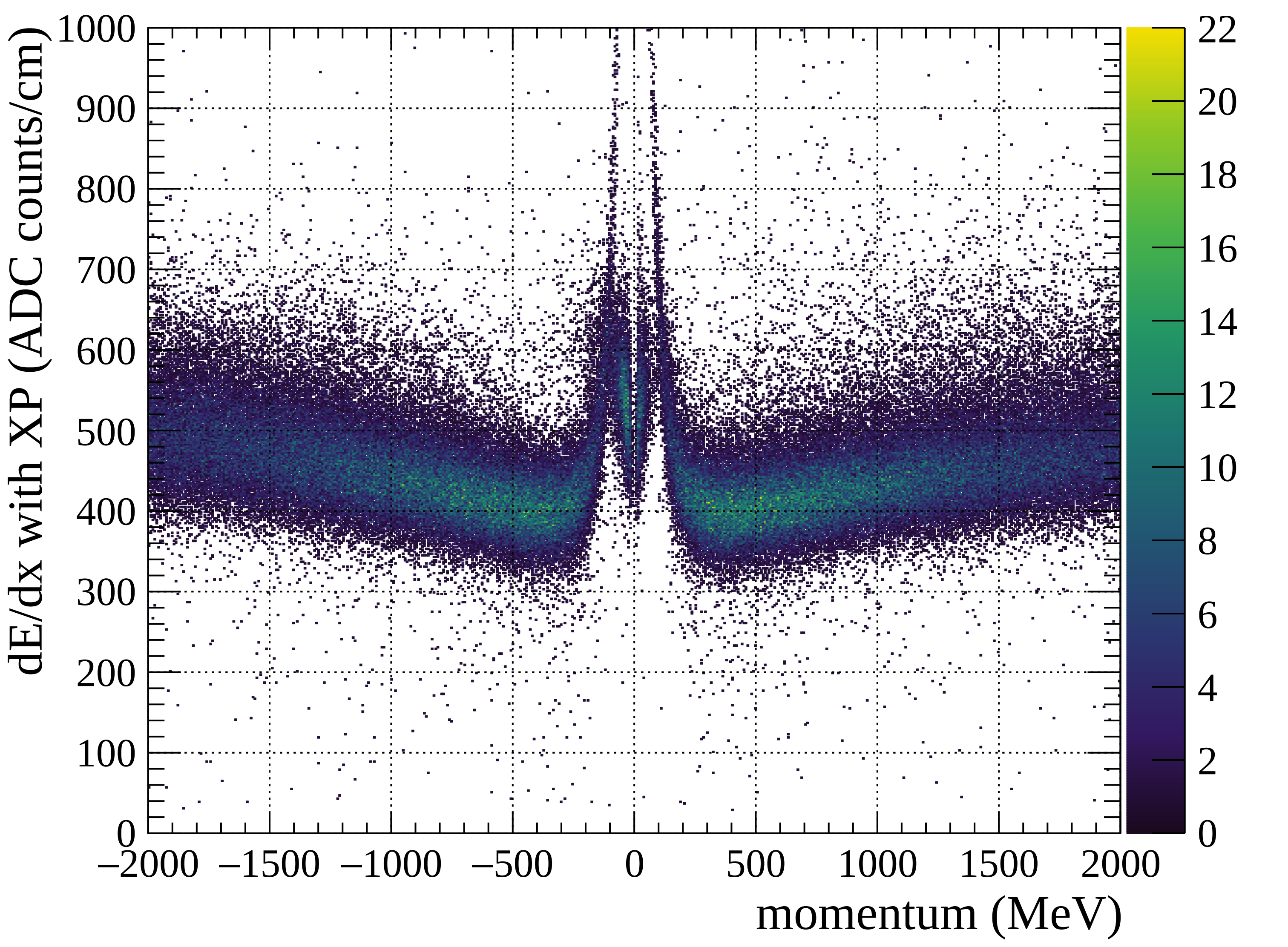

dE/dx with WF (ADC counts/cm)
1000
900
800
700
600
500
400
300
200
100
0
0
200
400
600
800
1000
1200
1400
1600
1800
2000
Absolute momentum (MeV/c)
100
80
60
40
20
0

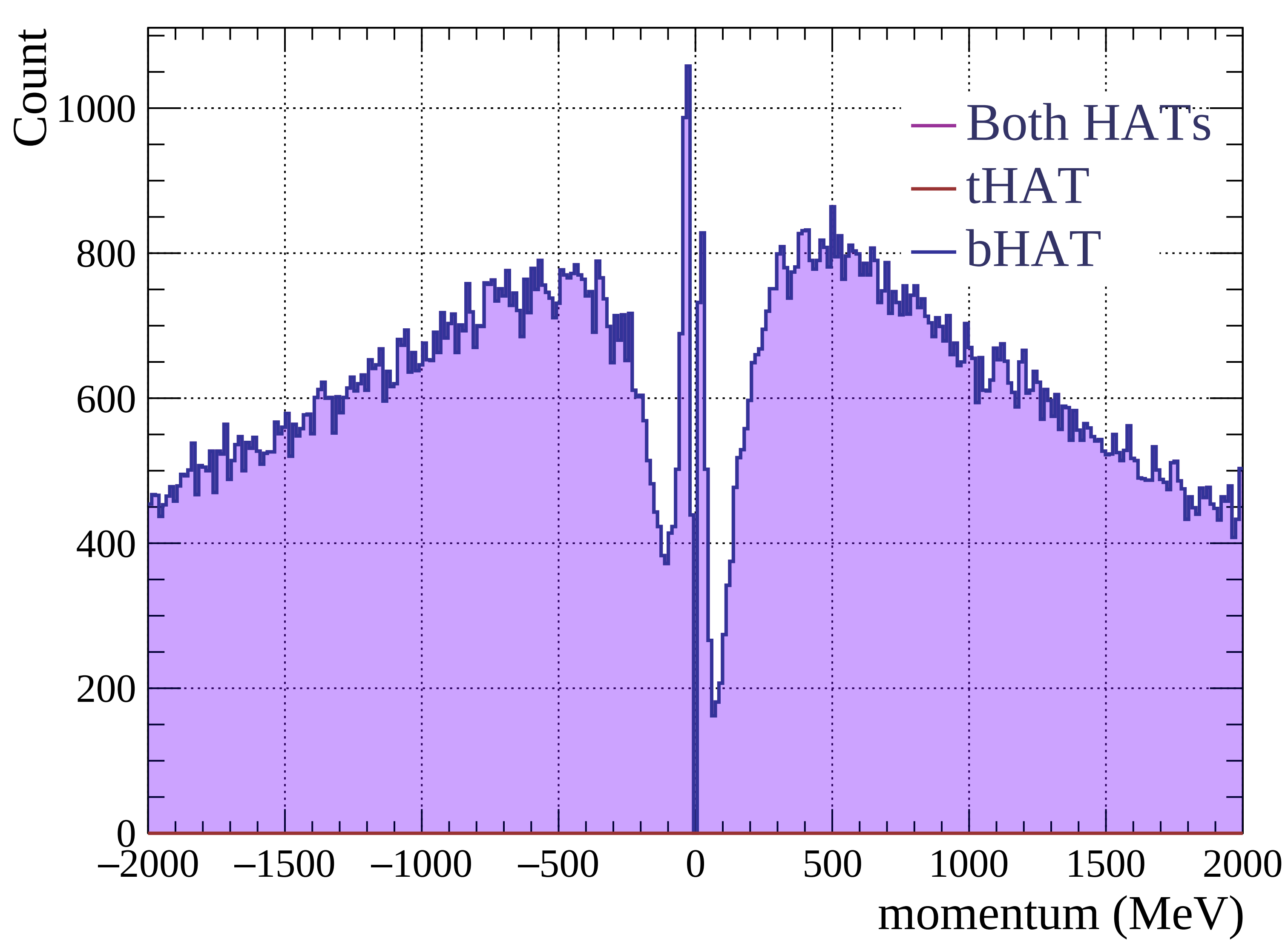


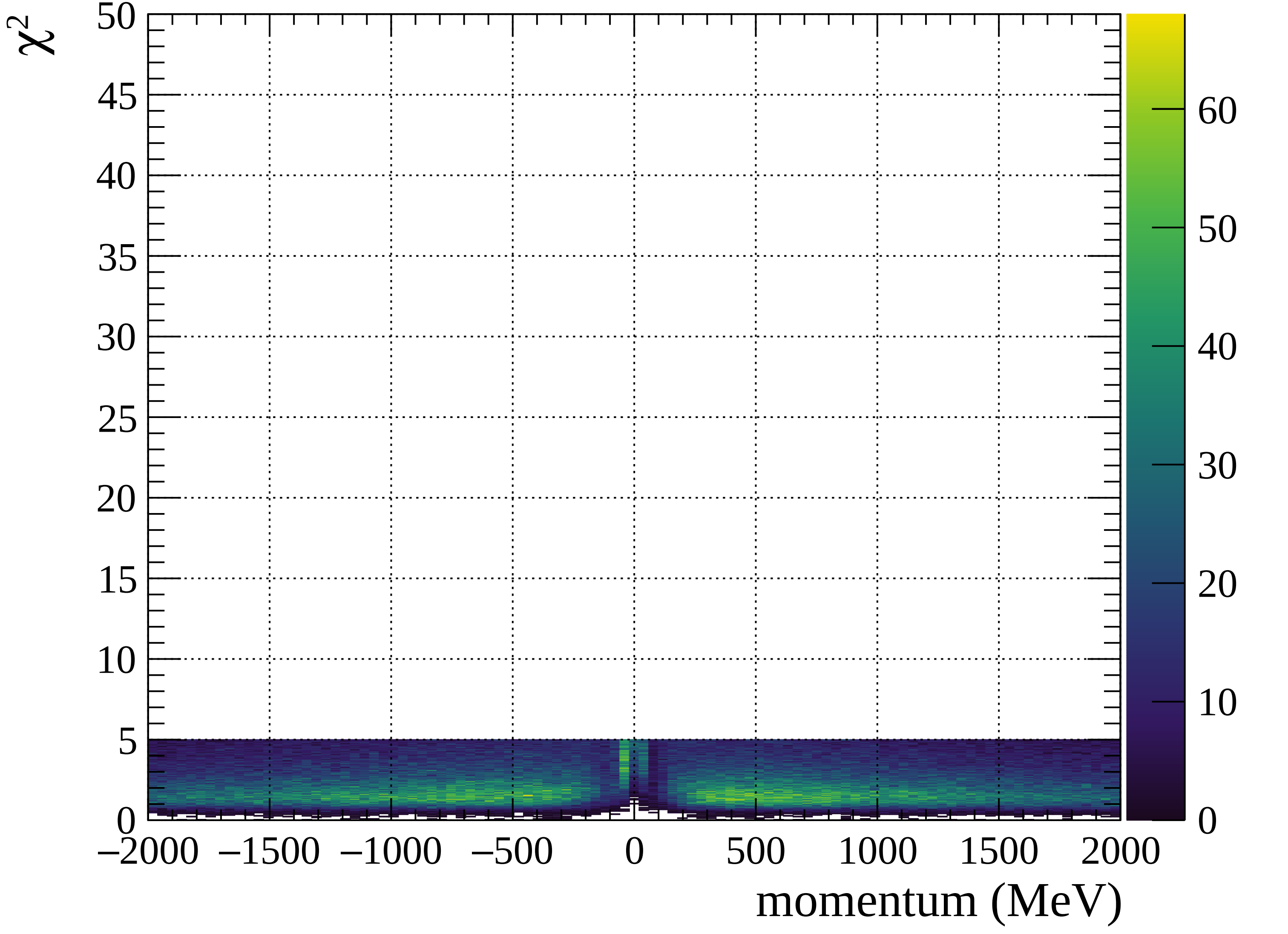
dE/dx resolution (%)
30
25
20
15
10
5
WF
XP
−2000
−1500
−1000
−500
0
500
1000
1500
2000
Momentum (MeV/c)


WF
XP
Mean (ADC counts/cm)
Momentum (MeV/c)
600
550
500
450
400
350
300
−2000
−1500
−1000
−500
0
500
1000
1500
2000

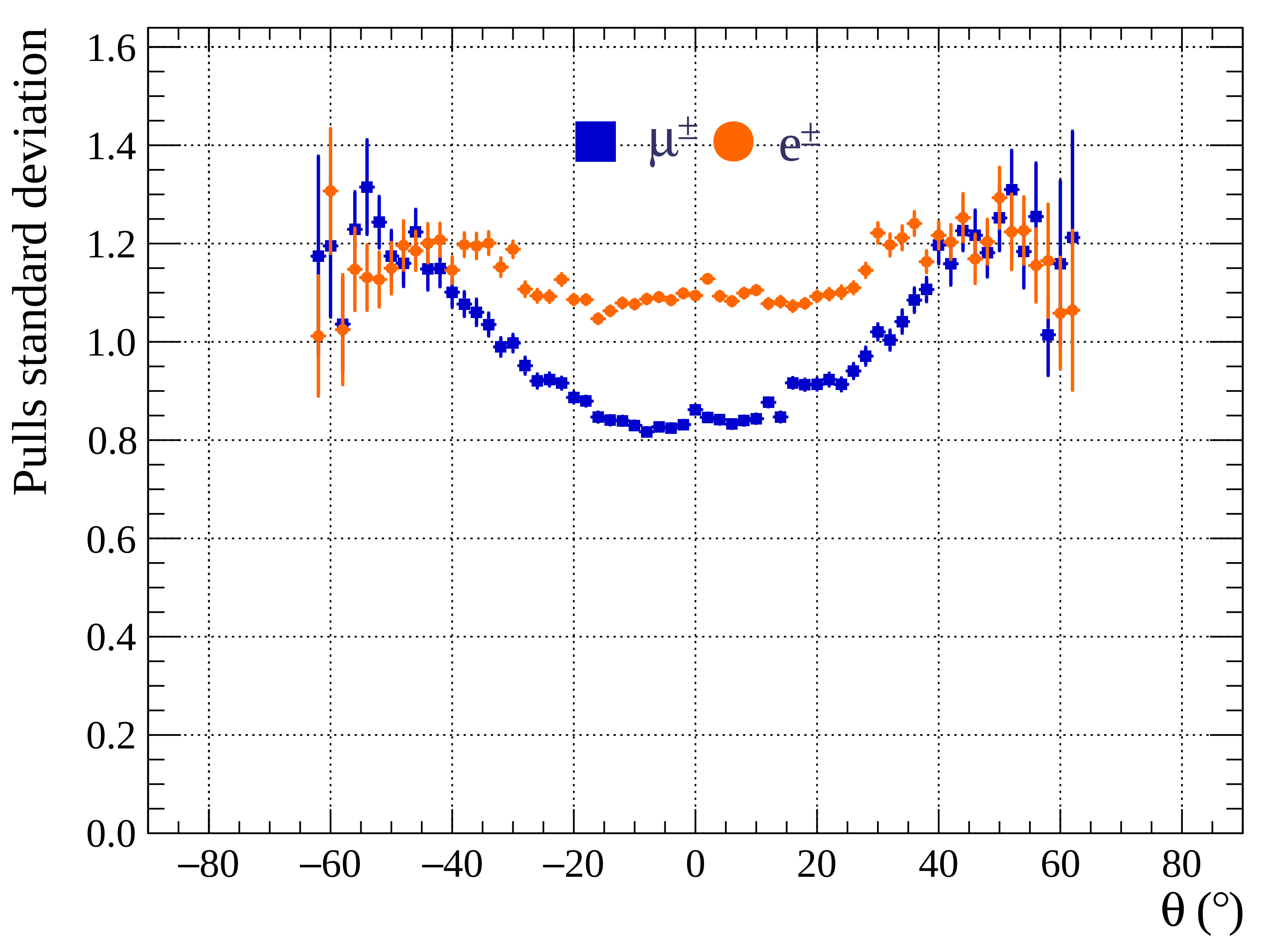

dE/dx with WF (ADC counts/cm)
momentum (MeV)
1000
900
800
700
600
500
400
300
200
100
0
−2000
−1500
−1000
−500
0
500
1000
1500
2000
20
18
16
14
12
10
8
6
4
2
0

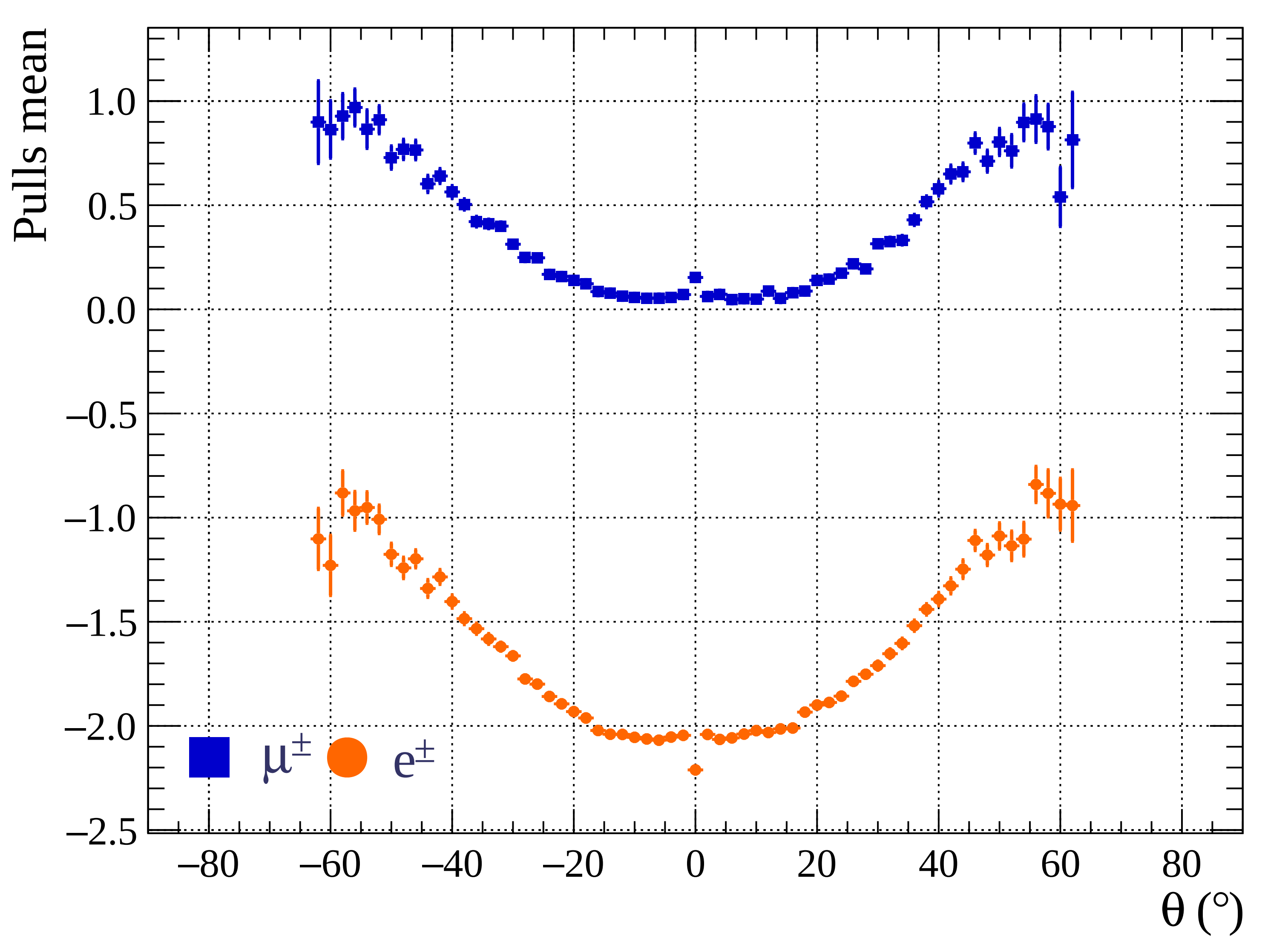

dE/dx with XP (ADC counts/cm)
momentum (MeV)
0
100
200
300
400
500
600
700
800
900
1000
−2000
−1500
−1000
−500
0
500
1000
1500
2000
0
2
4
6
8
10
12
14
16
18
20
22

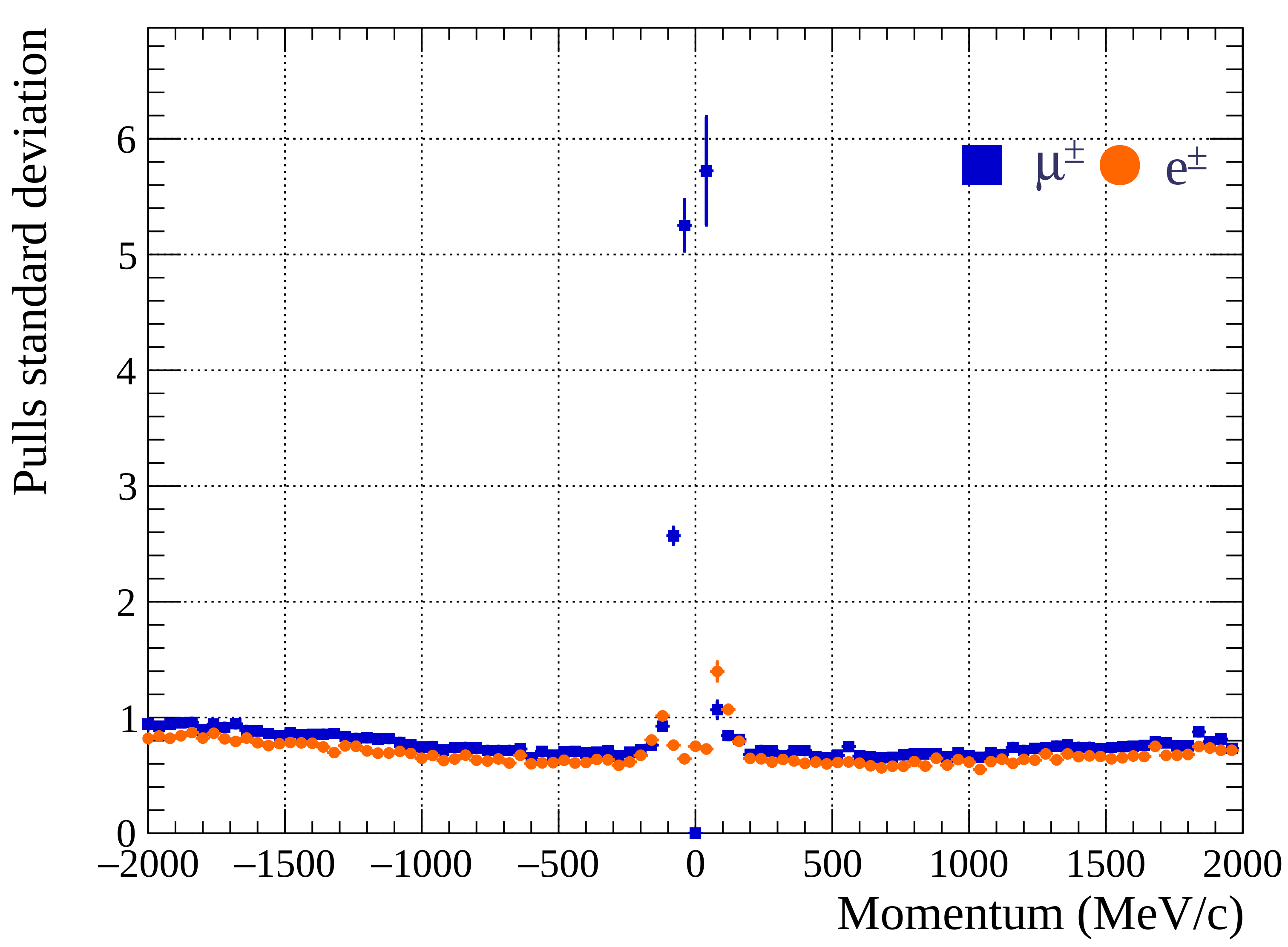

Count
1000
800
600
400
200
0
−2000
−1500
−1000
−500
0
500
1000
1500
2000
momentum (MeV)
Both HATs
tHAT
bHAT



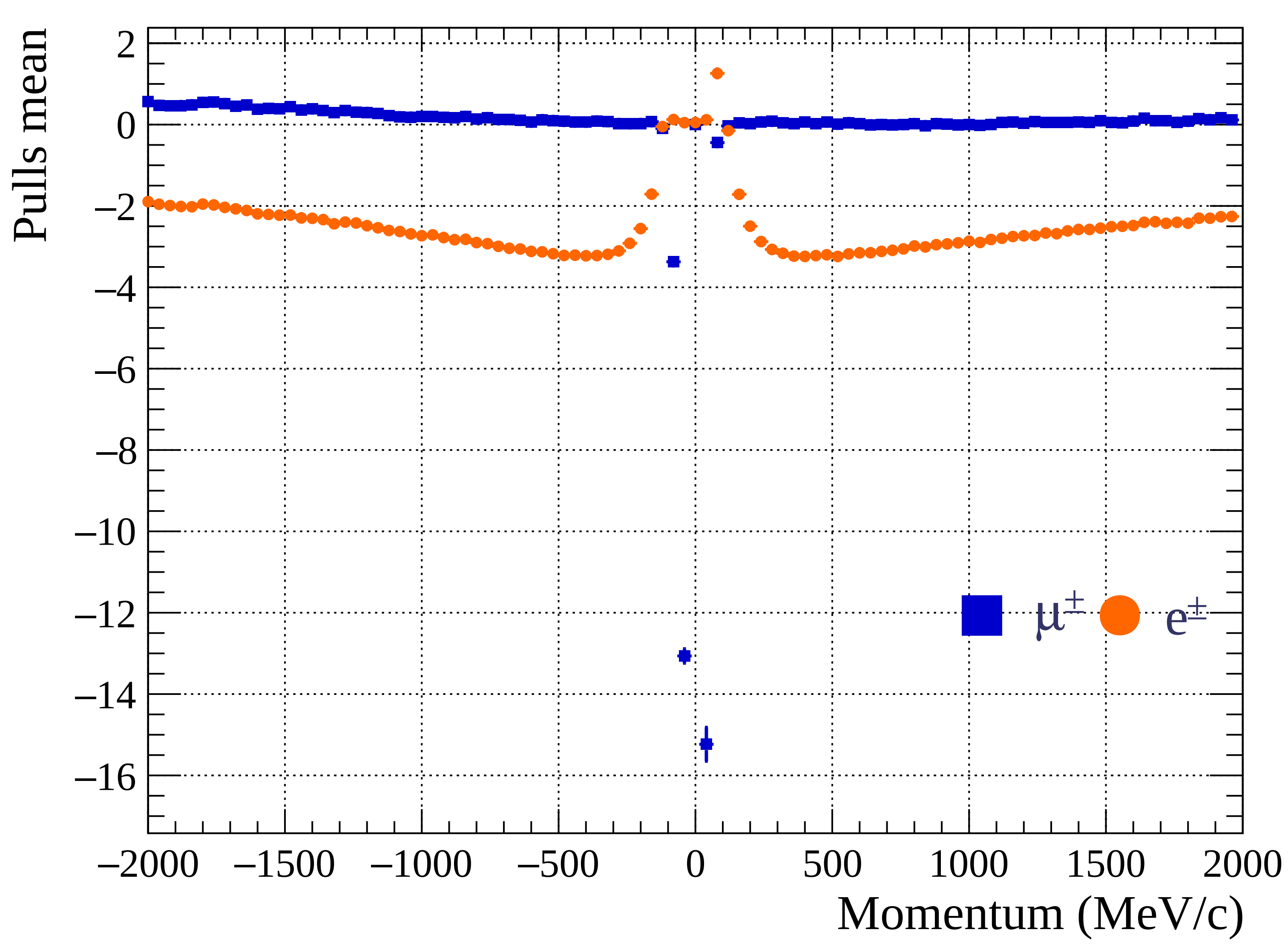
χ^2
50
45
40
35
30
25
20
15
10
5
0
−2000
−1500
−1000
−500
0
500
1000
1500
2000
momentum (MeV)
60
50
40
30
20
10
0


Pulls standard deviation
$\mu^{\pm}$
$e^{\pm}$
1.6
1.4
1.2
1.0
0.8
0.6
0.4
0.2
0.0
−80
−60
−40
−20
0
20
40
60
80
θ (°)

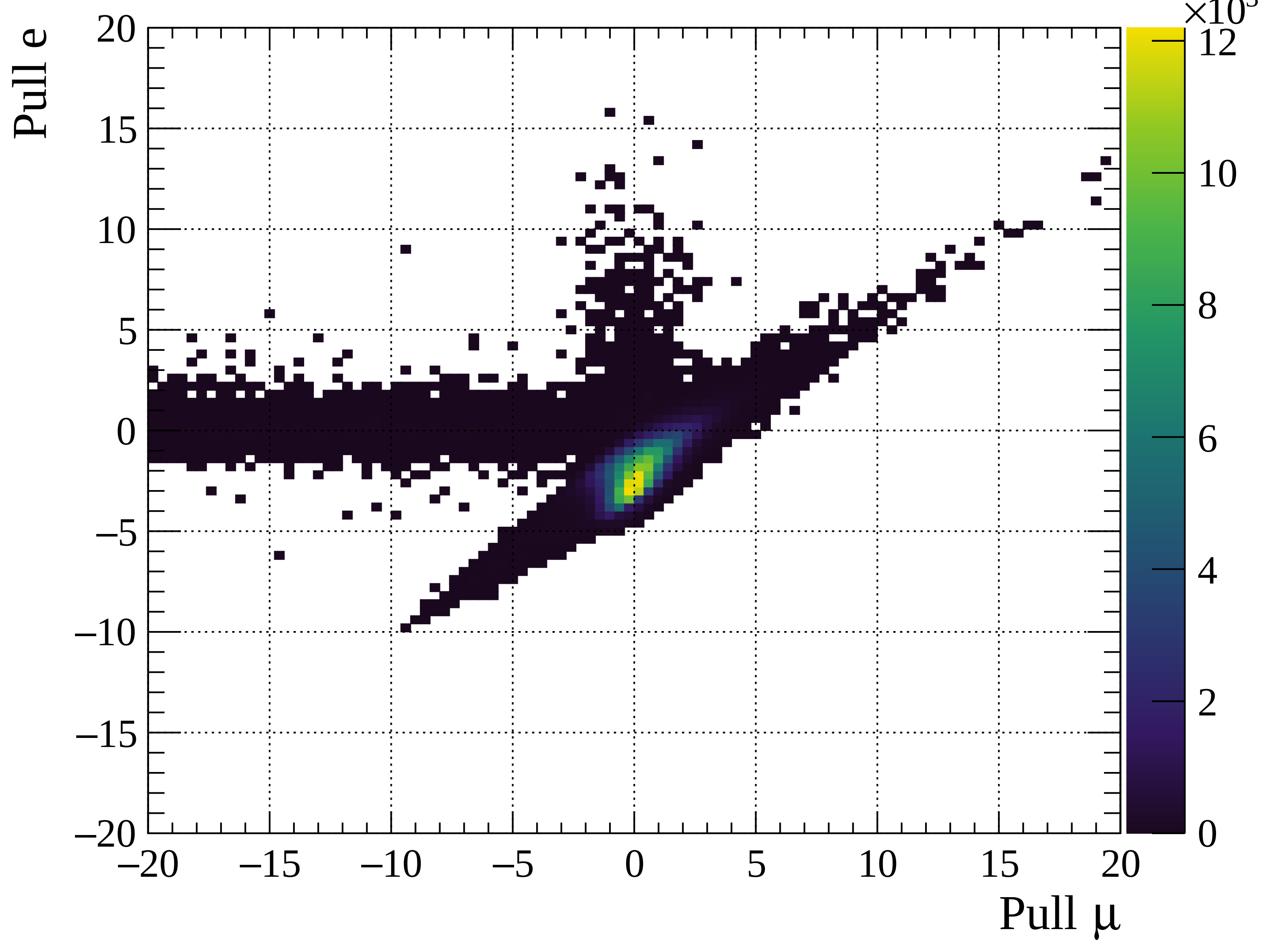

Pulls mean
1.0
0.5
0.0
−0.5
−1.0
−1.5
−2.0
−2.5
−80
−60
−40
−20
0
20
40
60
80
θ (°)
μ±
e±

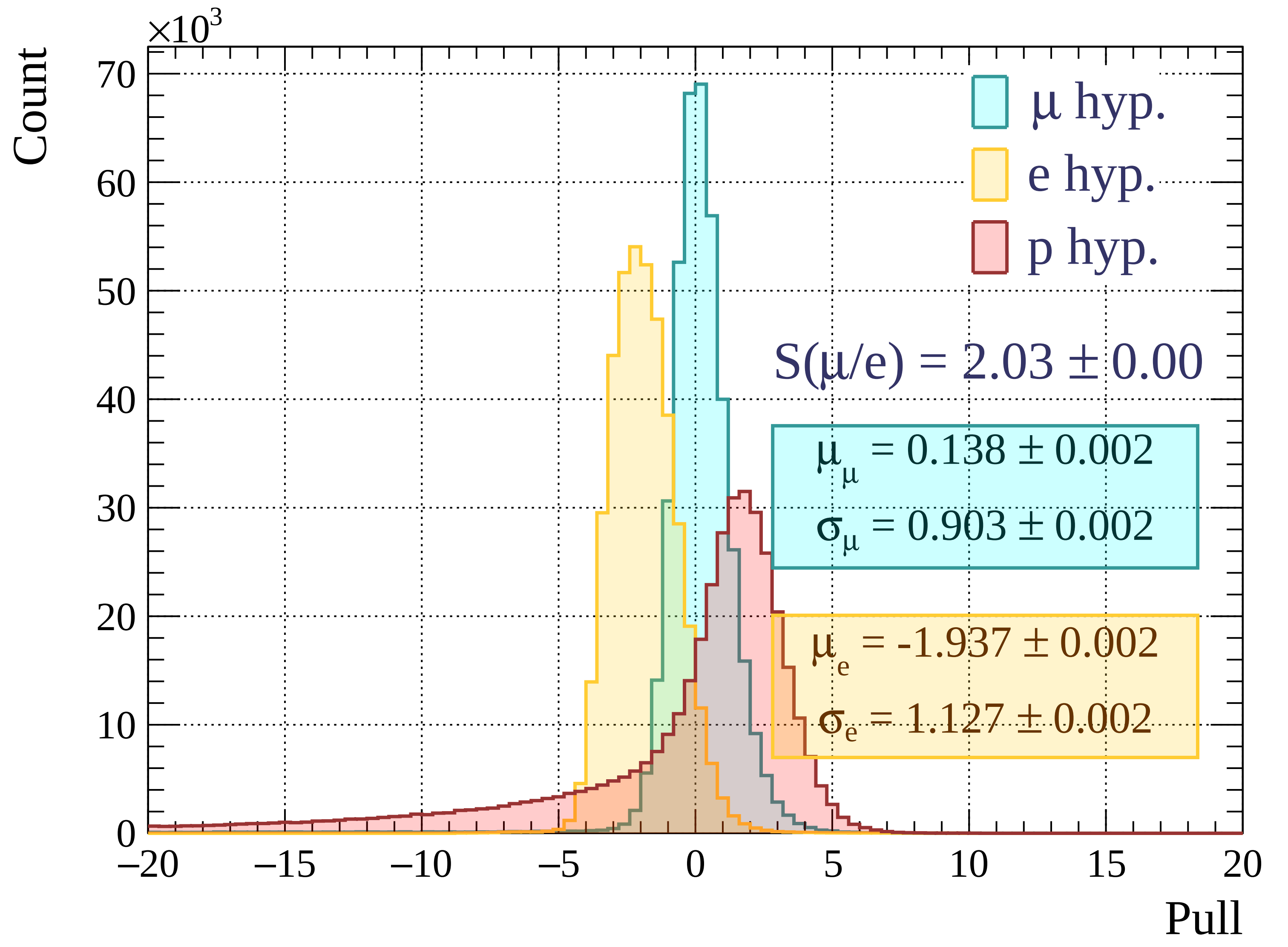


Pulls standard deviation
0
1
2
3
4
5
6
−2000
−1500
−1000
−500
0
500
1000
1500
2000
Momentum (MeV/c)
μ±
e±

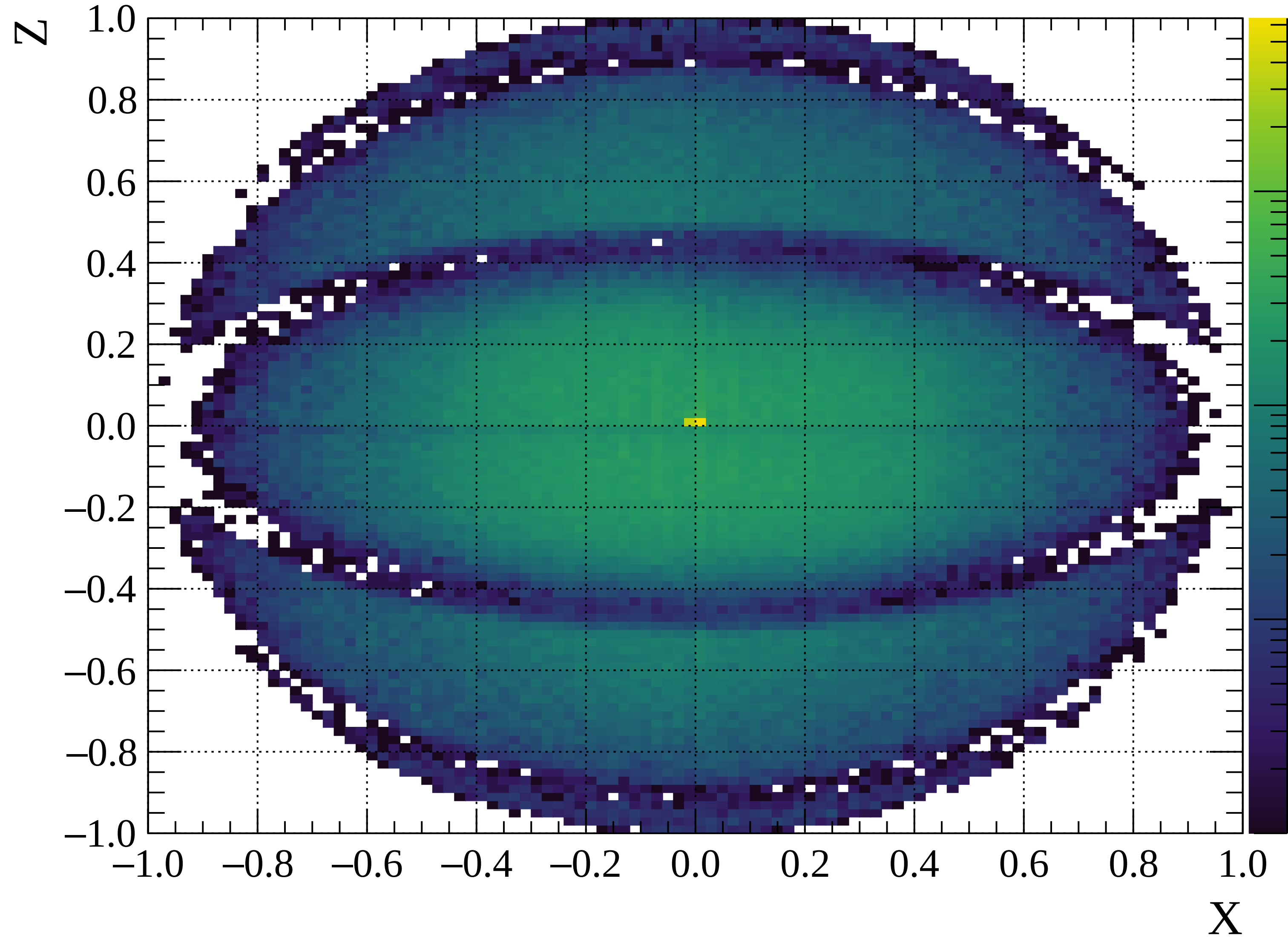


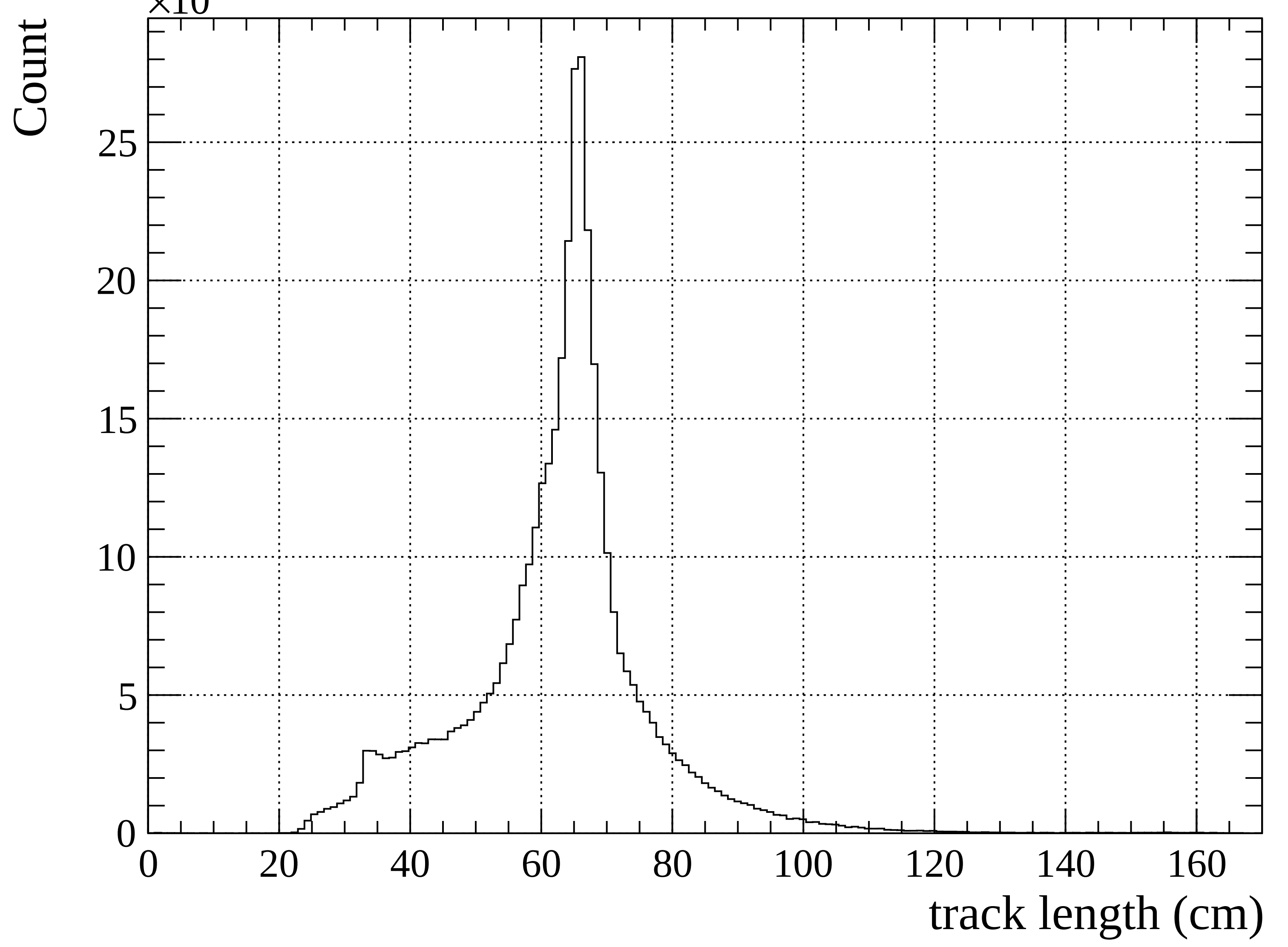
Pulls mean
2
0
−2
−4
−6
−8
−10
−12
−14
−16
−2000
−1500
−1000
−500
0
500
1000
1500
2000
Momentum (MeV/c)
μ±
e±


Pull e
20
15
10
5
0
−5
−10
−15
−20
−20
−15
−10
−5
0
5
10
15
20
Pull μ
×10³
12
10
8
6
4
2
0

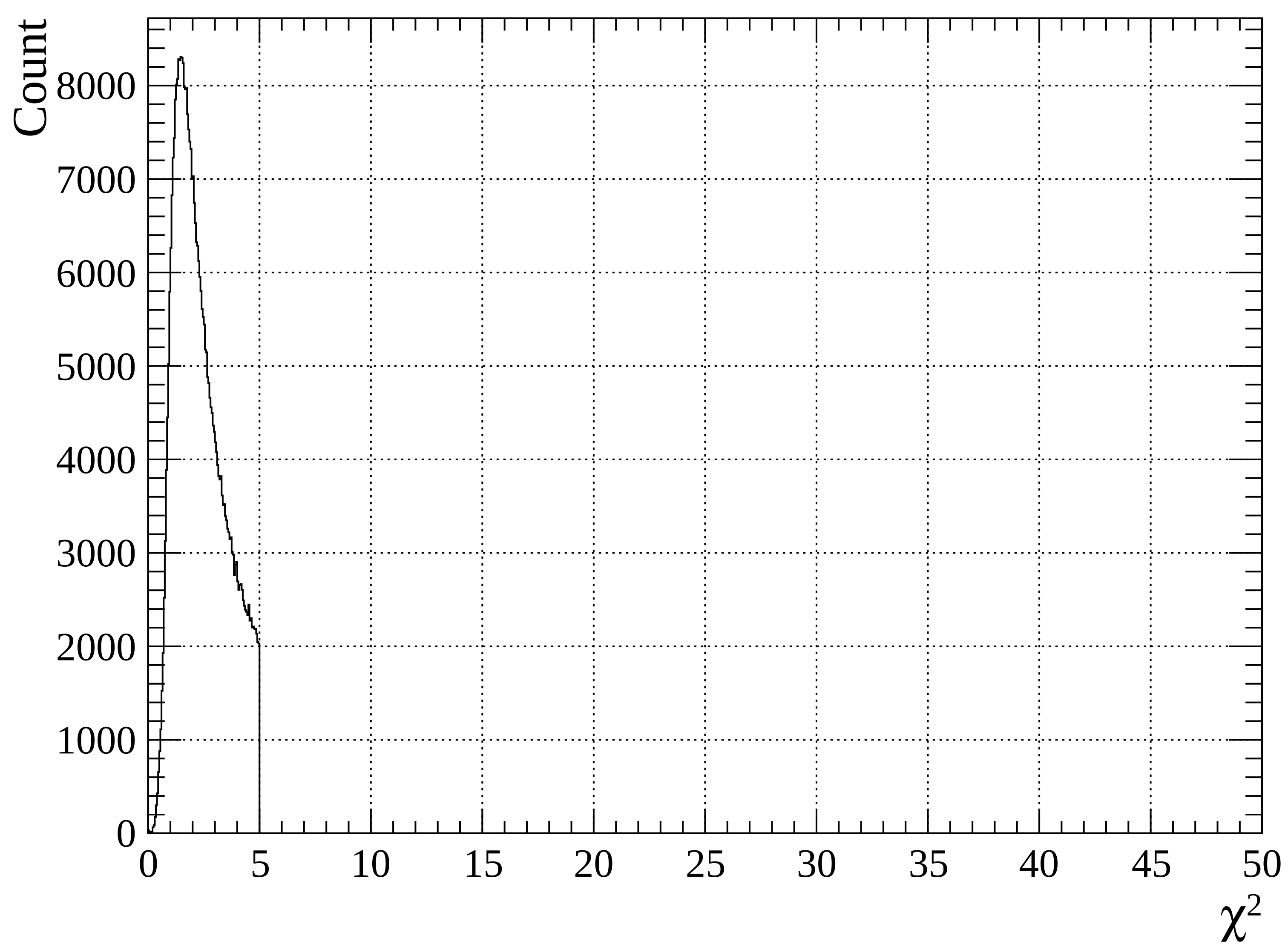


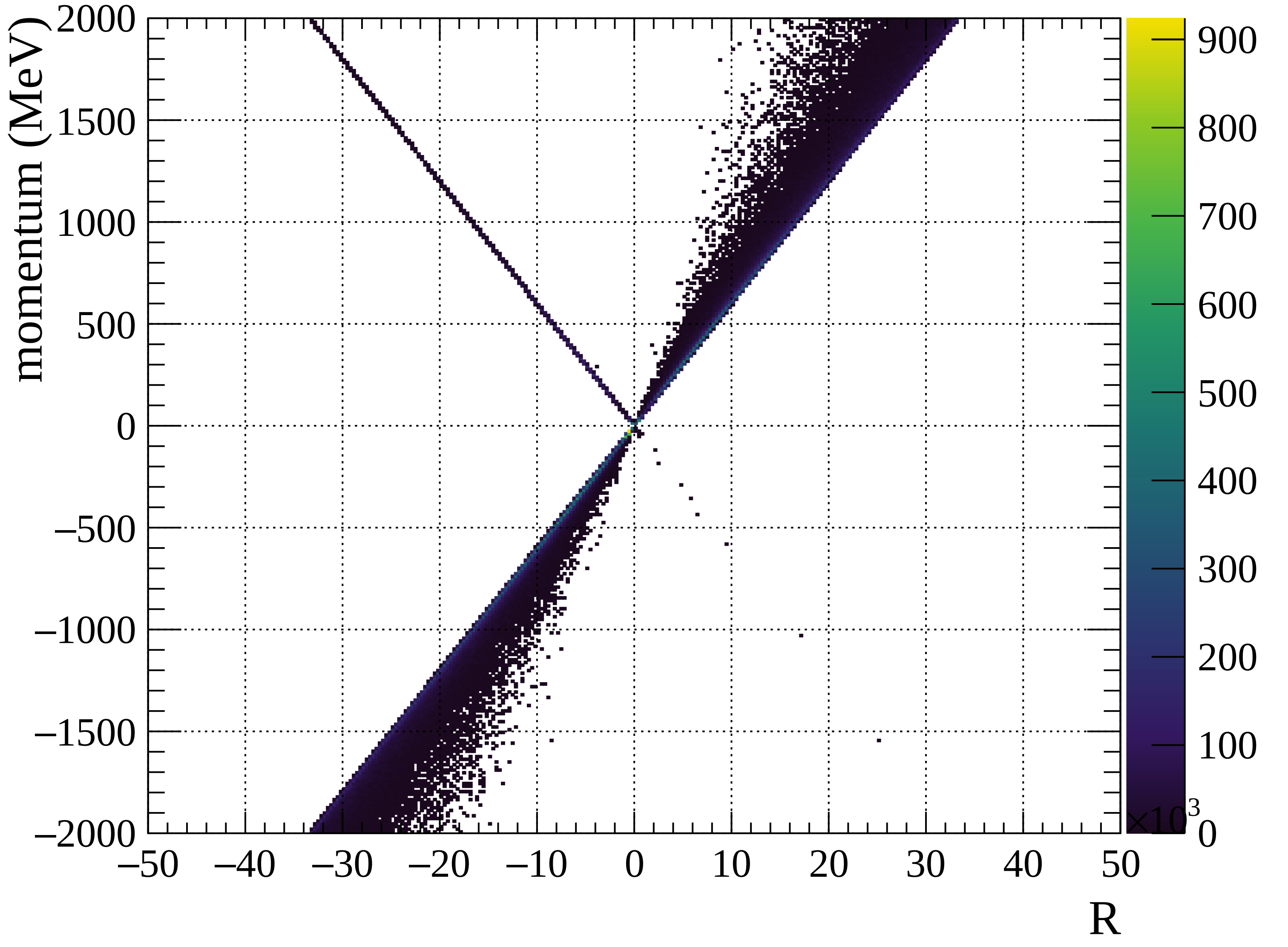
×10^3
Count
70
60
50
40
30
20
10
0
−20
−15
−10
−5
0
5
10
15
20
Pull
μ hyp.
e hyp.
p hyp.
S(μ/e) = 2.03 ± 0.00
μ_μ = 0.138 ± 0.002
σ_μ = 0.903 ± 0.002
μ_e = -1.937 ± 0.002
σ_e = 1.127 ± 0.002

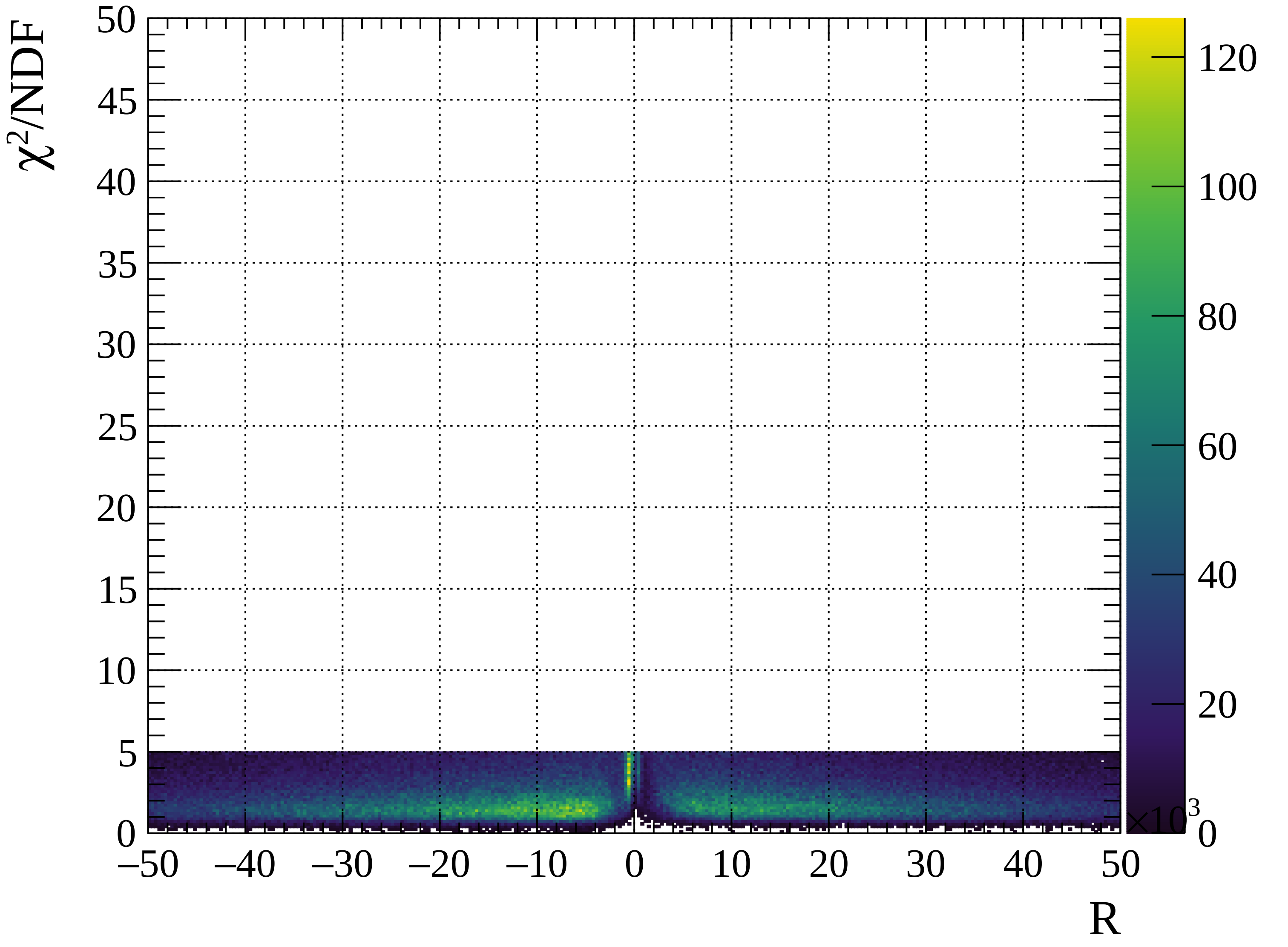

Z
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
−0.8
−1.0
−1.0
−0.8
−0.6
−0.4
−0.2
0.0
0.2
0.4
0.6
0.8
1.0
X



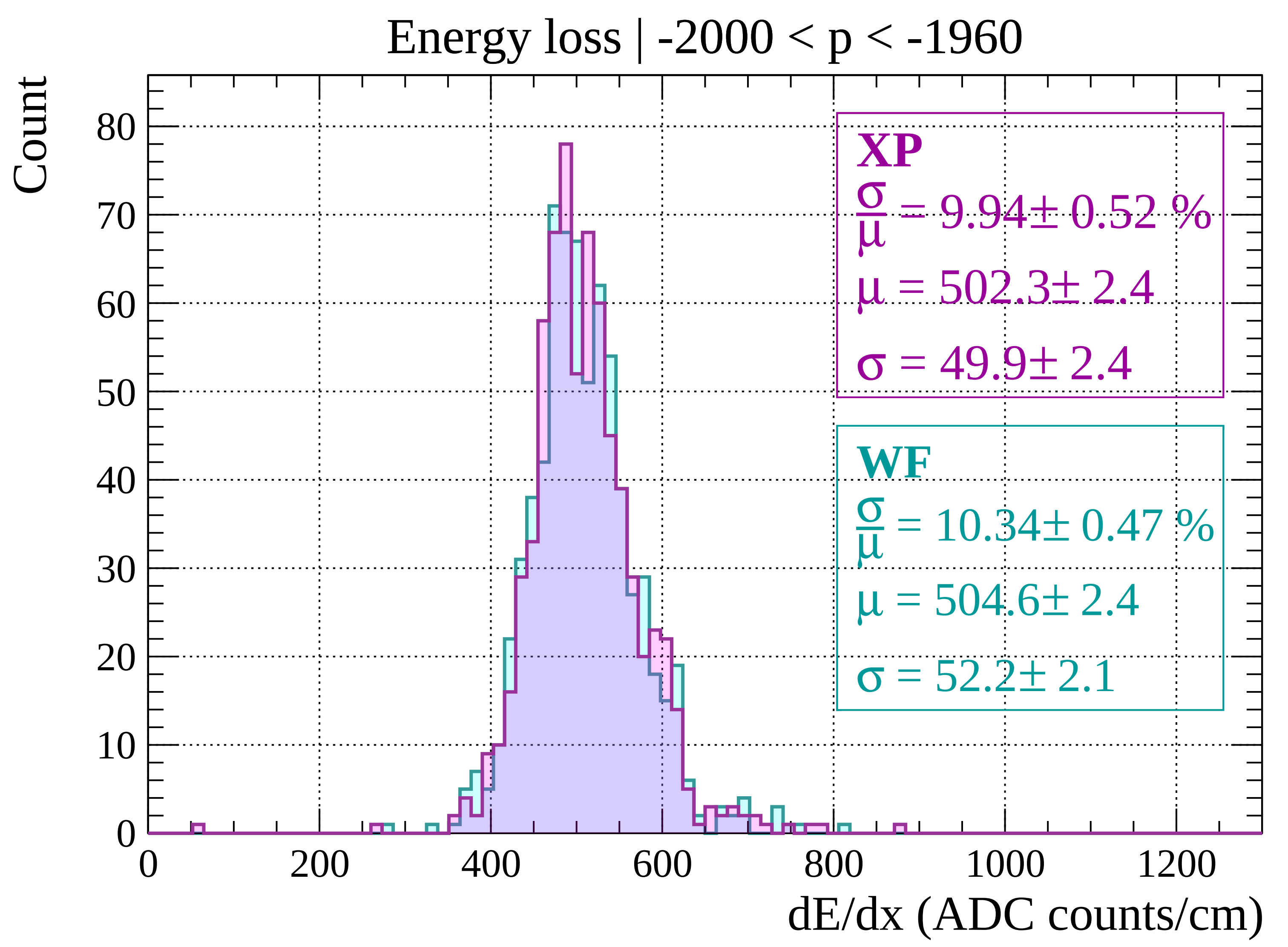
×10
Count
0
5
10
15
20
25
0
20
40
60
80
100
120
140
160
track length (cm)


Count
8000
7000
6000
5000
4000
3000
2000
1000
0
0
5
10
15
20
25
30
35
40
45
50
χ^2

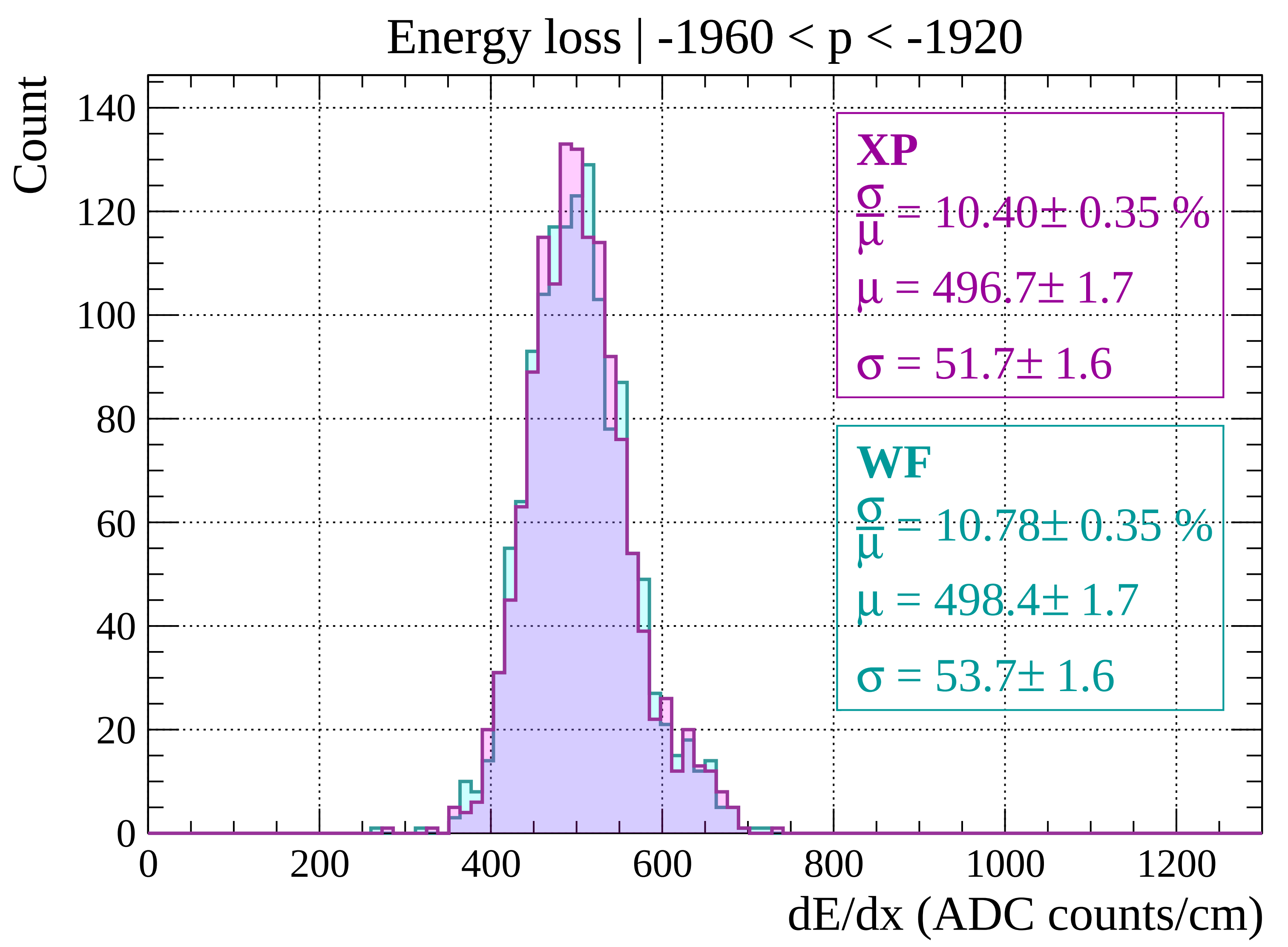

momentum (MeV)
2000
1500
1000
500
0
−500
−1000
−1500
−2000
−50
−40
−30
−20
−10
0
10
20
30
40
50
R
900
800
700
600
500
400
300
200
100
0
×10^3



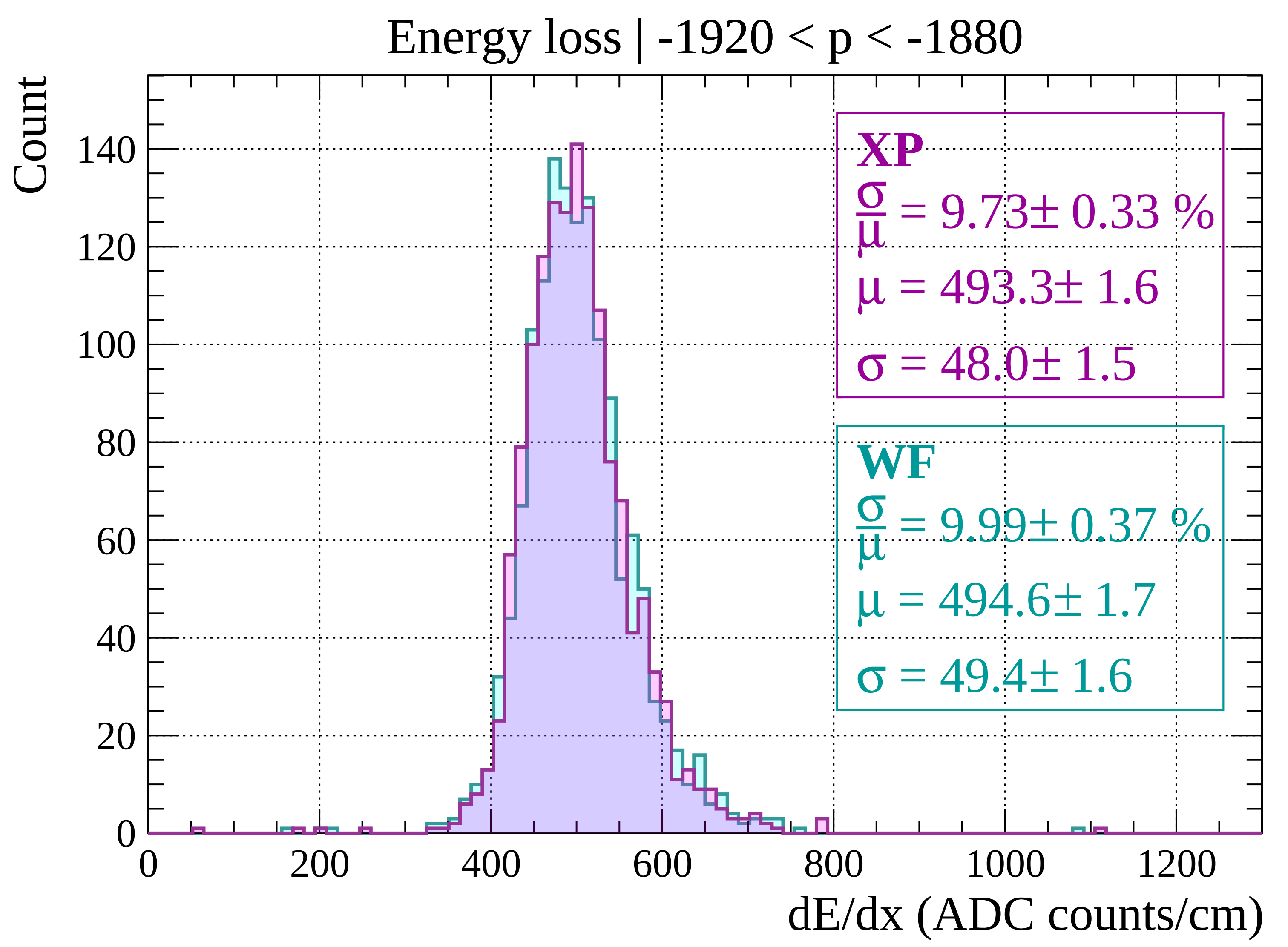
χ^2/NDF
50
45
40
35
30
25
20
15
10
5
0
−50
−40
−30
−20
−10
0
10
20
30
40
50
R
120
100
80
60
40
20
0
$\times 10^3$

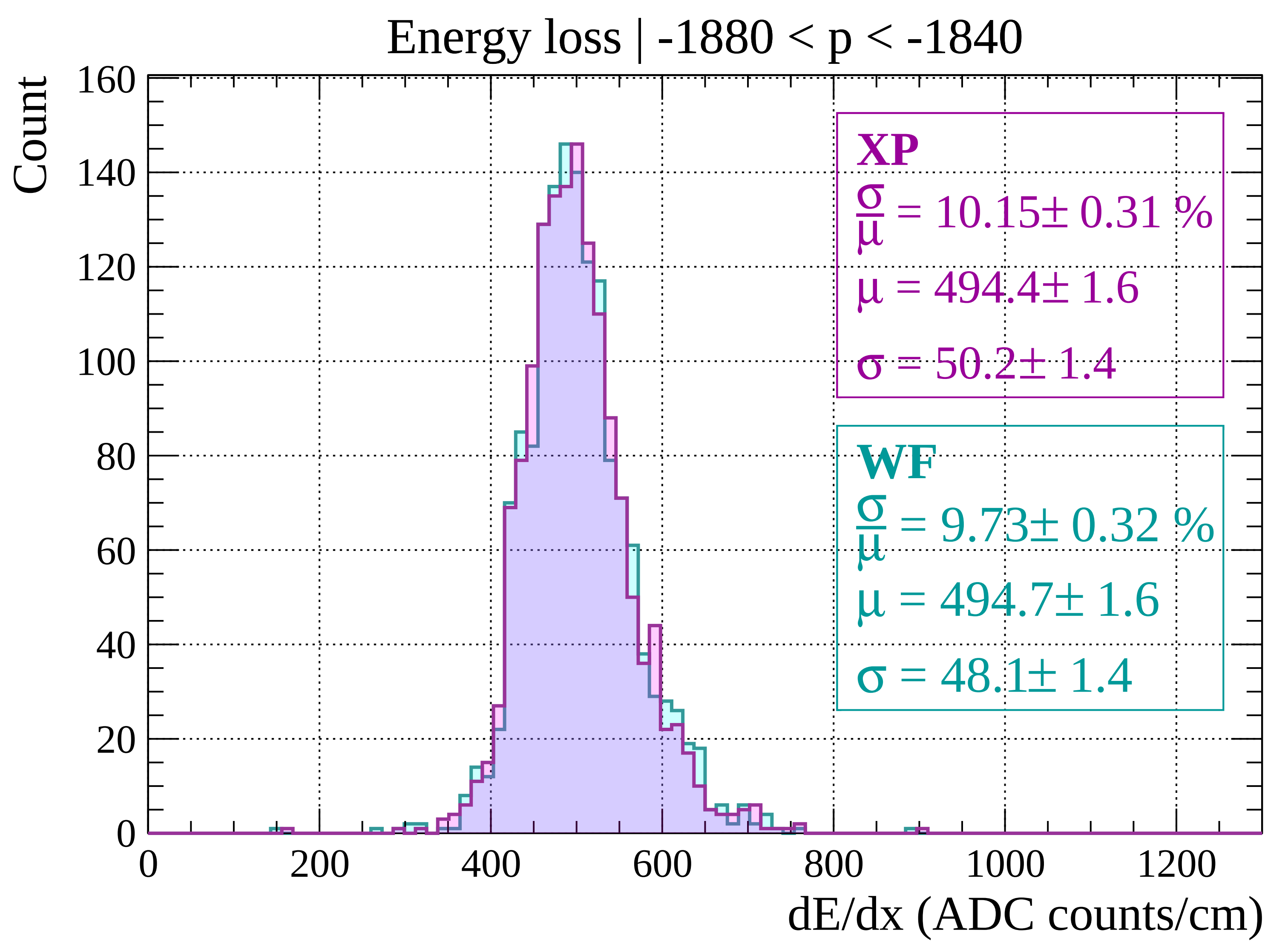


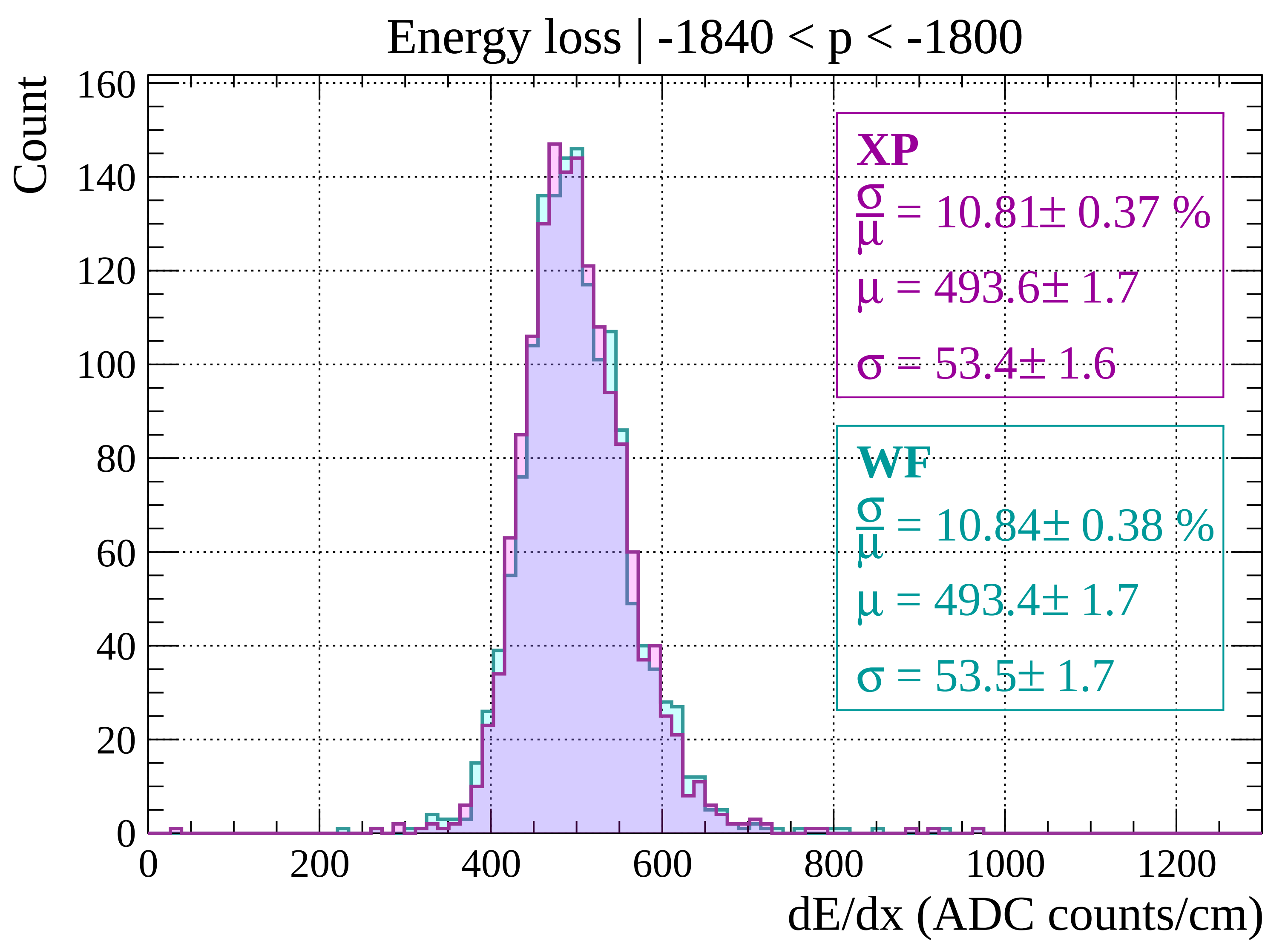
Energy loss | -2000 < p < -1960
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
10
20
30
40
50
60
70
80
XP
σ/μ = 9.94± 0.52 %
μ = 502.3± 2.4
σ = 49.9± 2.4
WF
σ/μ = 10.34± 0.47 %
μ = 504.6± 2.4
σ = 52.2± 2.1



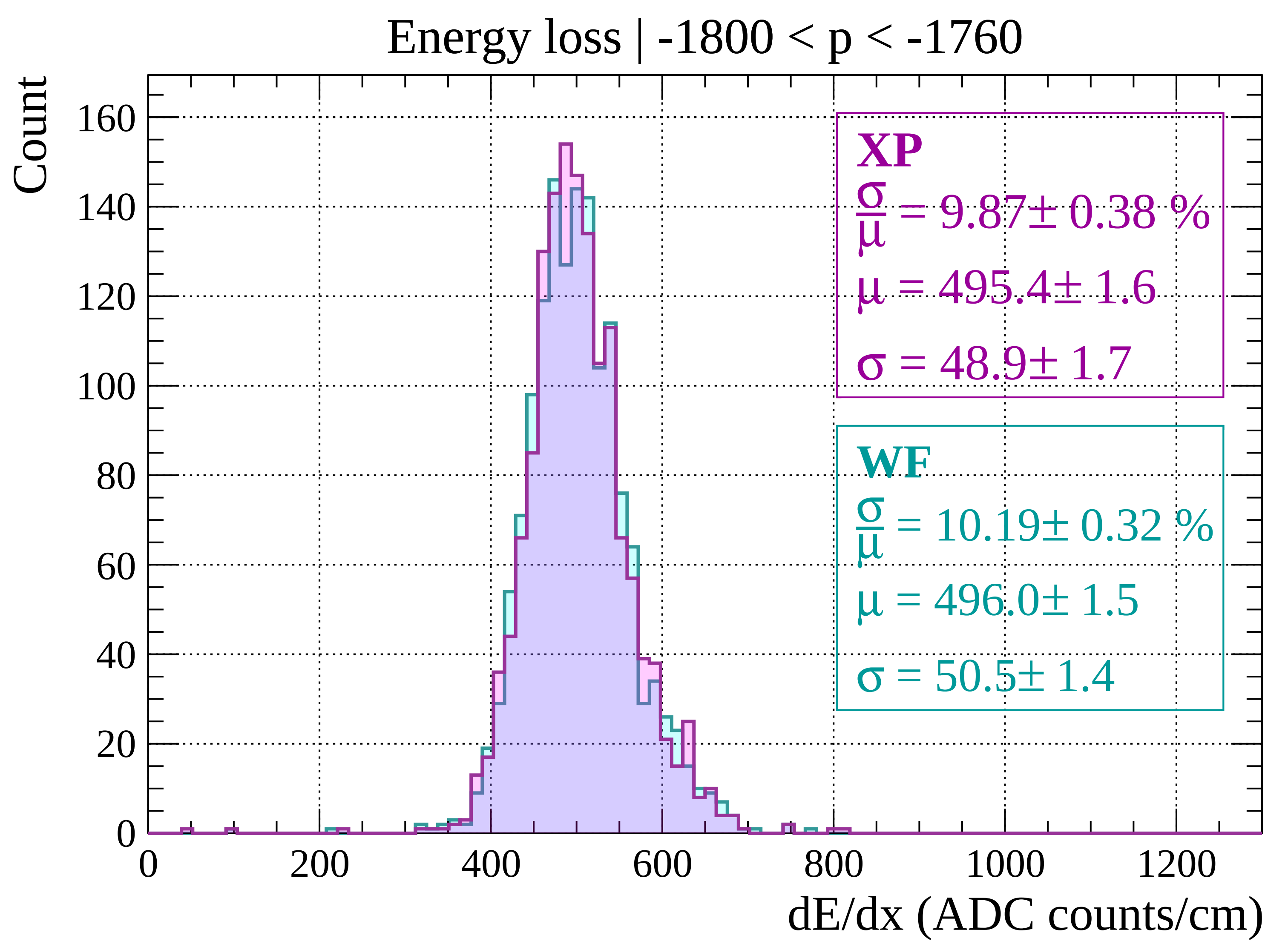
Energy loss | -1960 < p < -1920
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 10.40± 0.35 %
μ = 496.7± 1.7
σ = 51.7± 1.6
WF
σ/μ = 10.78± 0.35 %
μ = 498.4± 1.7
σ = 53.7± 1.6
0
200
400
600
800
1000
1200
20
40
60
80
100
120
140


Energy loss | -1920 < p < -1880
Count
140
120
100
80
60
40
20
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 9.73± 0.33 %
μ = 493.3± 1.6
σ = 48.0± 1.5
WF
σ/μ = 9.99± 0.37 %
μ = 494.6± 1.7
σ = 49.4± 1.6

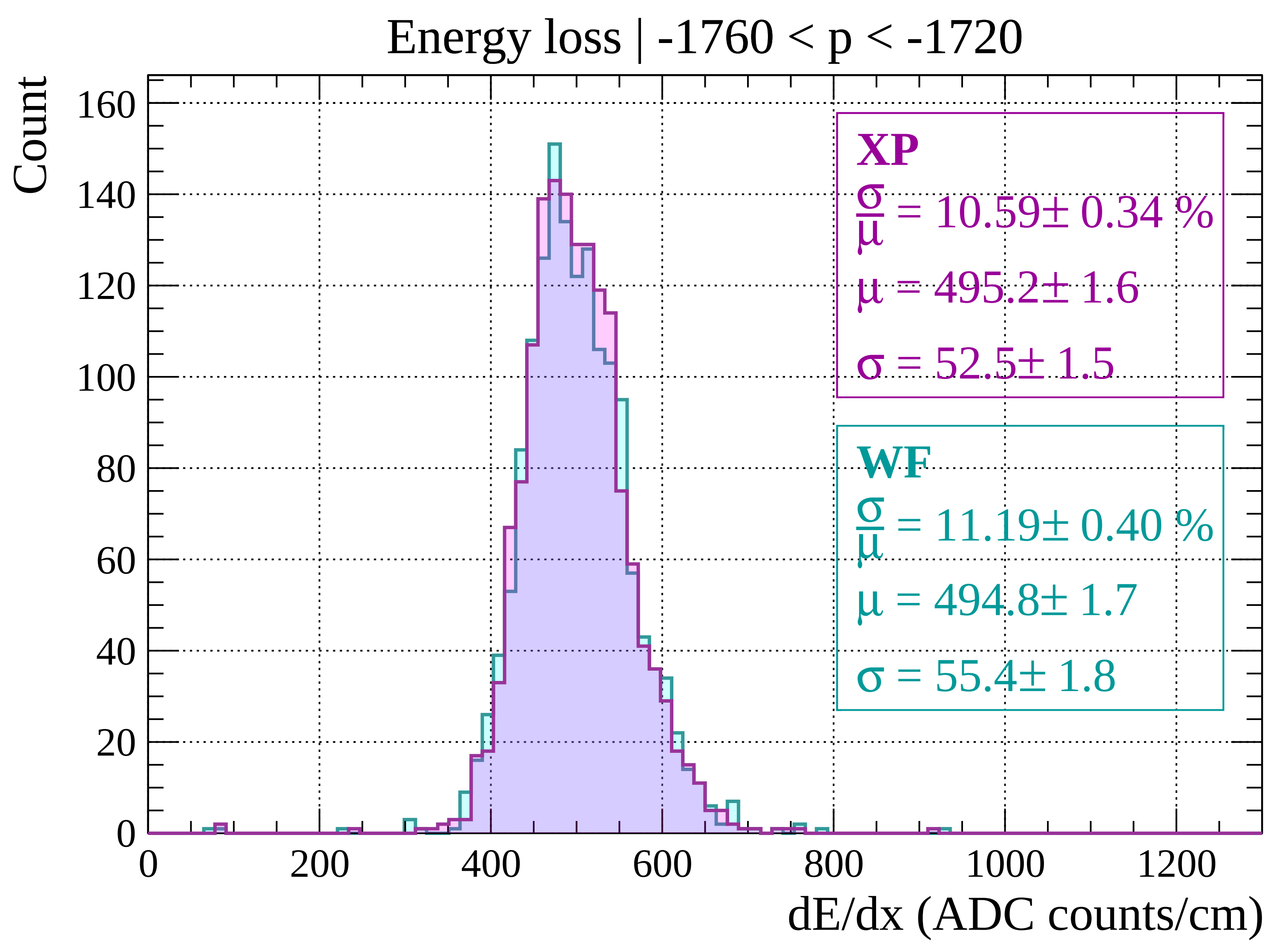

Energy loss | -1880 < p < -1840
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
0
200
400
600
800
1000
1200
XP
σ/μ = 10.15± 0.31 %
μ = 494.4± 1.6
σ = 50.2± 1.4
WF
σ/μ = 9.73± 0.32 %
μ = 494.7± 1.6
σ = 48.1± 1.4

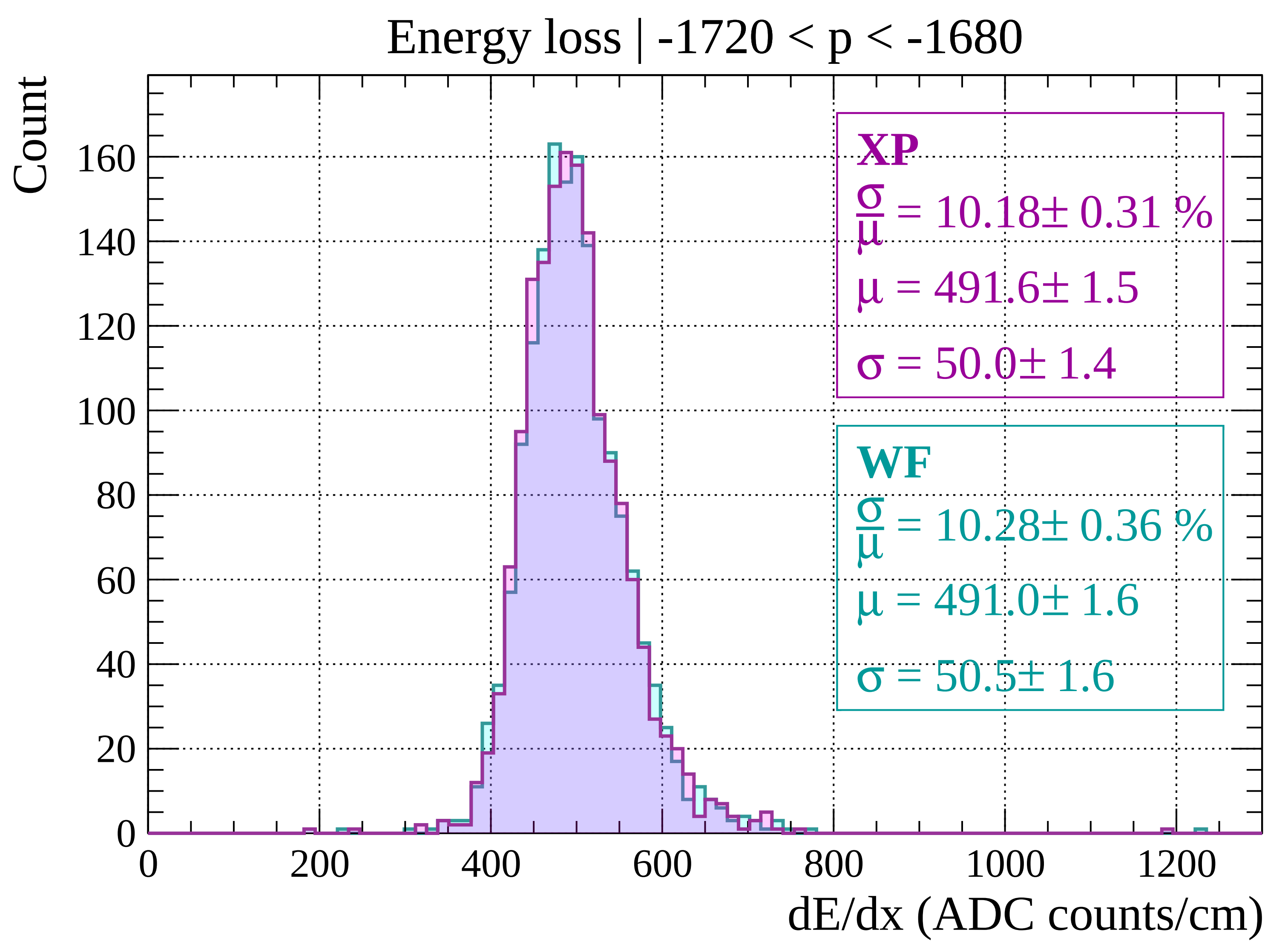


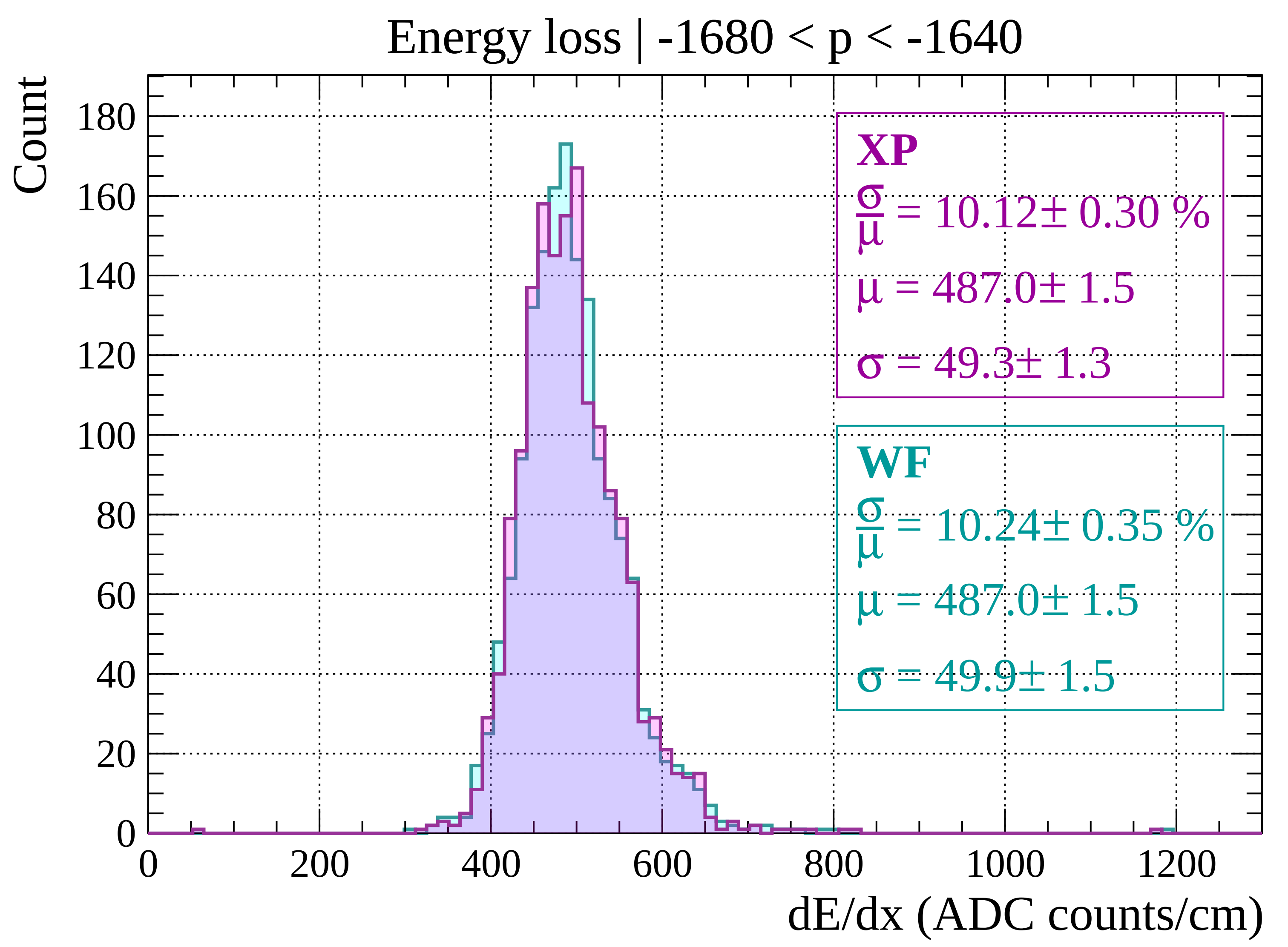
Energy loss | -1840 < p < -1800
Count
160
140
120
100
80
60
40
20
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 10.81± 0.37 %
μ = 493.6± 1.7
σ = 53.4± 1.6
WF
σ/μ = 10.84± 0.38 %
μ = 493.4± 1.7
σ = 53.5± 1.7

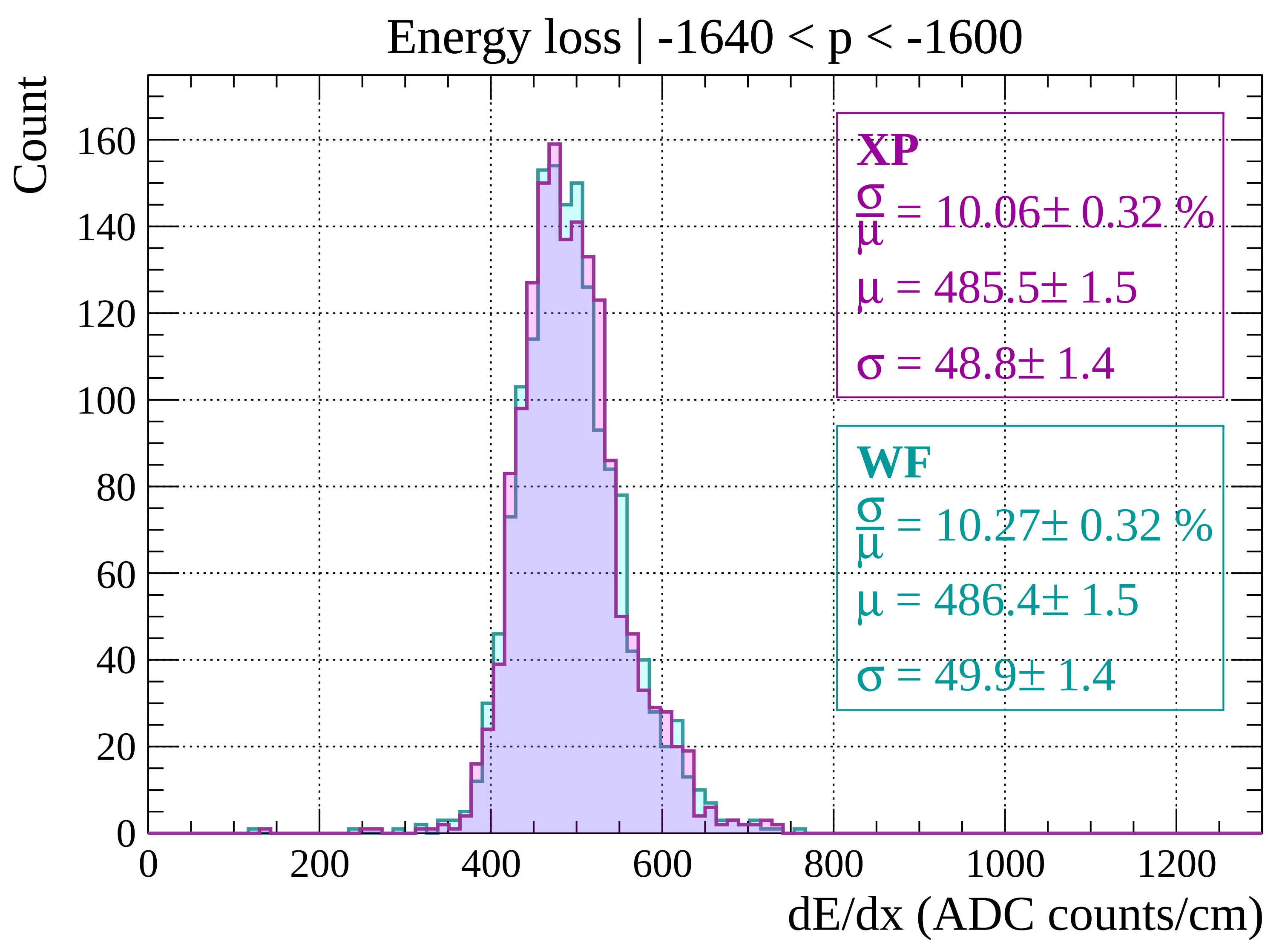


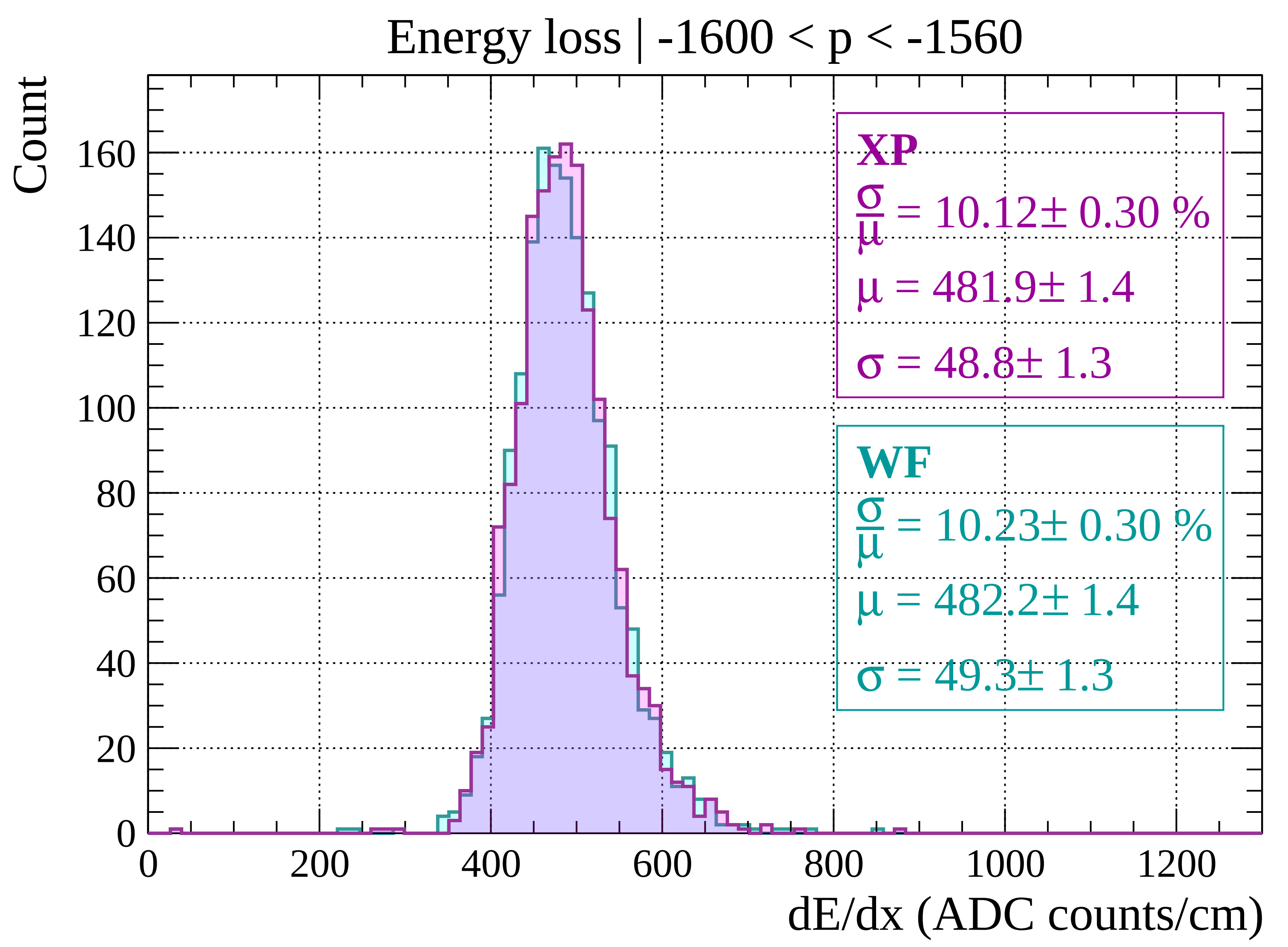
Energy loss | -1800 < p < -1760
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 9.87± 0.38 %
μ = 495.4± 1.6
σ = 48.9± 1.7
WF
$\frac{\sigma}{\mu}$ = 10.19± 0.32 %
μ = 496.0± 1.5
σ = 50.5± 1.4


Energy loss | -1760 < p < -1720
Count
160
140
120
100
80
60
40
20
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 10.59± 0.34 %
μ = 495.2± 1.6
σ = 52.5± 1.5
WF
σ/μ = 11.19± 0.40 %
μ = 494.8± 1.7
σ = 55.4± 1.8

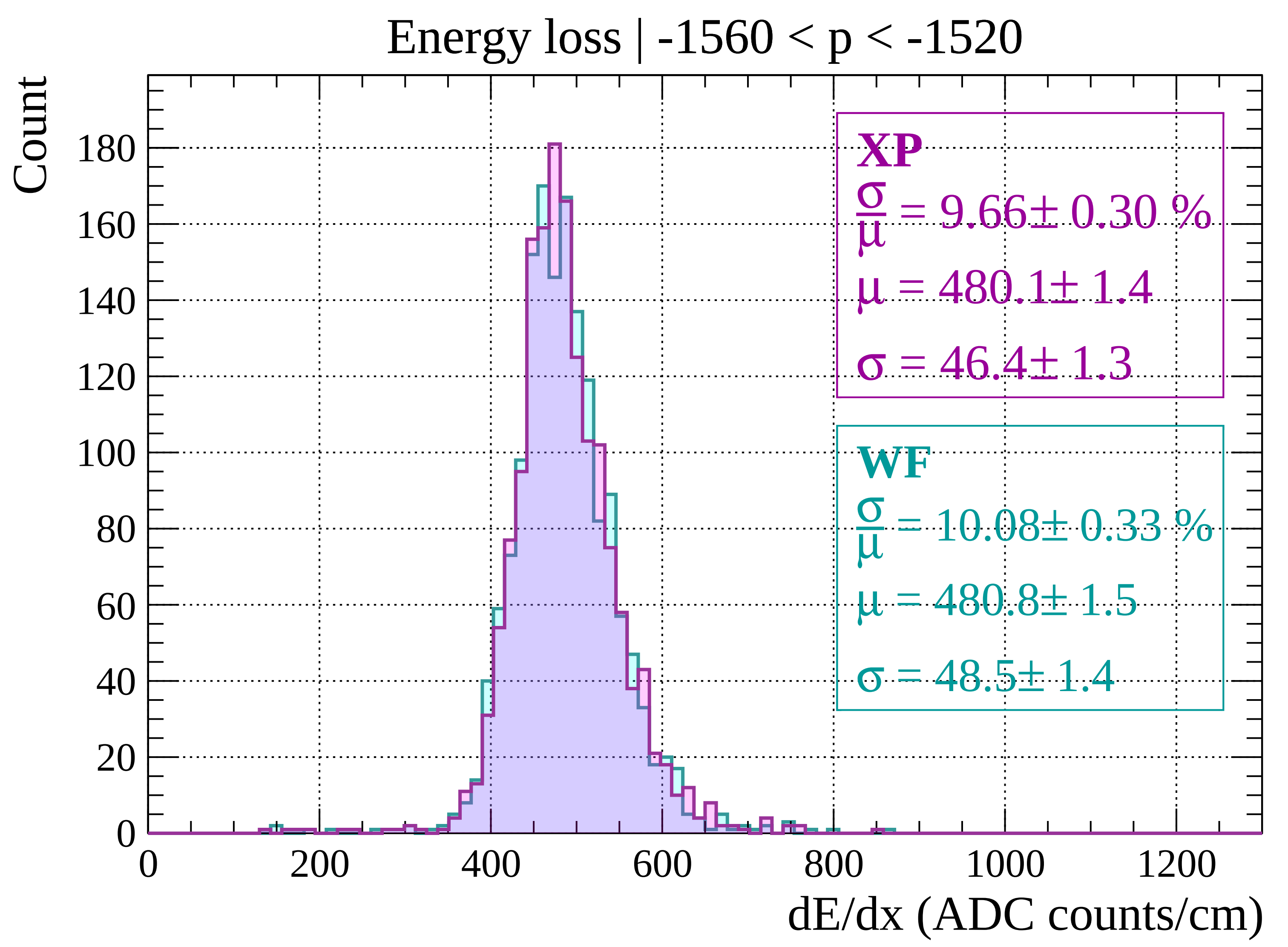

Energy loss | -1720 < p < -1680
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
0
200
400
600
800
1000
1200
XP
σ/μ = 10.18± 0.31 %
μ = 491.6± 1.5
σ = 50.0± 1.4
WF
σ/μ = 10.28± 0.36 %
μ = 491.0± 1.6
σ = 50.5± 1.6

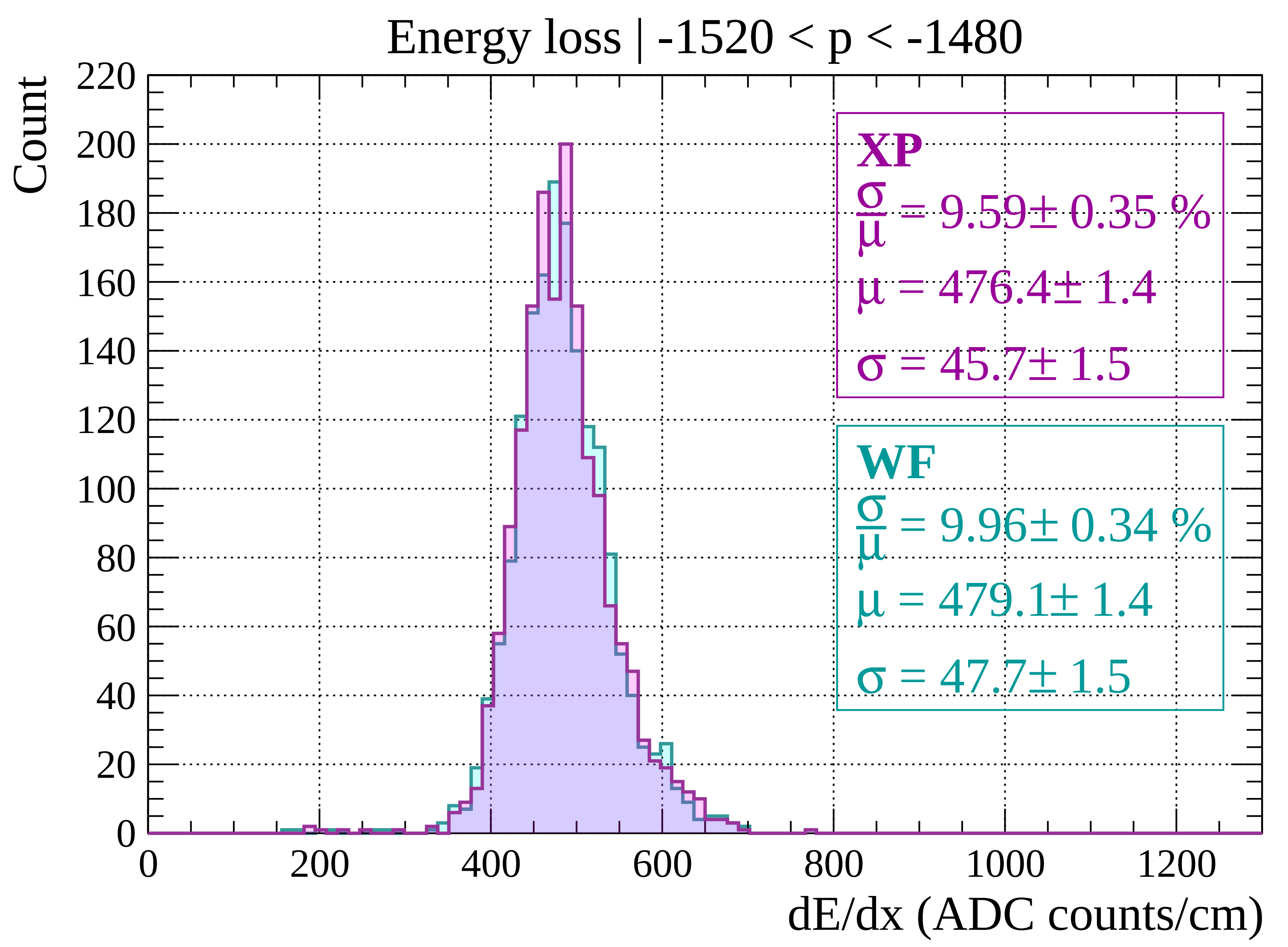

Energy loss | -1680 < p < -1640
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 10.12± 0.30 %
μ = 487.0± 1.5
σ = 49.3± 1.3
WF
σ/μ = 10.24± 0.35 %
μ = 487.0± 1.5
σ = 49.9± 1.5
0
20
40
60
80
100
120
140
160
180
0
200
400
600
800
1000
1200


Energy loss | -1640 < p < -1600
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
0
200
400
600
800
1000
1200
XP
σ/μ = 10.06± 0.32 %
μ = 485.5± 1.5
σ = 48.8± 1.4
WF
σ/μ = 10.27± 0.32 %
μ = 486.4± 1.5
σ = 49.9± 1.4

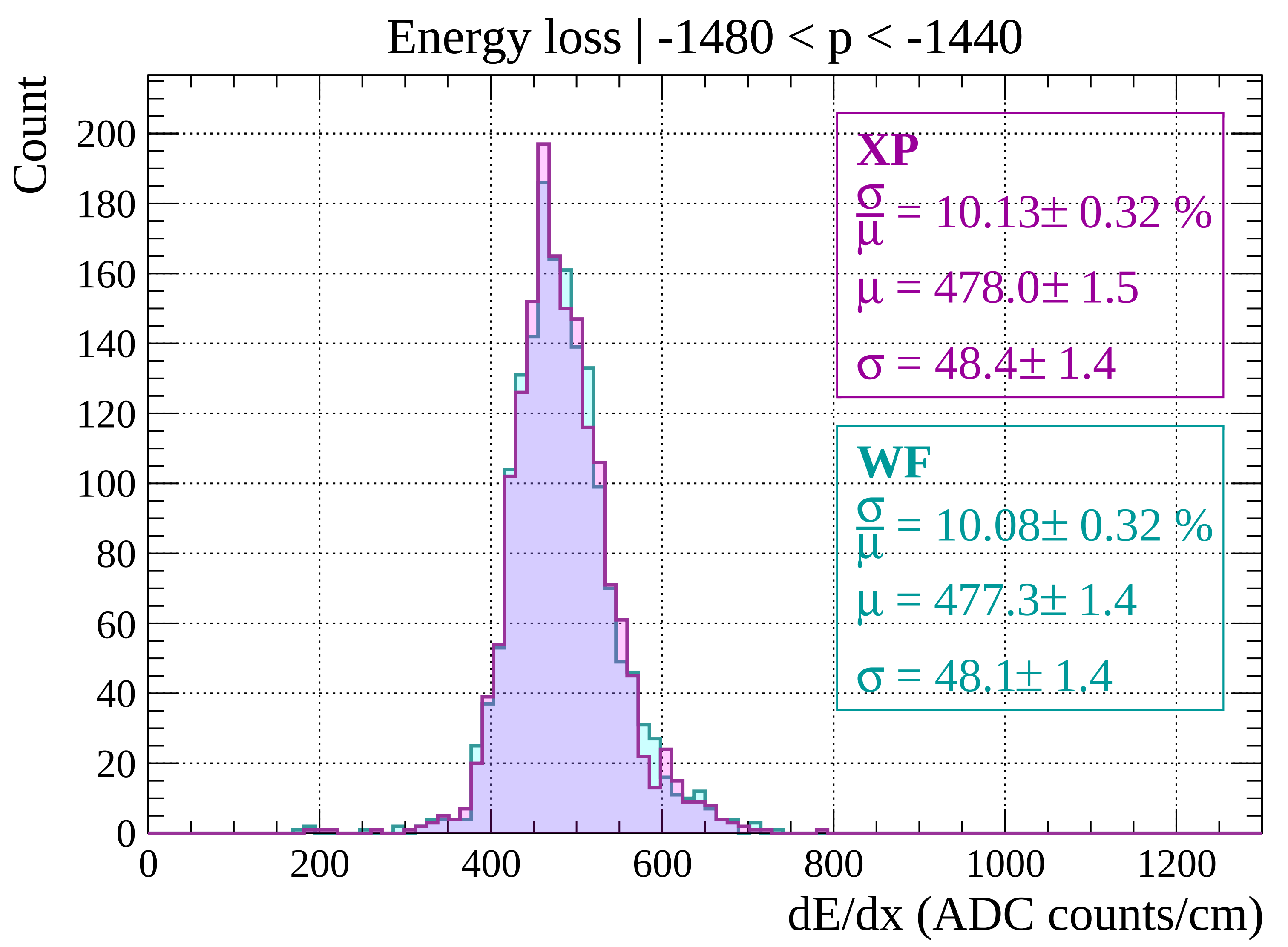


Energy loss | -1600 < p < -1560
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
0
200
400
600
800
1000
1200
XP
σ/μ = 10.12± 0.30 %
μ = 481.9± 1.4
σ = 48.8± 1.3
WF
σ/μ = 10.23± 0.30 %
μ = 482.2± 1.4
σ = 49.3± 1.3

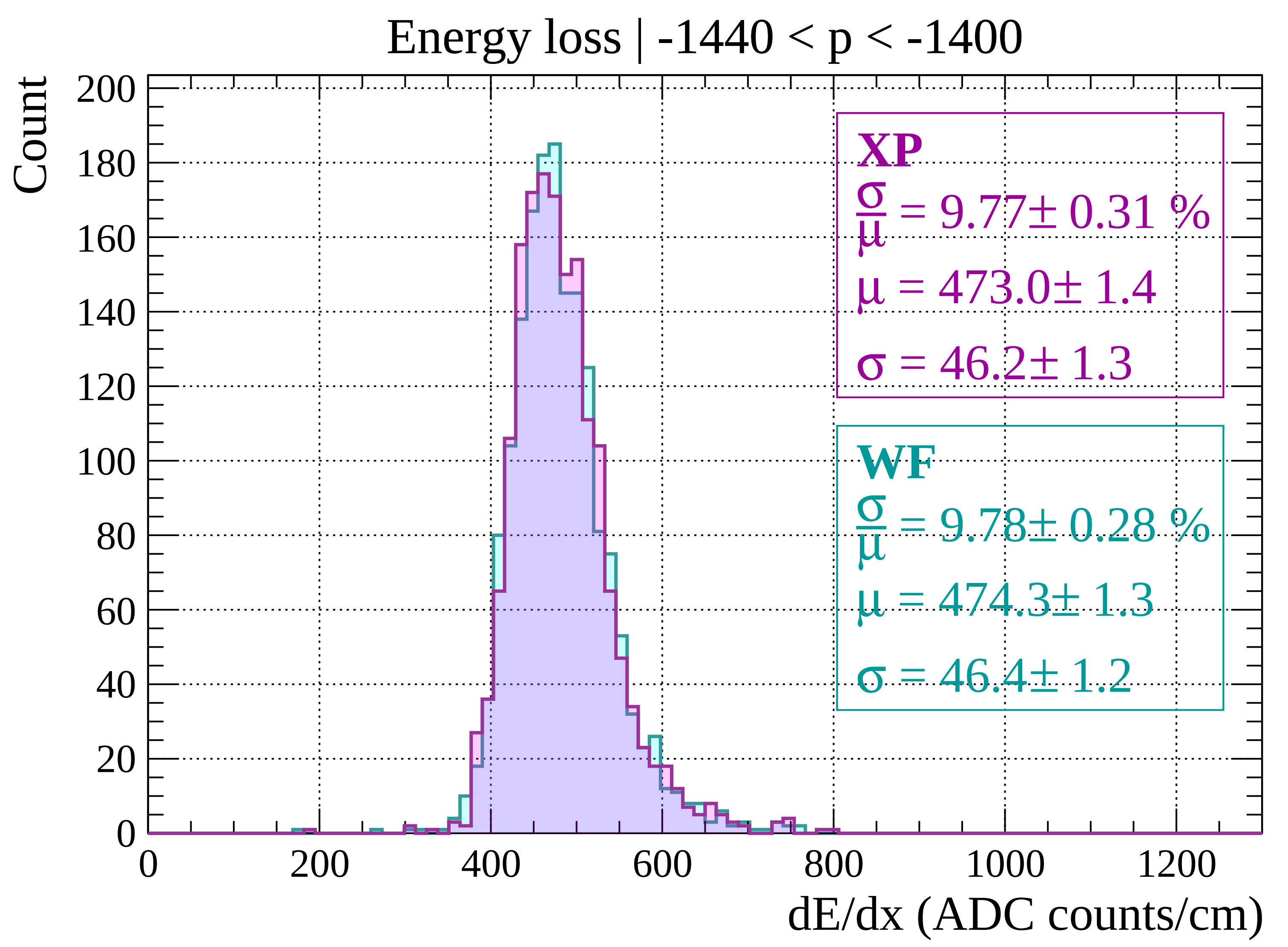

Energy loss | -1560 < p < -1520
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 9.66± 0.30 %
μ = 480.1± 1.4
σ = 46.4± 1.3
WF
$\frac{\sigma}{\mu}$ = 10.08± 0.33 %
μ = 480.8± 1.5
σ = 48.5± 1.4
0
20
40
60
80
100
120
140
160
180
0
200
400
600
800
1000
1200



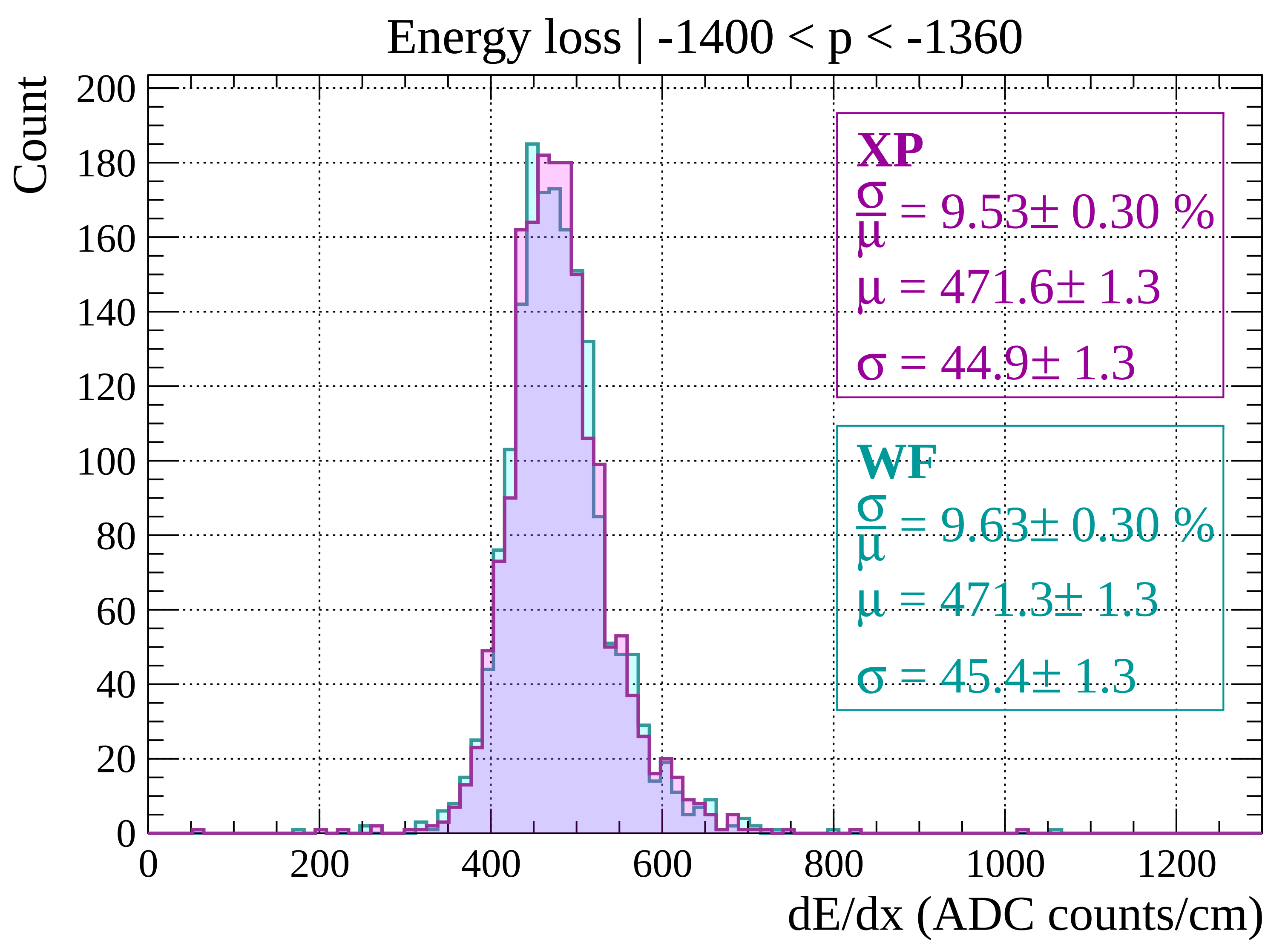
Energy loss | -1520 < p < -1480
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
120
140
160
180
200
220
XP
σ/μ = 9.59± 0.35 %
μ = 476.4± 1.4
σ = 45.7± 1.5
WF
σ/μ = 9.96± 0.34 %
μ = 479.1± 1.4
σ = 47.7± 1.5



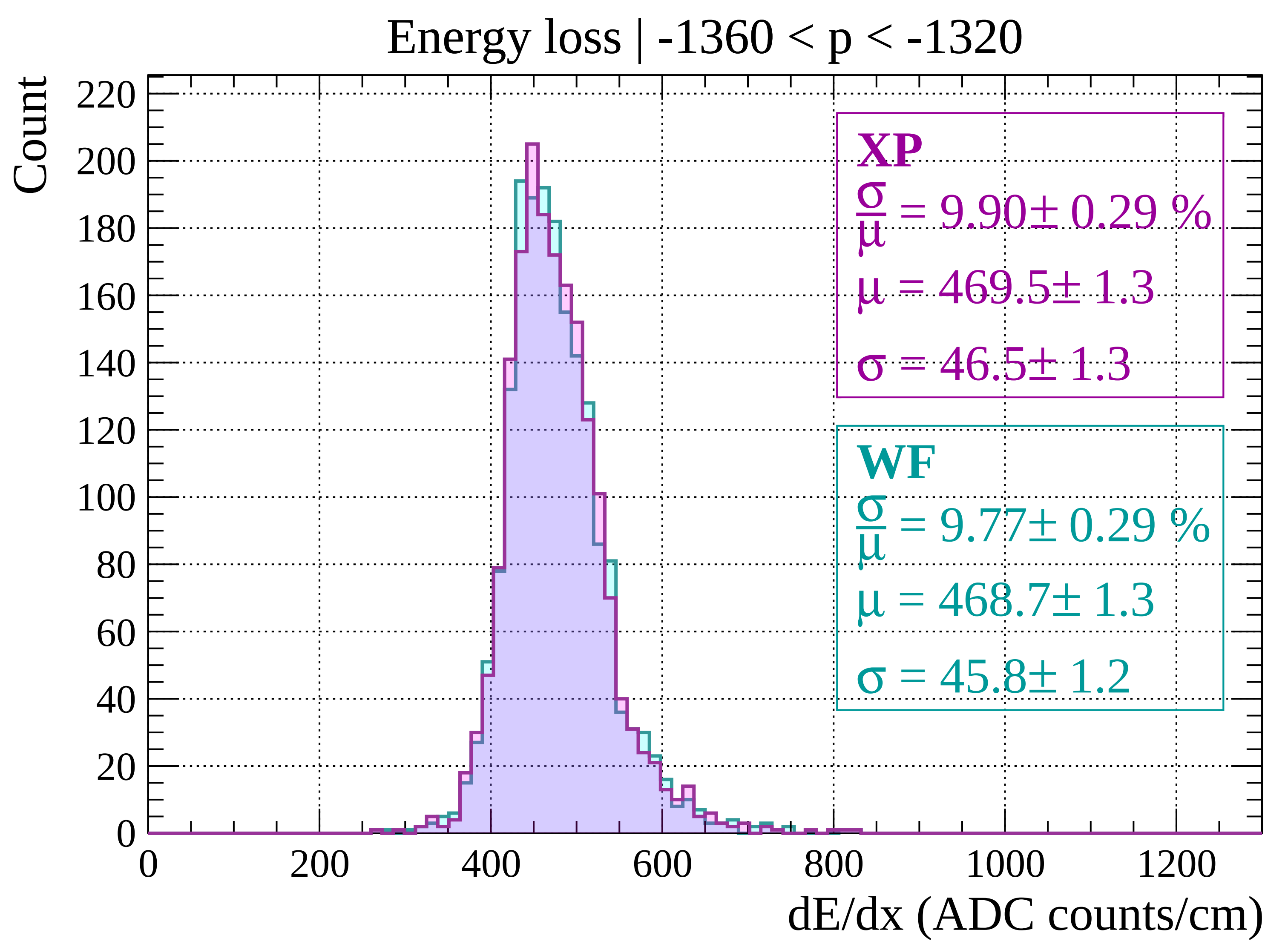
Energy loss | -1480 < p < -1440
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
120
140
160
180
XP
σ/μ = 10.13± 0.32 %
μ = 478.0± 1.5
σ = 48.4± 1.4
WF
σ/μ = 10.08± 0.32 %
μ = 477.3± 1.4
σ = 48.1± 1.4



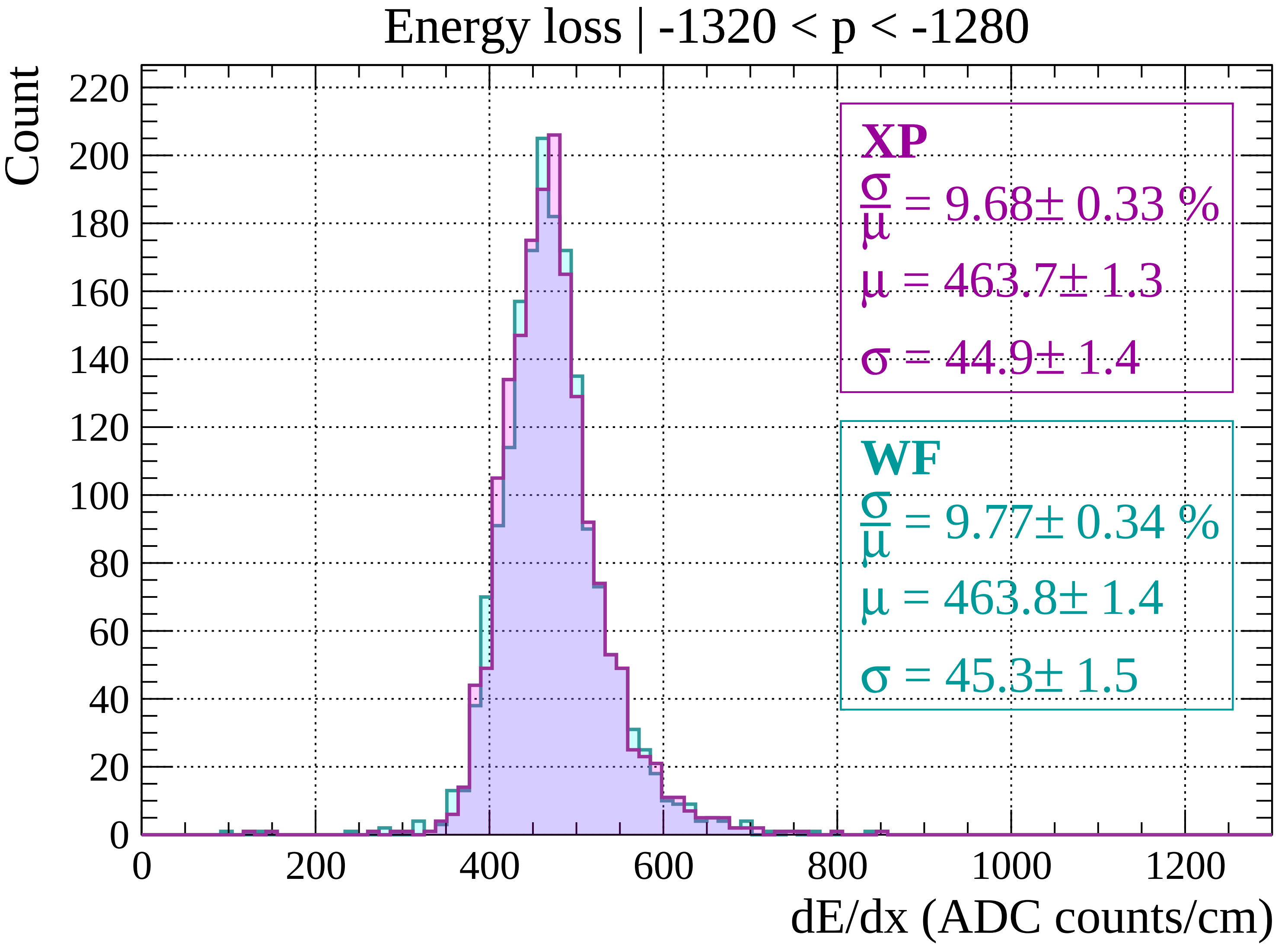
Energy loss | -1440 < p < -1400
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 9.77± 0.31 %
μ = 473.0± 1.4
σ = 46.2± 1.3
WF
σ/μ = 9.78± 0.28 %
μ = 474.3± 1.3
σ = 46.4± 1.2



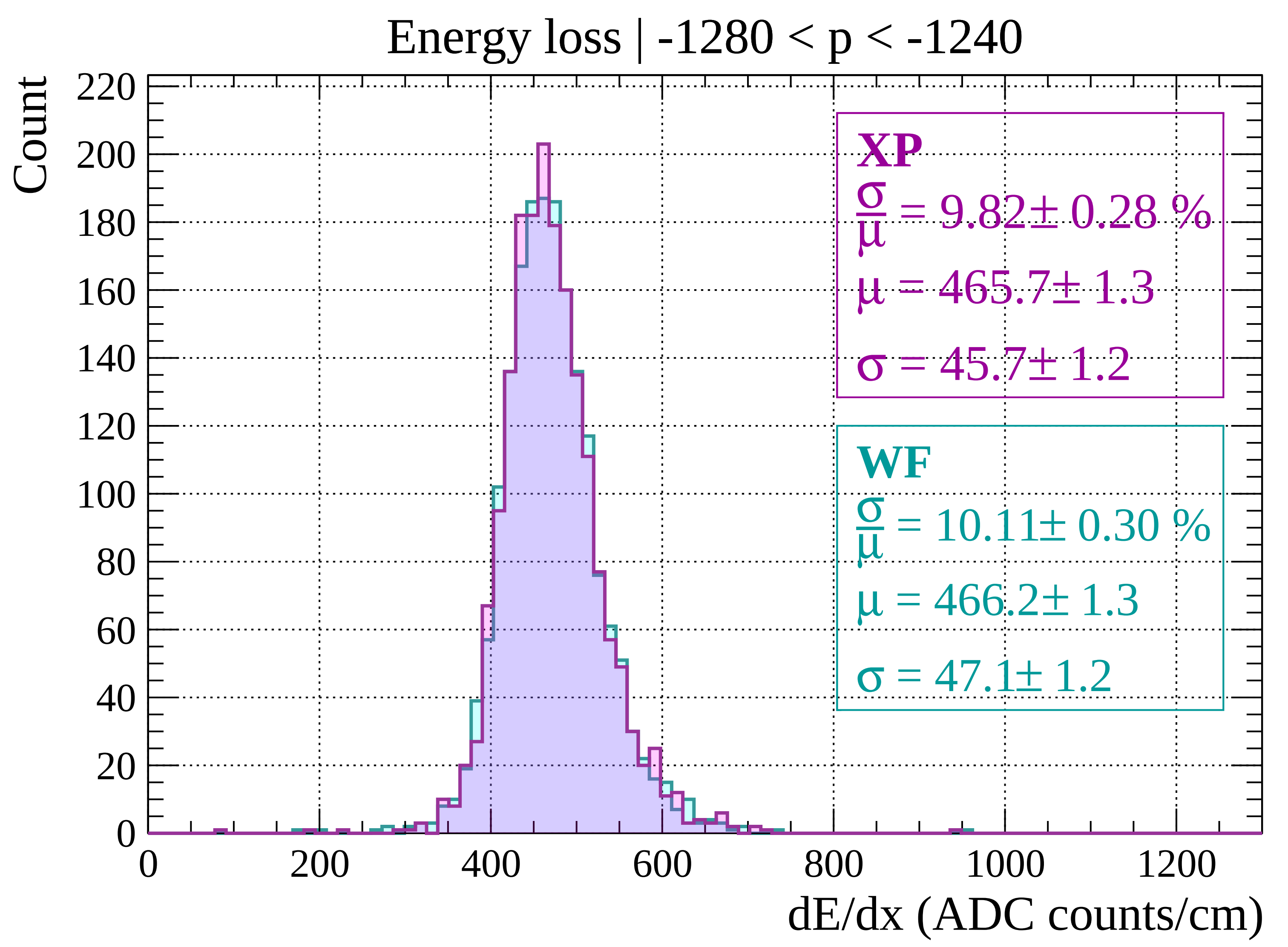
Energy loss | -1400 < p < -1360
Count
200
180
160
140
120
100
80
60
40
20
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 9.53± 0.30 %
μ = 471.6± 1.3
σ = 44.9± 1.3
WF
σ/μ = 9.63± 0.30 %
μ = 471.3± 1.3
σ = 45.4± 1.3



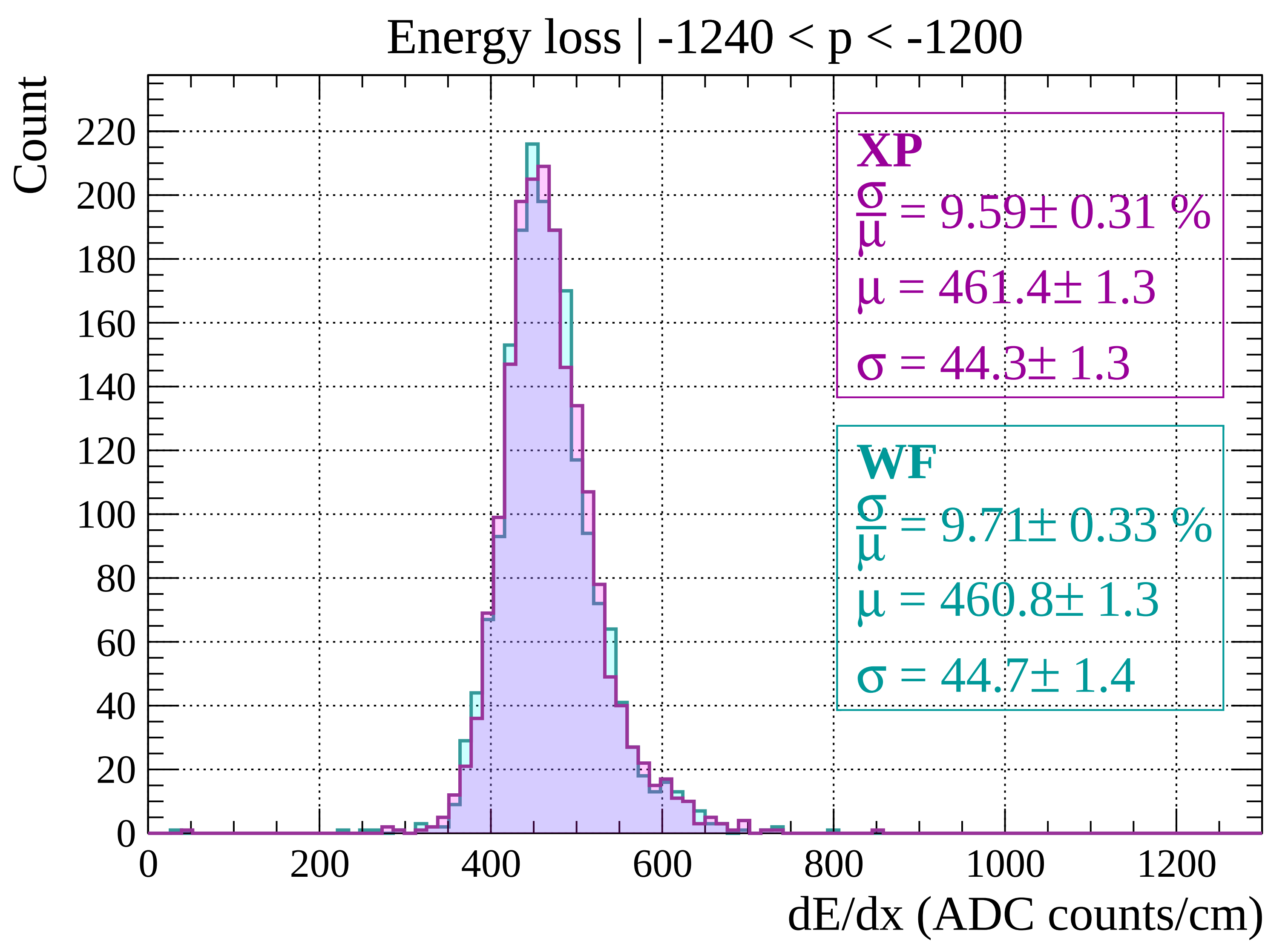
Energy loss | -1360 < p < -1320
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 9.90± 0.29 %
μ = 469.5± 1.3
σ = 46.5± 1.3
WF
$\frac{\sigma}{\mu}$ = 9.77± 0.29 %
μ = 468.7± 1.3
σ = 45.8± 1.2



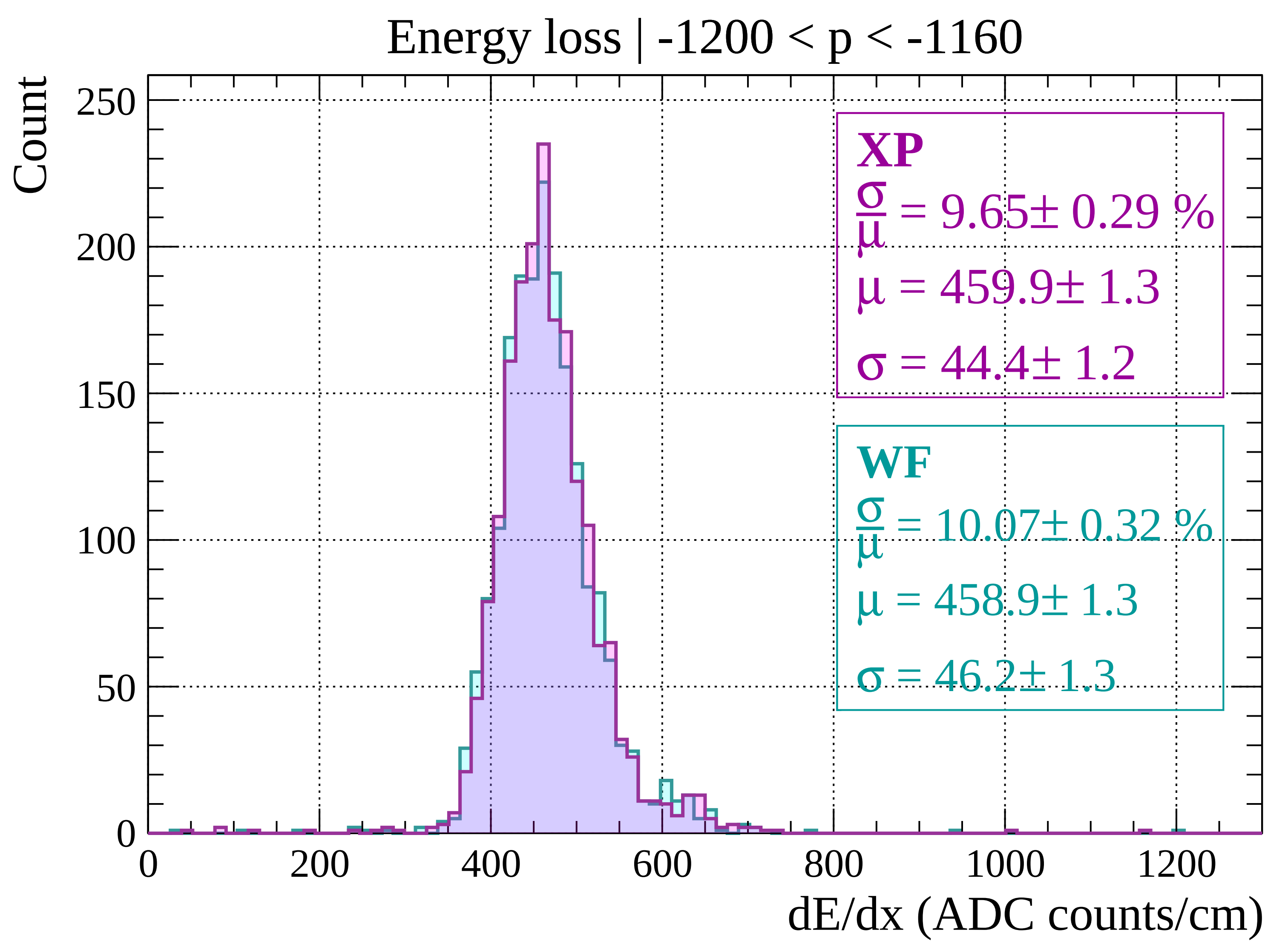
Energy loss | -1320 < p < -1280
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
180
200
220
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 9.68± 0.33 %
μ = 463.7± 1.3
σ = 44.9± 1.4
WF
$\frac{\sigma}{\mu}$ = 9.77± 0.34 %
μ = 463.8± 1.4
σ = 45.3± 1.5

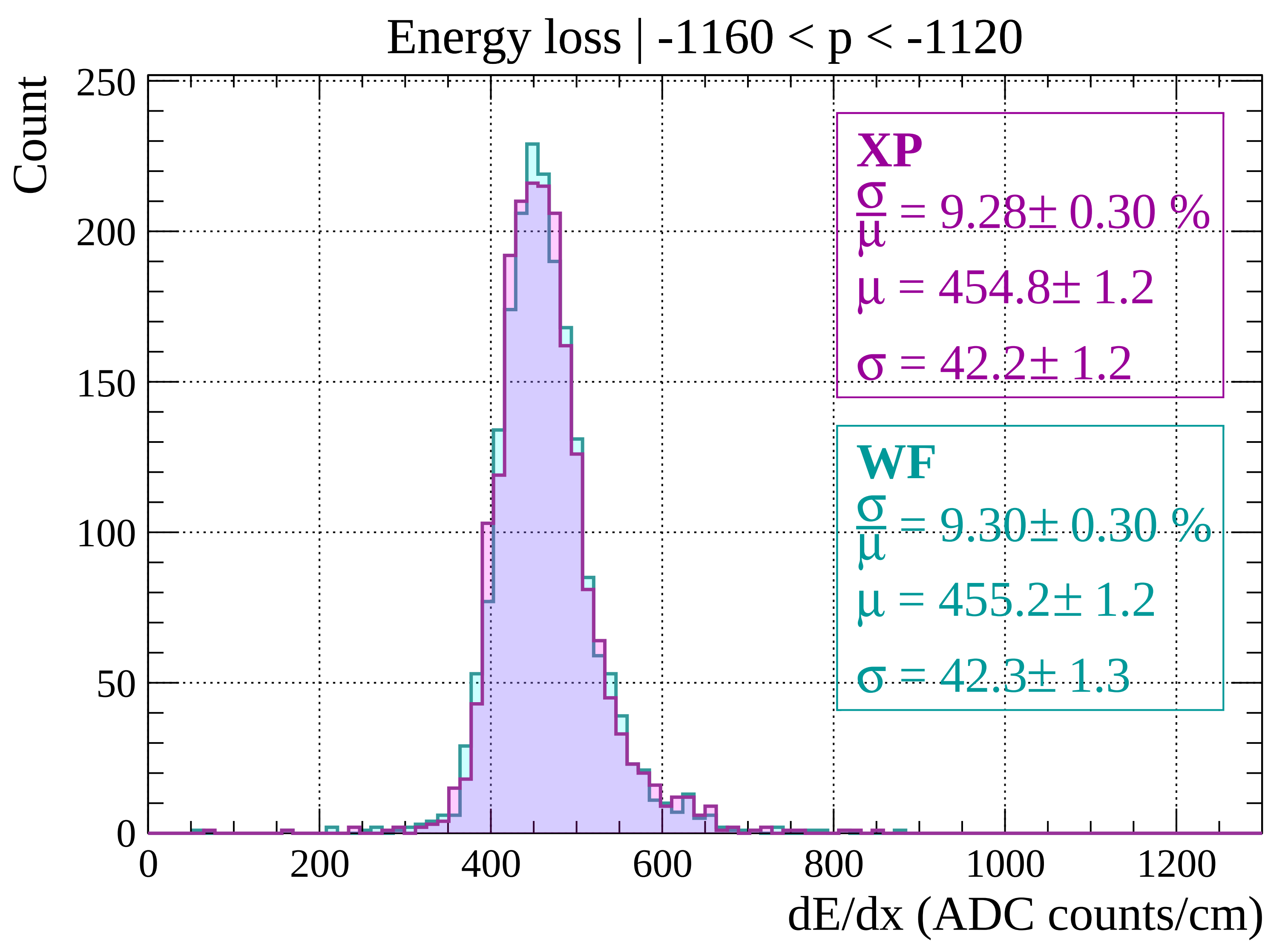


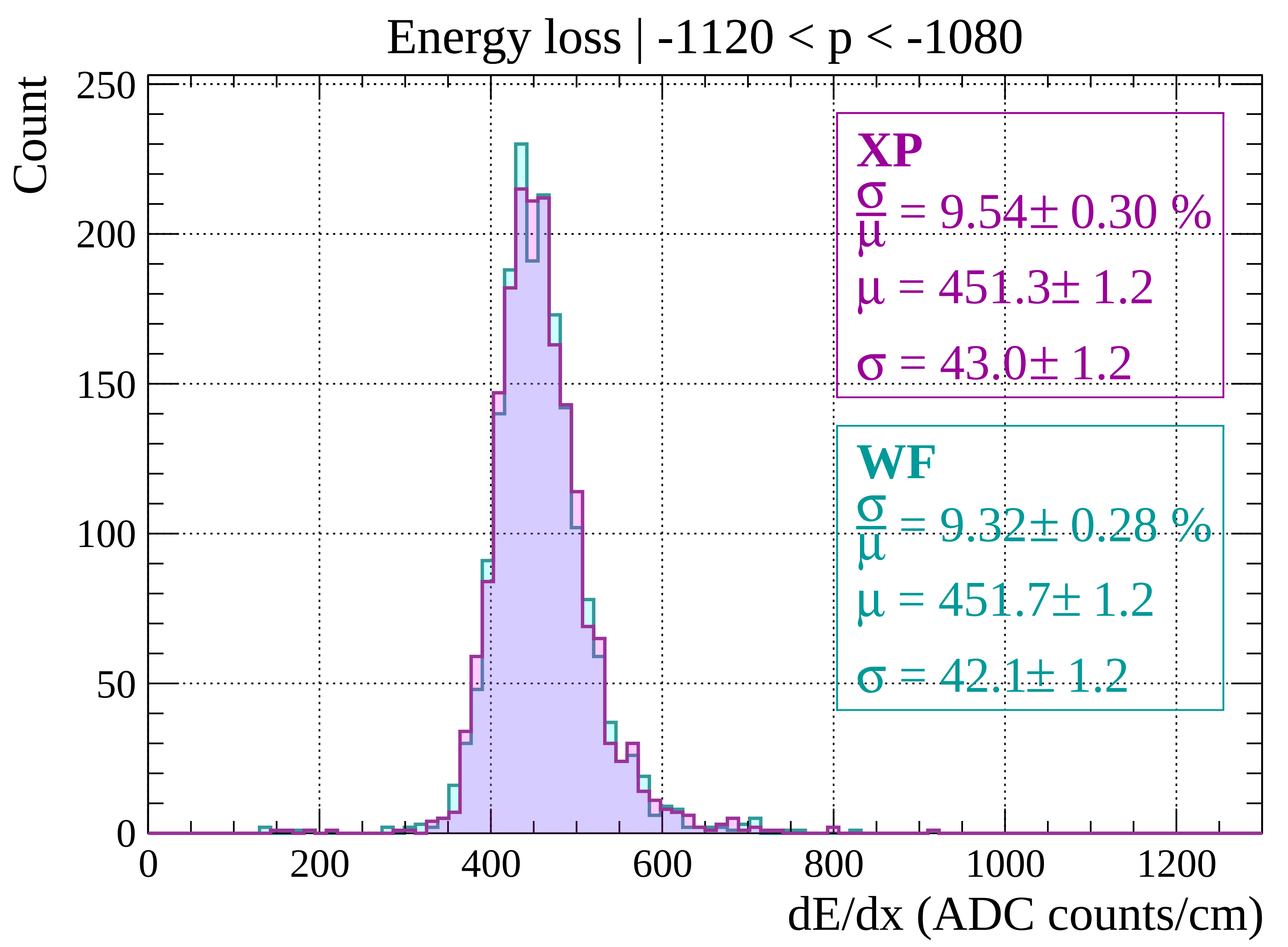
Energy loss | -1280 < p < -1240
Count
220
200
180
160
140
120
100
80
60
40
20
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 9.82± 0.28 %
μ = 465.7± 1.3
σ = 45.7± 1.2
WF
σ/μ = 10.11± 0.30 %
μ = 466.2± 1.3
σ = 47.1± 1.2

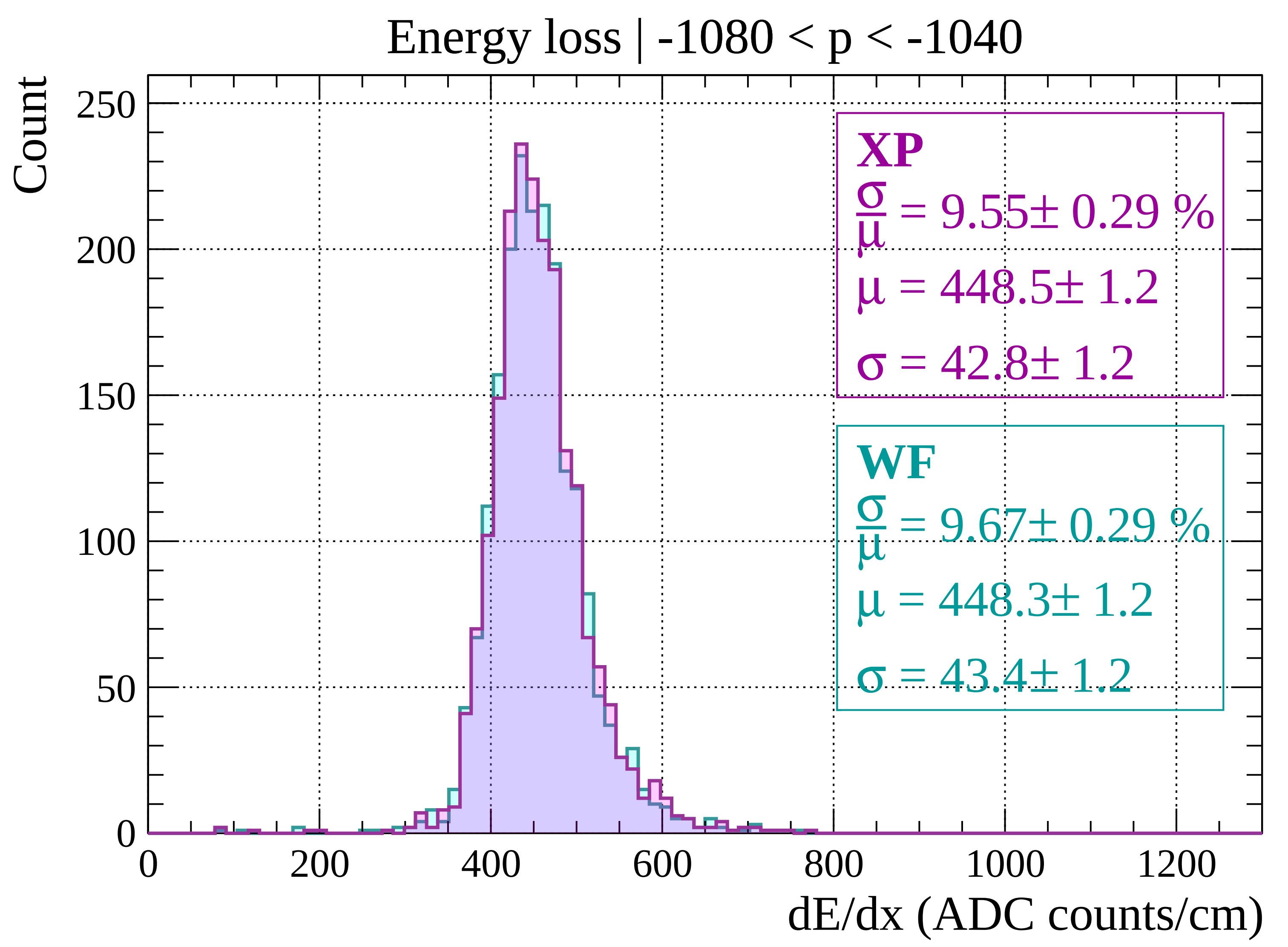


Energy loss | -1240 < p < -1200
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 9.59± 0.31 %
μ = 461.4± 1.3
σ = 44.3± 1.3
WF
σ/μ = 9.71± 0.33 %
μ = 460.8± 1.3
σ = 44.7± 1.4

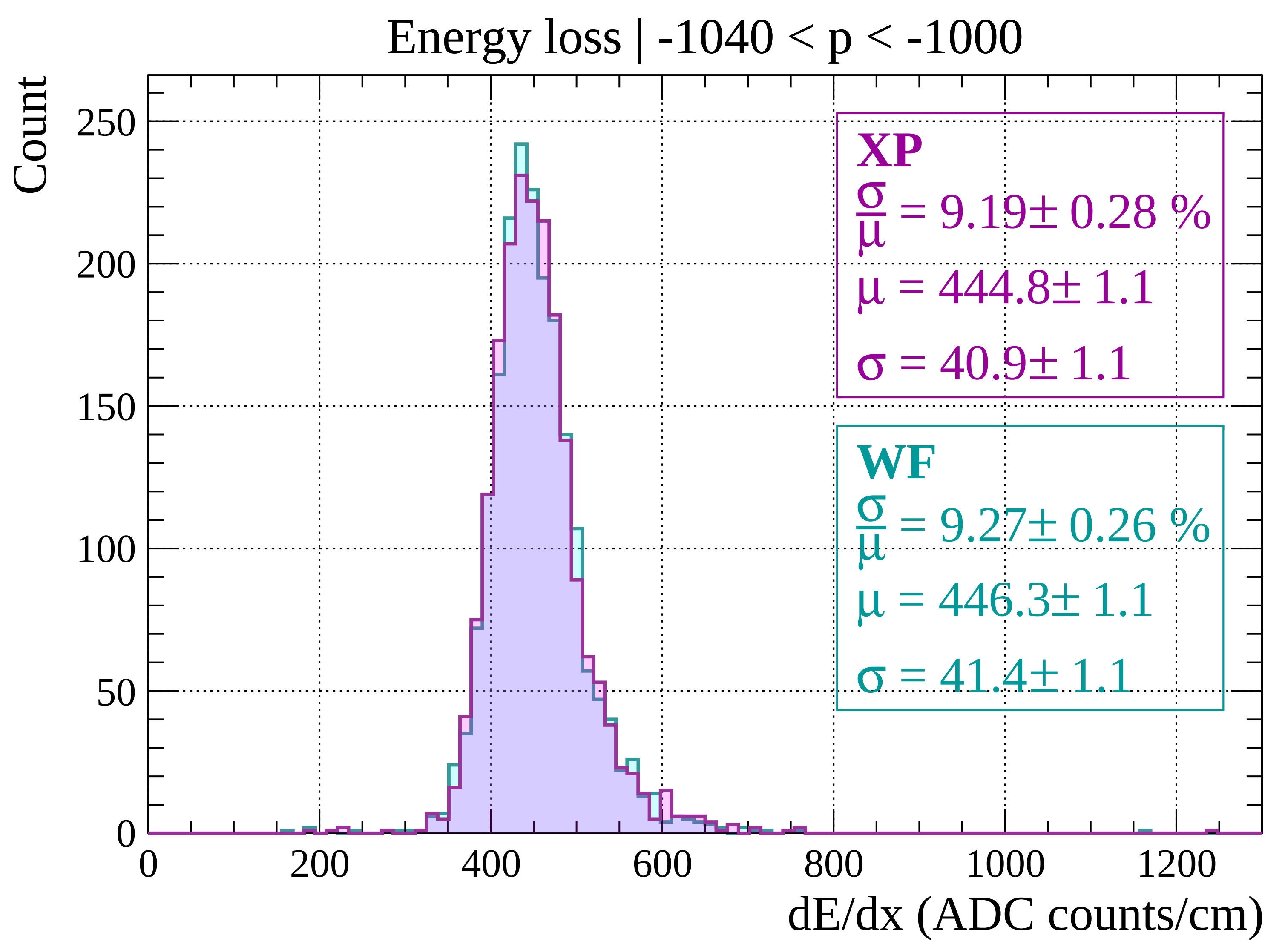

Energy loss | -1200 < p < -1160
Count
250
200
150
100
50
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 9.65± 0.29 %
μ = 459.9± 1.3
σ = 44.4± 1.2
WF
$\frac{\sigma}{\mu}$ = 10.07± 0.32 %
μ = 458.9± 1.3
σ = 46.2± 1.3


Energy loss | -1160 < p < -1120
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
0
200
400
600
800
1000
1200
XP
σ/μ = 9.28± 0.30 %
μ = 454.8± 1.2
σ = 42.2± 1.2
WF
σ/μ = 9.30± 0.30 %
μ = 455.2± 1.2
σ = 42.3± 1.3

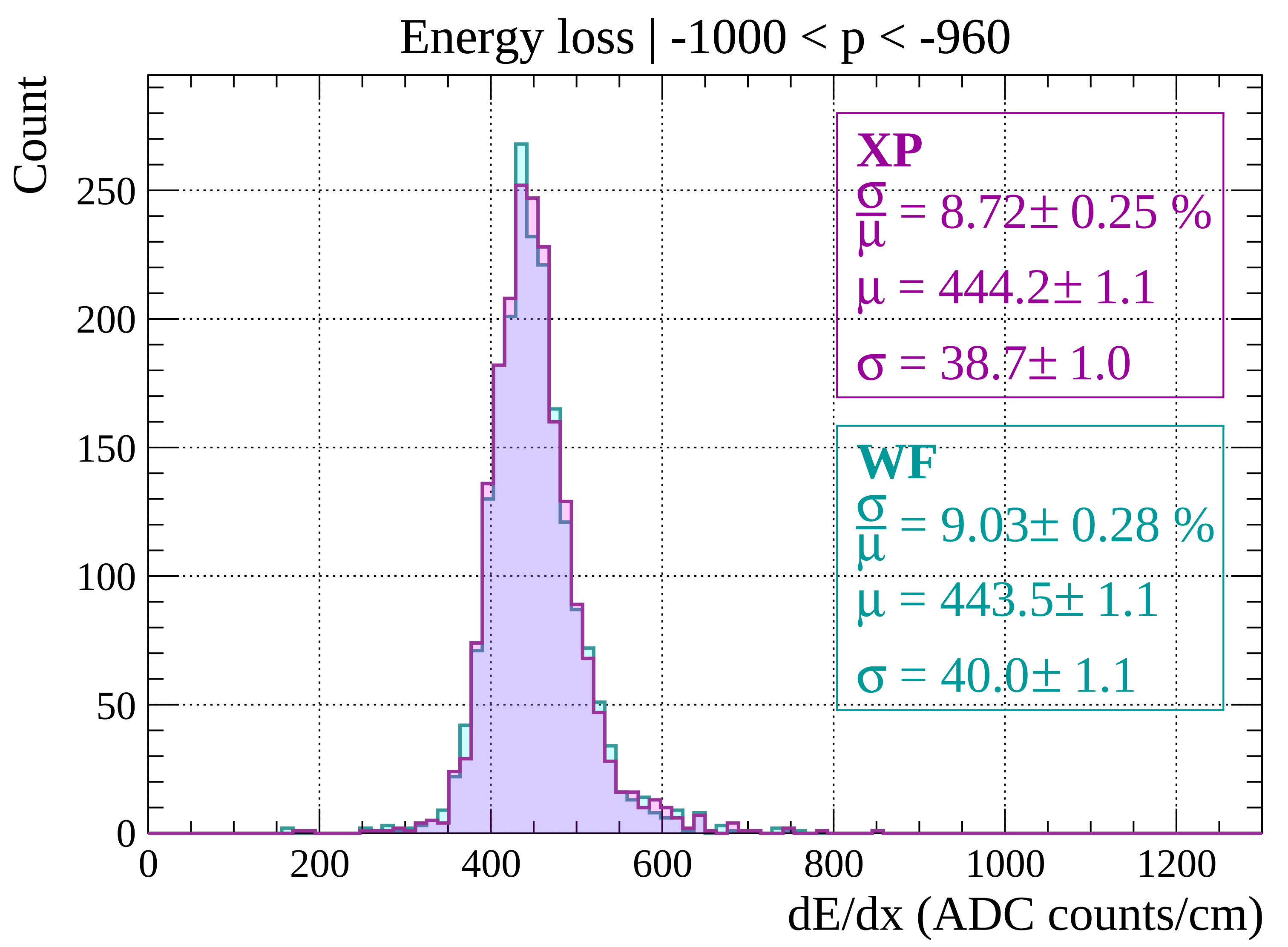

Energy loss | -1120 < p < -1080
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 9.54 ± 0.30 %
μ = 451.3 ± 1.2
σ = 43.0 ± 1.2
WF
$\frac{\sigma}{\mu}$ = 9.32 ± 0.28 %
μ = 451.7 ± 1.2
σ = 42.1 ± 1.2



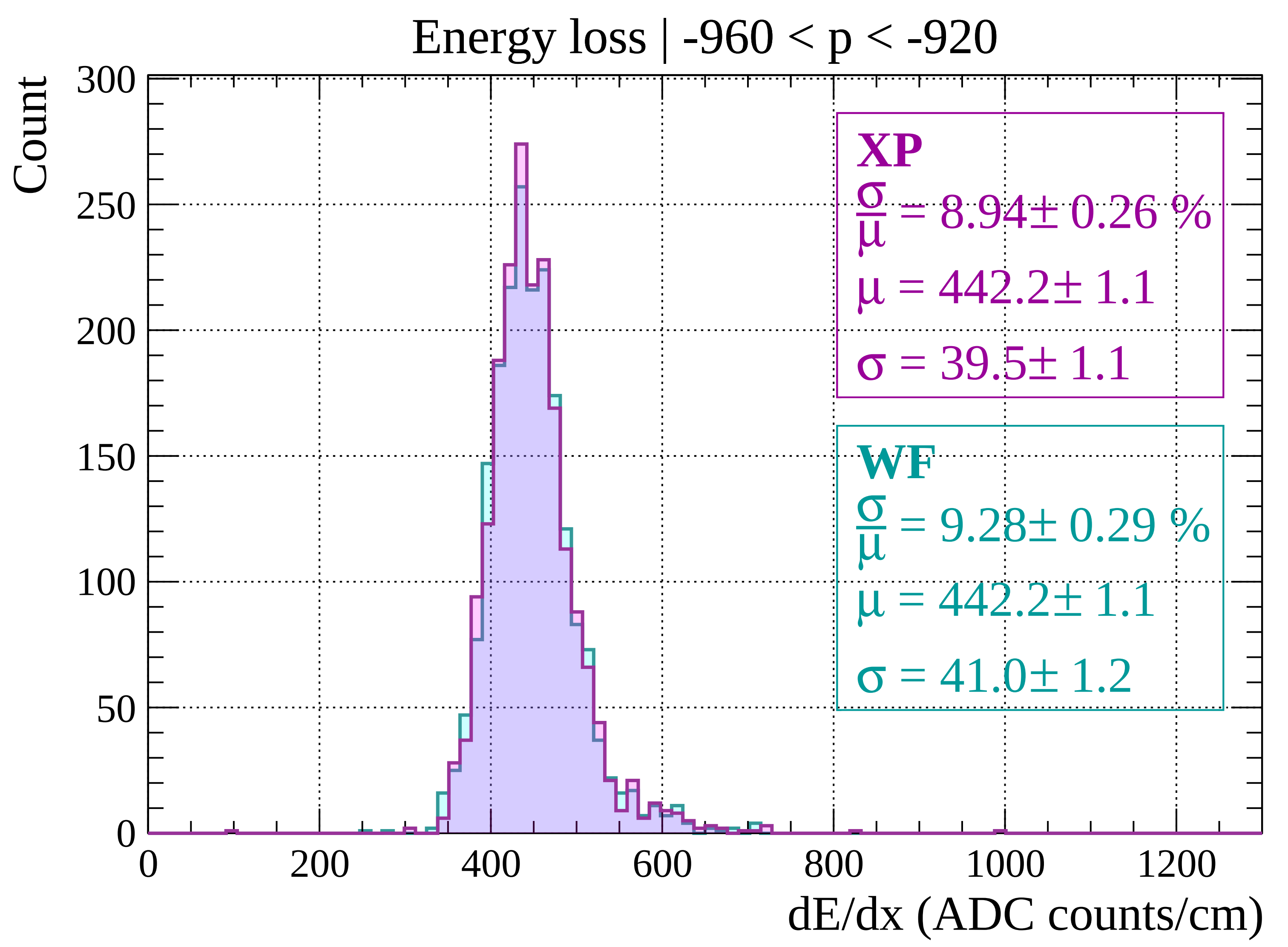
Energy loss | -1080 < p < -1040
Count
250
200
150
100
50
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 9.55± 0.29 %
μ = 448.5± 1.2
σ = 42.8± 1.2
WF
σ/μ = 9.67± 0.29 %
μ = 448.3± 1.2
σ = 43.4± 1.2



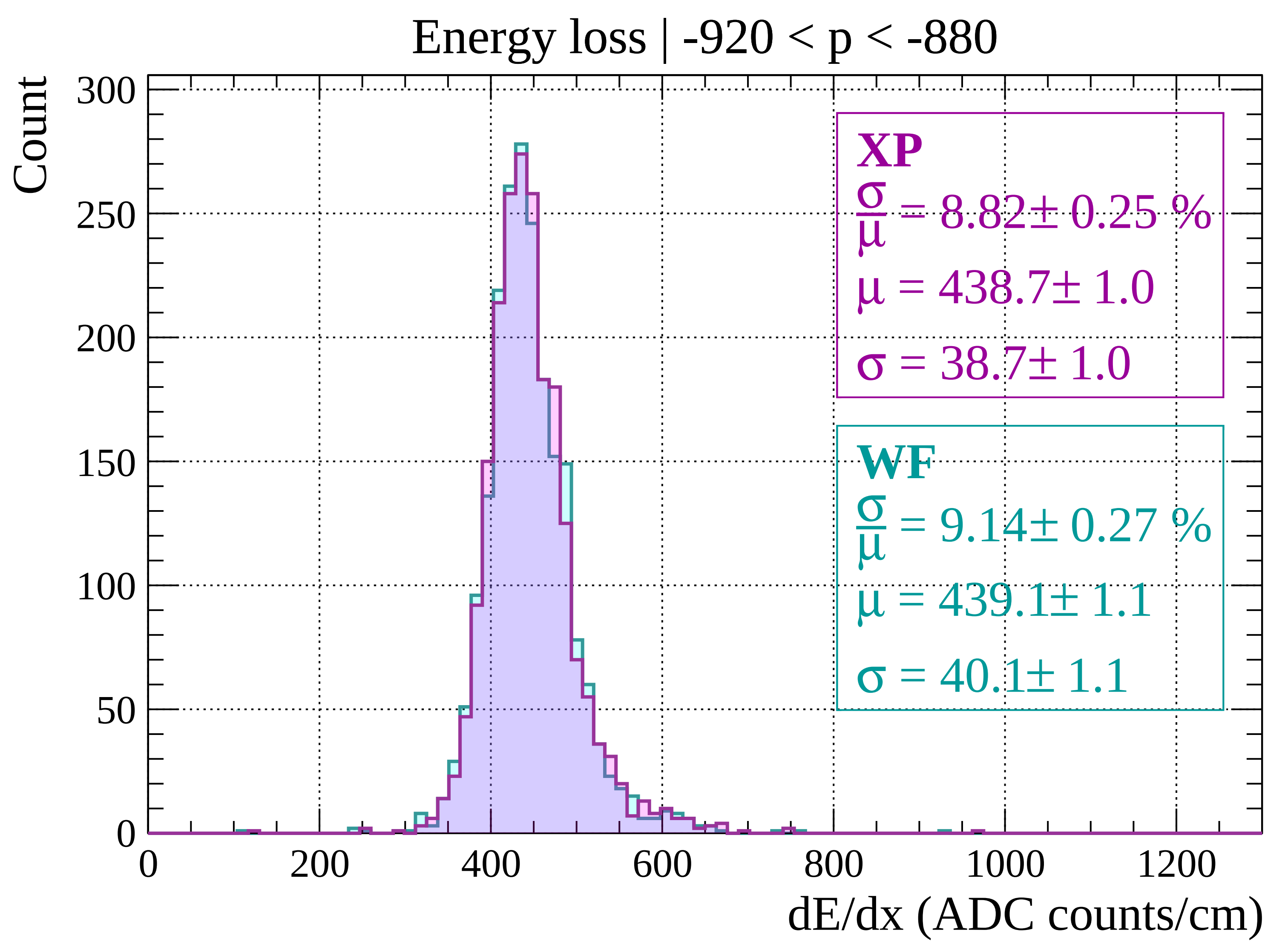
Energy loss | -1040 < p < -1000
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
50
100
150
200
250
XP
$\frac{\sigma}{\mu}$ = 9.19± 0.28 %
μ = 444.8± 1.1
σ = 40.9± 1.1
WF
$\frac{\sigma}{\mu}$ = 9.27± 0.26 %
μ = 446.3± 1.1
σ = 41.4± 1.1

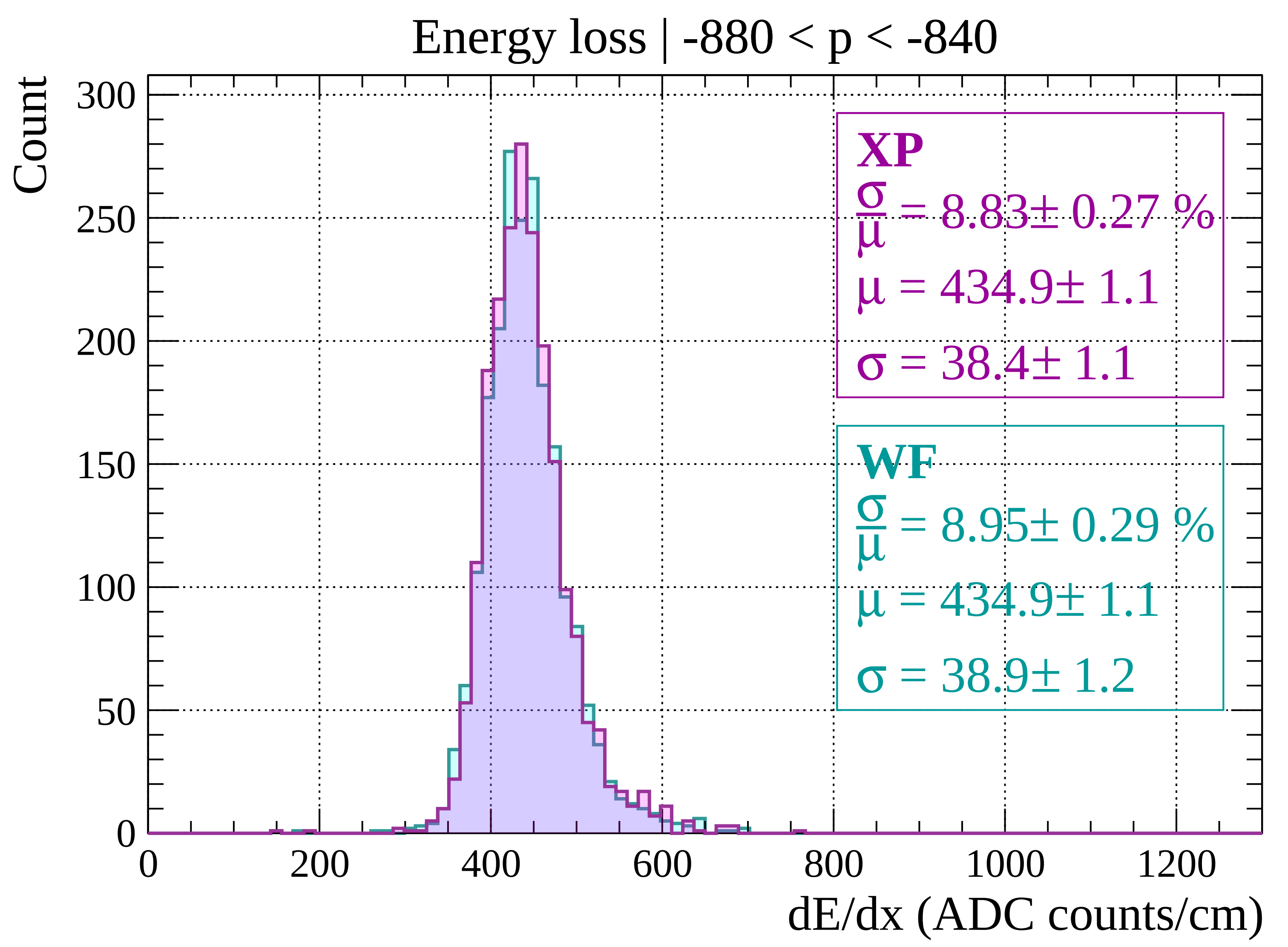


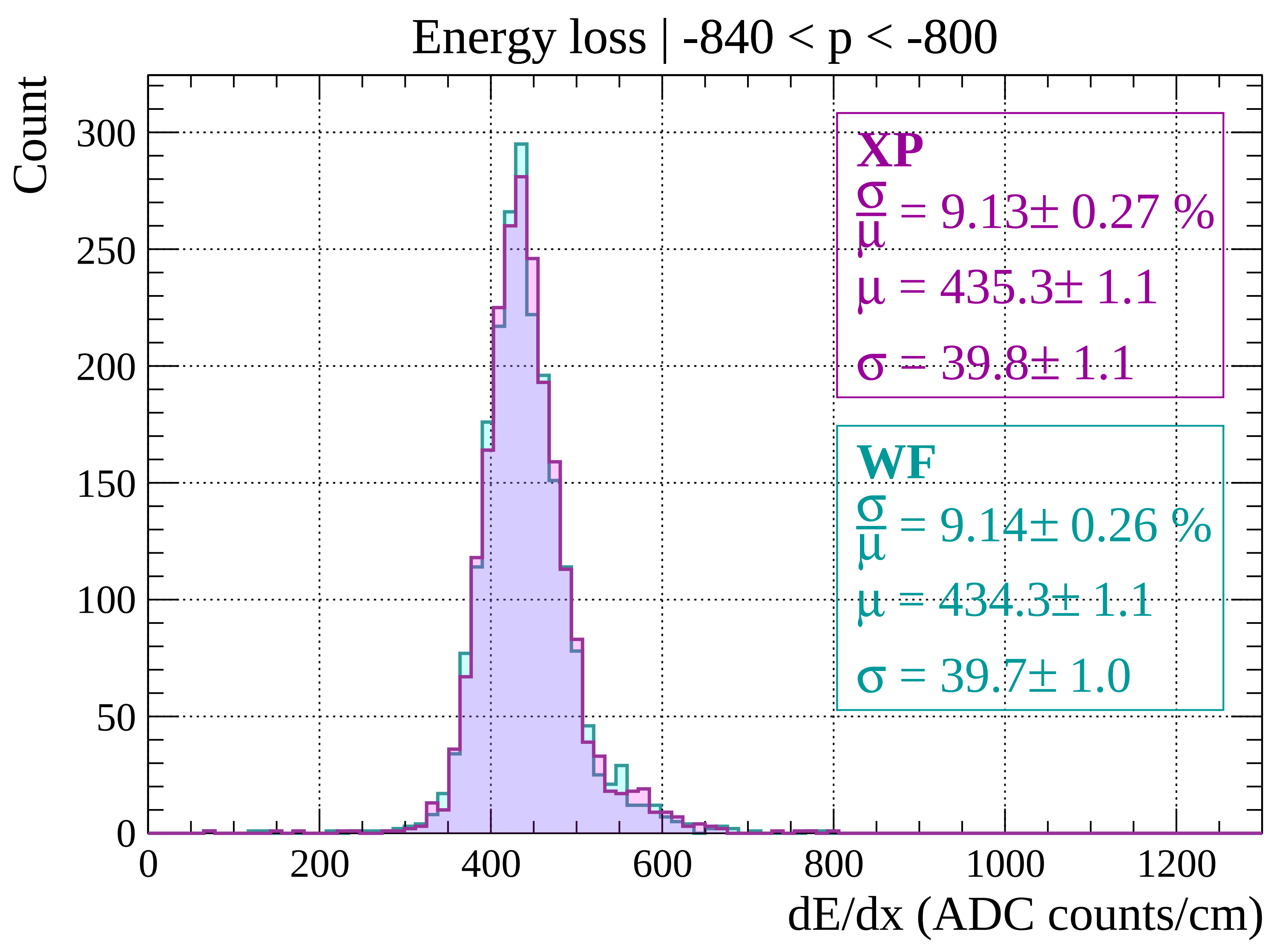
Energy loss | -1000 < p < -960
Count
0
50
100
150
200
250
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 8.72± 0.25 %
μ = 444.2± 1.1
σ = 38.7± 1.0
WF
$\frac{\sigma}{\mu}$ = 9.03± 0.28 %
μ = 443.5± 1.1
σ = 40.0± 1.1



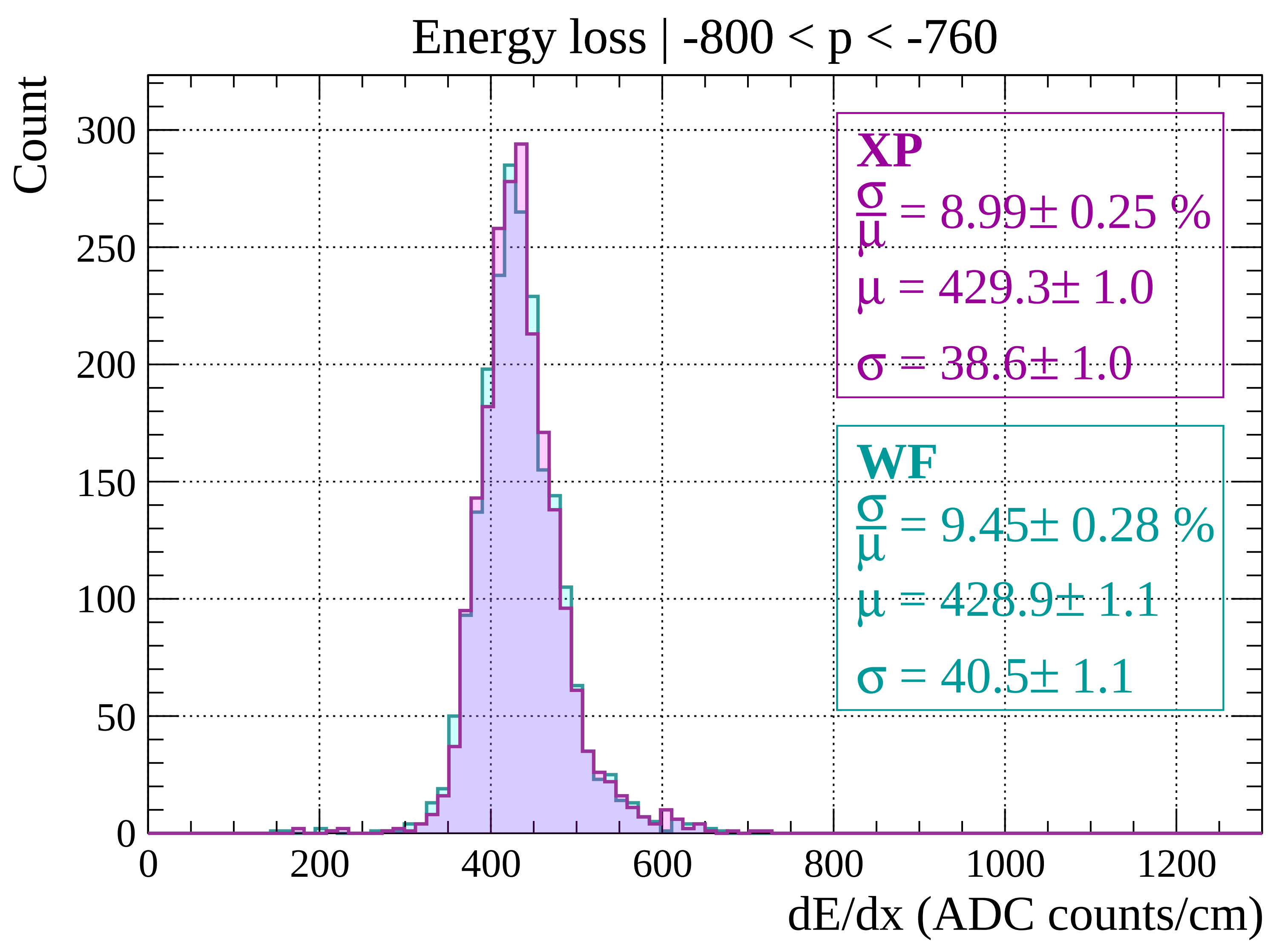
Energy loss | -960 < p < -920
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 8.94± 0.26 %
μ = 442.2± 1.1
σ = 39.5± 1.1
WF
$\frac{\sigma}{\mu}$ = 9.28± 0.29 %
μ = 442.2± 1.1
σ = 41.0± 1.2
0
50
100
150
200
250
300
0
200
400
600
800
1000
1200

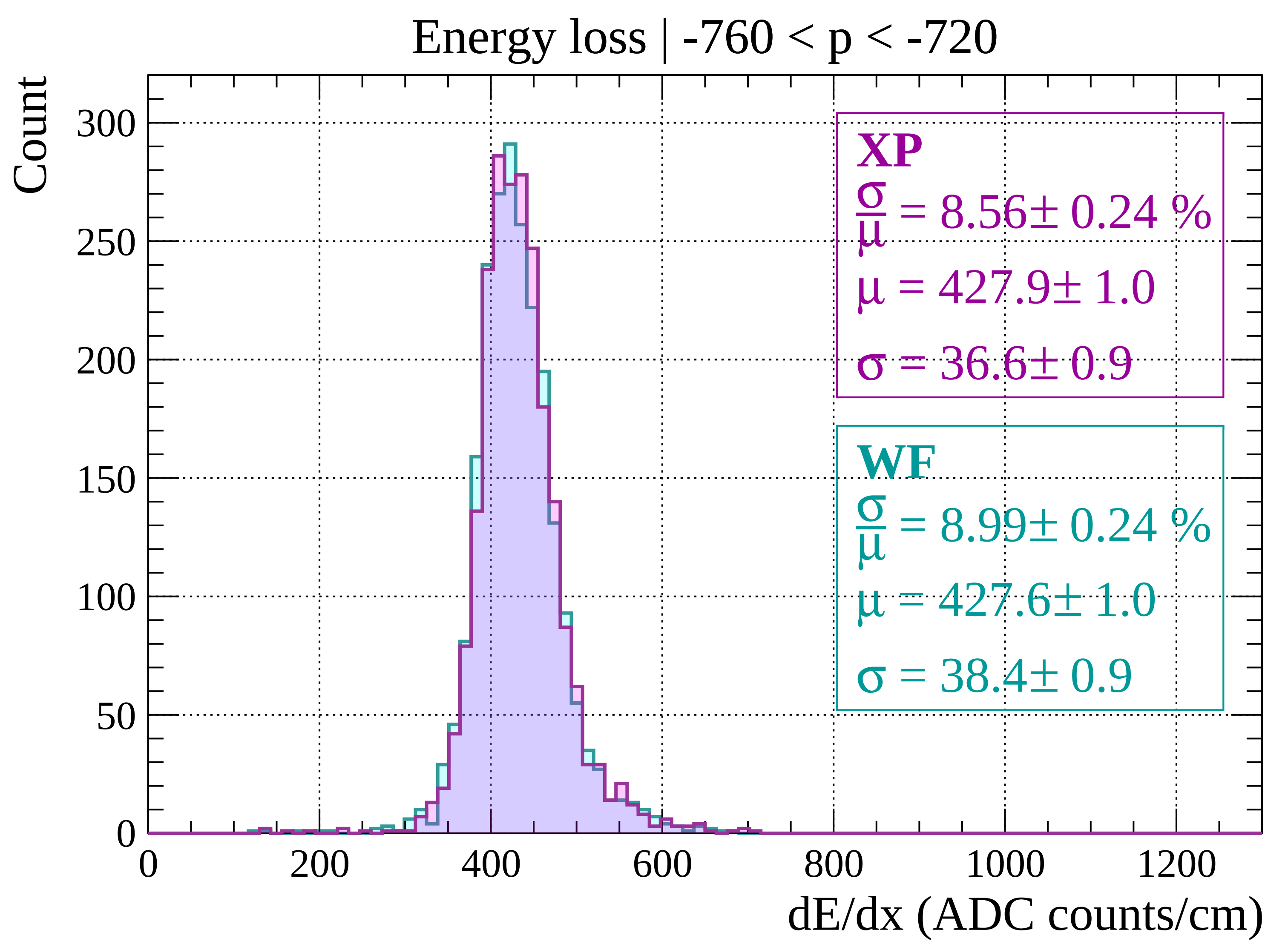

Energy loss | -920 < p < -880
Count
300
250
200
150
100
50
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 8.82± 0.25 %
μ = 438.7± 1.0
σ = 38.7± 1.0
WF
σ/μ = 9.14± 0.27 %
μ = 439.1± 1.1
σ = 40.1± 1.1

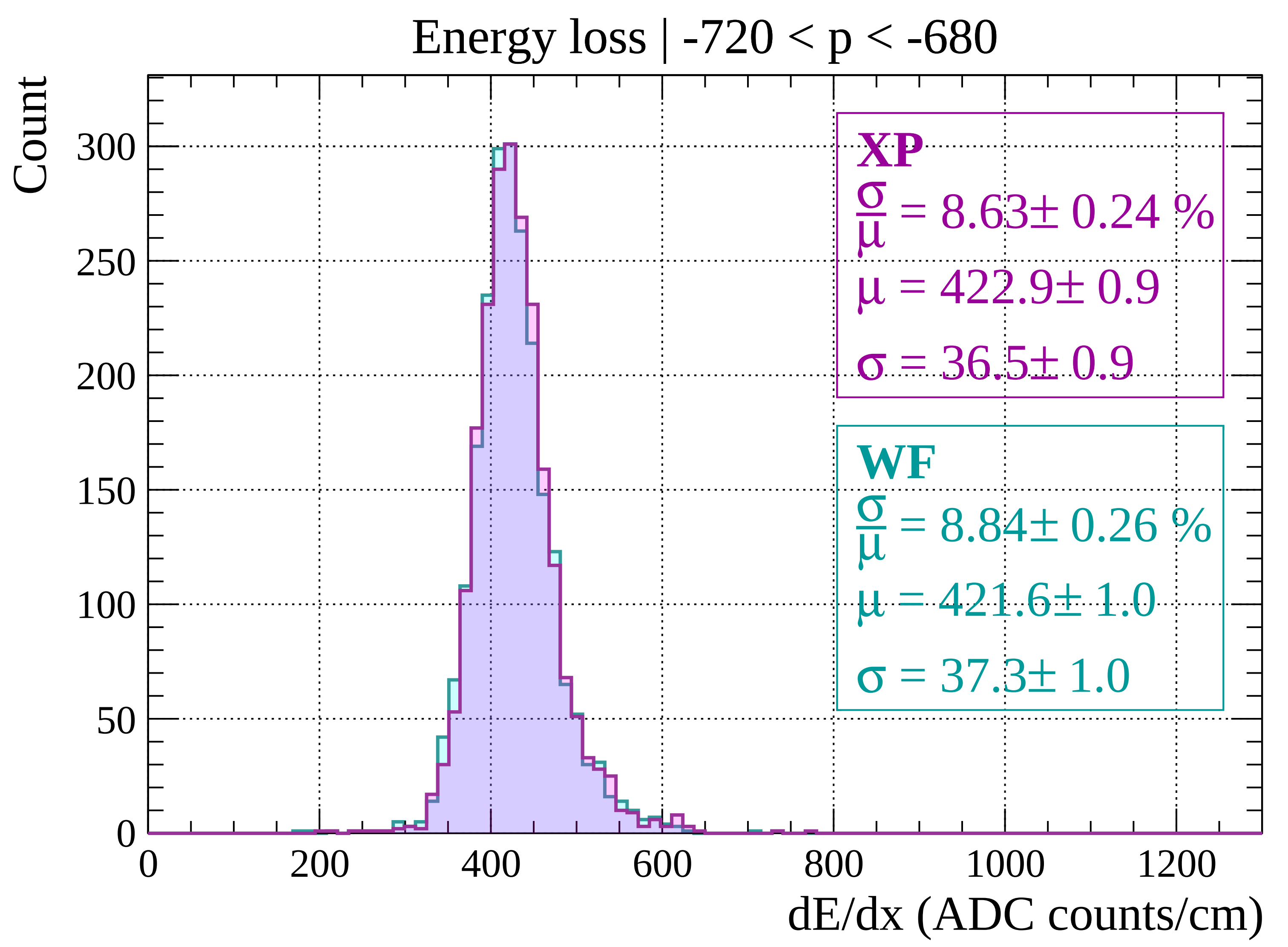

Energy loss | -880 < p < -840
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
0
200
400
600
800
1000
1200
XP
σ/μ = 8.83± 0.27 %
μ = 434.9± 1.1
σ = 38.4± 1.1
WF
σ/μ = 8.95± 0.29 %
μ = 434.9± 1.1
σ = 38.9± 1.2



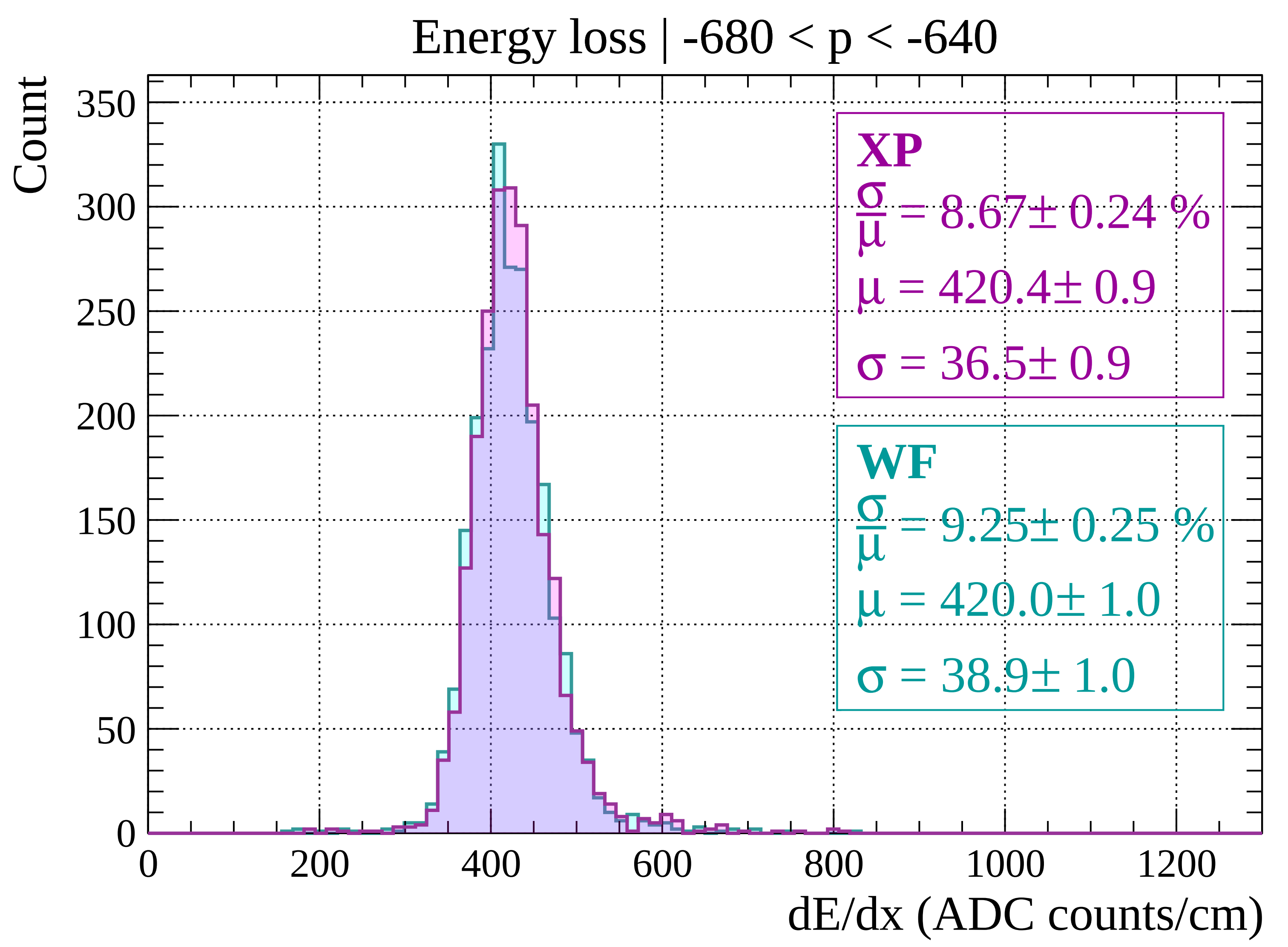
Energy loss | -840 < p < -800
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
50
100
150
200
250
300
XP
$\frac{\sigma}{\mu}$ = 9.13± 0.27 %
μ = 435.3± 1.1
σ = 39.8± 1.1
WF
$\frac{\sigma}{\mu}$ = 9.14± 0.26 %
μ = 434.3± 1.1
σ = 39.7± 1.0

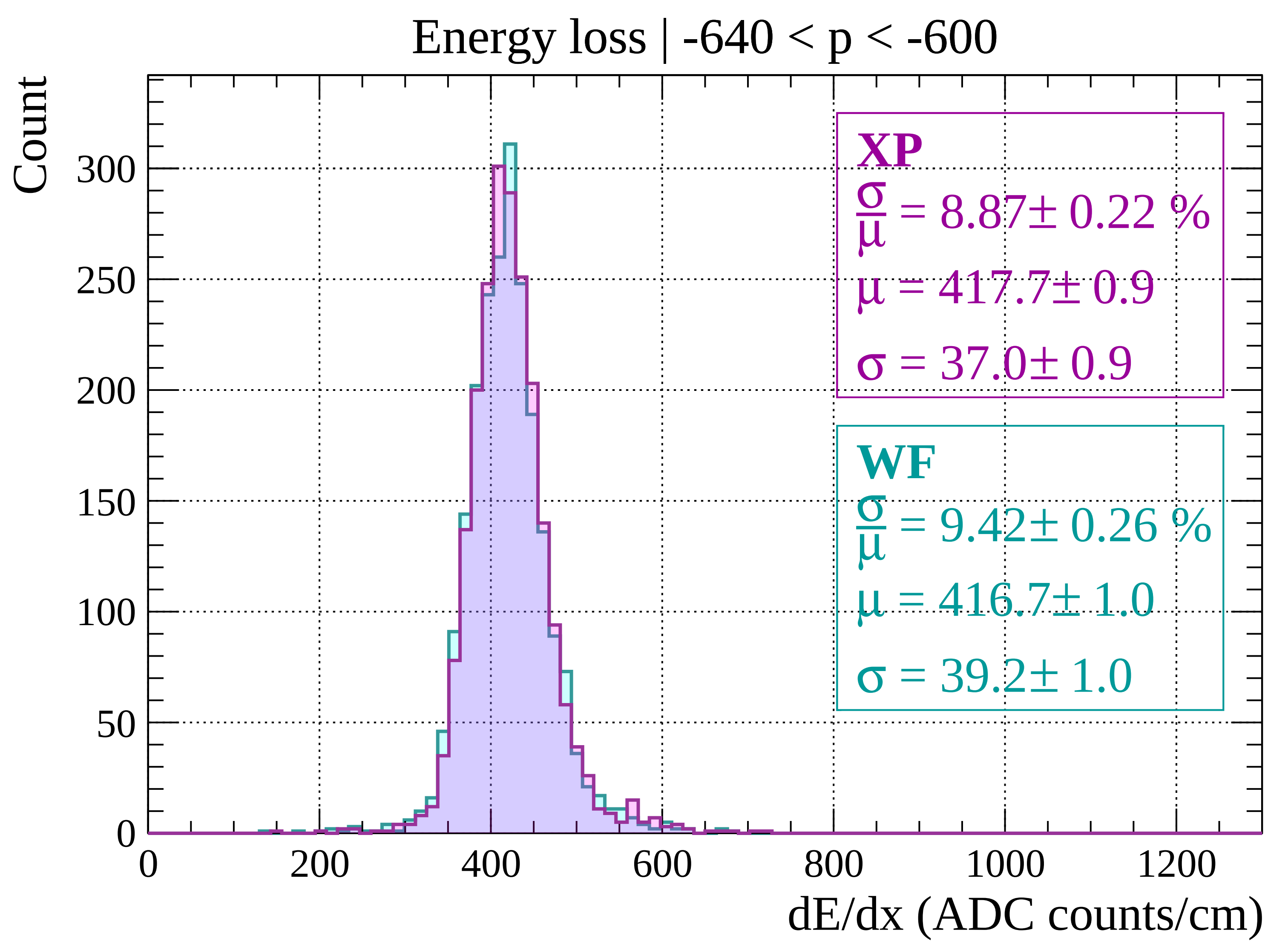

Energy loss | -800 < p < -760
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
50
100
150
200
250
300
XP
σ/μ = 8.99± 0.25 %
μ = 429.3± 1.0
σ = 38.6± 1.0
WF
σ/μ = 9.45± 0.28 %
μ = 428.9± 1.1
σ = 40.5± 1.1

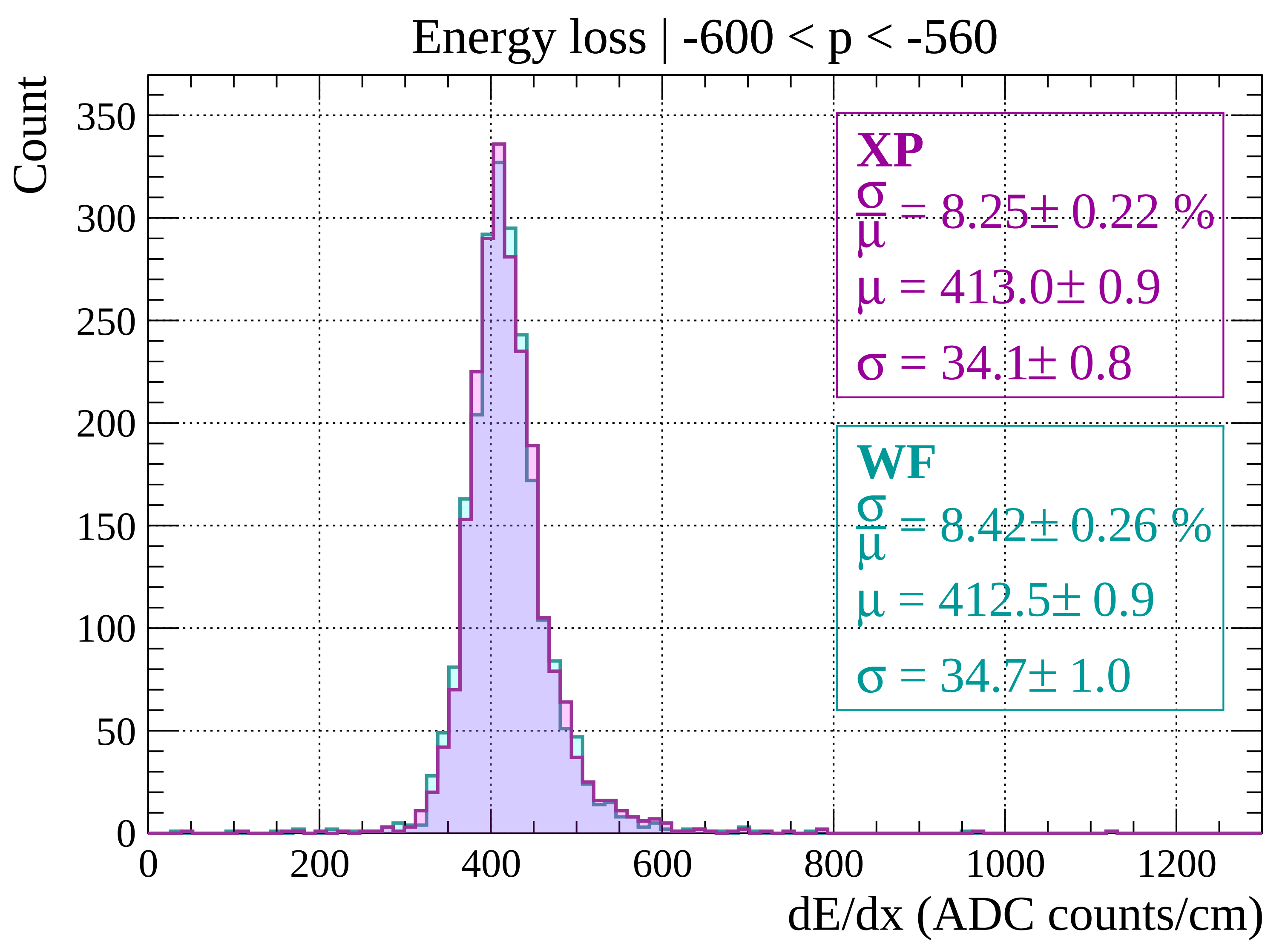

Energy loss | -760 < p < -720
Count
300
250
200
150
100
50
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 8.56± 0.24 %
μ = 427.9± 1.0
σ = 36.6± 0.9
WF
σ/μ = 8.99± 0.24 %
μ = 427.6± 1.0
σ = 38.4± 0.9

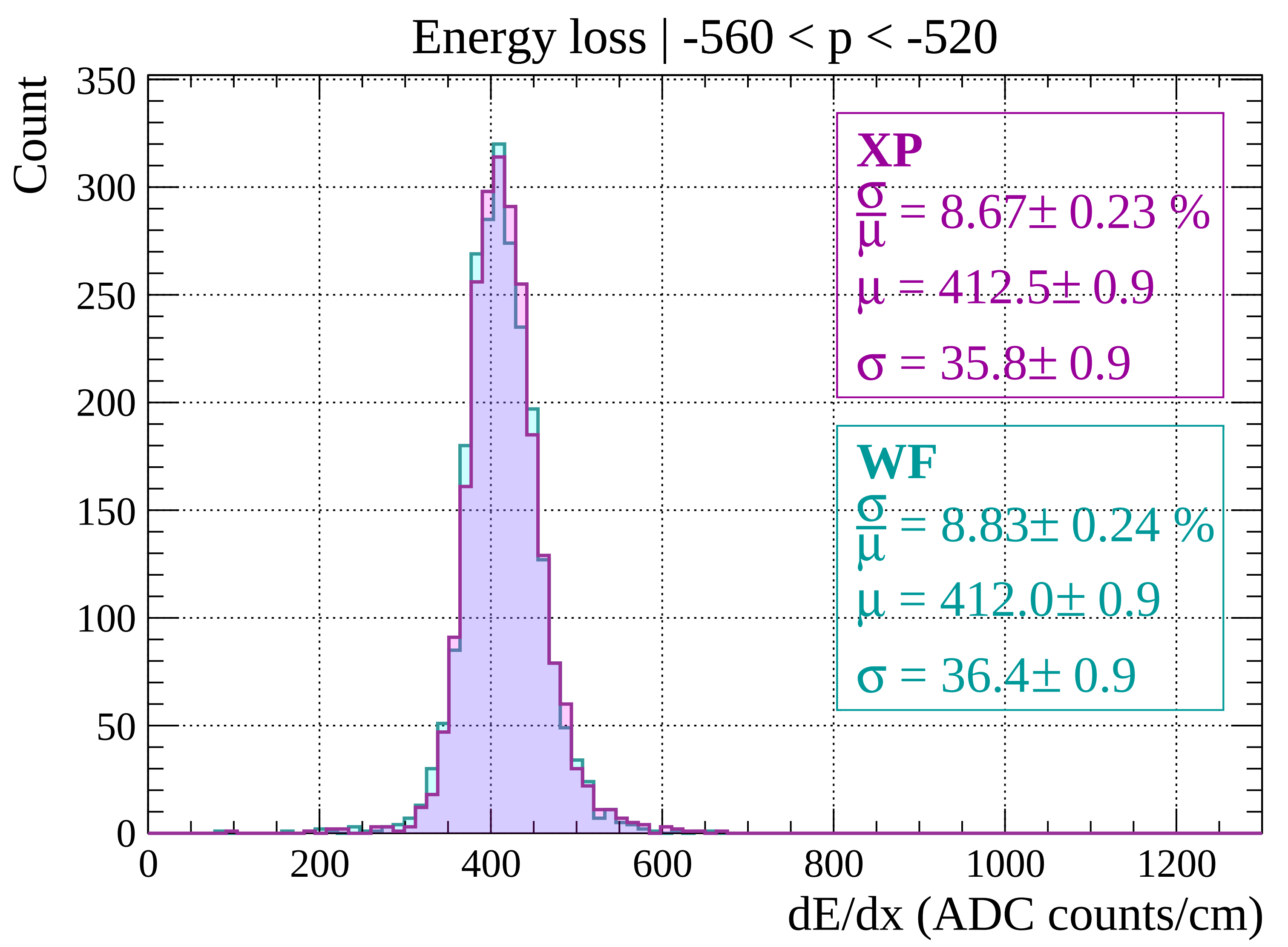


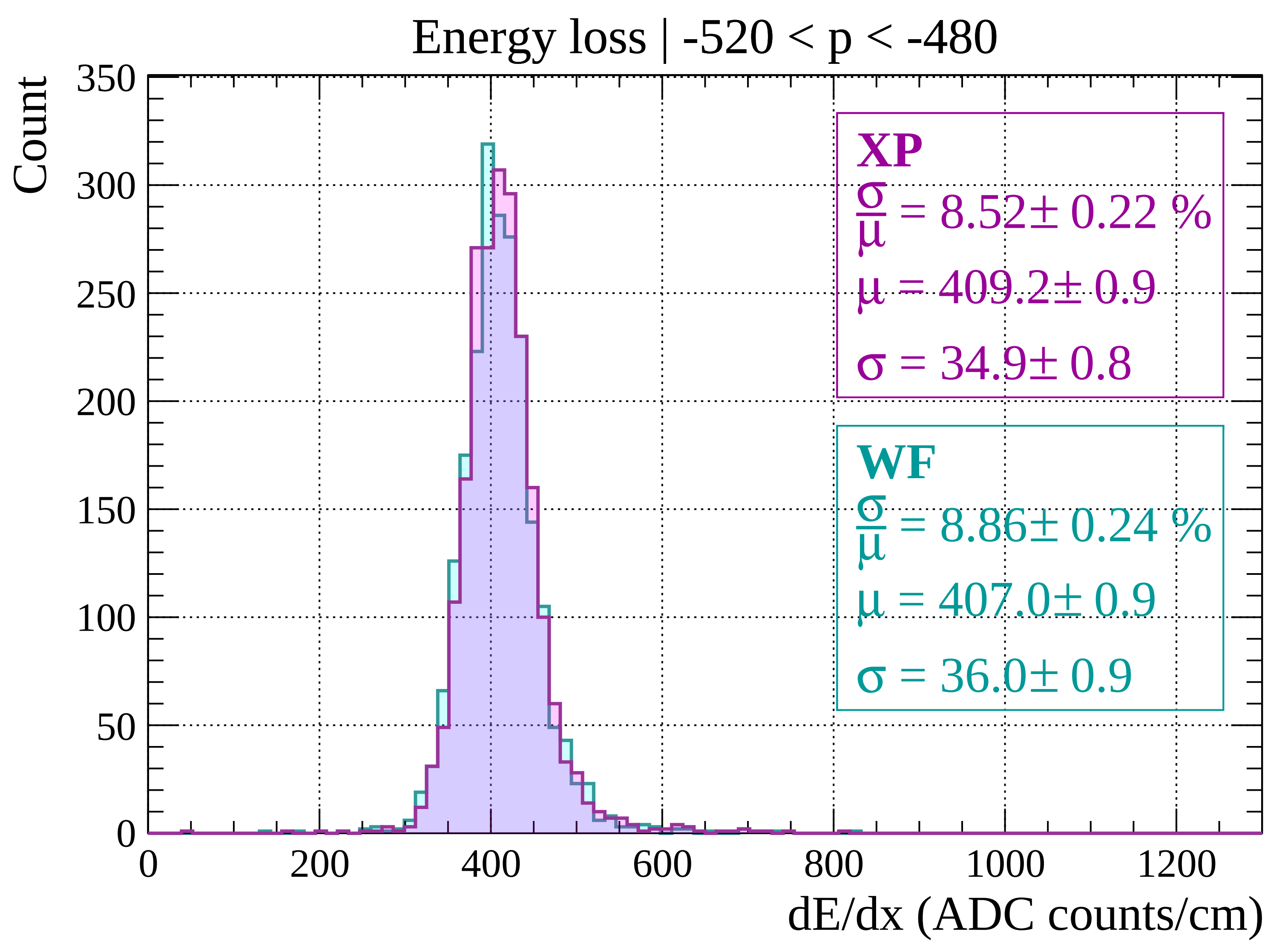
Energy loss | -720 < p < -680
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 8.63± 0.24 %
μ = 422.9± 0.9
σ = 36.5± 0.9
WF
σ/μ = 8.84± 0.26 %
μ = 421.6± 1.0
σ = 37.3± 1.0

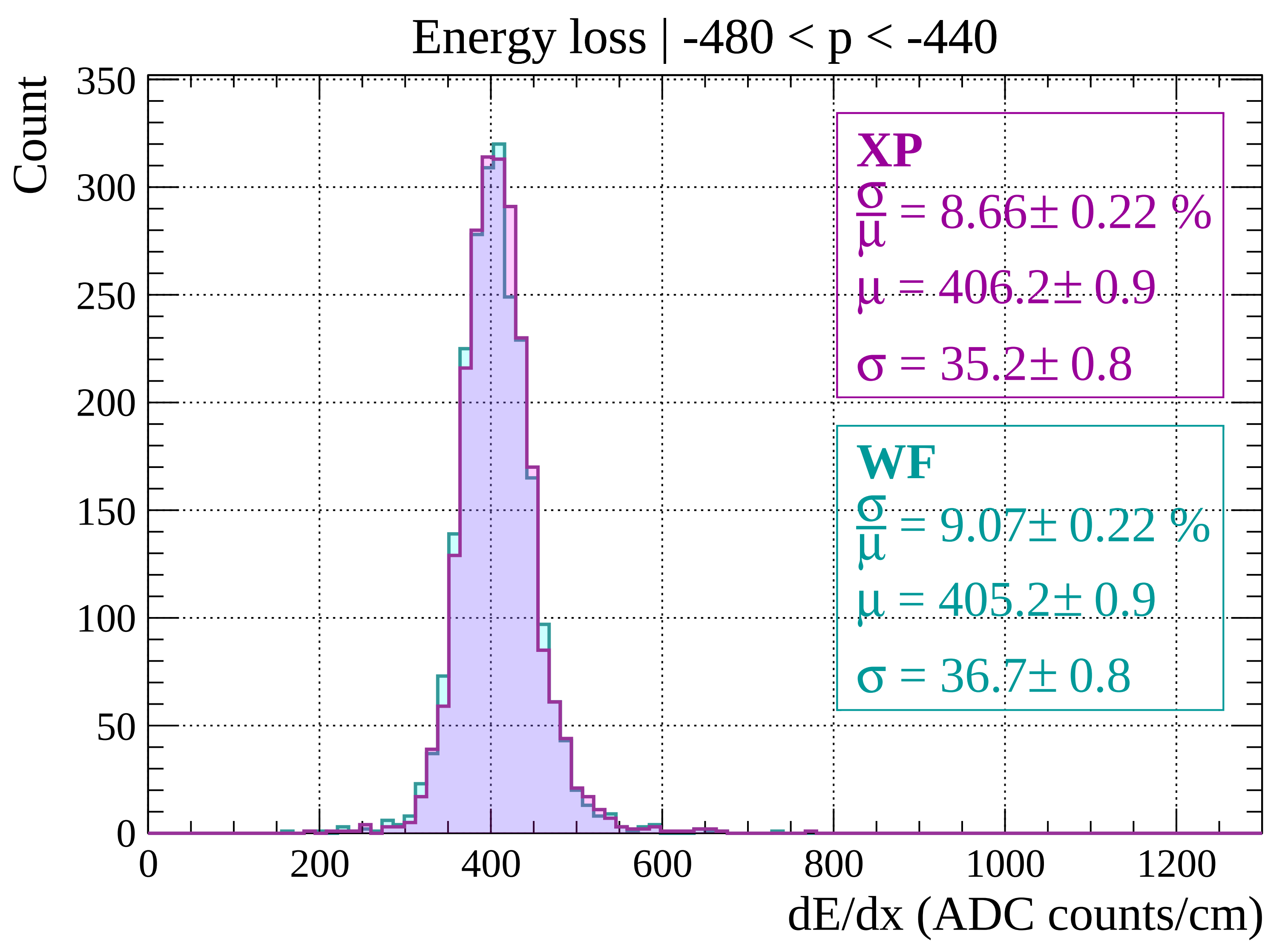

Energy loss | -680 < p < -640
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 8.67± 0.24 %
μ = 420.4± 0.9
σ = 36.5± 0.9
WF
σ/μ = 9.25± 0.25 %
μ = 420.0± 1.0
σ = 38.9± 1.0

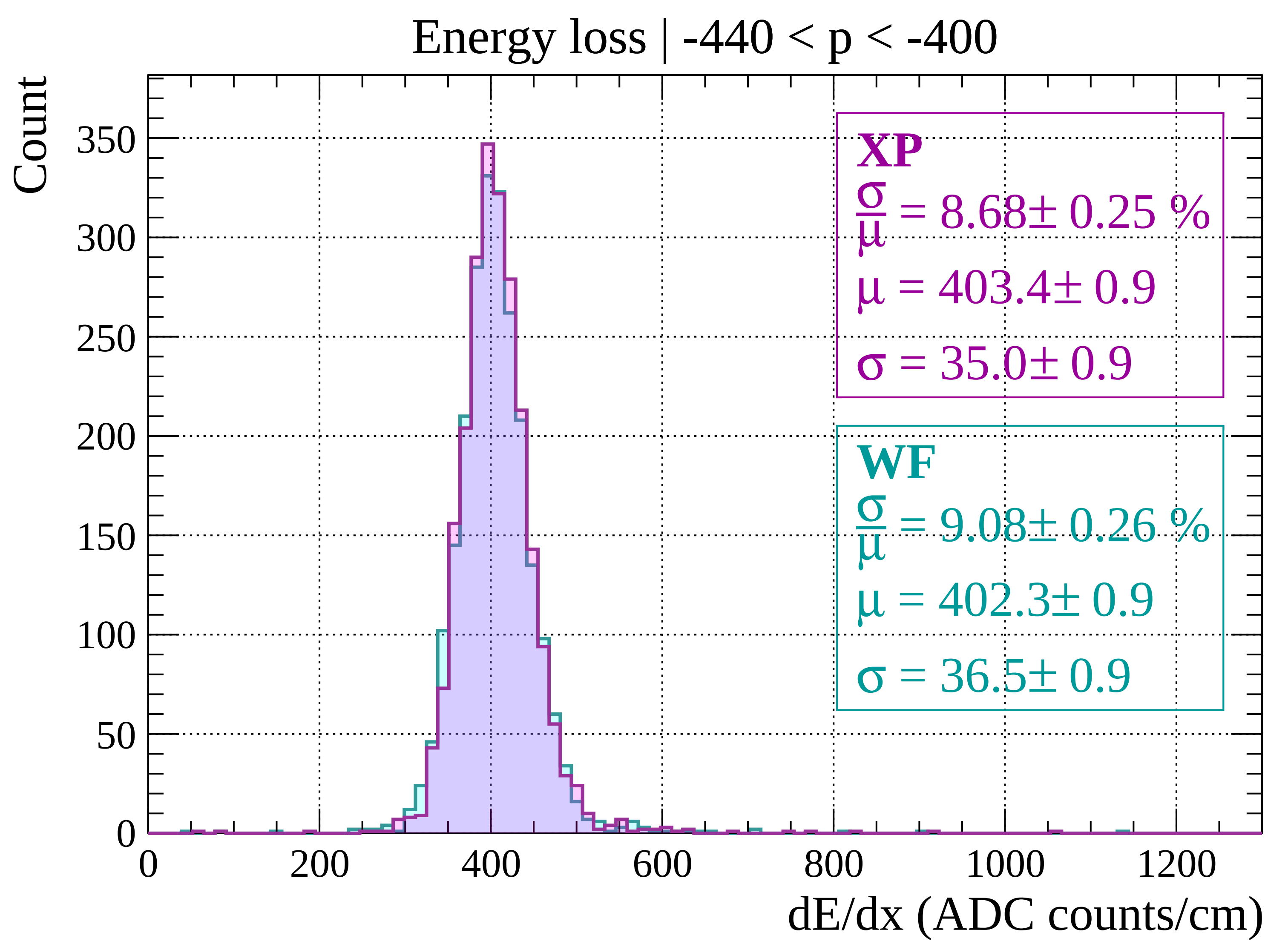

Energy loss | -640 < p < -600
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
0
200
400
600
800
1000
1200
XP
σ/μ = 8.87± 0.22 %
μ = 417.7± 0.9
σ = 37.0± 0.9
WF
σ/μ = 9.42± 0.26 %
μ = 416.7± 1.0
σ = 39.2± 1.0

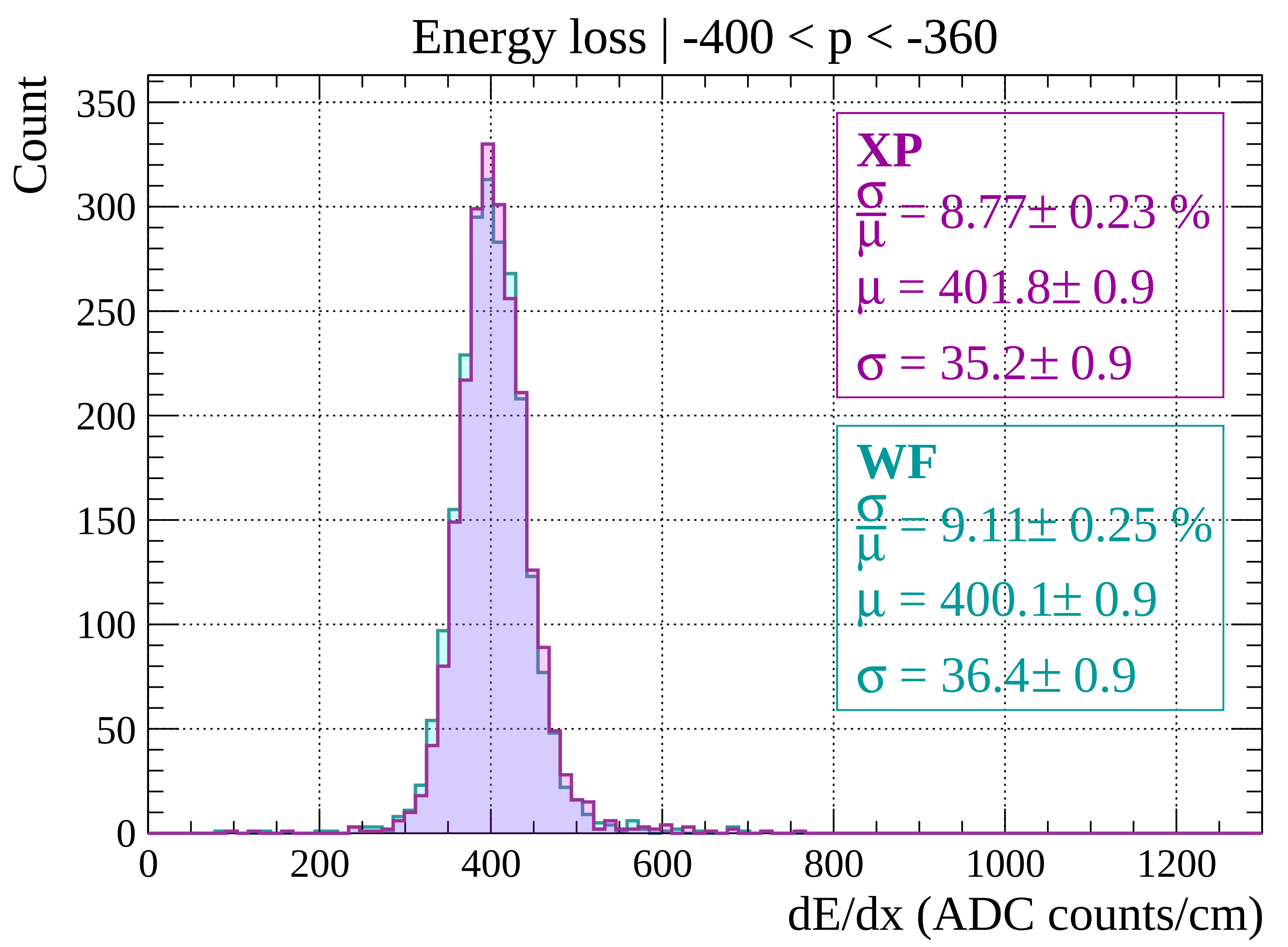

Energy loss | -600 < p < -560
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 8.25± 0.22 %
μ = 413.0± 0.9
σ = 34.1± 0.8
WF
$\frac{\sigma}{\mu}$ = 8.42± 0.26 %
μ = 412.5± 0.9
σ = 34.7± 1.0
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200


Energy loss | -560 < p < -520
Count
350
300
250
200
150
100
50
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 8.67± 0.23 %
μ = 412.5± 0.9
σ = 35.8± 0.9
WF
σ/μ = 8.83± 0.24 %
μ = 412.0± 0.9
σ = 36.4± 0.9

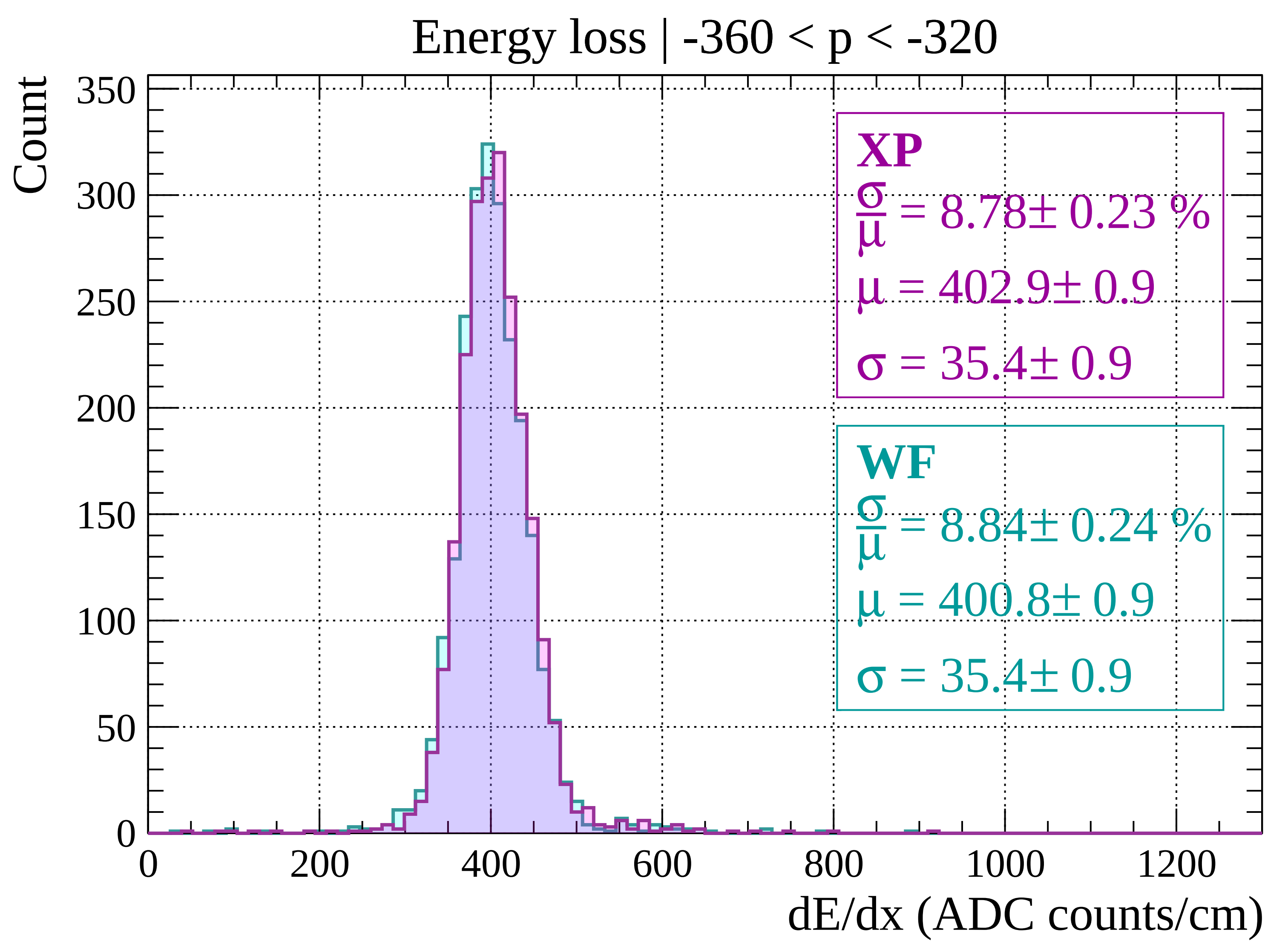

Energy loss | -520 < p < -480
Count
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu} = 8.52 \pm 0.22$ %
$\mu = 409.2 \pm 0.9$
$\sigma = 34.9 \pm 0.8$
WF
$\frac{\sigma}{\mu} = 8.86 \pm 0.24$ %
$\mu = 407.0 \pm 0.9$
$\sigma = 36.0 \pm 0.9$


Energy loss | -480 < p < -440
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200
XP
σ/μ = 8.66± 0.22 %
μ = 406.2± 0.9
σ = 35.2± 0.8
WF
σ/μ = 9.07± 0.22 %
μ = 405.2± 0.9
σ = 36.7± 0.8

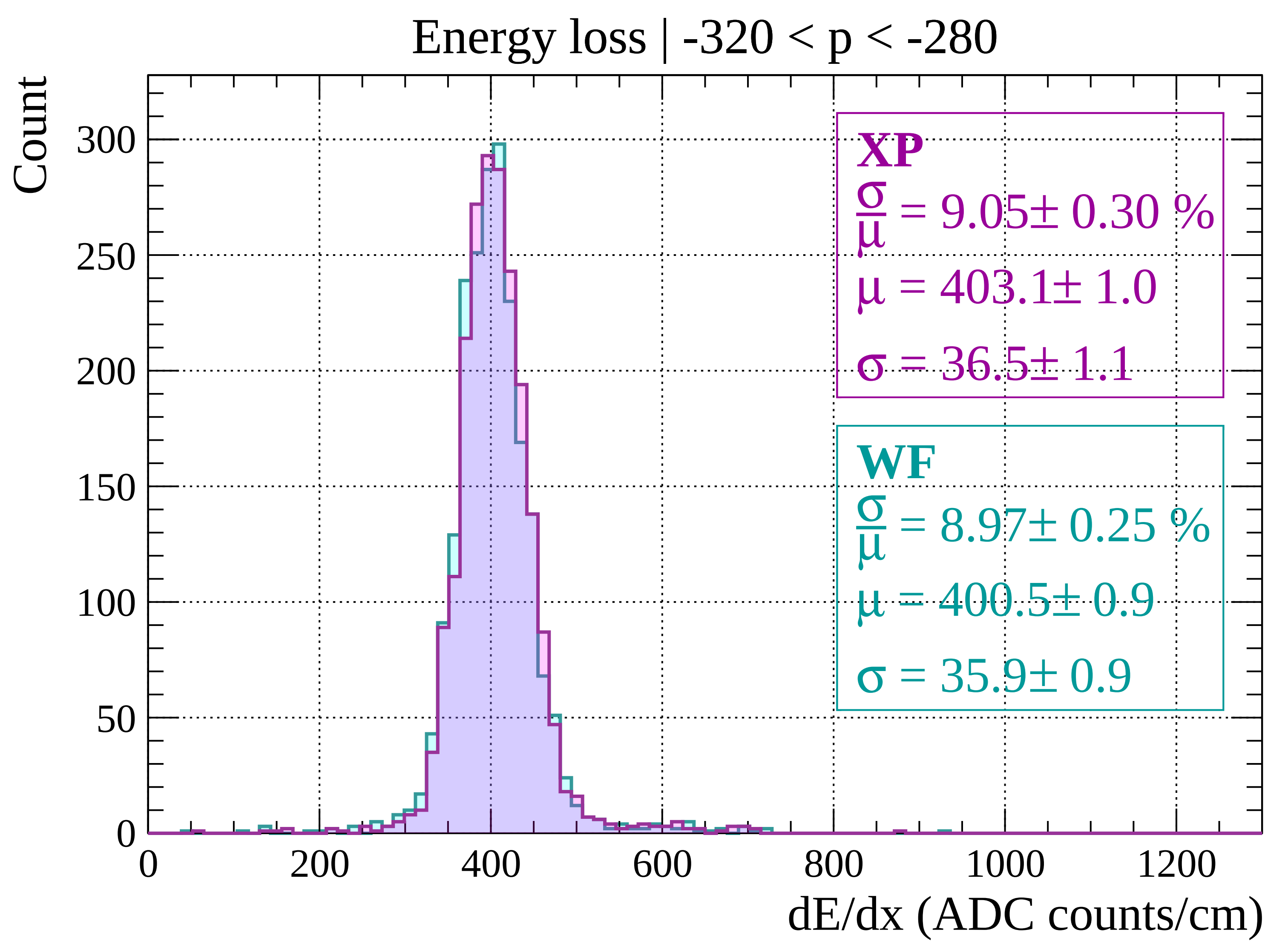

Energy loss | -440 < p < -400
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
50
100
150
200
250
300
350
XP
σ/μ = 8.68± 0.25 %
μ = 403.4± 0.9
σ = 35.0± 0.9
WF
σ/μ = 9.08± 0.26 %
μ = 402.3± 0.9
σ = 36.5± 0.9

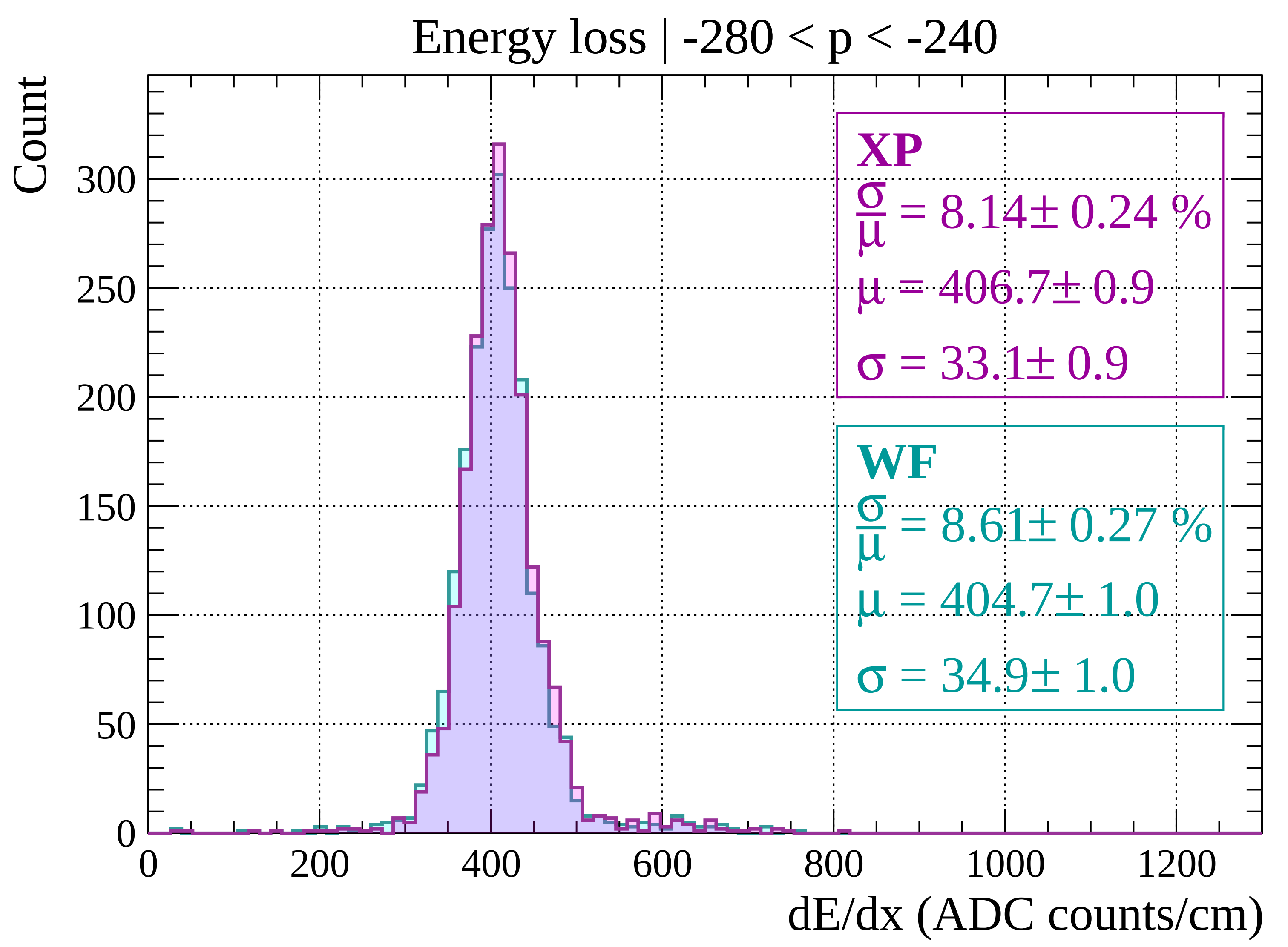


Energy loss | -400 < p < -360
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 8.77± 0.23 %
μ = 401.8± 0.9
σ = 35.2± 0.9
WF
σ/μ = 9.11± 0.25 %
μ = 400.1± 0.9
σ = 36.4± 0.9

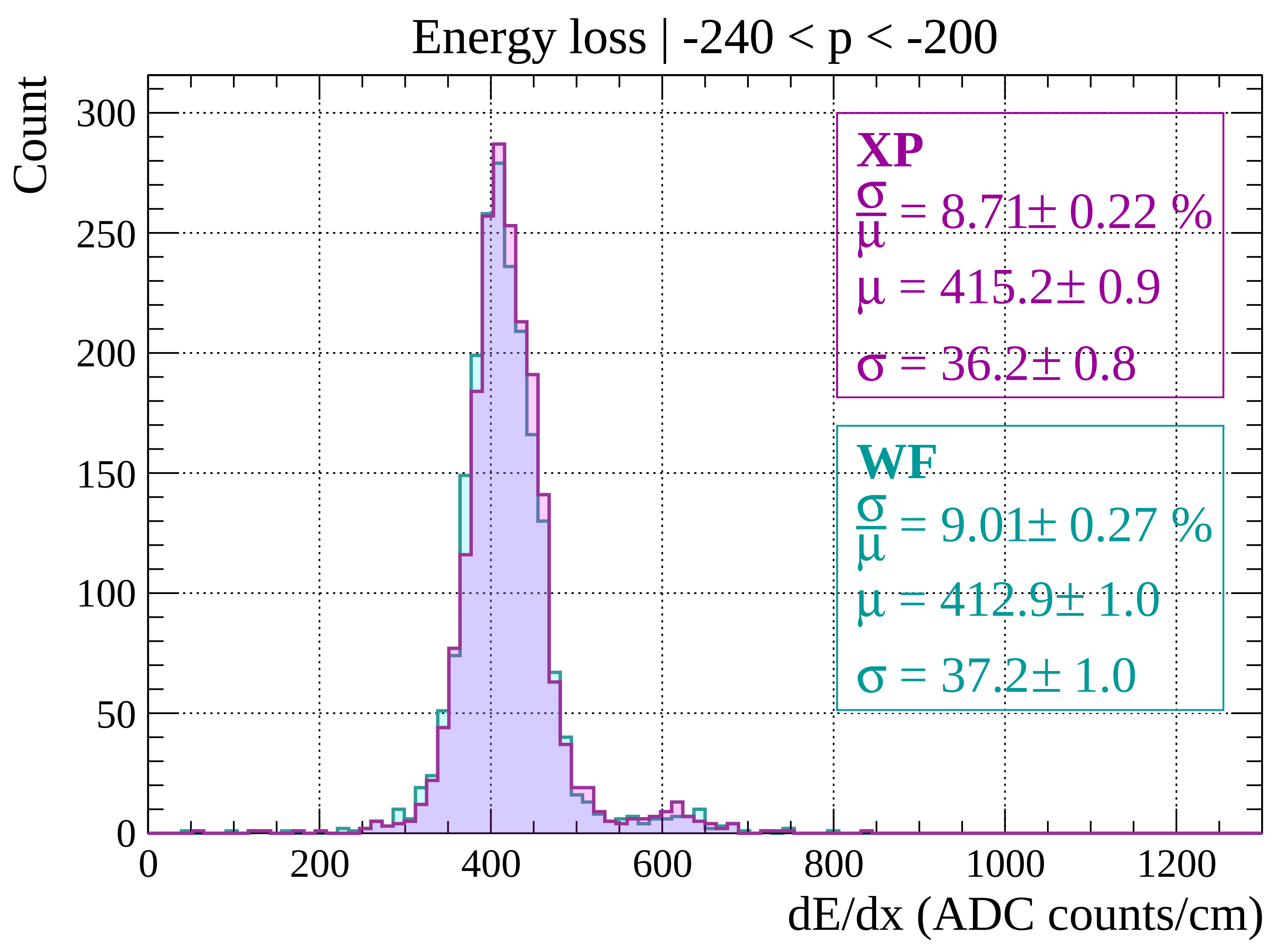


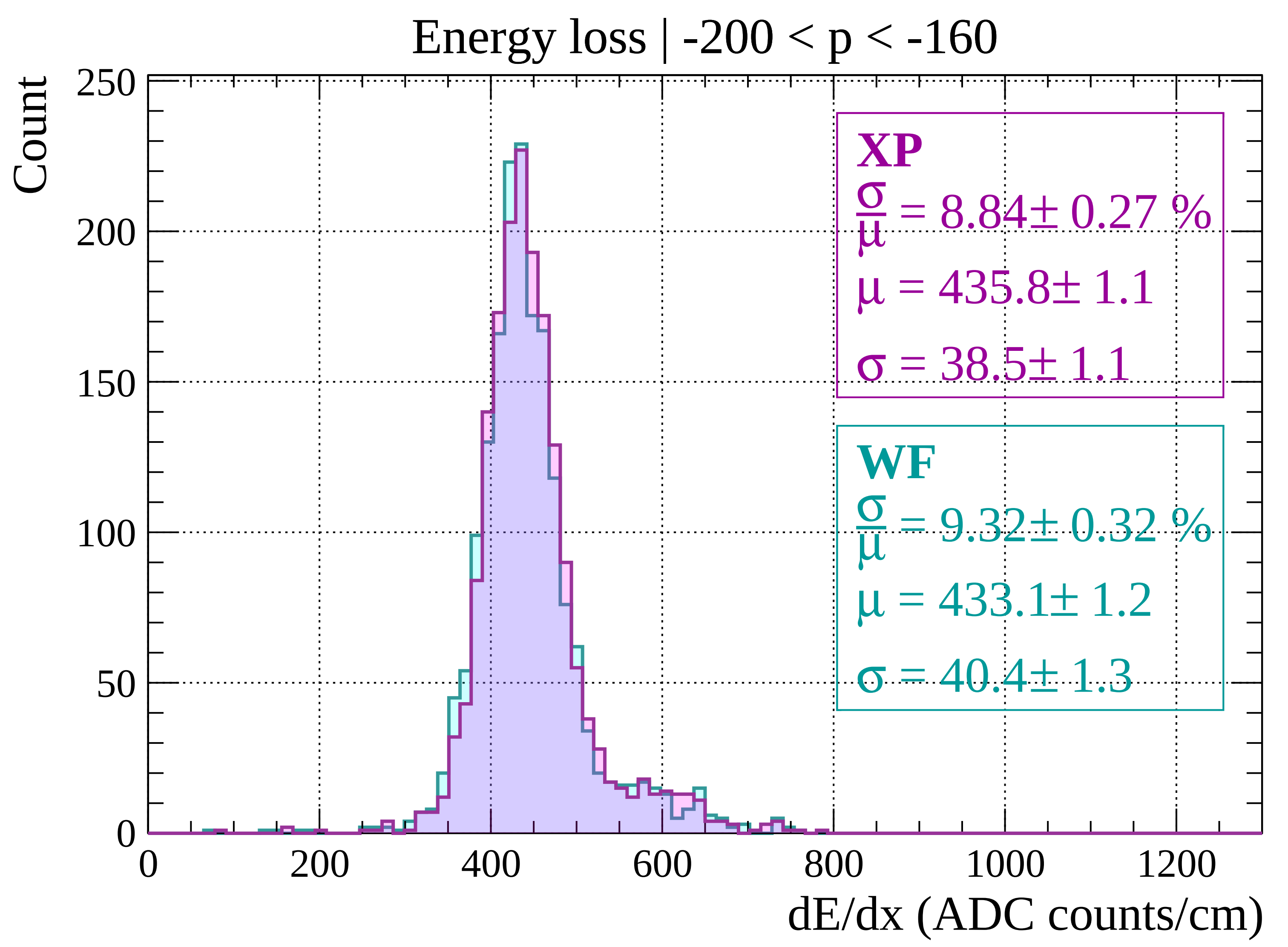
Energy loss | -360 < p < -320
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 8.78± 0.23 %
μ = 402.9± 0.9
σ = 35.4± 0.9
WF
$\frac{\sigma}{\mu}$ = 8.84± 0.24 %
μ = 400.8± 0.9
σ = 35.4± 0.9

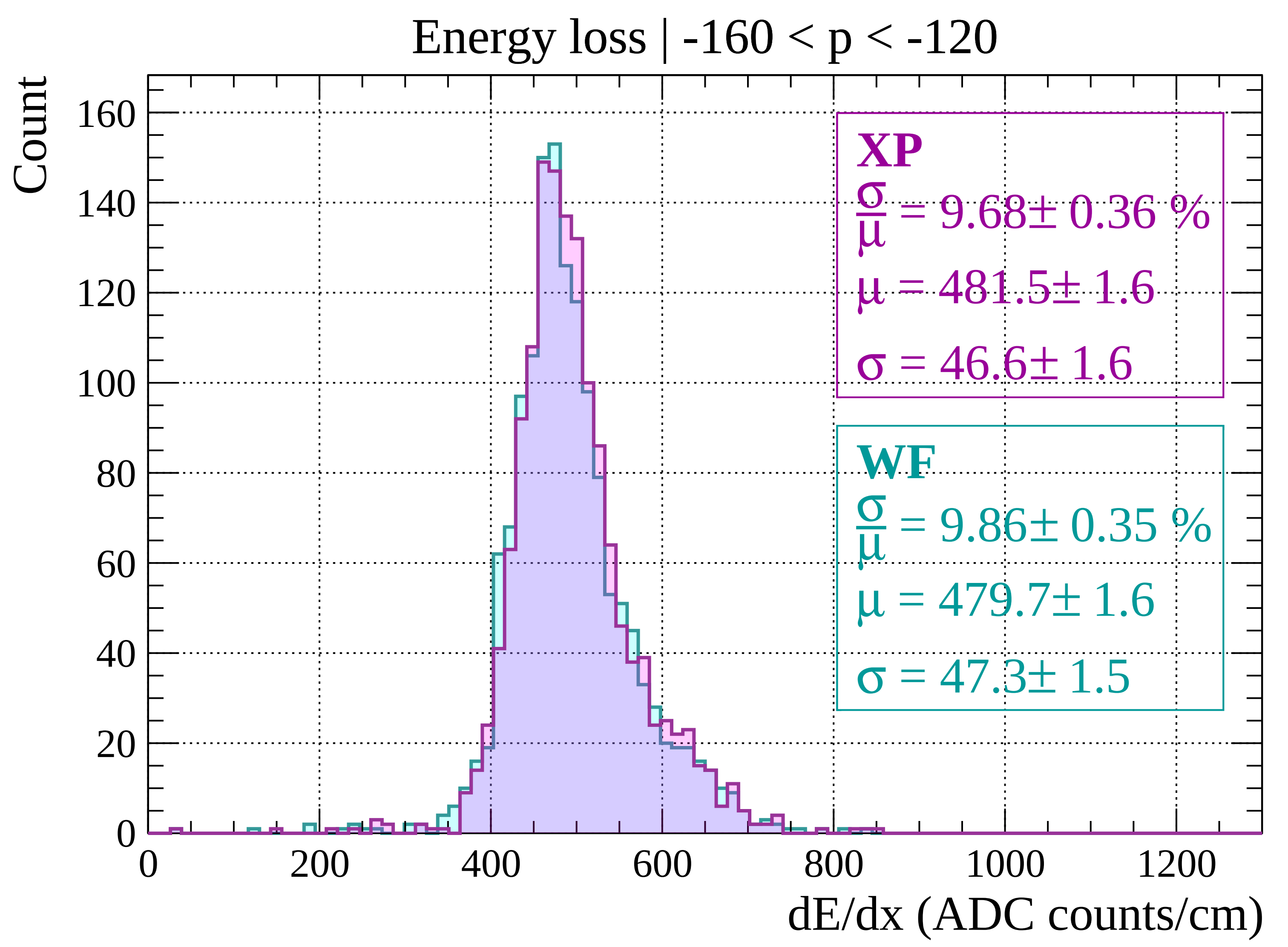

Energy loss | -320 < p < -280
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 9.05± 0.30 %
μ = 403.1± 1.0
σ = 36.5± 1.1
WF
σ/μ = 8.97± 0.25 %
μ = 400.5± 0.9
σ = 35.9± 0.9

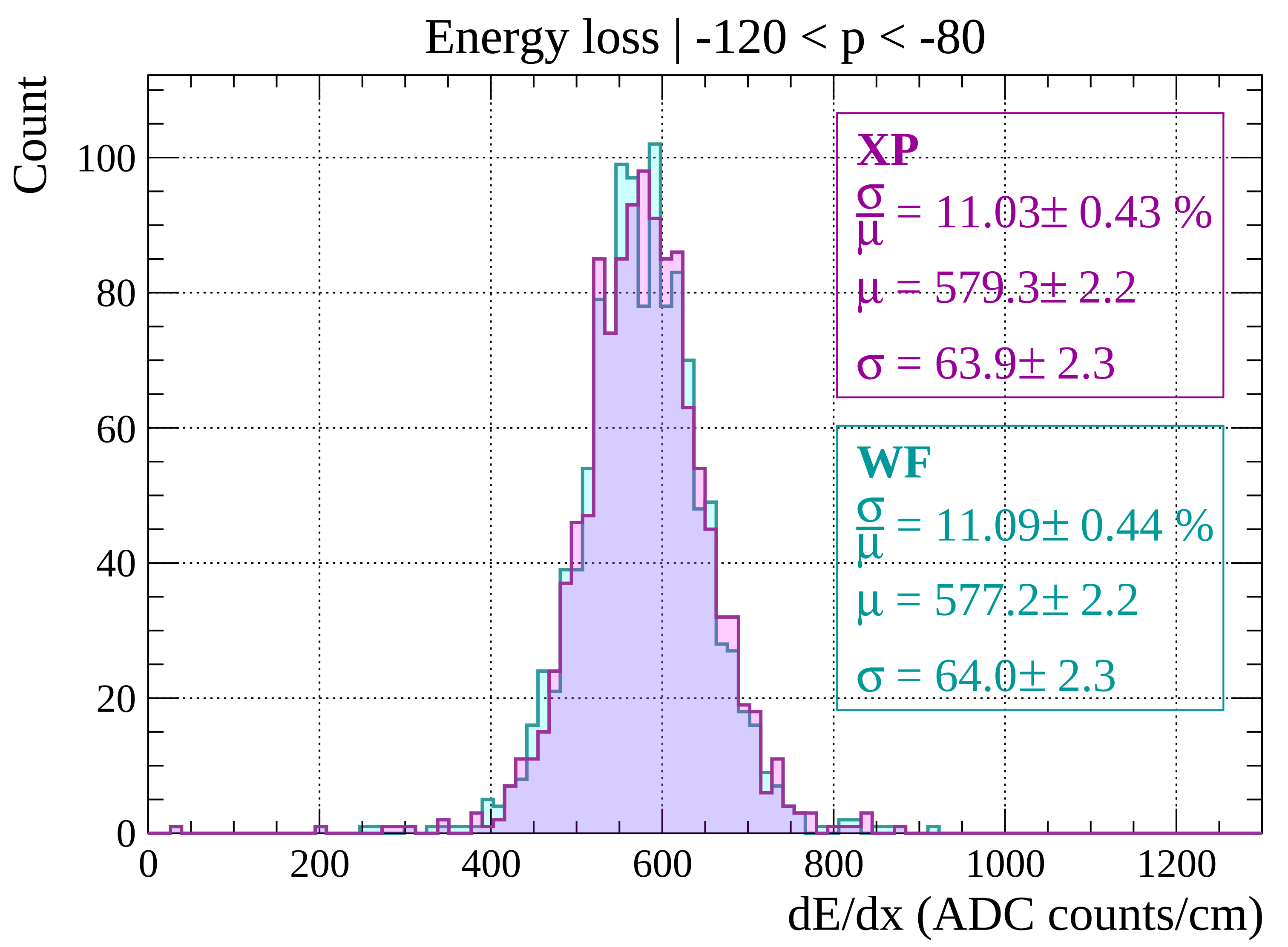


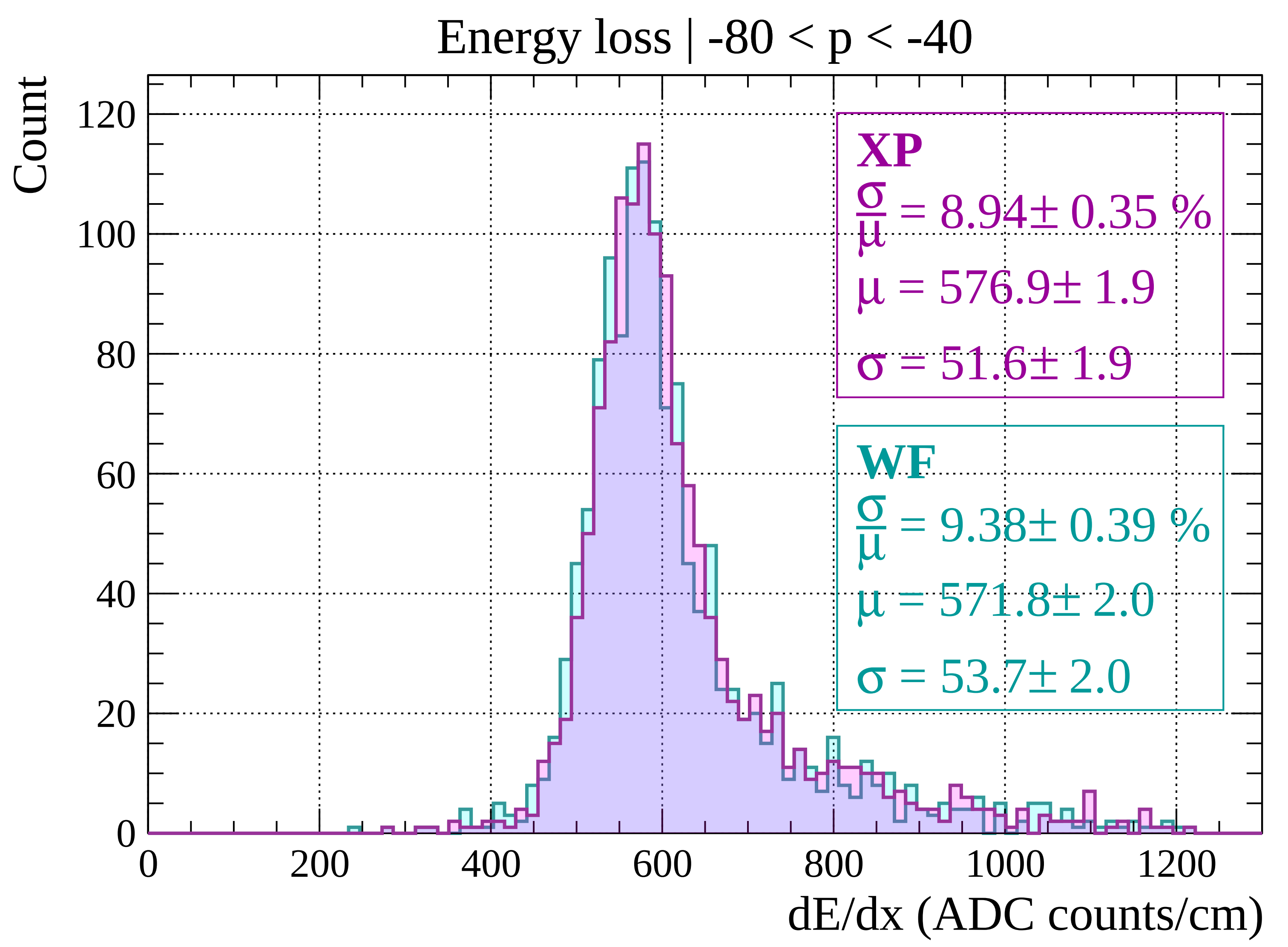
Energy loss | -280 < p < -240
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
0
200
400
600
800
1000
1200
XP
σ/μ = 8.14± 0.24 %
μ = 406.7± 0.9
σ = 33.1± 0.9
WF
σ/μ = 8.61± 0.27 %
μ = 404.7± 1.0
σ = 34.9± 1.0

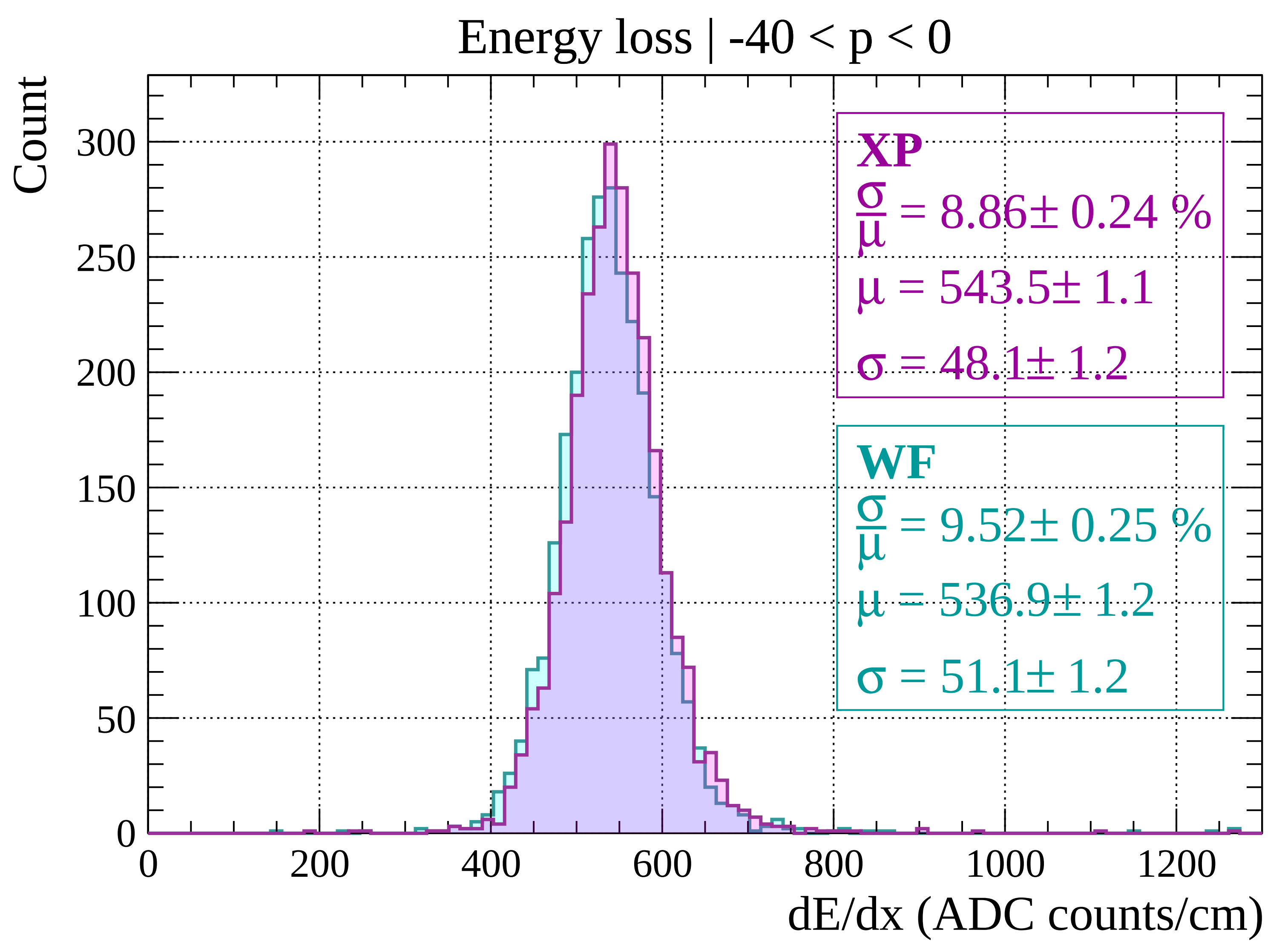

Energy loss | -240 < p < -200
Count
300
250
200
150
100
50
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 8.71± 0.22 %
μ = 415.2± 0.9
σ = 36.2± 0.8
WF
σ/μ = 9.01± 0.27 %
μ = 412.9± 1.0
σ = 37.2± 1.0

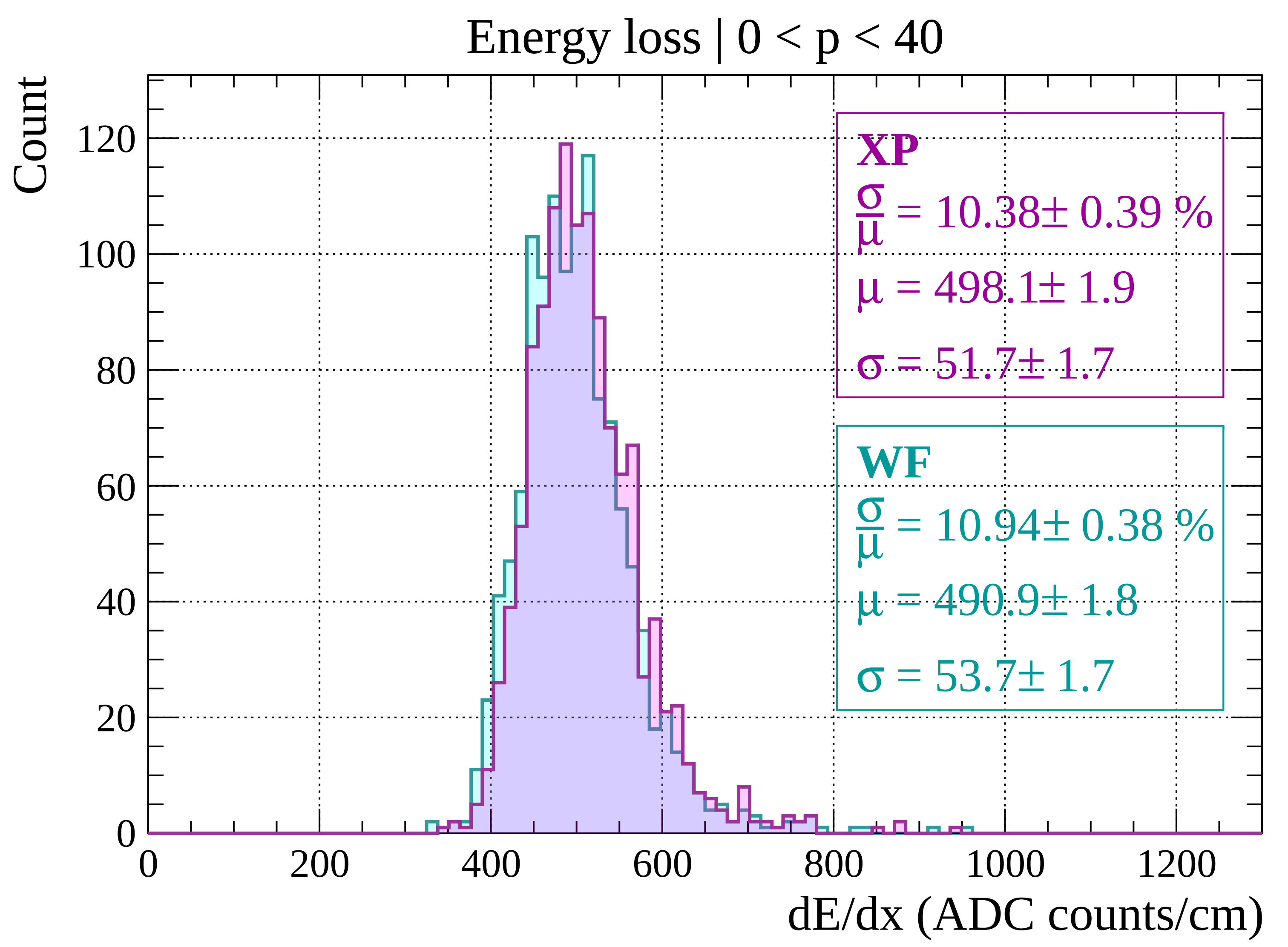

Energy loss | -200 < p < -160
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
0
200
400
600
800
1000
1200
XP
σ/μ = 8.84± 0.27 %
μ = 435.8± 1.1
σ = 38.5± 1.1
WF
σ/μ = 9.32± 0.32 %
μ = 433.1± 1.2
σ = 40.4± 1.3


Energy loss | -160 < p < -120
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
0
200
400
600
800
1000
1200
XP
σ/μ = 9.68± 0.36 %
μ = 481.5± 1.6
σ = 46.6± 1.6
WF
σ/μ = 9.86± 0.35 %
μ = 479.7± 1.6
σ = 47.3± 1.5

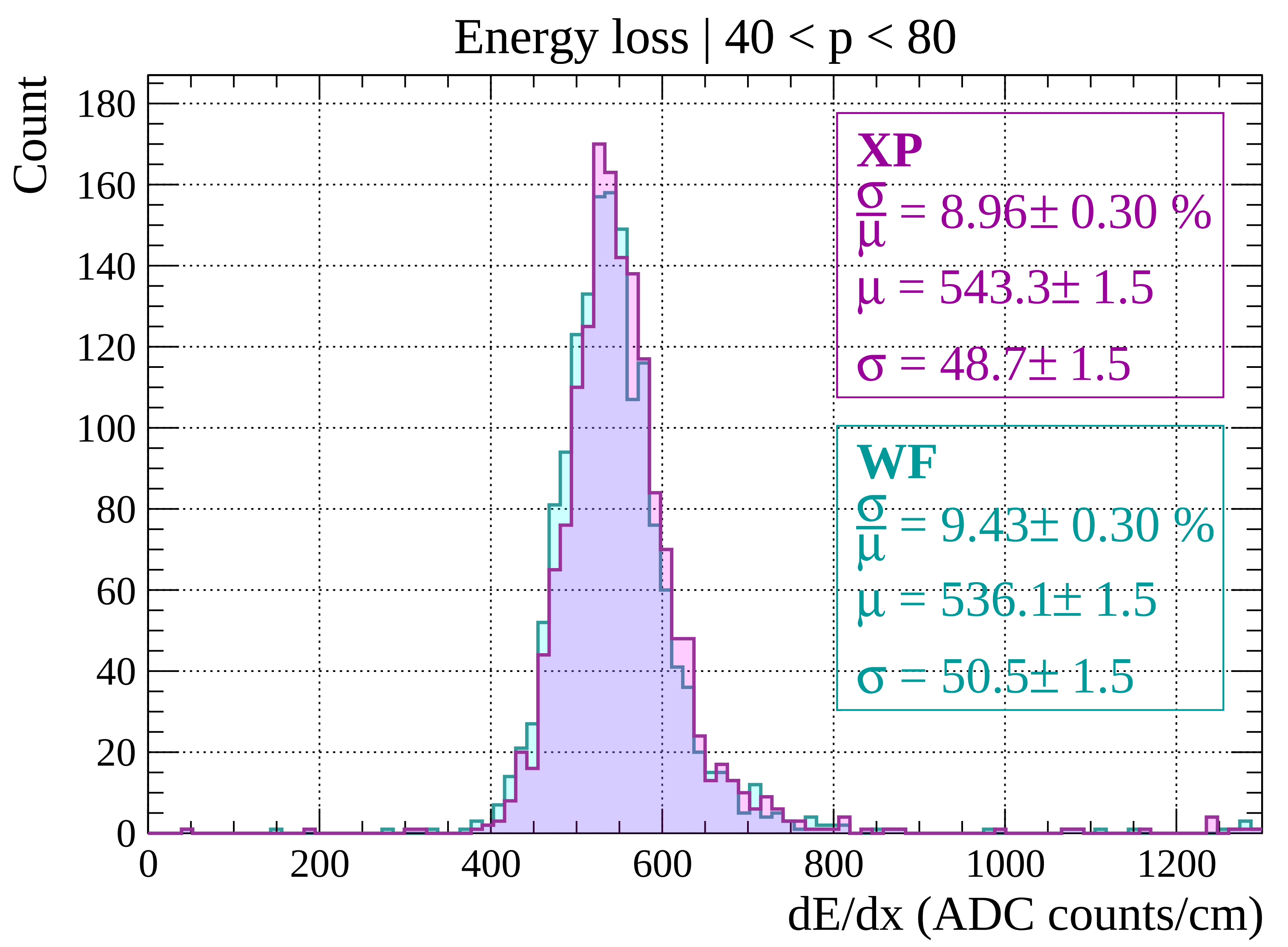

Energy loss | -120 < p < -80
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 11.03± 0.43 %
μ = 579.3± 2.2
σ = 63.9± 2.3
WF
σ/μ = 11.09± 0.44 %
μ = 577.2± 2.2
σ = 64.0± 2.3
0
20
40
60
80
100
0
200
400
600
800
1000
1200


Energy loss | -80 < p < -40
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
0
200
400
600
800
1000
1200
XP
σ/μ = 8.94± 0.35 %
μ = 576.9± 1.9
σ = 51.6± 1.9
WF
σ/μ = 9.38± 0.39 %
μ = 571.8± 2.0
σ = 53.7± 2.0

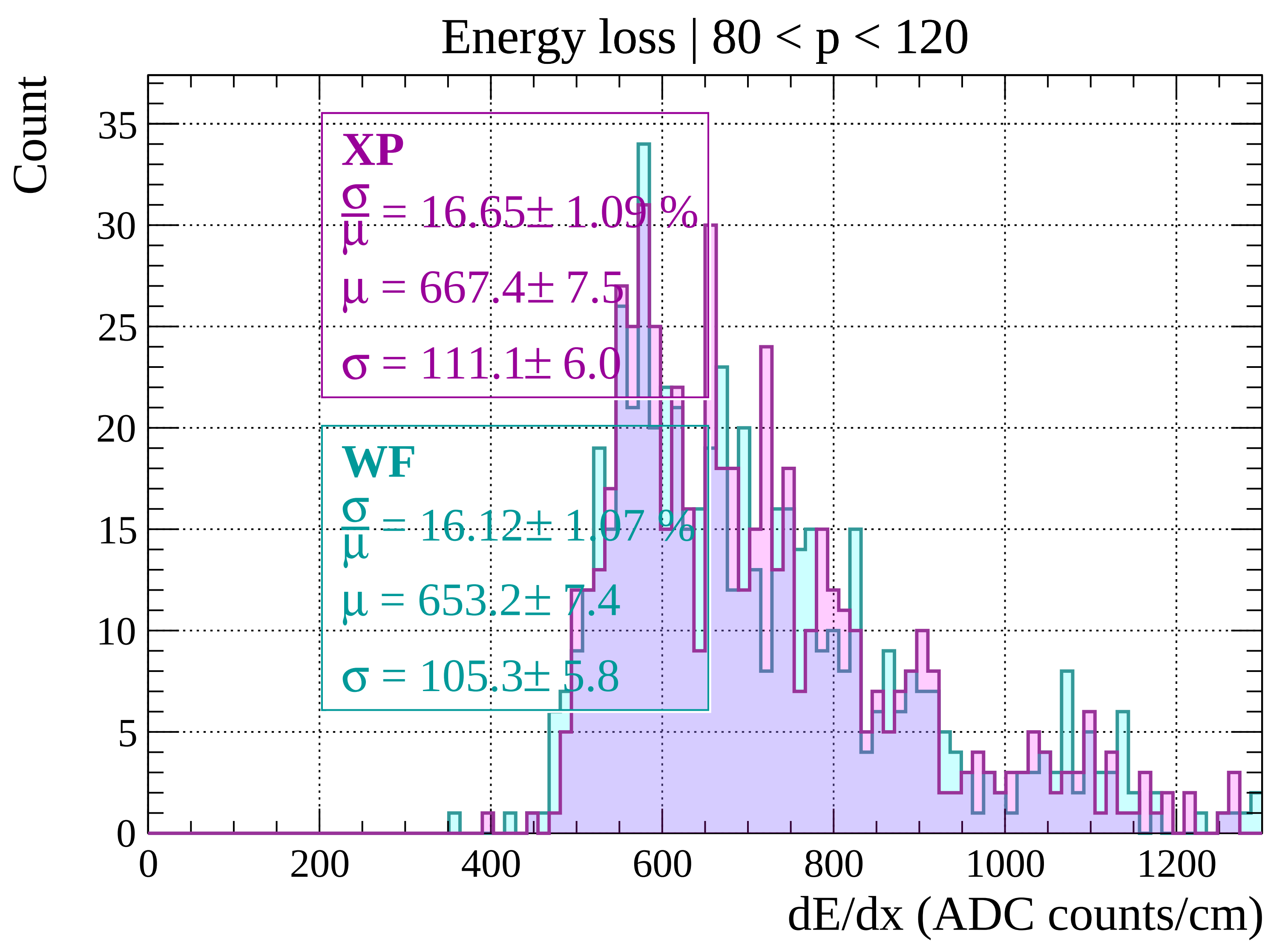

Energy loss | -40 < p < 0
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 8.86± 0.24 %
μ = 543.5± 1.1
σ = 48.1± 1.2
WF
σ/μ = 9.52± 0.25 %
μ = 536.9± 1.2
σ = 51.1± 1.2
0
50
100
150
200
250
300
0
200
400
600
800
1000
1200

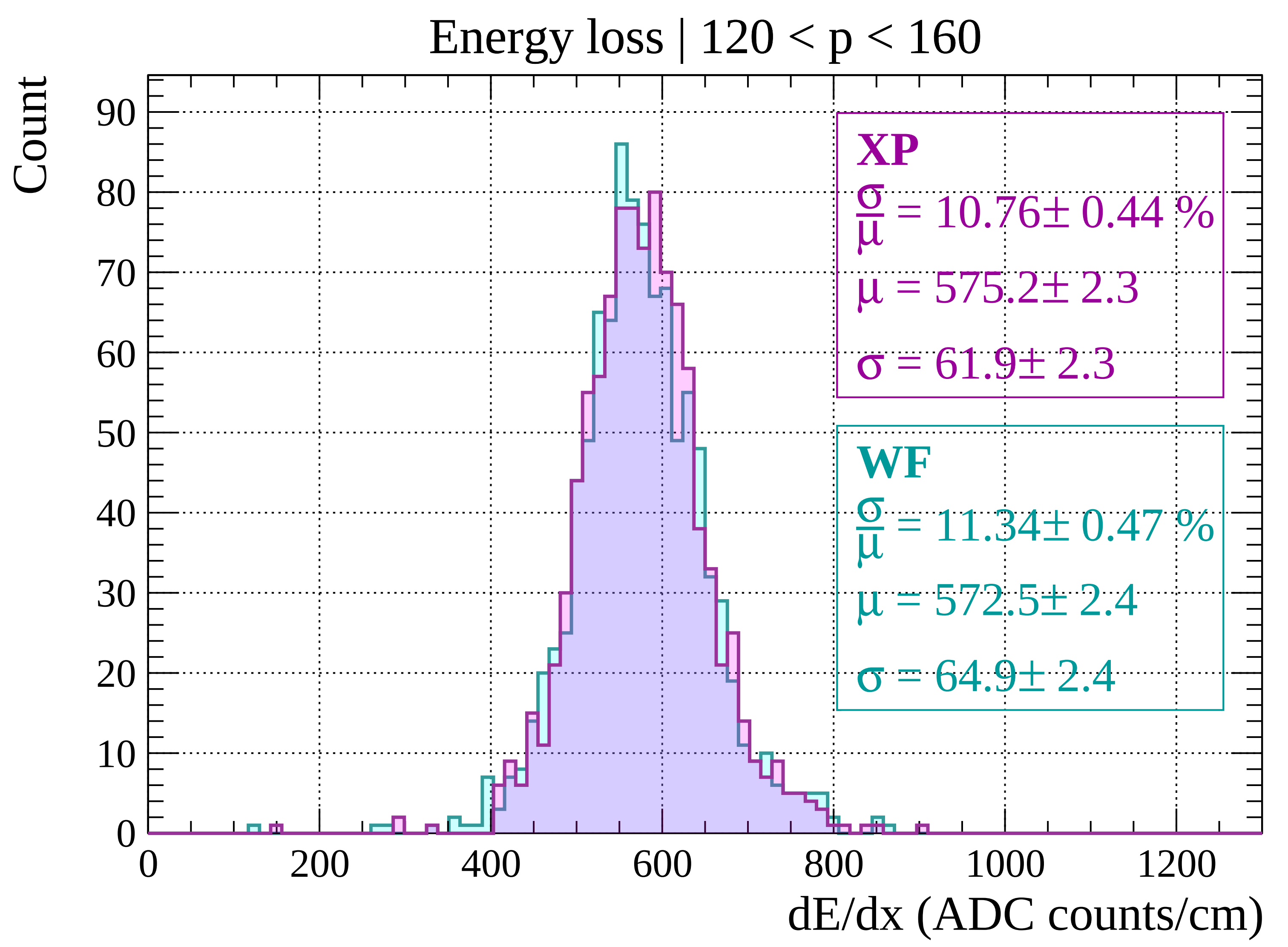


Energy loss | 0 < p < 40
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
120
XP
σ/μ = 10.38± 0.39 %
μ = 498.1± 1.9
σ = 51.7± 1.7
WF
σ/μ = 10.94± 0.38 %
μ = 490.9± 1.8
σ = 53.7± 1.7

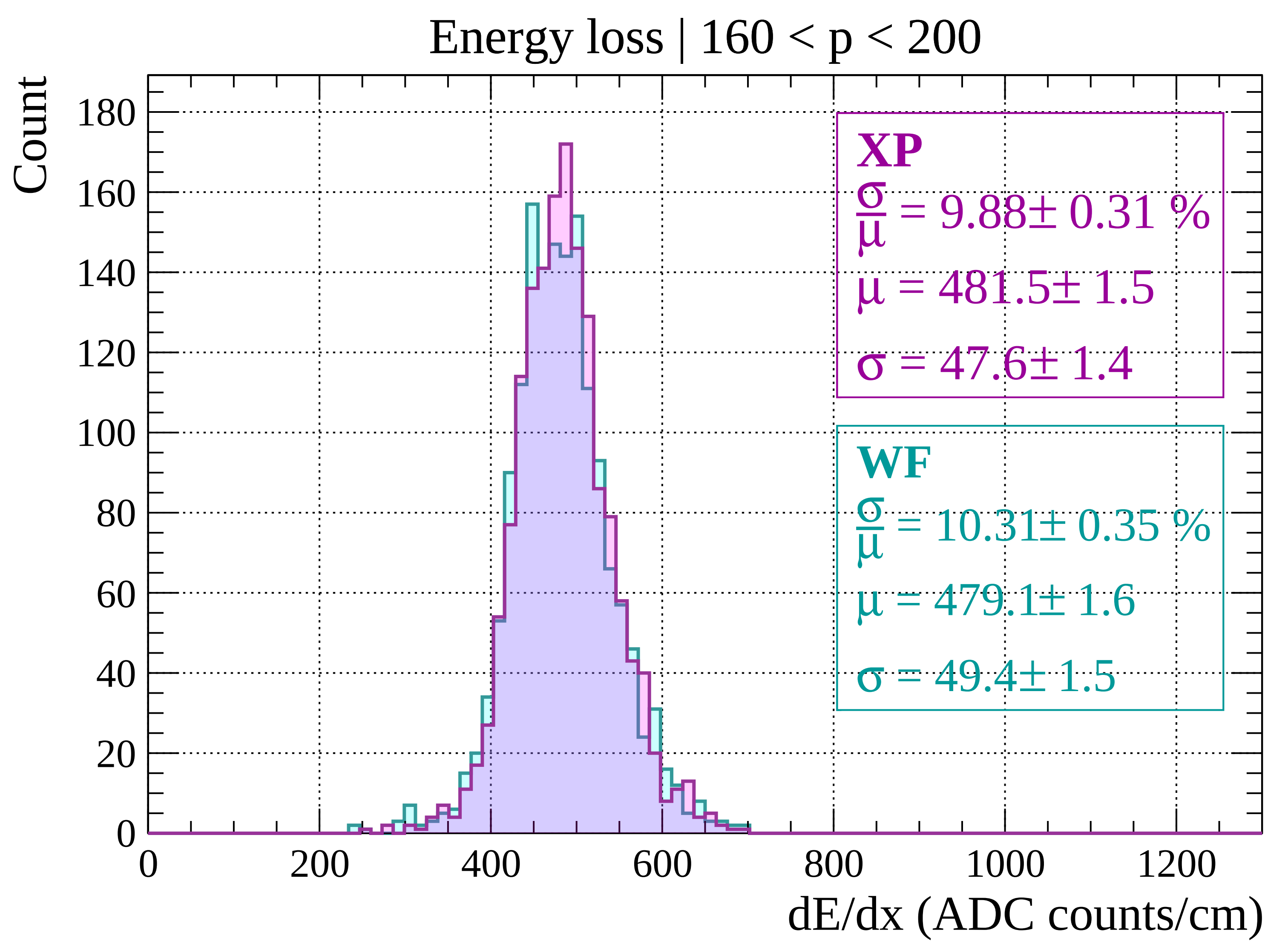


Energy loss | 40 < p < 80
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
120
140
160
180
XP
$\frac{\sigma}{\mu}$ = 8.96± 0.30 %
μ = 543.3± 1.5
σ = 48.7± 1.5
WF
$\frac{\sigma}{\mu}$ = 9.43± 0.30 %
μ = 536.1± 1.5
σ = 50.5± 1.5

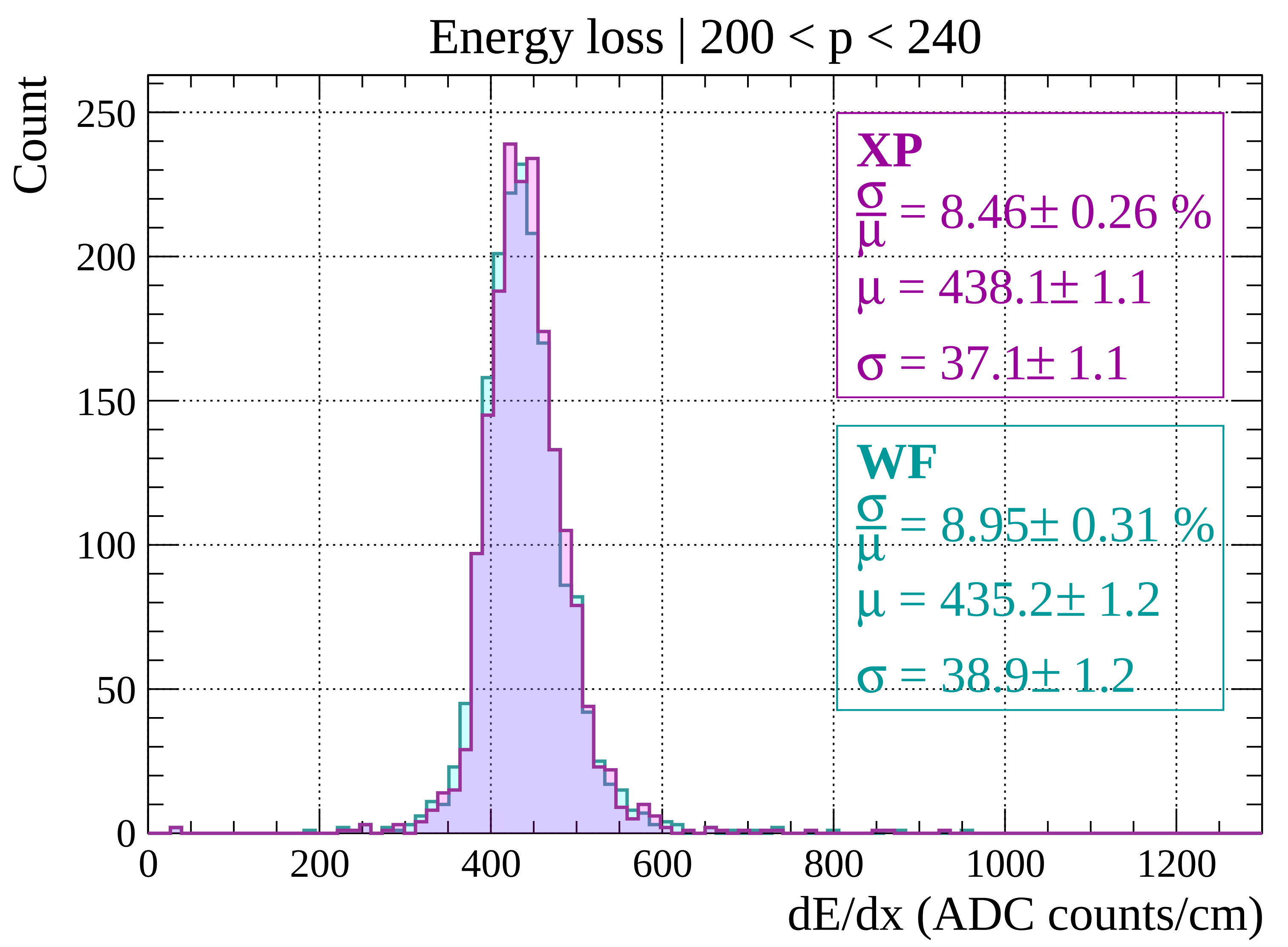

Energy loss | 80 < p < 120
Count
XP
σ/μ = 16.65± 1.09 %
μ = 667.4± 7.5
σ = 111.1± 6.0
WF
σ/μ = 16.12± 1.07 %
μ = 653.2± 7.4
σ = 105.3± 5.8
0
5
10
15
20
25
30
35
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)

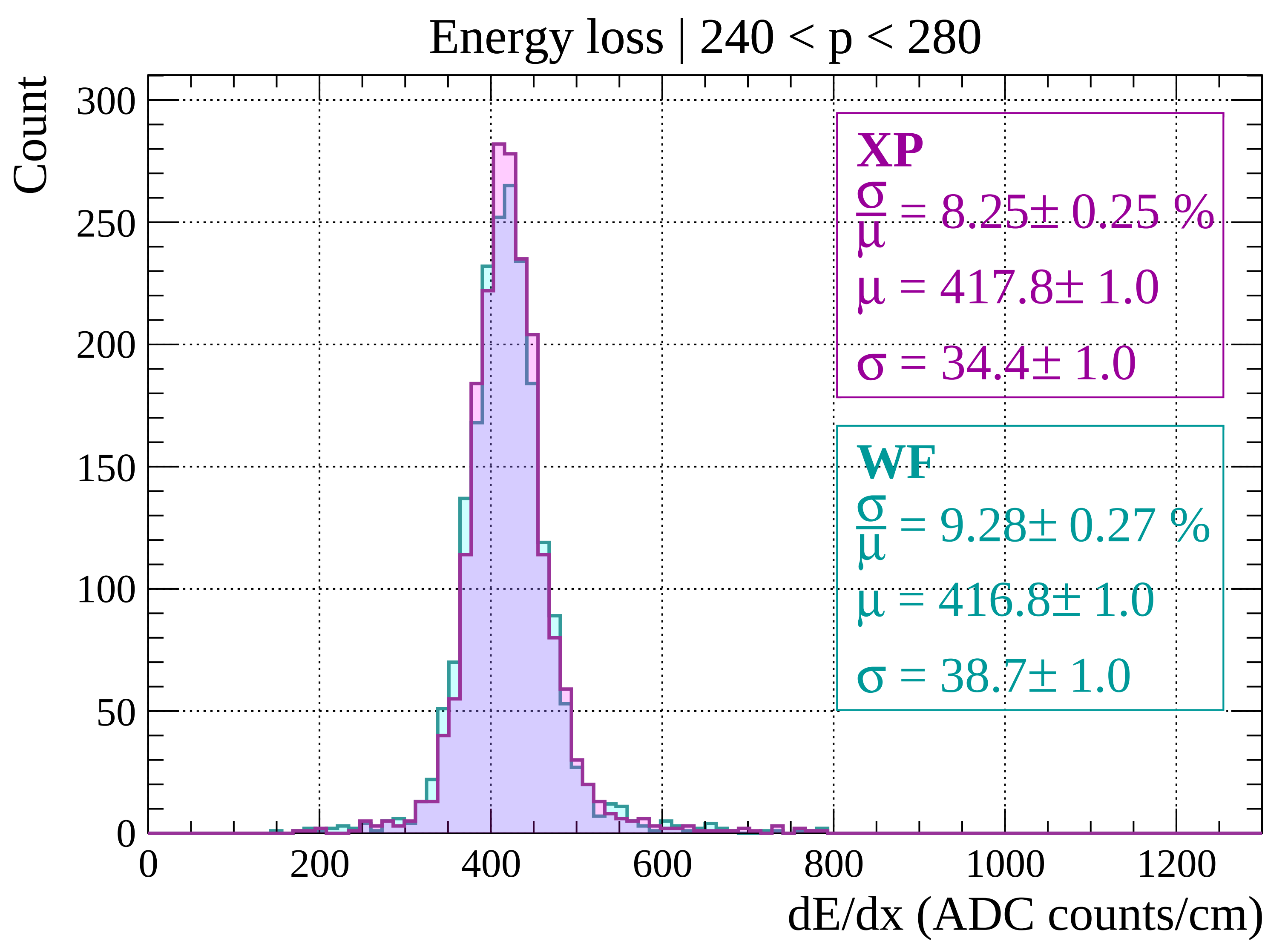


Energy loss | 120 < p < 160
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
70
80
90
0
200
400
600
800
1000
1200
XP
σ/μ = 10.76± 0.44 %
μ = 575.2± 2.3
σ = 61.9± 2.3
WF
σ/μ = 11.34± 0.47 %
μ = 572.5± 2.4
σ = 64.9± 2.4

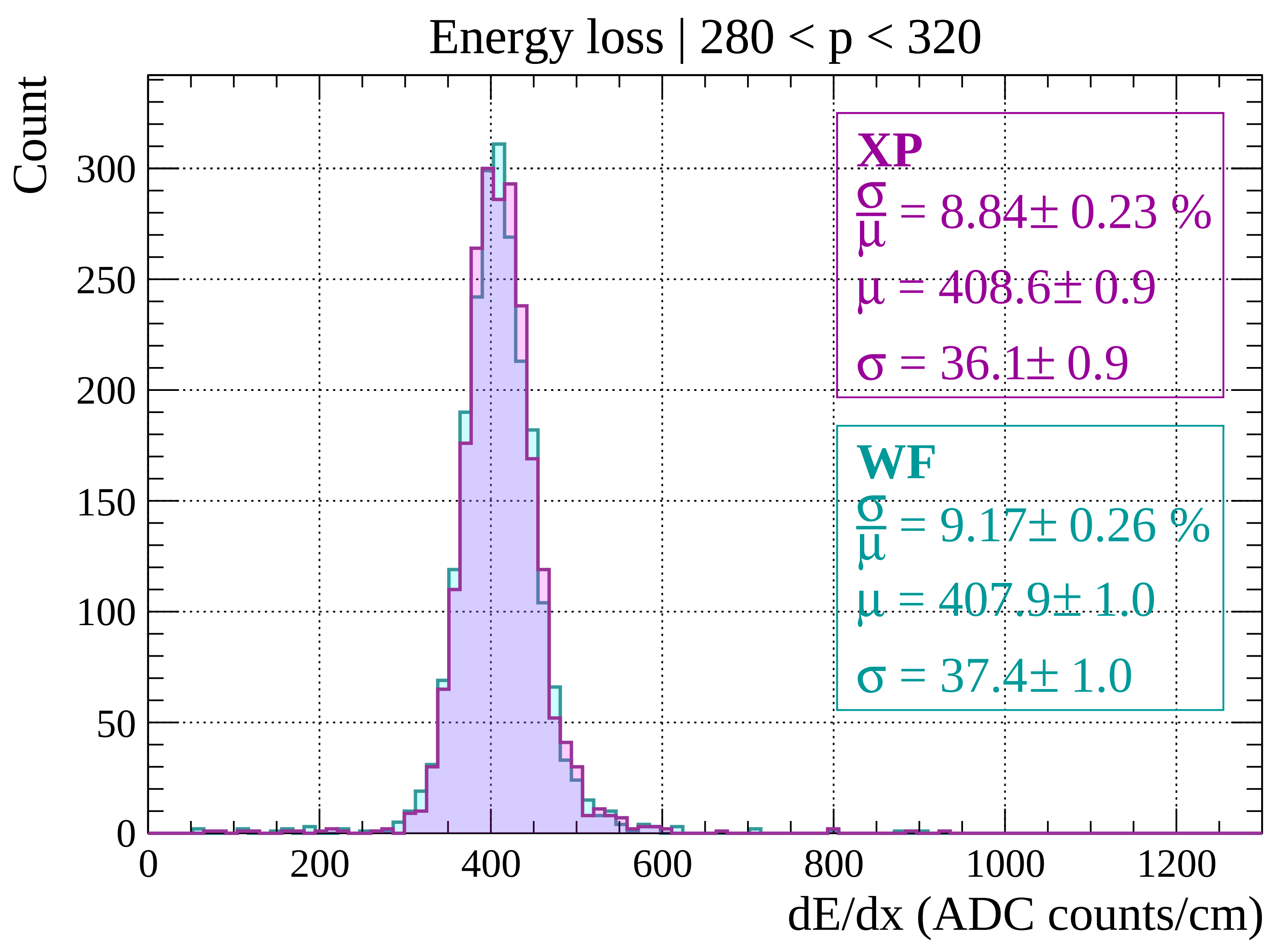


Energy loss | 160 < p < 200
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
180
200
400
600
800
1000
1200
XP
σ/μ = 9.88± 0.31 %
μ = 481.5± 1.5
σ = 47.6± 1.4
WF
σ/μ = 10.31± 0.35 %
μ = 479.1± 1.6
σ = 49.4± 1.5

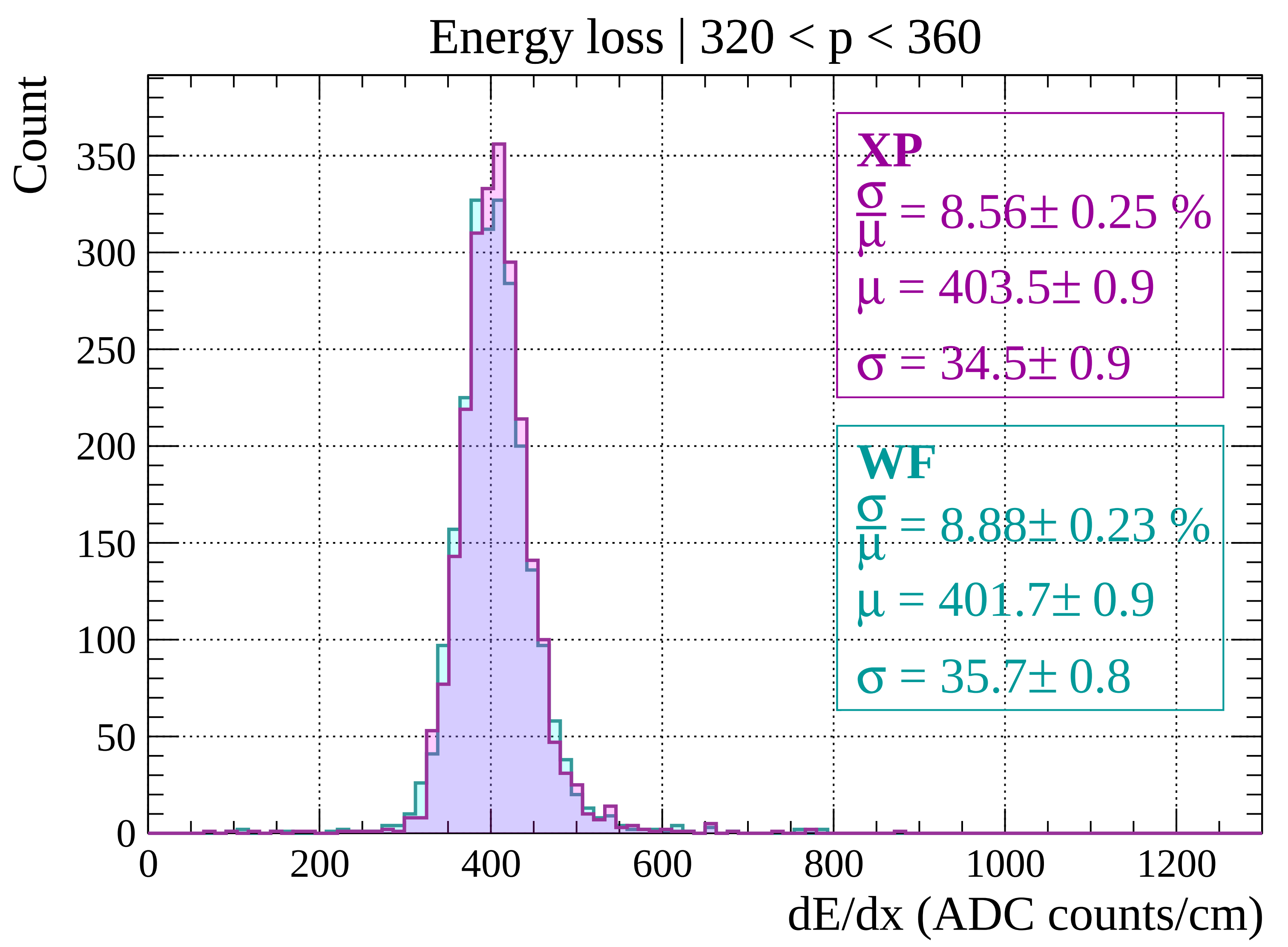

Energy loss | 200 < p < 240
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
0
200
400
600
800
1000
1200
XP
σ/μ = 8.46± 0.26 %
μ = 438.1± 1.1
σ = 37.1± 1.1
WF
σ/μ = 8.95± 0.31 %
μ = 435.2± 1.2
σ = 38.9± 1.2



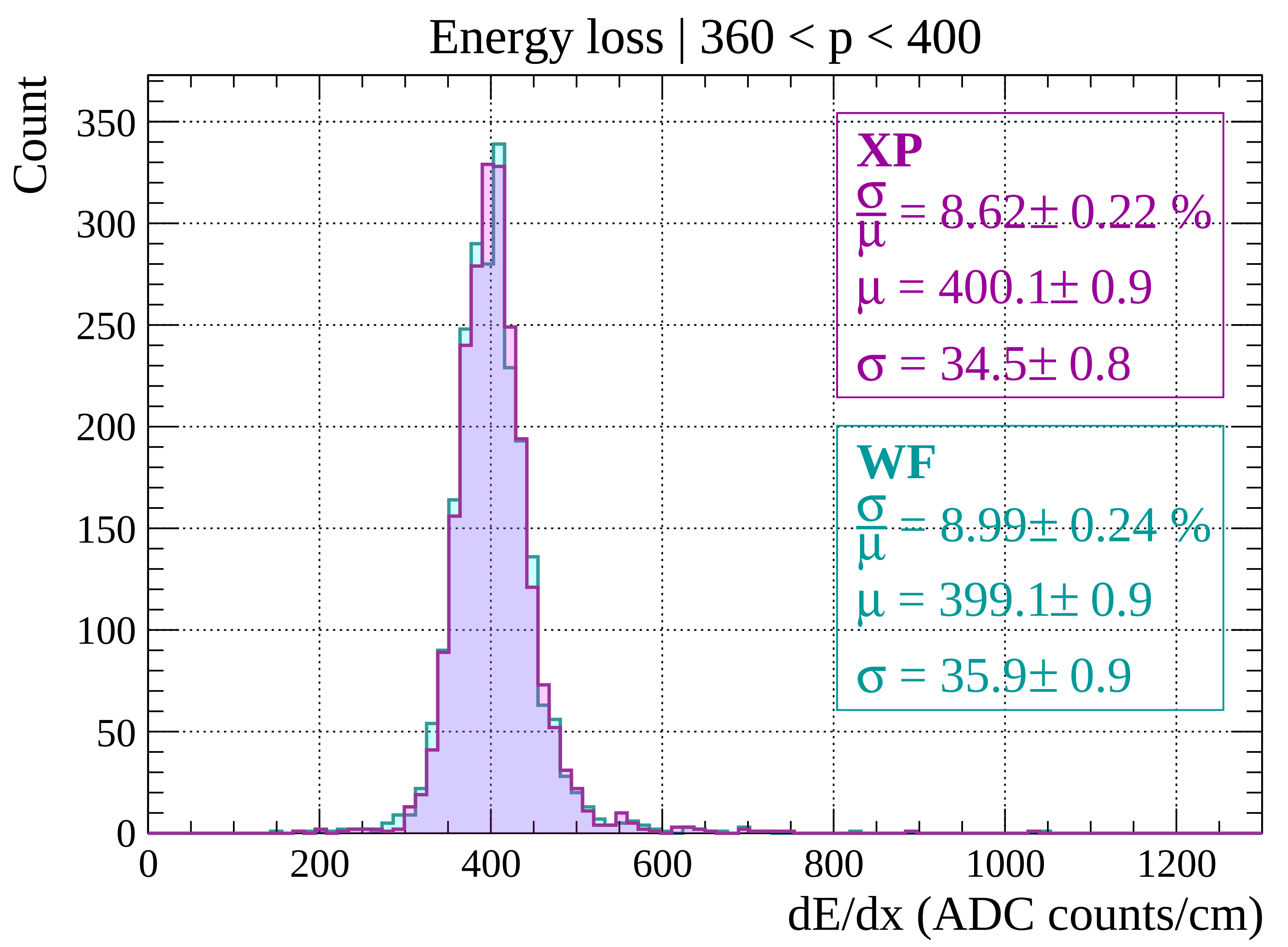
Energy loss | 240 < p < 280
Count
300
250
200
150
100
50
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 8.25± 0.25 %
μ = 417.8± 1.0
σ = 34.4± 1.0
WF
σ/μ = 9.28± 0.27 %
μ = 416.8± 1.0
σ = 38.7± 1.0

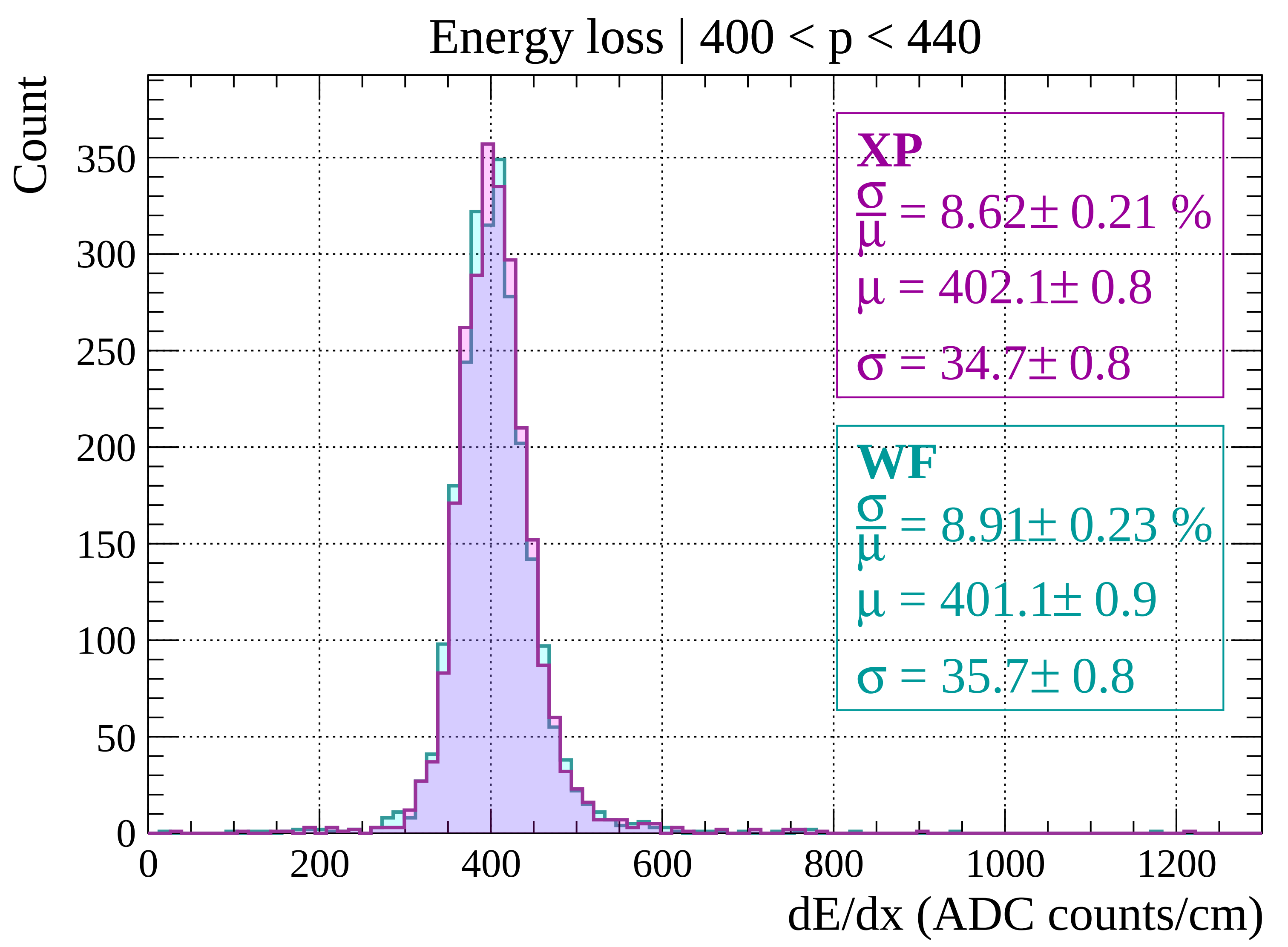


Energy loss | 280 < p < 320
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
0
200
400
600
800
1000
1200
XP
σ/μ = 8.84± 0.23 %
μ = 408.6± 0.9
σ = 36.1± 0.9
WF
σ/μ = 9.17± 0.26 %
μ = 407.9± 1.0
σ = 37.4± 1.0

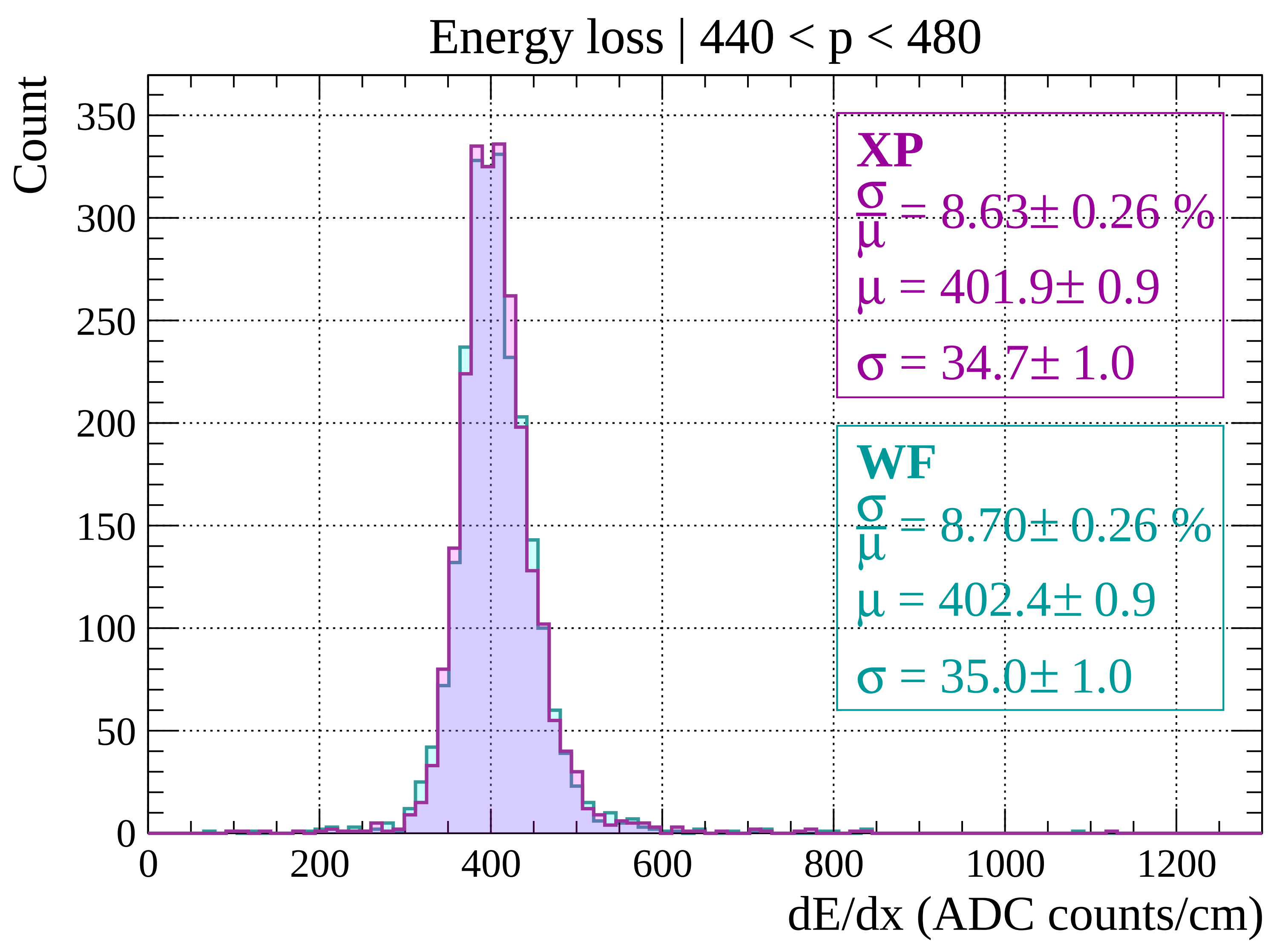


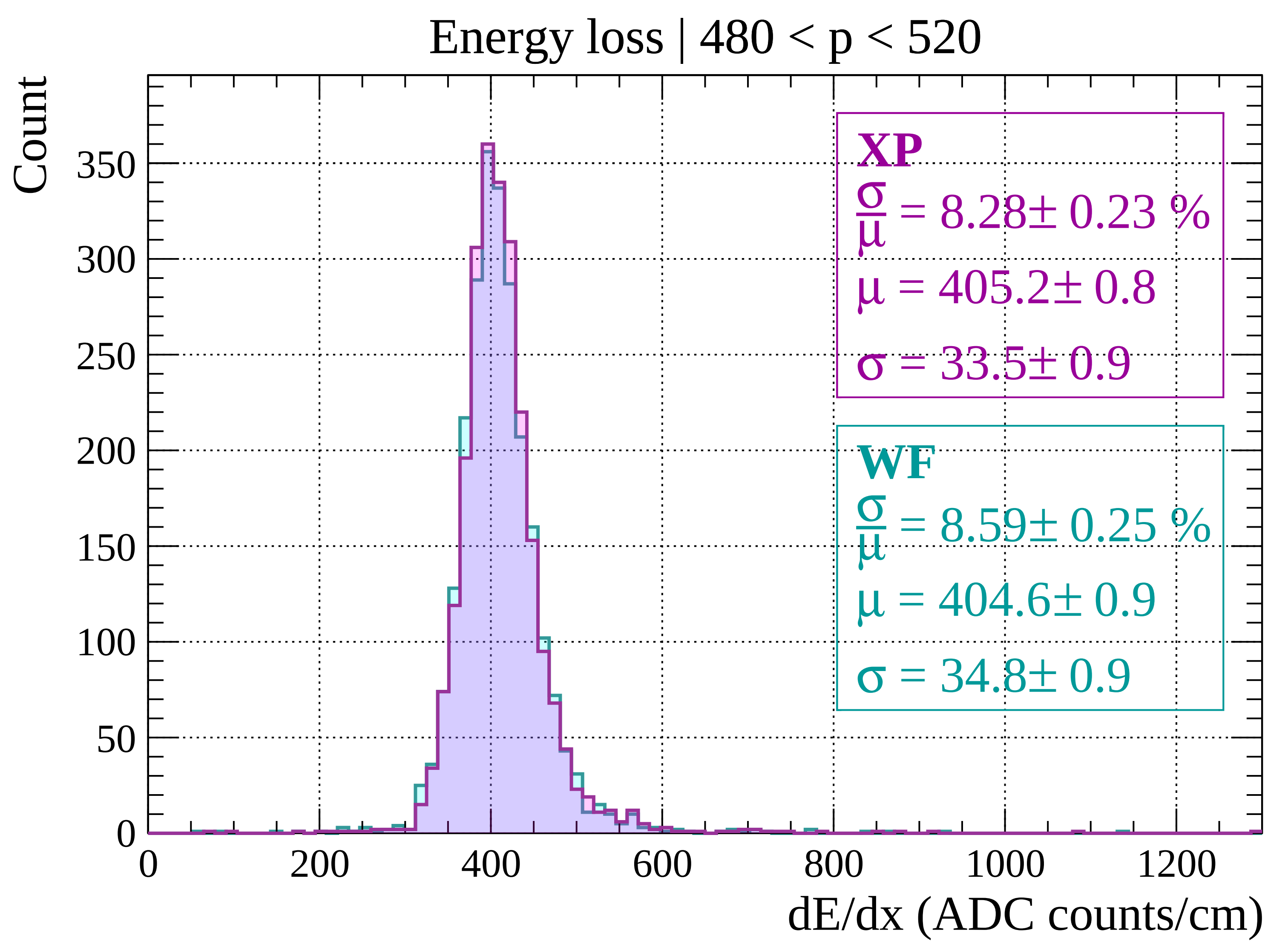
Energy loss | 320 < p < 360
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 8.56± 0.25 %
μ = 403.5± 0.9
σ = 34.5± 0.9
WF
$\frac{\sigma}{\mu}$ = 8.88± 0.23 %
μ = 401.7± 0.9
σ = 35.7± 0.8

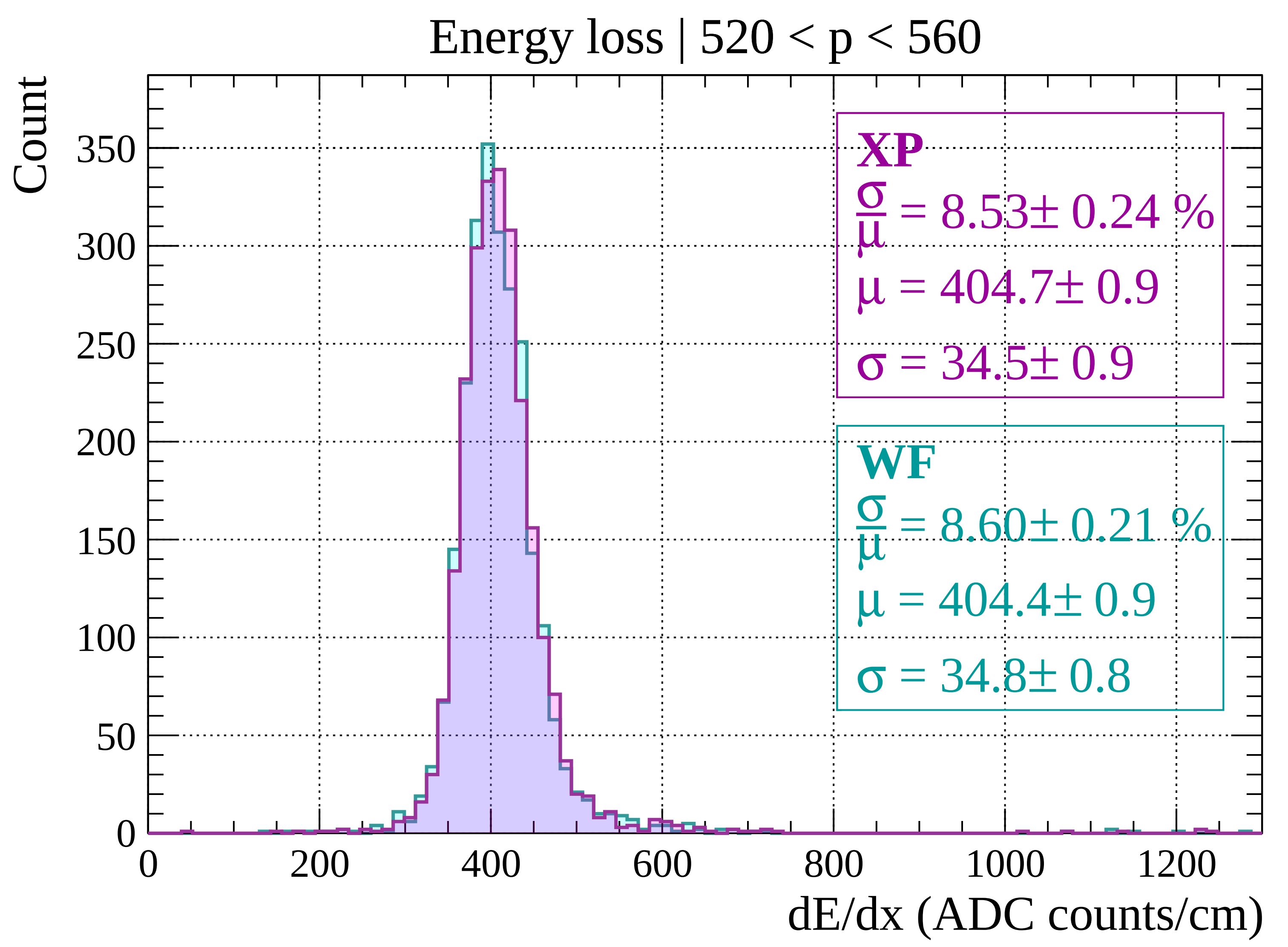

Energy loss | 360 < p < 400
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu} = 8.62 \pm 0.22$ %
$\mu = 400.1 \pm 0.9$
$\sigma = 34.5 \pm 0.8$
WF
$\frac{\sigma}{\mu} = 8.99 \pm 0.24$ %
$\mu = 399.1 \pm 0.9$
$\sigma = 35.9 \pm 0.9$



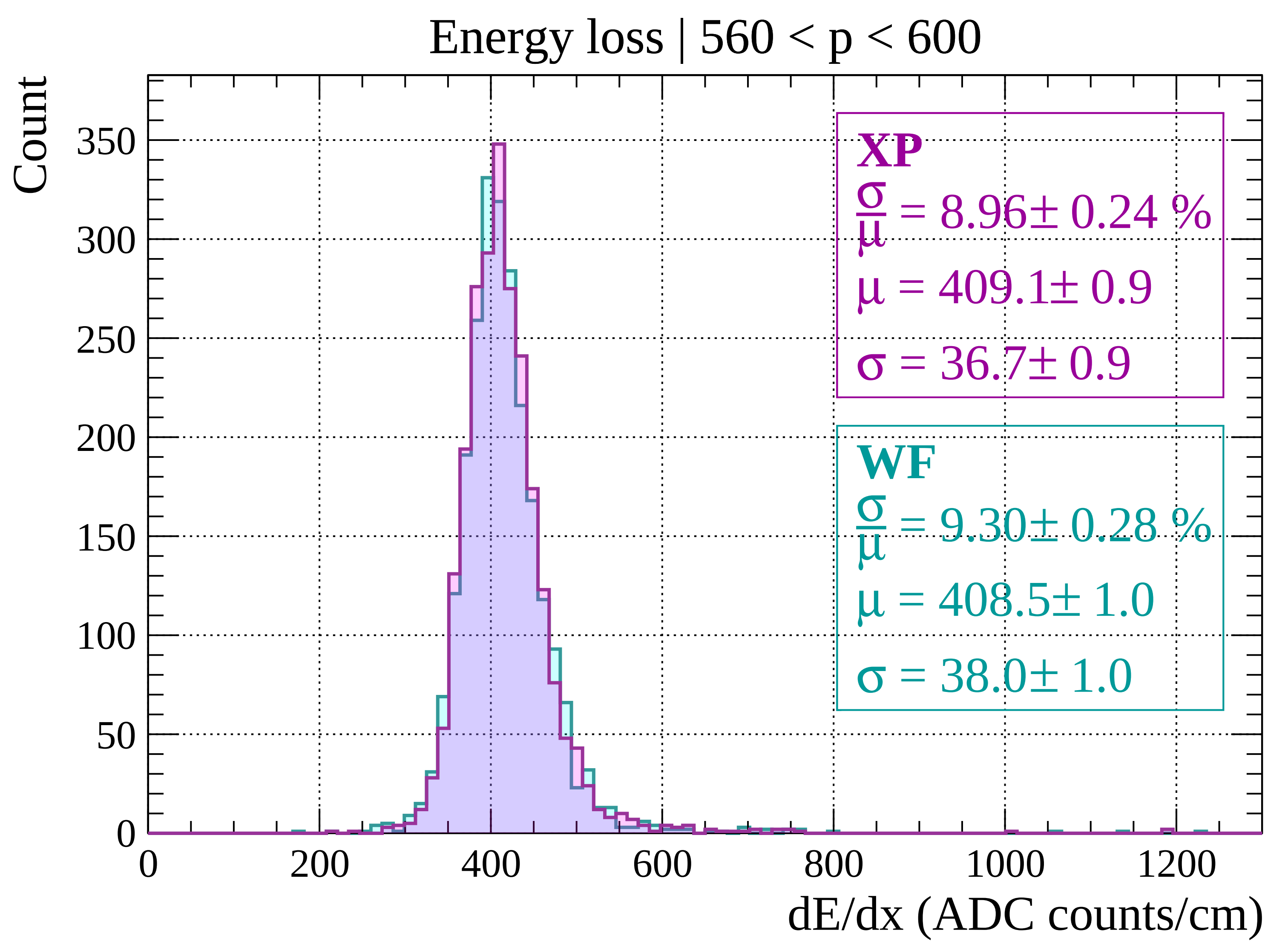
Energy loss | 400 < p < 440
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 8.62± 0.21 %
μ = 402.1± 0.8
σ = 34.7± 0.8
WF
σ/μ = 8.91± 0.23 %
μ = 401.1± 0.9
σ = 35.7± 0.8


Energy loss | 440 < p < 480
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
50
100
150
200
250
300
350
XP
$\frac{\sigma}{\mu}$ = 8.63± 0.26 %
μ = 401.9± 0.9
σ = 34.7± 1.0
WF
$\frac{\sigma}{\mu}$ = 8.70± 0.26 %
μ = 402.4± 0.9
σ = 35.0± 1.0

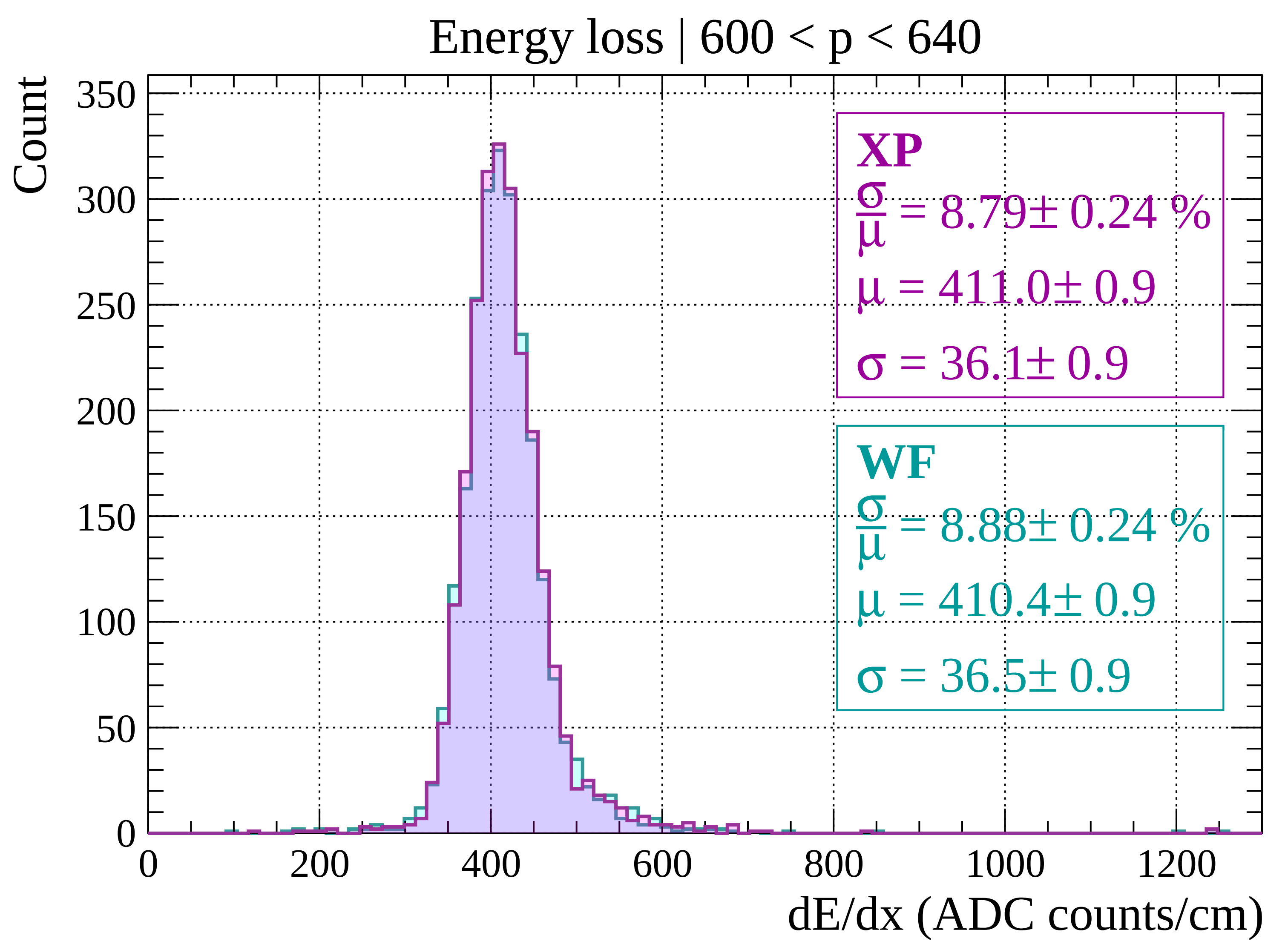

Energy loss | 480 < p < 520
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
50
100
150
200
250
300
350
XP
σ/μ = 8.28± 0.23 %
μ = 405.2± 0.8
σ = 33.5± 0.9
WF
σ/μ = 8.59± 0.25 %
μ = 404.6± 0.9
σ = 34.8± 0.9

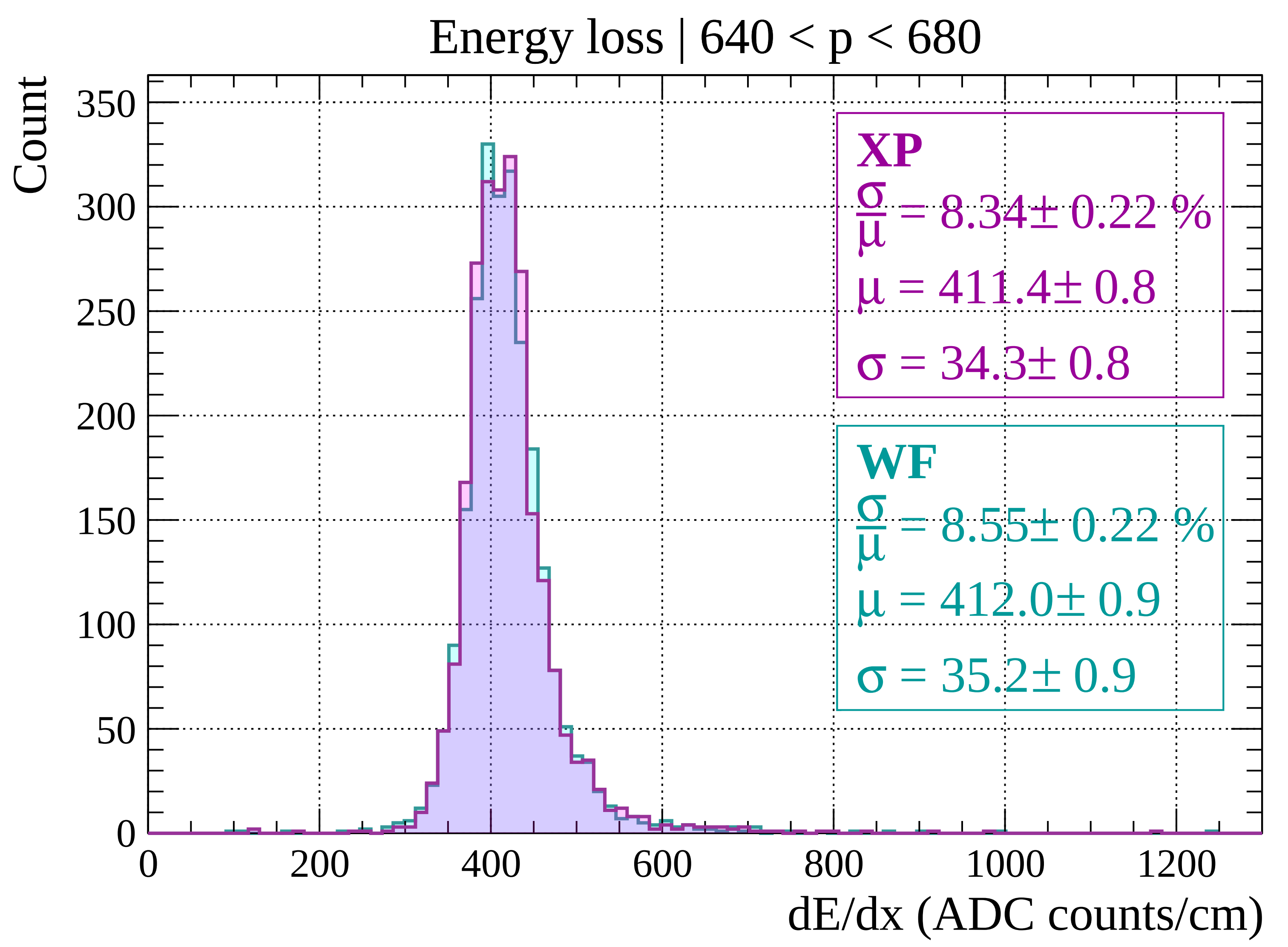


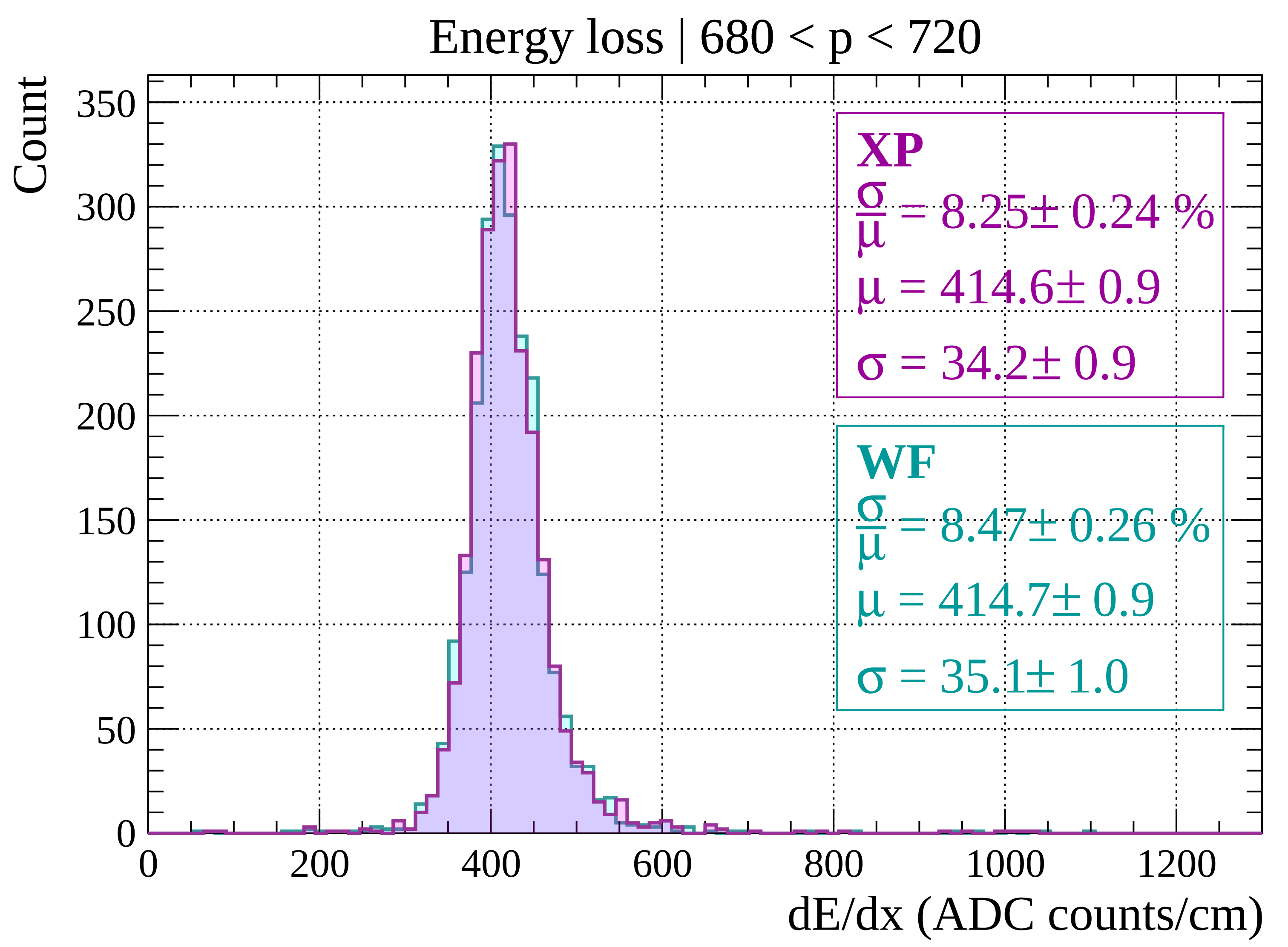
Energy loss | 520 < p < 560
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200
XP
σ/μ = 8.53± 0.24 %
μ = 404.7± 0.9
σ = 34.5± 0.9
WF
σ/μ = 8.60± 0.21 %
μ = 404.4± 0.9
σ = 34.8± 0.8

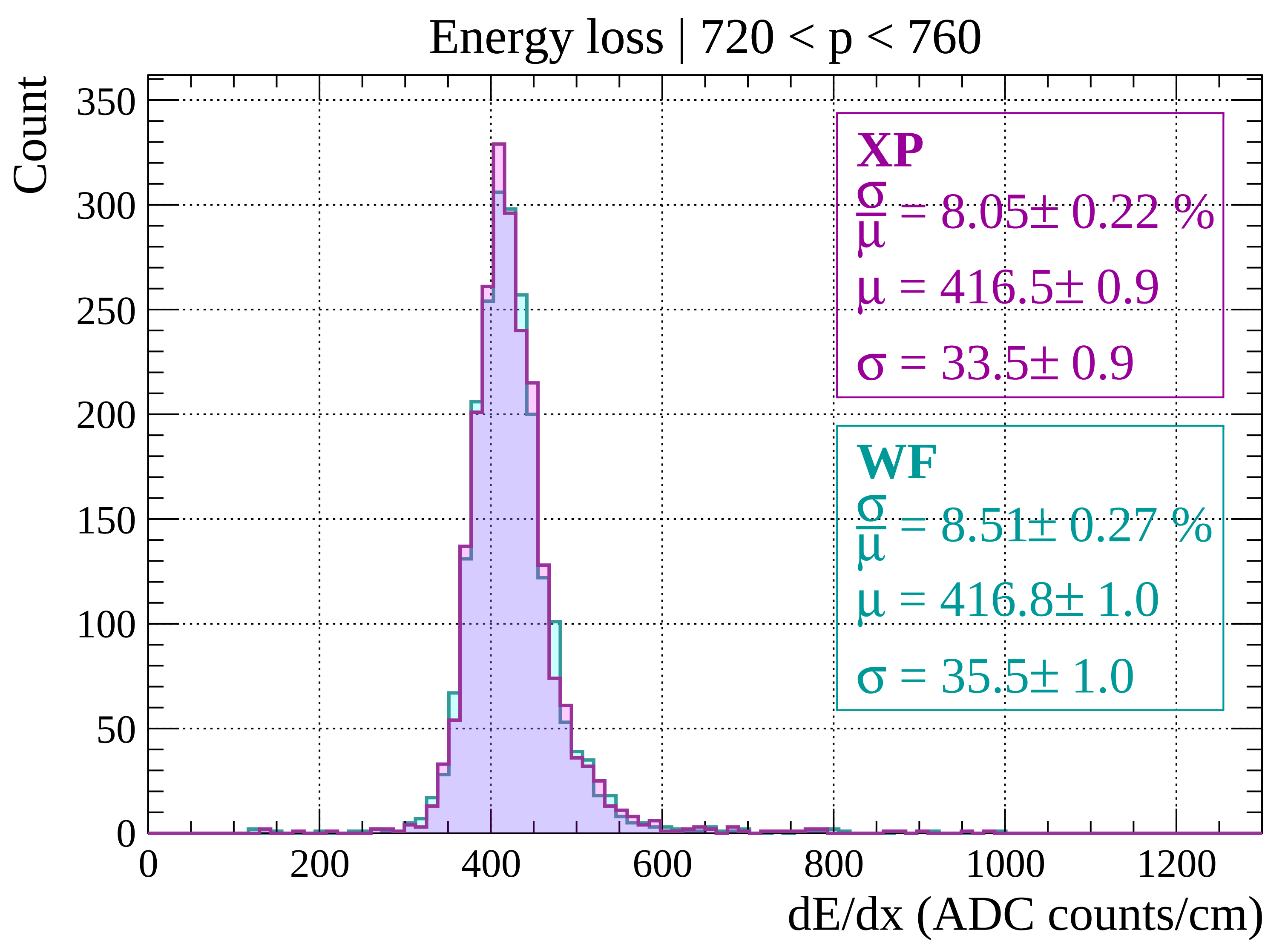


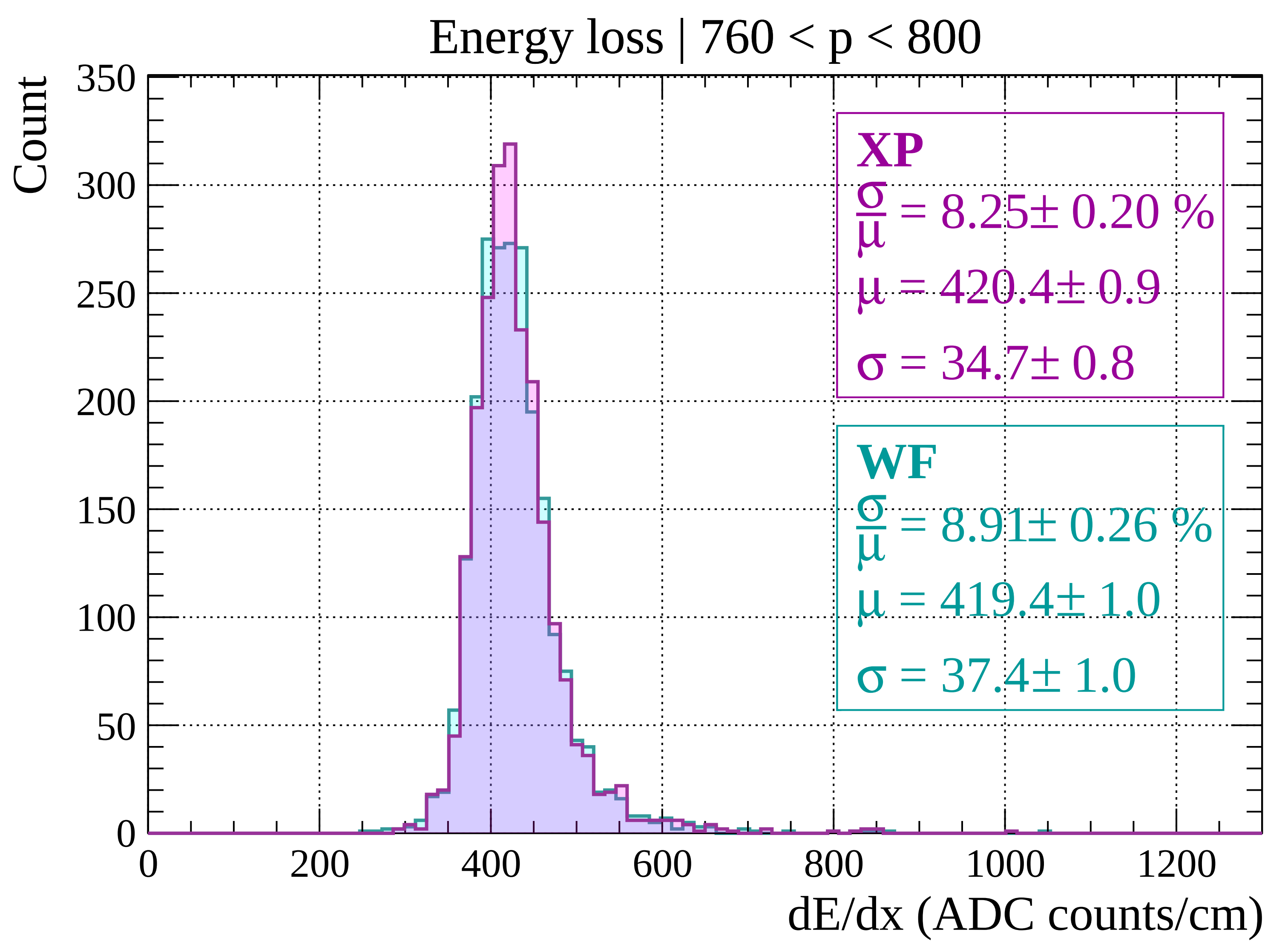
Energy loss | 560 < p < 600
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200
XP
σ/μ = 8.96± 0.24 %
μ = 409.1± 0.9
σ = 36.7± 0.9
WF
σ/μ = 9.30± 0.28 %
μ = 408.5± 1.0
σ = 38.0± 1.0

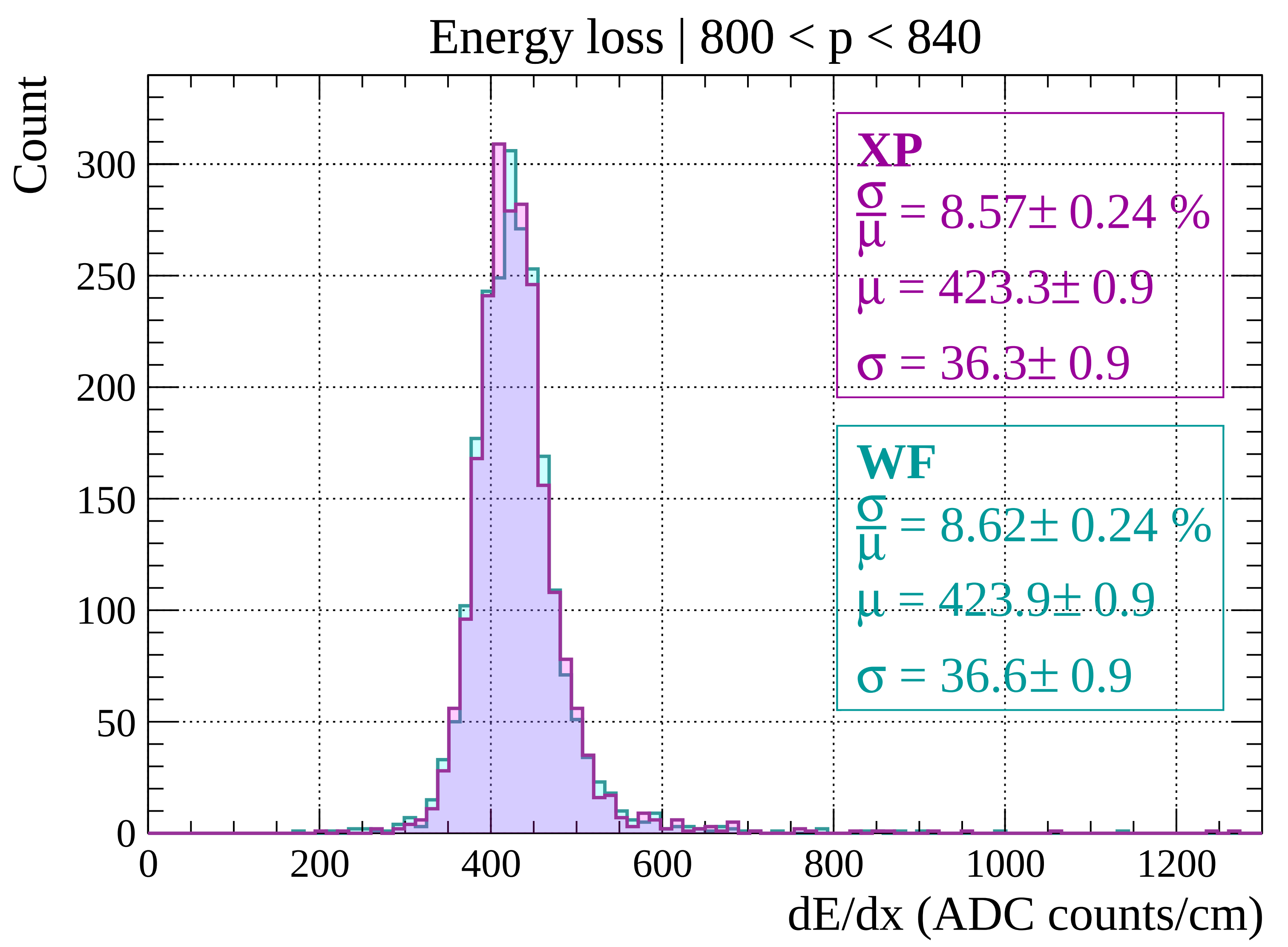

Energy loss | 600 < p < 640
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 8.79± 0.24 %
μ = 411.0± 0.9
σ = 36.1± 0.9
WF
σ/μ = 8.88± 0.24 %
μ = 410.4± 0.9
σ = 36.5± 0.9
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200

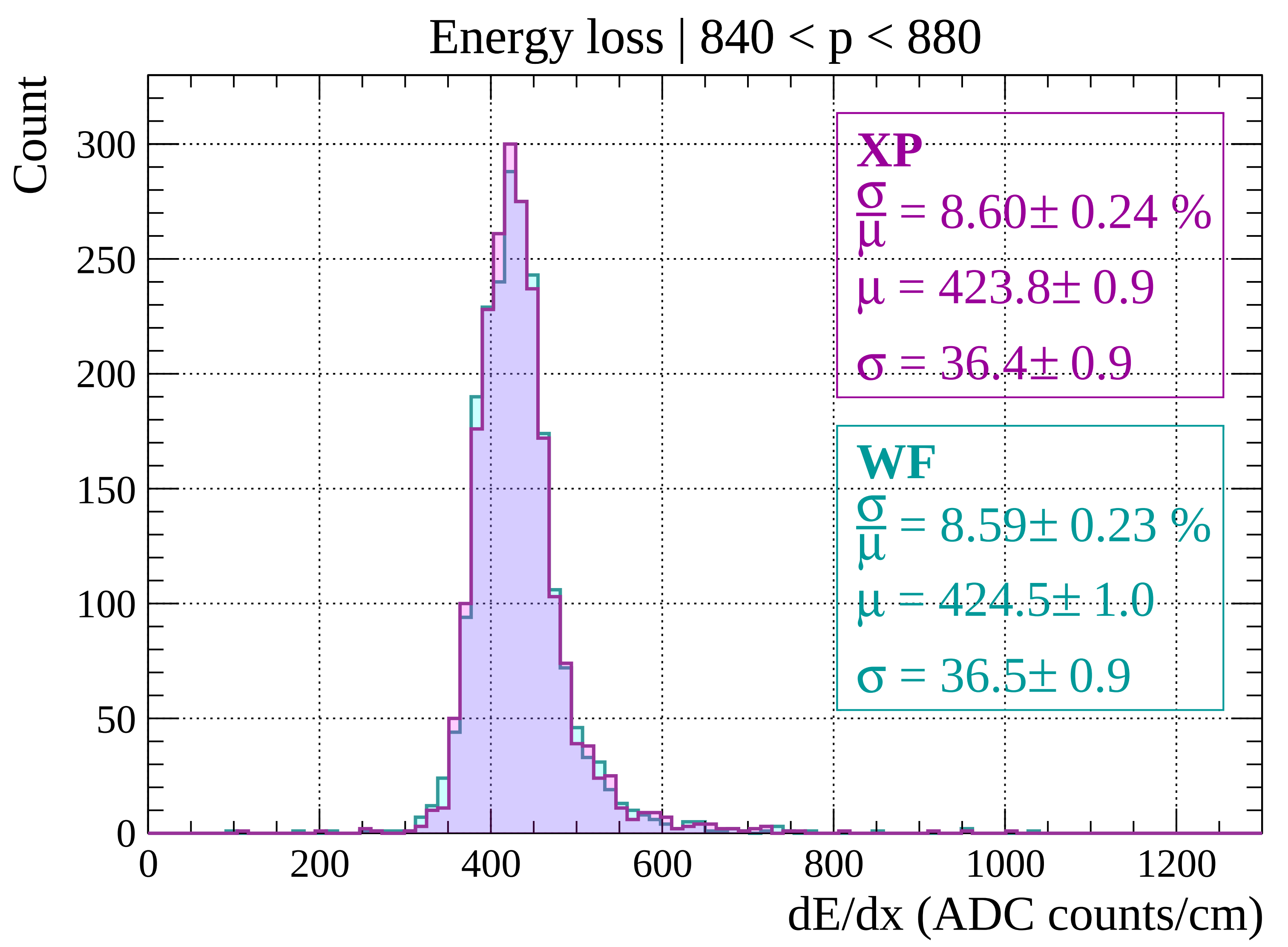

Energy loss | 640 < p < 680
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
50
100
150
200
250
300
350
XP
σ/μ = 8.34± 0.22 %
μ = 411.4± 0.8
σ = 34.3± 0.8
WF
σ/μ = 8.55± 0.22 %
μ = 412.0± 0.9
σ = 35.2± 0.9


Energy loss | 680 < p < 720
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200
XP
σ/μ = 8.25± 0.24 %
μ = 414.6± 0.9
σ = 34.2± 0.9
WF
σ/μ = 8.47± 0.26 %
μ = 414.7± 0.9
σ = 35.1± 1.0

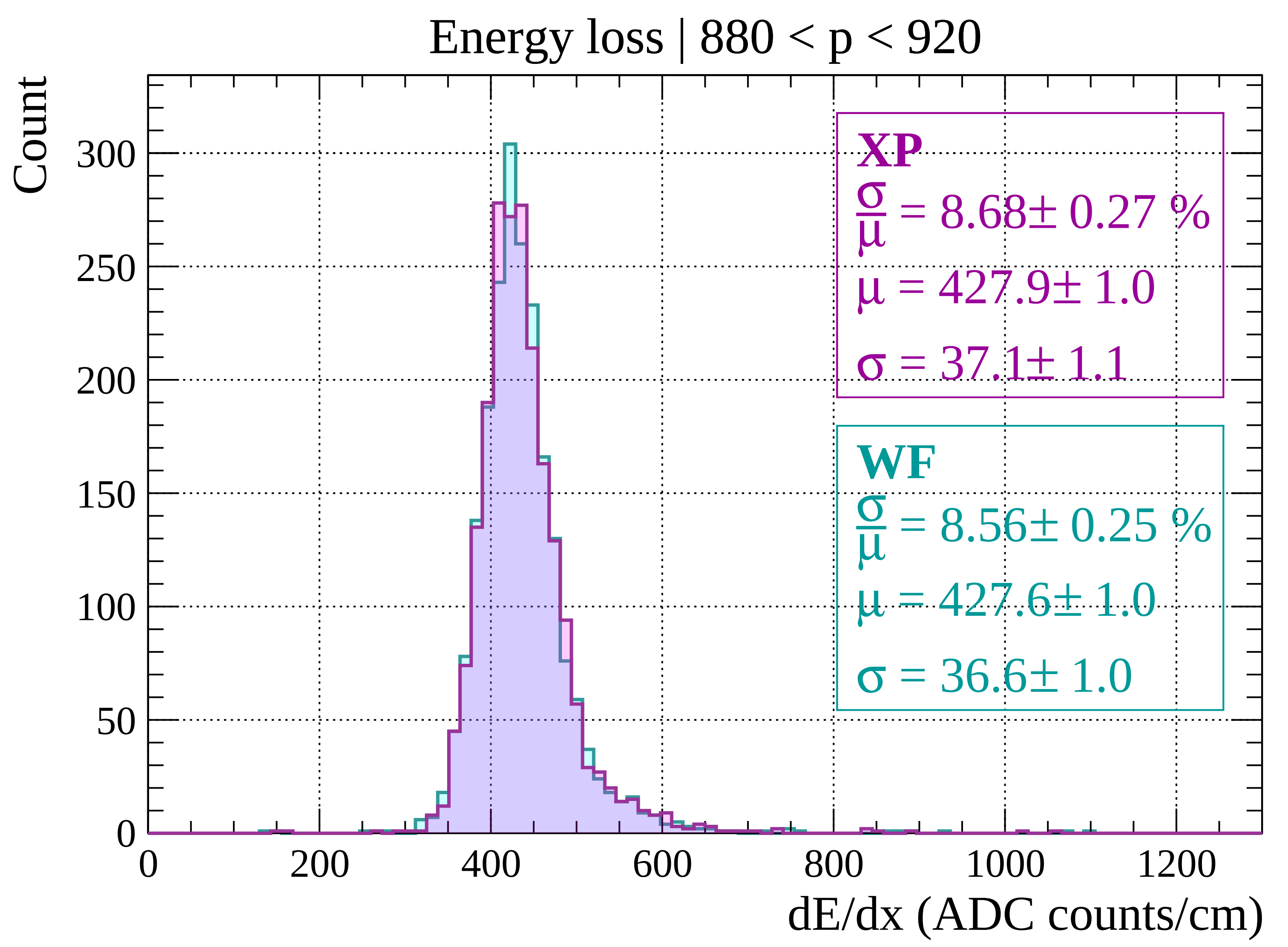

Energy loss | 720 < p < 760
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 8.05± 0.22 %
μ = 416.5± 0.9
σ = 33.5± 0.9
WF
$\frac{\sigma}{\mu}$ = 8.51± 0.27 %
μ = 416.8± 1.0
σ = 35.5± 1.0
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200


Energy loss | 760 < p < 800
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
50
100
150
200
250
300
350
XP
σ/μ = 8.25± 0.20 %
μ = 420.4± 0.9
σ = 34.7± 0.8
WF
σ/μ = 8.91± 0.26 %
μ = 419.4± 1.0
σ = 37.4± 1.0

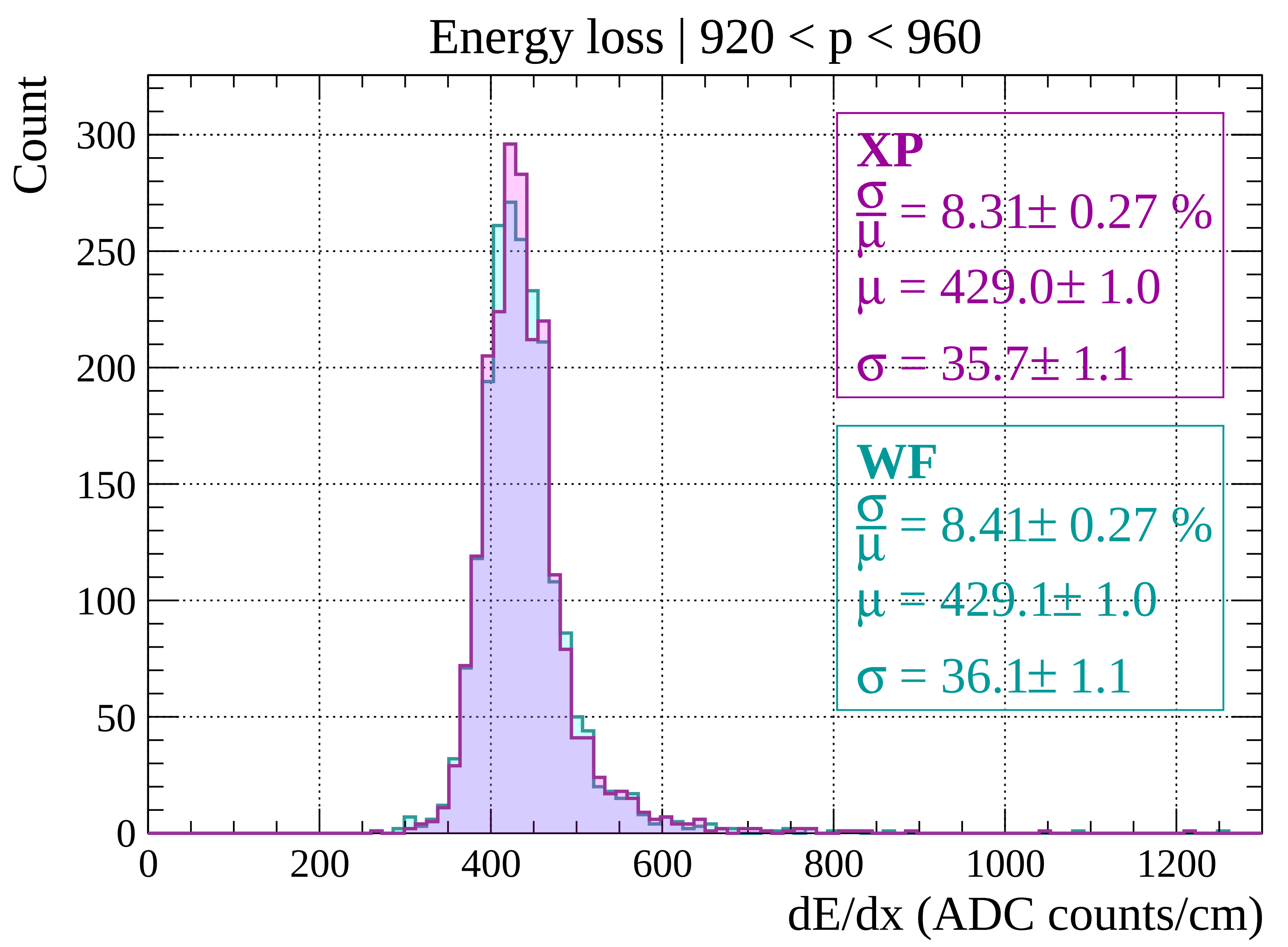

Energy loss | 800 < p < 840
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
50
100
150
200
250
300
XP
$\frac{\sigma}{\mu}$ = 8.57± 0.24 %
μ = 423.3± 0.9
σ = 36.3± 0.9
WF
$\frac{\sigma}{\mu}$ = 8.62± 0.24 %
μ = 423.9± 0.9
σ = 36.6± 0.9



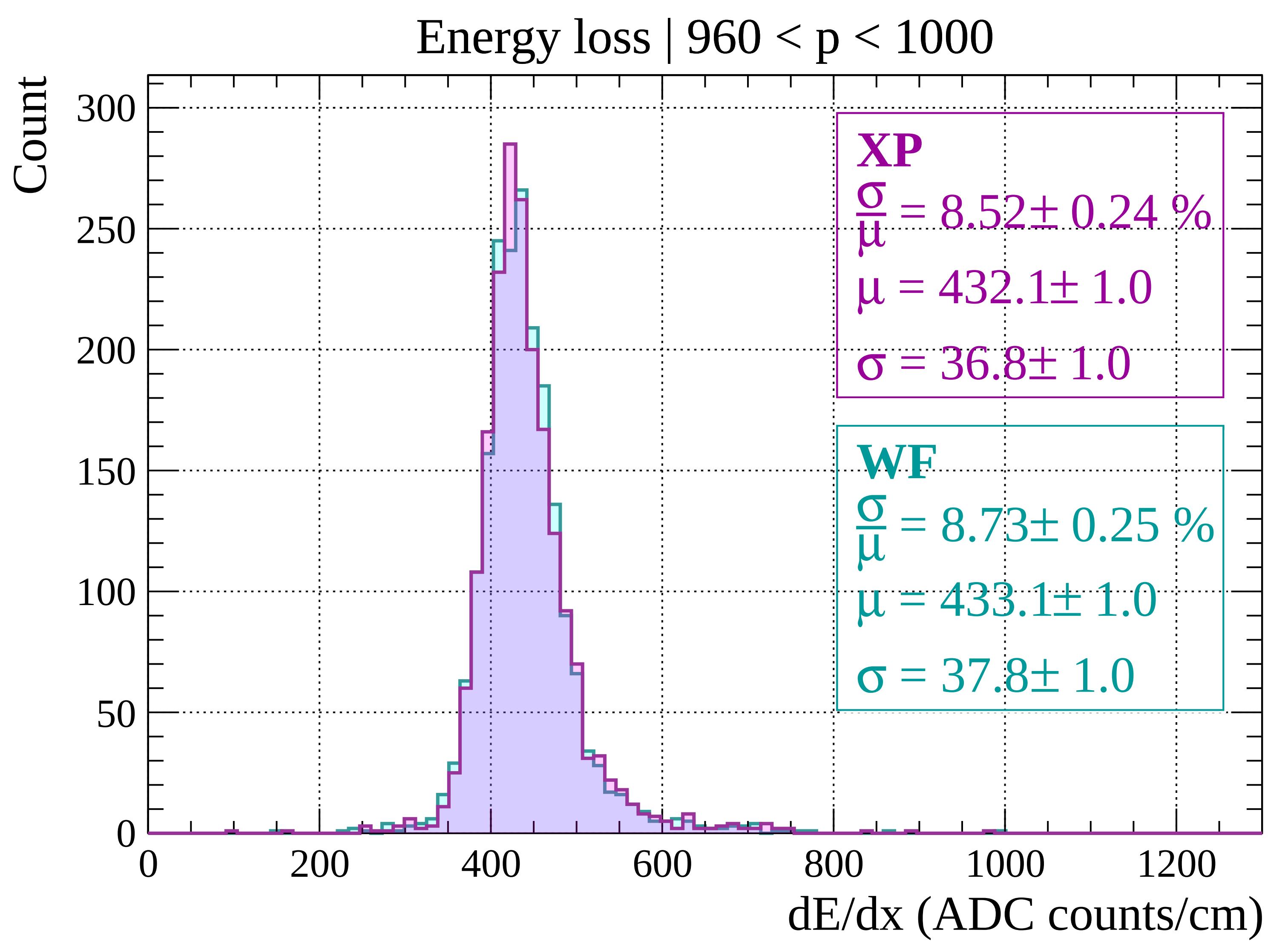
Energy loss | 840 < p < 880
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
0
200
400
600
800
1000
1200
XP
σ/μ = 8.60± 0.24 %
μ = 423.8± 0.9
σ = 36.4± 0.9
WF
σ/μ = 8.59± 0.23 %
μ = 424.5± 1.0
σ = 36.5± 0.9



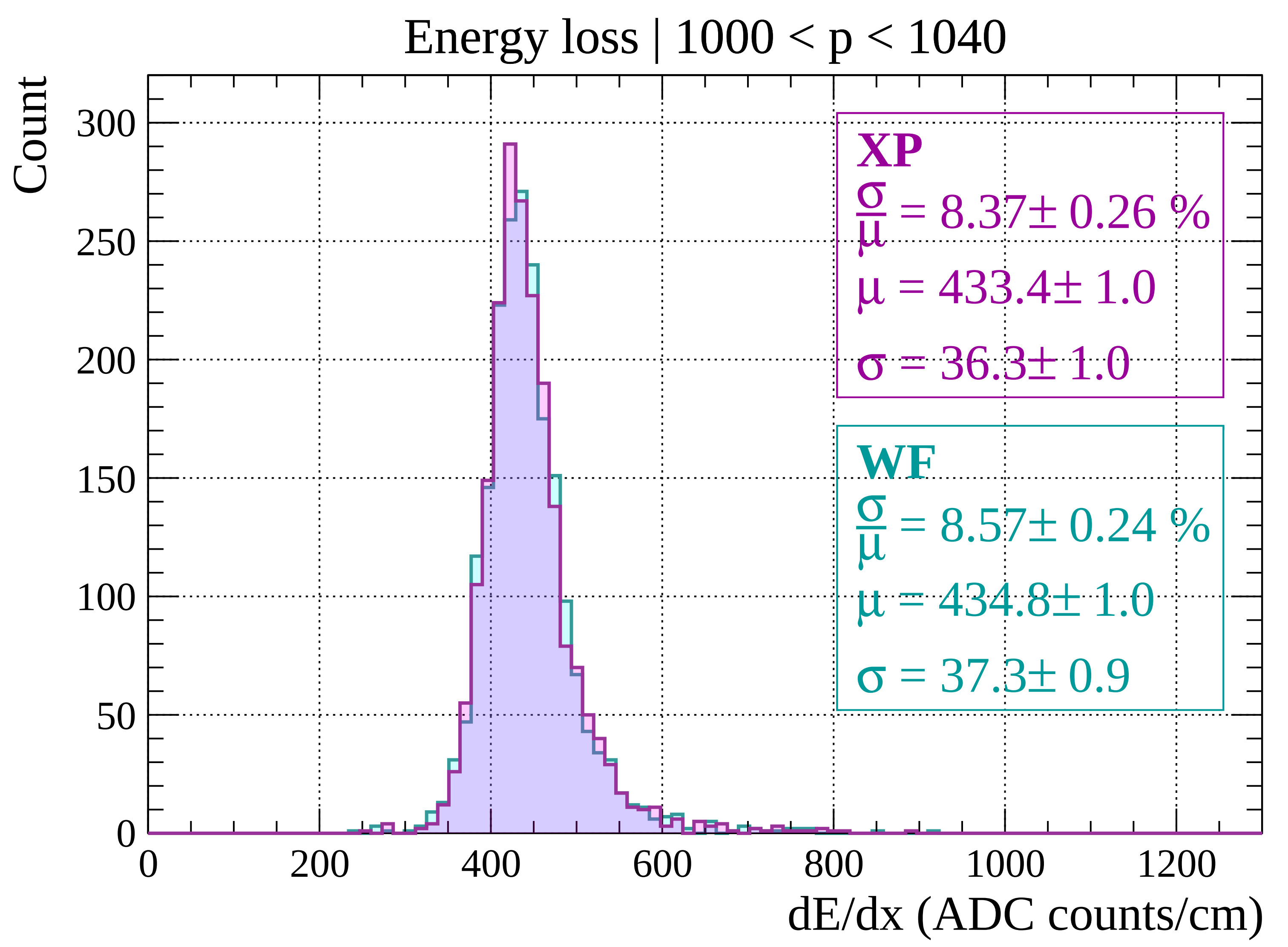
Energy loss | 880 < p < 920
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
0
200
400
600
800
1000
1200
XP
σ/μ = 8.68± 0.27 %
μ = 427.9± 1.0
σ = 37.1± 1.1
WF
σ/μ = 8.56± 0.25 %
μ = 427.6± 1.0
σ = 36.6± 1.0

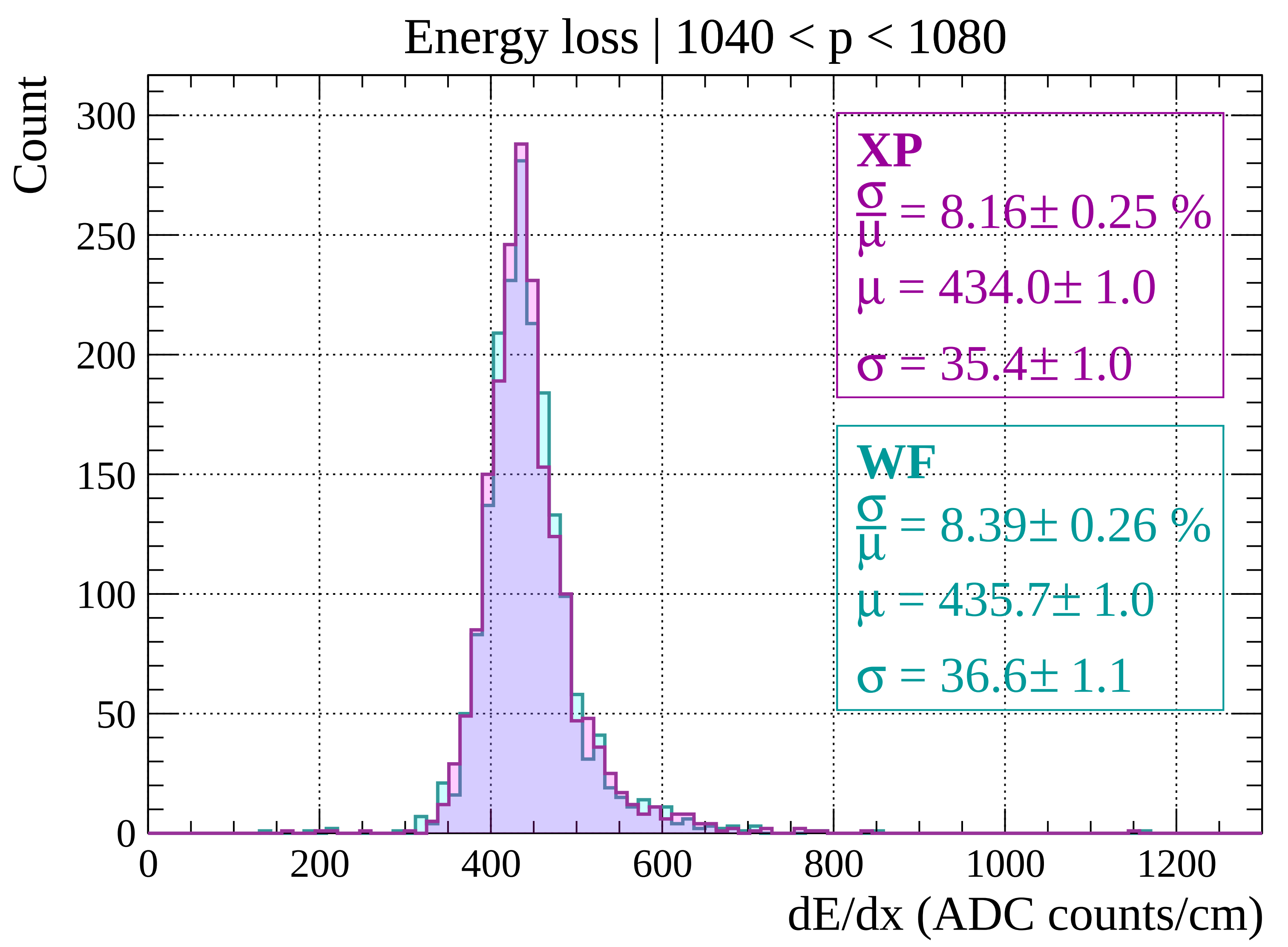


Energy loss | 920 < p < 960
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 8.31± 0.27 %
μ = 429.0± 1.0
σ = 35.7± 1.1
WF
$\frac{\sigma}{\mu}$ = 8.41± 0.27 %
μ = 429.1± 1.0
σ = 36.1± 1.1

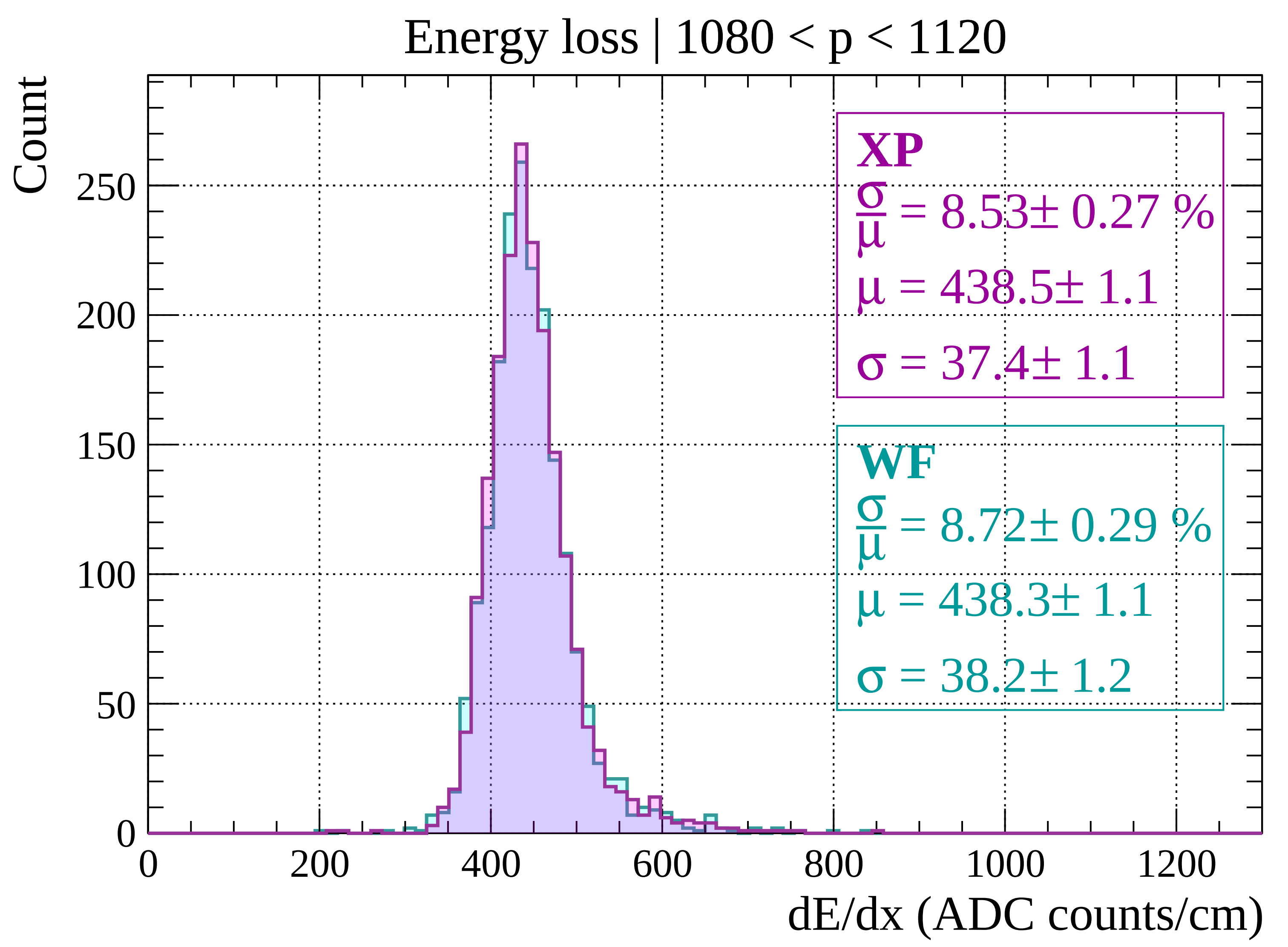


Energy loss | 960 < p < 1000
Count
0
50
100
150
200
250
300
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 8.52± 0.24 %
μ = 432.1± 1.0
σ = 36.8± 1.0
WF
$\frac{\sigma}{\mu}$ = 8.73± 0.25 %
μ = 433.1± 1.0
σ = 37.8± 1.0

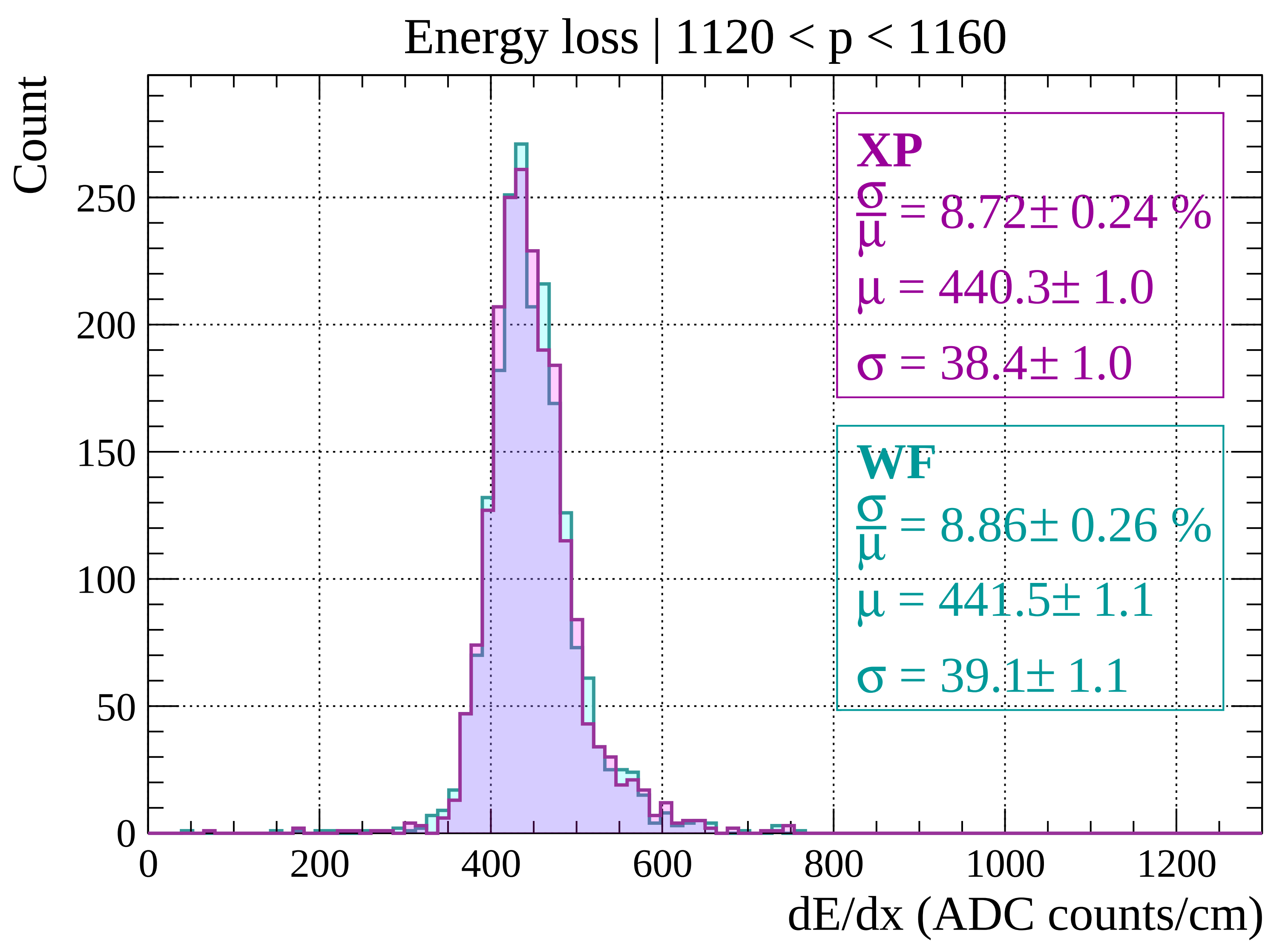

Energy loss | 1000 < p < 1040
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
0
200
400
600
800
1000
1200
XP
σ/μ = 8.37± 0.26 %
μ = 433.4± 1.0
σ = 36.3± 1.0
WF
σ/μ = 8.57± 0.24 %
μ = 434.8± 1.0
σ = 37.3± 0.9



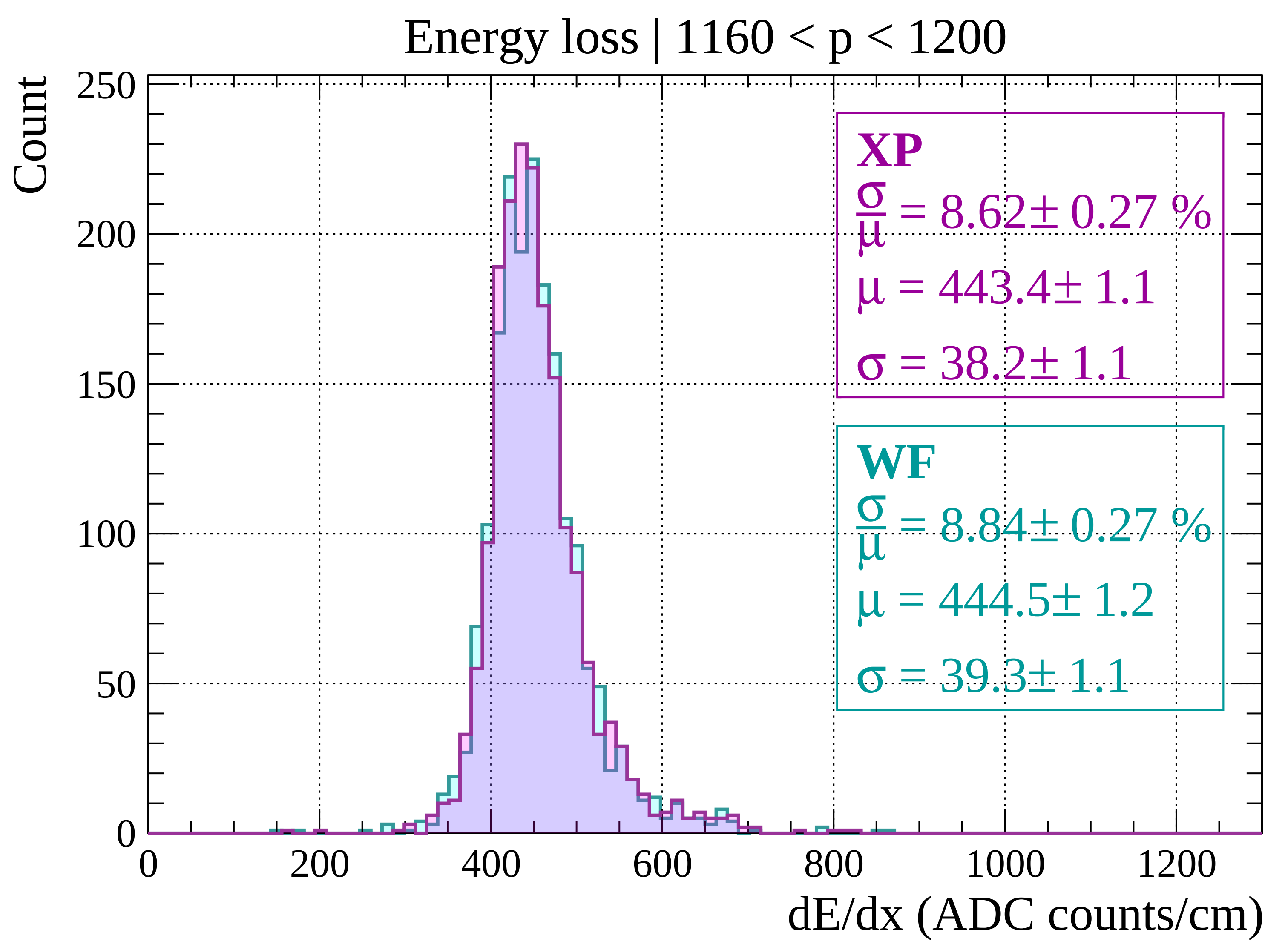
Energy loss | 1040 < p < 1080
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 8.16± 0.25 %
μ = 434.0± 1.0
σ = 35.4± 1.0
WF
$\frac{\sigma}{\mu}$ = 8.39± 0.26 %
μ = 435.7± 1.0
σ = 36.6± 1.1

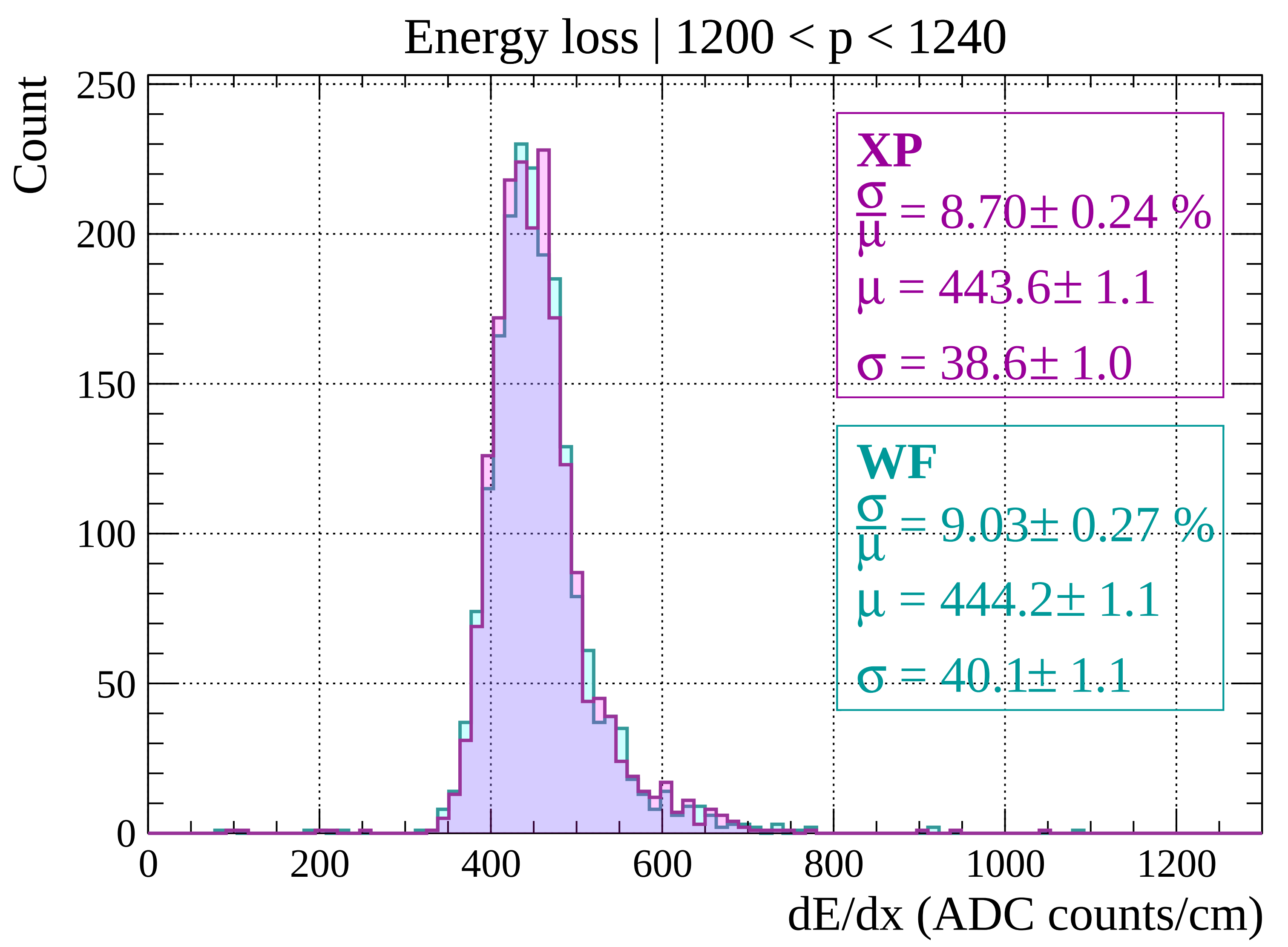


Energy loss | 1080 < p < 1120
Count
0
50
100
150
200
250
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 8.53± 0.27 %
μ = 438.5± 1.1
σ = 37.4± 1.1
WF
σ/μ = 8.72± 0.29 %
μ = 438.3± 1.1
σ = 38.2± 1.2

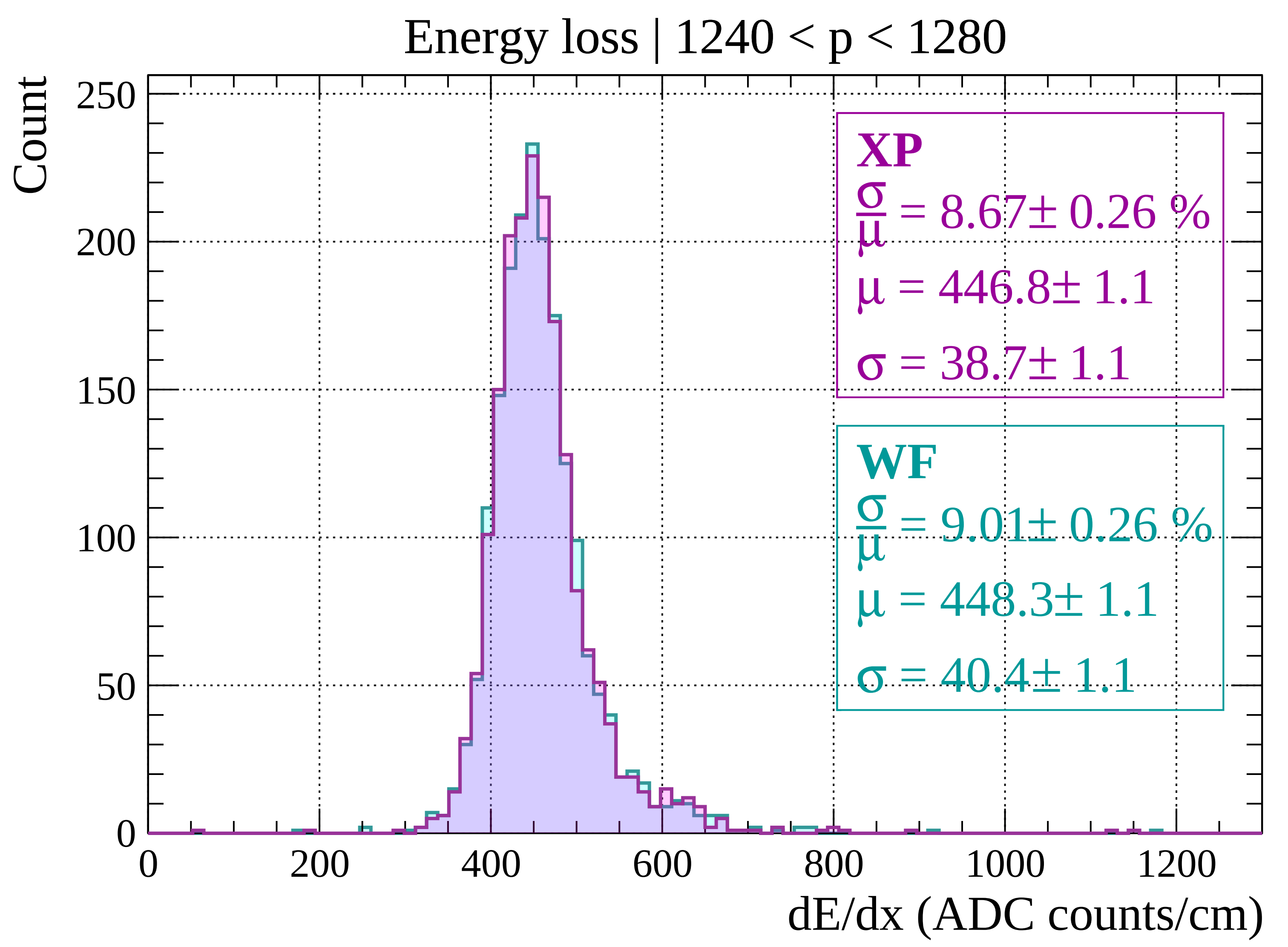


Energy loss | 1120 < p < 1160
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
50
100
150
200
250
XP
$\frac{\sigma}{\mu}$ = 8.72± 0.24 %
μ = 440.3± 1.0
σ = 38.4± 1.0
WF
$\frac{\sigma}{\mu}$ = 8.86± 0.26 %
μ = 441.5± 1.1
σ = 39.1± 1.1

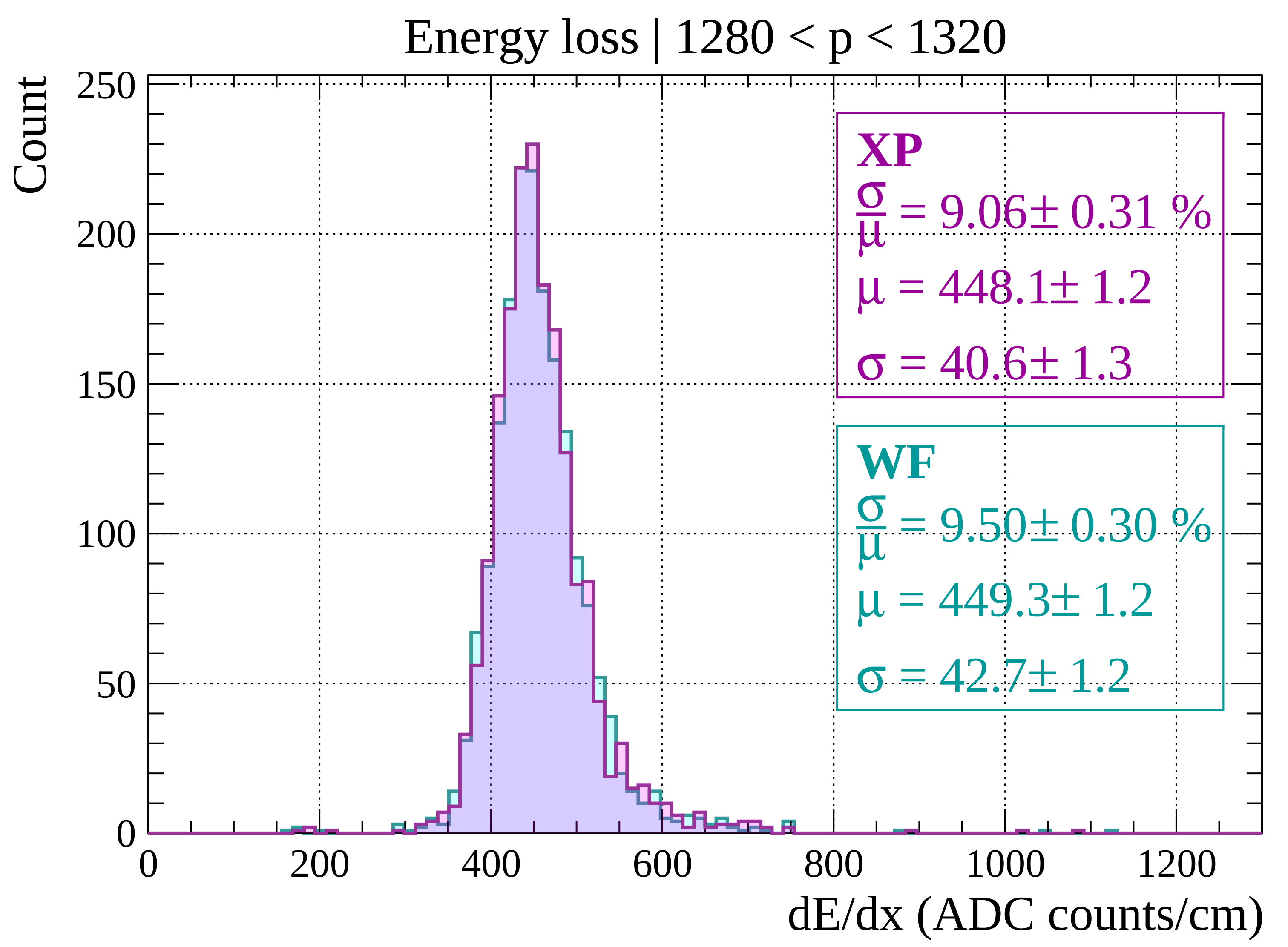

Energy loss | 1160 < p < 1200
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 8.62± 0.27 %
μ = 443.4± 1.1
σ = 38.2± 1.1
WF
σ/μ = 8.84± 0.27 %
μ = 444.5± 1.2
σ = 39.3± 1.1
0
50
100
150
200
250
0
200
400
600
800
1000
1200

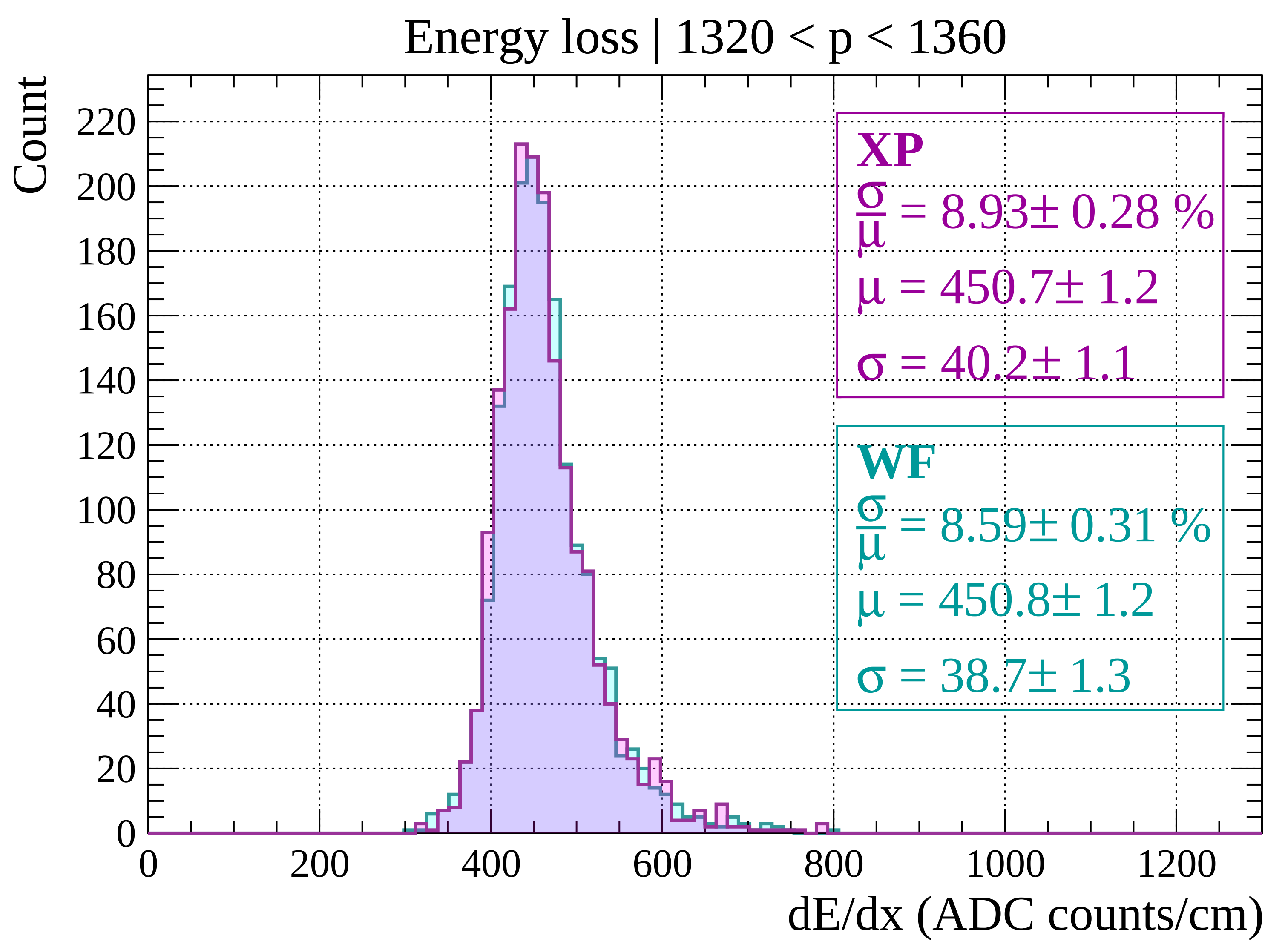


Energy loss | 1200 < p < 1240
Count
250
200
150
100
50
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 8.70± 0.24 %
μ = 443.6± 1.1
σ = 38.6± 1.0
WF
$\frac{\sigma}{\mu}$ = 9.03± 0.27 %
μ = 444.2± 1.1
σ = 40.1± 1.1

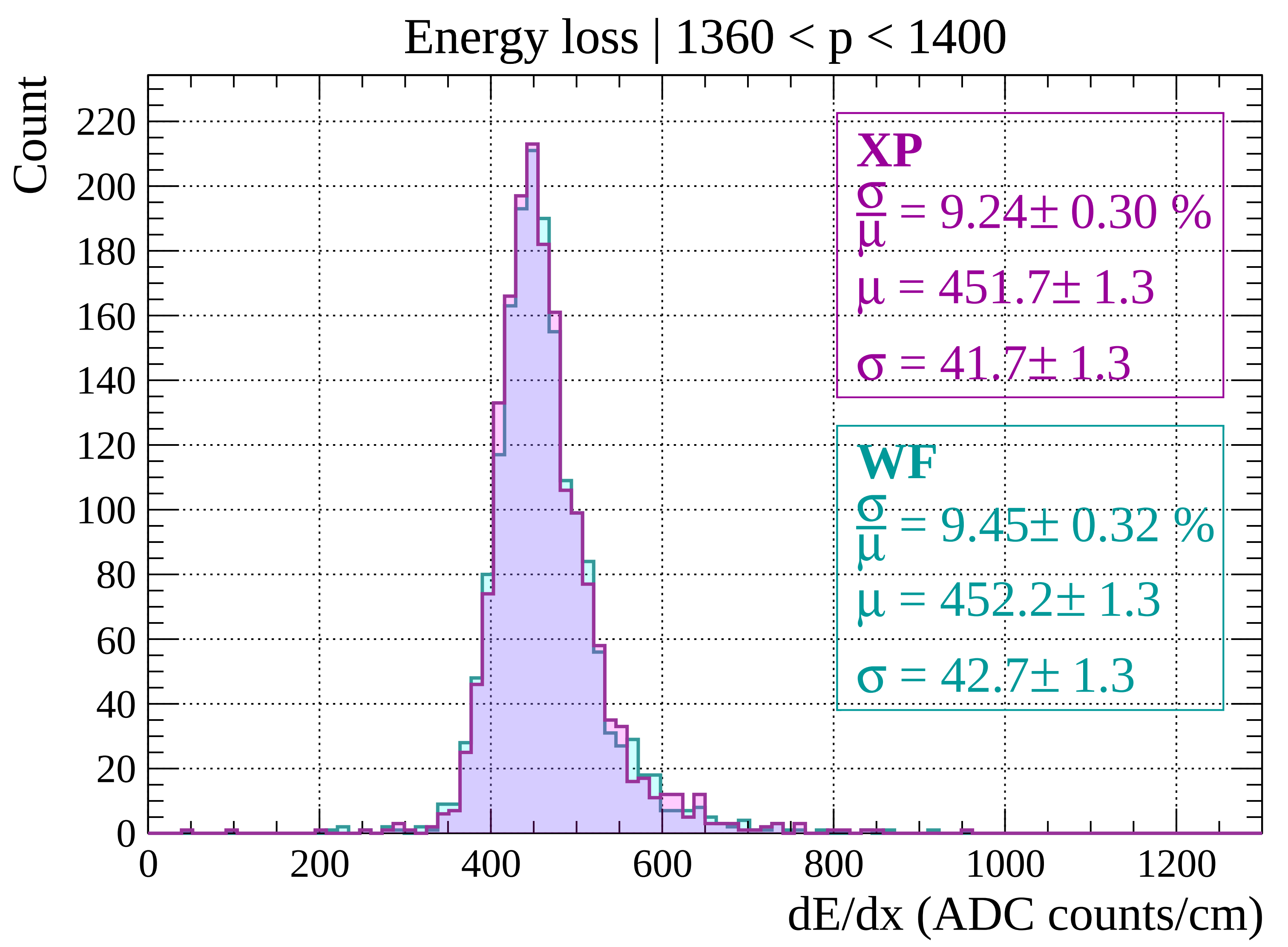


Energy loss | 1240 < p < 1280
Count
250
200
150
100
50
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 8.67± 0.26 %
μ = 446.8± 1.1
σ = 38.7± 1.1
WF
$\frac{\sigma}{\mu}$ = 9.01± 0.26 %
μ = 448.3± 1.1
σ = 40.4± 1.1

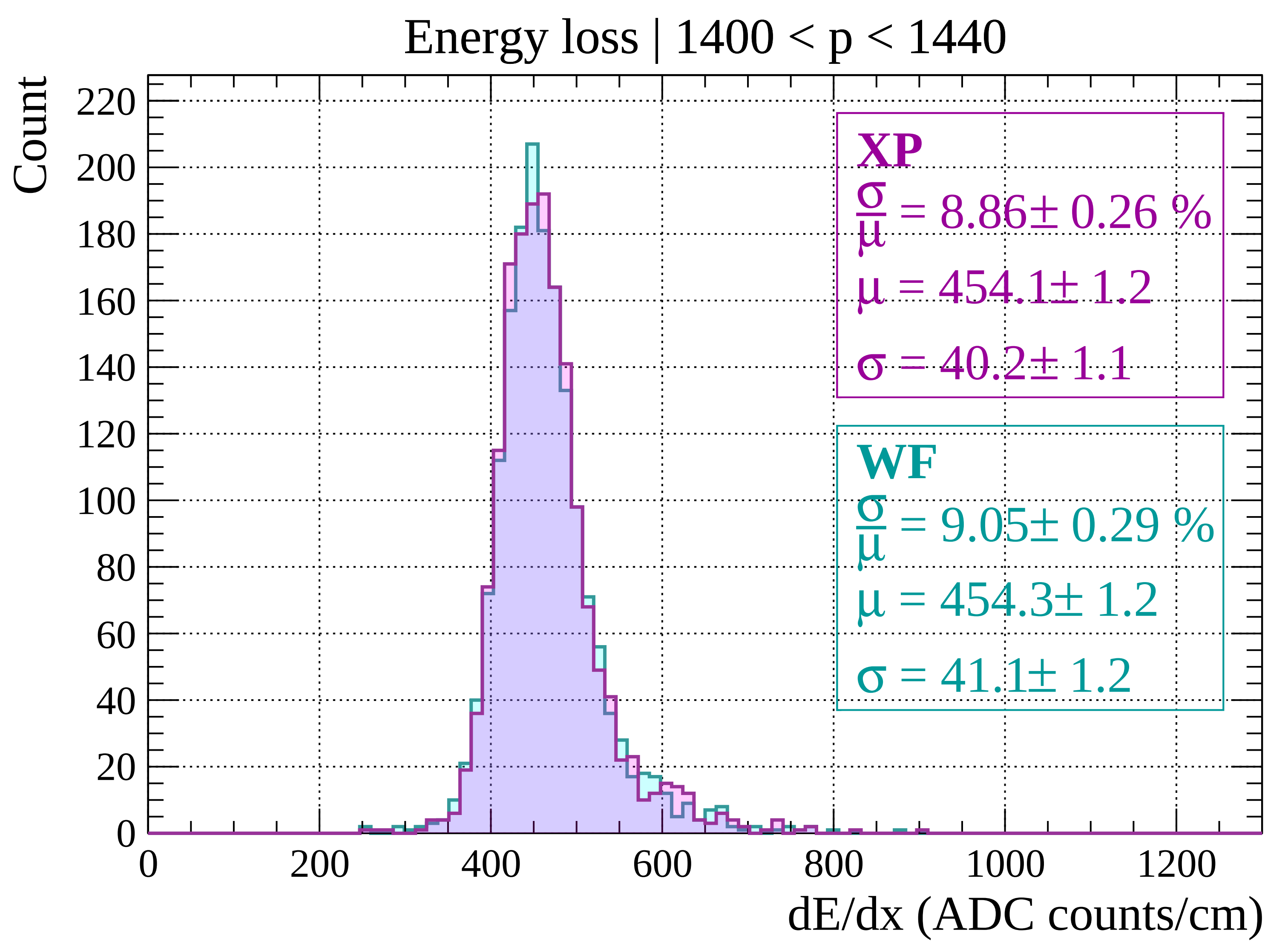

Energy loss | 1280 < p < 1320
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu} = 9.06 \pm 0.31$ %
$\mu = 448.1 \pm 1.2$
$\sigma = 40.6 \pm 1.3$
WF
$\frac{\sigma}{\mu} = 9.50 \pm 0.30$ %
$\mu = 449.3 \pm 1.2$
$\sigma = 42.7 \pm 1.2$

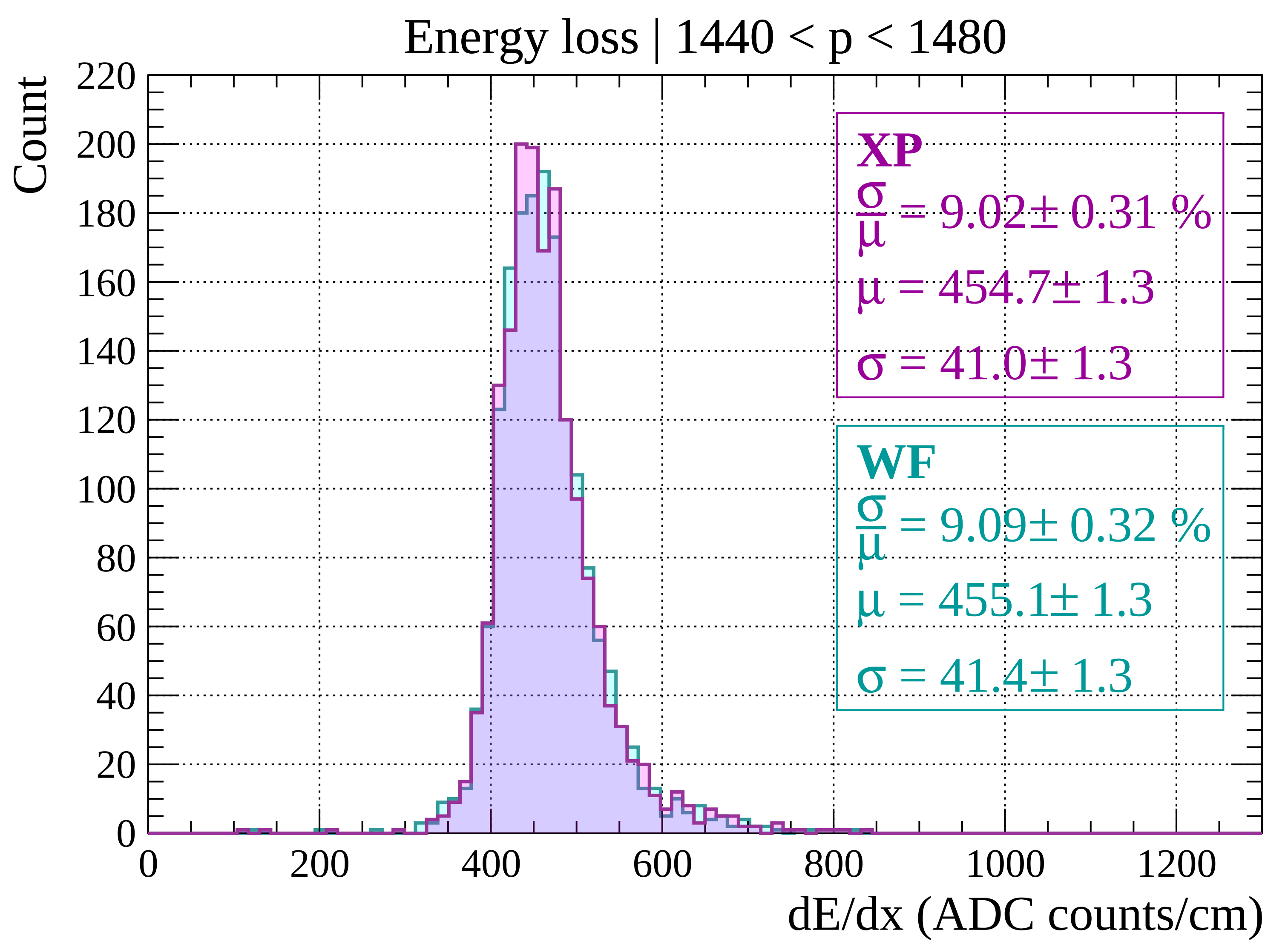


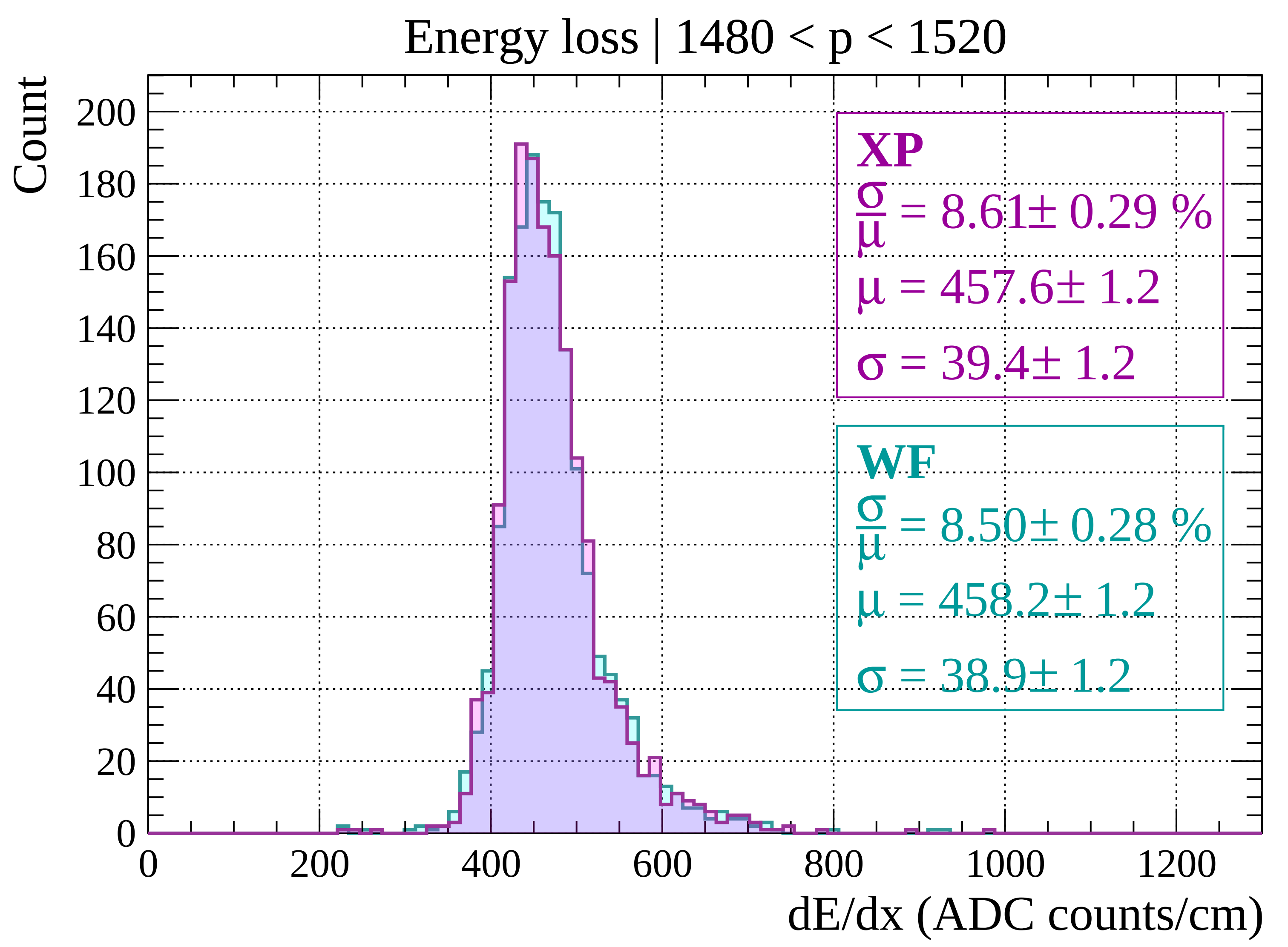
Energy loss | 1320 < p < 1360
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
120
140
160
180
200
220
XP
$\frac{\sigma}{\mu} = 8.93 \pm 0.28$ %
$\mu = 450.7 \pm 1.2$
$\sigma = 40.2 \pm 1.1$
WF
$\frac{\sigma}{\mu} = 8.59 \pm 0.31$ %
$\mu = 450.8 \pm 1.2$
$\sigma = 38.7 \pm 1.3$

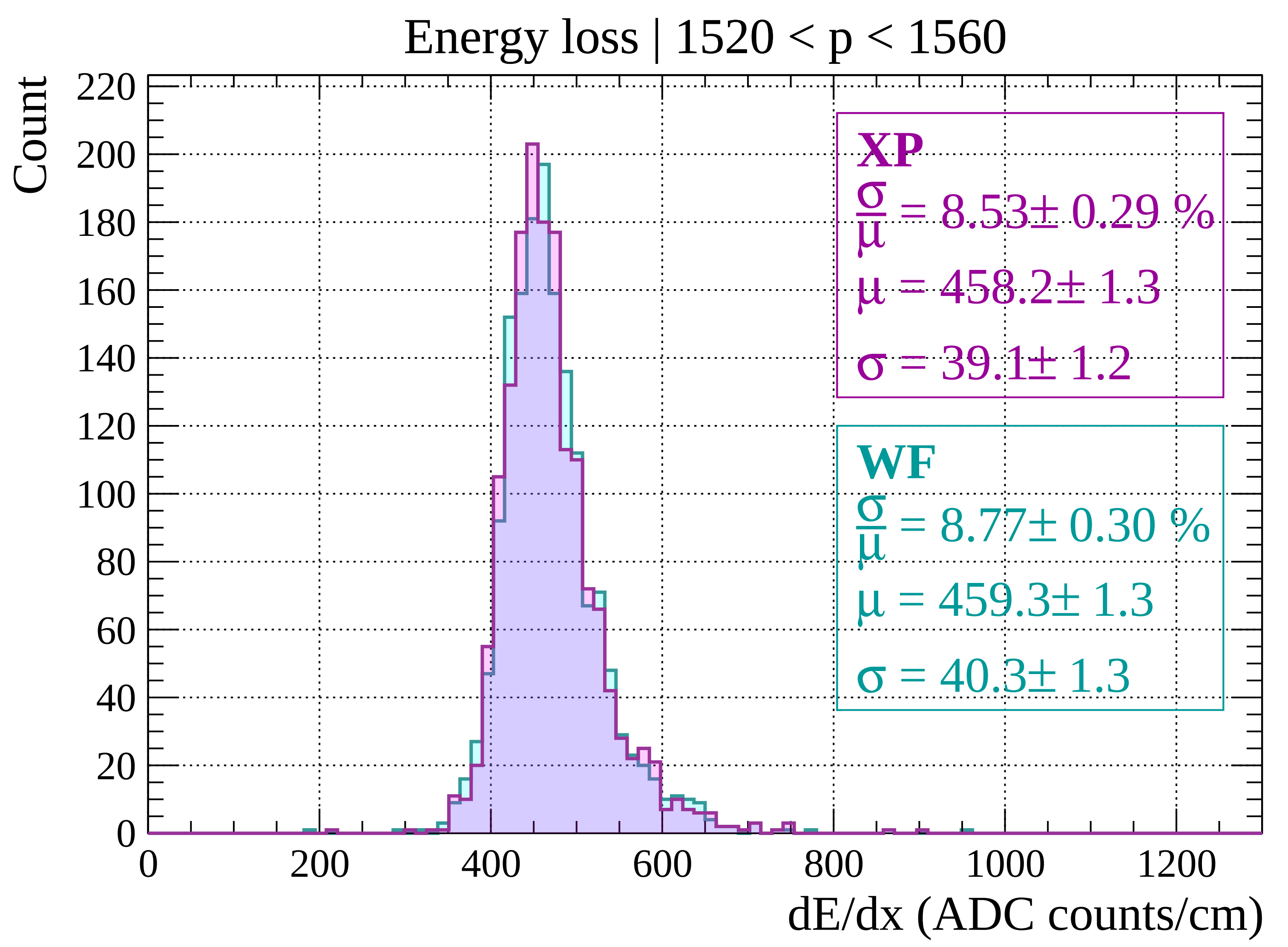


Energy loss | 1360 < p < 1400
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
180
200
220
0
200
400
600
800
1000
1200
XP
σ/μ = 9.24± 0.30 %
μ = 451.7± 1.3
σ = 41.7± 1.3
WF
σ/μ = 9.45± 0.32 %
μ = 452.2± 1.3
σ = 42.7± 1.3

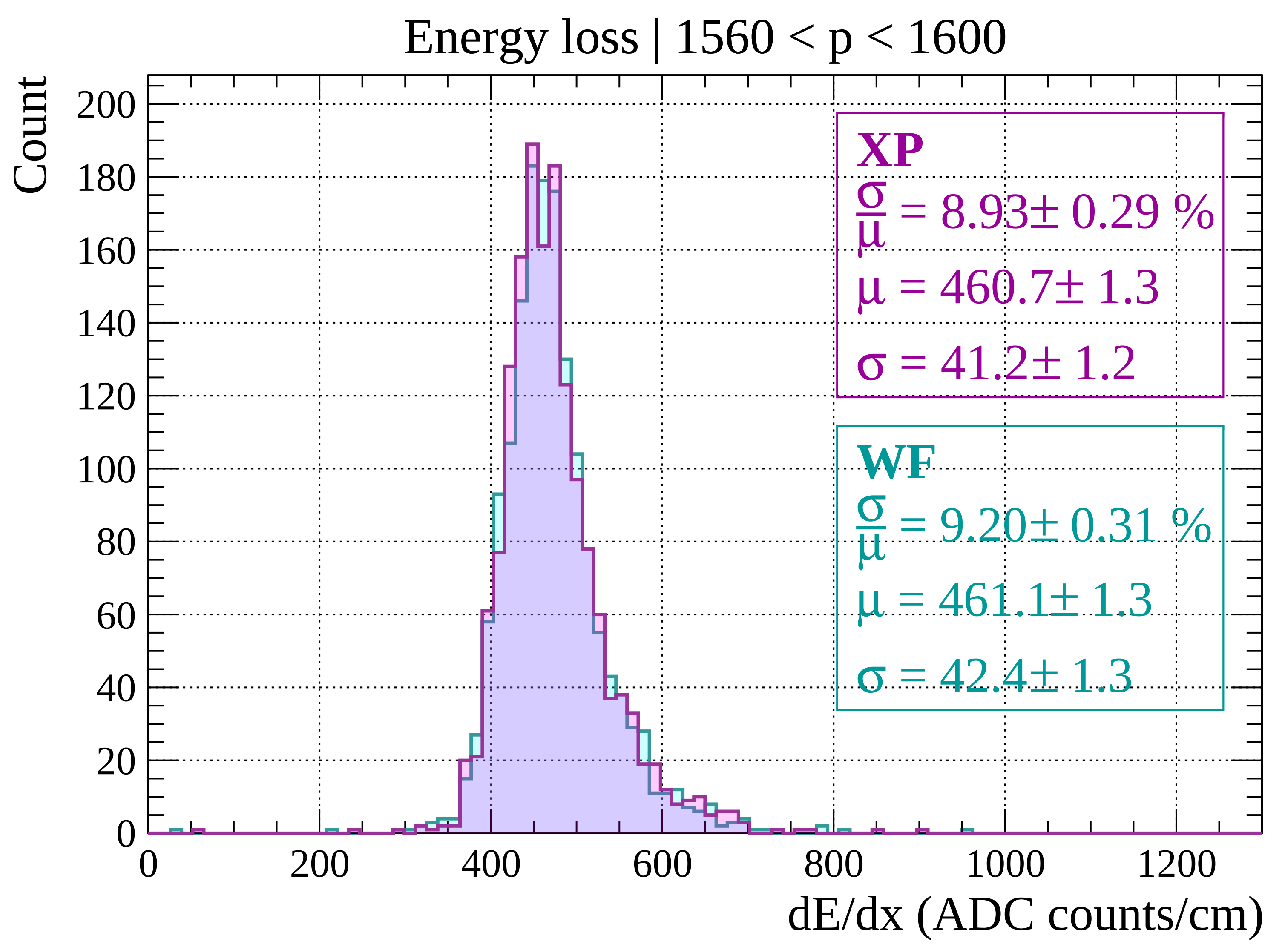

Energy loss | 1400 < p < 1440
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
180
200
220
0
200
400
600
800
1000
1200
XP
σ/μ = 8.86± 0.26 %
μ = 454.1± 1.2
σ = 40.2± 1.1
WF
σ/μ = 9.05± 0.29 %
μ = 454.3± 1.2
σ = 41.1± 1.2

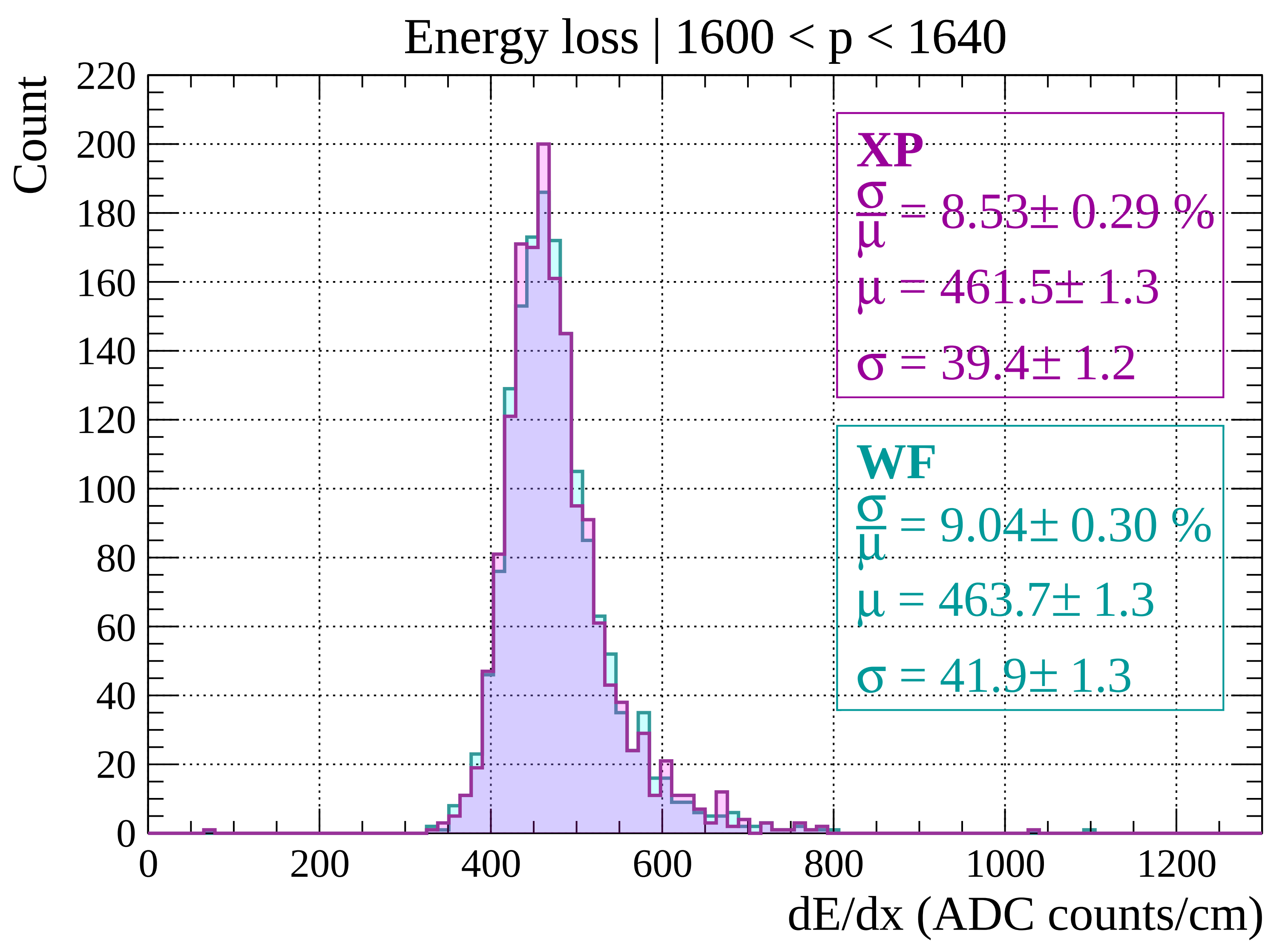

Energy loss | 1440 < p < 1480
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 9.02± 0.31 %
μ = 454.7± 1.3
σ = 41.0± 1.3
WF
σ/μ = 9.09± 0.32 %
μ = 455.1± 1.3
σ = 41.4± 1.3
0
20
40
60
80
100
120
140
160
180
200
220
0
200
400
600
800
1000
1200

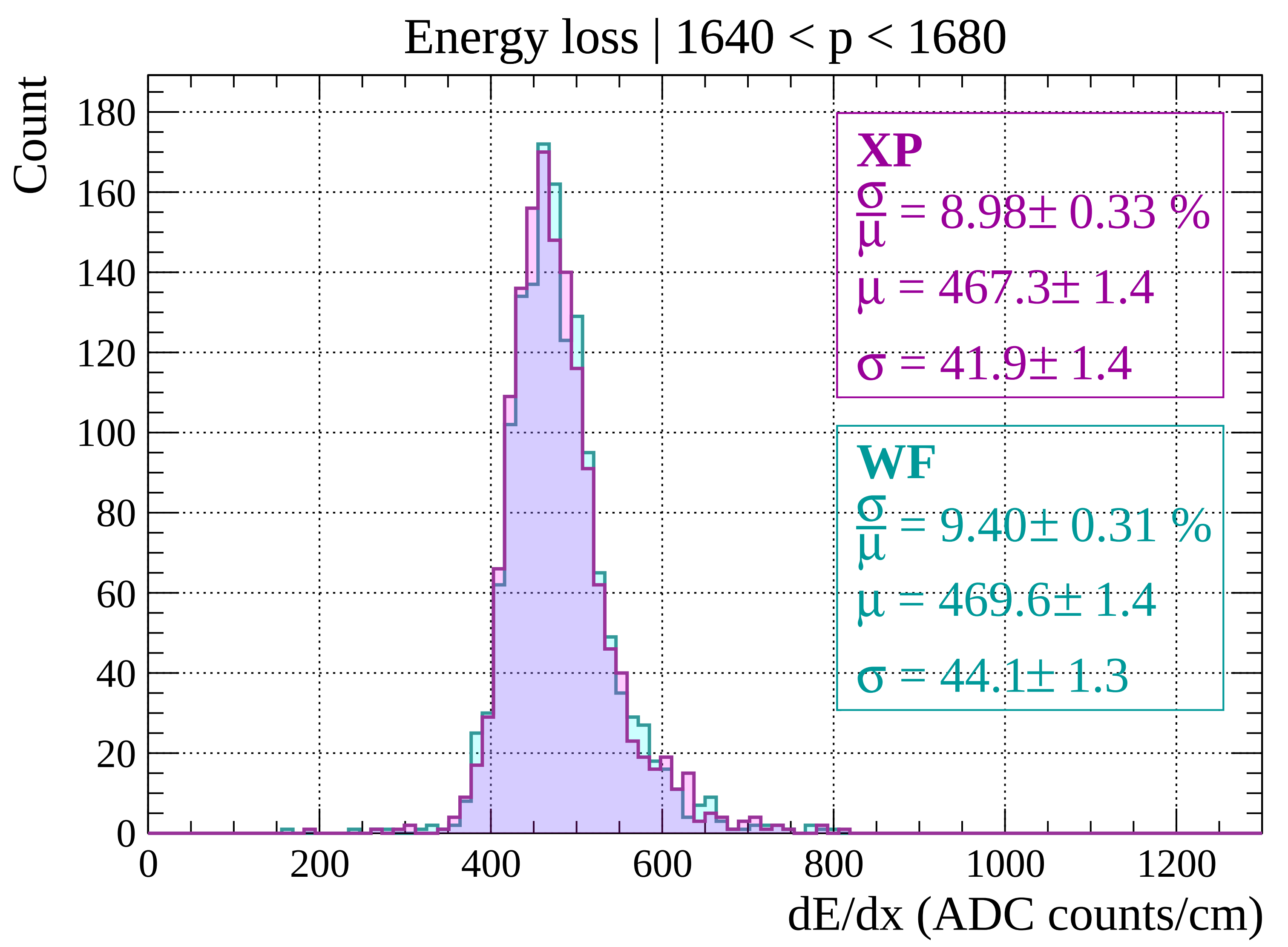

Energy loss | 1480 < p < 1520
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 8.61± 0.29 %
μ = 457.6± 1.2
σ = 39.4± 1.2
WF
$\frac{\sigma}{\mu}$ = 8.50± 0.28 %
μ = 458.2± 1.2
σ = 38.9± 1.2

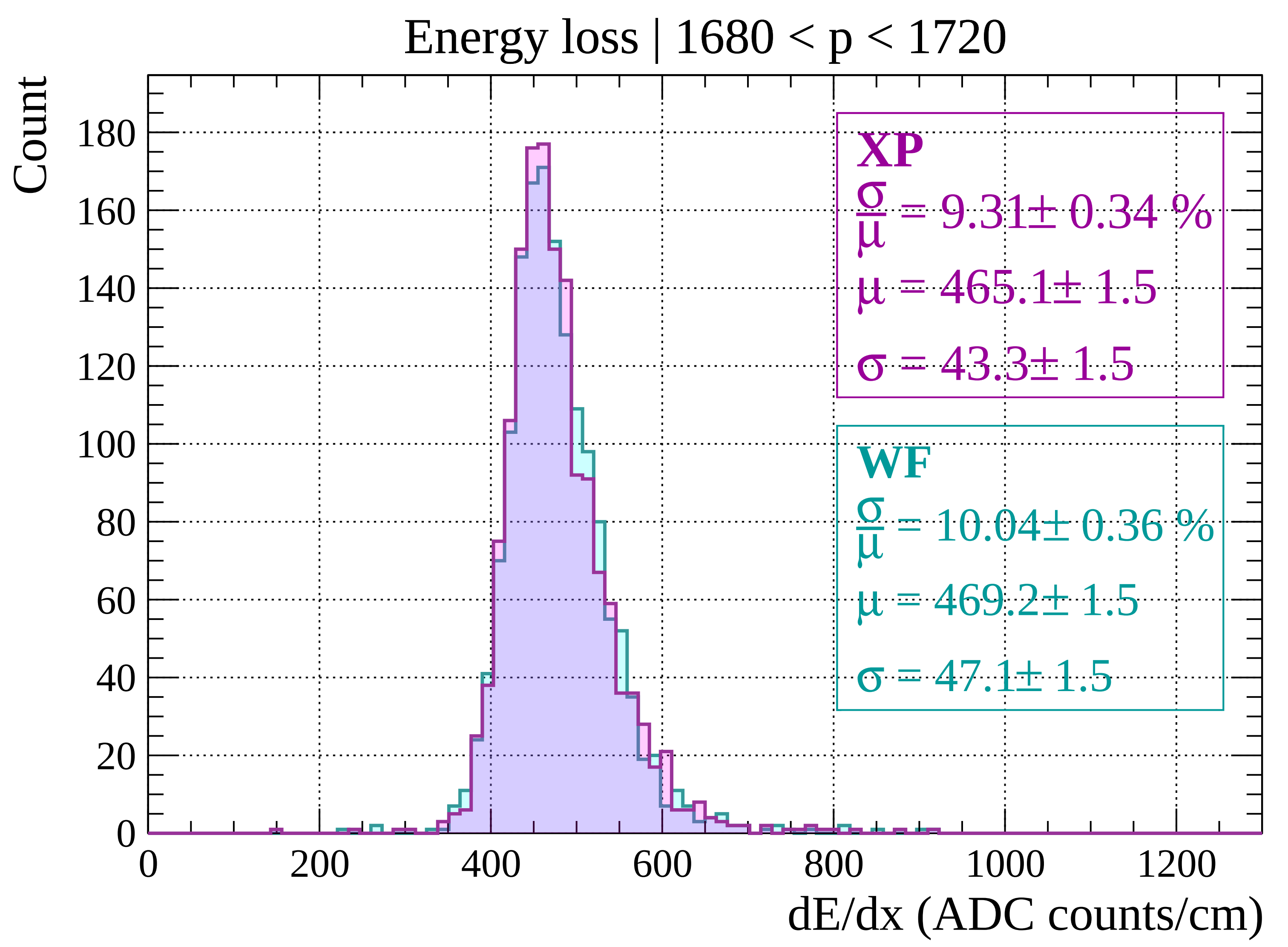


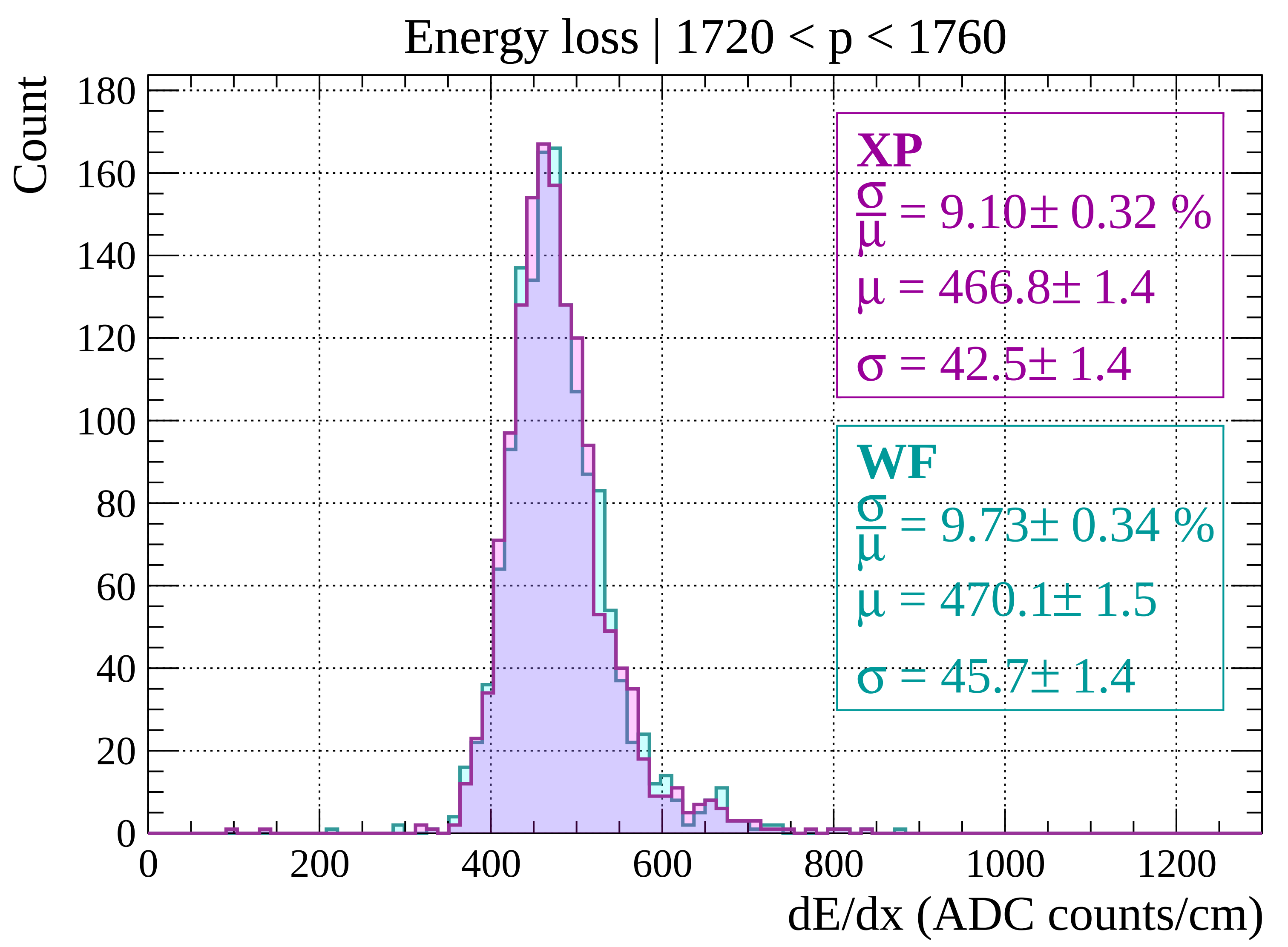
Energy loss | 1520 < p < 1560
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
180
200
220
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 8.53± 0.29 %
μ = 458.2± 1.3
σ = 39.1± 1.2
WF
$\frac{\sigma}{\mu}$ = 8.77± 0.30 %
μ = 459.3± 1.3
σ = 40.3± 1.3

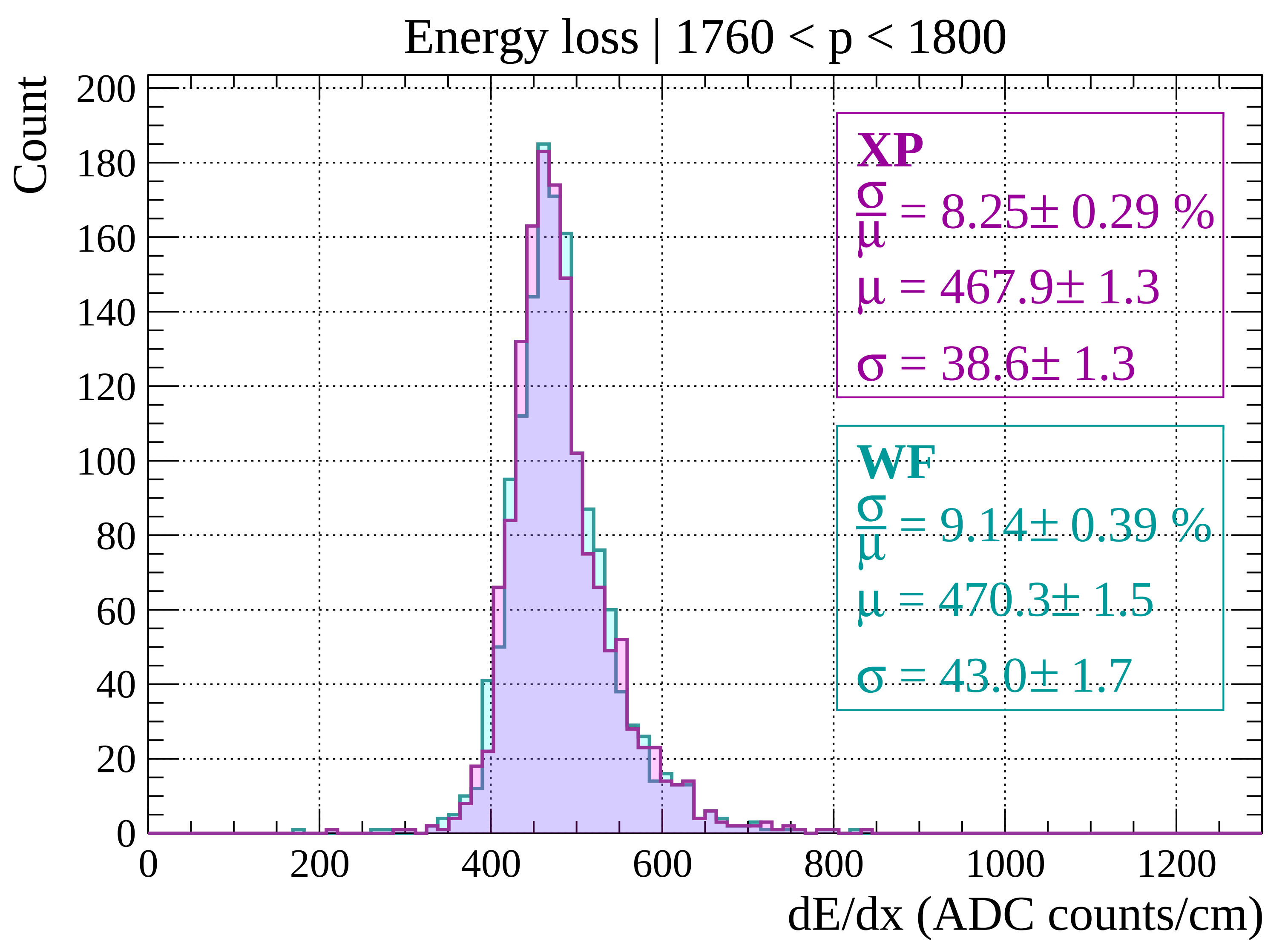

Energy loss | 1560 < p < 1600
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 8.93± 0.29 %
μ = 460.7± 1.3
σ = 41.2± 1.2
WF
σ/μ = 9.20± 0.31 %
μ = 461.1± 1.3
σ = 42.4± 1.3
0
20
40
60
80
100
120
140
160
180
200
0
200
400
600
800
1000
1200

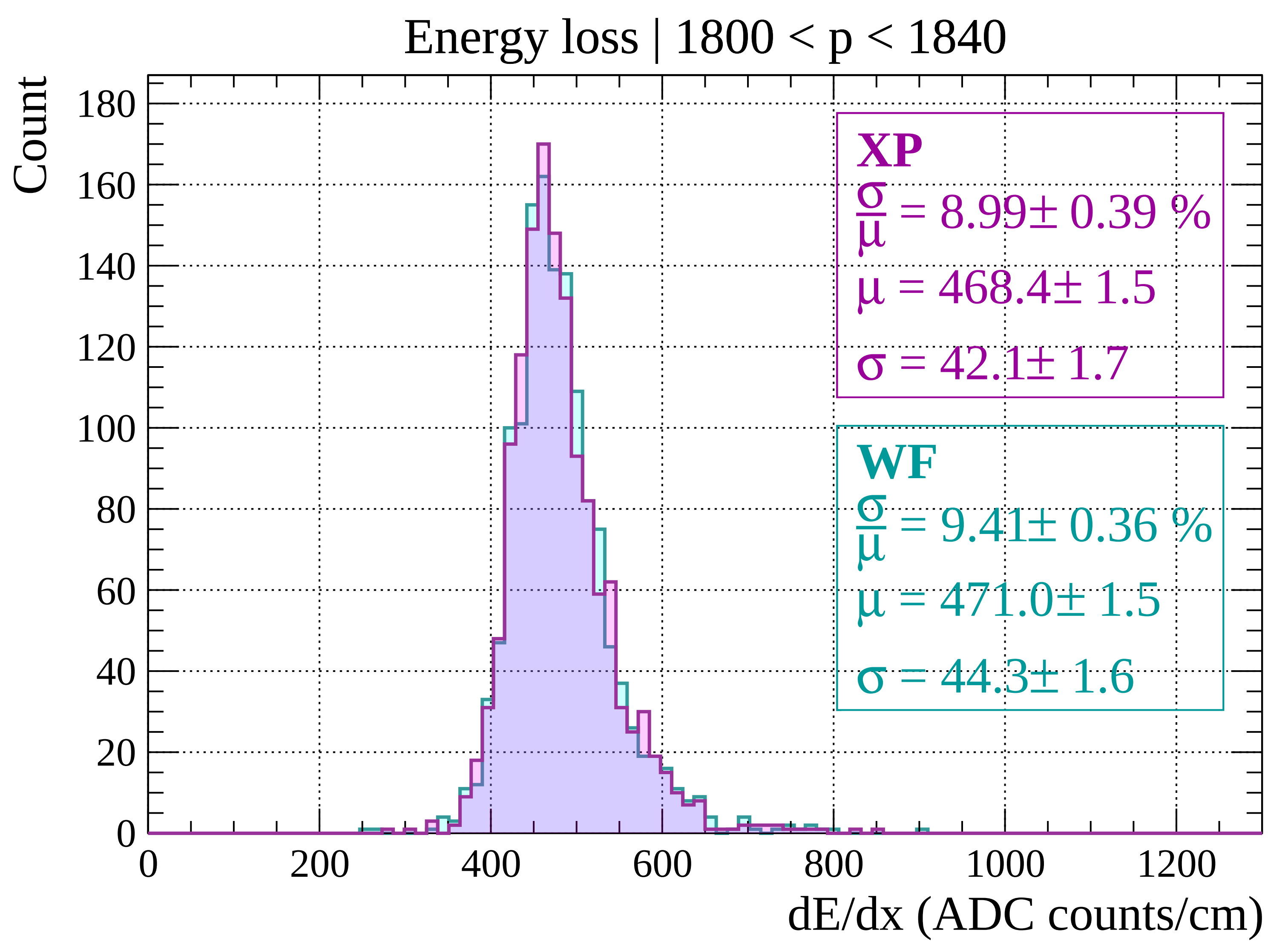

Energy loss | 1600 < p < 1640
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 8.53± 0.29 %
μ = 461.5± 1.3
σ = 39.4± 1.2
WF
σ/μ = 9.04± 0.30 %
μ = 463.7± 1.3
σ = 41.9± 1.3
0
200
400
600
800
1000
1200
20
40
60
80
100
120
140
160
180
200
220


Energy loss | 1640 < p < 1680
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 8.98± 0.33 %
μ = 467.3± 1.4
σ = 41.9± 1.4
WF
$\frac{\sigma}{\mu}$ = 9.40± 0.31 %
μ = 469.6± 1.4
σ = 44.1± 1.3
0
20
40
60
80
100
120
140
160
180
0
200
400
600
800
1000
1200

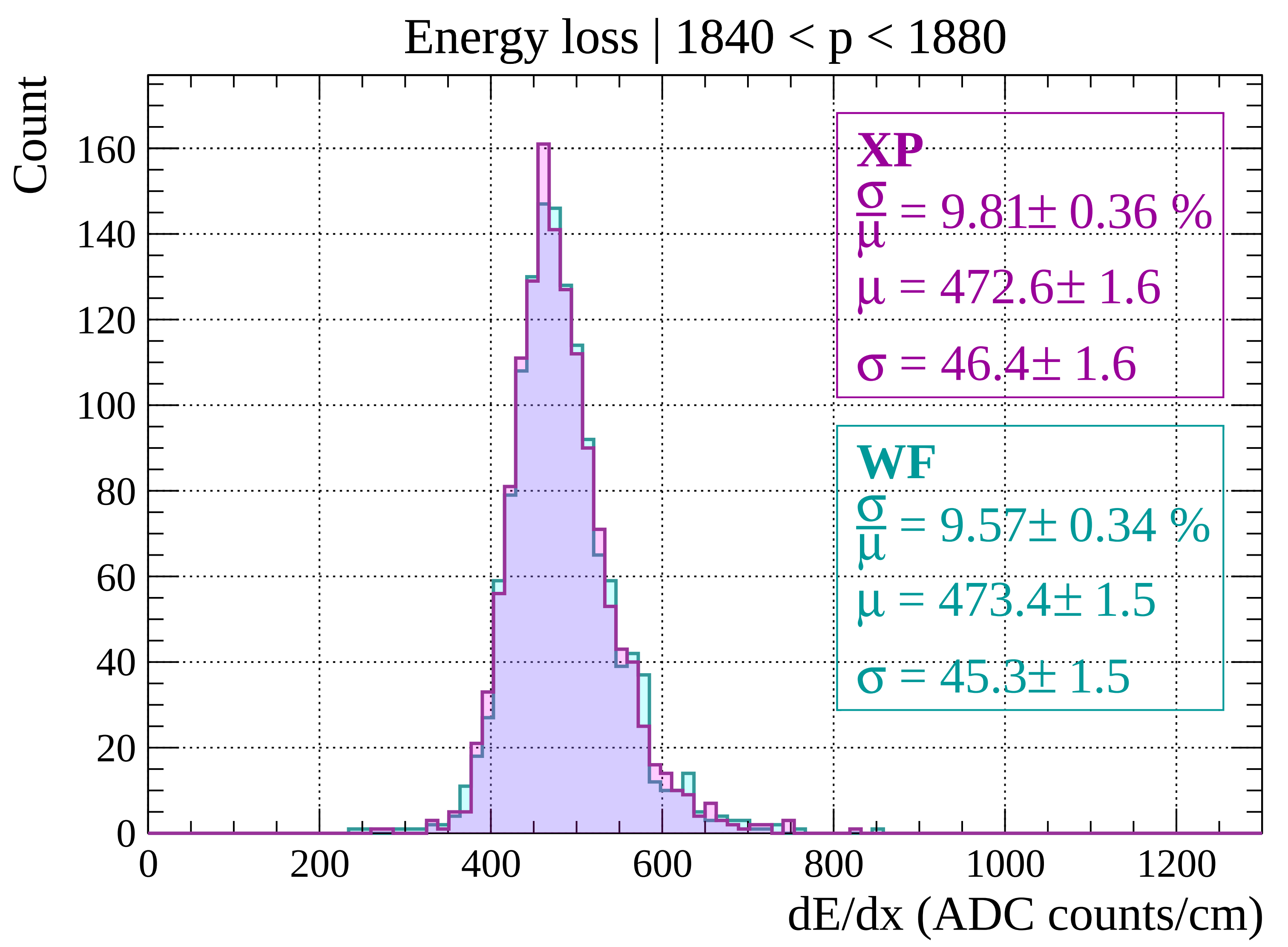

Energy loss | 1680 < p < 1720
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 9.31± 0.34 %
μ = 465.1± 1.5
σ = 43.3± 1.5
WF
σ/μ = 10.04± 0.36 %
μ = 469.2± 1.5
σ = 47.1± 1.5
0
20
40
60
80
100
120
140
160
180
0
200
400
600
800
1000
1200


Energy loss | 1720 < p < 1760
Count
180
160
140
120
100
80
60
40
20
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 9.10± 0.32 %
μ = 466.8± 1.4
σ = 42.5± 1.4
WF
σ/μ = 9.73± 0.34 %
μ = 470.1± 1.5
σ = 45.7± 1.4

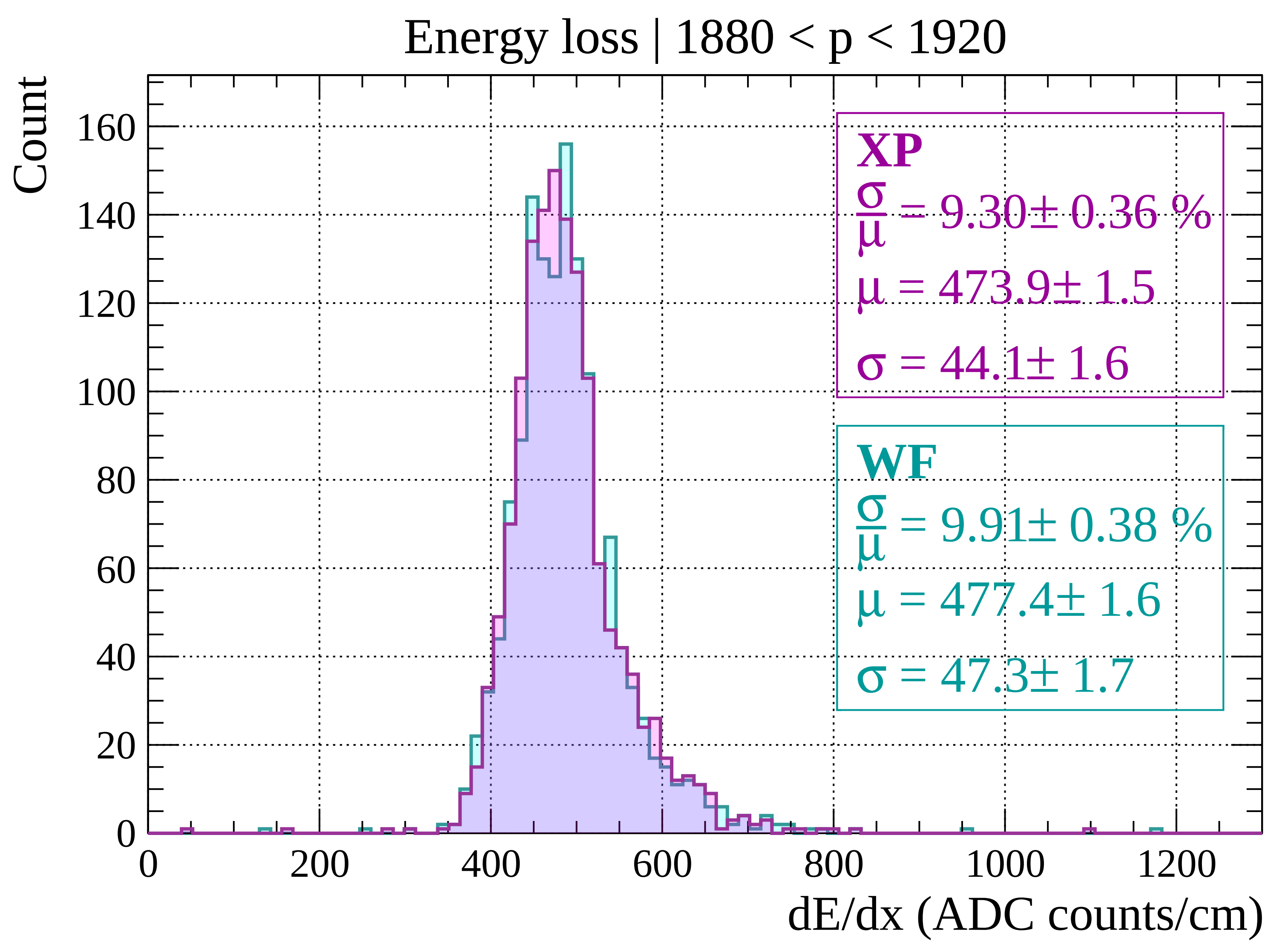

Energy loss | 1760 < p < 1800
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 8.25± 0.29 %
μ = 467.9± 1.3
σ = 38.6± 1.3
WF
σ/μ = 9.14± 0.39 %
μ = 470.3± 1.5
σ = 43.0± 1.7
0
20
40
60
80
100
120
140
160
180
200
0
200
400
600
800
1000
1200



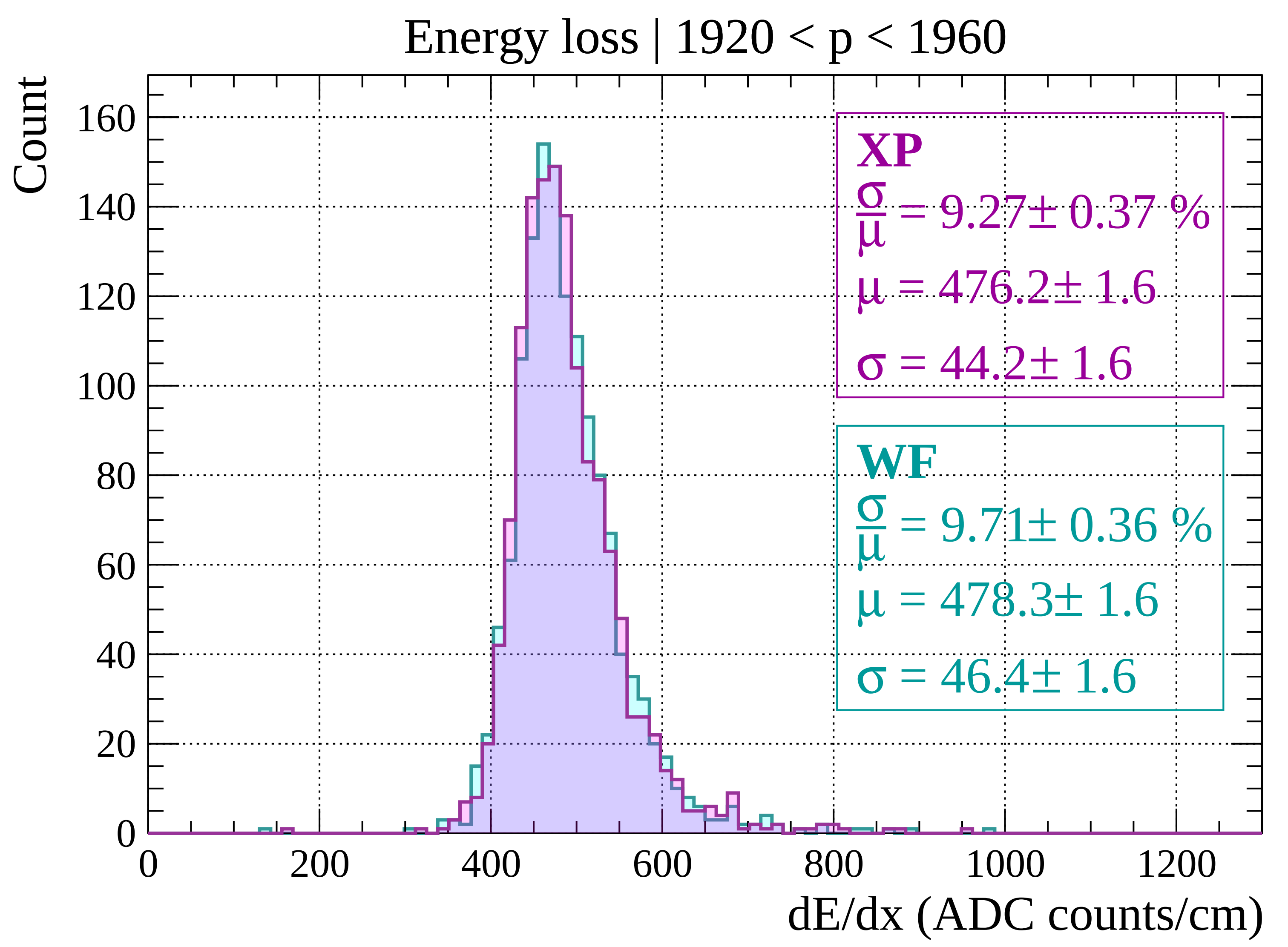
Energy loss | 1800 < p < 1840
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
120
140
160
180
XP
σ/μ = 8.99± 0.39 %
μ = 468.4± 1.5
σ = 42.1± 1.7
WF
σ/μ = 9.41± 0.36 %
μ = 471.0± 1.5
σ = 44.3± 1.6



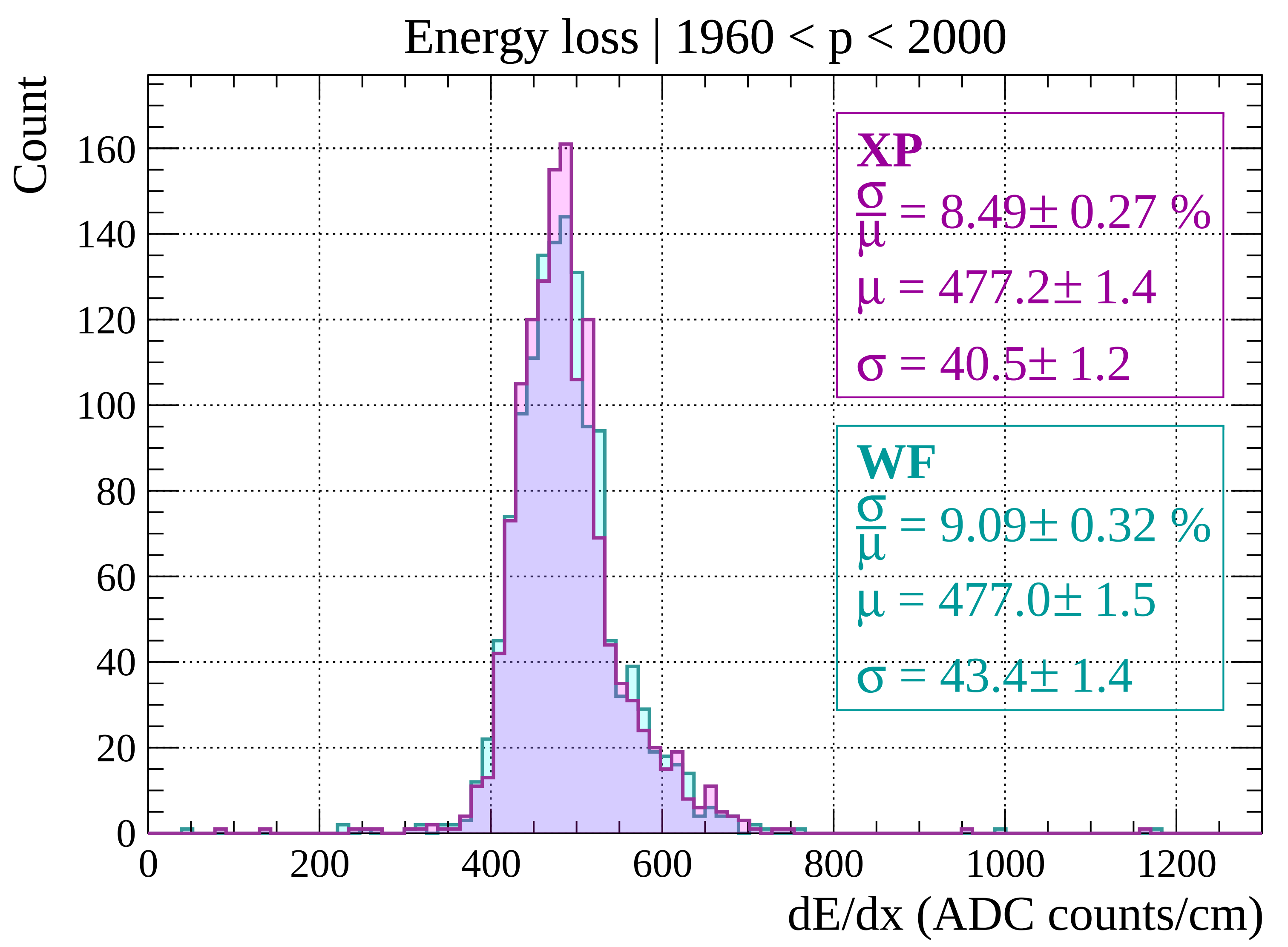
Energy loss | 1840 < p < 1880
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
120
140
160
XP
σ/μ = 9.81± 0.36 %
μ = 472.6± 1.6
σ = 46.4± 1.6
WF
σ/μ = 9.57± 0.34 %
μ = 473.4± 1.5
σ = 45.3± 1.5



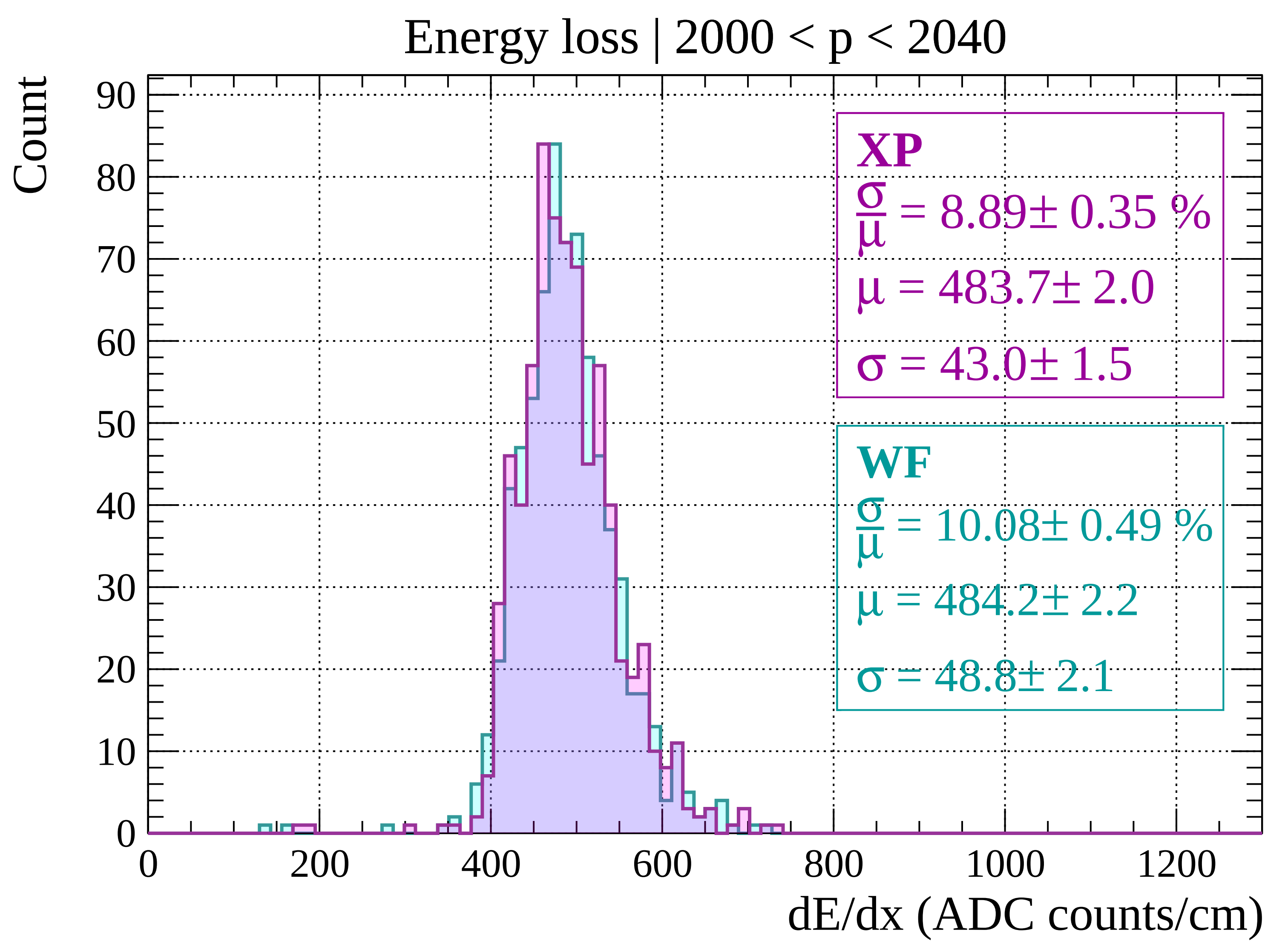
Energy loss | 1880 < p < 1920
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
120
140
160
XP
σ/μ = 9.30± 0.36 %
μ = 473.9± 1.5
σ = 44.1± 1.6
WF
σ/μ = 9.91± 0.38 %
μ = 477.4± 1.6
σ = 47.3± 1.7

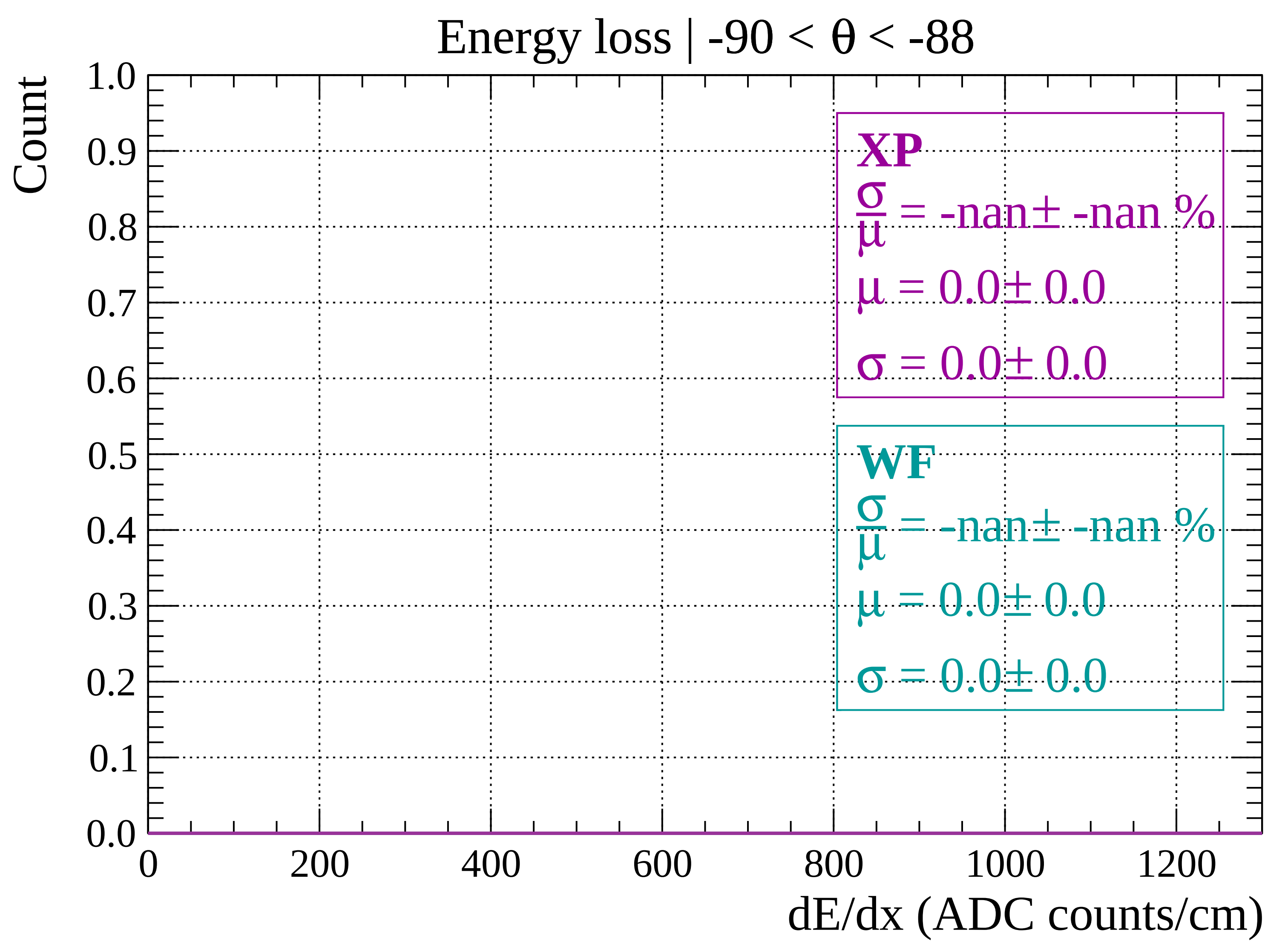


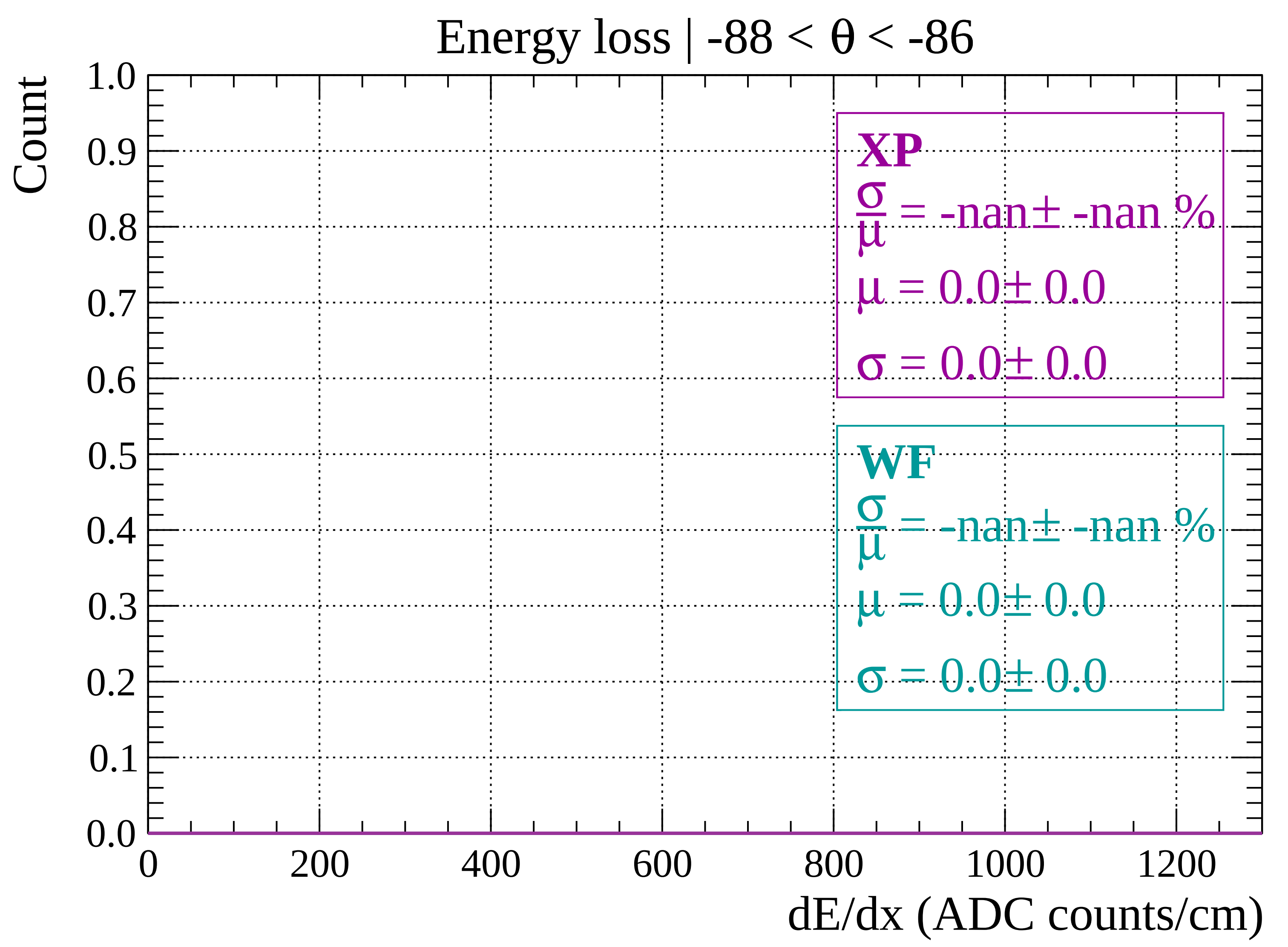
Energy loss | 1920 < p < 1960
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
120
140
160
XP
σ/μ = 9.27± 0.37 %
μ = 476.2± 1.6
σ = 44.2± 1.6
WF
σ/μ = 9.71± 0.36 %
μ = 478.3± 1.6
σ = 46.4± 1.6

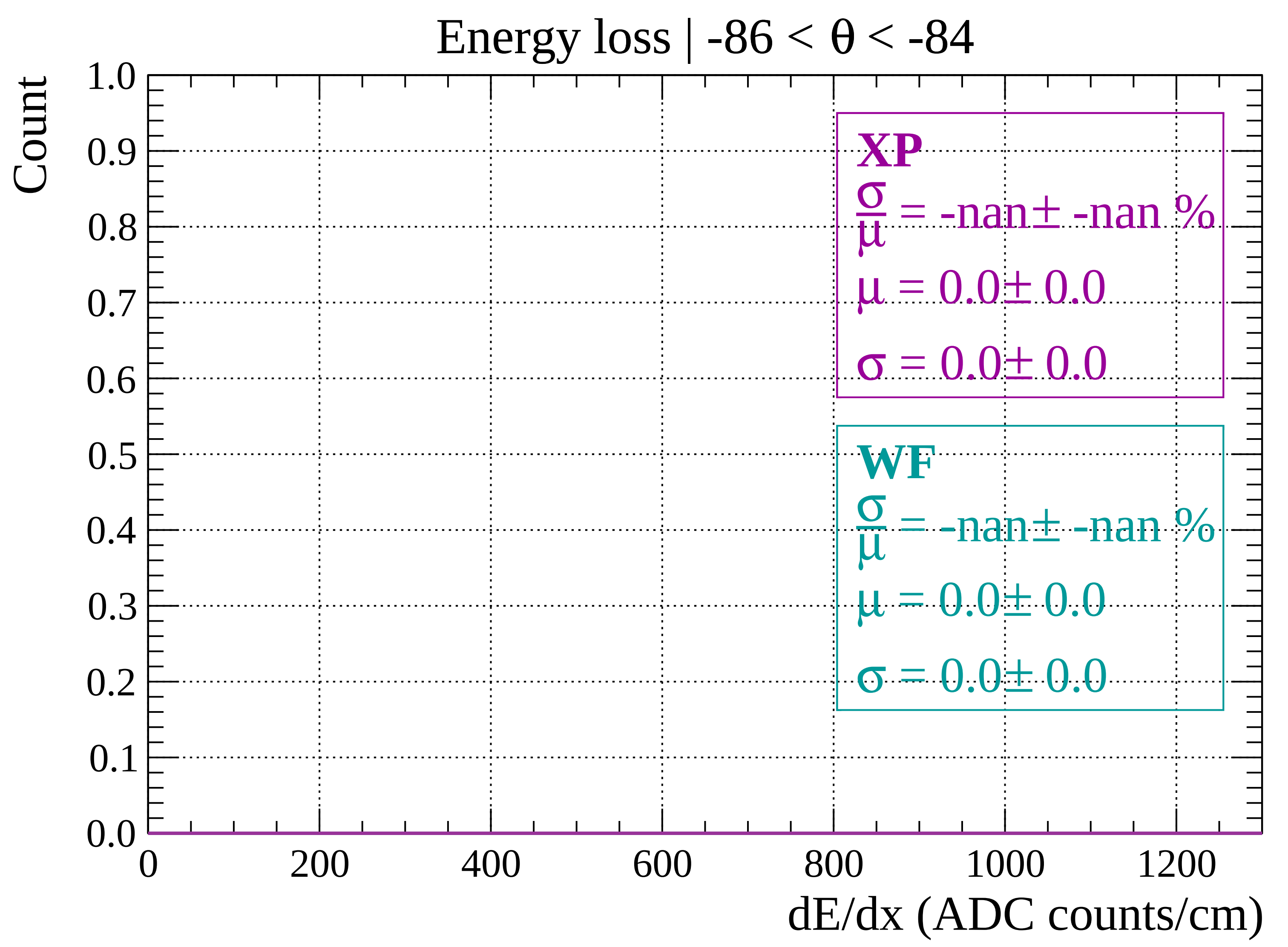


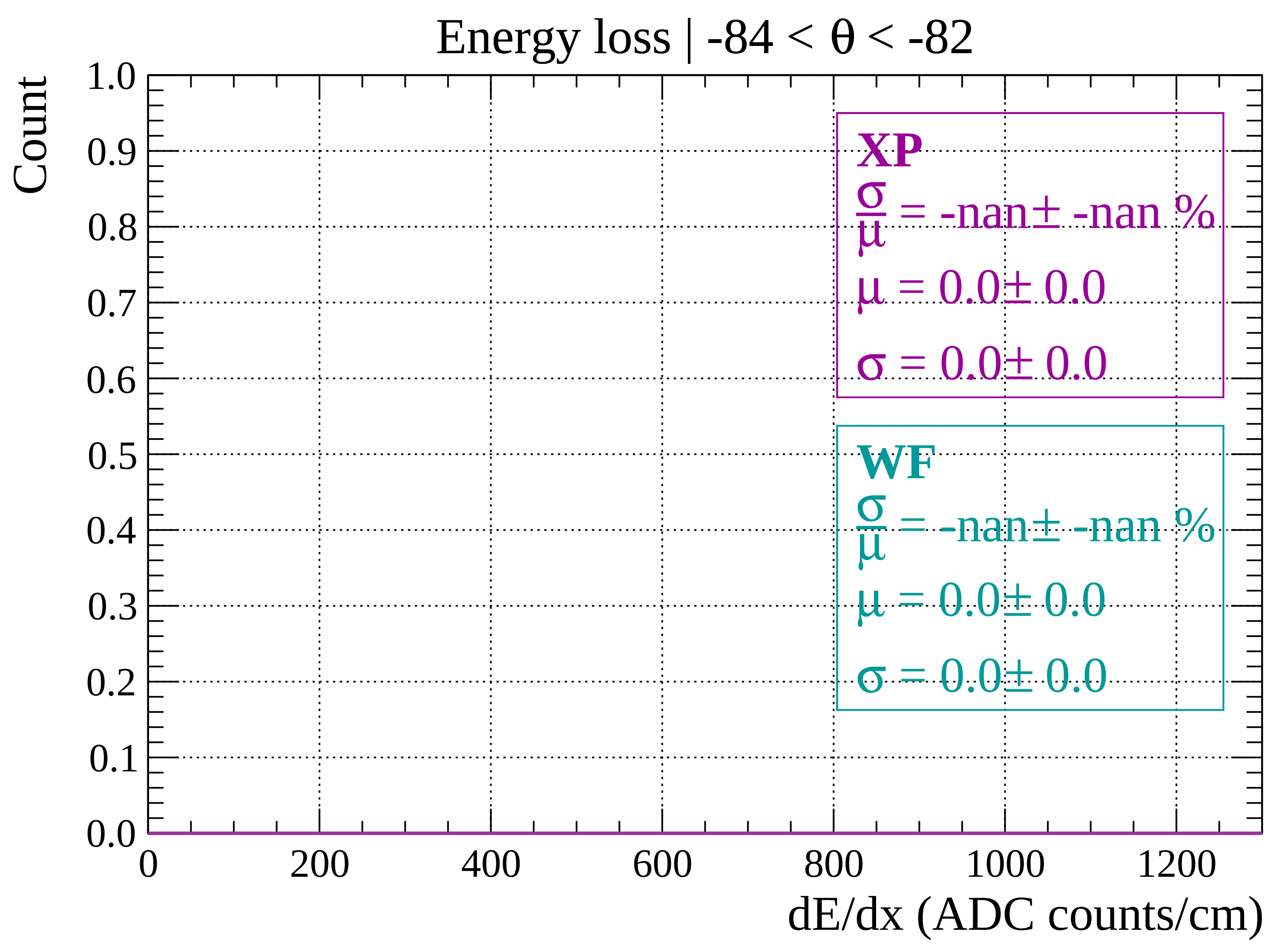
Energy loss | 1960 < p < 2000
Count
0
20
40
60
80
100
120
140
160
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 8.49± 0.27 %
μ = 477.2± 1.4
σ = 40.5± 1.2
WF
$\frac{\sigma}{\mu}$ = 9.09± 0.32 %
μ = 477.0± 1.5
σ = 43.4± 1.4


Energy loss | 2000 < p < 2040
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
10
20
30
40
50
60
70
80
90
XP
$\frac{\sigma}{\mu}$ = 8.89± 0.35 %
μ = 483.7± 2.0
σ = 43.0± 1.5
WF
$\frac{\sigma}{\mu}$ = 10.08± 0.49 %
μ = 484.2± 2.2
σ = 48.8± 2.1

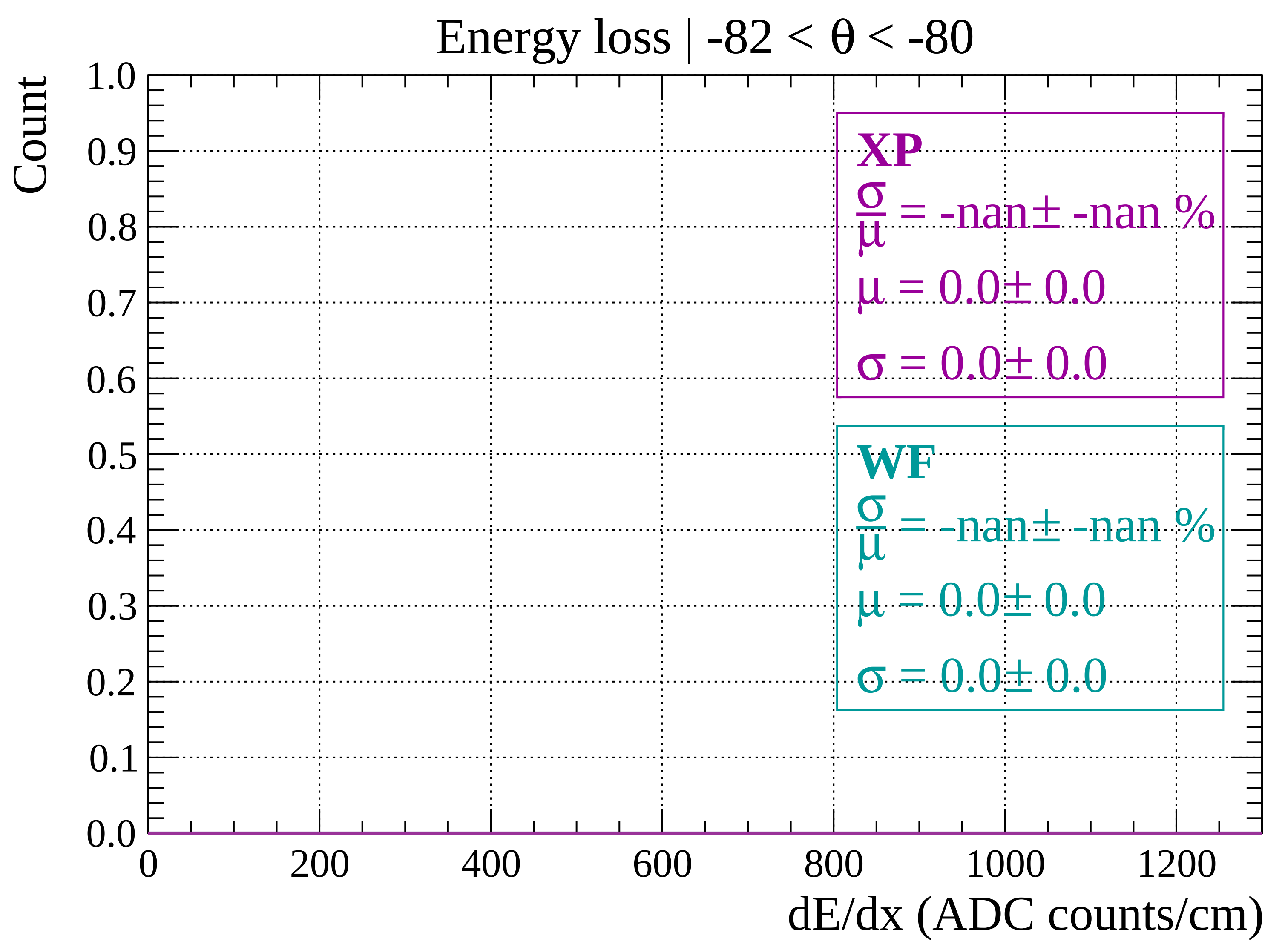

Energy loss | -90 < θ < -88
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

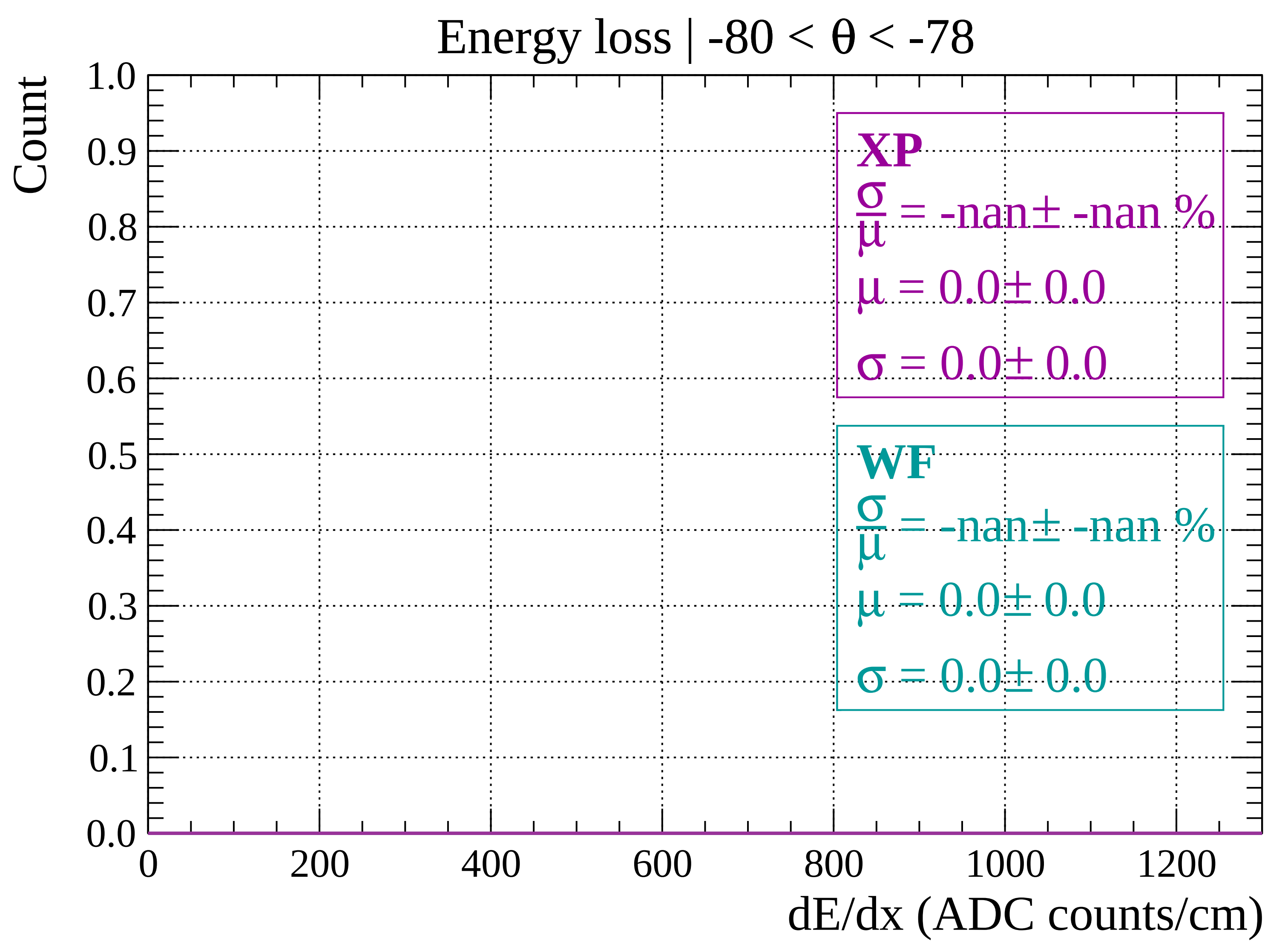

Energy loss | -88 < θ < -86
Count
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

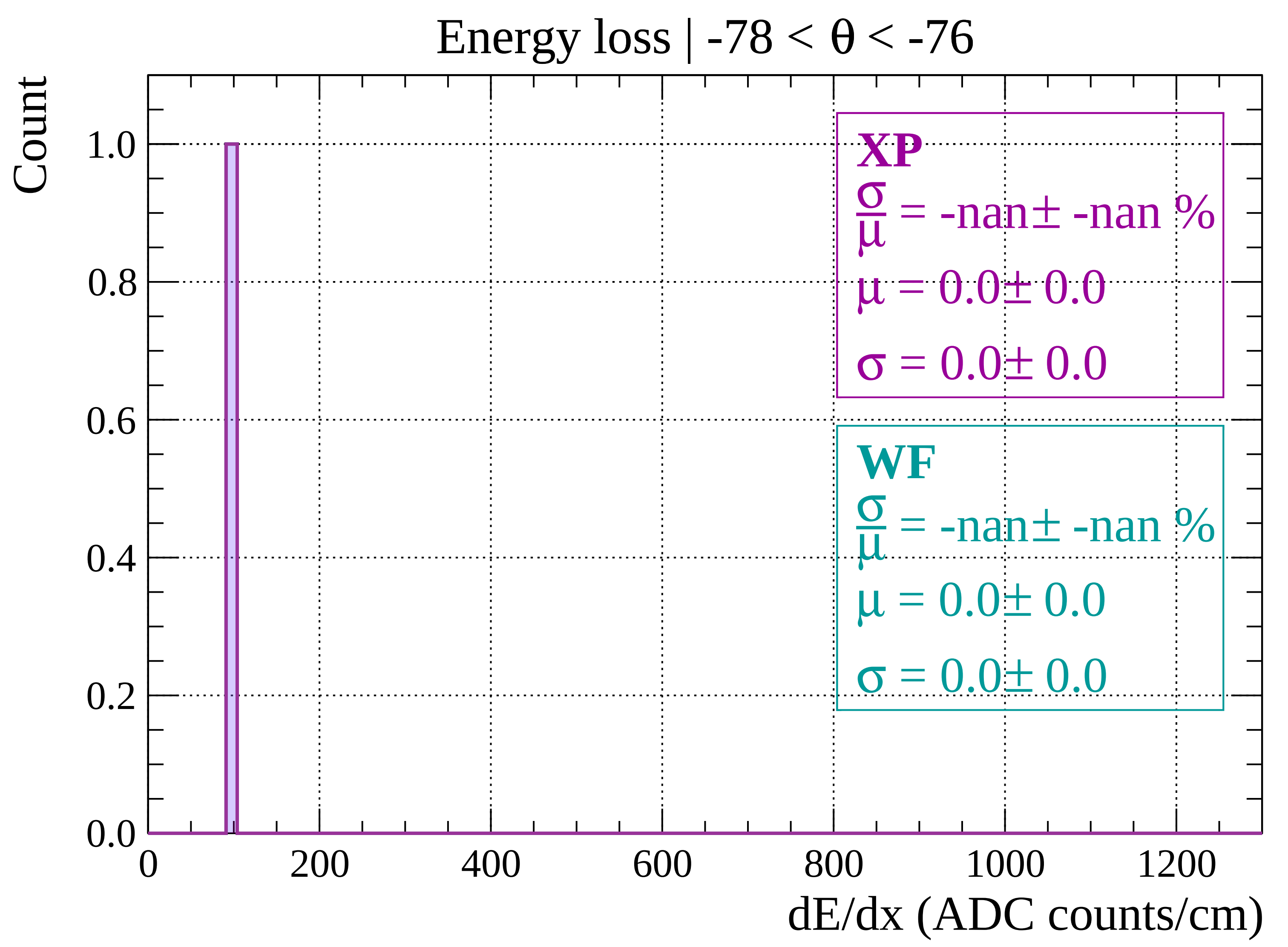

Energy loss | -86 < θ < -84
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

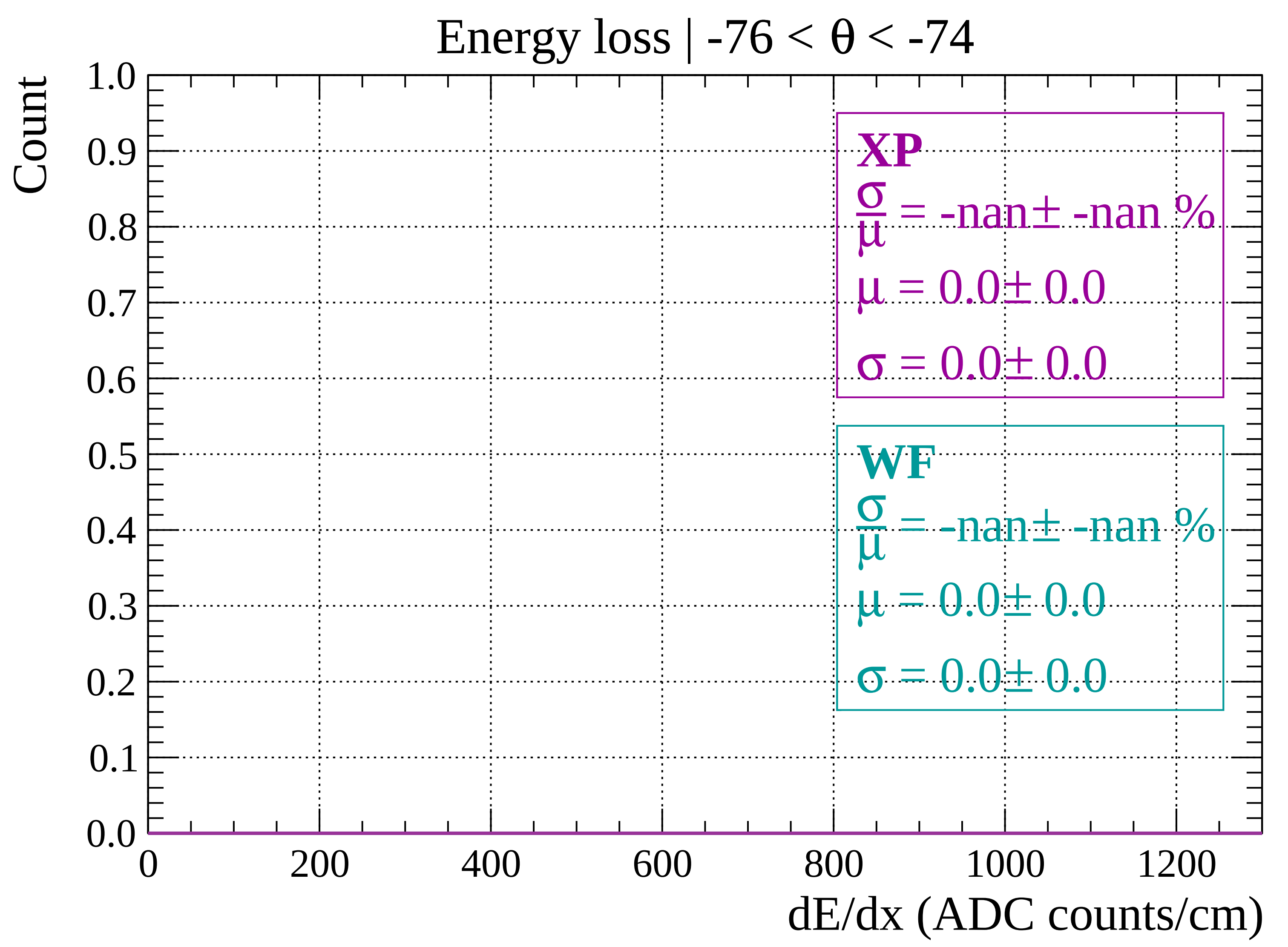


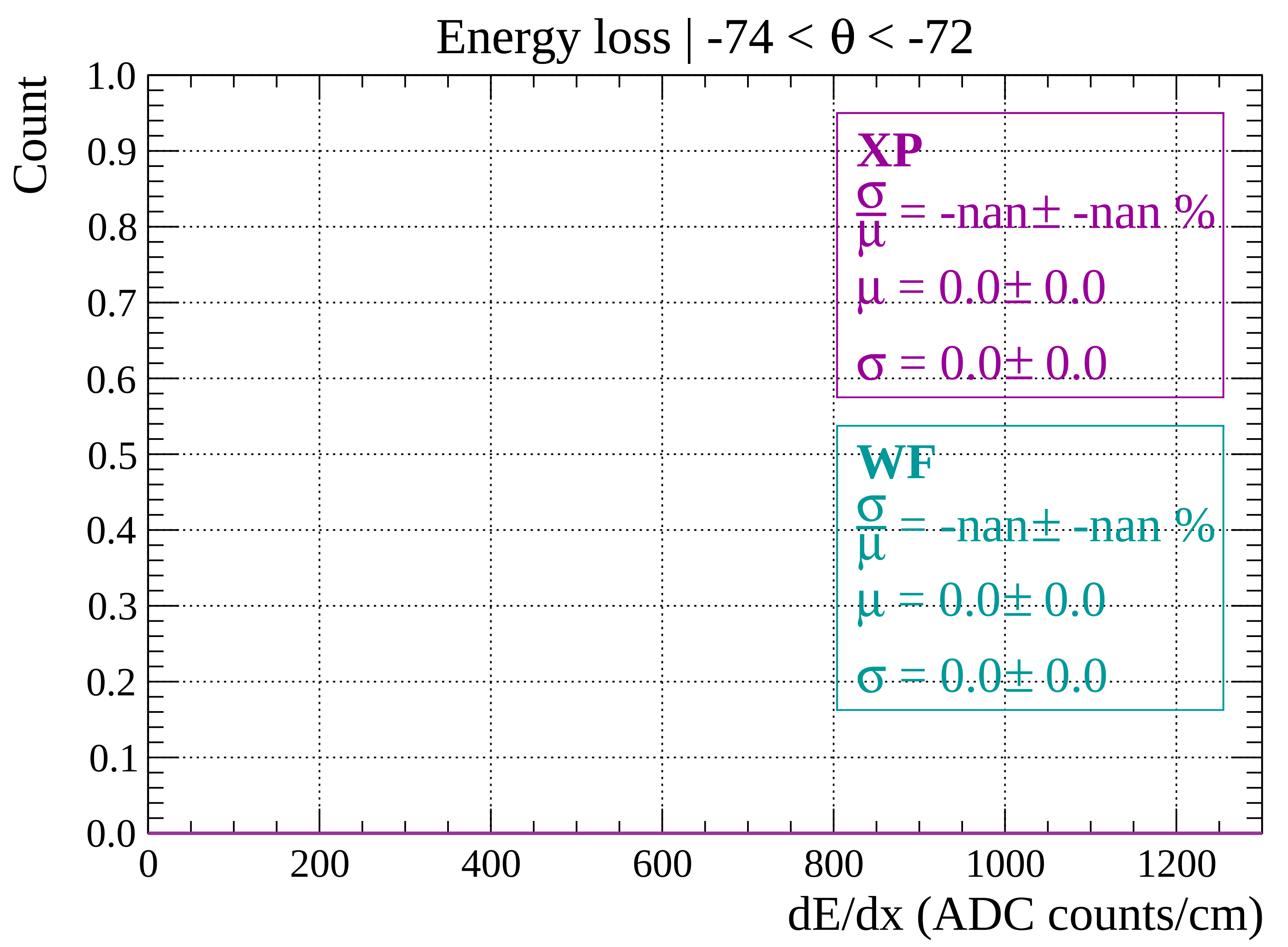
Energy loss | -84 < θ < -82
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

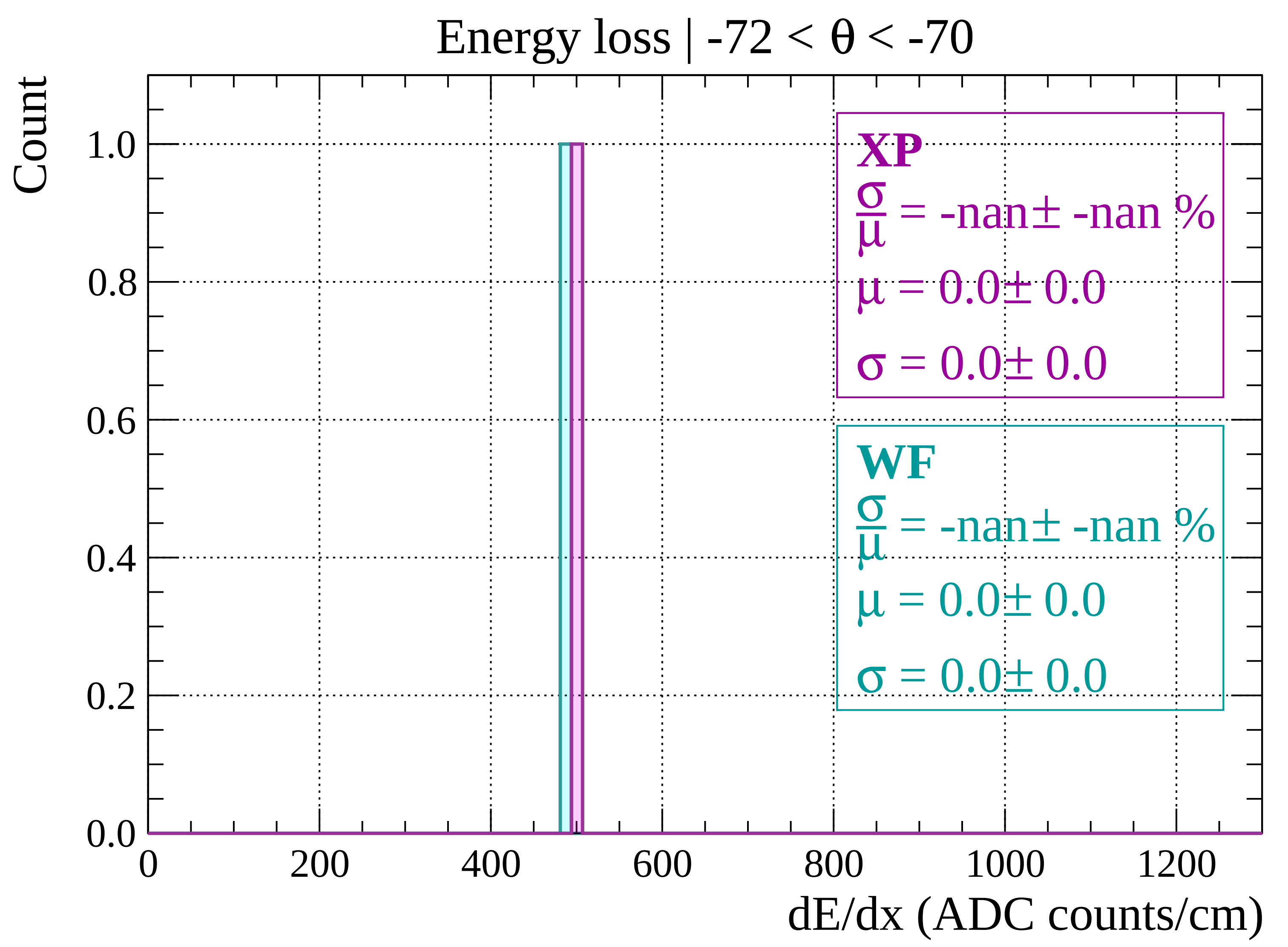

Energy loss | -82 < θ < -80
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

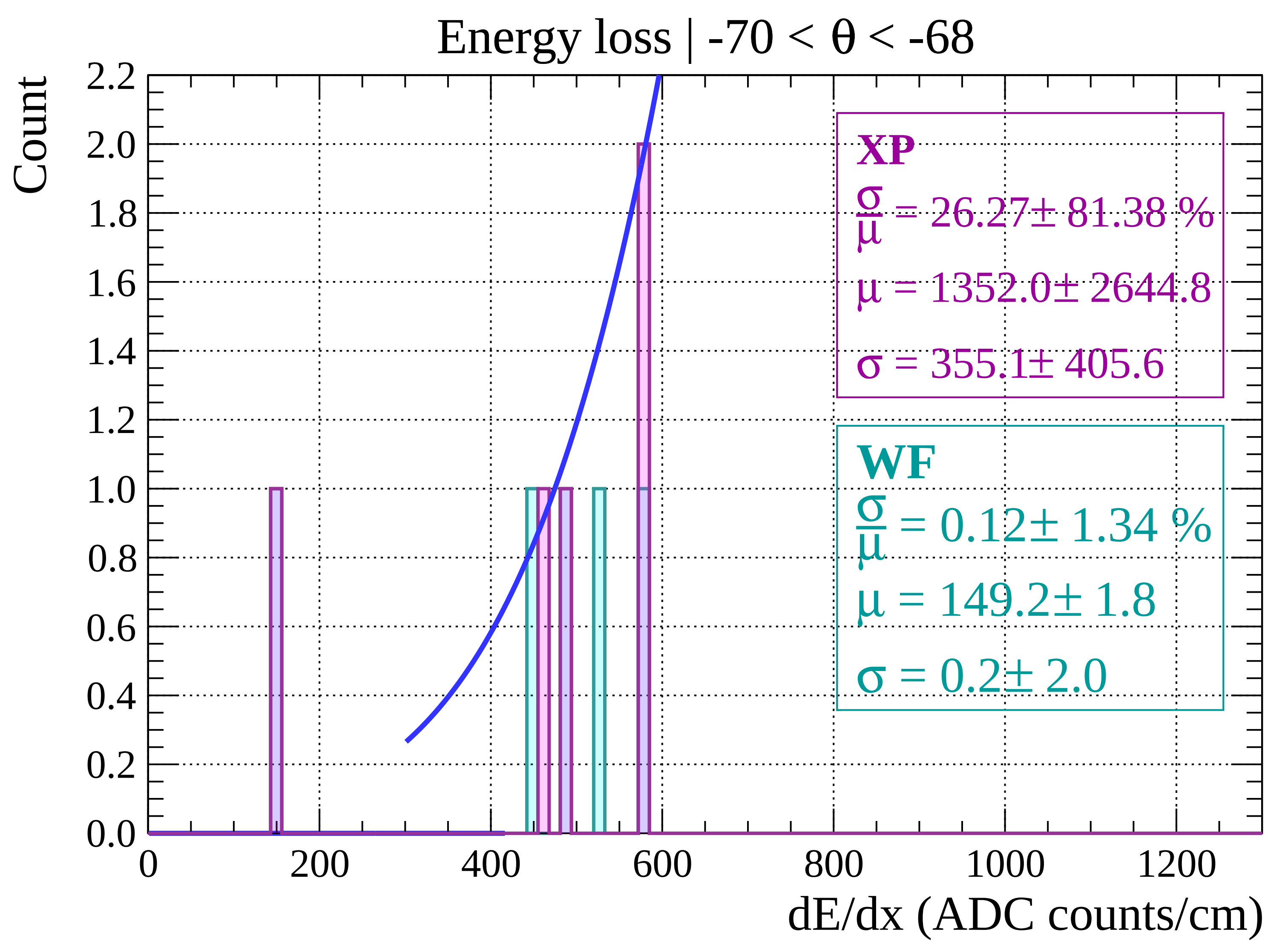

Energy loss | -80 < θ < -78
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0


Energy loss | -78 < θ < -76
Count
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
0.0
0.2
0.4
0.6
0.8
1.0
0
200
400
600
800
1000
1200

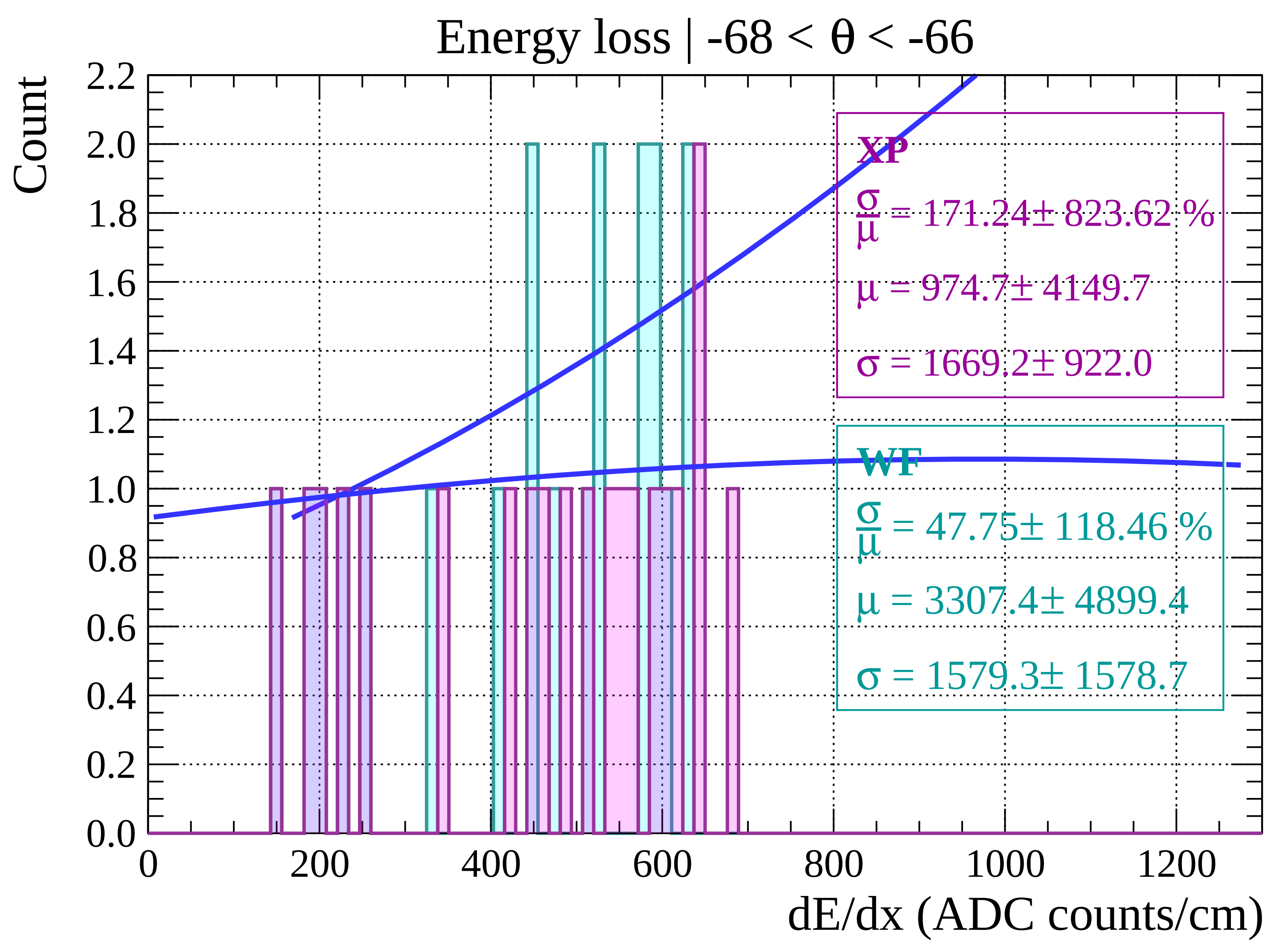

Energy loss | -76 < θ < -74
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0


Energy loss | -74 < θ < -72
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

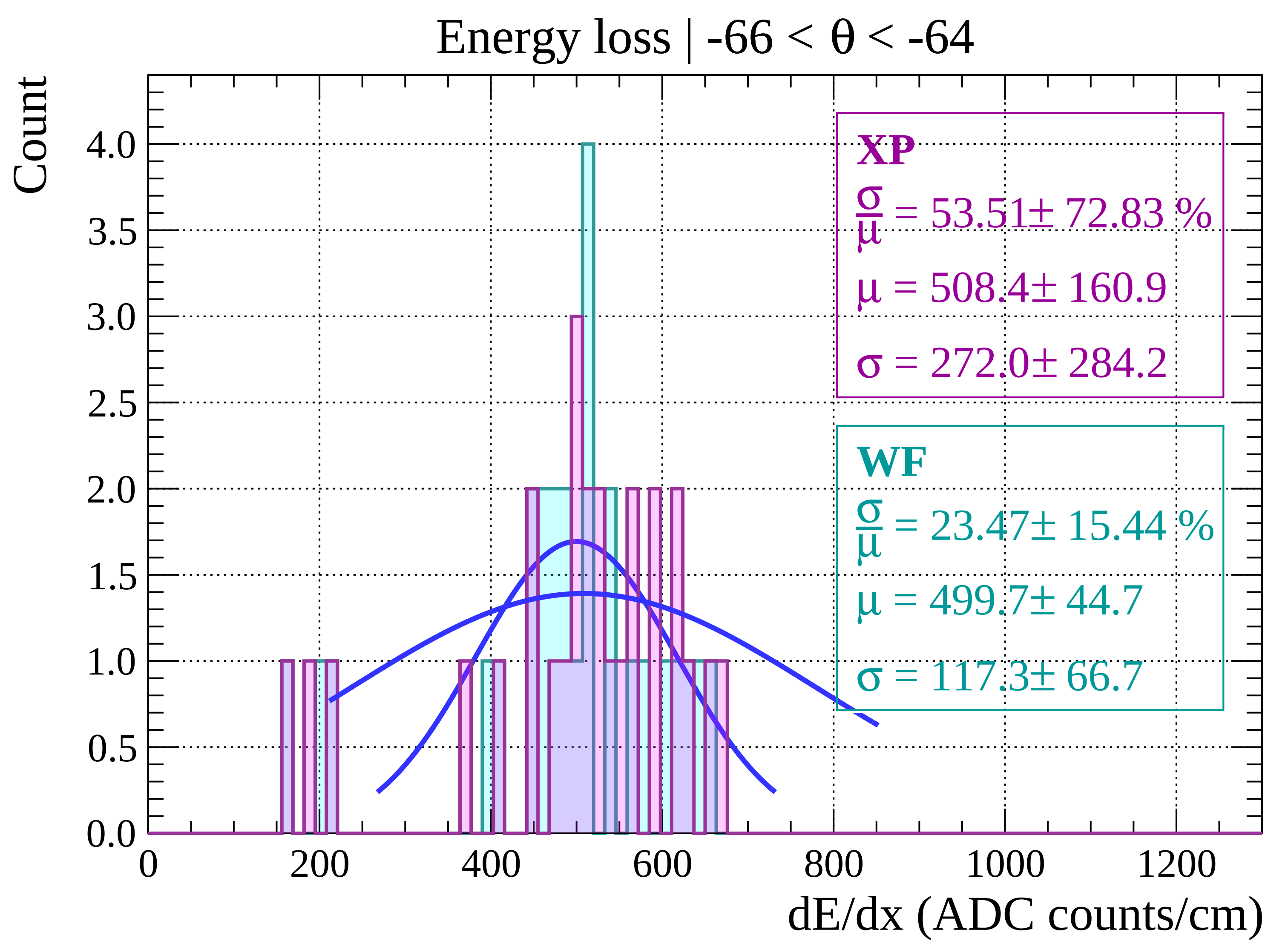

Energy loss | -72 < θ < -70
Count
1.0
0.8
0.6
0.4
0.2
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



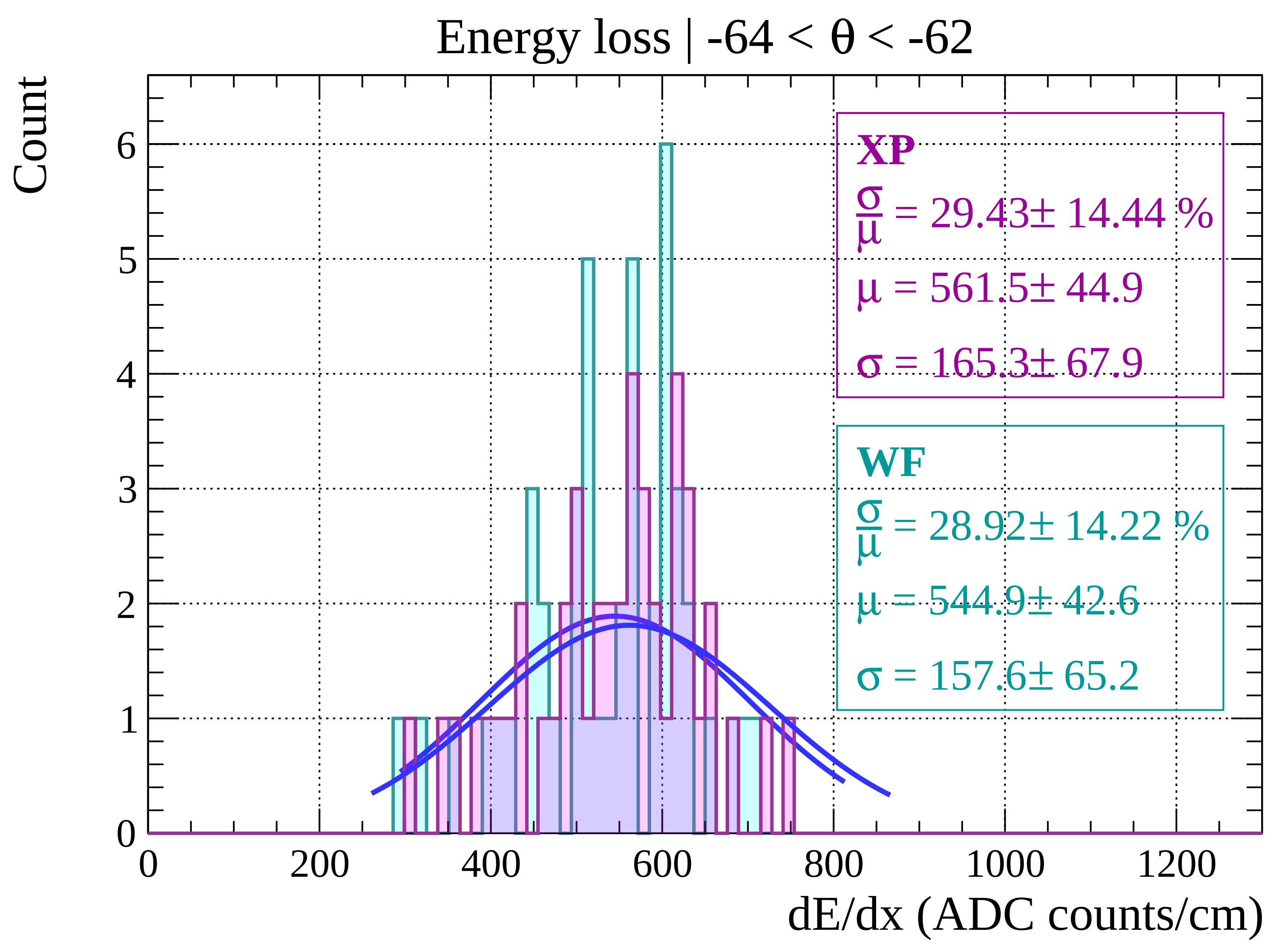
Energy loss | -70 < θ < -68
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 26.27± 81.38 %
μ = 1352.0± 2644.8
σ = 355.1± 405.6
WF
σ/μ = 0.12± 1.34 %
μ = 149.2± 1.8
σ = 0.2± 2.0
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
1.6
1.8
2.0
2.2
0
200
400
600
800
1000
1200

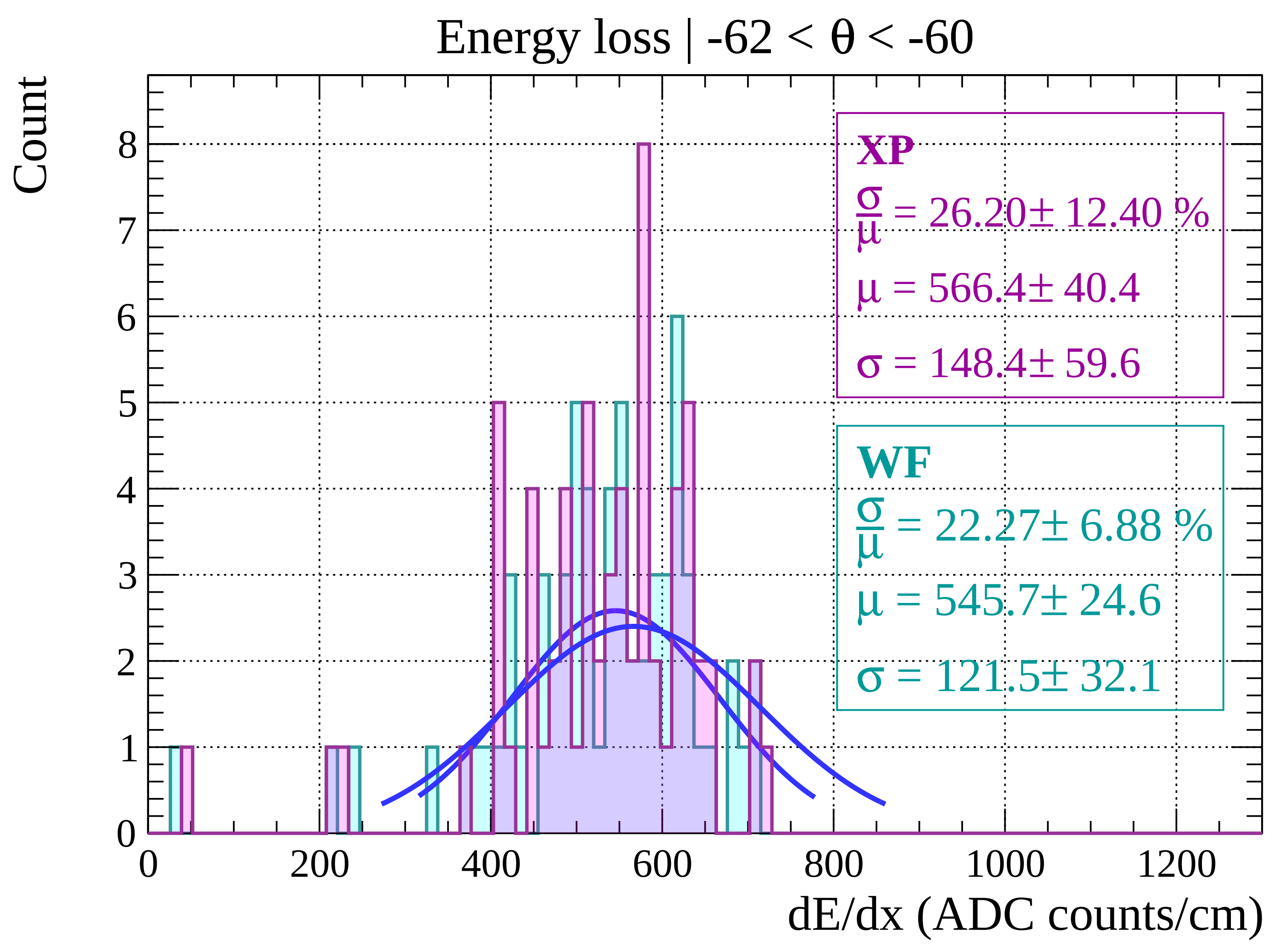


Energy loss | -68 < θ < -66
Count
2.2
2.0
1.8
1.6
1.4
1.2
1.0
0.8
0.6
0.4
0.2
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 171.24± 823.62 %
μ = 974.7± 4149.7
σ = 1669.2± 922.0
WF
σ/μ = 47.75± 118.46 %
μ = 3307.4± 4899.4
σ = 1579.3± 1578.7

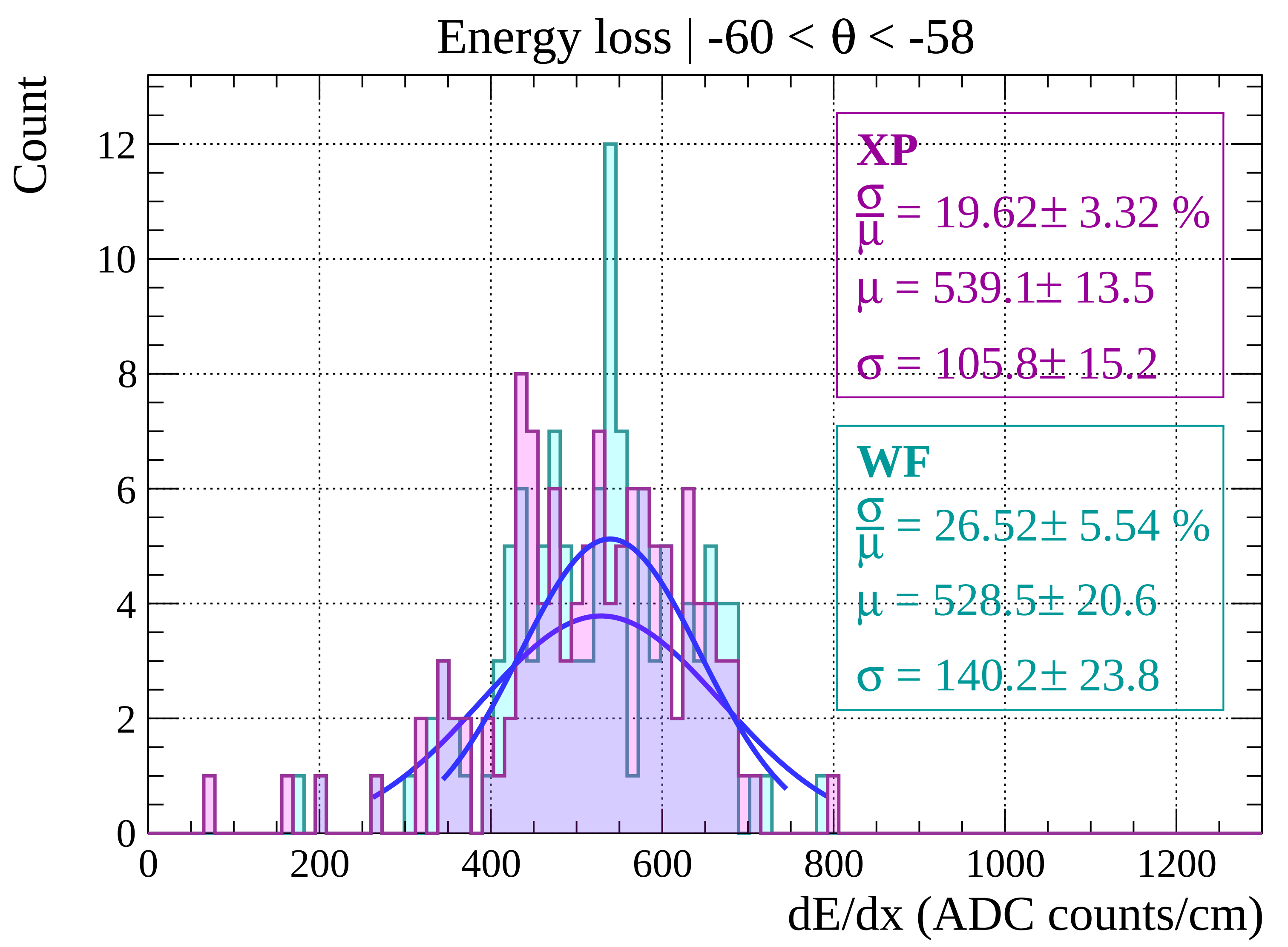


Energy loss | -66 < θ < -64
Count
dE/dx (ADC counts/cm)
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
0
200
400
600
800
1000
1200
XP
σ/μ = 53.51± 72.83 %
μ = 508.4± 160.9
σ = 272.0± 284.2
WF
σ/μ = 23.47± 15.44 %
μ = 499.7± 44.7
σ = 117.3± 66.7

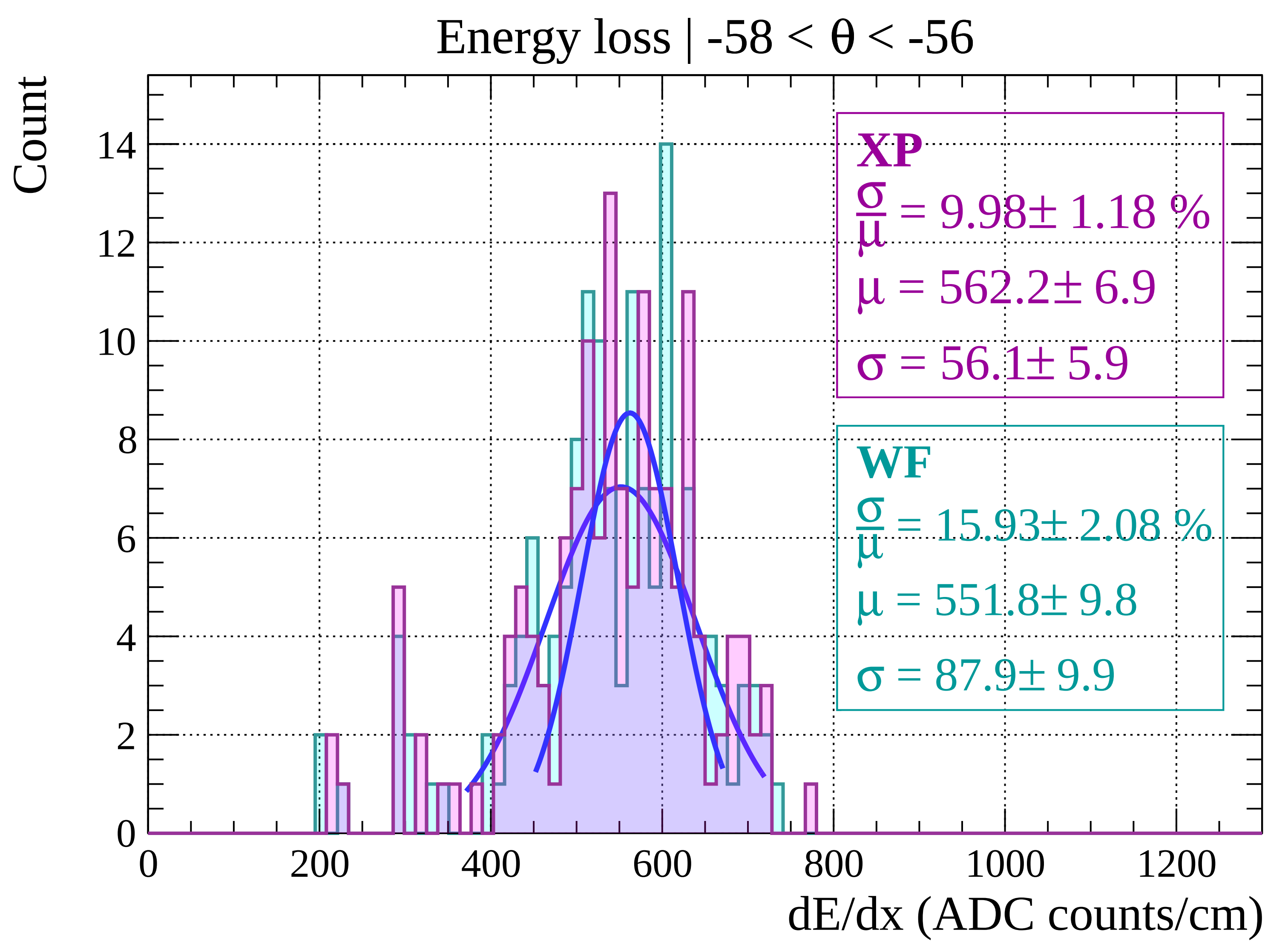

Energy loss | -64 < θ < -62
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
1
2
3
4
5
6
XP
σ/μ = 29.43± 14.44 %
μ = 561.5± 44.9
σ = 165.3± 67.9
WF
σ/μ = 28.92± 14.22 %
μ = 544.9± 42.6
σ = 157.6± 65.2



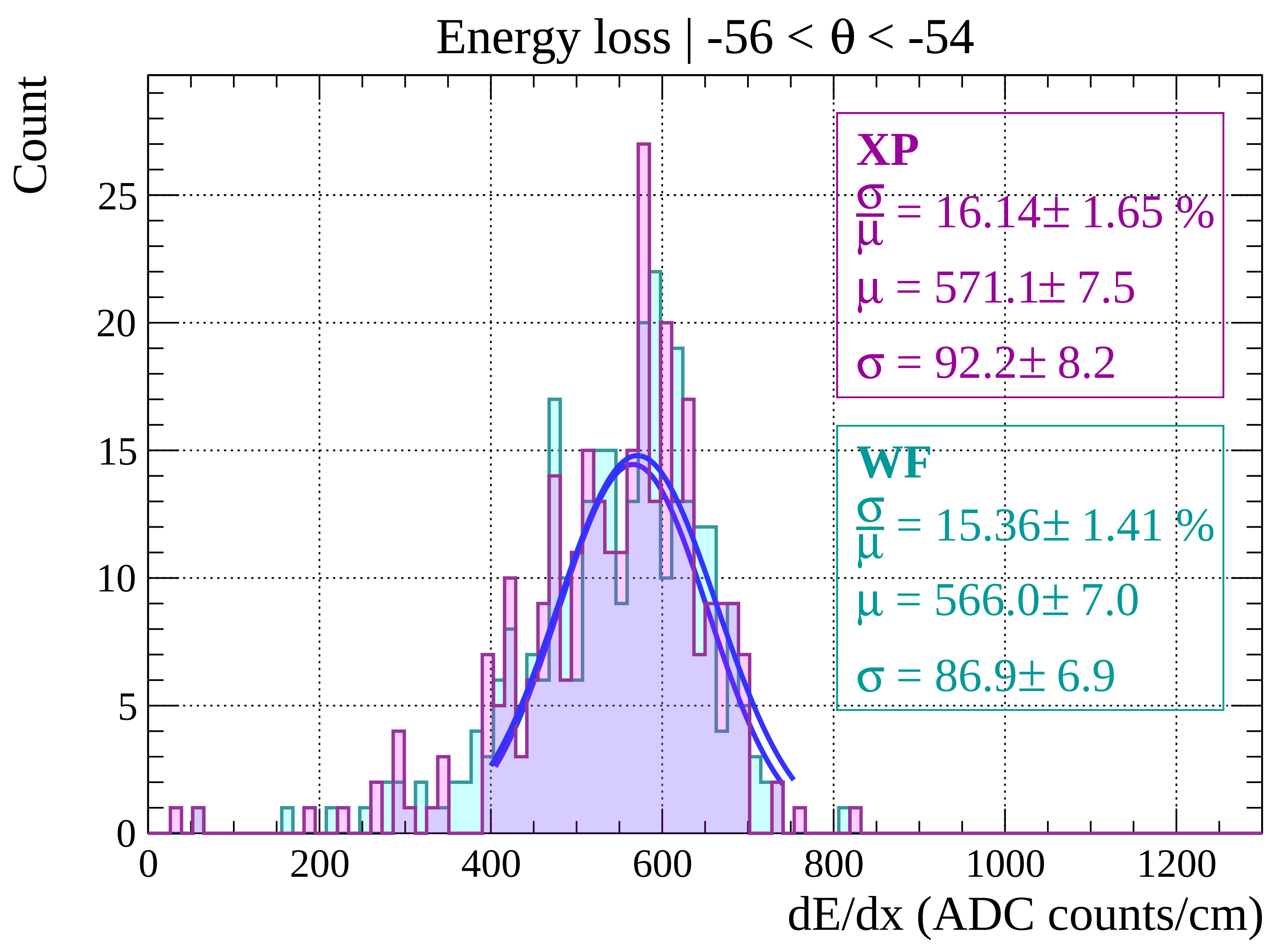
Energy loss | -62 < θ < -60
Count
dE/dx (ADC counts/cm)
0
1
2
3
4
5
6
7
8
0
200
400
600
800
1000
1200
XP
σ/μ = 26.20± 12.40 %
μ = 566.4± 40.4
σ = 148.4± 59.6
WF
σ/μ = 22.27± 6.88 %
μ = 545.7± 24.6
σ = 121.5± 32.1



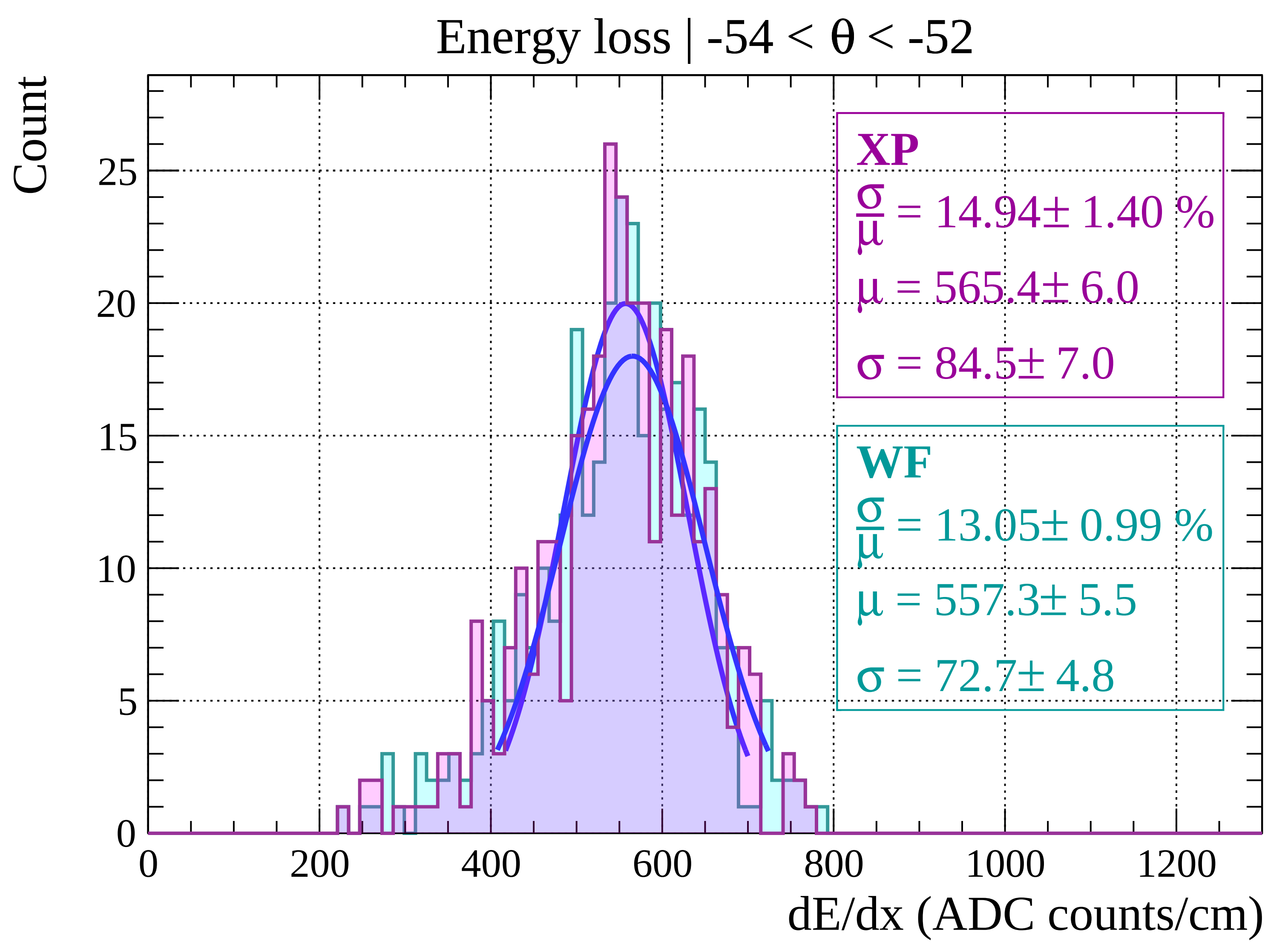
Energy loss | -60 < θ < -58
Count
dE/dx (ADC counts/cm)
0
2
4
6
8
10
12
0
200
400
600
800
1000
1200
XP
σ/μ = 19.62± 3.32 %
μ = 539.1± 13.5
σ = 105.8± 15.2
WF
σ/μ = 26.52± 5.54 %
μ = 528.5± 20.6
σ = 140.2± 23.8



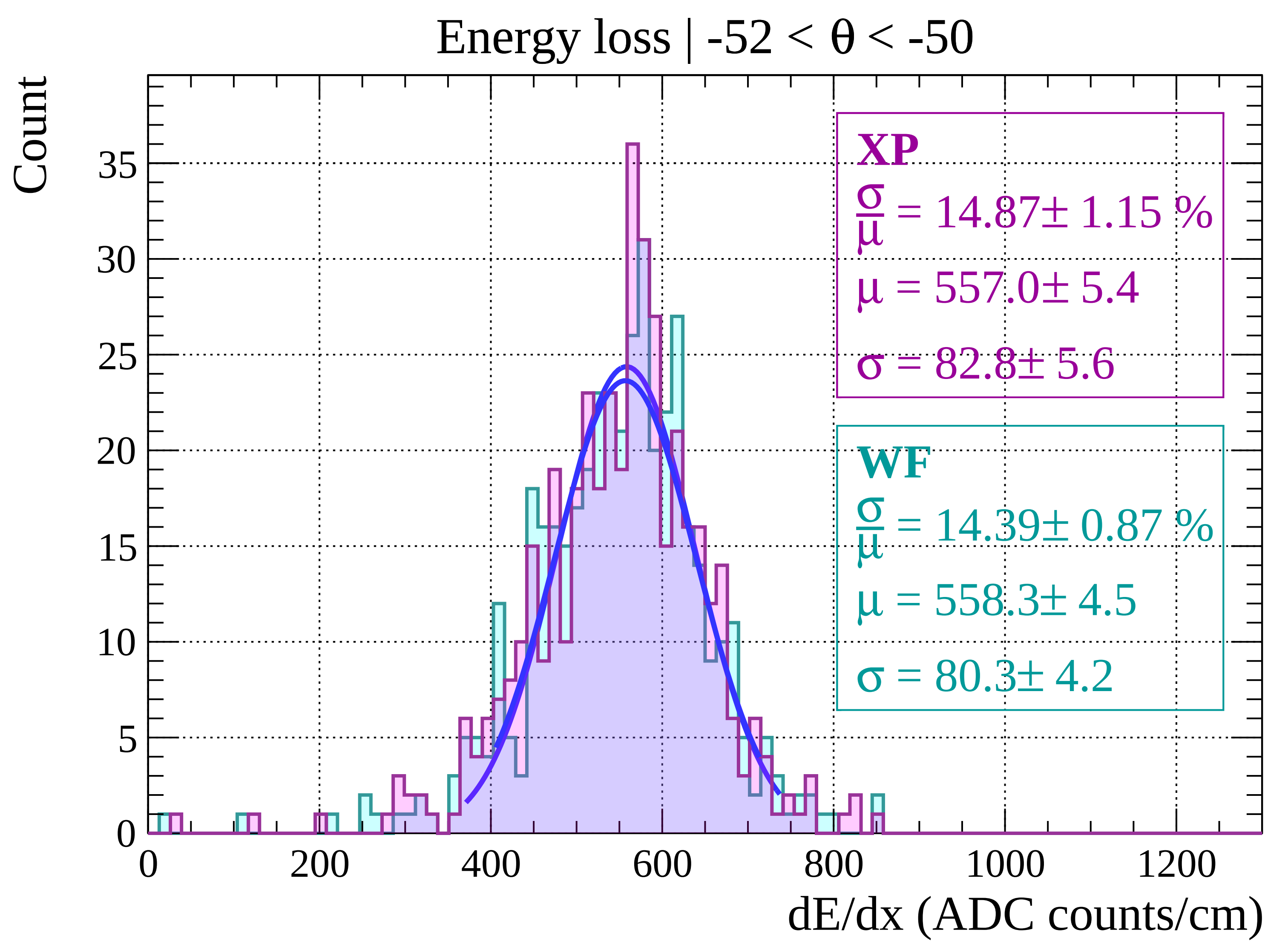
Energy loss | -58 < θ < -56
Count
dE/dx (ADC counts/cm)
0
2
4
6
8
10
12
14
0
200
400
600
800
1000
1200
XP
σ/μ = 9.98± 1.18 %
μ = 562.2± 6.9
σ = 56.1± 5.9
WF
σ/μ = 15.93± 2.08 %
μ = 551.8± 9.8
σ = 87.9± 9.9



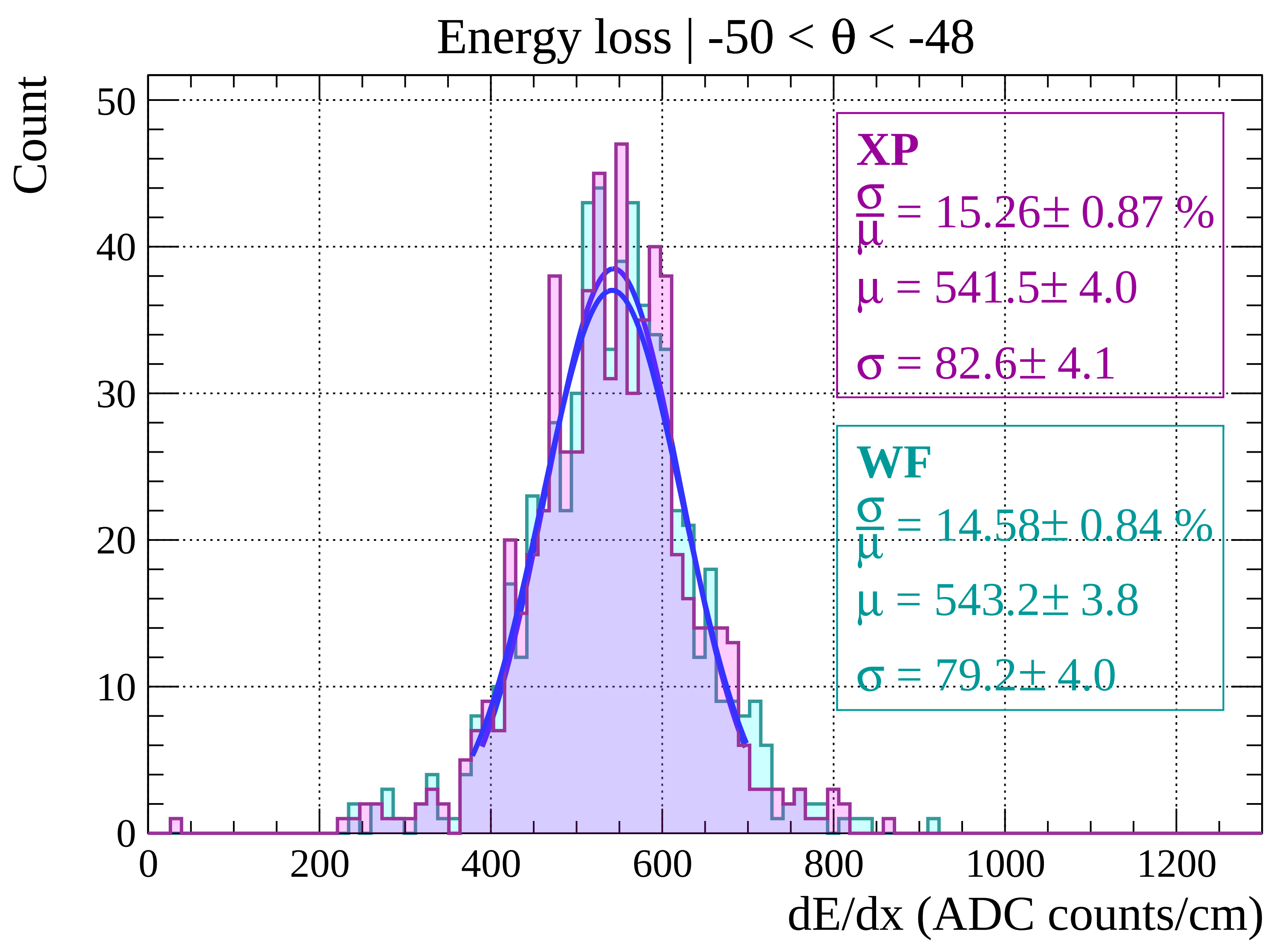
Energy loss | -56 < θ < -54
Count
dE/dx (ADC counts/cm)
0
5
10
15
20
25
0
200
400
600
800
1000
1200
XP
σ/μ = 16.14± 1.65 %
μ = 571.1± 7.5
σ = 92.2± 8.2
WF
σ/μ = 15.36± 1.41 %
μ = 566.0± 7.0
σ = 86.9± 6.9



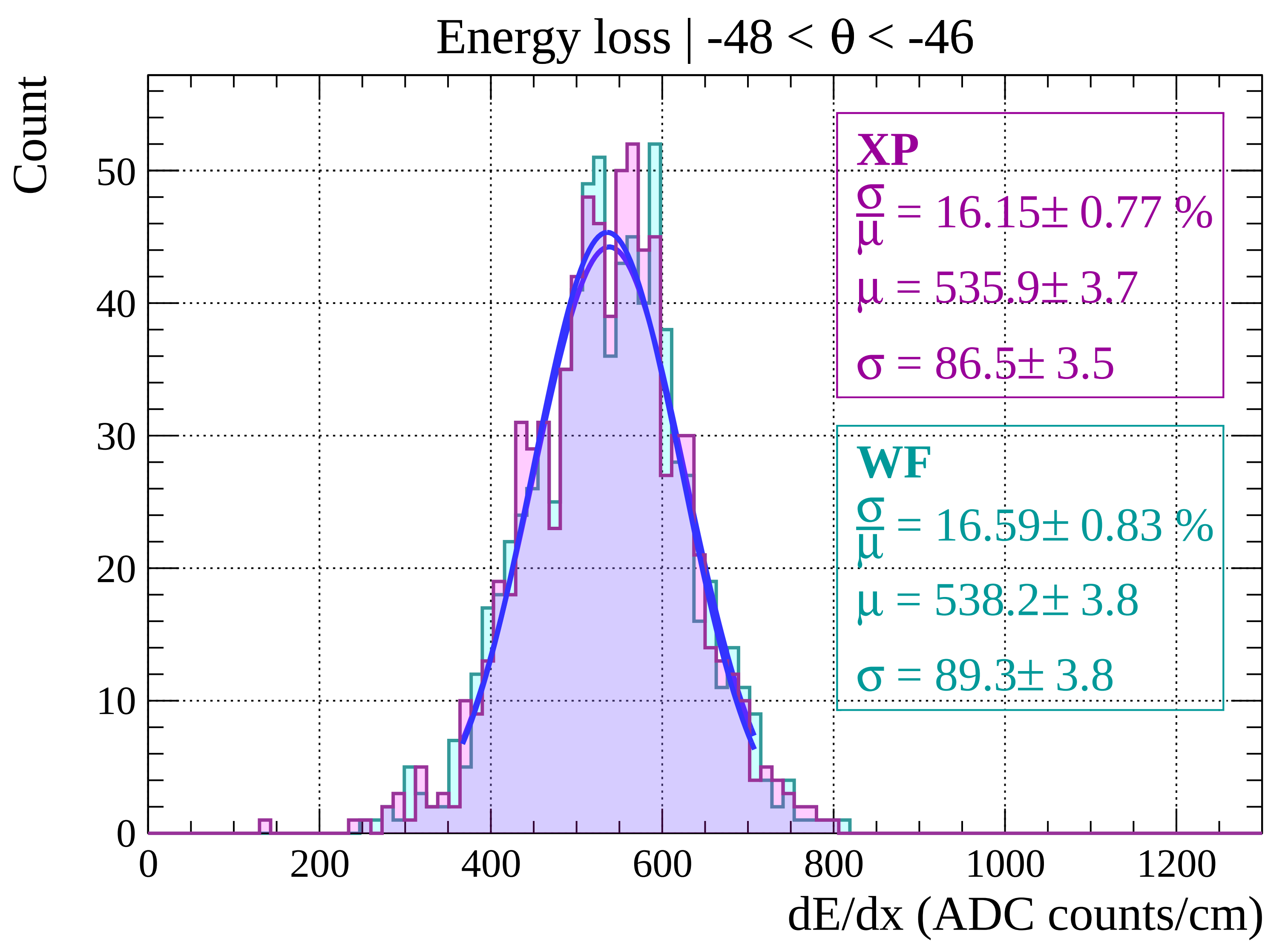
Energy loss | -54 < θ < -52
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
5
10
15
20
25
XP
σ/μ = 14.94± 1.40 %
μ = 565.4± 6.0
σ = 84.5± 7.0
WF
σ/μ = 13.05± 0.99 %
μ = 557.3± 5.5
σ = 72.7± 4.8



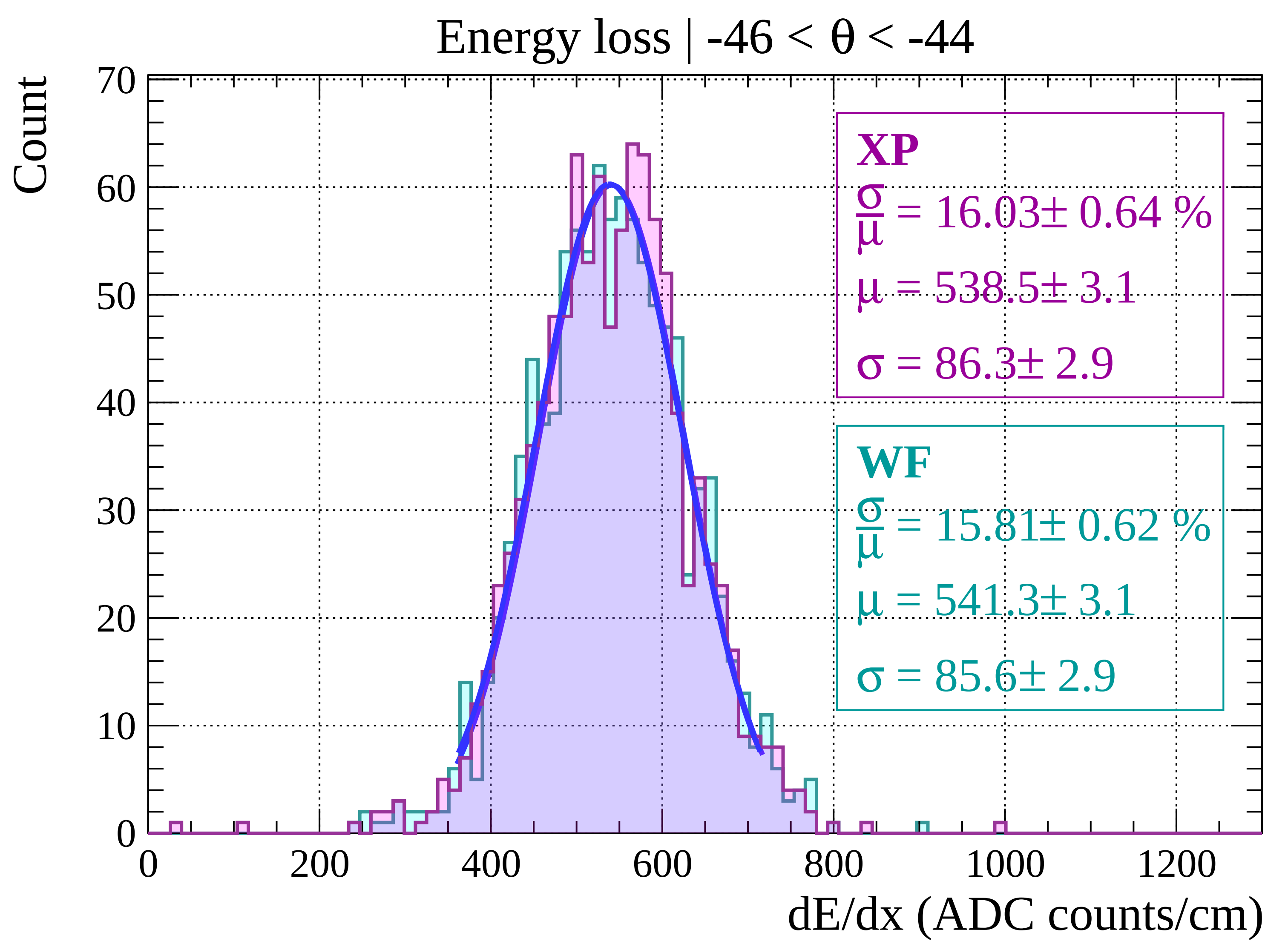
Energy loss | -52 < θ < -50
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 14.87± 1.15 %
μ = 557.0± 5.4
σ = 82.8± 5.6
WF
σ/μ = 14.39± 0.87 %
μ = 558.3± 4.5
σ = 80.3± 4.2
0
5
10
15
20
25
30
35
0
200
400
600
800
1000
1200



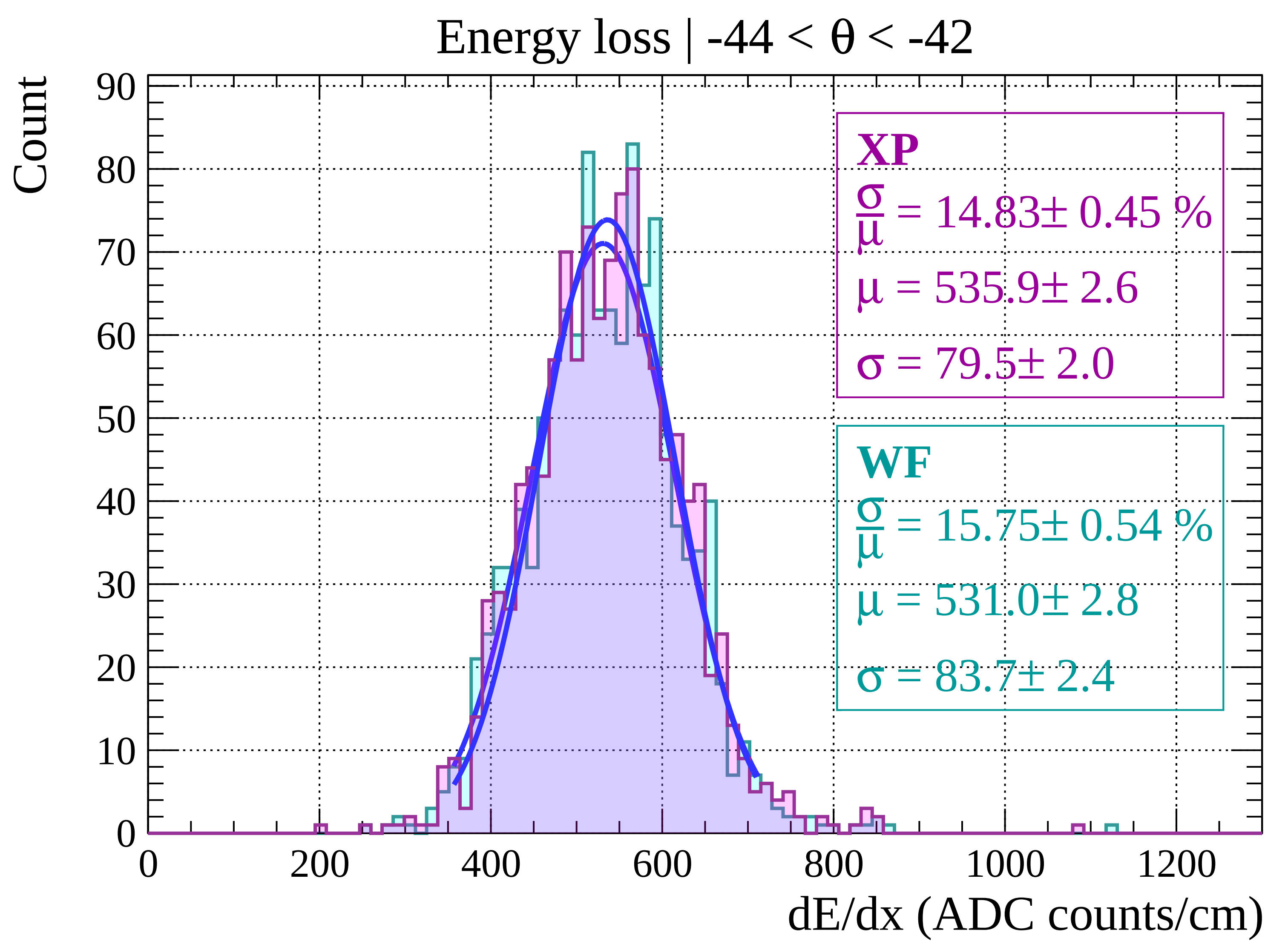
Energy loss | -50 < θ < -48
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
10
20
30
40
50
XP
$\frac{\sigma}{\mu}$ = 15.26± 0.87 %
μ = 541.5± 4.0
σ = 82.6± 4.1
WF
$\frac{\sigma}{\mu}$ = 14.58± 0.84 %
μ = 543.2± 3.8
σ = 79.2± 4.0



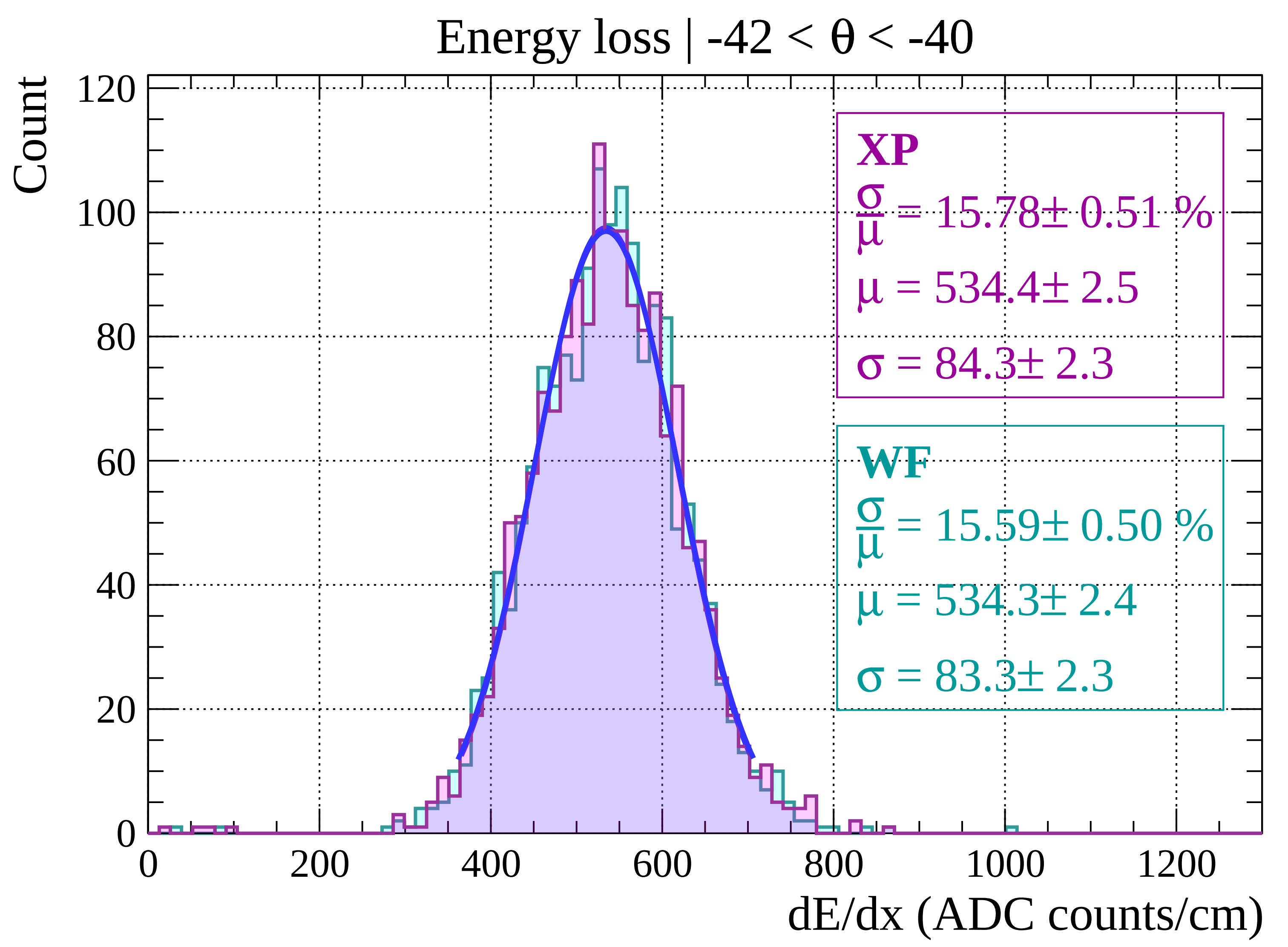
Energy loss | -48 < θ < -46
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
10
20
30
40
50
XP
σ/μ = 16.15± 0.77 %
μ = 535.9± 3.7
σ = 86.5± 3.5
WF
σ/μ = 16.59± 0.83 %
μ = 538.2± 3.8
σ = 89.3± 3.8

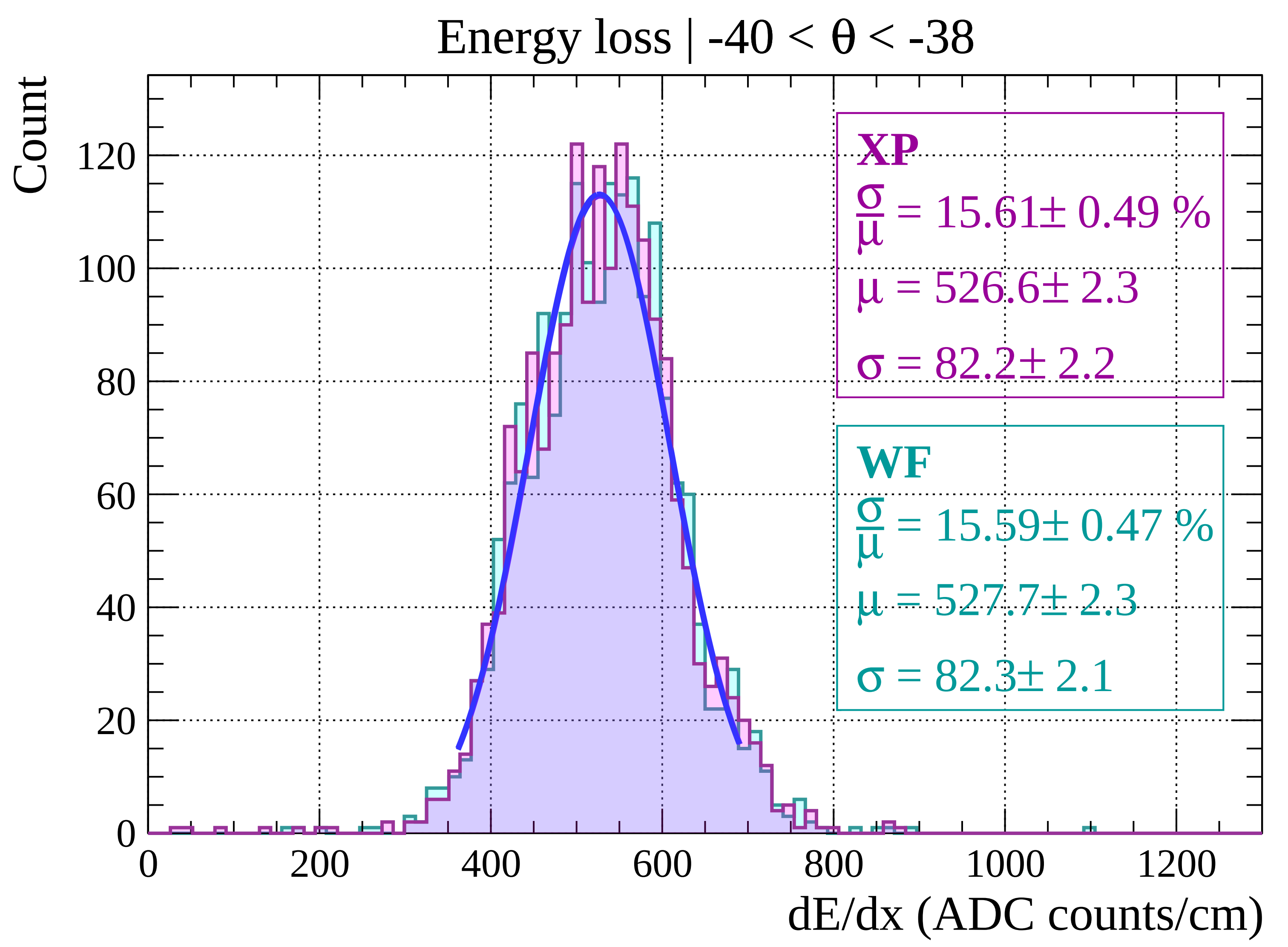


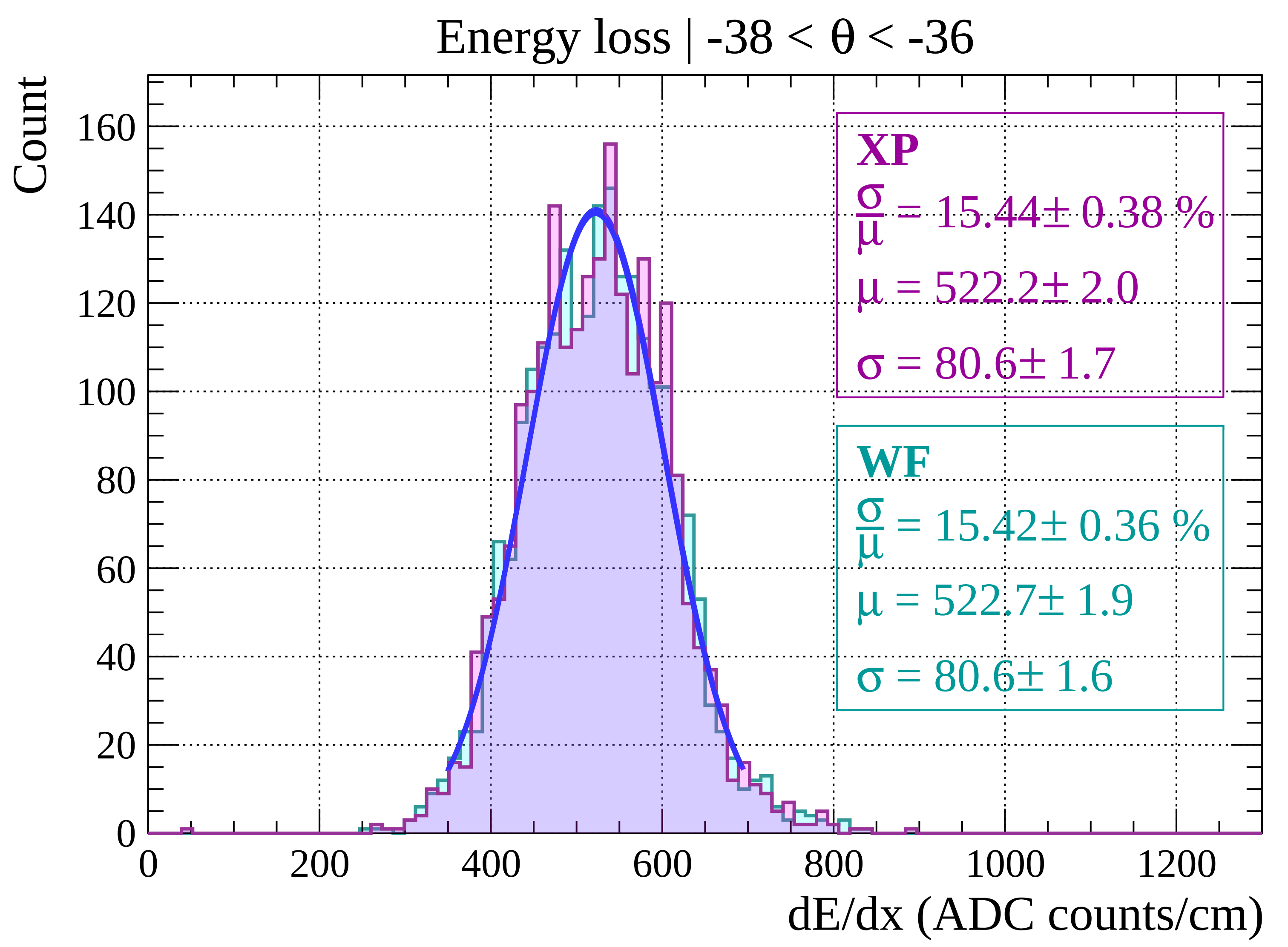
Energy loss | -46 < θ < -44
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 16.03± 0.64 %
μ = 538.5± 3.1
σ = 86.3± 2.9
WF
σ/μ = 15.81± 0.62 %
μ = 541.3± 3.1
σ = 85.6± 2.9
0
10
20
30
40
50
60
70
0
200
400
600
800
1000
1200

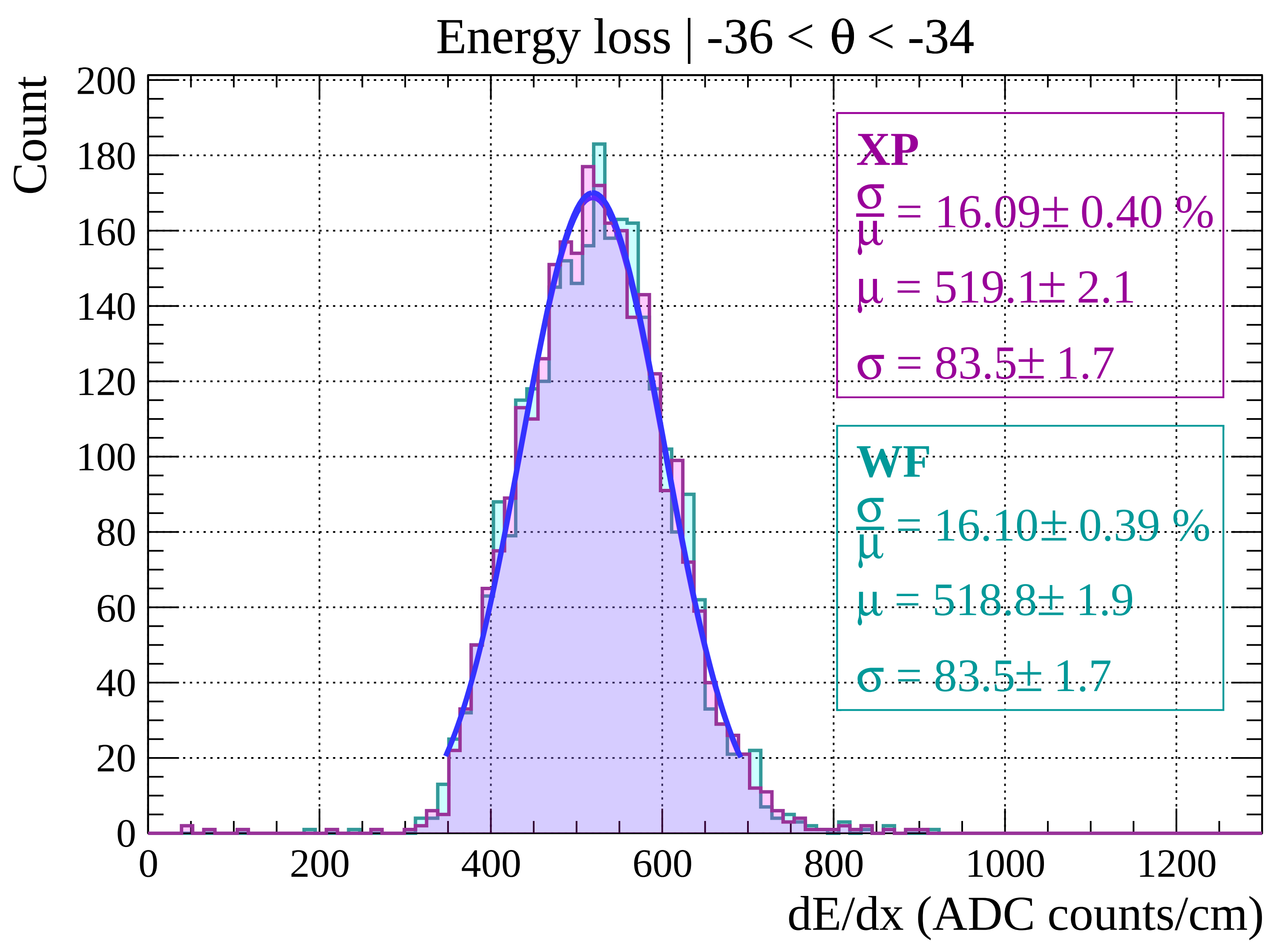


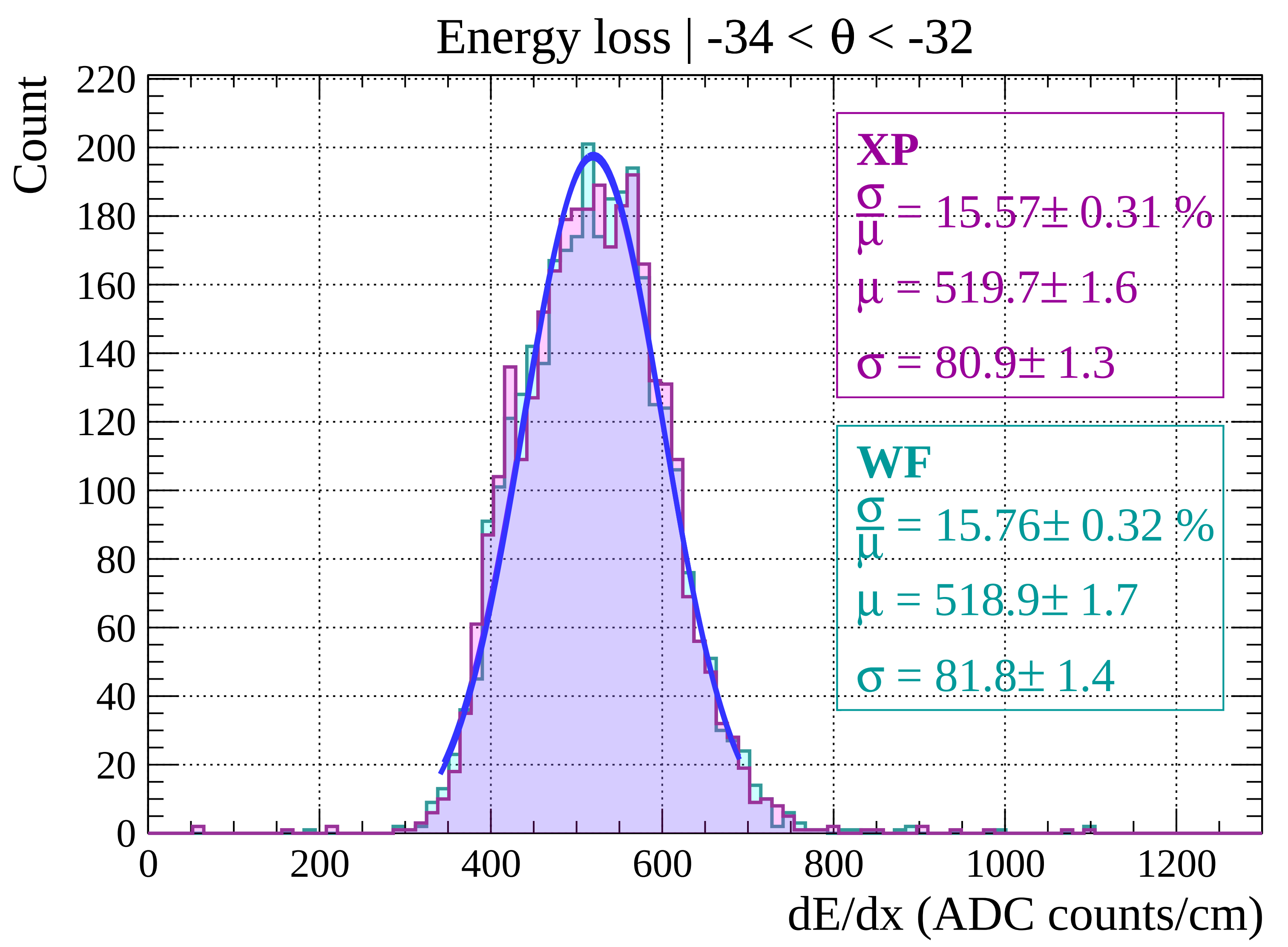
Energy loss | -44 < θ < -42
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
70
80
90
0
200
400
600
800
1000
1200
XP
σ/μ = 14.83± 0.45 %
μ = 535.9± 2.6
σ = 79.5± 2.0
WF
σ/μ = 15.75± 0.54 %
μ = 531.0± 2.8
σ = 83.7± 2.4


Energy loss | -42 < θ < -40
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 15.78± 0.51 %
μ = 534.4± 2.5
σ = 84.3± 2.3
WF
σ/μ = 15.59± 0.50 %
μ = 534.3± 2.4
σ = 83.3± 2.3
0
20
40
60
80
100
120
0
200
400
600
800
1000
1200

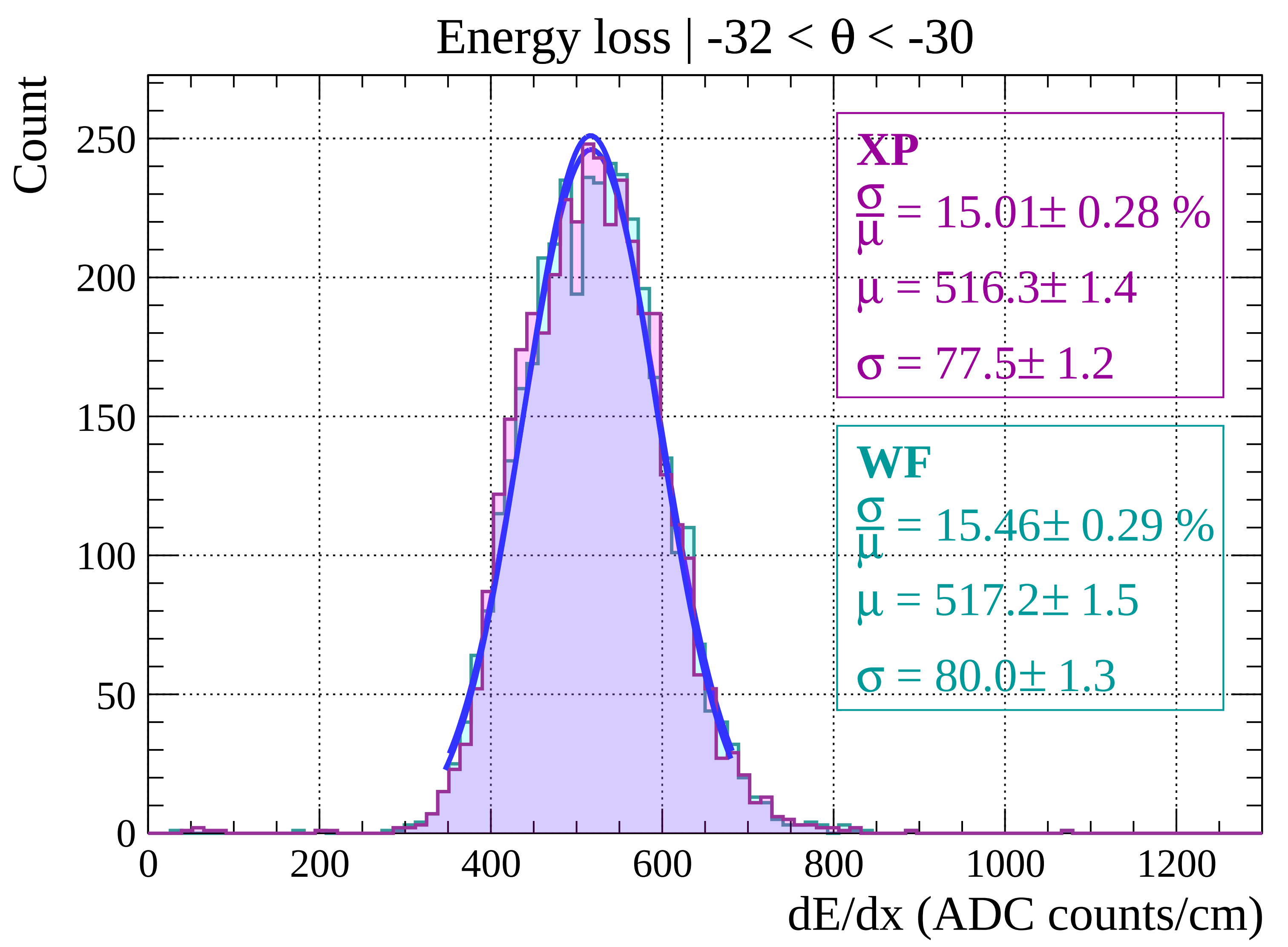

Energy loss | -40 < θ < -38
Count
120
100
80
60
40
20
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 15.61± 0.49 %
μ = 526.6± 2.3
σ = 82.2± 2.2
WF
σ/μ = 15.59± 0.47 %
μ = 527.7± 2.3
σ = 82.3± 2.1


Energy loss | -38 < θ < -36
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
120
140
160
XP
σ/μ = 15.44± 0.38 %
μ = 522.2± 2.0
σ = 80.6± 1.7
WF
σ/μ = 15.42± 0.36 %
μ = 522.7± 1.9
σ = 80.6± 1.6

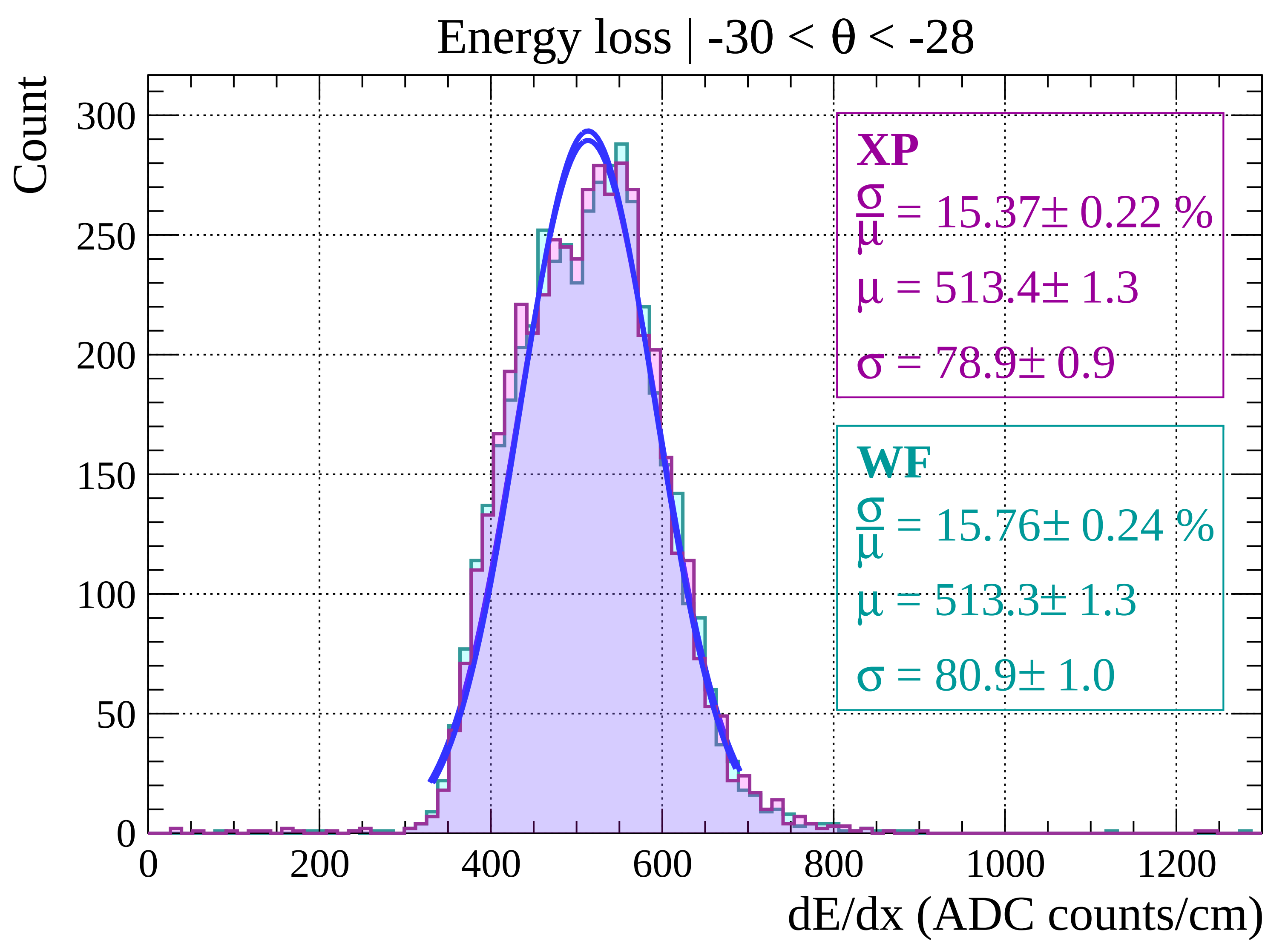

Energy loss | -36 < θ < -34
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
180
200
0
200
400
600
800
1000
1200
XP
σ/μ = 16.09± 0.40 %
μ = 519.1± 2.1
σ = 83.5± 1.7
WF
σ/μ = 16.10± 0.39 %
μ = 518.8± 1.9
σ = 83.5± 1.7

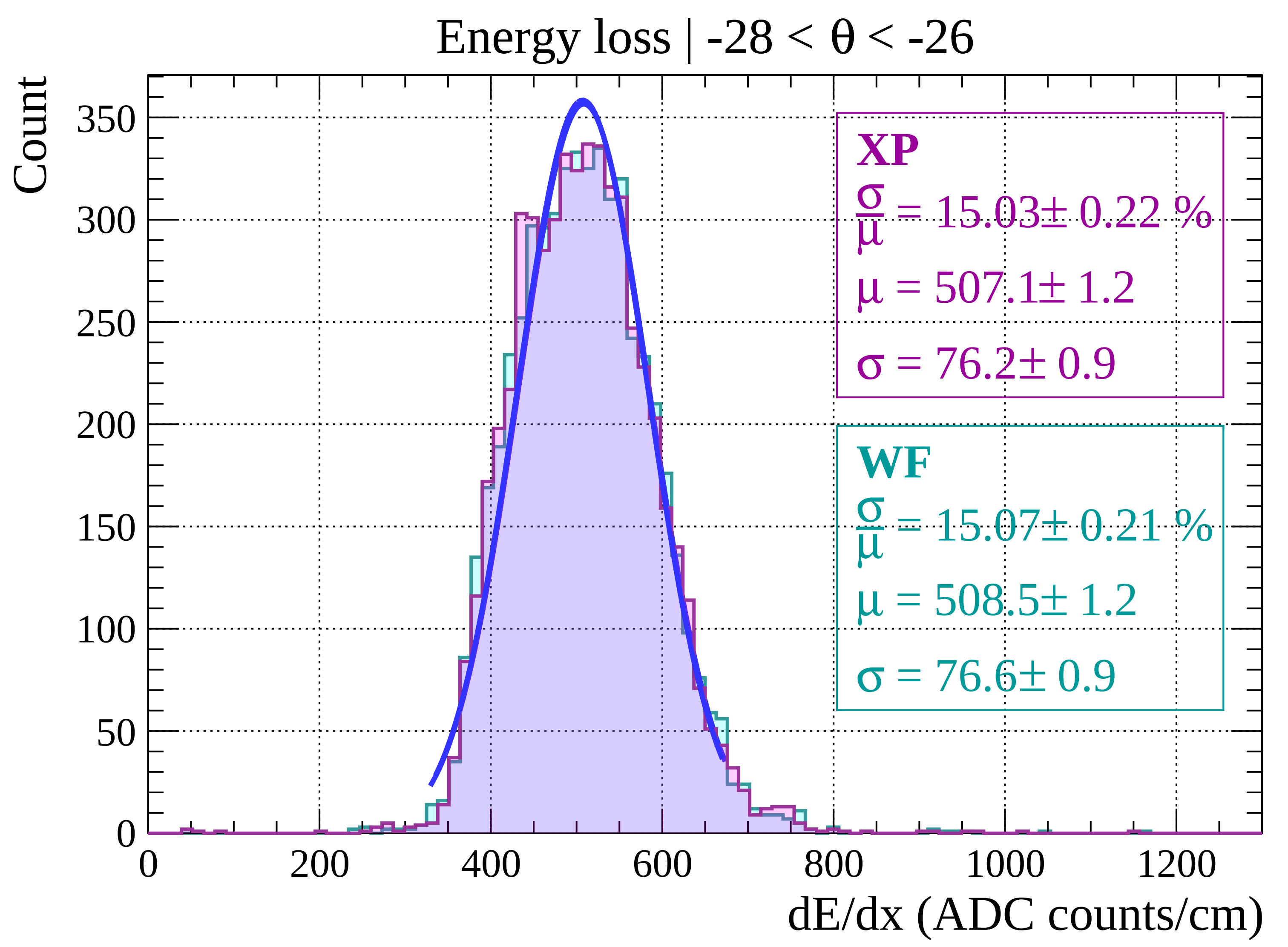


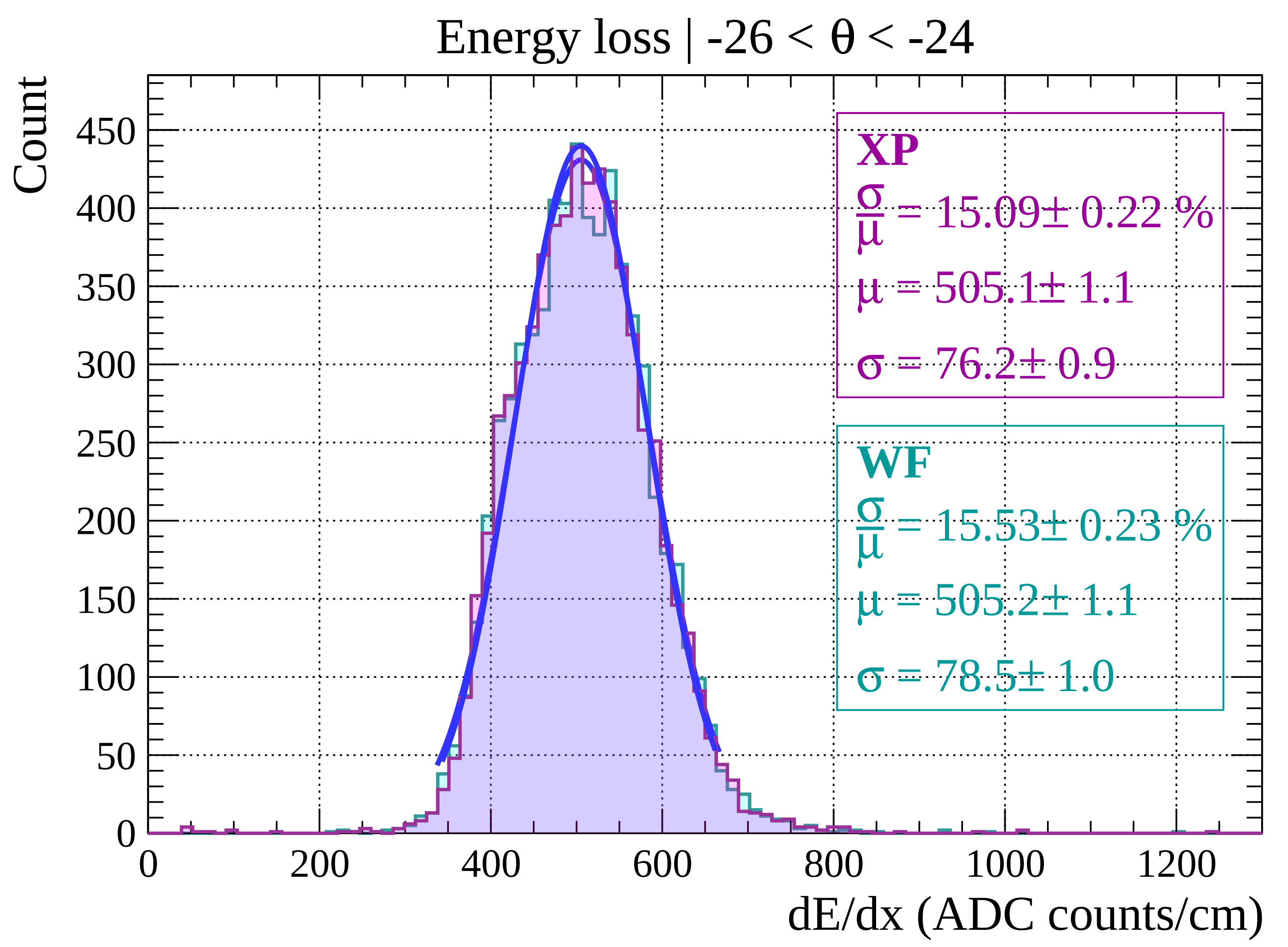
Energy loss | -34 < θ < -32
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 15.57± 0.31 %
μ = 519.7± 1.6
σ = 80.9± 1.3
WF
σ/μ = 15.76± 0.32 %
μ = 518.9± 1.7
σ = 81.8± 1.4
0
20
40
60
80
100
120
140
160
180
200
220
0
200
400
600
800
1000
1200

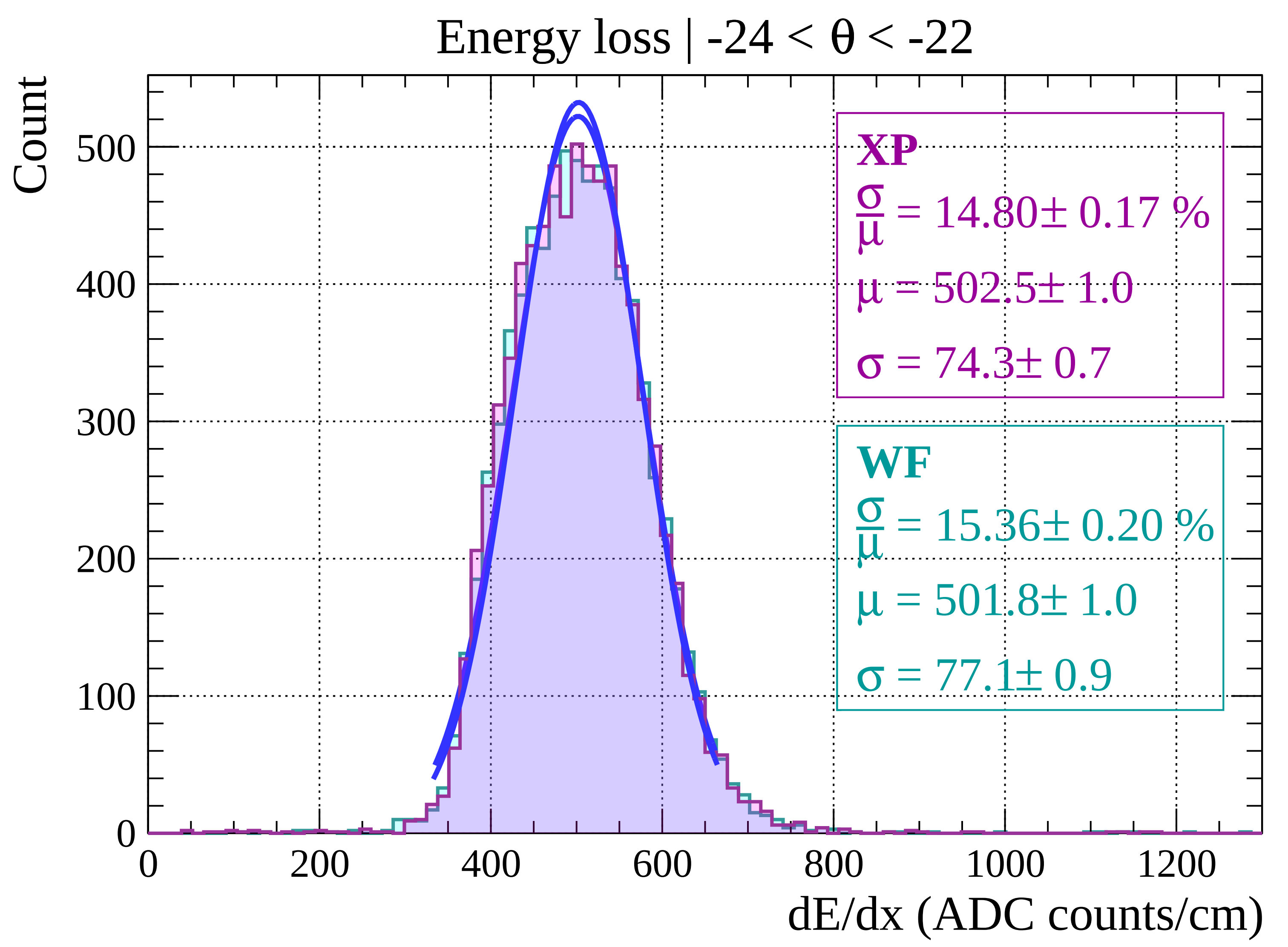


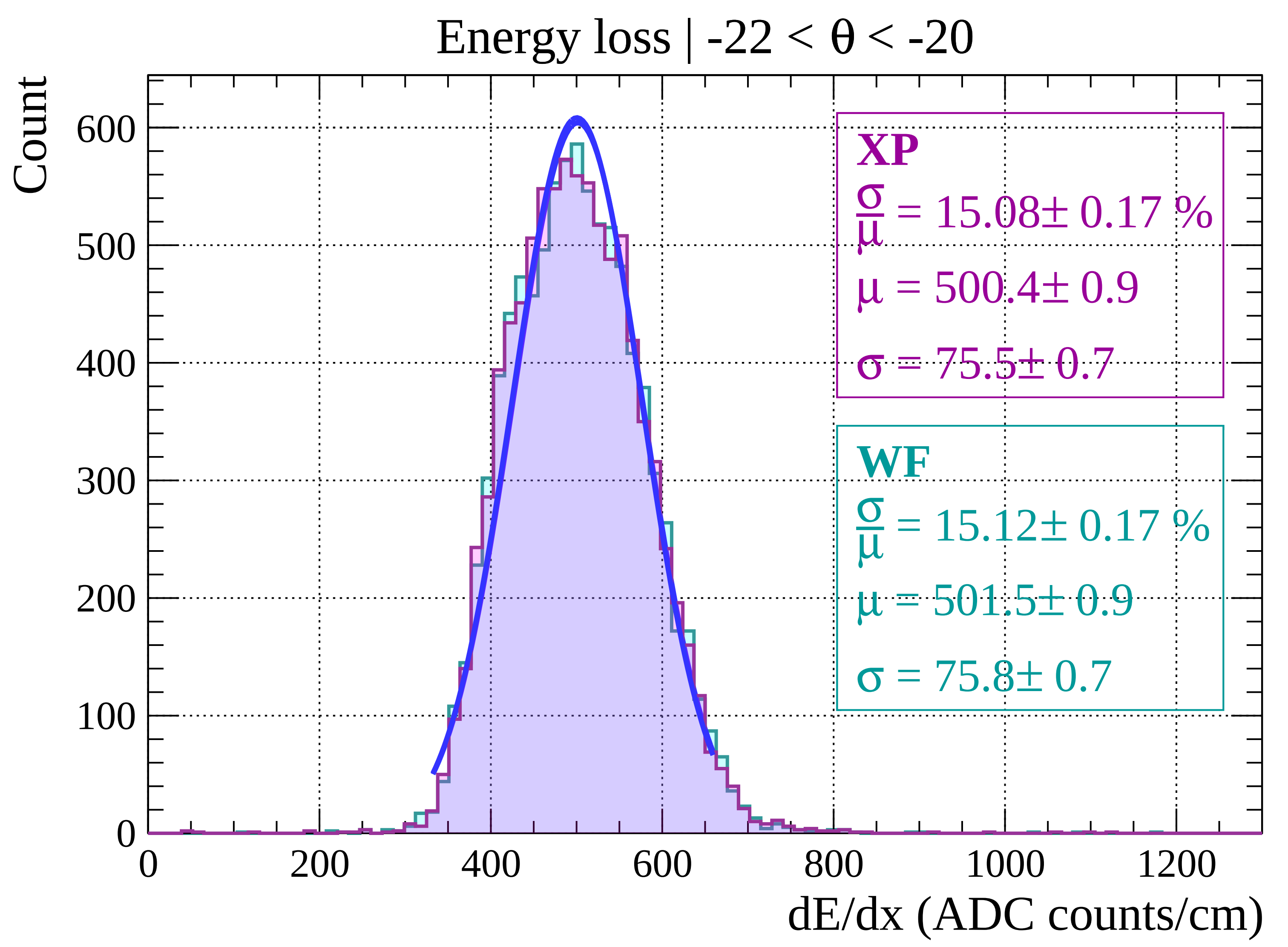
Energy loss | -32 < θ < -30
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 15.01± 0.28 %
μ = 516.3± 1.4
σ = 77.5± 1.2
WF
σ/μ = 15.46± 0.29 %
μ = 517.2± 1.5
σ = 80.0± 1.3

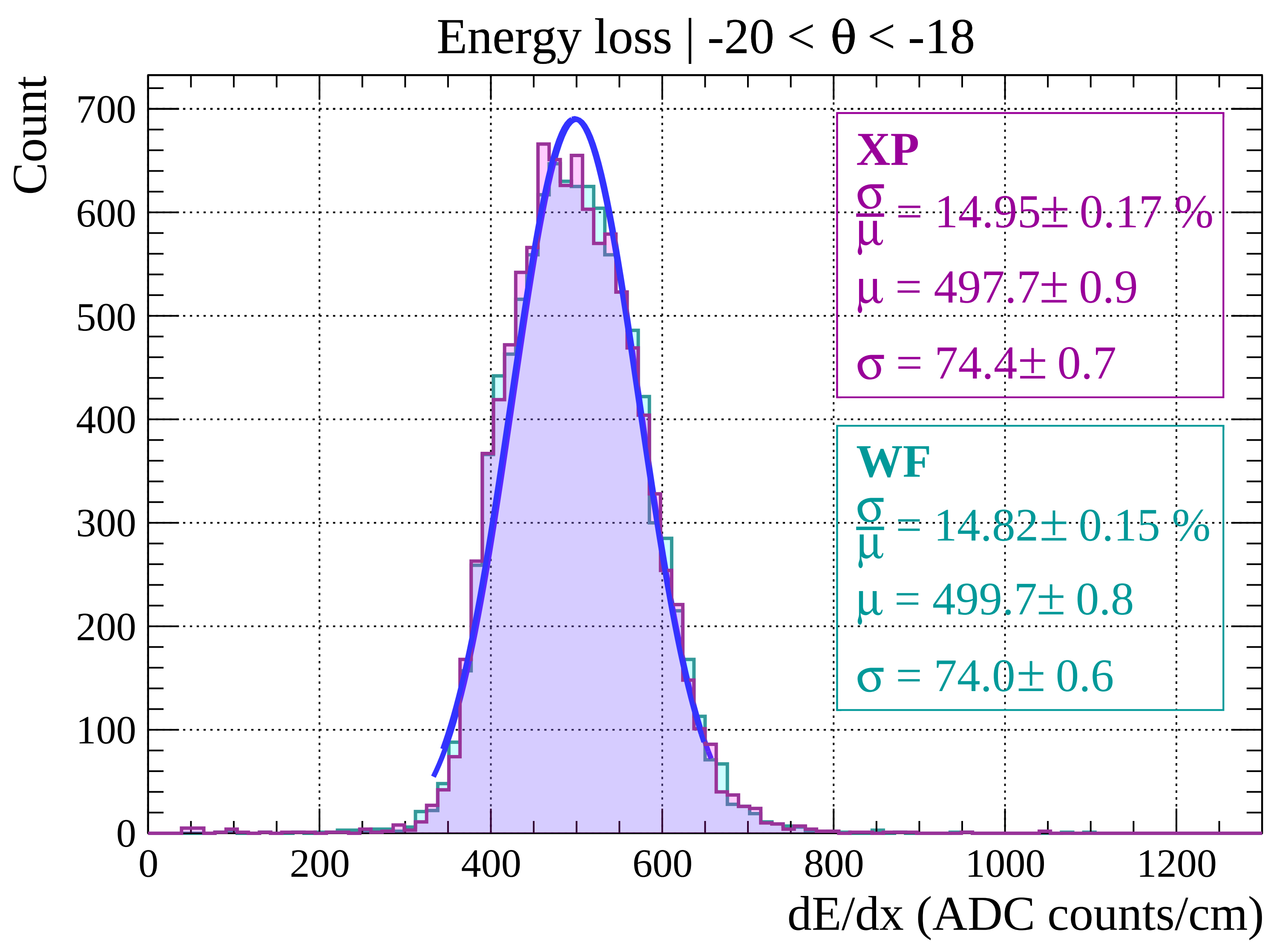

Energy loss | -30 < θ < -28
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 15.37± 0.22 %
μ = 513.4± 1.3
σ = 78.9± 0.9
WF
σ/μ = 15.76± 0.24 %
μ = 513.3± 1.3
σ = 80.9± 1.0
0
50
100
150
200
250
300
0
200
400
600
800
1000
1200

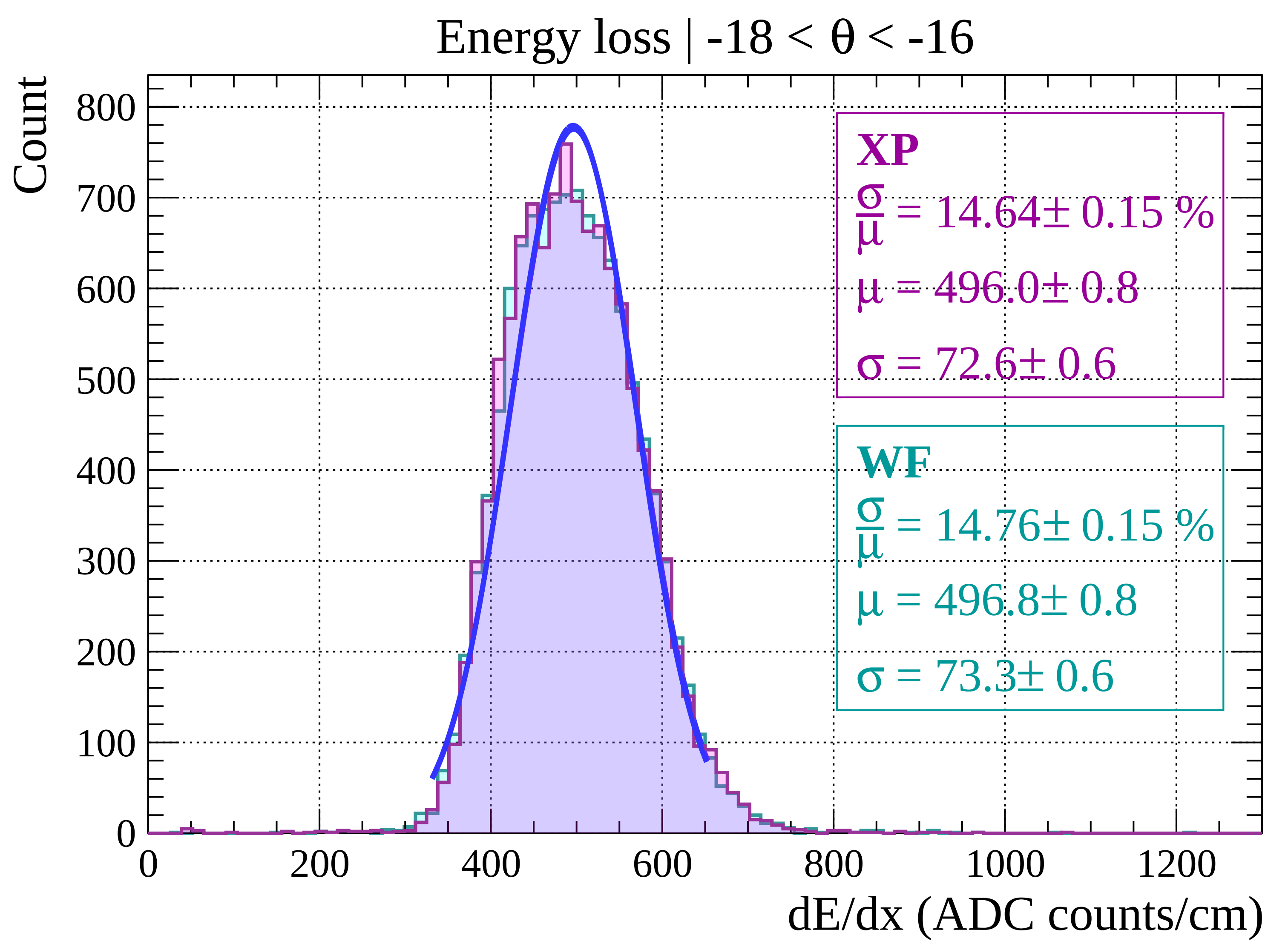

Energy loss | -28 < θ < -26
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 15.03 ± 0.22 %
μ = 507.1 ± 1.2
σ = 76.2 ± 0.9
WF
σ/μ = 15.07 ± 0.21 %
μ = 508.5 ± 1.2
σ = 76.6 ± 0.9
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200

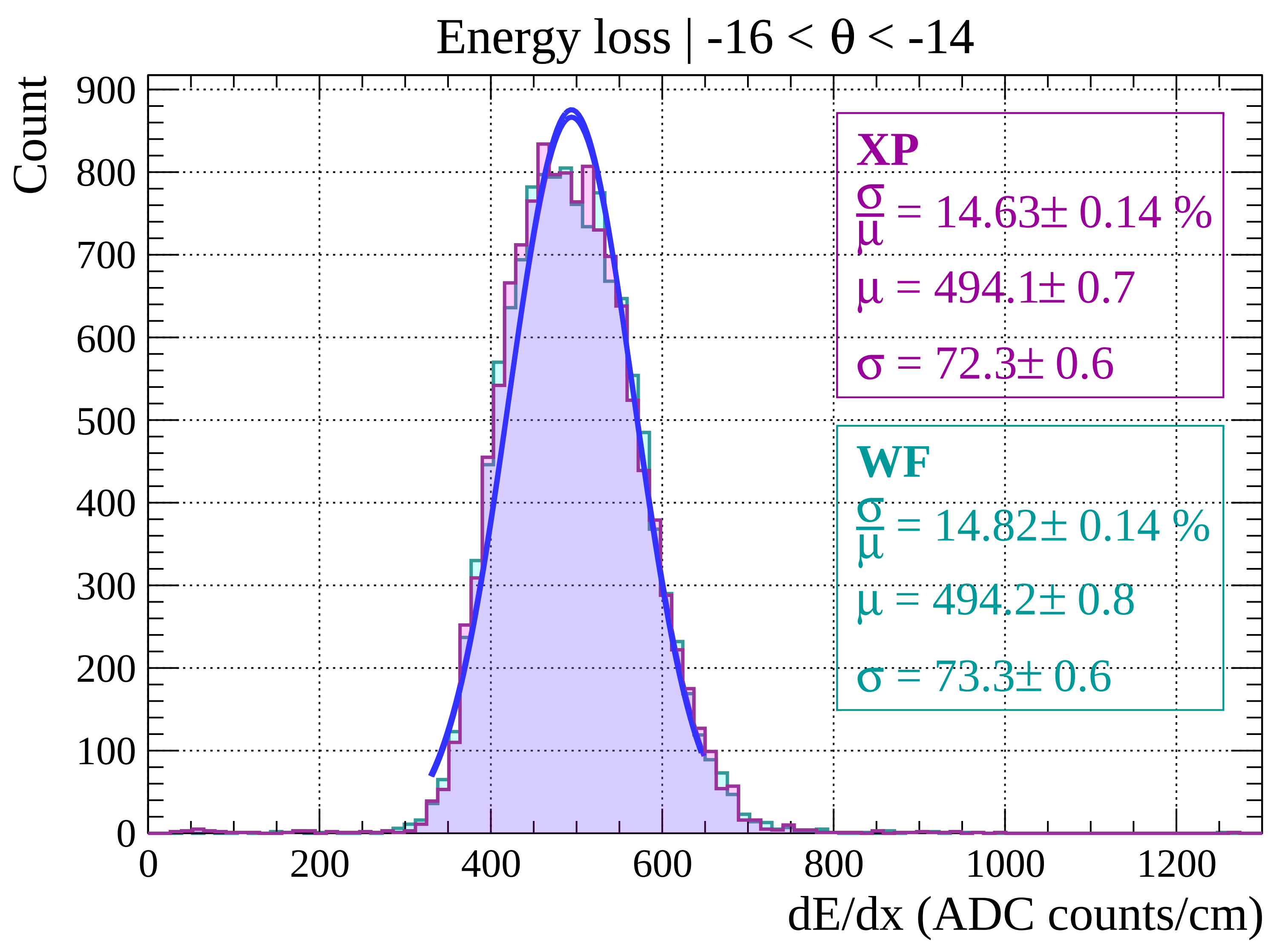

Energy loss | -26 < θ < -24
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 15.09± 0.22 %
μ = 505.1± 1.1
σ = 76.2± 0.9
WF
σ/μ = 15.53± 0.23 %
μ = 505.2± 1.1
σ = 78.5± 1.0
0
50
100
150
200
250
300
350
400
450
0
200
400
600
800
1000
1200


Energy loss | -24 < θ < -22
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
0
200
400
600
800
1000
1200
XP
σ/μ = 14.80± 0.17 %
μ = 502.5± 1.0
σ = 74.3± 0.7
WF
σ/μ = 15.36± 0.20 %
μ = 501.8± 1.0
σ = 77.1± 0.9

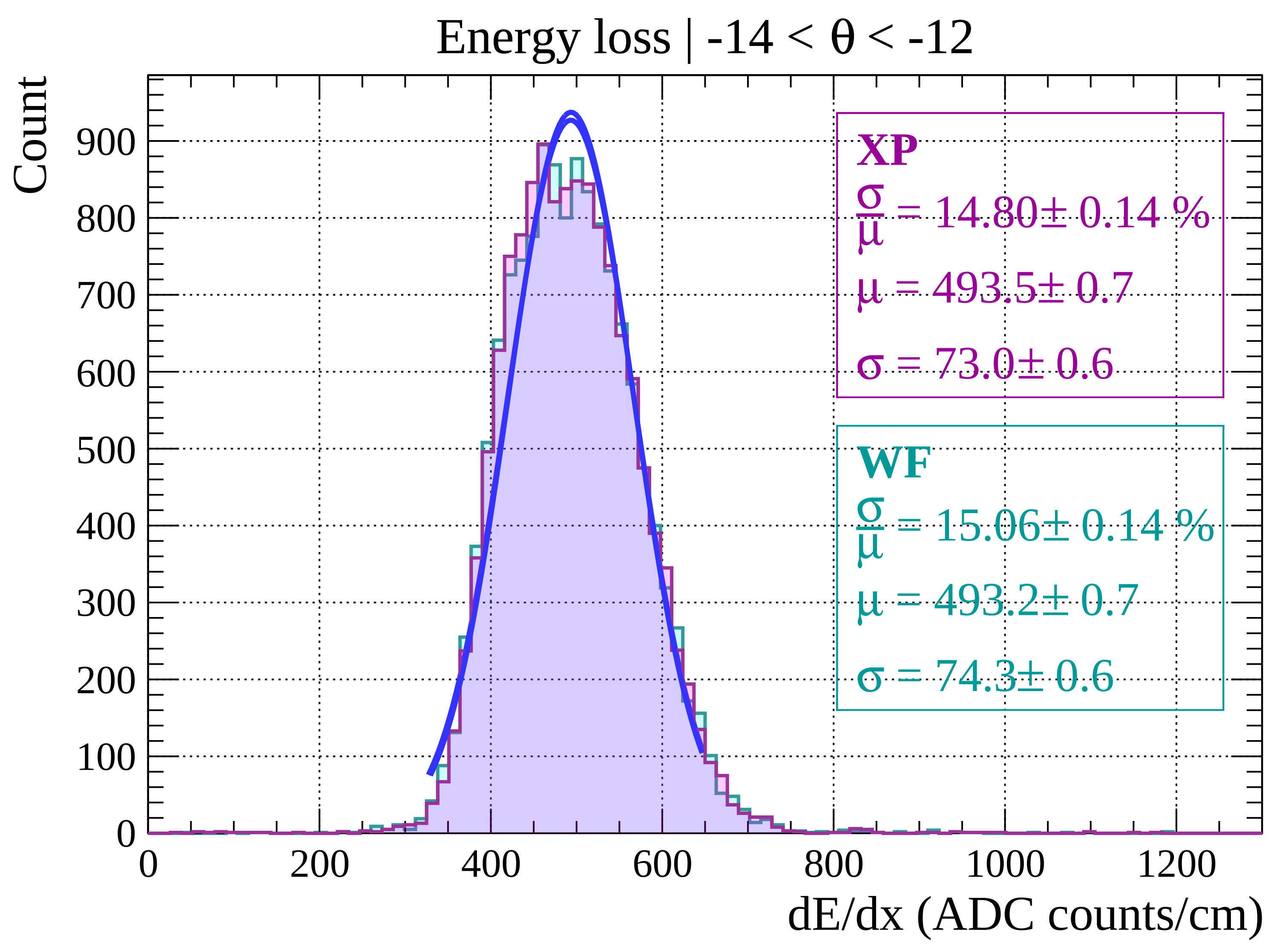

Energy loss | -22 < θ < -20
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 15.08± 0.17 %
μ = 500.4± 0.9
σ = 75.5± 0.7
WF
σ/μ = 15.12± 0.17 %
μ = 501.5± 0.9
σ = 75.8± 0.7
0
100
200
300
400
500
600
0
200
400
600
800
1000
1200

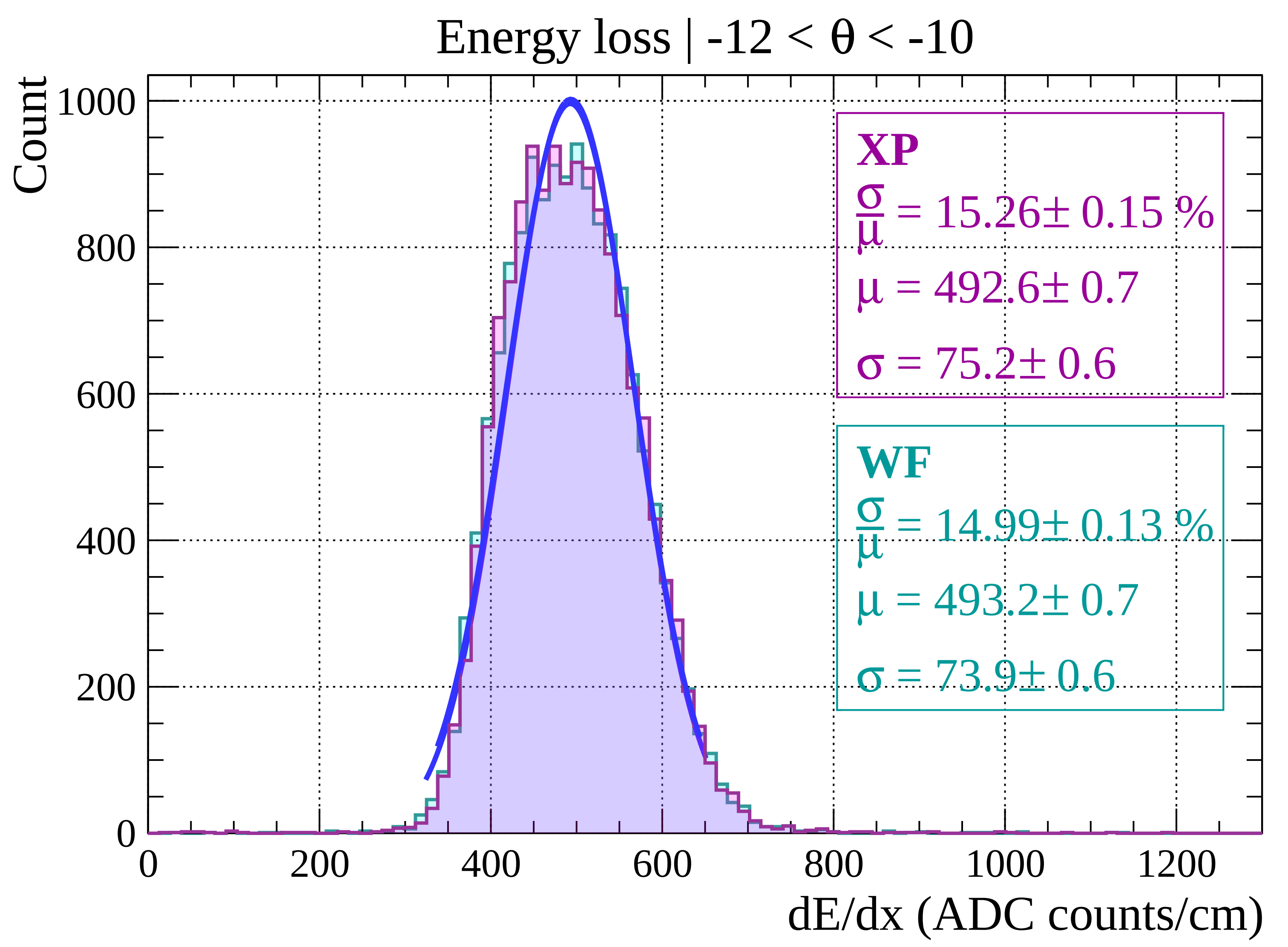

Energy loss | -20 < θ < -18
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 14.95± 0.17 %
μ = 497.7± 0.9
σ = 74.4± 0.7
WF
$\frac{\sigma}{\mu}$ = 14.82± 0.15 %
μ = 499.7± 0.8
σ = 74.0± 0.6

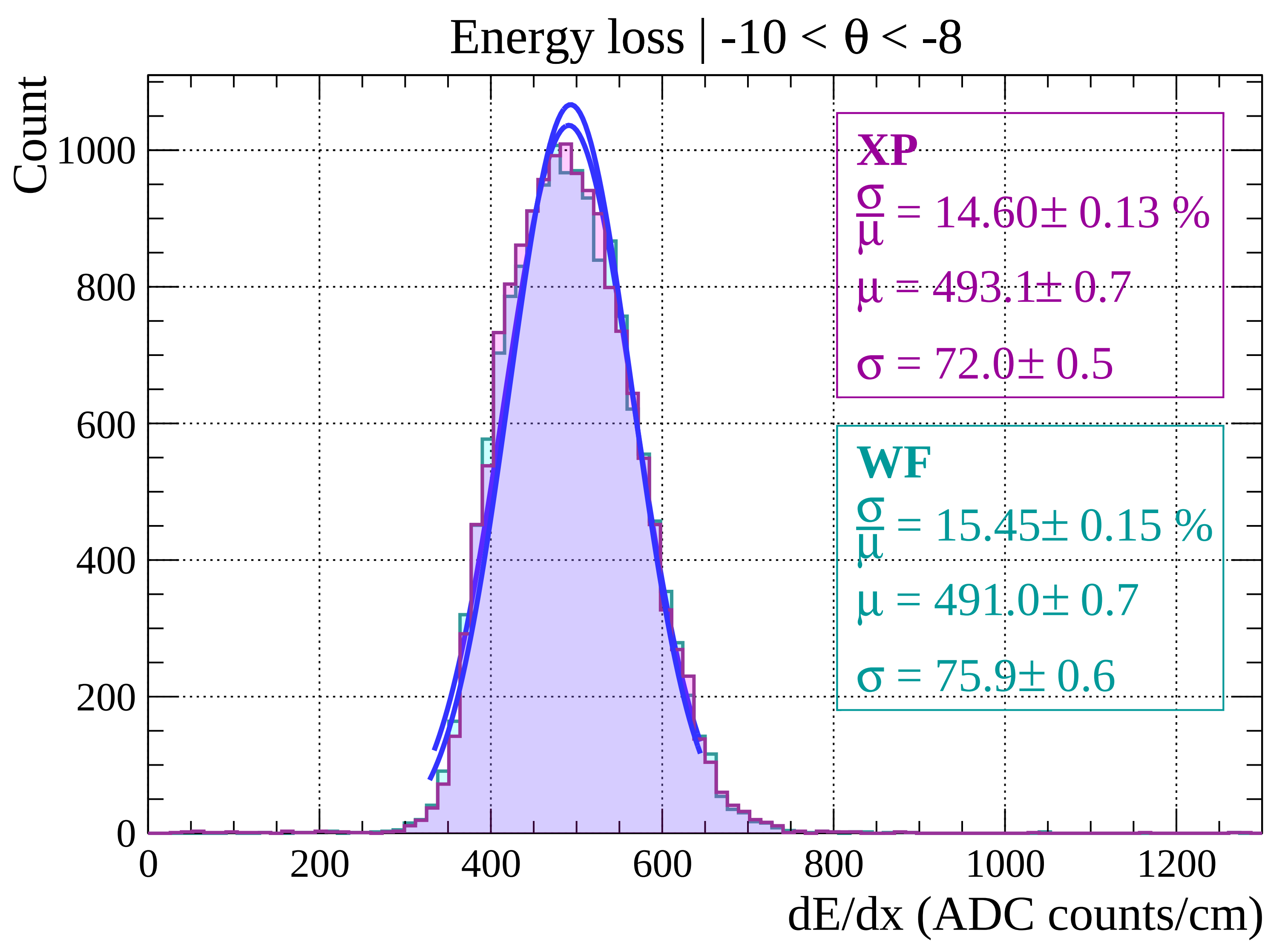

Energy loss | -18 < θ < -16
Count
XP
$\frac{\sigma}{\mu}$ = 14.64± 0.15 %
μ = 496.0± 0.8
σ = 72.6± 0.6
WF
$\frac{\sigma}{\mu}$ = 14.76± 0.15 %
μ = 496.8± 0.8
σ = 73.3± 0.6
dE/dx (ADC counts/cm)



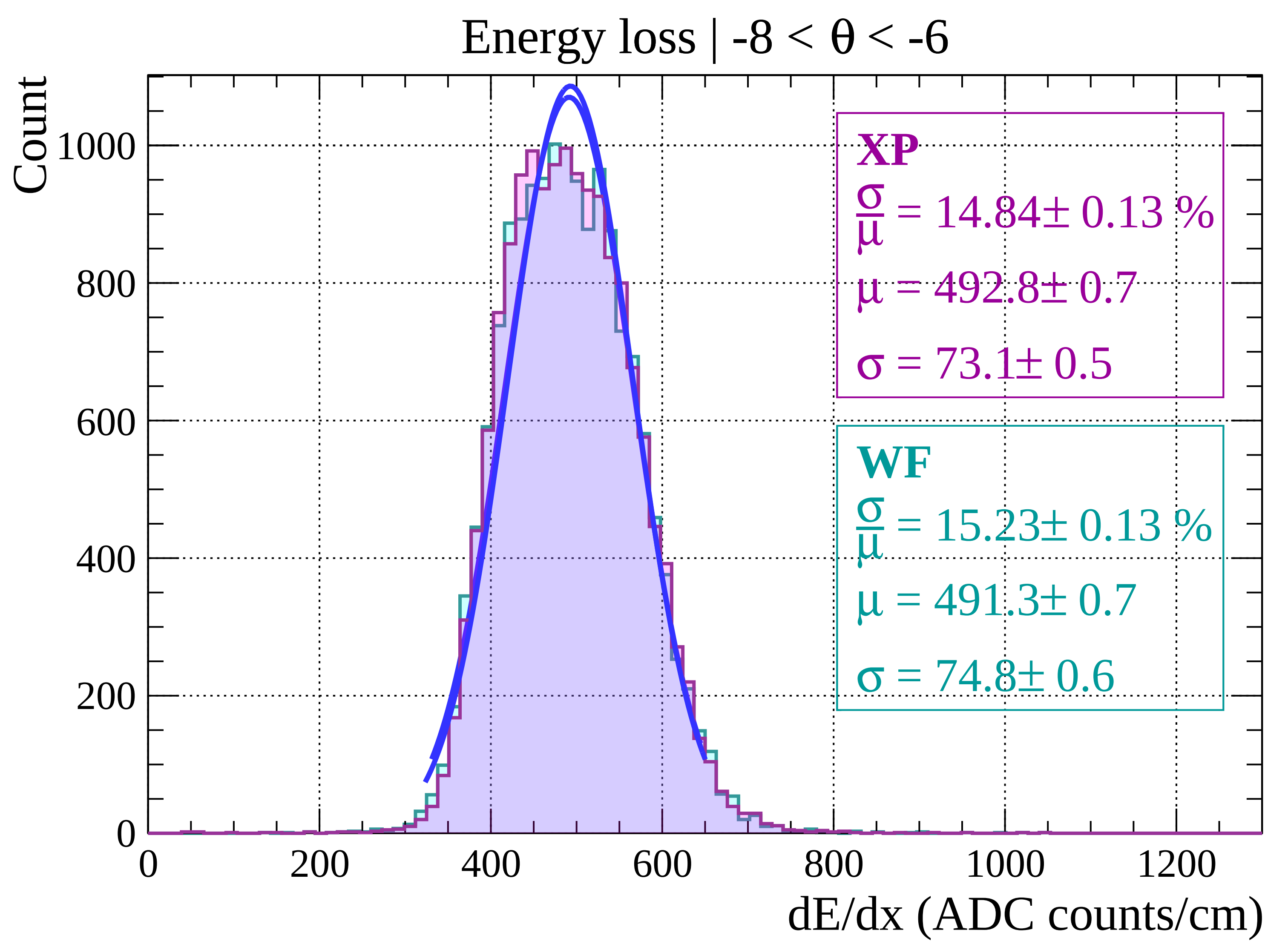
Energy loss | -16 < θ < -14
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
100
300
500
700
900
XP
σ/μ = 14.63± 0.14 %
μ = 494.1± 0.7
σ = 72.3± 0.6
WF
σ/μ = 14.82± 0.14 %
μ = 494.2± 0.8
σ = 73.3± 0.6


Energy loss | -14 < θ < -12
Count
900
800
700
600
500
400
300
200
100
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 14.80± 0.14 %
μ = 493.5± 0.7
σ = 73.0± 0.6
WF
σ/μ = 15.06± 0.14 %
μ = 493.2± 0.7
σ = 74.3± 0.6

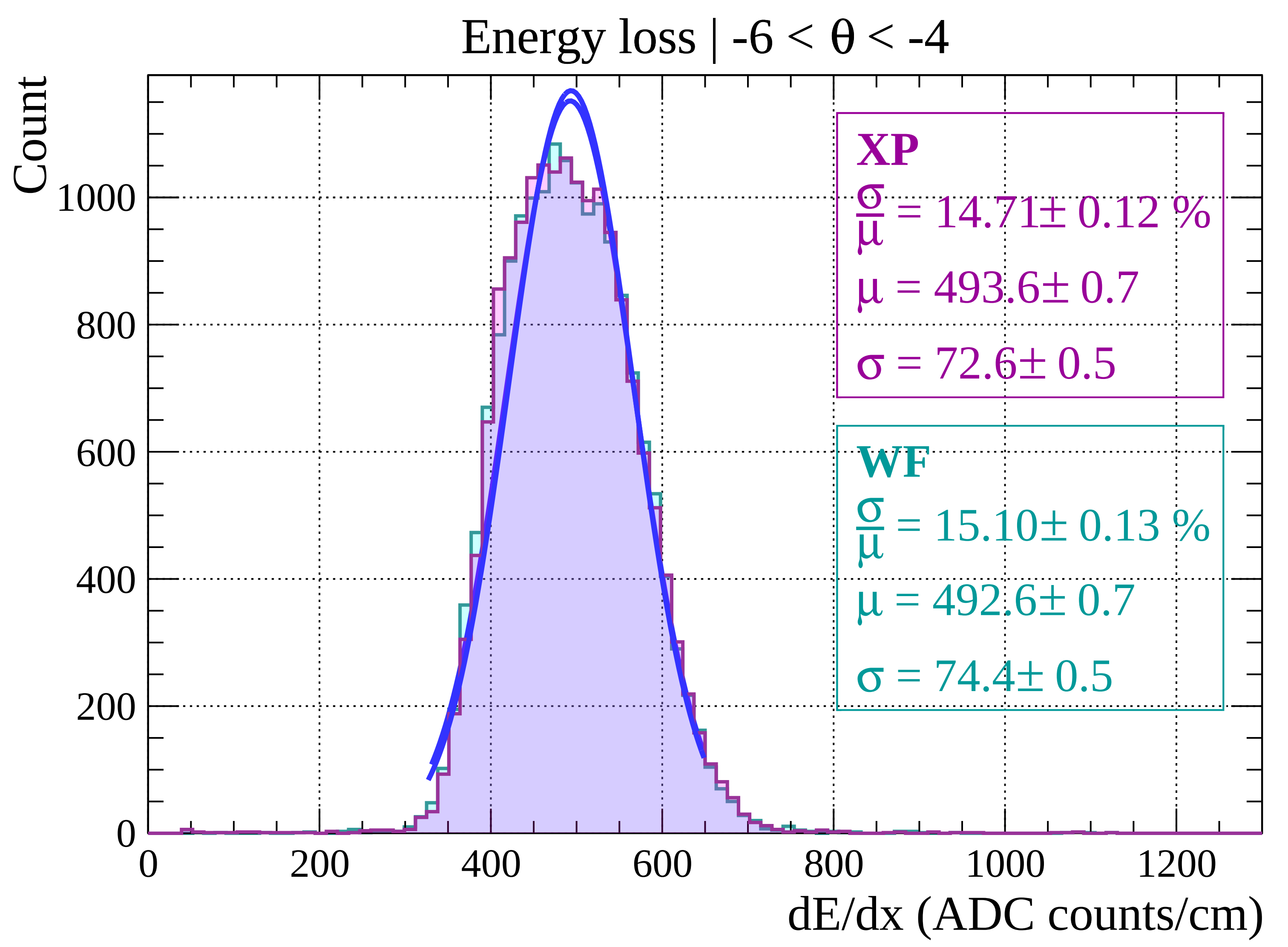

Energy loss | -12 < θ < -10
Count
0
200
400
600
800
1000
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 15.26± 0.15 %
μ = 492.6± 0.7
σ = 75.2± 0.6
WF
σ/μ = 14.99± 0.13 %
μ = 493.2± 0.7
σ = 73.9± 0.6



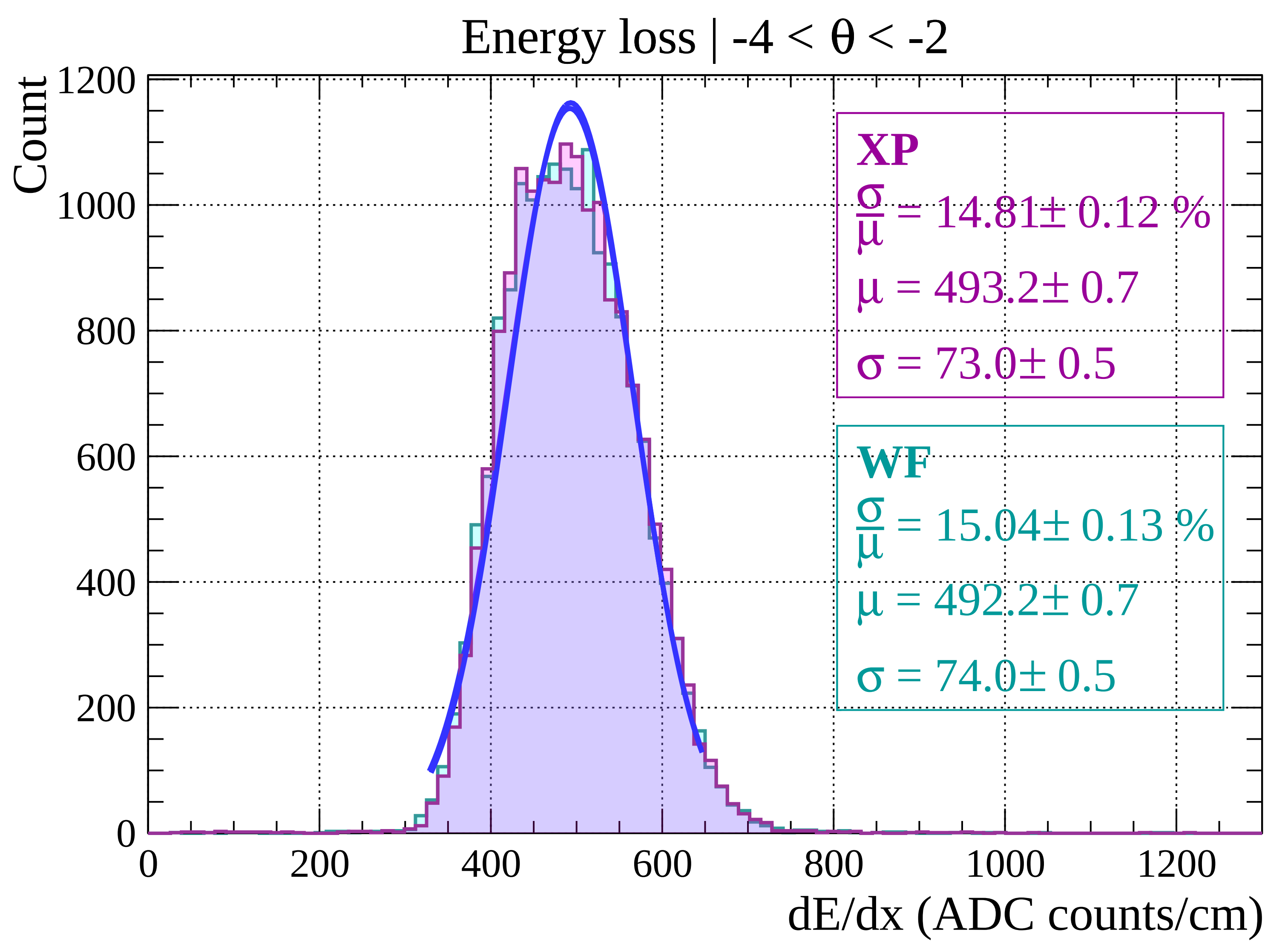
Energy loss | -10 < θ < -8
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 14.60± 0.13 %
μ = 493.1± 0.7
σ = 72.0± 0.5
WF
σ/μ = 15.45± 0.15 %
μ = 491.0± 0.7
σ = 75.9± 0.6
0
200
400
600
800
1000
1200



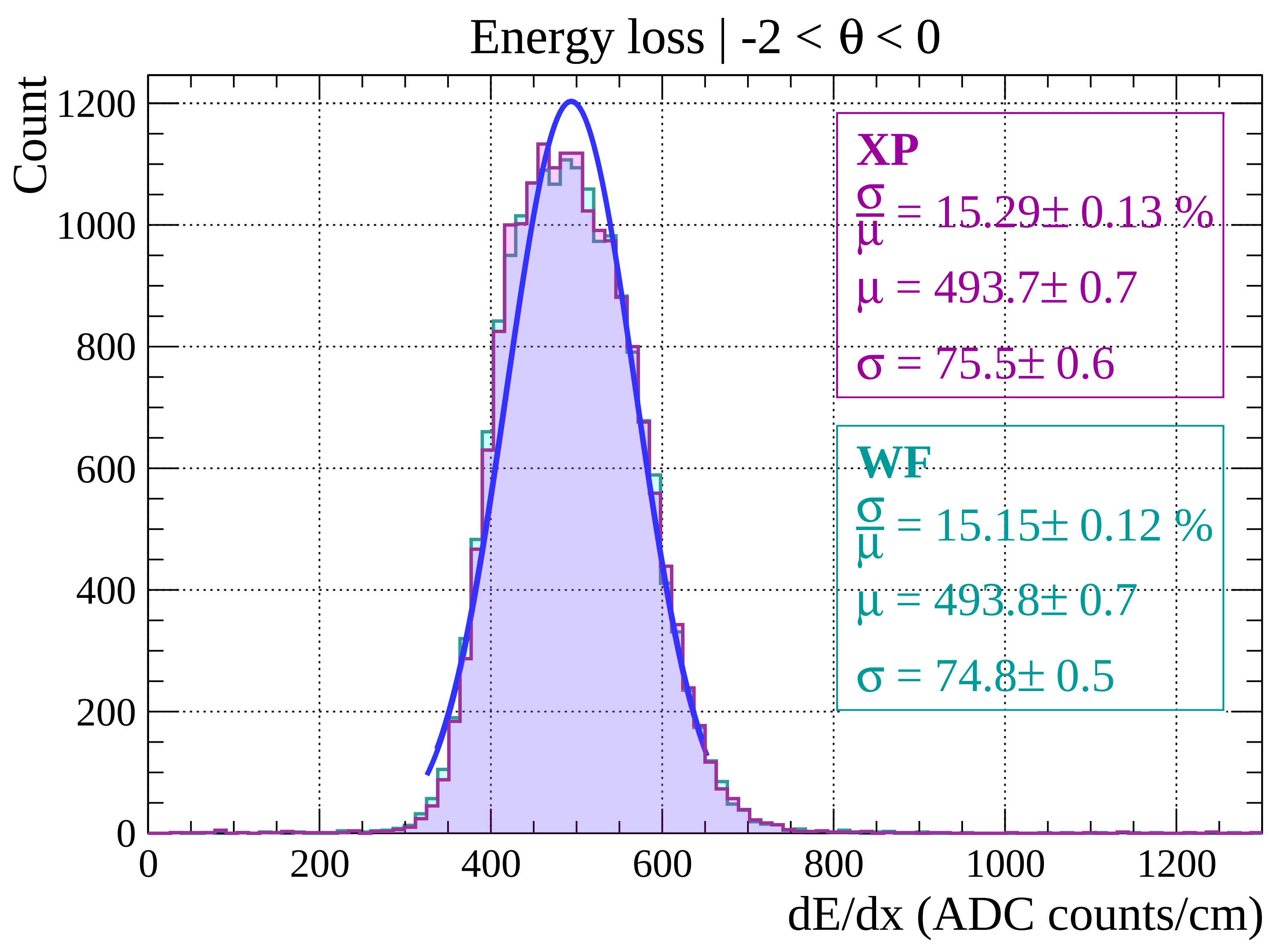
Energy loss | -8 < θ < -6
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
XP
σ/μ = 14.84± 0.13 %
μ = 492.8± 0.7
σ = 73.1± 0.5
WF
σ/μ = 15.23± 0.13 %
μ = 491.3± 0.7
σ = 74.8± 0.6



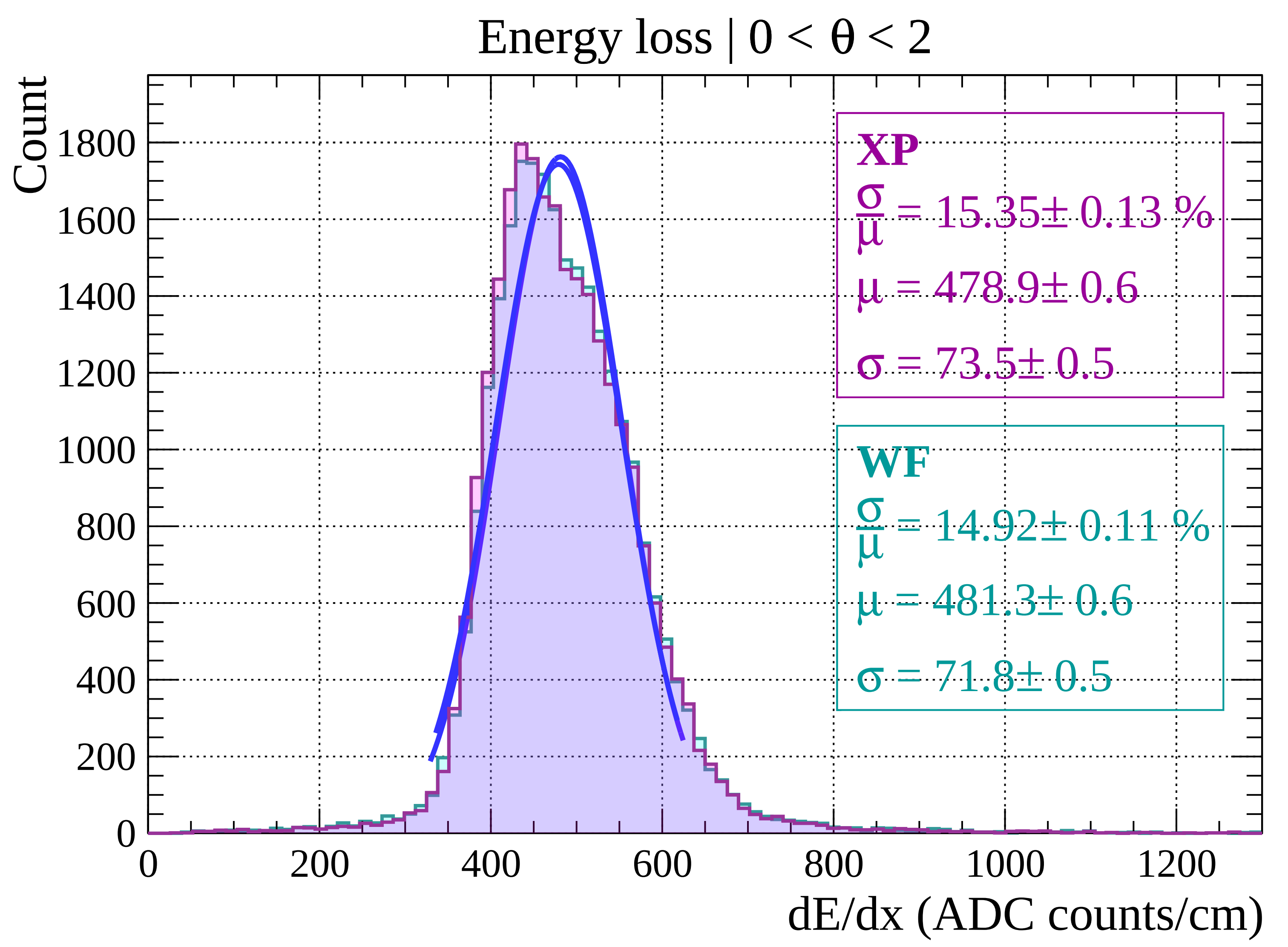
Energy loss | -6 < θ < -4
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
XP
σ/μ = 14.71± 0.12 %
μ = 493.6± 0.7
σ = 72.6± 0.5
WF
σ/μ = 15.10± 0.13 %
μ = 492.6± 0.7
σ = 74.4± 0.5



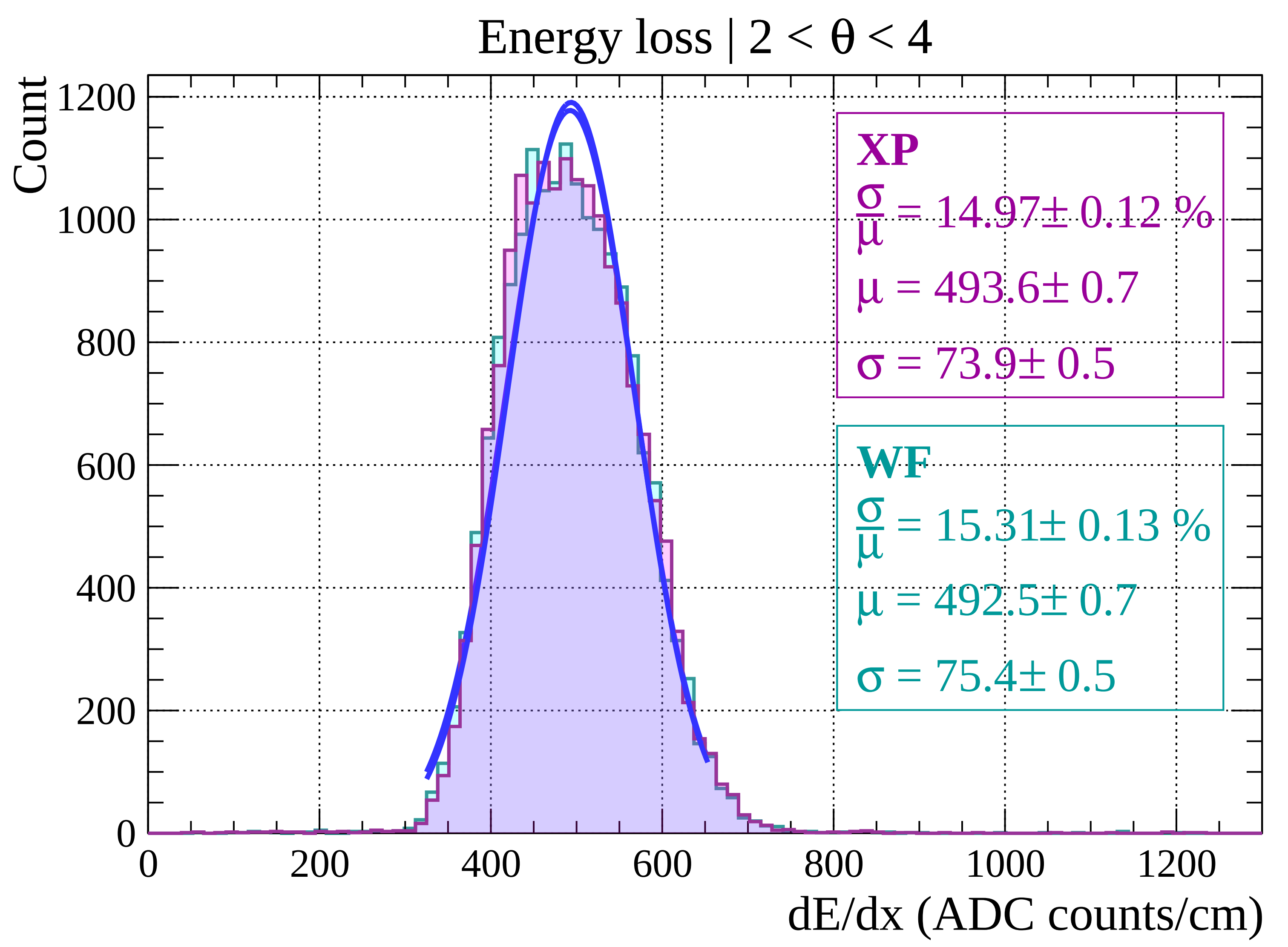
Energy loss | -4 < θ < -2
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
XP
σ/μ = 14.81± 0.12 %
μ = 493.2± 0.7
σ = 73.0± 0.5
WF
σ/μ = 15.04± 0.13 %
μ = 492.2± 0.7
σ = 74.0± 0.5

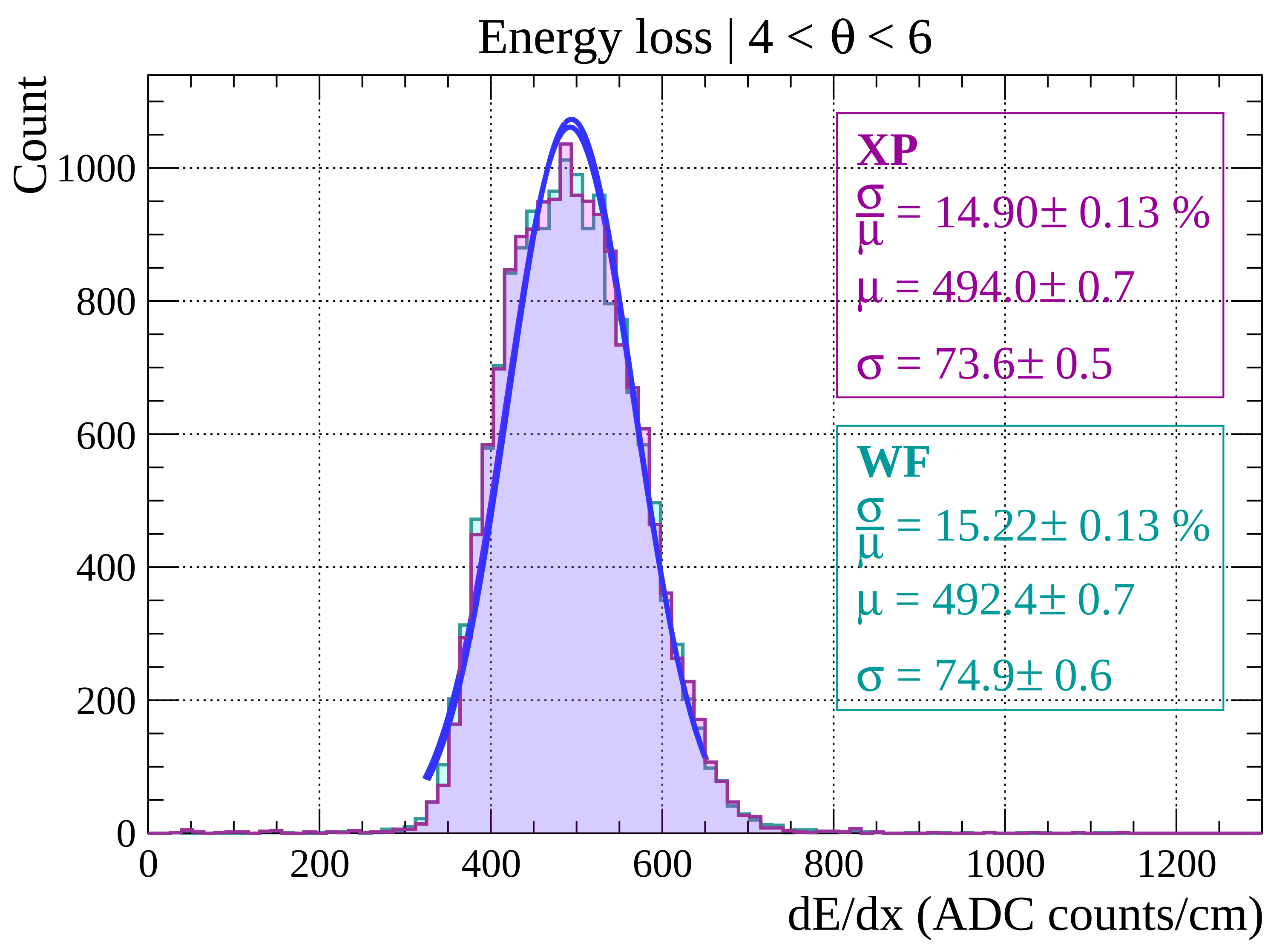


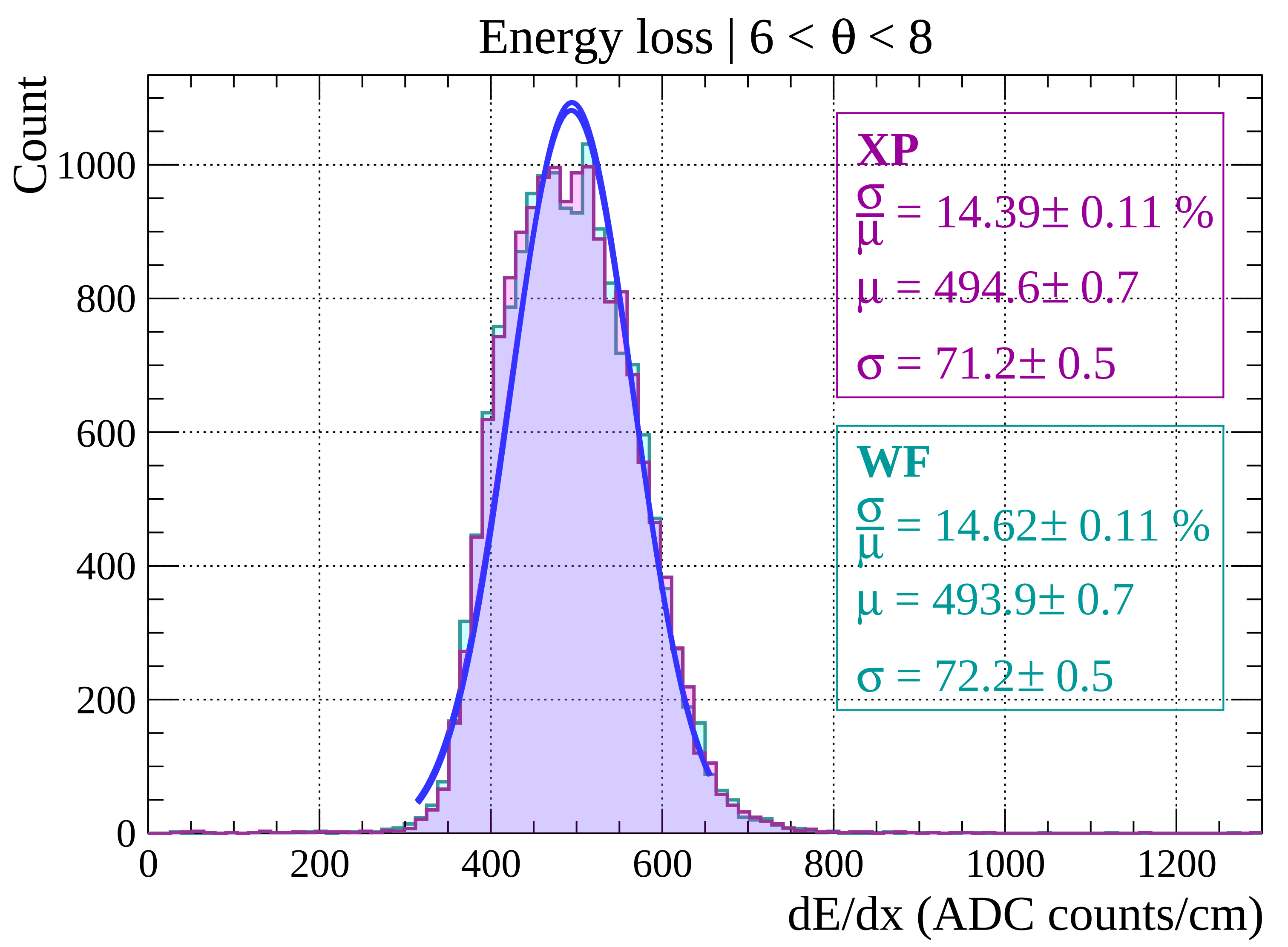
Energy loss | -2 < θ < 0
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
XP
σ/μ = 15.29± 0.13 %
μ = 493.7± 0.7
σ = 75.5± 0.6
WF
σ/μ = 15.15± 0.12 %
μ = 493.8± 0.7
σ = 74.8± 0.5



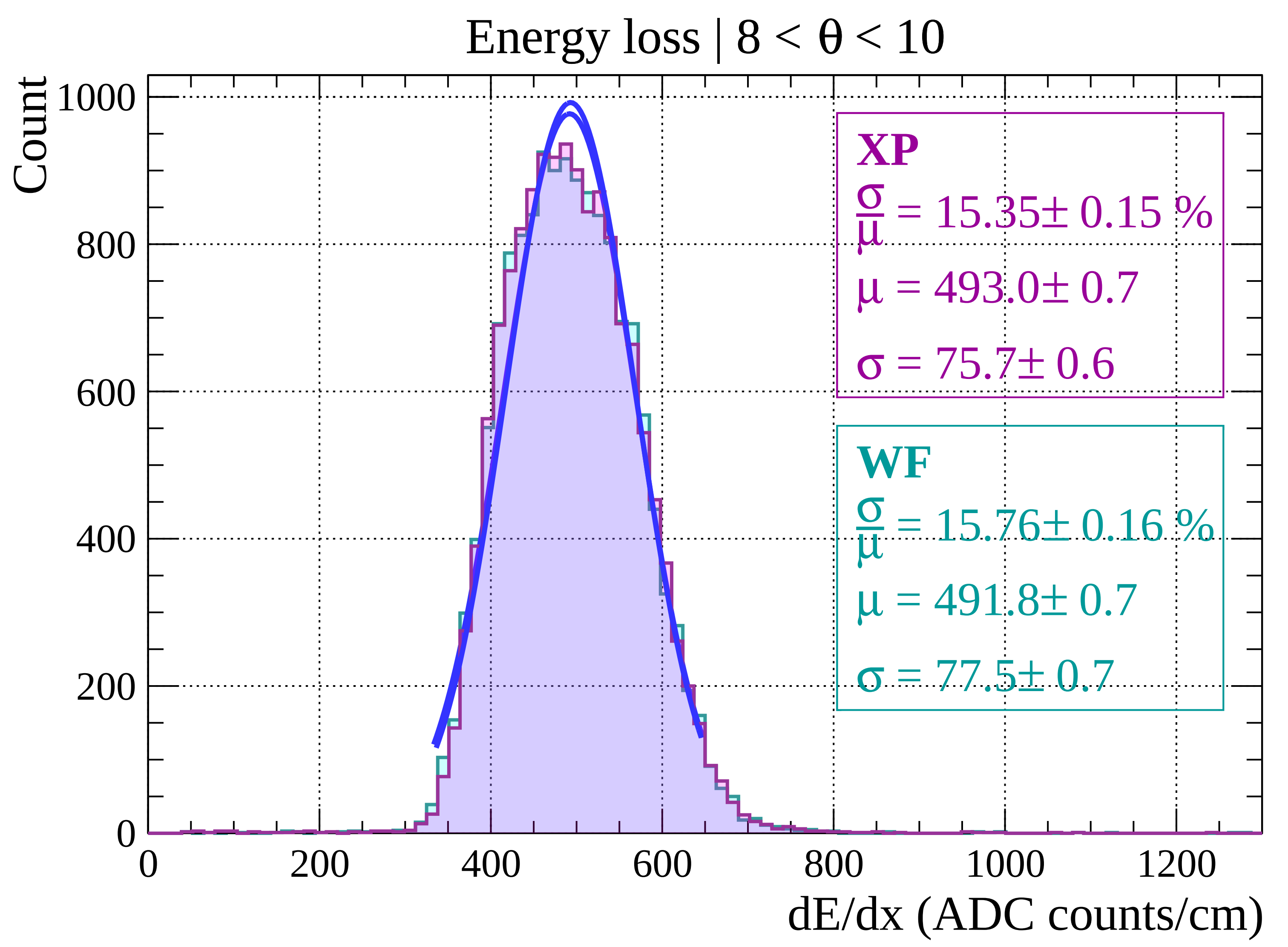
Energy loss | 0 < θ < 2
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
1400
1600
1800
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu} = 15.35 \pm 0.13$ %
$\mu = 478.9 \pm 0.6$
$\sigma = 73.5 \pm 0.5$
WF
$\frac{\sigma}{\mu} = 14.92 \pm 0.11$ %
$\mu = 481.3 \pm 0.6$
$\sigma = 71.8 \pm 0.5$


Energy loss | 2 < θ < 4
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 14.97± 0.12 %
μ = 493.6± 0.7
σ = 73.9± 0.5
WF
$\frac{\sigma}{\mu}$ = 15.31± 0.13 %
μ = 492.5± 0.7
σ = 75.4± 0.5

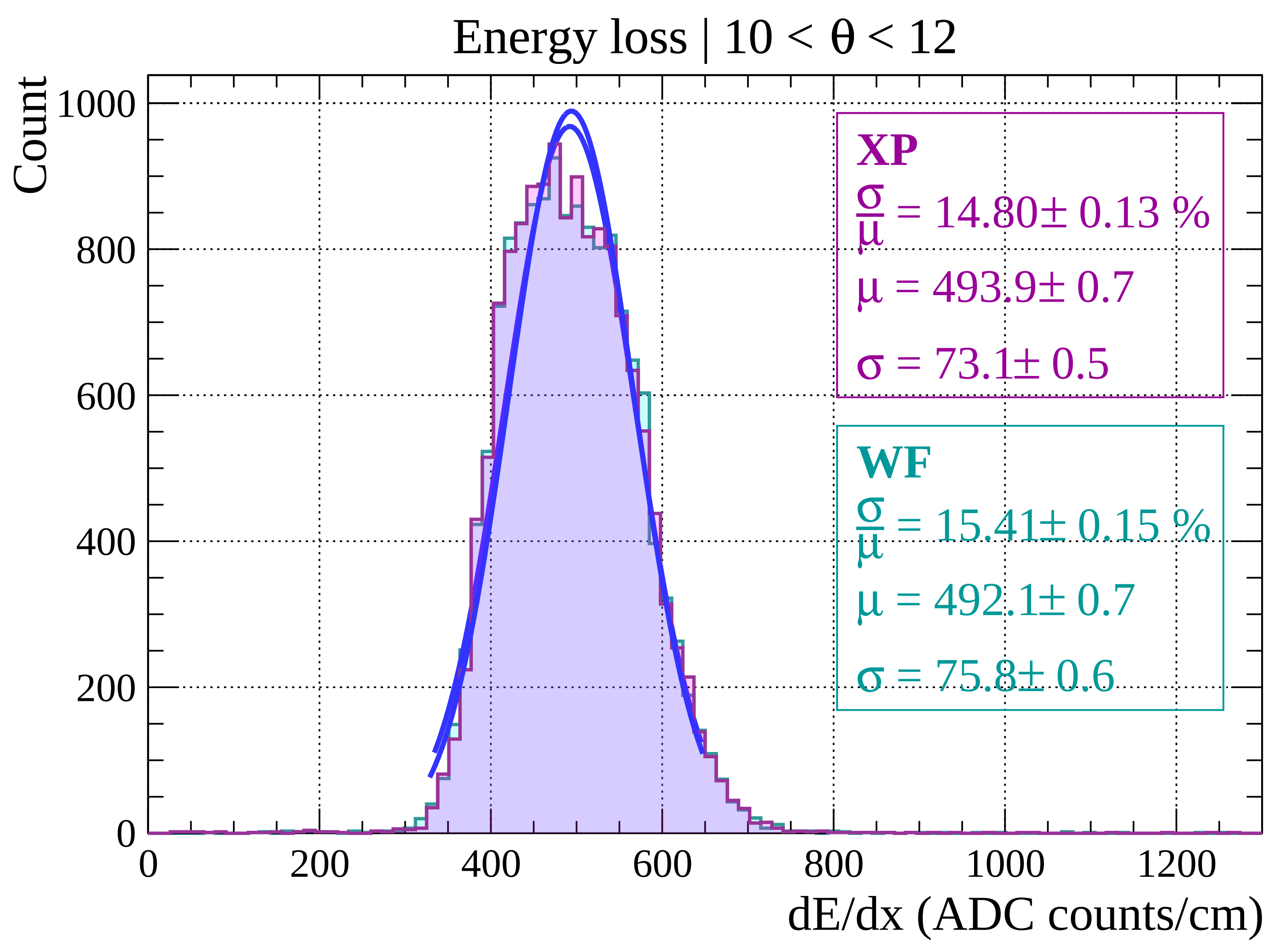

Energy loss | 4 < θ < 6
Count
0
200
400
600
800
1000
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 14.90± 0.13 %
μ = 494.0± 0.7
σ = 73.6± 0.5
WF
σ/μ = 15.22± 0.13 %
μ = 492.4± 0.7
σ = 74.9± 0.6



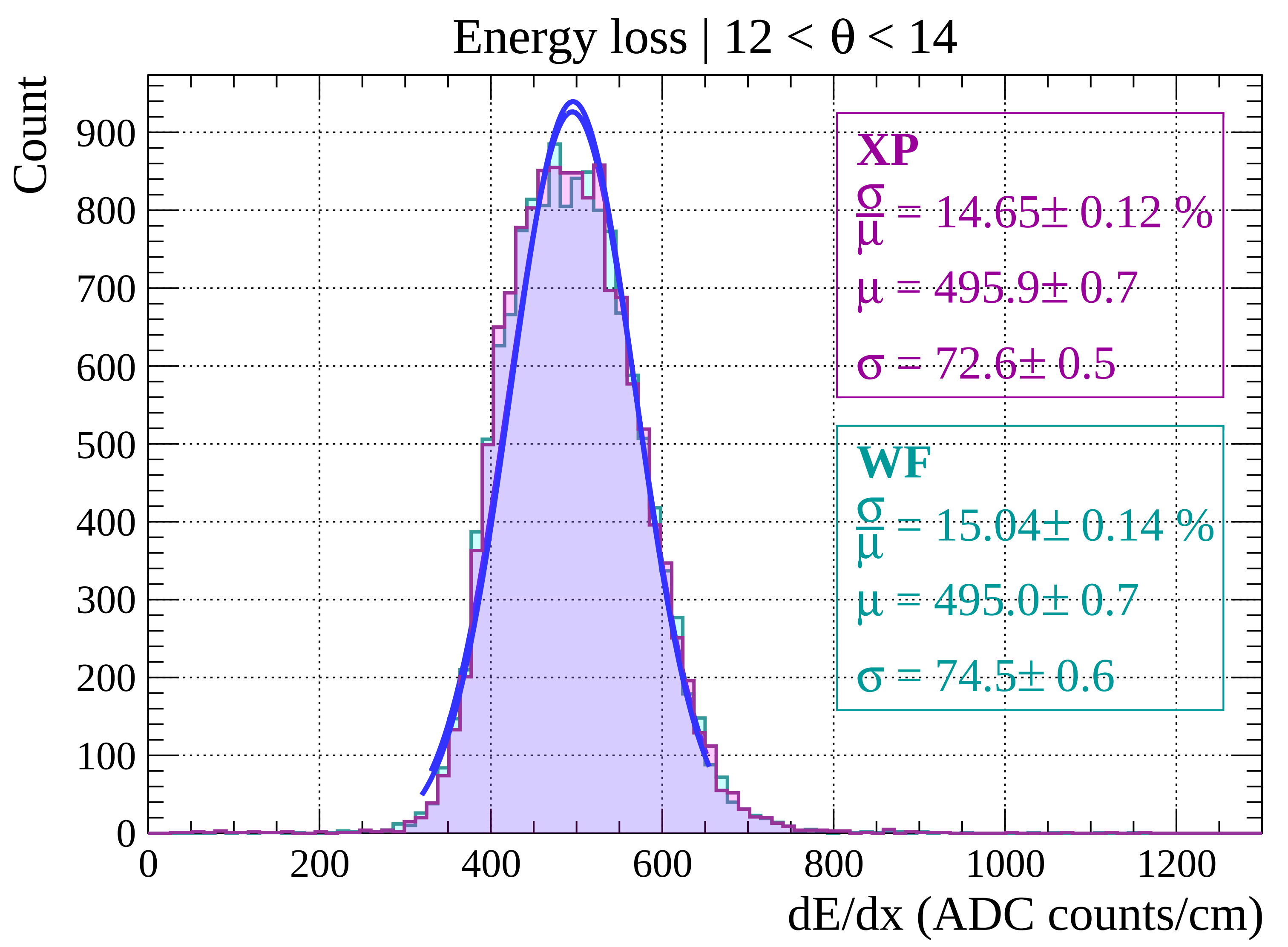
Energy loss | 6 < θ < 8
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
XP
σ/μ = 14.39 ± 0.11 %
μ = 494.6 ± 0.7
σ = 71.2 ± 0.5
WF
σ/μ = 14.62 ± 0.11 %
μ = 493.9 ± 0.7
σ = 72.2 ± 0.5

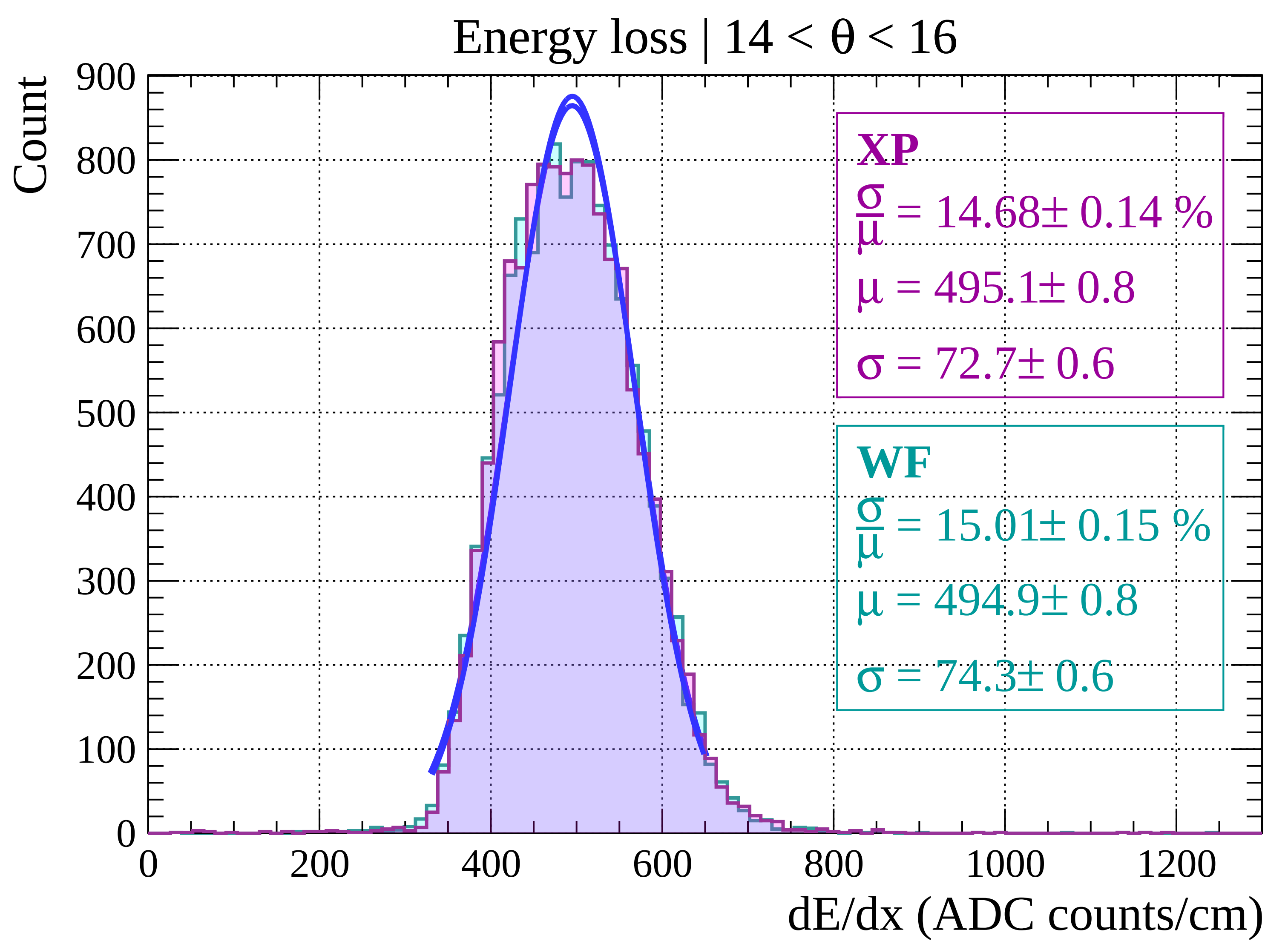


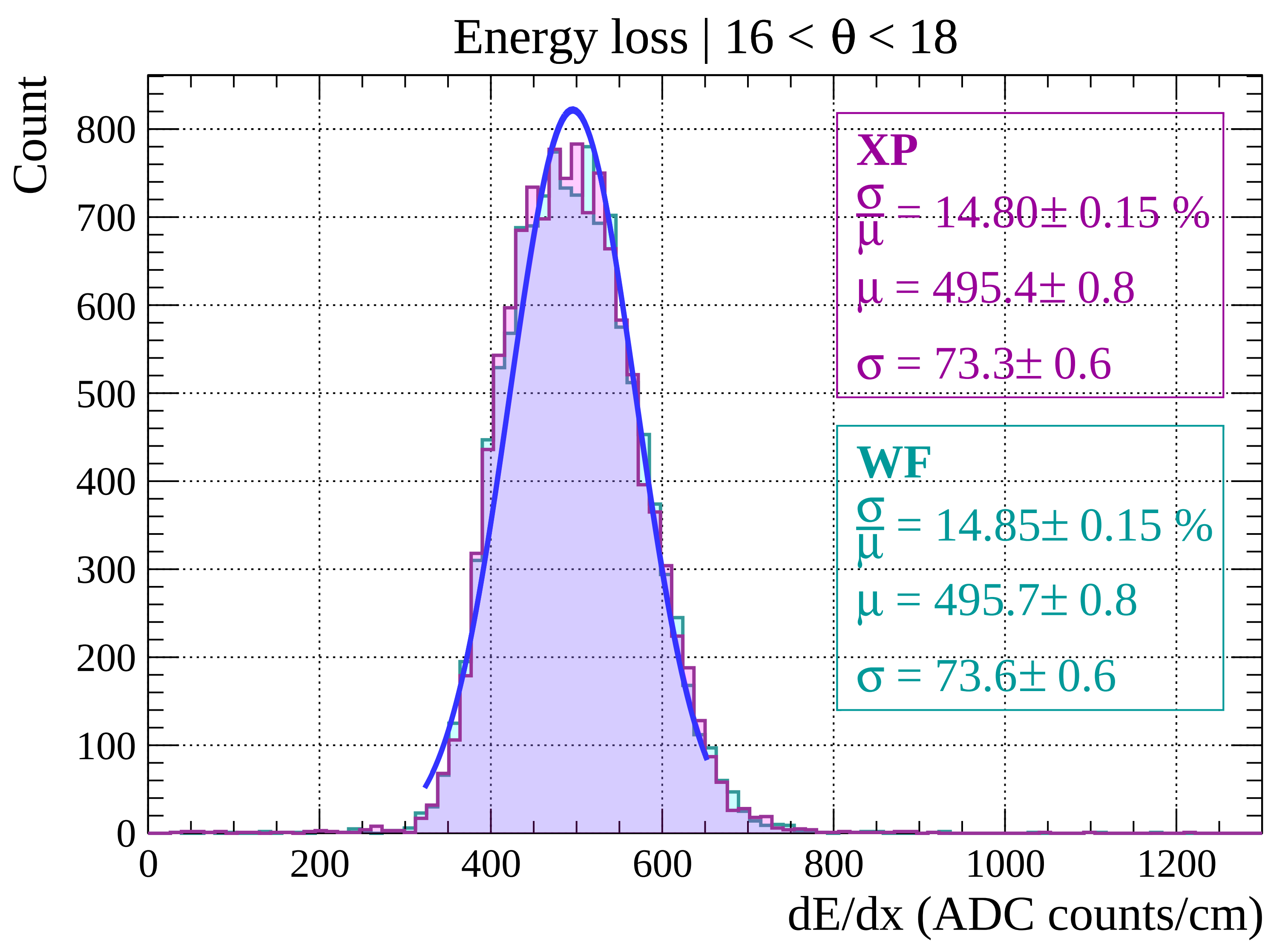
Energy loss | 8 < θ < 10
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
XP
σ/μ = 15.35± 0.15 %
μ = 493.0± 0.7
σ = 75.7± 0.6
WF
σ/μ = 15.76± 0.16 %
μ = 491.8± 0.7
σ = 77.5± 0.7



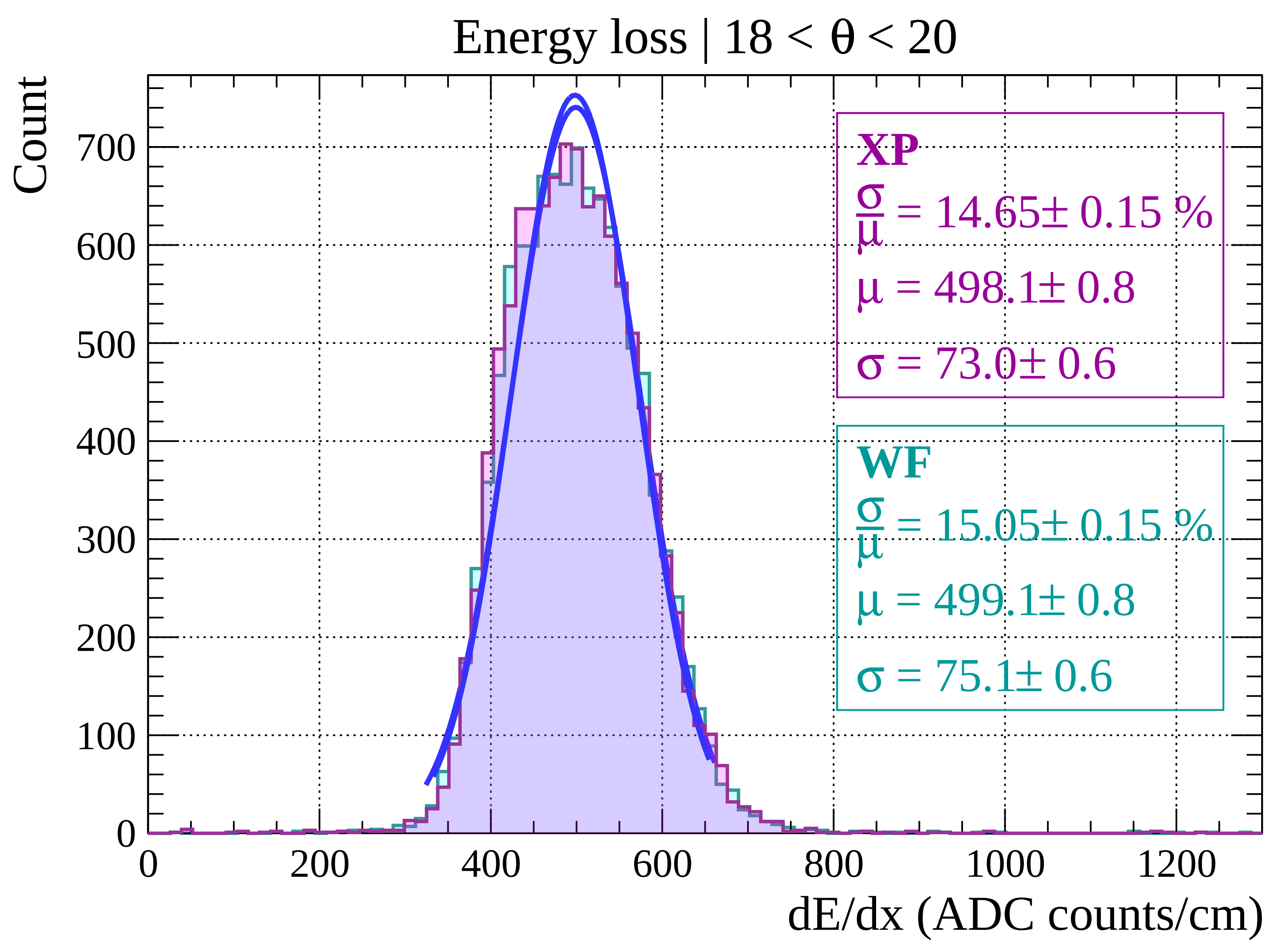
Energy loss | 10 < θ < 12
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
XP
σ/μ = 14.80± 0.13 %
μ = 493.9± 0.7
σ = 73.1± 0.5
WF
σ/μ = 15.41± 0.15 %
μ = 492.1± 0.7
σ = 75.8± 0.6

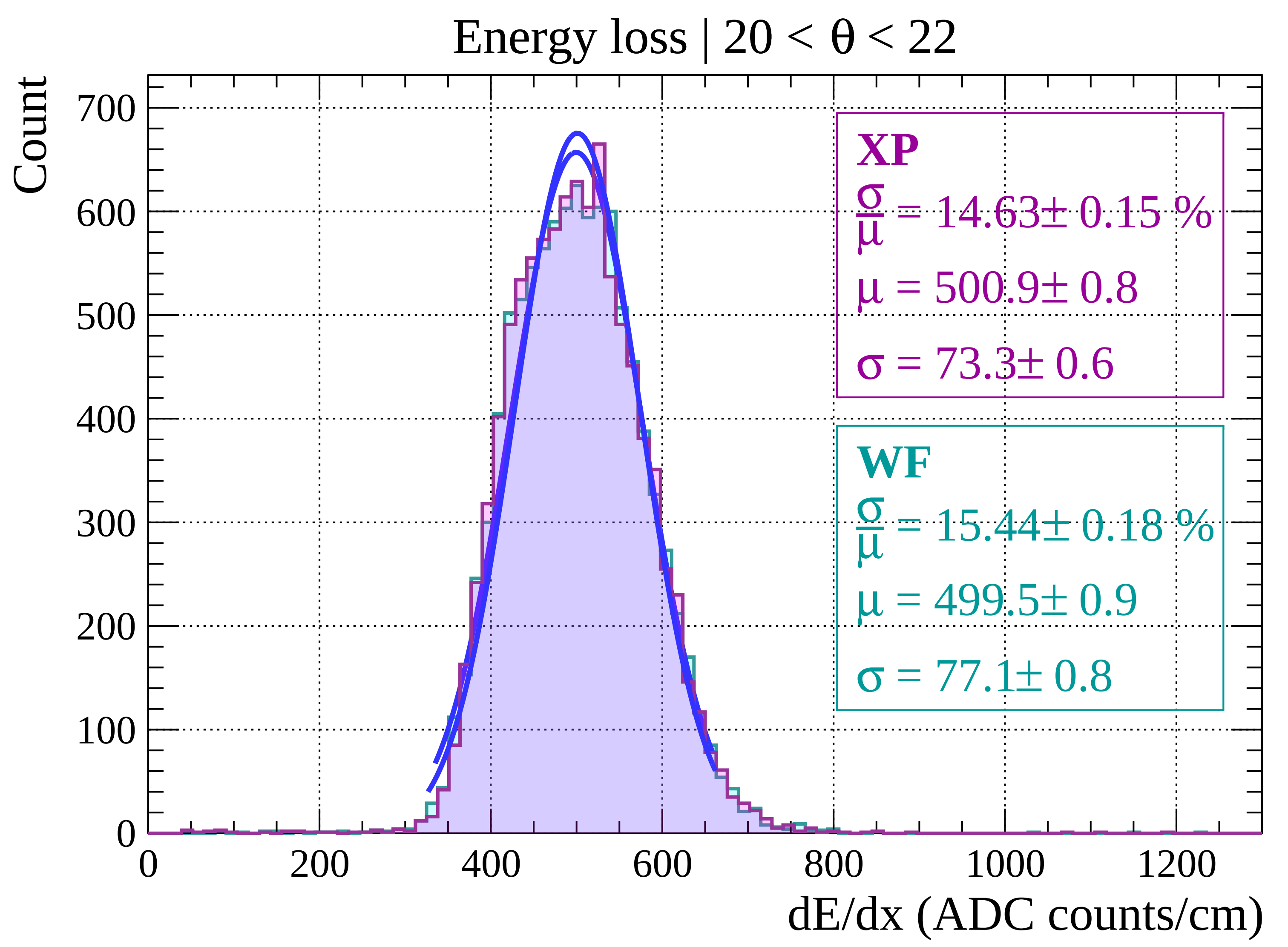

Energy loss | 12 < θ < 14
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
800
900
0
200
400
600
800
1000
1200
XP
σ/μ = 14.65± 0.12 %
μ = 495.9± 0.7
σ = 72.6± 0.5
WF
σ/μ = 15.04± 0.14 %
μ = 495.0± 0.7
σ = 74.5± 0.6


Energy loss | 14 < θ < 16
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
800
900
0
200
400
600
800
1000
1200
XP
σ/μ = 14.68± 0.14 %
μ = 495.1± 0.8
σ = 72.7± 0.6
WF
σ/μ = 15.01± 0.15 %
μ = 494.9± 0.8
σ = 74.3± 0.6

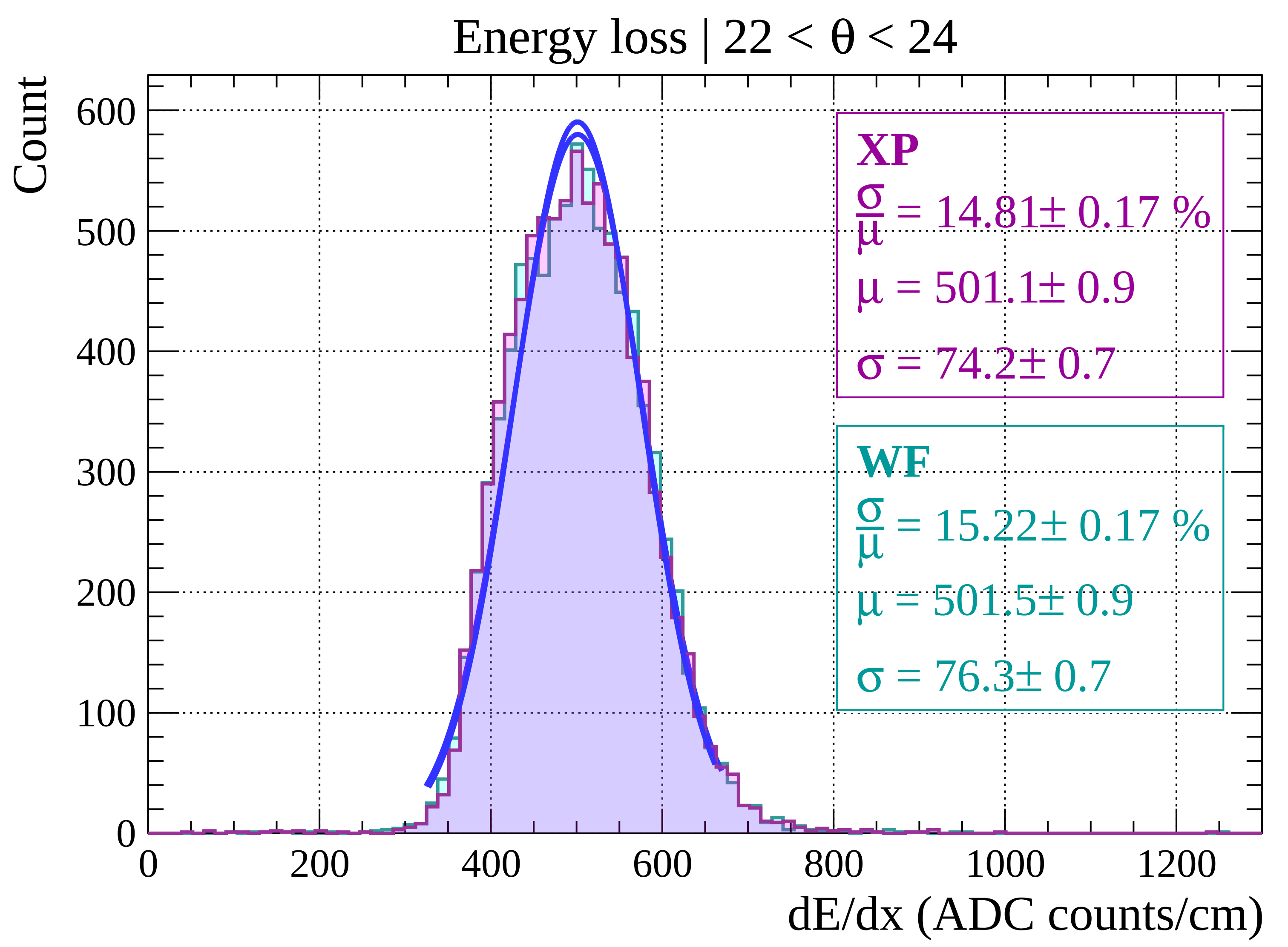


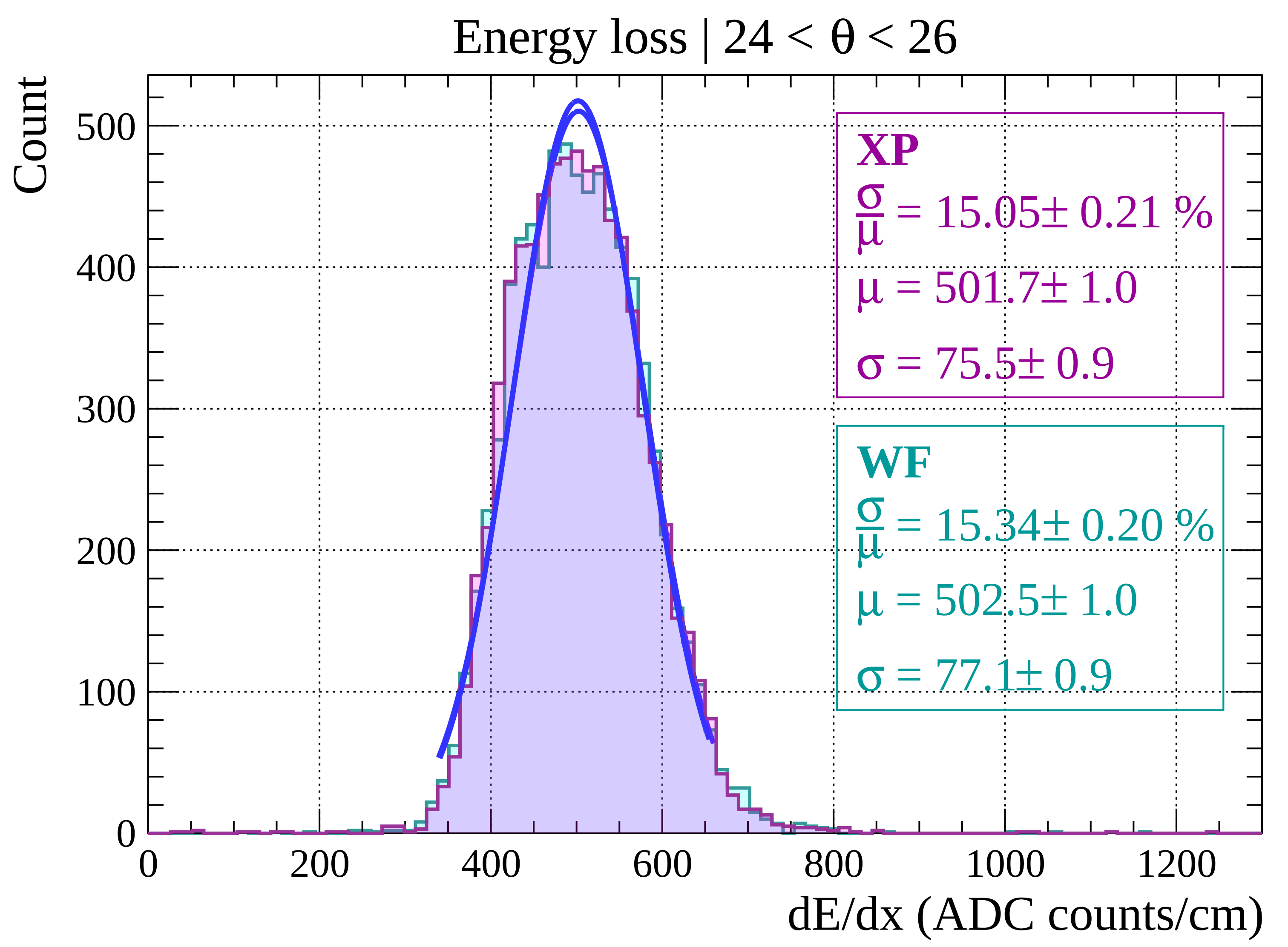
Energy loss | 16 < θ < 18
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 14.80± 0.15 %
μ = 495.4± 0.8
σ = 73.3± 0.6
WF
σ/μ = 14.85± 0.15 %
μ = 495.7± 0.8
σ = 73.6± 0.6
0
100
200
300
400
500
600
700
800
0
200
400
600
800
1000
1200


Energy loss | 18 < θ < 20
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
0
200
400
600
800
1000
1200
XP
σ/μ = 14.65± 0.15 %
μ = 498.1± 0.8
σ = 73.0± 0.6
WF
σ/μ = 15.05± 0.15 %
μ = 499.1± 0.8
σ = 75.1± 0.6

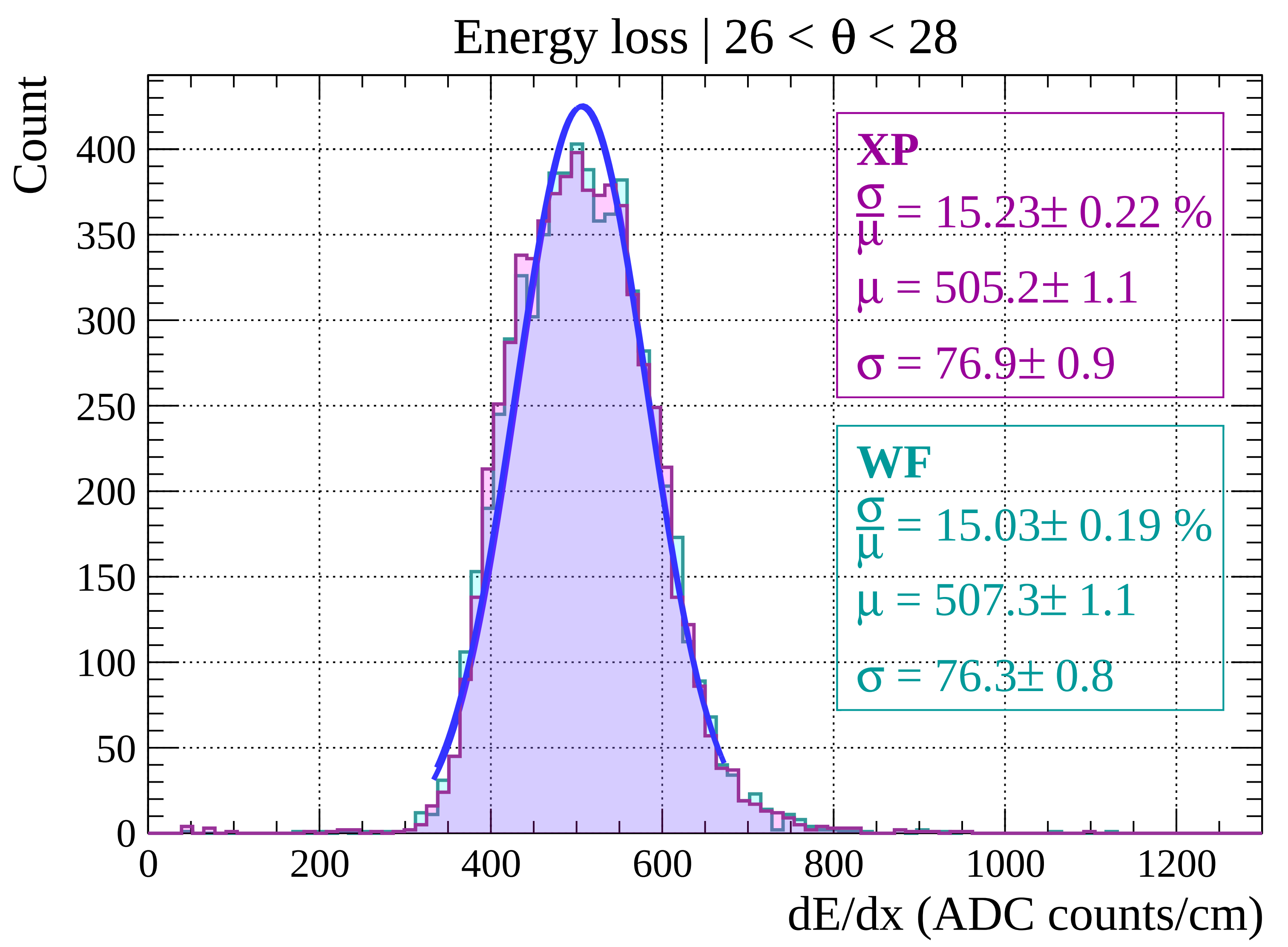


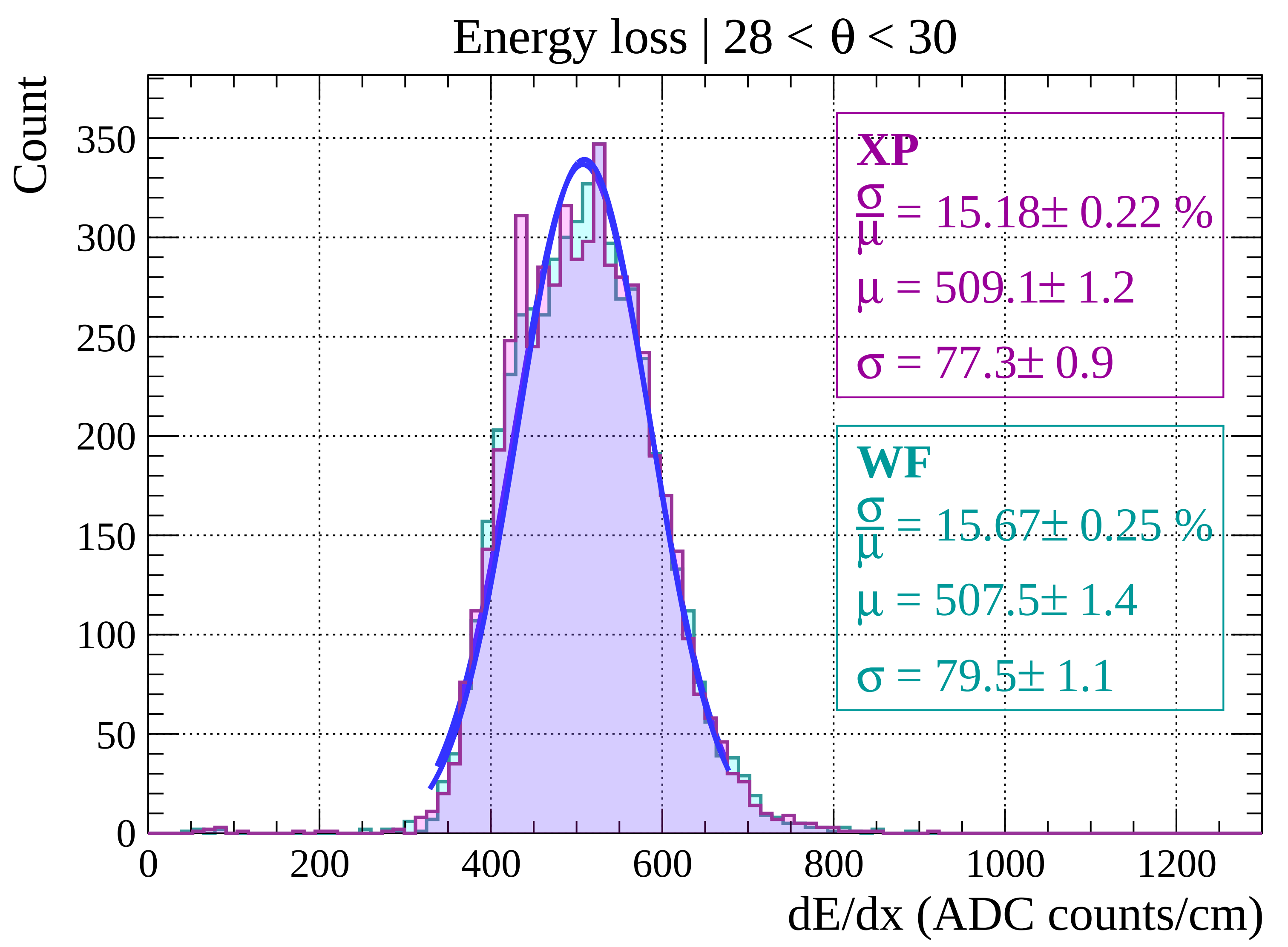
Energy loss | 20 < θ < 22
Count
700
600
500
400
300
200
100
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 14.63± 0.15 %
μ = 500.9± 0.8
σ = 73.3± 0.6
WF
σ/μ = 15.44± 0.18 %
μ = 499.5± 0.9
σ = 77.1± 0.8

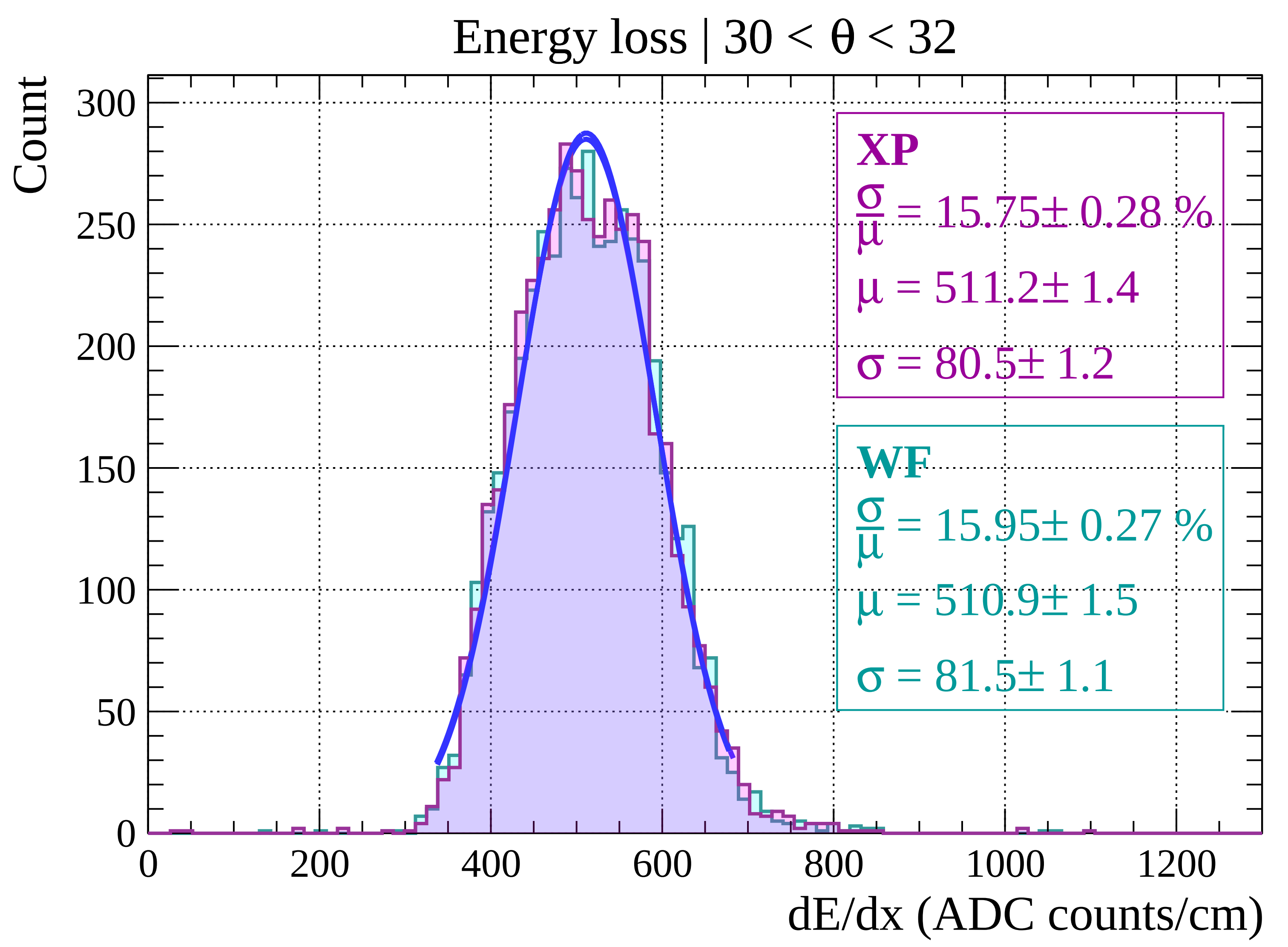

Energy loss | 22 < θ < 24
Count
600
500
400
300
200
100
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 14.81 ± 0.17 %
μ = 501.1 ± 0.9
σ = 74.2 ± 0.7
WF
σ/μ = 15.22 ± 0.17 %
μ = 501.5 ± 0.9
σ = 76.3 ± 0.7

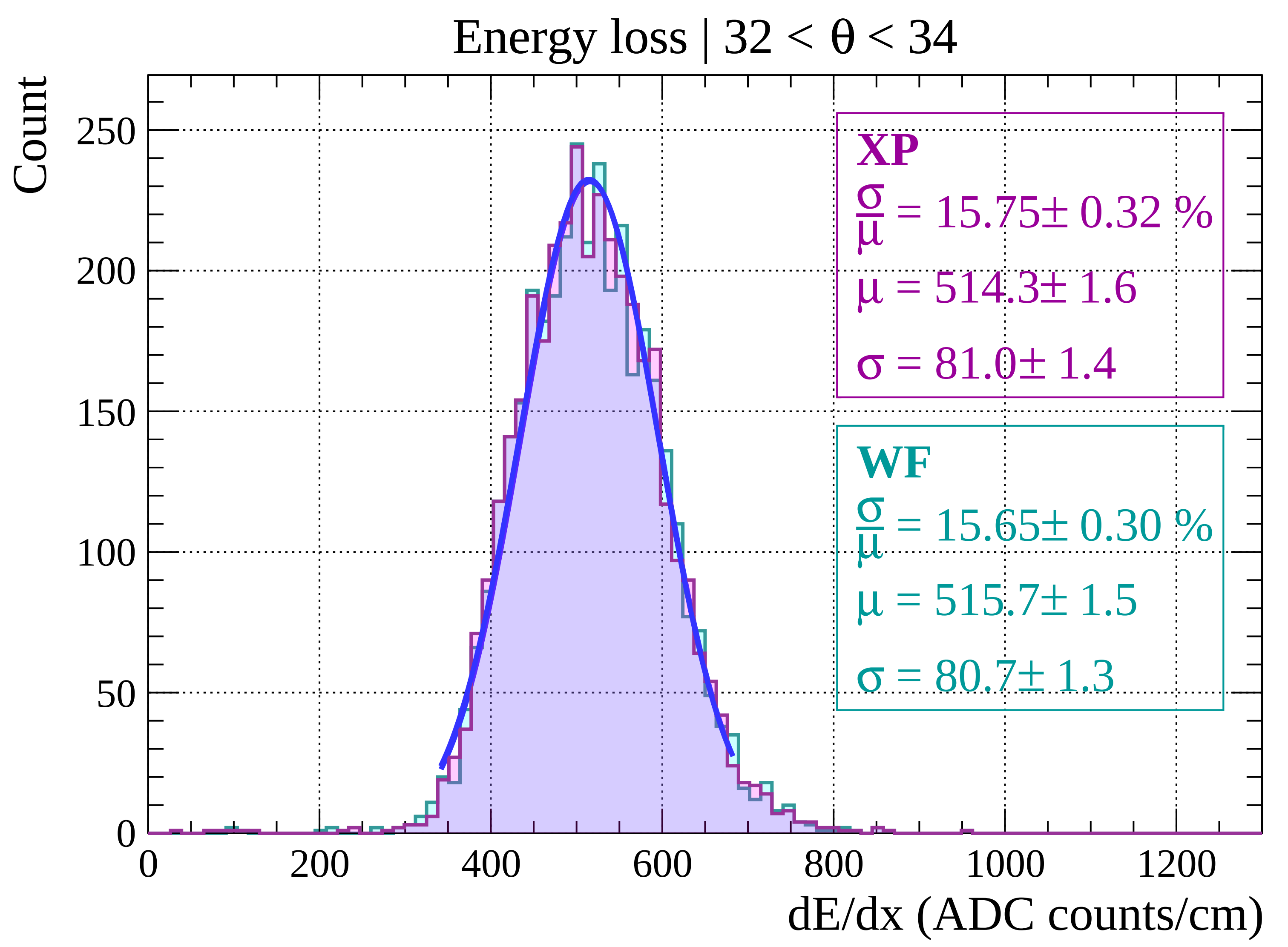


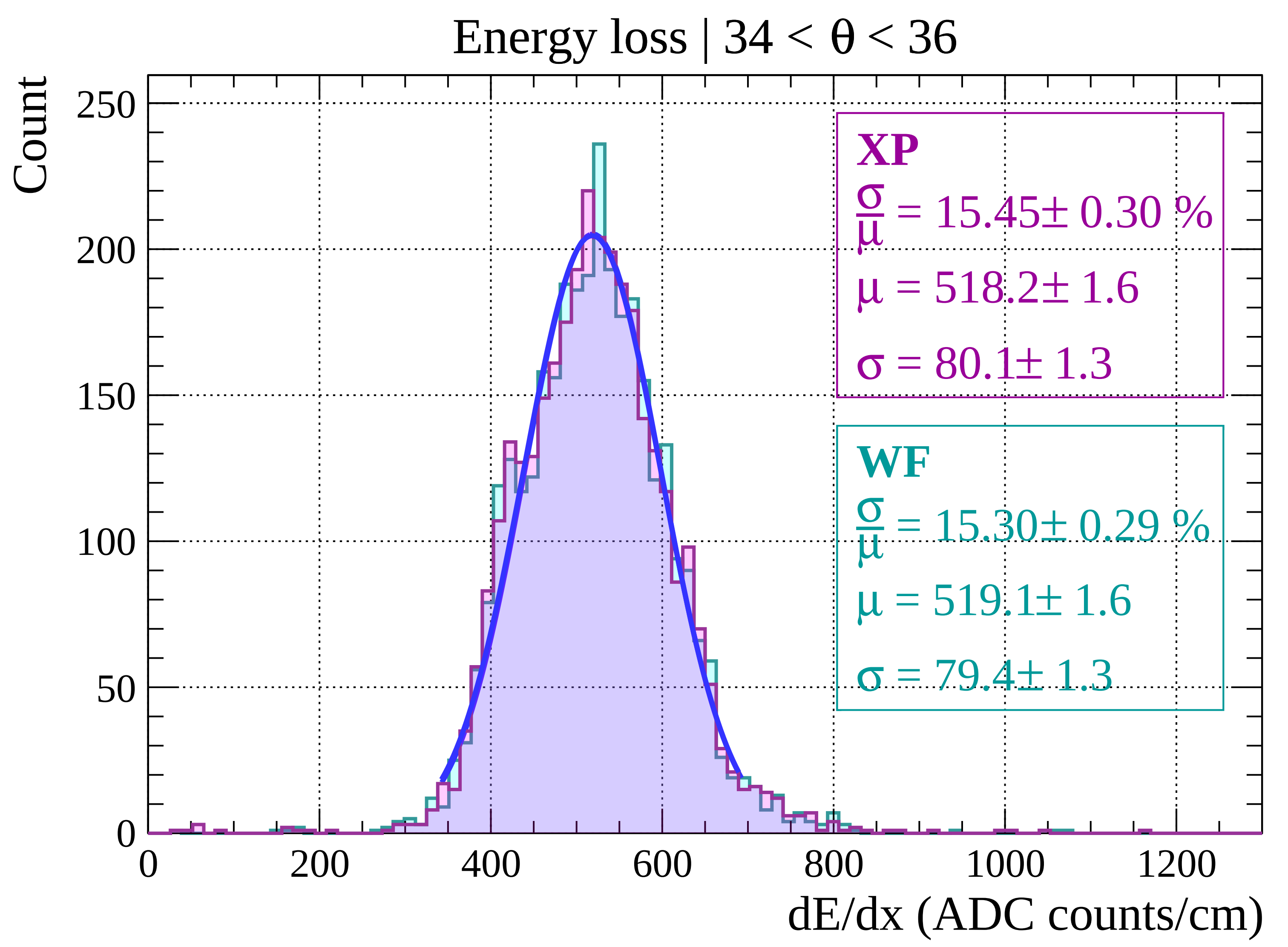
Energy loss | 24 < θ < 26
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
0
200
400
600
800
1000
1200
XP
σ/μ = 15.05± 0.21 %
μ = 501.7± 1.0
σ = 75.5± 0.9
WF
σ/μ = 15.34± 0.20 %
μ = 502.5± 1.0
σ = 77.1± 0.9

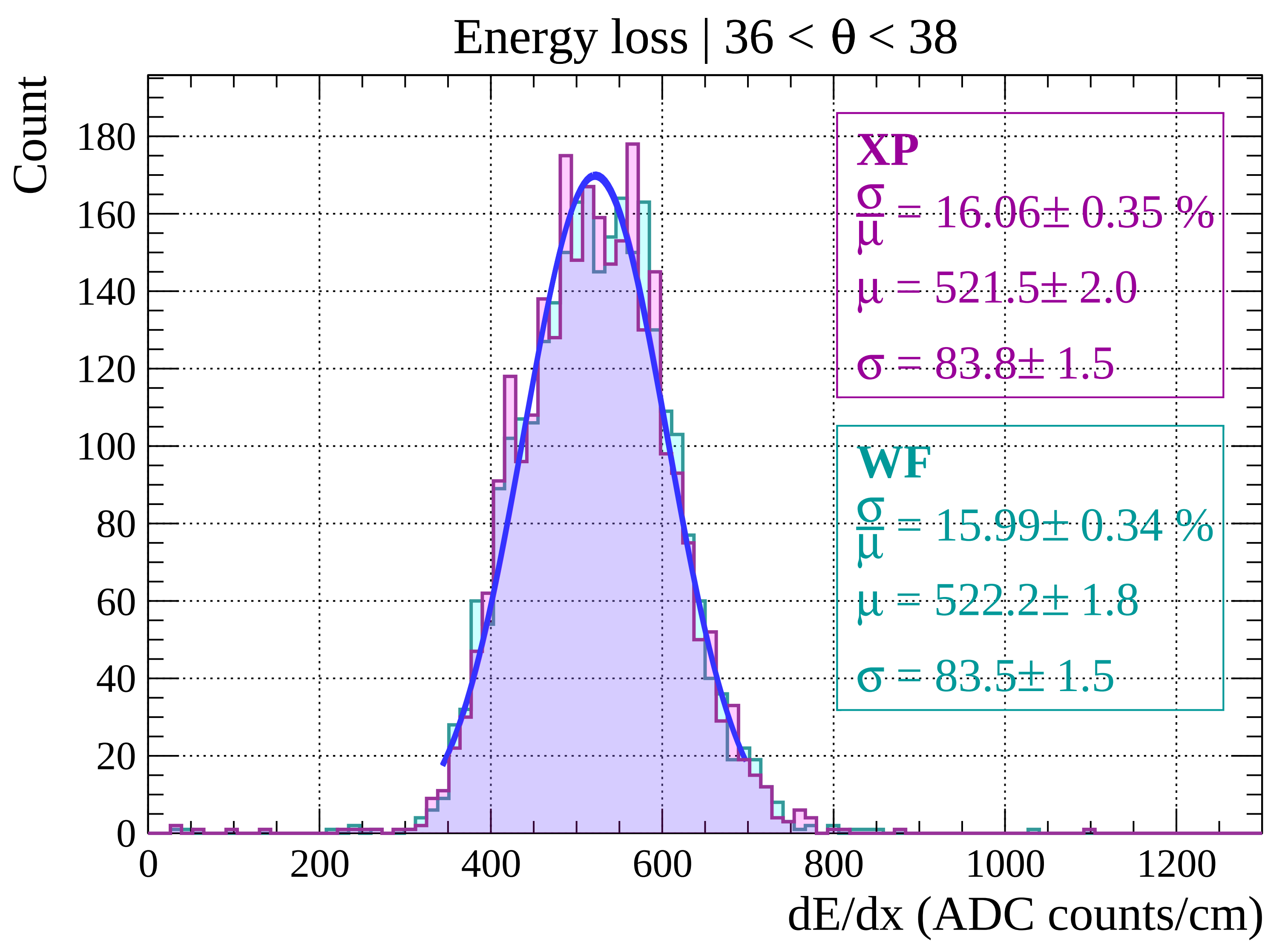

Energy loss | 26 < θ < 28
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
400
0
200
400
600
800
1000
1200
XP
σ/μ = 15.23± 0.22 %
μ = 505.2± 1.1
σ = 76.9± 0.9
WF
σ/μ = 15.03± 0.19 %
μ = 507.3± 1.1
σ = 76.3± 0.8


Energy loss | 28 < θ < 30
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 15.18± 0.22 %
μ = 509.1± 1.2
σ = 77.3± 0.9
WF
σ/μ = 15.67± 0.25 %
μ = 507.5± 1.4
σ = 79.5± 1.1
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200

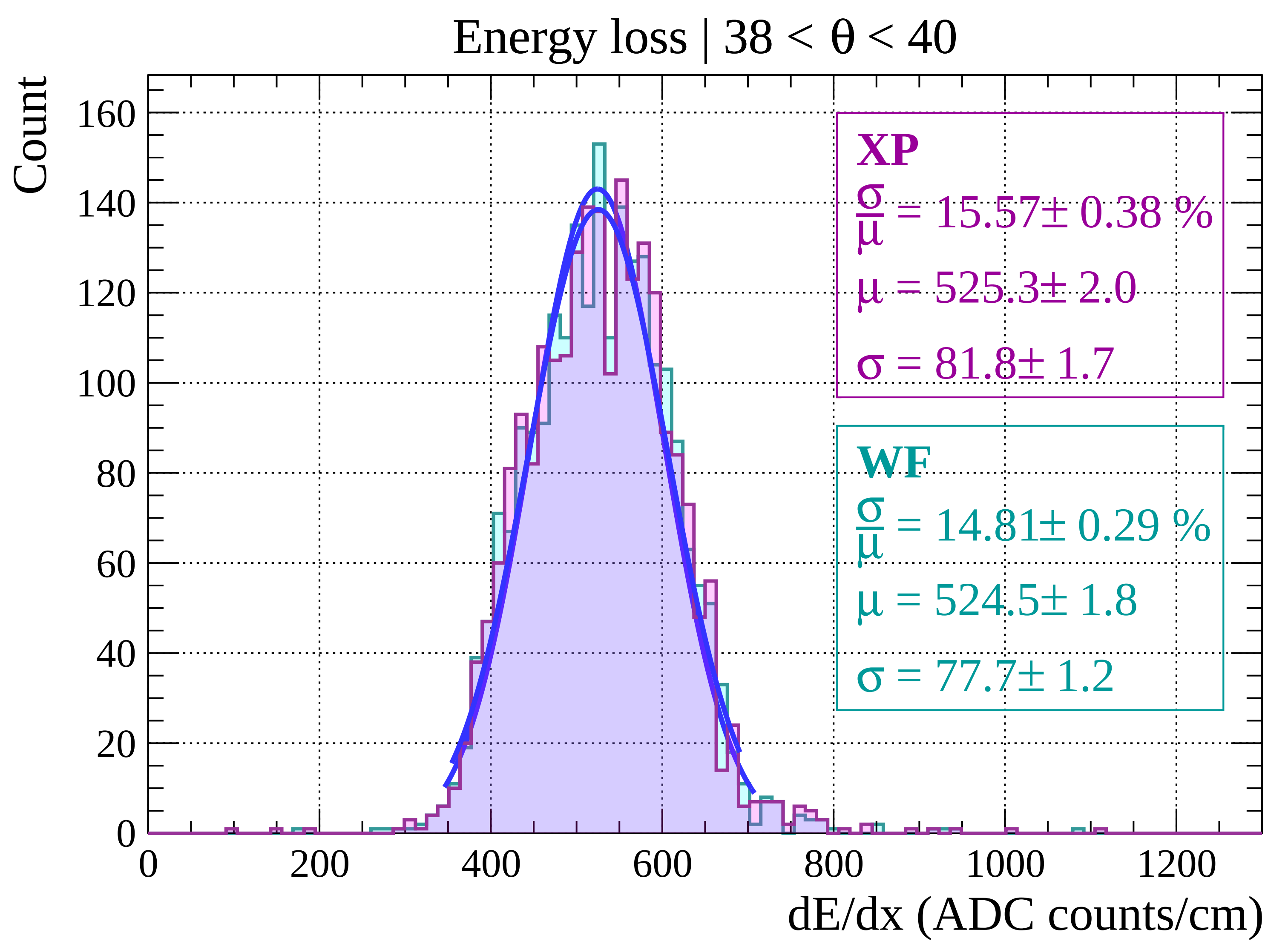

Energy loss | 30 < θ < 32
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
0
200
400
600
800
1000
1200
XP
σ/μ = 15.75± 0.28 %
μ = 511.2± 1.4
σ = 80.5± 1.2
WF
σ/μ = 15.95± 0.27 %
μ = 510.9± 1.5
σ = 81.5± 1.1

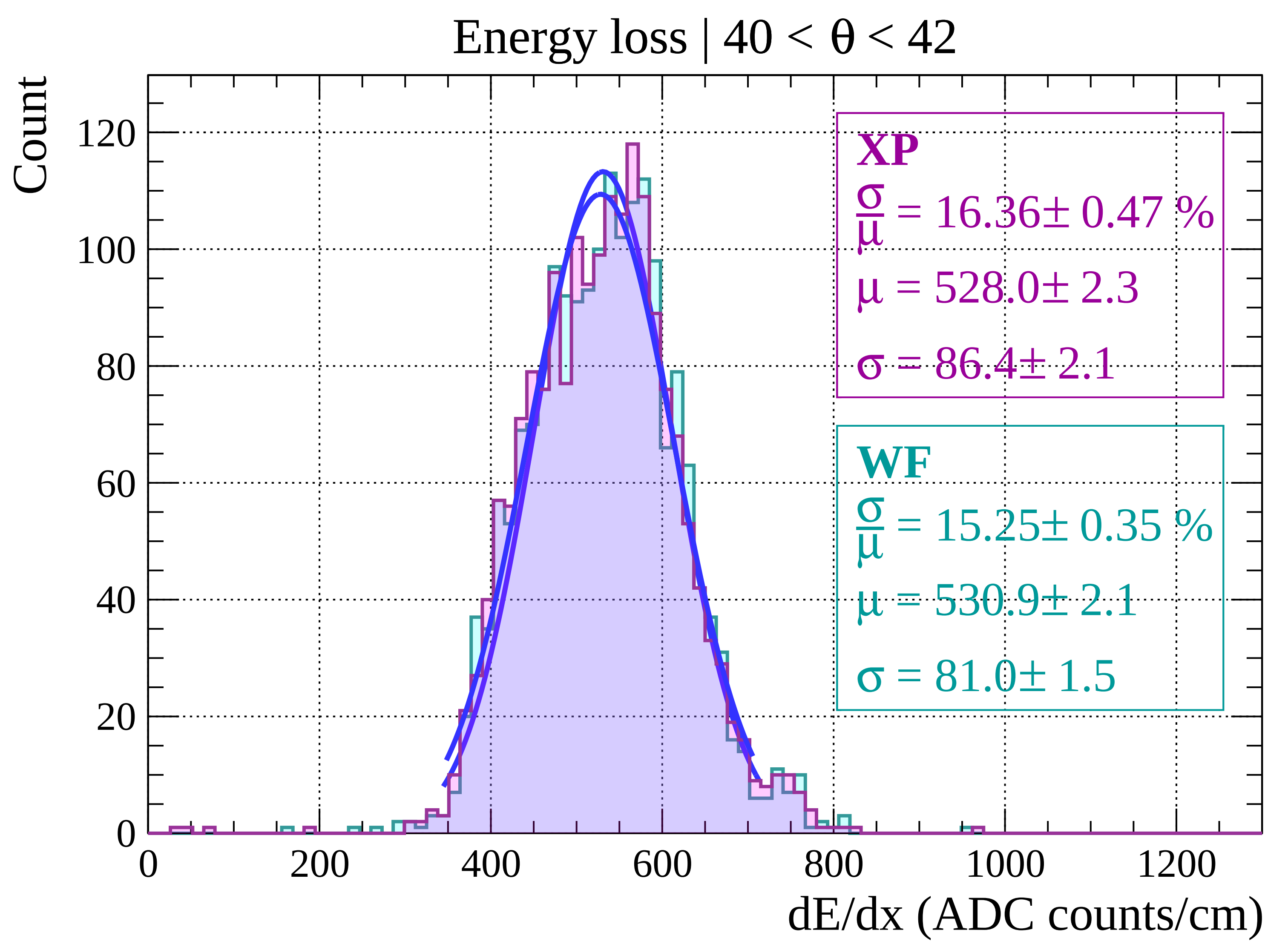

Energy loss | 32 < θ < 34
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 15.75± 0.32 %
μ = 514.3± 1.6
σ = 81.0± 1.4
WF
$\frac{\sigma}{\mu}$ = 15.65± 0.30 %
μ = 515.7± 1.5
σ = 80.7± 1.3

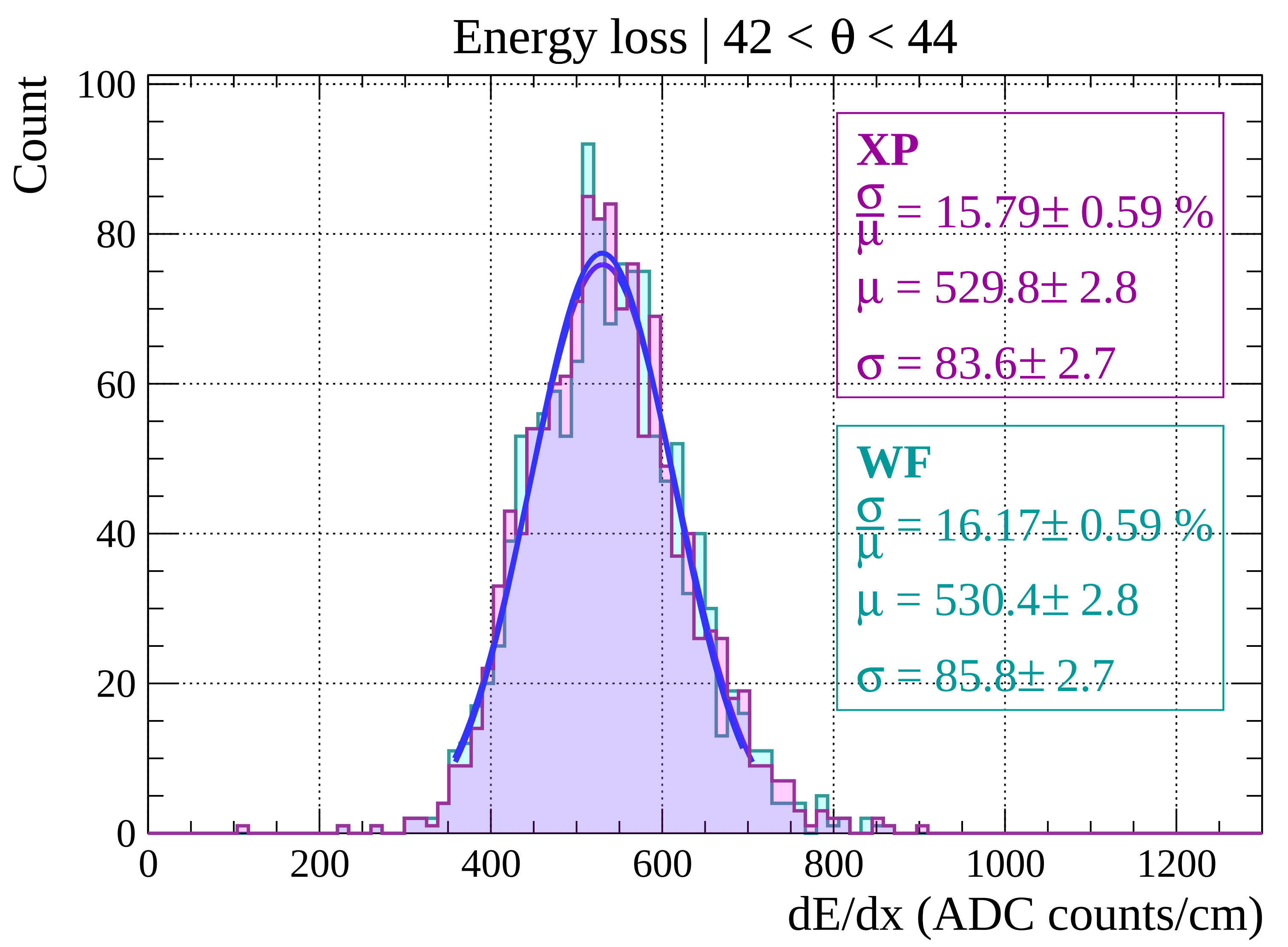

Energy loss | 34 < θ < 36
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 15.45± 0.30 %
μ = 518.2± 1.6
σ = 80.1± 1.3
WF
$\frac{\sigma}{\mu}$ = 15.30± 0.29 %
μ = 519.1± 1.6
σ = 79.4± 1.3
0
200
400
600
800
1000
1200
0
50
100
150
200
250



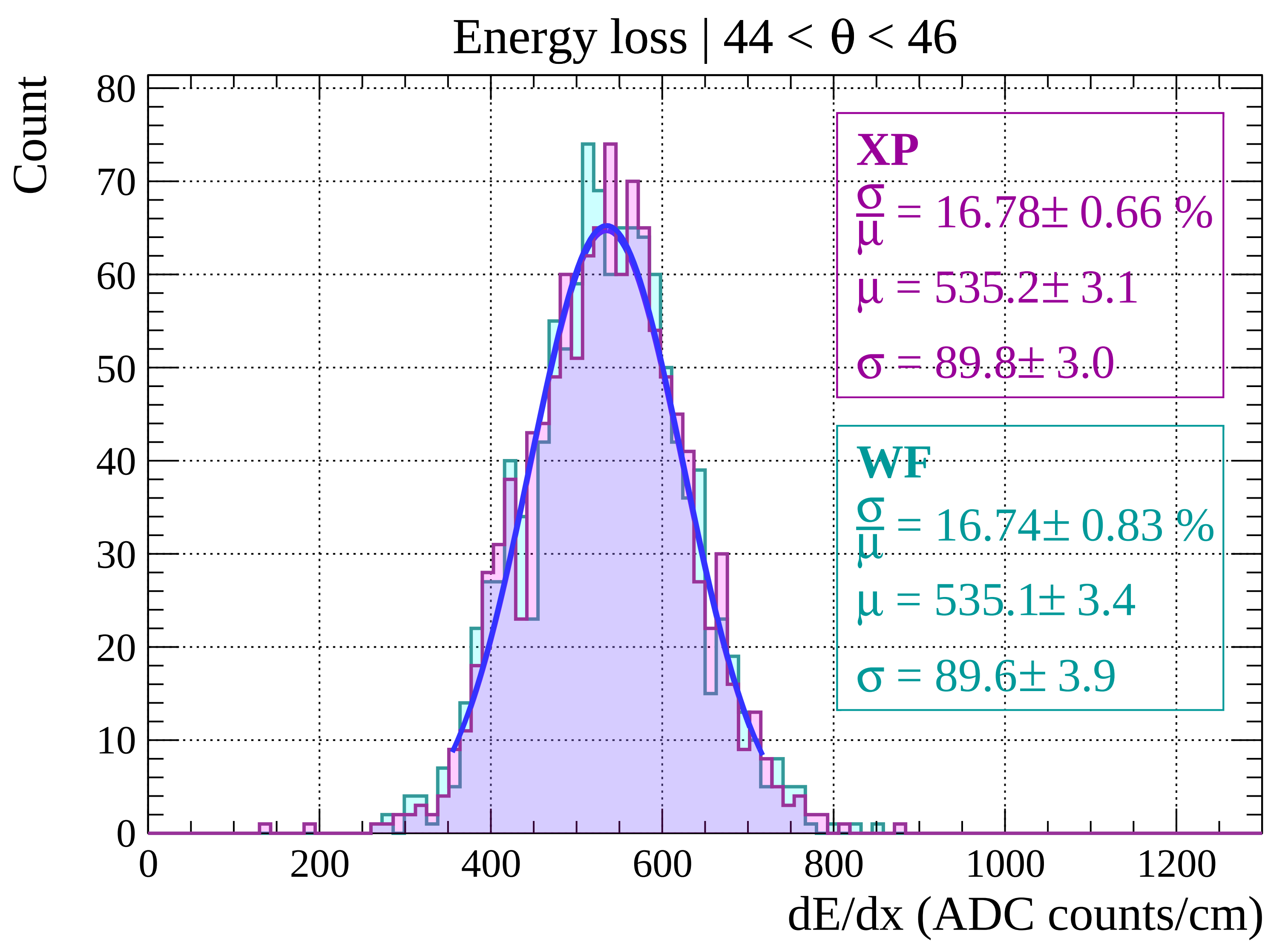
Energy loss | 36 < θ < 38
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
180
0
200
400
600
800
1000
1200
XP
σ/μ = 16.06± 0.35 %
μ = 521.5± 2.0
σ = 83.8± 1.5
WF
σ/μ = 15.99± 0.34 %
μ = 522.2± 1.8
σ = 83.5± 1.5

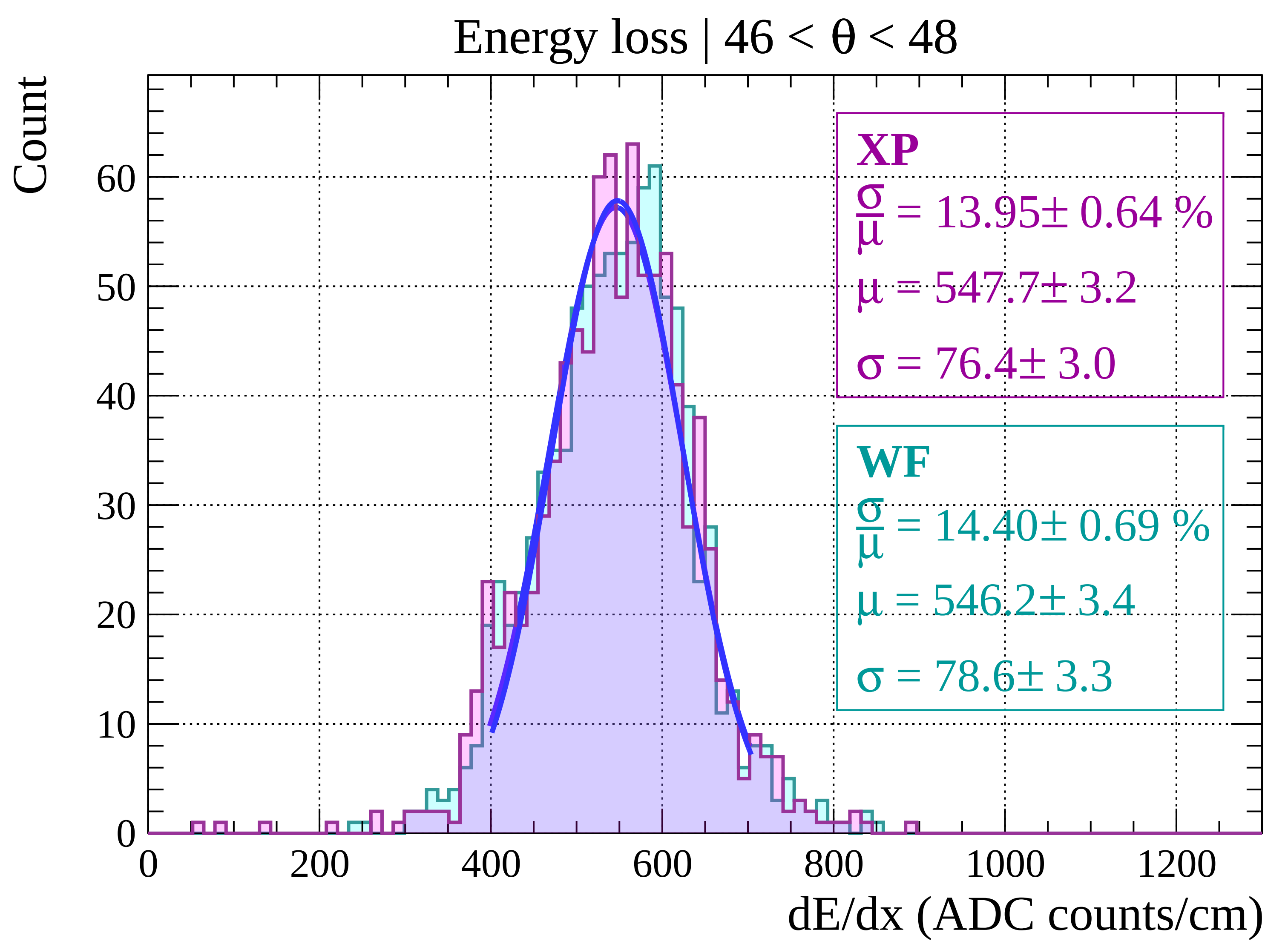

Energy loss | 38 < θ < 40
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
0
200
400
600
800
1000
1200
XP
σ/μ = 15.57± 0.38 %
μ = 525.3± 2.0
σ = 81.8± 1.7
WF
σ/μ = 14.81± 0.29 %
μ = 524.5± 1.8
σ = 77.7± 1.2

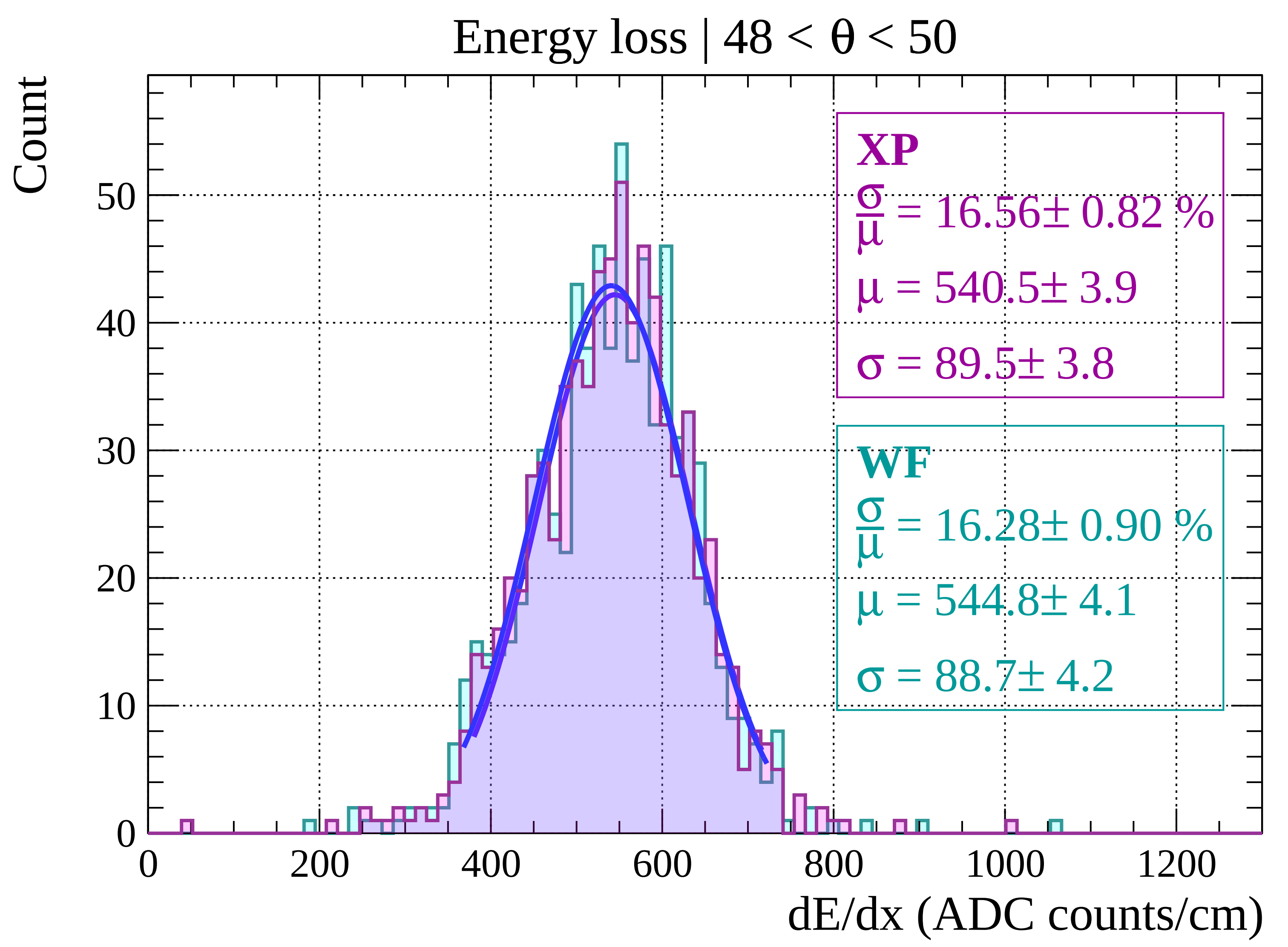

Energy loss | 40 < θ < 42
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 16.36± 0.47 %
μ = 528.0± 2.3
σ = 86.4± 2.1
WF
σ/μ = 15.25± 0.35 %
μ = 530.9± 2.1
σ = 81.0± 1.5



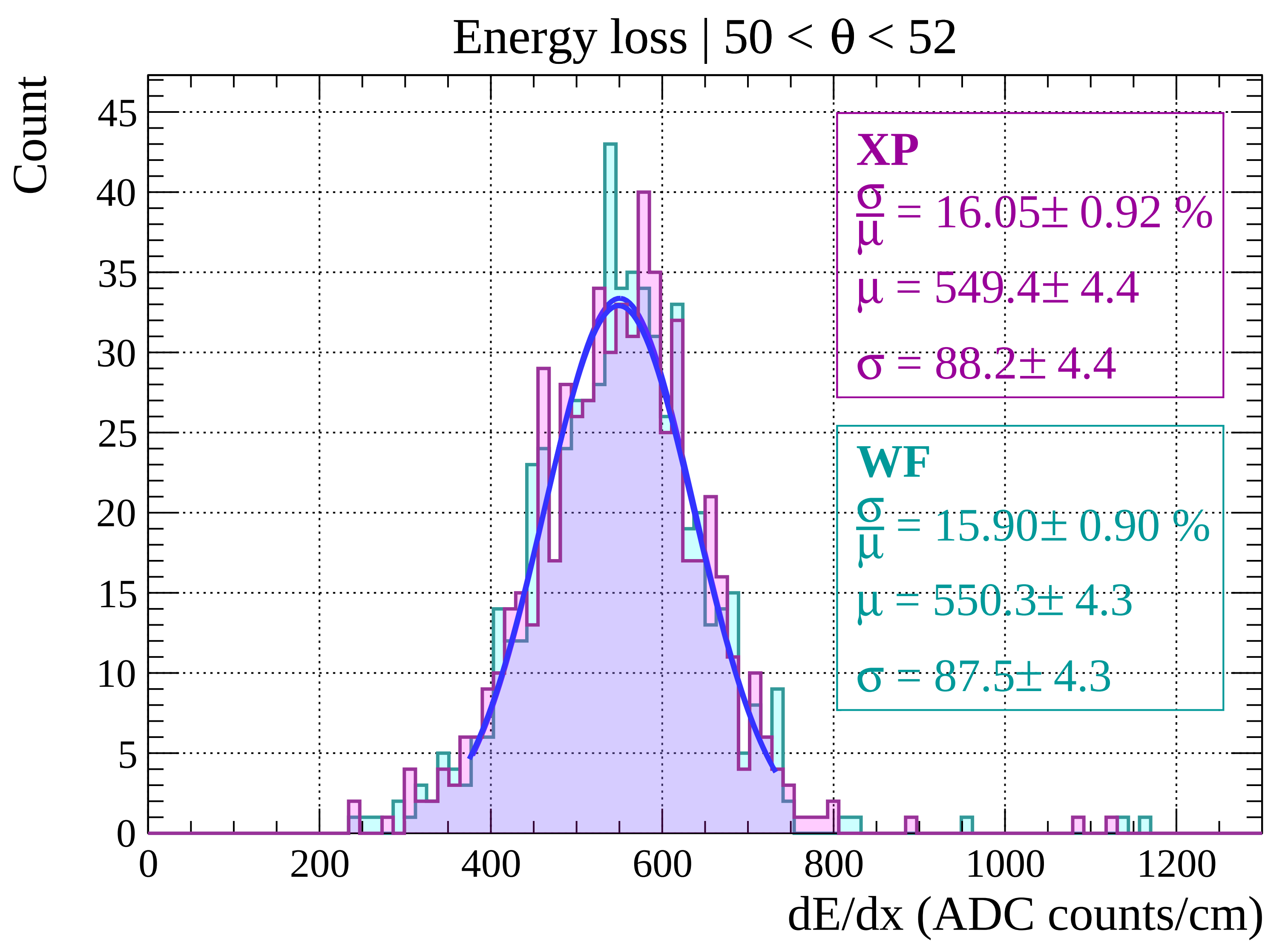
Energy loss | 42 < θ < 44
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
XP
σ/μ = 15.79± 0.59 %
μ = 529.8± 2.8
σ = 83.6± 2.7
WF
σ/μ = 16.17± 0.59 %
μ = 530.4± 2.8
σ = 85.8± 2.7


Energy loss | 44 < θ < 46
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 16.78± 0.66 %
μ = 535.2± 3.1
σ = 89.8± 3.0
WF
σ/μ = 16.74± 0.83 %
μ = 535.1± 3.4
σ = 89.6± 3.9
0
10
20
30
40
50
60
70
80
0
200
400
600
800
1000
1200

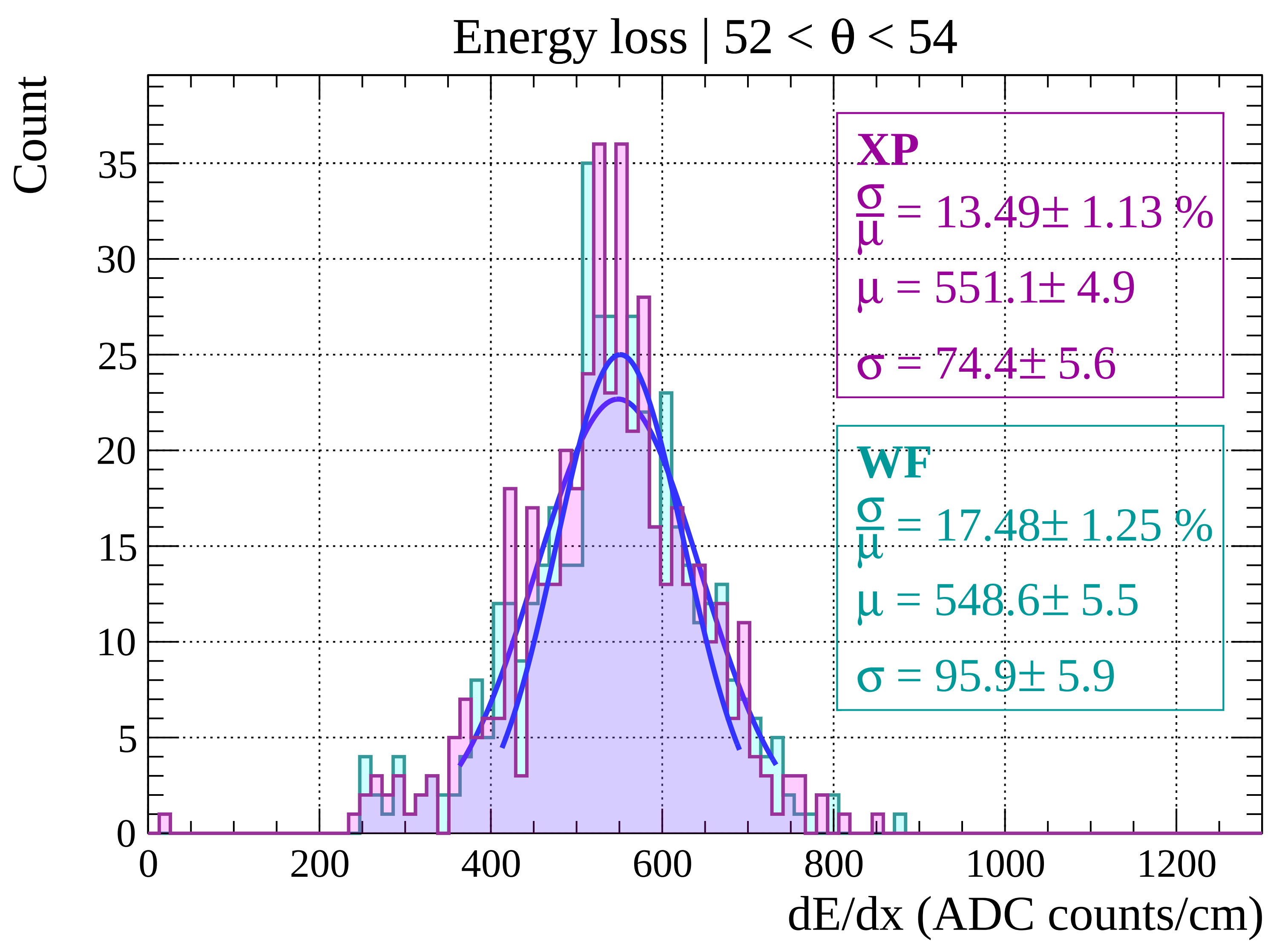

Energy loss | 46 < θ < 48
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
0
200
400
600
800
1000
1200
XP
σ/μ = 13.95± 0.64 %
μ = 547.7± 3.2
σ = 76.4± 3.0
WF
σ/μ = 14.40± 0.69 %
μ = 546.2± 3.4
σ = 78.6± 3.3



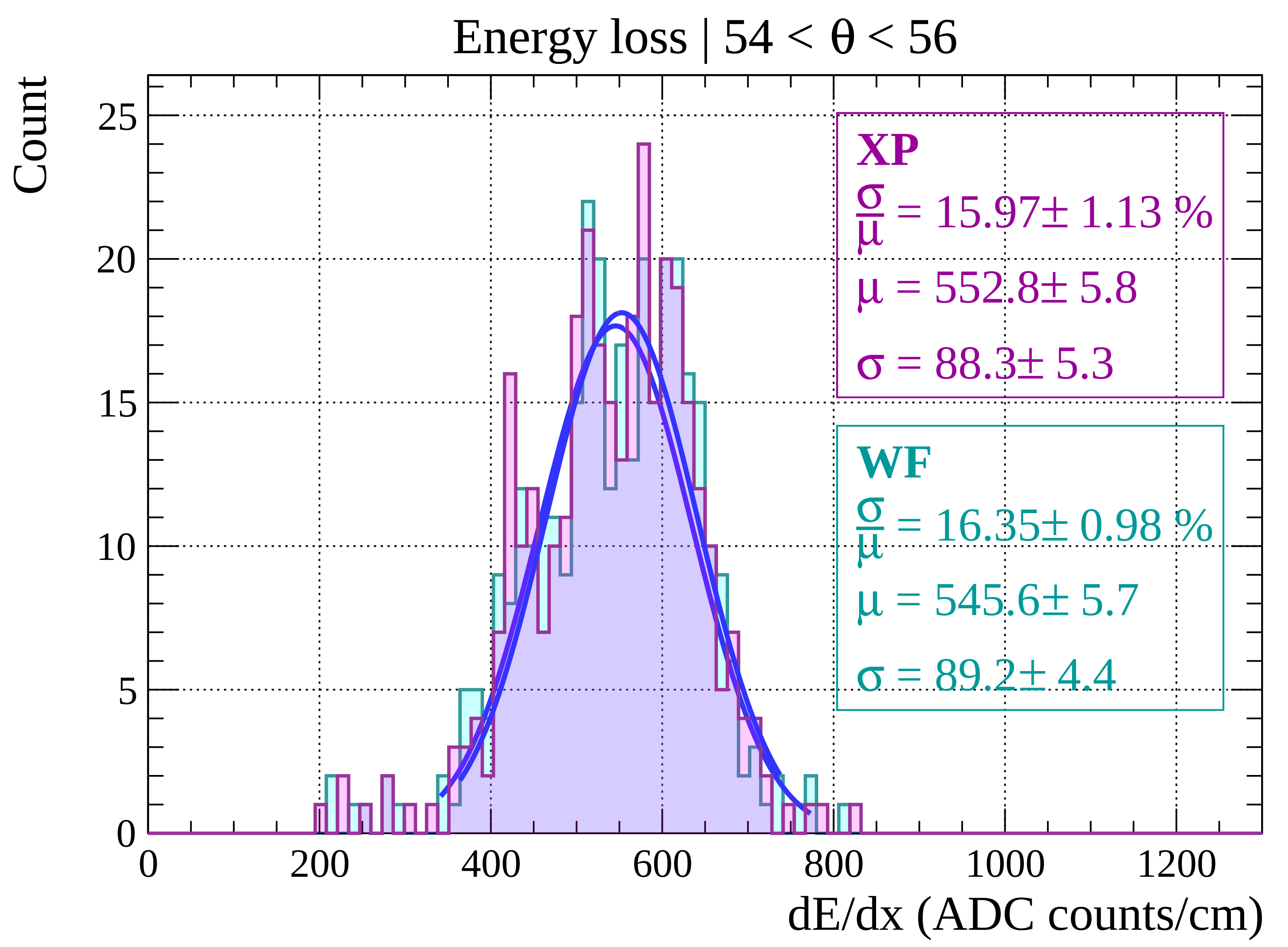
Energy loss | 48 < θ < 50
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
0
200
400
600
800
1000
1200
XP
σ/μ = 16.56± 0.82 %
μ = 540.5± 3.9
σ = 89.5± 3.8
WF
σ/μ = 16.28± 0.90 %
μ = 544.8± 4.1
σ = 88.7± 4.2



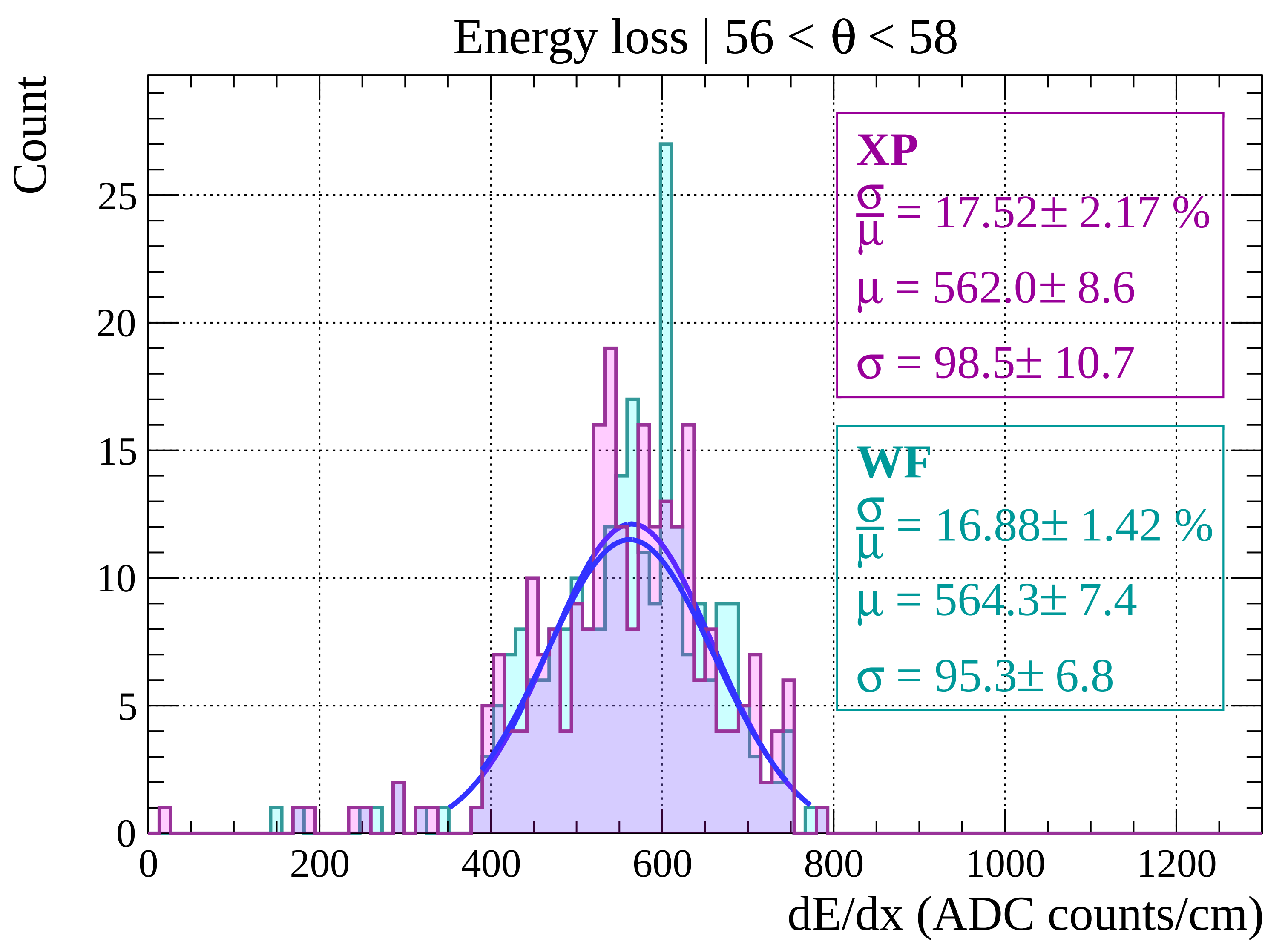
Energy loss | 50 < θ < 52
Count
dE/dx (ADC counts/cm)
0
5
10
15
20
25
30
35
40
45
0
200
400
600
800
1000
1200
XP
σ/μ = 16.05± 0.92 %
μ = 549.4± 4.4
σ = 88.2± 4.4
WF
σ/μ = 15.90± 0.90 %
μ = 550.3± 4.3
σ = 87.5± 4.3



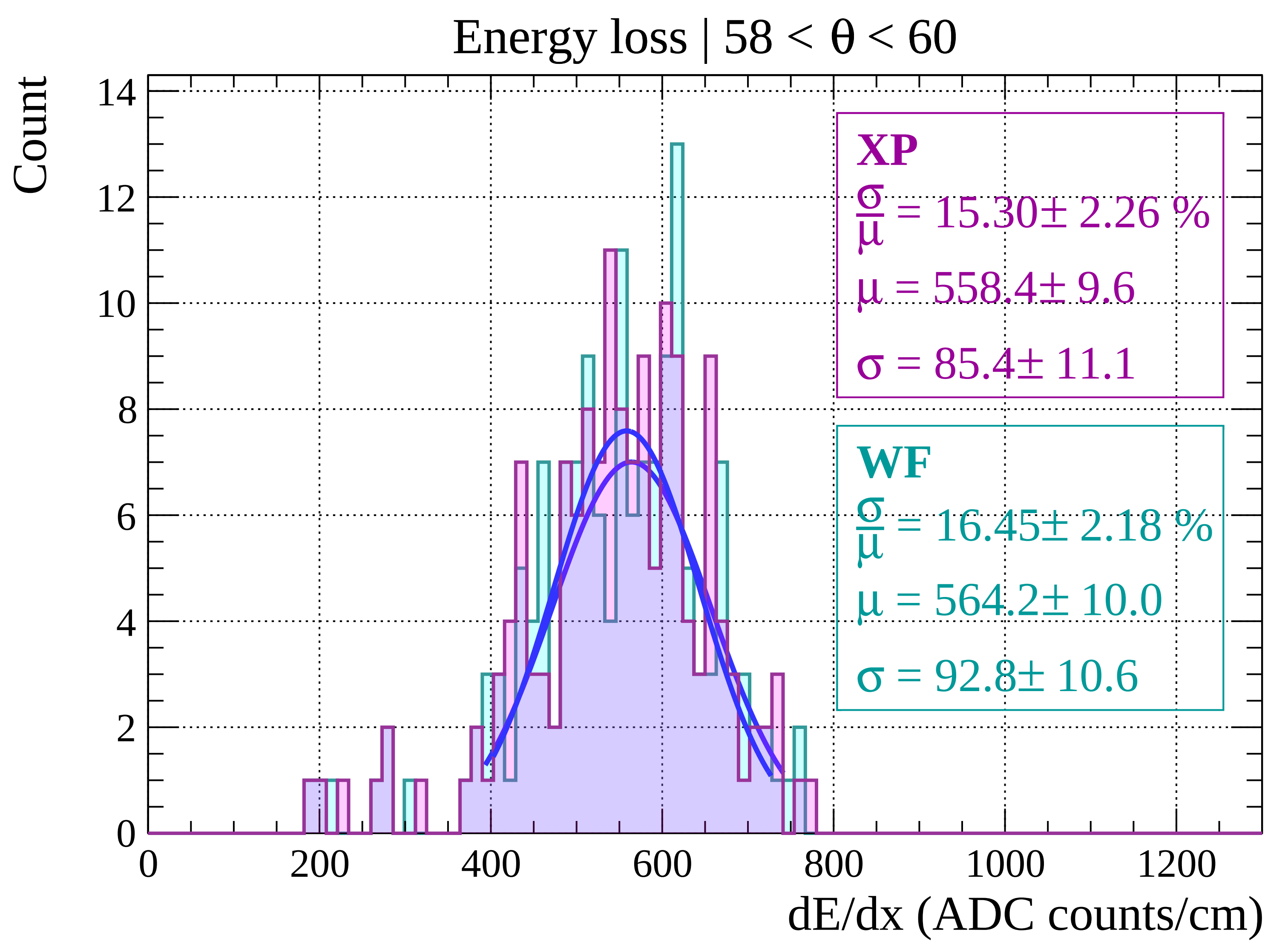
Energy loss | 52 < θ < 54
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
5
10
15
20
25
30
35
XP
σ/μ = 13.49± 1.13 %
μ = 551.1± 4.9
σ = 74.4± 5.6
WF
σ/μ = 17.48± 1.25 %
μ = 548.6± 5.5
σ = 95.9± 5.9

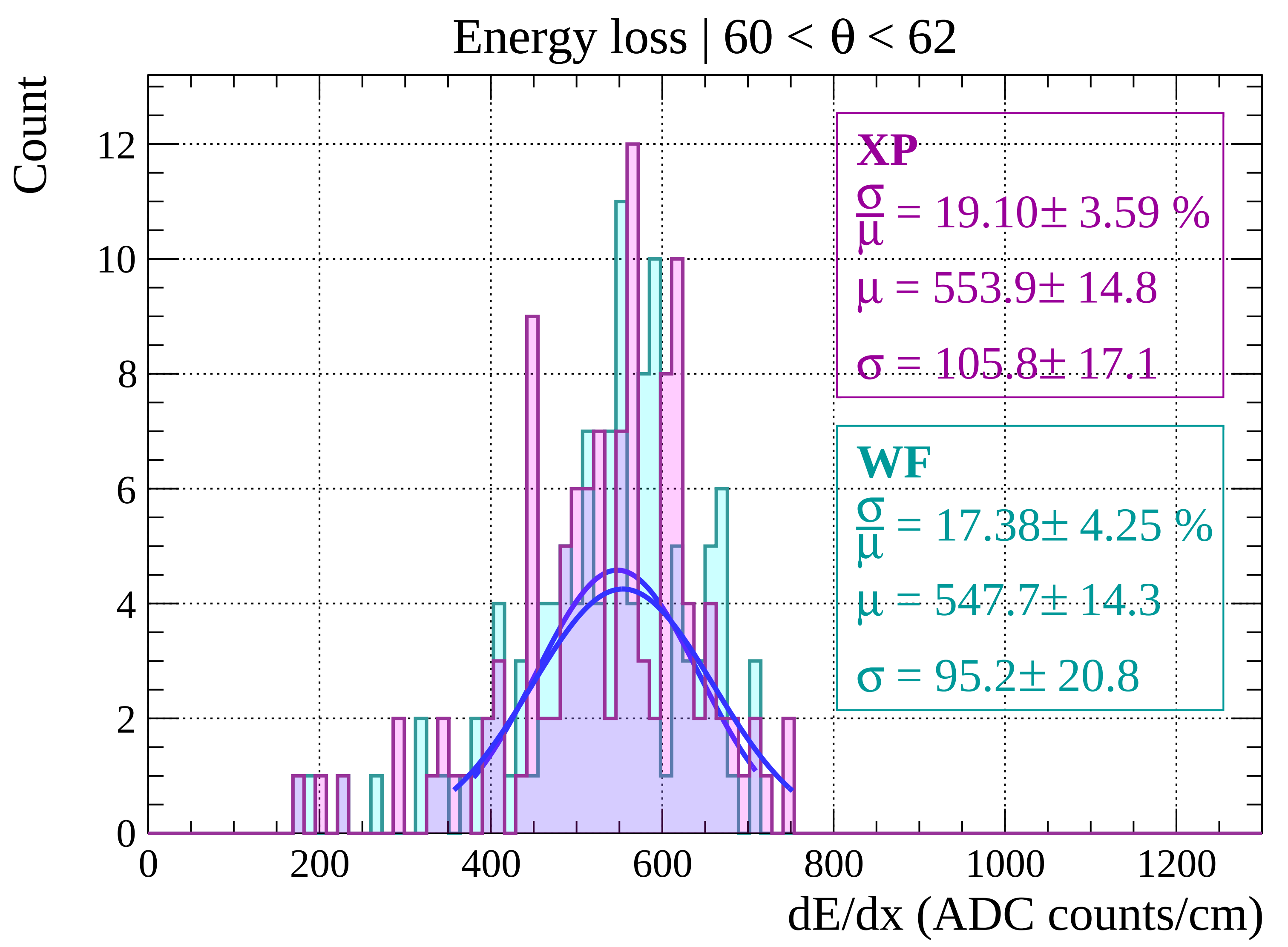


Energy loss | 54 < θ < 56
Count
25
20
15
10
5
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 15.97± 1.13 %
μ = 552.8± 5.8
σ = 88.3± 5.3
WF
σ/μ = 16.35± 0.98 %
μ = 545.6± 5.7
σ = 89.2± 4.4

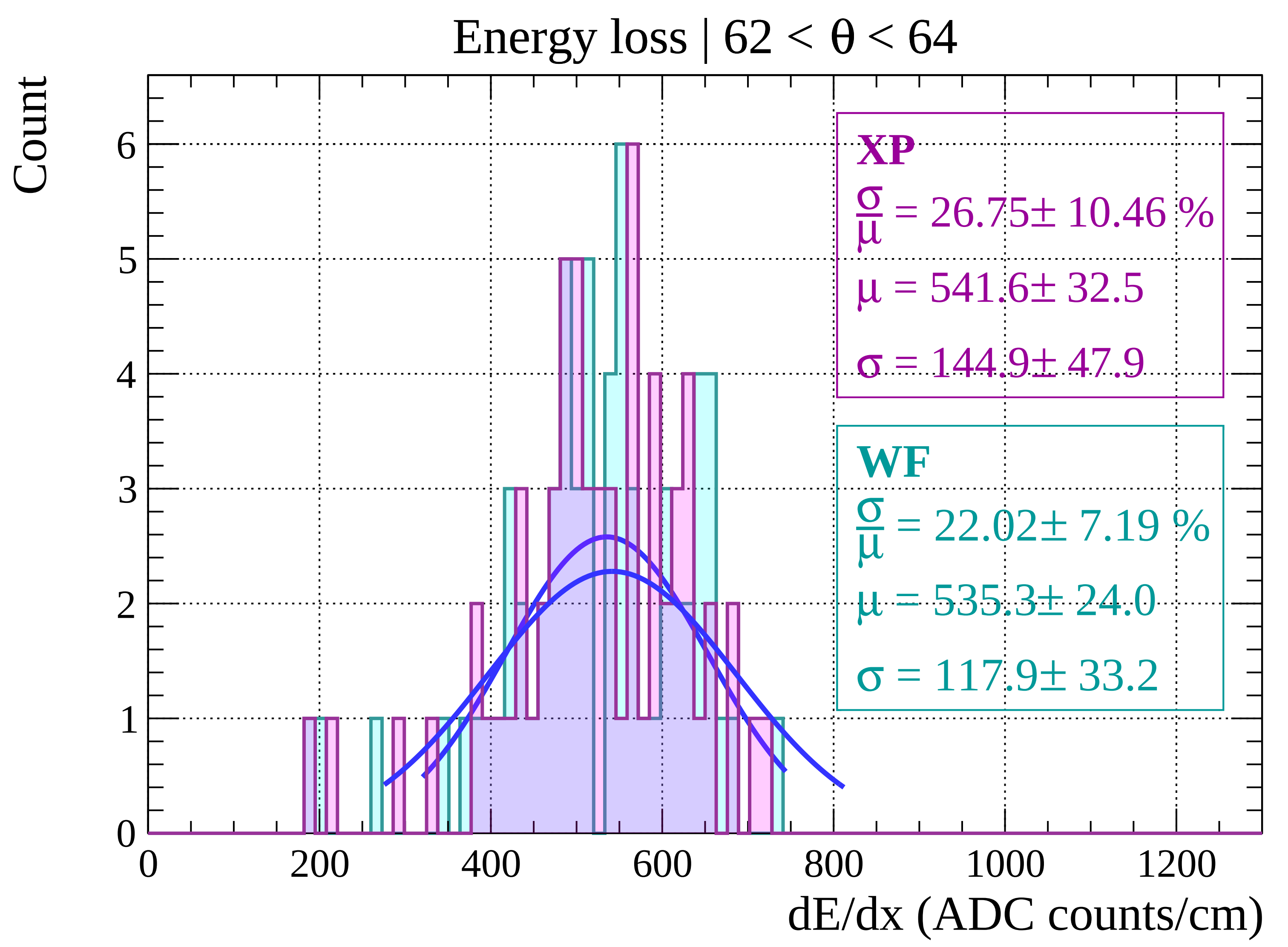


Energy loss | 56 < θ < 58
Count
dE/dx (ADC counts/cm)
0
5
10
15
20
25
0
200
400
600
800
1000
1200
XP
σ/μ = 17.52± 2.17 %
μ = 562.0± 8.6
σ = 98.5± 10.7
WF
σ/μ = 16.88± 1.42 %
μ = 564.3± 7.4
σ = 95.3± 6.8

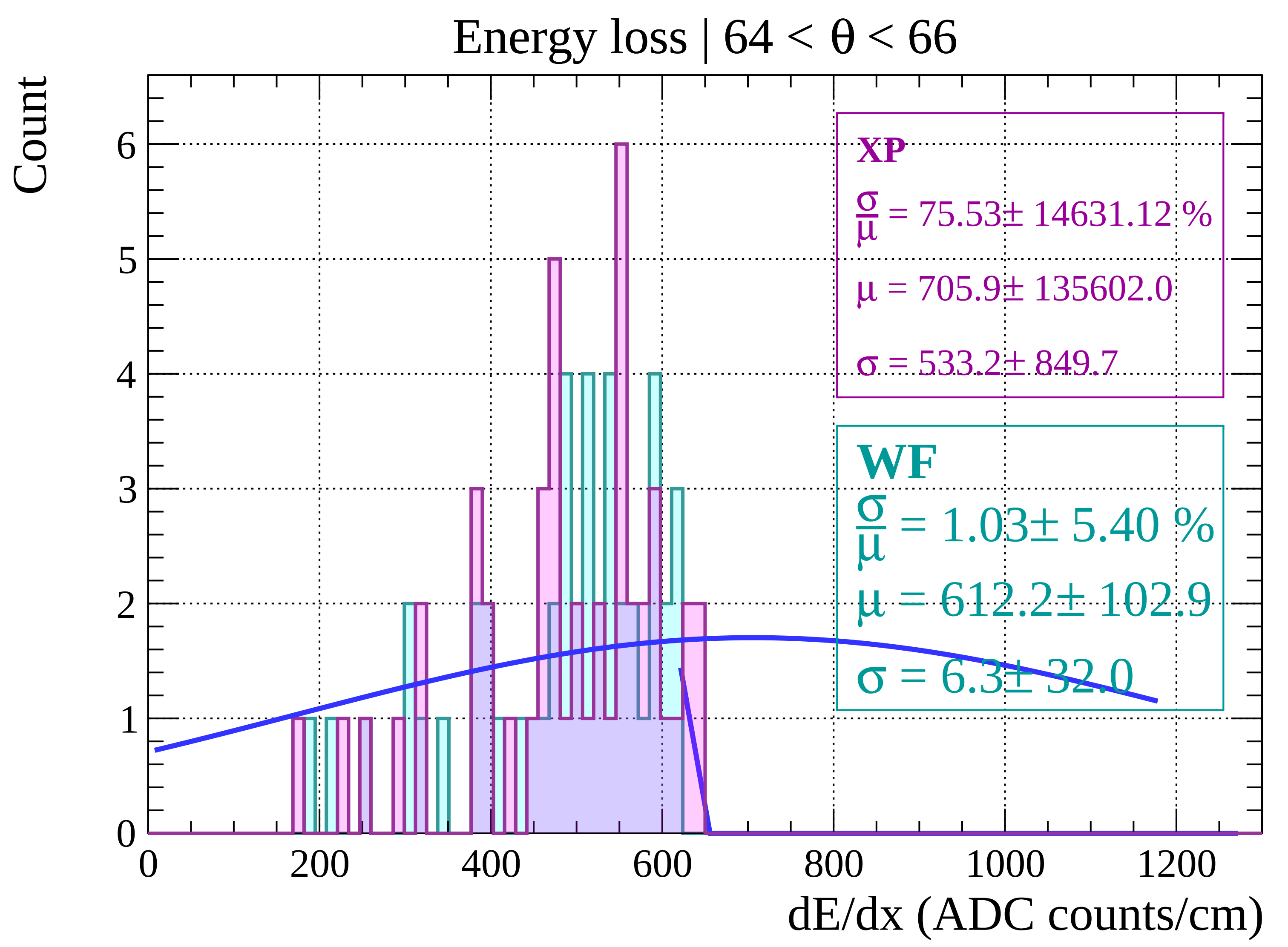

Energy loss | 58 < θ < 60
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
2
4
6
8
10
12
14
XP
$\frac{\sigma}{\mu}$ = 15.30± 2.26 %
μ = 558.4± 9.6
σ = 85.4± 11.1
WF
$\frac{\sigma}{\mu}$ = 16.45± 2.18 %
μ = 564.2± 10.0
σ = 92.8± 10.6



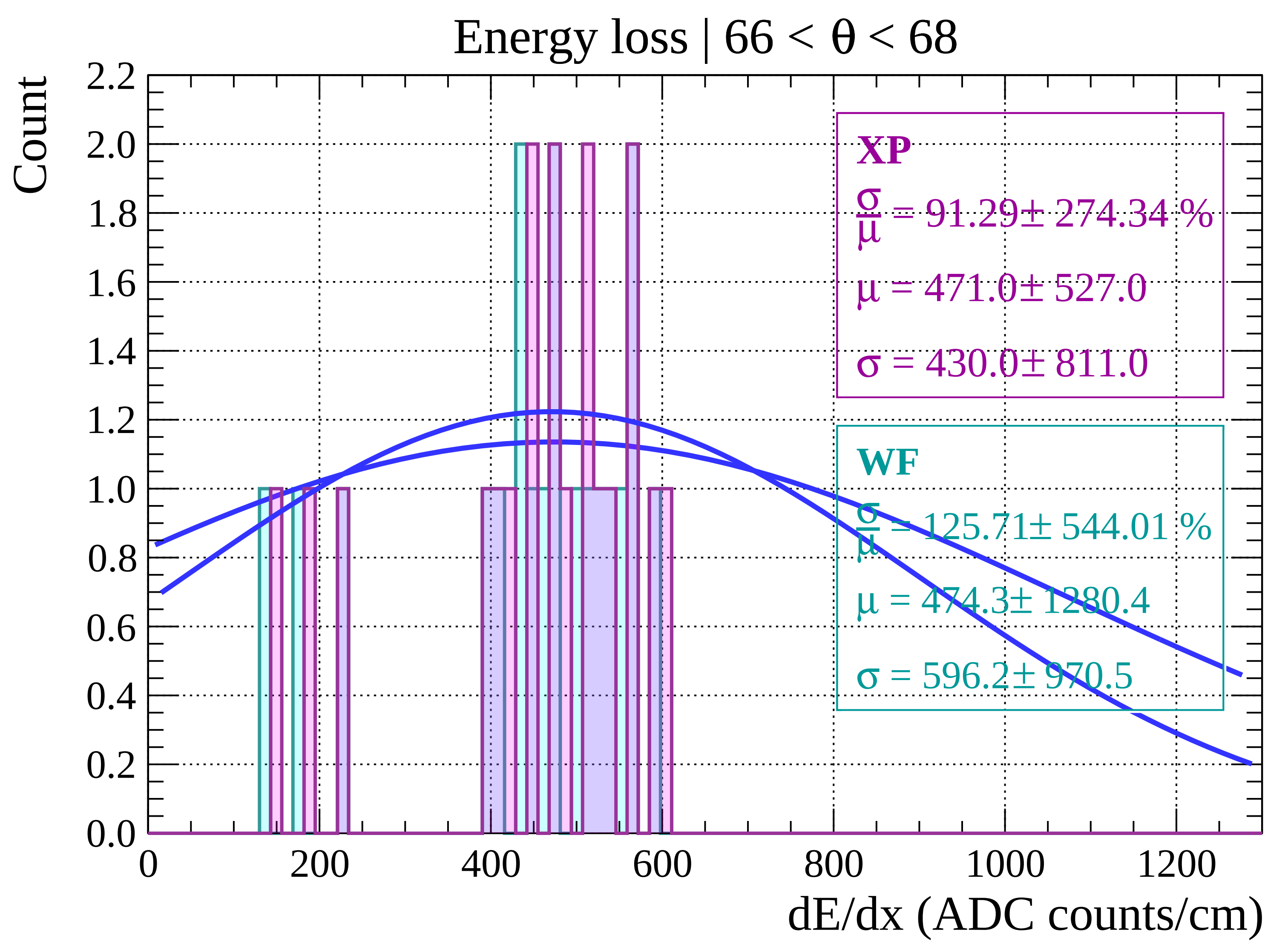
Energy loss | 60 < θ < 62
Count
dE/dx (ADC counts/cm)
0
2
4
6
8
10
12
0
200
400
600
800
1000
1200
XP
σ/μ = 19.10± 3.59 %
μ = 553.9± 14.8
σ = 105.8± 17.1
WF
σ/μ = 17.38± 4.25 %
μ = 547.7± 14.3
σ = 95.2± 20.8



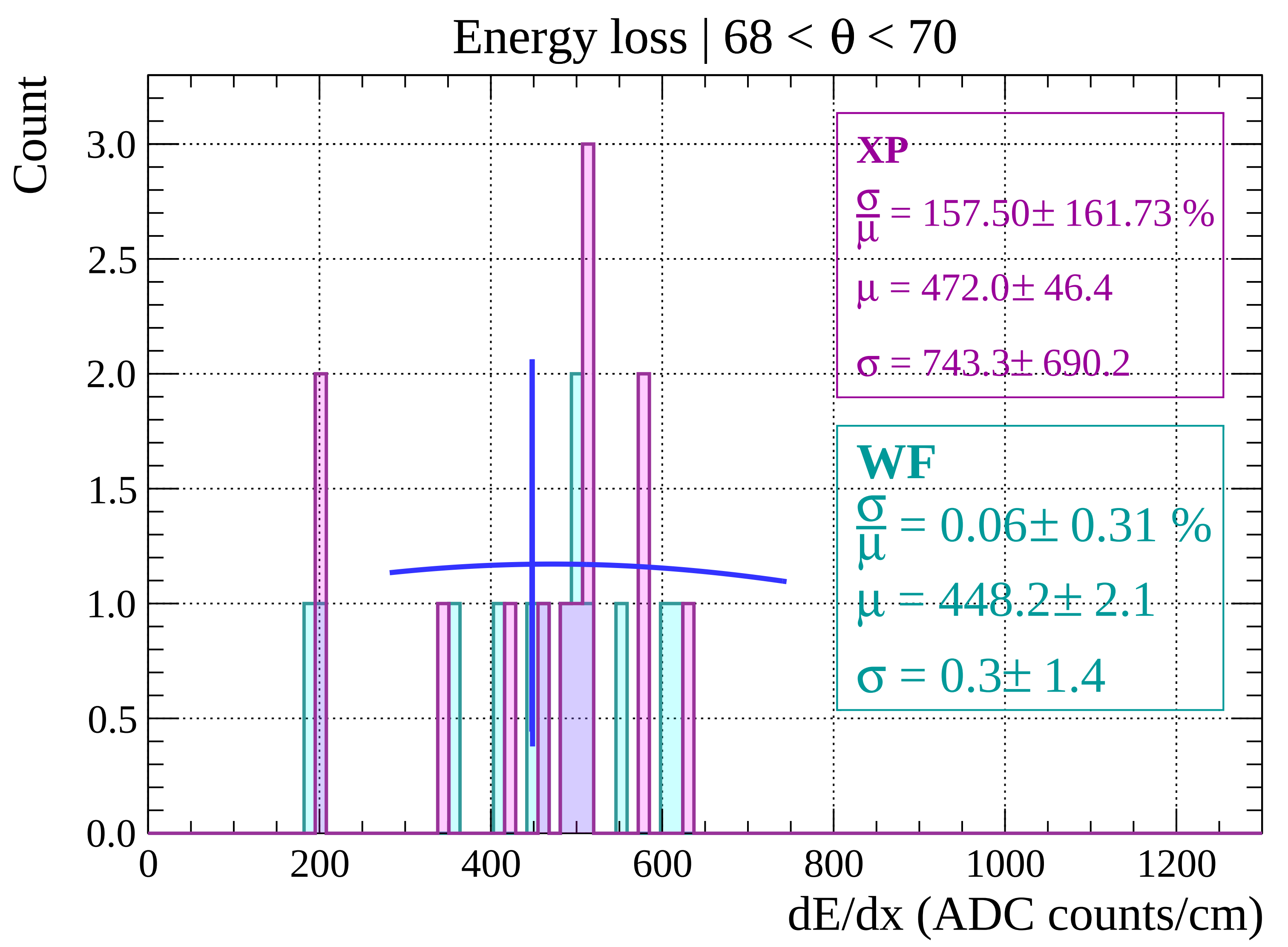
Energy loss | 62 < θ < 64
Count
dE/dx (ADC counts/cm)
0
1
2
3
4
5
6
0
200
400
600
800
1000
1200
XP
σ/μ = 26.75± 10.46 %
μ = 541.6± 32.5
σ = 144.9± 47.9
WF
σ/μ = 22.02± 7.19 %
μ = 535.3± 24.0
σ = 117.9± 33.2



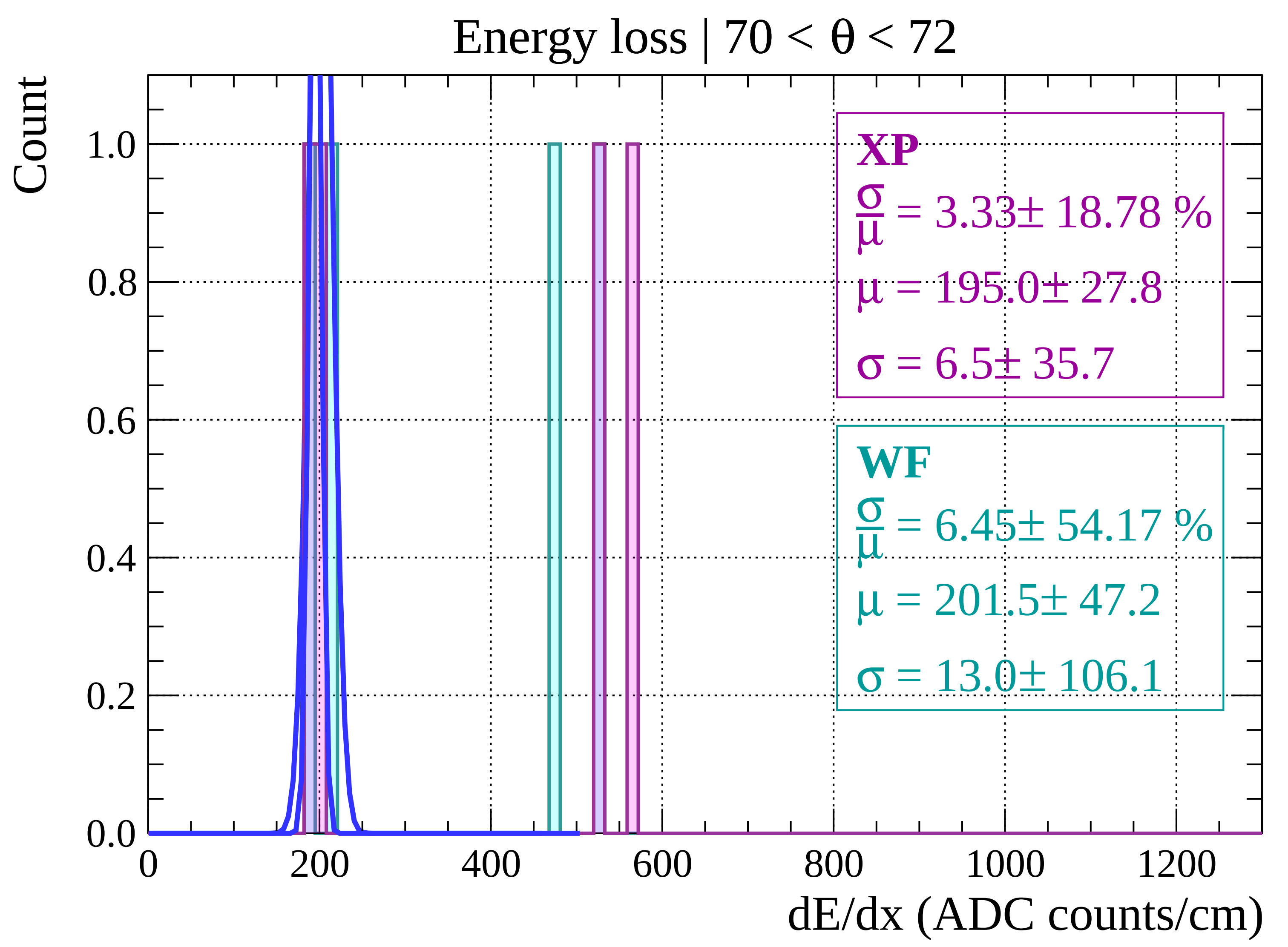
Energy loss | 64 < θ < 66
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
1
2
3
4
5
6
XP
σ/μ = 75.53± 14631.12 %
μ = 705.9± 135602.0
σ = 533.2± 849.7
WF
σ/μ = 1.03± 5.40 %
μ = 612.2± 102.9
σ = 6.3± 32.0

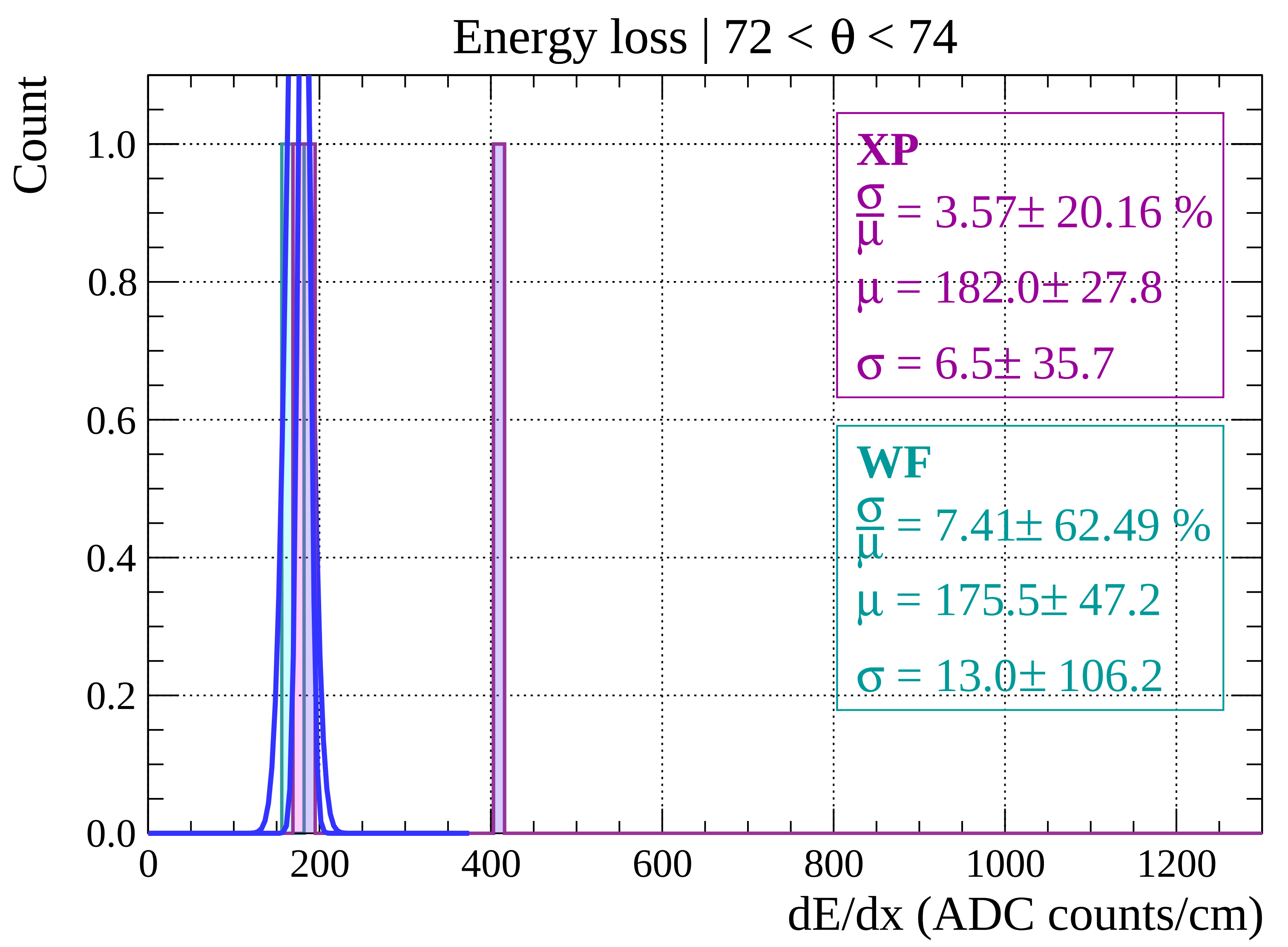


Energy loss | 66 < θ < 68
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 91.29± 274.34 %
μ = 471.0± 527.0
σ = 430.0± 811.0
WF
σ/μ = 125.71± 544.01 %
μ = 474.3± 1280.4
σ = 596.2± 970.5

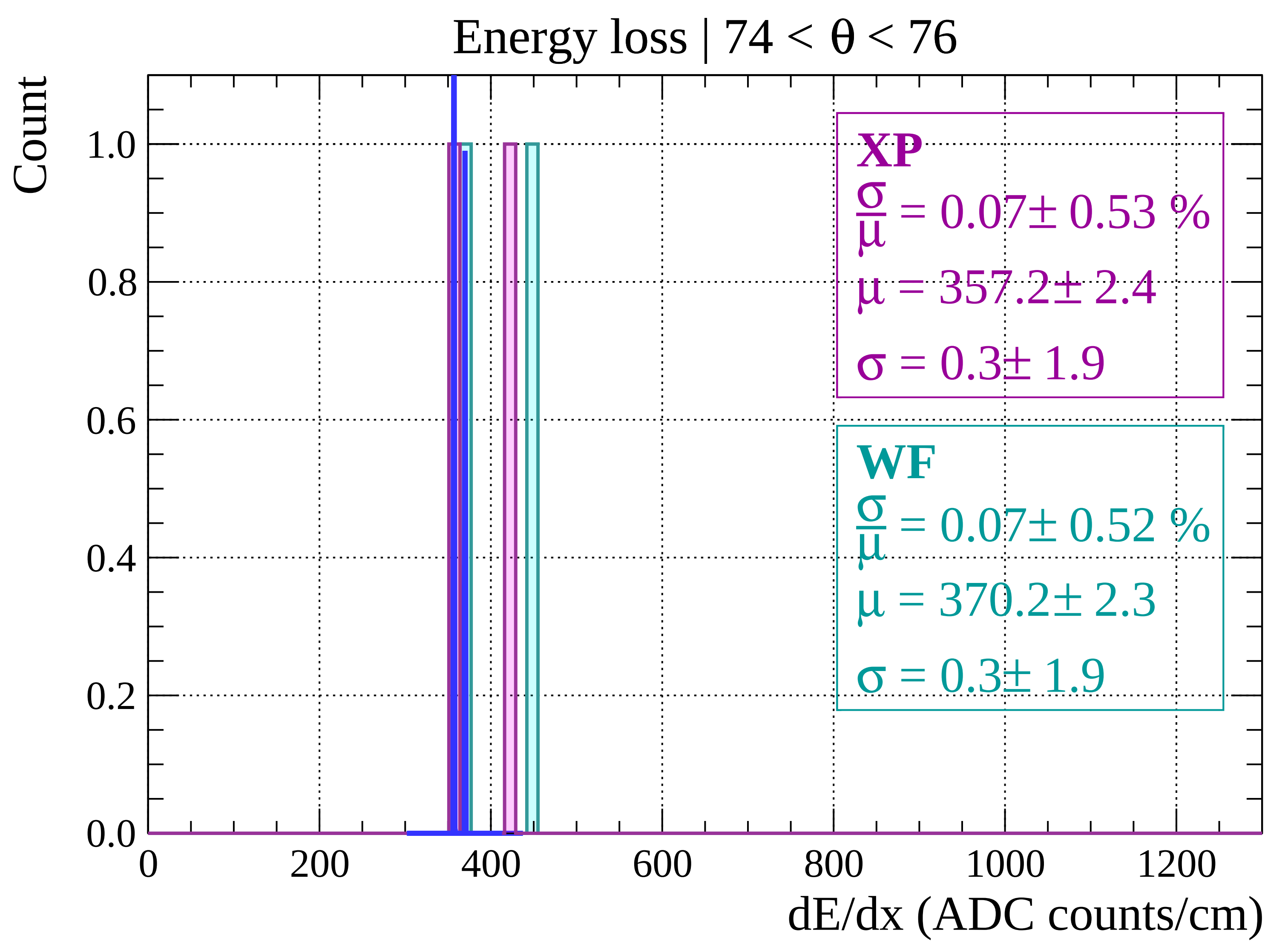


Energy loss | 68 < θ < 70
Count
3.0
2.5
2.0
1.5
1.0
0.5
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 157.50± 161.73 %
μ = 472.0± 46.4
σ = 743.3± 690.2
WF
σ/μ = 0.06± 0.31 %
μ = 448.2± 2.1
σ = 0.3± 1.4

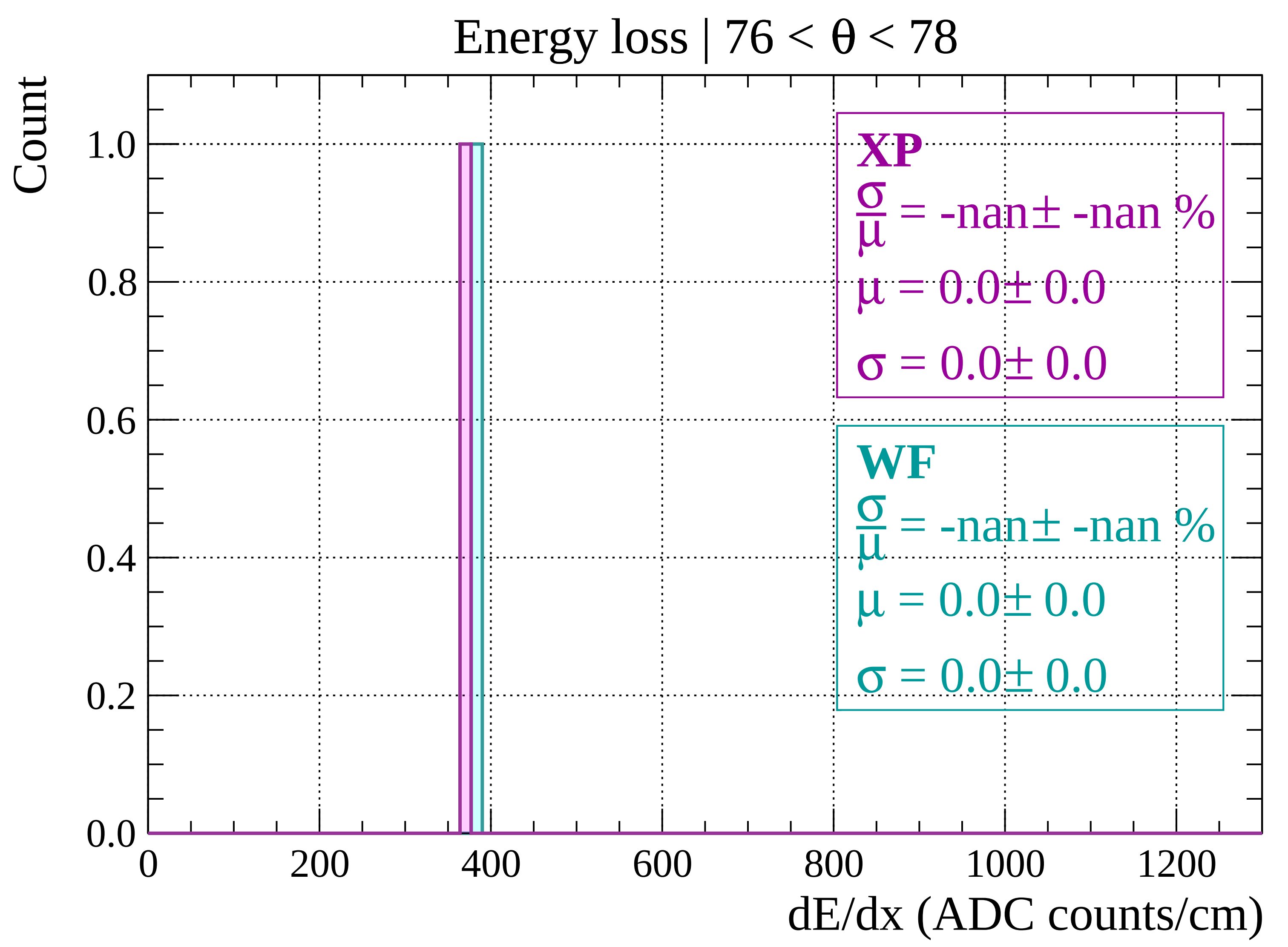

Energy loss | 70 < θ < 72
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0.0
0.2
0.4
0.6
0.8
1.0
XP
σ/μ = 3.33± 18.78 %
μ = 195.0± 27.8
σ = 6.5± 35.7
WF
σ/μ = 6.45± 54.17 %
μ = 201.5± 47.2
σ = 13.0± 106.1



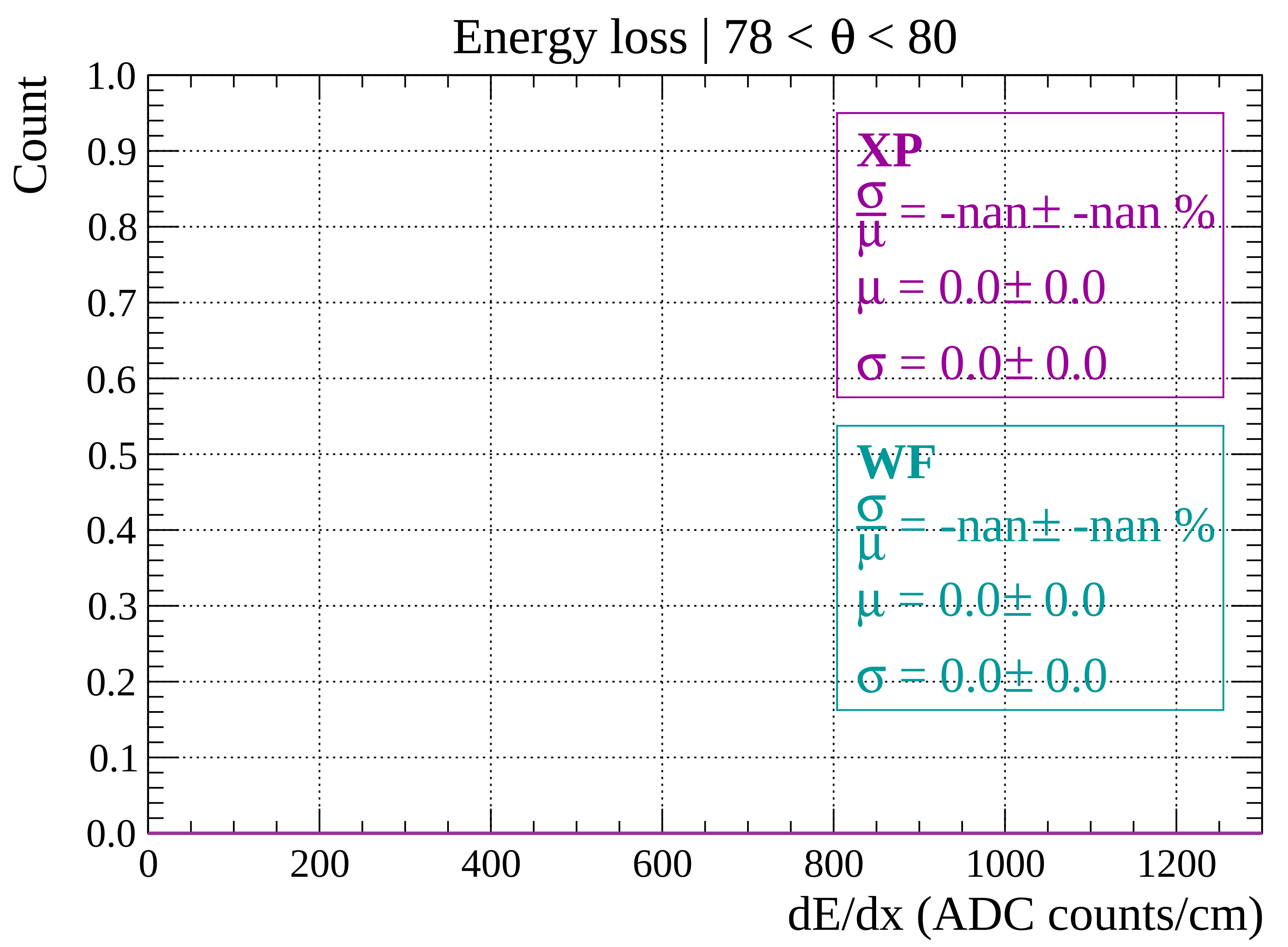
Energy loss | 72 < θ < 74
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 3.57± 20.16 %
μ = 182.0± 27.8
σ = 6.5± 35.7
WF
σ/μ = 7.41± 62.49 %
μ = 175.5± 47.2
σ = 13.0± 106.2
0.0
0.2
0.4
0.6
0.8
1.0
0
200
400
600
800
1000
1200

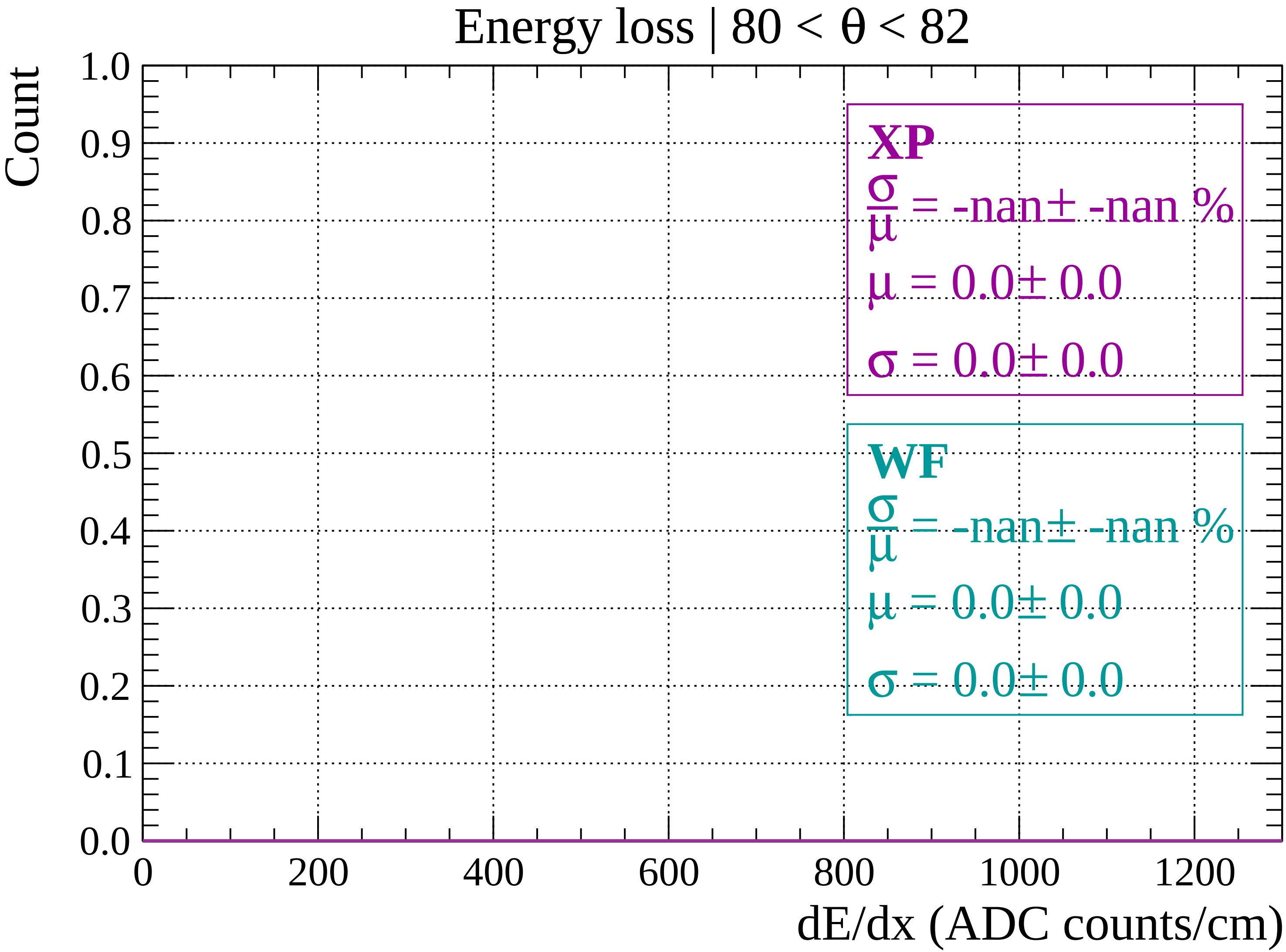


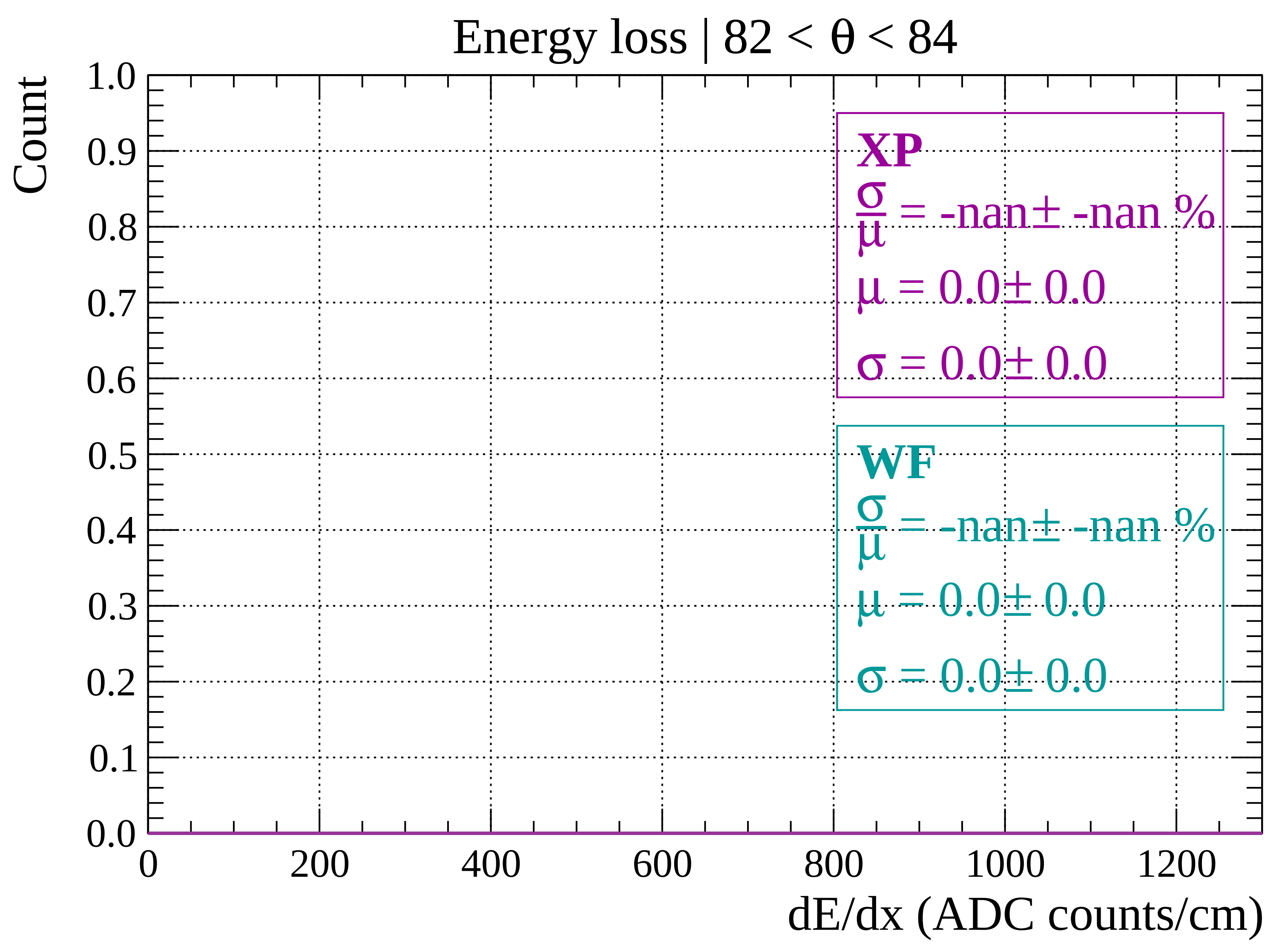
Energy loss | 74 < θ < 76
Count
dE/dx (ADC counts/cm)
0.0
0.2
0.4
0.6
0.8
1.0
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 0.07± 0.53 %
μ = 357.2± 2.4
σ = 0.3± 1.9
WF
$\frac{\sigma}{\mu}$ = 0.07± 0.52 %
μ = 370.2± 2.3
σ = 0.3± 1.9

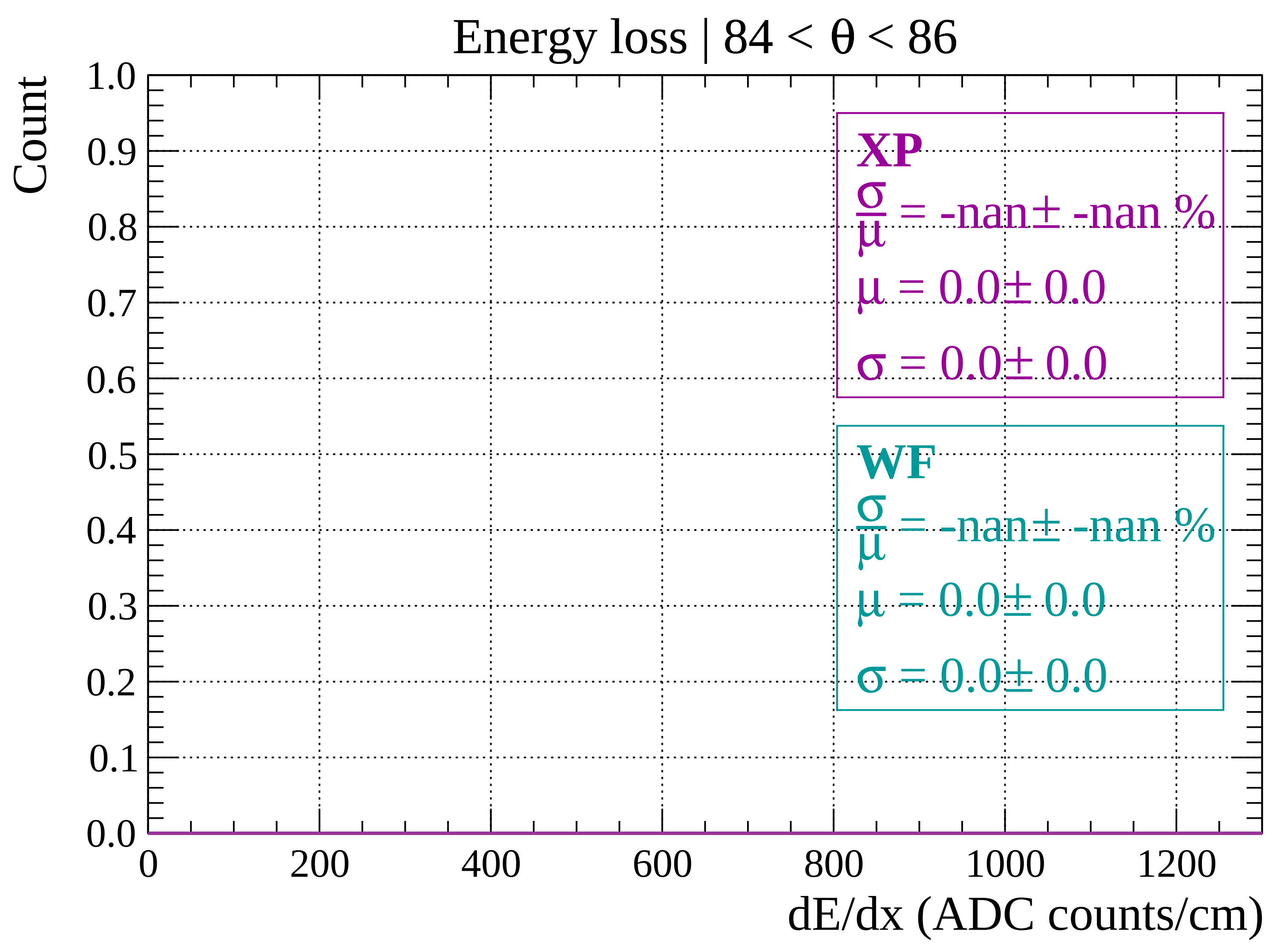


Energy loss | 76 < θ < 78
Count
1.0
0.8
0.6
0.4
0.2
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

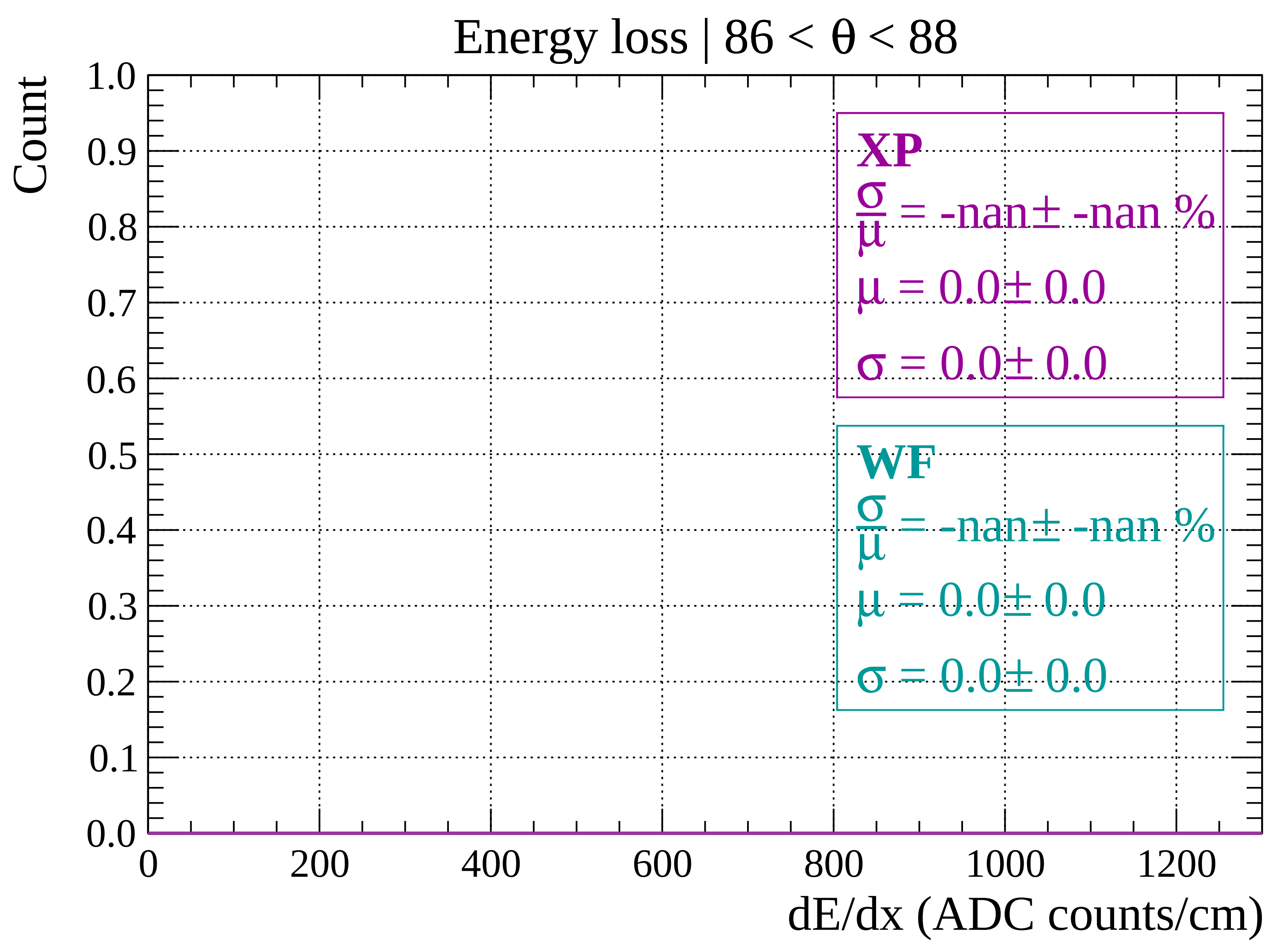

Energy loss | 78 < θ < 80
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

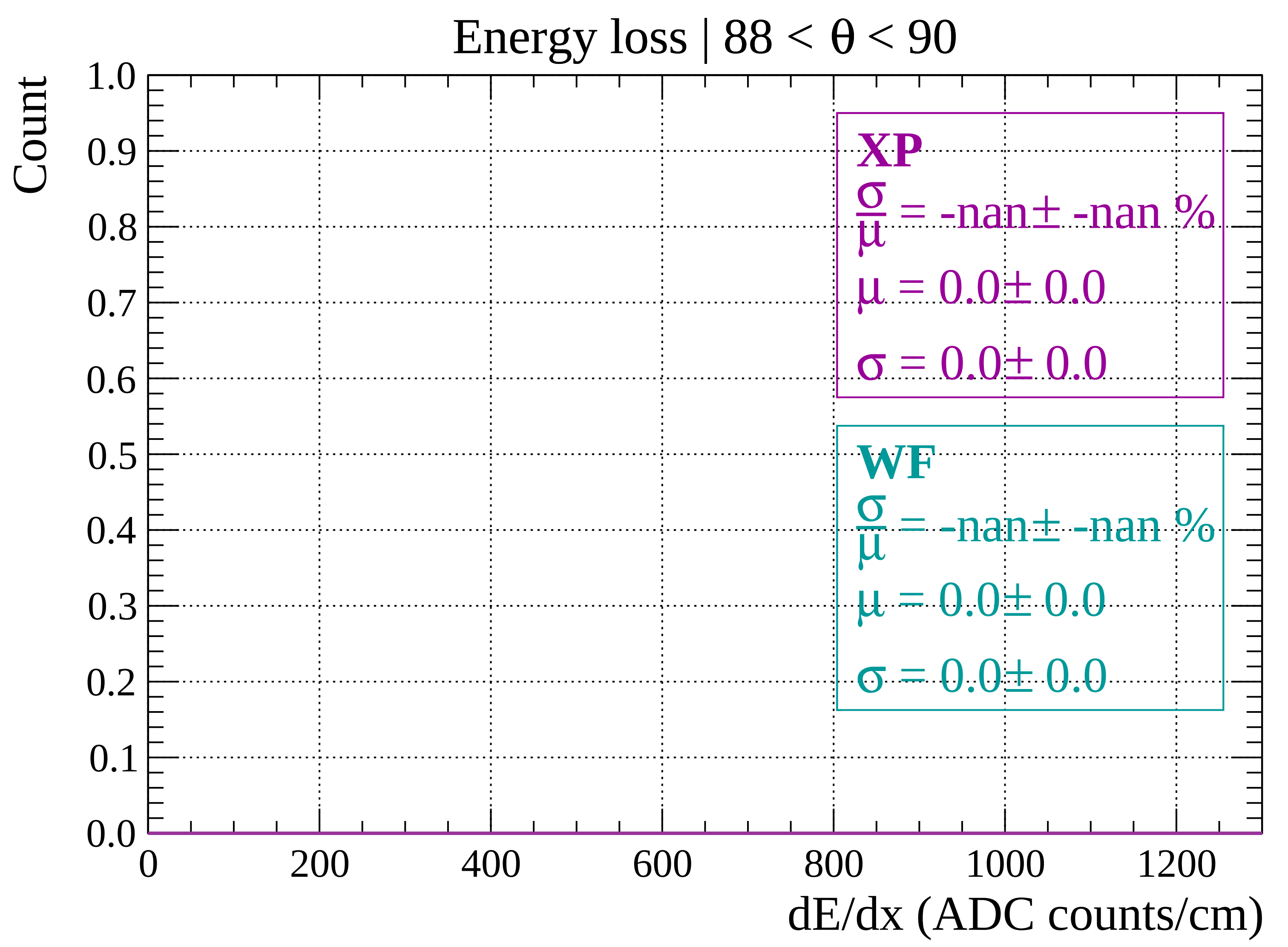

Energy loss | 80 < θ < 82
Count
dE/dx (ADC counts/cm)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
0
200
400
600
800
1000
1200
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

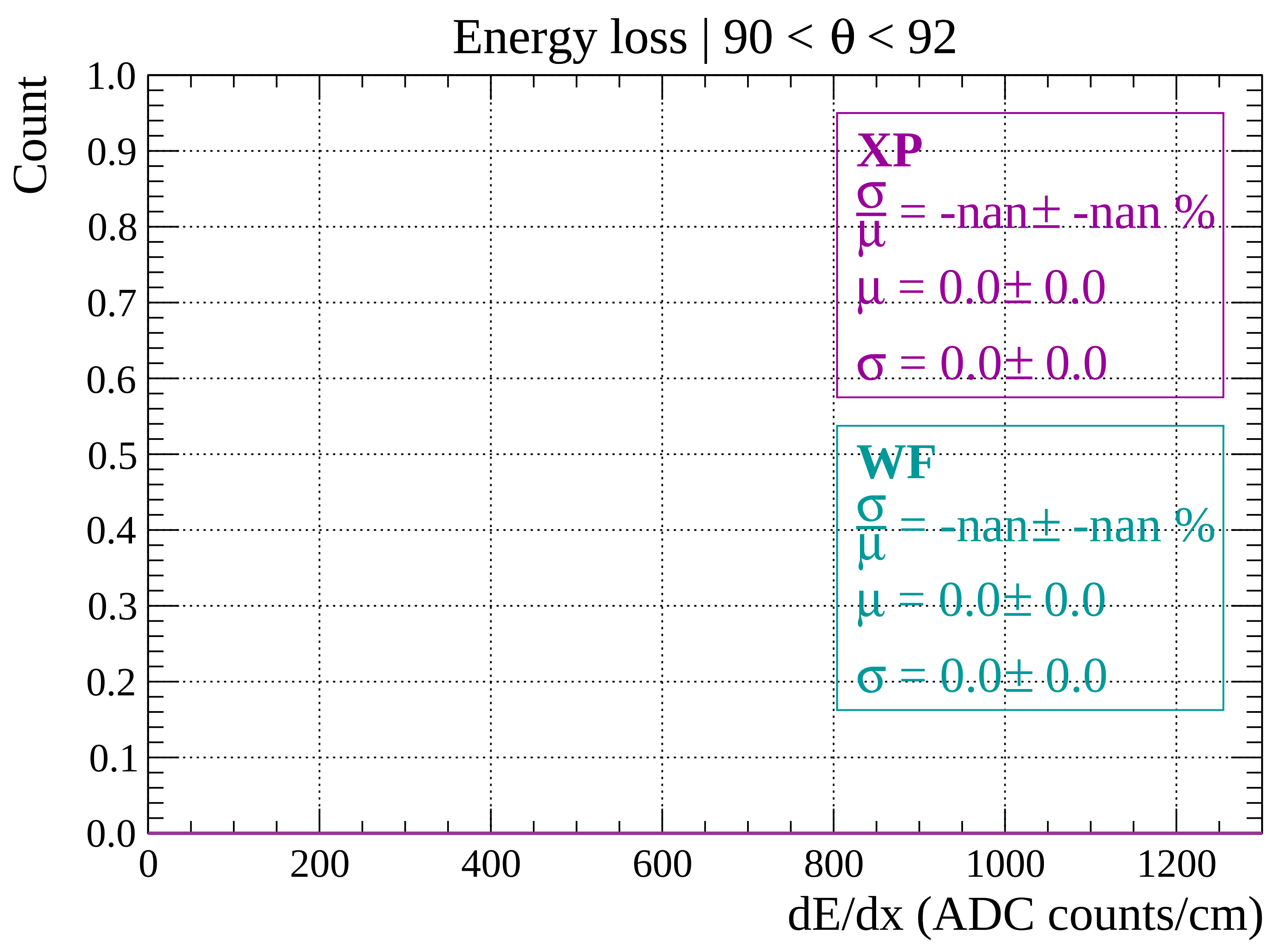

Energy loss | 82 < θ < 84
Count
dE/dx (ADC counts/cm)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
0
200
400
600
800
1000
1200
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

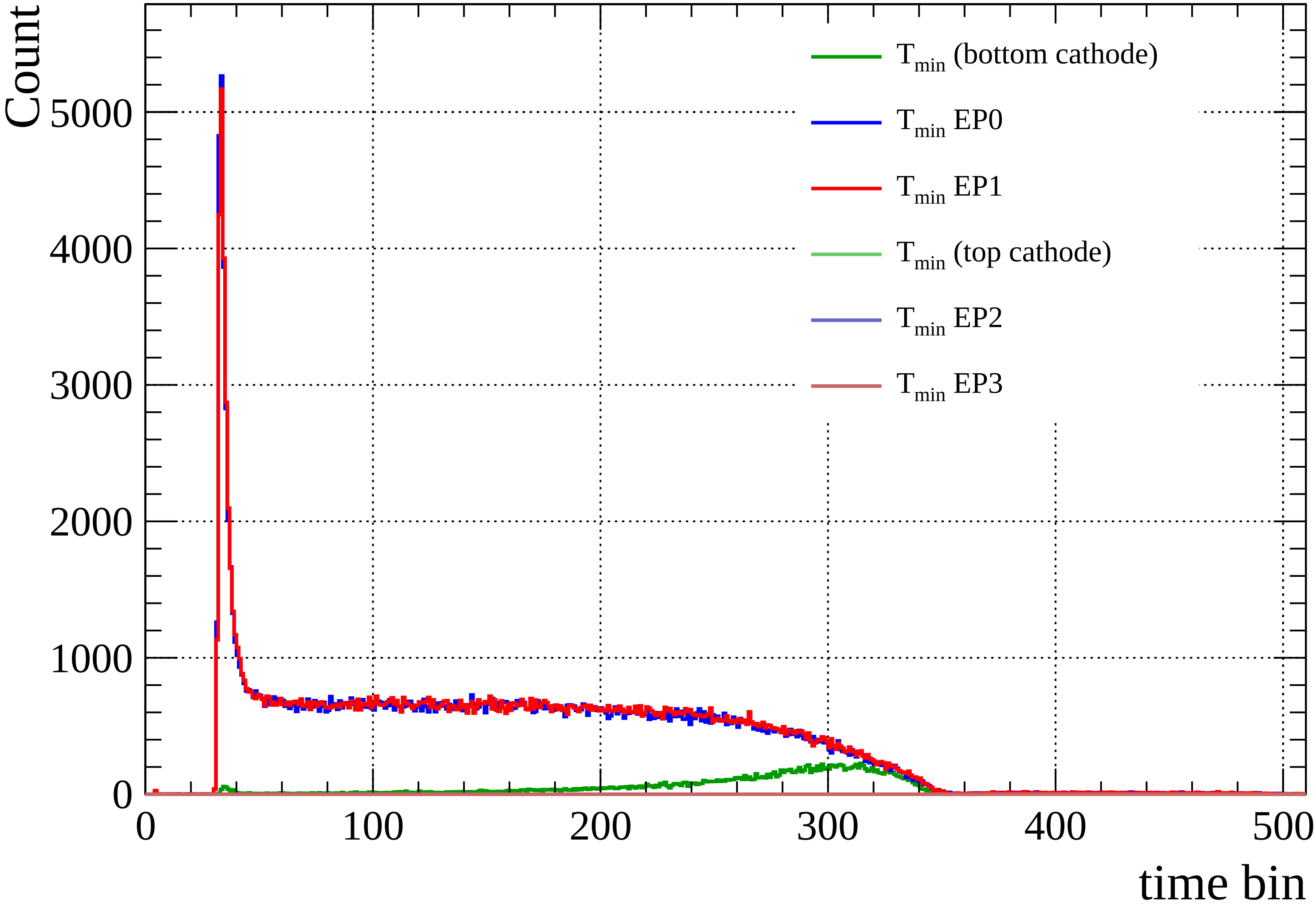


Energy loss | 84 < θ < 86
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

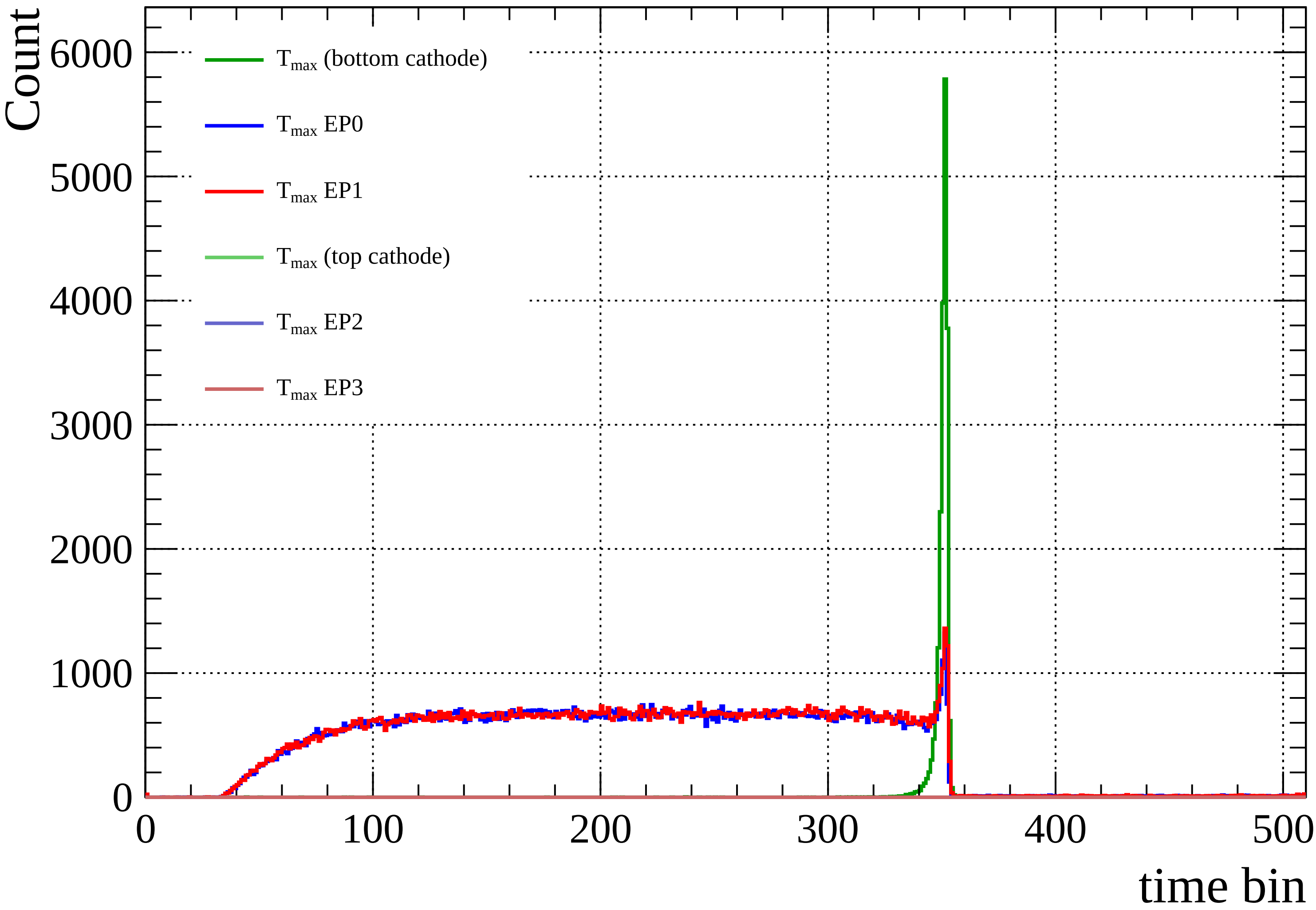

Energy loss | 86 < θ < 88
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0


Energy loss | 88 < θ < 90
Count
dE/dx (ADC counts/cm)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
0
200
400
600
800
1000
1200
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0


Energy loss | 90 < θ < 92
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



Count
5000
4000
3000
2000
1000
0
0
100
200
300
400
500
time bin
T_{min} (bottom cathode)
T_{min} EP0
T_{min} EP1
T_{min} (top cathode)
T_{min} EP2
T_{min} EP3


Count
time bin
0
1000
2000
3000
4000
5000
6000
0
100
200
300
400
500
T max (bottom cathode)
T max EP0
T max EP1
T max (top cathode)
T max EP2
T max EP3